Supplemental Material: Finite spectral triples as towerless intervals

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Contents

Part 03 **3**

Part 05 **10**

Part 06 **17**

Part 07 **23**

Part 08 **25**

Part 09 **33**

Part 10 **41**

Part 11 **42**

Part 12 **46**

Part 13 **55**

Part 14 **63**

Part 15 **75**

Part 16 **81**

Part 17 **86**

Part 18 **92**

Part 19 **105**

Part 20 **107**

Part 21 **119**

Part 22 **121**

Part 23 **127**

Part 24 **134**

Part 25 **147**

Part 26 **151**

Part 27 **158**

Part 28 **162**

Part 29 **170**

Part 30 **177**

Part 11D (README, referee pending) **182**

Part 11D (01_interval_and_eleventh_dimension, referee pending) **189**

Part 11D (02_witten_spaces_hypercharge_quotient, referee pending) **193**

Part 11D (03_branes_hypercharge_and_the_84, referee pending) **196**

Part 11D (04_two_families_m2_m5, referee pending) **200**

Part 11D (05_colour_family_exchange, referee pending) **204**

Part 11D (06_main_point_heterotic_route, referee pending) **208**

Part 11D (07_dualities_9_11, referee pending) **211**

Part 11D (08_orientifold_sign, referee pending) **214**

Part 11D (09_dilaton_spectral_and_so32, referee pending) **216**

Part 11D (10_dilaton_and_second_scale, referee pending) **220**

Part 11D (11_so30_fluctuations, referee pending) **224**

Part L (literature check and record of searches) **227**

Part R1 (assessment of Farsi-Latremoliere, with the counterexample to their Theorem 3.4) **241**

Part R2 (referee report, lemmas 08, 13, 14) **249**

Part R3 (referee report, lemmas 15, 16) **256**

Part R4 (referee report, lemma 18, two sheets, framework note) **263**

Part R5 (referee report, main draft and note) **273**

Part R6 (referee report, no-go results and lemma 19) **282**

Part R7 (referee report, lemmas 20 and 21, Proposition I) **288**

Part R8 (referee report, lemma 22) **294**

Part R9 (referee report, lemma 23) **300**

Part R10 (referee report, lemma 25) **306**

Part R11 (referee report, lemma 24) **310**

Part R12 (referee report, lemma 26) **314**

Part R13 (referee report, lemma 27 and additions to lemma 26) **320**

Part R14 (Astra validation of the ledger, citations and steer note) **324**

Part R15 (Terra referee, lemma 28) **336**

Part R16 (Terra referee, lemma 29) **341**

Part R17 (Terra referee, lemma 30 and the joint-limit note) **346**

Part R18 (assessment of the Astra audit) **352**

Part 03

Part A. Statement.

For a constant finite operator commuting with the gauge connection, the Yang–Mills term is

$$\boxed{S_{\mathrm{YM}} = -\frac{1}{24\pi^2}\int_M \sum_j f(m_j^2/\Lambda^2)\ \mathrm{tr}_{H_j}(F_{\mu\nu}F^{\mu\nu})\,dV.}$$

Here H_j is an eigenspace of D_F^2. The trace includes its entire finite multiplicity; the spin trace has already been taken.

Hypotheses.

Use Hermitian Euclidean gamma matrices, $\{\gamma^\mu,\gamma^\nu\} = 2g^{\mu\nu}$, and skew-Hermitian gauge curvature

$$F_{\mu\nu} = [\nabla_\mu + A_\mu, \nabla_\nu + A_\nu]_{\mathrm{gauge}}.$$

The spectral action is the literal trace on $L^2(M,S)\otimes H_F$, without a subsequent chirality or reality projection.

To distinguish the Laplace measure from the mass ratios, write

$$f(z) = \int_{(0,\infty)} e^{-sz}\,d\nu(s), \qquad 0 < \nu((0,\infty)) < \infty, \qquad \int (s^{-2}+s^2)\,d\nu(s) < \infty.$$

This class includes $f(z) = e^{-z}$. In particular, $f(0) < \infty$, $f(\infty) = 0$, and the spectral action is trace class.

Assume the finite-sector decomposition is fixed as Λ varies, A is smooth and independent of Λ, and

$$\partial_\mu D_F = 0, \qquad [A_\mu, D_F] = 0.$$

Proof.

Put $\mathcal{D} = D_{M,A}$, suppressing identity factors. Chirality satisfies

$$\{\mathcal{D},\gamma_5\} = 0, \qquad \gamma_5^2 = 1.$$

Consequently,

$$(\mathcal{D} + \gamma_5 D_F)^2 = \mathcal{D}^2 + D_F^2 + \gamma_5[D_F,\mathcal{D}] = \mathcal{D}^2 + D_F^2.$$

The two summands strongly commute. Since A preserves every eigenspace H_j of D_F^2,

$$\boxed{\mathrm{Tr}\,e^{-tD_A^2} = \sum_j e^{-tm_j^2}\,\mathrm{Tr}\,e^{-tD_{M,A_j}^2}.}$$

Let $P_j = D^2_{M,A_j}$, and use integrated heat coefficients:

$$\operatorname{Tr} e^{-tP_j} = t^{-2}a_0(P_j) + t^{-1}a_2(P_j) + a_4(P_j) + R_j(t).$$

For fixed smooth backgrounds, $|R_j(t)| \le Ct$ for all $t > 0$: the small-t estimate is the heat expansion, and the large-t estimate follows from $P_j \ge 0$.

Define

$$x_j = m_j^2/\Lambda^2, \qquad \Phi_k(x) = \int s^{-k} e^{-sx}\, d\nu(s).$$

Integration of the exact factorization gives

$$S = \sum_j \left[\Lambda^4 \Phi_2(x_j) a_0(P_j) + \Lambda^2 \Phi_1(x_j) a_2(P_j) + f(x_j) a_4(P_j)\right] + O(\Lambda^{-2}).$$

The remainder is bounded by

$$C\Lambda^{-2} \int s\, d\nu(s),$$

uniformly for all $x_j \ge 0$. It is also uniform over compact smooth families of connections. Thus fixed mass ratios require no expansion in m_j/Λ.

It remains to compute the gauge part of a_4. With $\gamma^{\mu\nu} = \frac{1}{2}[\gamma^\mu, \gamma^\nu]$, the gauge-dependent endomorphism in the convention $P = -(\nabla^2 + E)$ is

$$E_F = \tfrac{1}{2}\gamma^{\mu\nu} F_{\mu\nu}.$$

The relevant terms are

$$a_4(P)\big|_{F^2} = (4\pi)^{-2} \int_M \operatorname{tr}_{S,F} \left(\frac{1}{12} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} E_F^2\right) dV.$$

Using

$$\operatorname{tr}_S 1 = 4, \qquad \operatorname{tr}_S(\gamma^{\mu\nu}\gamma^{\rho\sigma}) = 4(g^{\mu\sigma}g^{\nu\rho} - g^{\mu\rho}g^{\nu\sigma}),$$

the two contributions are $+\frac{1}{3}\operatorname{tr}_F F^2$ and $-\operatorname{tr}_F F^2$. Hence

$$a_4(P)\big|_{F^2} = -\frac{1}{24\pi^2} \int_M \operatorname{tr}_F F^2\, dV.$$

This proves the statement.

For $F_{\mu\nu} = F^a_{\mu\nu} T_a$, with skew-Hermitian T_a, the coefficient matrix is therefore

$$-\frac{1}{24\pi^2} \sum_j f(x_j) \operatorname{tr}_{H_j}(T_a T_b).$$

It is positive semidefinite. The normalization is per full four-component spin Dirac operator, with the finite representation traced explicitly.

The three regimes are:

Heavy mass ratio	Weight	Consequence for the kinetic term
$m_h/\Lambda \to 0$	$f(0)$	Full inherited finite trace
$m_h/\Lambda \to c \in (0,\infty)$	$f(c^2)$	Partial suppression
$m_h/\Lambda \to \infty$	0	Trace over the light sectors

Absolute divergence $m_h \to \infty$ does not distinguish these regimes.

Part B. Statement.

Write

$$\ell = f(0), \qquad h = f(\mu^2).$$

For N generations with exactly one heavy up-type quark, the coefficients on the full finite space are

$$\boxed{\begin{aligned}\kappa_{\rm em}^{(N)} &= \frac{1}{24\pi^2}\frac{16}{3}[(2N-1)\ell + h],\\ \kappa_s^{(N)} &= \frac{1}{24\pi^2}2[(2N-1)\ell + h].\end{aligned}}$$

Here $S_{\rm YM} = \int (\kappa_{\rm em} F_{\rm em}^2 + \kappa_s \sum_a G_a^2)\, dV$; there is no $1/4$ in this definition.

Thus

$$\boxed{\frac{\kappa_{\rm em}^{(N)}(\mu)}{\kappa_{\rm em}^{(N)}(0)} = \frac{\kappa_s^{(N)}(\mu)}{\kappa_s^{(N)}(0)} = \frac{(2N-1)f(0) + f(\mu^2)}{2Nf(0)}.}$$

Hypotheses.

Use $Q = T_3 + Y$ and the following particle states:

Multiplet	Number of states	Y	T_3	Q
(u_L, d_L)	6	$1/6$	$+1/2, -1/2$	$2/3, -1/3$
u_R	3	$2/3$	0	$2/3$
d_R	3	$-1/3$	0	$-1/3$
(ν_L, e_L)	2	$-1/2$	$+1/2, -1/2$	$0, -1$
ν_R	1	0	0	0
e_R	1	-1	0	-1

Charge conjugates double this to the standard 32-dimensional finite space. For color, use Hermitian generators with

$$\mathrm{tr}_{\mathbf{3}}(t_a t_b) = \tfrac{1}{2}\delta_{ab}.$$

All three colors of the selected up-type quark have the same Dirac mass. Photon and gluon generators commute with these masses.

Proof.

On the particle space, both chiralities contribute

	u	d	e
$\mathrm{tr}\, Q^2$	8/3	2/3	2
color index	1	1	0.

Therefore, for one generation,

$$K_{\rm em}^{\rm particle} = \frac{8}{3}(h + \ell), \qquad K_s^{\rm particle} = h + \ell.$$

Charge conjugation doubles both expressions. Each additional light generation adds $32\ell/3$ and 4ℓ, respectively, on the full finite space. Part A now gives the stated coefficients.

In particular:

Heavy sector	Both kinetic-coefficient ratios	$\mu \to \infty$
Up quark in one generation	$\dfrac{f(0)+f(\mu^2)}{2f(0)}$	1/2
Top only in three generations	$\dfrac{5f(0)+f(\mu^2)}{6f(0)}$	5/6

The common suppression implies

$$\boxed{\kappa_{\mathrm{em}}/\kappa_s = 8/3}$$

throughout this particular family. The absolute kinetic form changes, while its photon-to-gluon ratio remains unchanged.

The same charge table gives, on the particle space,

$$K_W = 3 \cdot \tfrac{1}{2} + \tfrac{1}{2} = 2,$$

$$K_Y = 6(1/6)^2 + 3(2/3)^2 + 3(1/3)^2 + 2(1/2)^2 + 1 = \frac{10}{3},$$

reproducing the conventions in section 14.3.

Part C. Statement.

For broken generators, a mass-dependent "F^2 coefficient" requires a prescription for separating gauge and Higgs derivative terms. I use the **integrated quadratic transverse kinetic coefficient**: expand about zero gauge field with D_F held constant, remove the gauge-field mass term, and extract the coefficient of $q^2|a_T(q)|^2$. Since

$$F_{\mu\nu}(q)^* F_{\mu\nu}(q) = 2q^2|a_T(q)|^2,$$

this specifies the quadratic F^2 coefficient, including all Higgs contributions to it.

For a vectorial doublet with nonnegative masses m_1, m_2, put

$$x_i = m_i^2/\Lambda^2, \quad z(u) = (1-u)x_1 + ux_2, \quad b(u) = u(1-u).$$

The normalized weight is

$$\boxed{w_f(x_1,x_2) = \frac{3}{2}\int_0^1 \left[(1-2b)f(z) + b(\sqrt{x_1}-\sqrt{x_2})^2 f'(z)\right] du.}$$

It is symmetric and satisfies $w_f(x,x) = f(x)$.

For the heat cutoff,

$$\boxed{w_{\exp}(x_1,x_2) = \frac{3}{2}\int_0^1 e^{-z(u)}\left[1-(2+(\sqrt{x_1}-\sqrt{x_2})^2)b(u)\right] du.}$$

This is generally more complicated than a divided difference of an antiderivative.

Hypotheses and scope.

Compute the local, gravity-independent coefficient in flat coordinates, with fixed m_i/Λ and $q/\Lambda \to 0$. The result concerns the quadratic action; longitudinal derivative terms and the full covariant Higgs-operator expansion remain distinct.

For an odd finite operator, the vectorial model can be realized as

$$D_F = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}, \qquad M = \mathrm{diag}(m_1, m_2), \qquad T = \begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix}.$$

The Standard Model instead has a left doublet and right singlets. That representation changes the answer, as calculated below.

This distinction is necessary: prescribing D_F^2 alone does not prescribe $[T, D_F]$. Moreover,

$$[\nabla_\mu, \nabla_\nu] D_F = [F_{\mu\nu}, D_F],$$

so a noncommuting curvature cannot coexist with a covariantly constant finite operator. A separate "pure curvature" coefficient obtained by discarding Higgs derivative operators would require an additional operator-basis convention.

Proof by Duhamel expansion.

Without the commuting assumption,

$$\boxed{D_A^2 = D_{M,A}^2 + D_F^2 + i\gamma^\mu\gamma_5[A_\mu, D_F].}$$

The last term is essential at fixed D_F/Λ.

First take $f(z) = e^{-z}$, put $\tau = \Lambda^{-2}$, and diagonalize

$$D_F = \mathrm{diag}(\lambda_i).$$

Use $A_\mu = i\, a_\mu T$, with T Hermitian. The free squared operator is

$$P_0 = -\partial^2 + D_F^2.$$

The linear vertex between internal states j and i, carrying momentum q, is

$$(V_1)_{ij} = T_{ij}\left[(2p+q)\cdot a + \gamma^{\mu\nu} q_\mu a_\nu + (\lambda_i - \lambda_j)\gamma^\mu\gamma_5 a_\mu\right].$$

The quadratic vertex has no external momentum and therefore makes no contribution to the q^2 coefficient.

The second-order heat trace is

$$\frac{\tau^2}{2}\int_0^1 \mathrm{Tr}\,\left(e^{-(1-u)\tau P_0} V_1 e^{-u\tau P_0} V_1\right) du,$$

together with the quadratic-vertex term. Completing the Gaussian square gives the factor

$$(4\pi\tau)^{-2} e^{-\tau[(1-u)\lambda_i^2 + u\lambda_j^2 + bq^2]}.$$

For $q\cdot a = 0$, the Gaussian-averaged spin trace of the vertex product is

$$4\left[\frac{2}{\tau} + (\lambda_i - \lambda_j)^2 + q^2\right]|a|^2.$$

Expanding to order q^2, and writing $r_i = \lambda_i/\Lambda$, yields

$$\frac{q^2|a|^2}{8\pi^2}\sum_{ij}|T_{ij}|^2\int_0^1 e^{-[(1-u)r_i^2 + u r_j^2]}\left[1 - 2b - b(r_i - r_j)^2\right] du.$$

Comparison with $F^2 = 2q^2|a|^2$ and Part A's normalization proves the heat-cutoff formula.

Integration over the Laplace measure replaces

$$e^{-z} \mapsto f(z), \qquad -se^{-sz} \mapsto f'(z),$$

giving the stated general formula. The moment assumptions justify these operations. The small-momentum expansion is uniform for mass ratios in compact sets.

For a vectorial positive-mass pair, the two relevant signed-mass branches both have $(r_i - r_j)^2 = (\sqrt{x_1} - \sqrt{x_2})^2$. At equal masses, that term vanishes and

$$\frac{3}{2}\int_0^1 (1 - 2u(1-u))\,du = 1,$$

which recovers Part A.

The divided difference does occur as

$$\int_0^1 f(z(u))\,du = \frac{\mathcal{F}(x_1) - \mathcal{F}(x_2)}{x_1 - x_2}, \qquad \mathcal{F}' = f.$$

The orbital vertices and finite-operator commutator supply the additional terms in w_f.

Standard Model doublet.

For the left-handed weak generator,

$$T = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}, \qquad [T, D_F] = \begin{pmatrix} 0 & tM \\ -Mt & 0 \end{pmatrix}.$$

For an off-diagonal t, summing these matrix elements replaces

$$(\sqrt{x_1} - \sqrt{x_2})^2 \quad \text{by} \quad x_1 + x_2.$$

Thus

$$w_L(x_1, x_2) = \frac{3}{2}\int_0^1 e^{-z(u)}[1 - (2 + x_1 + x_2)b(u)]\,du.$$

In particular,

$$w_L(x, x) = e^{-x}(1 - x/2).$$

Equal masses do not make a left-only weak generator commute with D_F, because its right partners are weak singlets. The conjectured diagonal identity therefore applies to the vectorial model, but not to this chiral representation.

For a heavy up quark and massless down quark, both models give

$$\boxed{w(x) := w_L(x, 0) = 3\int_0^1 u^2 e^{-xu}\,du = \frac{3[2 - (x^2 + 2x + 2)e^{-x}]}{x^3}, \qquad x = \mu^2.}$$

The value at zero is defined continuously. In particular,

$$w(x) = 1 - \frac{3}{4}x + \frac{3}{10}x^2 + O(x^3), \qquad w(x) \sim \frac{6}{x^3} \quad (x \to \infty).$$

The divided-difference candidate $(1 - e^{-x})/x$ already disagrees at first order.

Weak-angle consequences.

Use particle traces below; charge conjugation doubles the entire kinetic matrix. Let $N = 1$ or 3, with one heavy up-type quark, $x = \mu^2$, and $r = e^{-x}$. The charged and neutral transverse coefficients are

$$\begin{aligned} K_W &= 2N - \frac{3}{2} + \frac{3}{2}w(x), \\ K_{33} &= 2N - \frac{3}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{10N}{3} - \frac{17}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}$$

The neutral action contains $K_{33}F_3^2 + 2K_{3Y}F_3F_Y + K_{YY}F_Y^2$.

For completeness, a Dirac species of mass ratio squared x, with neutral charges q_L^a, q_R^a, contributes

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right].$$

Applying this formula to the three up colors proves the neutral entries above. The commutator term is responsible for the terms proportional to xr.

At $x = 0$,

$$K_W = K_{33} = 2N, \qquad K_{YY} = 10N/3, \qquad K_{3Y} = 0,$$

so the usual matching angle is $5/8$.

For $x > 0$, generally $K_W \neq K_{33}$ and $K_{3Y} \neq 0$. Consequently, the usual electroweak relations no longer define one common weak angle. Two precise continuations are:

1. **Continue the charged-W/hypercharge matching prescription.** Define

$$\boxed{\cos^2\theta_{\text{match}}(x) := \frac{K_{YY}}{K_W + K_{YY}} = \frac{80N - 34 + (34 - 9x)e^{-x}}{128N - 70 + (34 - 9x)e^{-x} + 36w(x)}.}$$

   Then

$$\cos^2\theta_{\text{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\text{match}}(\infty) = \begin{cases} 23/29, & N = 1, \\ 103/157, & N = 3. \end{cases}$$

   This prescription alone does not diagonalize the neutral kinetic terms.

2. **Define the angle from the neutral current.** When the neutral mass term has the Standard Model direction $(W_3 - B)^2$, the photon direction is $(1, 1)$. Its kinetic-orthogonal massive direction is proportional to

$$(K_{YY} + K_{3Y},\ -K_{33} - K_{3Y}).$$

   Writing its current as $T_3 - \sin^2\theta_{\text{NC}} Q$ gives

$$\cos^2\theta_{\text{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}}.$$

   Here

$$K_{33} + K_{3Y} = 2N - 1 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(2N - 1 + r),$$

   and therefore

$$\boxed{\cos^2\theta_{\text{NC}}(x) = 5/8.}$$

   This equality is special to the chosen heavy-up family. It does not restore a common $SU(2)$ kinetic coefficient or the usual relations among charged and neutral couplings.

As a consistency check,

$$K_{33} + 2K_{3Y} + K_{YY} = \frac{8}{3}(2N - 1 + e^{-x}),$$

exactly reproducing Part B's photon coefficient on the particle space.

Verdict on corpus claim.

Item 1 states: "The centralizer and compressed resolvent do not determine the surviving gauge kinetic terms." The calculation confirms that qualification.

- **The full inherited kinetic form survives** the joint limit $m_h \to \infty$, $\Lambda \to \infty$, $m_h/\Lambda \to 0$.

- **Finite nonzero ratios produce threshold weights.** When $m_h/\Lambda \to \infty$, the residual kinetic form is the light-sector trace. For the specified up-quark removals, photon and gluon coefficients retain their ratio but acquire the common reductions 1/2 or 5/6.
-

The corpus's $5/8$ is a valid electroweak matching condition when the full module is light relative to the matching cutoff, or when that condition is imposed before taking the heavy limit. The broken action does not retain one universal weak angle. Its neutral-current angle happens to remain $5/8$ in this family, whereas the continued charged-W/hypercharge matching ratio changes as above. At the strict $SU(3)\times U(1)_{\mathrm{em}}$ endpoint, a weak angle requires reference to such an earlier matching prescription.

Independent checks (Claude, 2 Nov 2026)

- Part A: (1/12) tr_S 1 + (1/2)(1/4) tr_S(gamma^{mu nu} gamma^{rho sigma}) F F gives (1/3 - 1) tr F^2 = -(2/3) tr F^2. Dividing by 16 pi^2 gives -1/(24 pi^2), the standard spectral-action Yang-Mills coefficient.
- Part B: the per-generation traces tr Q^2 = 8/3 (u), 2/3 (d), 2 (e) and color index 1, 1, 0 were rechecked by hand.
- Part C: w_L(x, 0) = 3 int_0^1 u^2 e^{-xu} du = 3[2 - (x^2+2x+2) e^{-x}]/x^3 agrees by quadrature to 1e-10 at x = 0, 0.3, 1, 4, 12. The limits cos^2 theta_match(infinity) = (80N-34)/(128N-70) give 23/29 (N=1) and 103/157 (N=3). The identity (K_YY + K_3Y) = (5/3)(K_33 + K_3Y), hence cos^2 theta_NC = 5/8 for every x, was verified algebraically.

Part 05

The geometric identification exists: the minimal grading-preserving spectral truncation of a suitable chiral interval is exactly Connes–Lott, with

$$|m| = \frac{\pi}{2l}, \qquad d_{\mathrm{trunc}} = \frac{2l}{\pi}.$$

The truncated endpoint states are smeared states on the original interval. Moreover, this interval has no zero mode, which limits the physical interpretation of both extremes.

Part 1. Deconstructed chain

Statement. There is a natural family with opposite endpoint chiralities, an exact Connes–Lott member at $N = 2$, and endpoint distance converging to l. The answer depends on the representation of the site algebra; the familiar lattice-distance anomaly is not universal.

Hypotheses. We consider complex even spectral triples, without imposing additional real-structure or KO-dimension axioms. Let $N \geq 2$, $n = N - 1$, and

$$A_N = \mathbb{C}^N, \qquad H_N = \mathrm{span}\{L_1, \dots, L_{N-1}, R_2, \dots, R_N\}.$$

Thus interior sites carry both components, site 1 only L, and site N only R. Define

$$\pi_N(a) = \mathrm{diag}(a_1, \dots, a_{N-1}, a_2, \dots, a_N), \qquad \Gamma_N = \mathrm{diag}(I_n, -I_n).$$

If S_n is the lower shift, set

$$B_N = h_N(I_n - S_n), \qquad D_N = \begin{pmatrix} 0 & B_N \\ B_N^* & 0 \end{pmatrix}.$$

These are forward/backward differences:

$$(D_N\psi)_{L_i} = h_N(R_{i+1} - R_i), \qquad (D_N\psi)_{R_j} = h_N(L_{j-1} - L_j),$$

with the missing components interpreted as

$$R_1 = 0, \qquad L_N = 0.$$

There are nearest-neighbour couplings and onsite L-R terms at interior sites.

For compatibility with the spectral truncation in Part 3, choose

$$\theta_N = \frac{\pi}{4N-2}, \qquad h_N = \frac{\pi}{4l\sin\theta_N} \sim \frac{N}{l}. \tag{1}$$

This fixes the smallest positive eigenvalue to $\pi/(2l)$ at every N.

Proof.

The representation is faithful, $[\Gamma_N, \pi_N(a)] = 0$, and $\Gamma_N D_N = -D_N\Gamma_N$. At $N = 2$,

$$A_2 = \mathbb{C}^2, \quad H_2 = \mathbb{C}^2, \quad D_2 = \begin{pmatrix} 0 & h_2 \\ h_2 & 0 \end{pmatrix}, \quad h_2 = \frac{\pi}{2l}.$$

This is exactly the stated two-point triple; a diagonal unitary supplies any desired phase of m.

Ordering the basis as

$$L_1, R_2, L_2, R_3, \dots, L_{N-1}, R_N$$

makes D_N a path adjacency matrix with alternating signs. A diagonal unitary removes those signs. The usual sine eigenvectors of a path therefore give

$$\operatorname{spec}(D_N) = \left\{ 2h_N \cos\frac{j\pi}{2N-1} : 1 \le j \le 2N-2 \right\} = \{\pm\lambda_{N,r} : 1 \le r \le N-1\},$$

where

$$\boxed{\lambda_{N,r} = 2h_N \sin((2r-1)\theta_N).} \tag{2}$$

Consequently,

$$\boxed{K_N(t) = \operatorname{Tr} e^{-tD_N^2} = 2\sum_{r=1}^{N-1} e^{-4th_N^2 \sin^2((2r-1)\theta_N)}.} \tag{3}$$

For a real site function a, the onsite terms commute with $\pi_N(a)$, while

$$B_N\pi_R(a) - \pi_L(a)B_N = h_N \operatorname{diag}(a_2 - a_1, \dots, a_N - a_{N-1}).$$

Hence

$$\|[D_N, \pi_N(a)]\| = h_N \max_i |a_{i+1} - a_i|.$$

The triangle inequality bounds $|a_j - a_i|$ by $|j-i|/h_N$; the function $a_i = (i-1)/h_N$ attains this bound. Thus

$$\boxed{d_N(i,j) = \frac{|i-j|}{h_N}.} \tag{4}$$

In particular,

$$\boxed{d_N(1,N) = \frac{4l(N-1)}{\pi}\sin\frac{\pi}{4N-2} \longrightarrow l.} \tag{5}$$

One can instead choose the geometric normalization

$$h_N = \frac{N-1}{l}. \tag{6}$$

Then $d_N(1,N) = l$ exactly for every N, with the same continuum limit. However, its $N=2$ mass is $1/l$, rather than the continuum lowest eigenvalue $\pi/(2l)$. Equations (1) and (6) distinguish spectral calibration from metric calibration at finite resolution.

Two other natural choices clarify the literature:

- **One-component staggered path.** Take $A = \mathbb{C}^M$, $H = \mathbb{C}^M$, alternating grading, and $D = h$ times path adjacency. Opposite endpoint chiralities require even M. Its spectrum is $2h\cos(j\pi/(M+1))$. For $k = |i-j|$,

$$\boxed{d(i,j) = \begin{cases} (k+1)/(2h), & k \text{ odd}, \\ \sqrt{k(k+2)}/(2h), & k \text{ even}. \end{cases}} \tag{7}$$

  In particular $d(1,M) = M/(2h)$. Thus $h = M/(2l)$ gives endpoint distance l, whereas $h = M/l$ gives $l/2$.

  Here is a direct proof of (7). Restrict the commutator to the connecting subpath and extend the function constantly outside it. After removing phases, its norm is that of weighted adjacency, with dimensionless edge weights $w_i = h|\Delta a_i|$. If its two bipartition sizes are p, q, the off-diagonal block C satisfies

$$\sum_i w_i = \mathbf{1}_p^T C \mathbf{1}_q \le \sqrt{pq}\,\|C\|.$$

  For $p = q$, alternating weights $1, 0, 1, \ldots, 1$ attain the bound. For $p = q+1$, the weights

$$\frac{(q, 1, q-1, 2, \ldots, 1, q)}{\sqrt{q(q+1)}}$$

  attain it: the normalized constant vectors on the two bipartitions are singular vectors with singular value 1, and positivity gives the norm 1. This proves (7).

  These are the nonadditive distances studied by Bimonte–Lizzi–Sparano, §5 and given explicitly by Atzmon, Eq. (30). Reduction to the connecting path also agrees with Iochum–Krajewski–Martinetti, Proposition 4.

- **Vertex-edge incidence Dirac.** Let $\partial : \mathbb{C}^N \to \mathbb{C}^{N-1}$ be the path difference matrix and use

$$D_{\text{inc}} = h \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}.$$

  Its spectrum and heat trace are

$$\{0\} \cup \left\{ \pm 2h \sin \frac{k\pi}{2N} : 1 \le k < N \right\},$$

$$K_{\text{inc}}(t) = 1 + 2\sum_{k=1}^{N-1} e^{-4th^2 \sin^2(k\pi/(2N))}.$$

  These follow by substituting the cosine eigenvectors into the path Laplacian $\partial^*\partial$. Representing a site function on each edge by its value at the tail gives distance $|i-j|/h$. But H has dimension $2N-1$, has a zero mode, and both endpoint vertex components have the same grading. At $N=2$, its spectrum is $0, \pm\sqrt{2}h$, so it is not Connes–Lott.

Thus the quoted integer deconstruction tower belongs to the incidence construction. Opposite chiral boundary conditions produce the shifted tower (2).

Status: PROVED

Part 2. Continuum limit

Statement. The chain above converges to an interval Dirac operator in three precise senses: ordered low eigenvalues, heat traces locally uniformly for $t > 0$, and pure-state metric spaces in Gromov–Hausdorff distance. It also has a controlled one-dimensional heat window.

Hypotheses. On $H_l = L^2([0,l],\mathbb{C}^2)$, define

$$D_l = \begin{pmatrix} 0 & \partial_x \\ -\partial_x & 0 \end{pmatrix}, \qquad \operatorname{Dom} D_l = \{(L,R) \in H^1 : R(0) = 0,\ L(l) = 0\}, \tag{8}$$

with scalar multiplication by $C^\infty([0,l])$ and grading $\Gamma = \operatorname{diag}(1,-1)$.

These are opposite local chiral conditions. Reflection parity is Γ at 0 and $-\Gamma$ at l; their composition gives antiperiodicity on the doubled interval.

Proof.

Integration by parts gives the boundary form

$$[\overline{L}\,\widetilde{R} - \overline{R}\,\widetilde{L}]_0^l.$$

The conditions in (8) form a maximal isotropic boundary subspace, so D_l is self-adjoint. Its resolvent is compact by the compact embedding $H^1 \hookrightarrow L^2$.

Solving $D_l\psi = \lambda\psi$ gives

$$\psi_\lambda(x) = c(\cos\lambda x, \sin\lambda x), \qquad \cos(\lambda l) = 0.$$

Therefore

$$\boxed{\operatorname{spec}(D_l) = \left\{ \pm\frac{(r-\frac{1}{2})\pi}{l} : r = 1,2,... \right\}.} \tag{9}$$

The kernel is zero. Squaring gives mixed Neumann–Dirichlet conditions on L, and Dirichlet–Neumann conditions on R.

For scalar f,

$$\|[D_l, f]\| = \|f'\|_\infty, \qquad d_l(x,y) = |x-y|. \tag{10}$$

Under normalization (1), with $m_1 = \pi/(2l)$,

$$\lambda_{N,r} = m_1 \frac{\sin((2r-1)\theta_N)}{\sin\theta_N} \longrightarrow (2r-1)m_1. \tag{11}$$

This proves low-eigenvalue convergence. Since

$$\sin y \geq \frac{2y}{\pi} \quad (0 \leq y \leq \pi/2),$$

we also have

$$\lambda_{N,r} \geq \frac{2r-1}{l}. \tag{12}$$

A summable Gaussian majorant now proves, uniformly on every $t \in [\varepsilon, T]$,

$$K_N(t) \longrightarrow K_l(t) = 2\sum_{r=1}^{\infty} e^{-t((r-\frac{1}{2})\pi/l)^2}. \tag{13}$$

Poisson summation gives

$$\boxed{K_l(t) = \frac{l}{\sqrt{\pi t}}\left(1 + 2\sum_{q=1}^{\infty}(-1)^q e^{-q^2 l^2/t}\right).} \tag{14}$$

In particular, $K_l(t) \sim l/\sqrt{\pi t}$ as $t/l^2 \to 0$.

The finite-chain heat window has the following rigorous meaning. For every sequence $t_N > 0$ satisfying

$$\frac{t_N}{l^2} \to 0, \qquad \frac{N^2 t_N}{l^2} \to \infty, \tag{15}$$

one has

$$\boxed{K_N(t_N) \sim \frac{l}{\sqrt{\pi t_N}}, \qquad -2t_N \frac{K_N'(t_N)}{K_N(t_N)} \to 1.} \tag{16}$$

To prove this, put $u_r = \pi\sqrt{t_N}(r - \frac{1}{2})/l$. On bounded u_r-intervals, the sine argument in (11) tends uniformly to zero, so $\sqrt{t_N}\lambda_{N,r}/u_r \to 1$. The mesh tends to zero and its upper endpoint tends to infinity. Bound (12) controls the tails. Gaussian Riemann sums, and then the same sums weighted by $t_N\lambda_{N,r}^2$, prove both assertions.

Finally, identify site i with $x_i = (i-1)l/(N-1)$. Equation (4) gives

$$d_N(i,j) = \rho_N |x_i - x_j|, \qquad \rho_N = \frac{N-1}{l h_N} = 1 + O(N^{-1}).$$

The metric distortion and the grid mesh are both $O(l/N)$, proving Gromov–Hausdorff convergence to $[0, l]$.

The endpoint chirality here concerns allowed boundary components. It does not supply localized chiral zero modes.

Status: PROVED

Part 3. Minimal spectral truncation

Statement. The minimal nonzero grading-preserving spectral truncation of (8) is unitarily equivalent to Connes–Lott, with

$$\boxed{|m| = m_1 = \frac{\pi}{2l}, \qquad c = \frac{\pi}{2}.}$$

Its natural truncated endpoint states have distance $1/m_1 = 2l/\pi$. Literal endpoint evaluations do not descend through compression.

Hypotheses. Let

$$P = \mathbf{1}_{[0,E]}(|D_l|), \qquad m_1 \le E < 3m_1,$$

and give $S = P\,C([0,l])\,P$ its inherited operator-system structure. This is the compression construction described in Connes–van Suijlekom, §1.

Proof.

An orthonormal chiral basis of PH_l is

$$u_L(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(\pi x/(2l)) \\ 0 \end{pmatrix}, \qquad u_R(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(\pi x/(2l)) \end{pmatrix}.$$

In this basis,

$$PD_lP = \begin{pmatrix} 0 & m_1 \\ m_1 & 0 \end{pmatrix}, \qquad P\Gamma P = \mathrm{diag}(1,-1). \tag{17}$$

Compression of multiplication is

$$Q(f) = \begin{pmatrix} \frac{2}{l}\int_0^l f(x)\cos^2\frac{\pi x}{2l}\,dx & 0 \\ 0 & \frac{2}{l}\int_0^l f(x)\sin^2\frac{\pi x}{2l}\,dx \end{pmatrix}. \tag{18}$$

Since

$$Q(1) = I, \qquad Q(\cos(\pi x/l)) = \tfrac{1}{2}\,\mathrm{diag}(1,-1),$$

its range is exactly the full diagonal algebra:

$$\boxed{S = \mathbb{C} \oplus \mathbb{C}.} \tag{19}$$

Here the compressed operator system happens to be an algebra, although Q is not multiplicative.

Evaluation on the finite-dimensional space PH_l is bounded. At 0, only u_L has a nonzero boundary value; at l, only u_R does. The normalized evaluation vectors therefore define

$$\omega_0(\mathrm{diag}(a,b)) = a, \qquad \omega_l(\mathrm{diag}(a,b)) = b.$$

Using (17),

$$\|[PD_lP, \mathrm{diag}(a,b)]\| = m_1|a-b|,$$

and hence

$$\boxed{d_{\mathrm{trunc}}(\omega_0,\omega_l) = \frac{1}{m_1} = \frac{2l}{\pi} < l.} \tag{20}$$

Minimality follows because there is no zero mode and the grading exchanges the eigenspaces at $+m_1$ and $-m_1$. A nonzero grading-preserving spectral subspace must retain both.

There is an essential distinction between these states and literal evaluations. For

$$g(x) = \cos(3\pi x/l),$$

equation (18) gives $Q(g) = 0$, while $g(0) = 1$ and $g(l) = -1$. Thus neither endpoint evaluation factors through Q.

The pullbacks of the truncated endpoint states are instead the probability measures

$$d\mu_L = \frac{2}{l}\cos^2\frac{\pi x}{2l}\,dx, \qquad d\mu_R = \frac{2}{l}\sin^2\frac{\pi x}{2l}\,dx.$$

Their distance using the **original** continuum seminorm is

$$d_{D_l}(\mu_L,\mu_R) = \int_0^l \left|\int_0^x (d\mu_L - d\mu_R)\right| dx = \frac{4l}{\pi^2}. \tag{21}$$

The equality follows by integration by parts and is attained by $f = -x$. Equations (10), (20), and (21) concern three different state/metric comparisons.

Status: PROVED

Part 4. The two limits

Amendment (Terra 6.1, 3 Nov 2026; report). The statements of this part are correct for the stated bare chain, with one precision: the condition N/l -> infinity gives uniform convergence of the dispersion on bounded energy windows and of normalized heat traces and counting densities, and does not by itself give convergence of the unnormalized difference of heat traces (for N = floor(l^(5/4)) it diverges). Escapes for the empty light sector: a tuned bulk mass or a scaled boundary coupling.

Statement. At fixed finite rank, $m \to 0$ produces infinitely separated finite geometry, not an additional heat dimension. A resolved interval requires a growing tower. As $m \to \infty$, the metric collapses, but the light Hilbert space is zero.

Hypotheses. Identify

$$m = m_1 = \frac{\pi}{2l},$$

use normalization (1), and hold physical energies or heat times fixed unless stated otherwise.

Proof.

For fixed N, $D_N = mT_N$ with T_N independent of m. Therefore, as $m \to 0$,

$$D_N \to 0, \qquad K_N(t) \to 2(N-1), \qquad d_N(i,j) \to \infty \quad (i \neq j).$$

For the minimal truncation,

$$K_2(t) = 2e^{-tm^2}.$$

Every finite N has ultraviolet heat dimension zero. Making its finitely many modes light does not change that conclusion.

For the full interval, a fixed energy E is in the resolved one-dimensional regime when

$$\frac{1}{l} \ll E.$$

For its finite-chain approximation, one additionally needs

$$\boxed{\frac{1}{l} \ll E \ll \frac{N}{l}.} \tag{22}$$

Thus, as $l \to \infty$, an exact continuum limit throughout a fixed positive energy window follows if $N/l \to \infty$. Equivalently, as $m \to 0$, one needs $Nm \to \infty$.

The continuum counting function and heat density make the emergence explicit:

$$\#\{|\lambda| \le E\} = 2\left\lfloor \frac{lE}{\pi} + \frac{1}{2} \right\rfloor, \qquad \frac{K_l(t)}{l} \longrightarrow \frac{1}{\sqrt{\pi t}} \quad (l \to \infty). \tag{23}$$

Based at an endpoint, the metric limit is a half-line; the opposite endpoint recedes to infinity.

The order of limits matters even for the normalized heat trace:

$$\lim_{l\to\infty}\lim_{N\to\infty} \frac{K_N(t)}{l} = \frac{1}{\sqrt{\pi t}}, \qquad \lim_{N\to\infty}\lim_{l\to\infty} \frac{K_N(t)}{l} = 0. \tag{24}$$

Likewise, taking $t \to 0$ at fixed N gives dimension zero, whereas taking $N \to \infty$ first gives the interval's dimension one.

As $m \to \infty$, all distances tend to zero. At fixed N, the algebra elements with uniformly bounded commutators reduce to constants. But every eigenvalue has magnitude at least m, so

$$\|(D_N - i)^{-1}\| \le \frac{1}{\sqrt{m^2+1}} \longrightarrow 0. \tag{25}$$

The same holds for D_l, after identifying the varying intervals by dilation. Its heat trace tends to zero at every fixed $t > 0$.

Thus the metric limit is a point, while the embedded resolvent limit has

$$H_{\text{light}} = \ker T_N = 0.$$

There is no finite operator limit on the original nonzero Hilbert space, and no surviving fermion pair at fixed energy.

Finally, the two-state spectral truncation exists for

$$m_1 \le E < 3m_1.$$

A genuine spectral cutoff $E \ll 1/l$ retains no modes. The ratio between the first two positive levels is always 3, so the minimal pair has no parametrically large separation from the next level.

Status: PROVED

Verdict on hypothesis H (theorem T1)

Supported for T1's geometric assertion. Connes–Lott is exactly the minimal nonzero grading-preserving spectral compression of the opposite-chirality interval above, with

$$l = \frac{\pi}{2|m|}.$$

The chain recovers its spectrum, heat window, and metric.

The precise replacement for the stronger interpretation of H is:

> The two sheets are truncated endpoint evaluation states, whose pullbacks are smeared over the interval. Refinement restores the interval. Decompactification produces a one-dimensional heat regime only when the number of retained modes grows sufficiently rapidly. Collapse identifies the metric points but removes all fermionic modes of this particular interval at fixed energy.

Accordingly, a nonempty two-state sector at $E \ll 1/l$, or surviving light fermions after collapse, is disproved by this model. The construction supplies no KO-6 or Standard Model identification.

Independent numerical check (Claude, 2 Nov 2026)

For l = 1 and N = 2, 3, 5, 9 we checked the following. The spectrum of D_N agrees with +/- 2 h_N sin((2r-1) theta_N) to 1e-14. The lowest positive eigenvalue equals pi/(2l) for every N. For random real site functions, ||[D_N, pi_N(a)]|| equals h_N max abs(a_{i+1} - a_i), which is the identity behind d_N(i,j) = abs(i-j)/h_N. The endpoint distance d_N(1,N) = 0.6366, 0.7869, 0.8844, 0.9398 tends to l. For the minimal truncation, Q(cos(3 pi x/l)) = 0, and the smeared-state distance 0.405290 matches 4l/pi^2. The truncated distance is 2l/pi = 0.6366, identical to d_2(1,2).

Literature (../LITERATURE.md, section A). The Kaluza-Klein reading of the two-point space is in Connes-Lott (sections I and III: "two copies of Z separated by a distance L"); the two points as points of a fifth dimension are in Viet-Wali 1996; the two-point model as an approximation of a manifold, to be related to dimensional reduction, is in Dimakis-Muller-Hoissen 1994 (section III, Example 4, and section VII). Connes-van Suijlekom (section 1) note that compressions P A P are operator systems, of which the chiral interval is an exception. The truncation theorem with |m| = pi/(2l) was not found in print.

Part 06

The proposed uniqueness is false: **the same KO-6 triple can admit both reductions**. Moreover, under the stated definition, the strict heavy-limit triple is a doubling of KO-7, although it is not a doubling of KO-5. The distinction between the full real Hilbert space and its physical particle subspace is essential.

1. Lemma — Product convention and necessary correction of the "Clifford point."

Under ordinary spectral-triple conventions, a graded triple on `C^2` is not itself an odd KO-1 or KO-7 triple. The appropriate interpretation is an odd point

$$P_k = (\mathbb{C}, \mathbb{C}, 0, K), \qquad k = 1 \text{ or } 7,$$

together with the auxiliary `C^2` introduced by the odd-times-odd product. For real algebras, use the point algebra `R`, so that the product algebra remains `A`.

I use the following products, with the standard antilinear tensor product of real structures.

For even × even,

$$D = D_1 \otimes 1 + \gamma_1 \otimes D_2, \qquad \gamma = \gamma_1 \otimes \gamma_2,$$
$$J = J_1 \otimes (J_2 \gamma_2^b), \qquad b = (1 - \epsilon_1'')/2.$$

For even × odd, use the same formula for `D`, no grading, and

$$J = (J_1\gamma_1^a) \otimes J_2, \qquad (-1)^a = \epsilon_1''\epsilon_2'.$$

Odd × even is obtained by transporting the even-first construction through the tensor flip.

For odd × odd,

$$H = H_1 \otimes H_2 \otimes \mathbb{C}^2, \quad D = D_1 \otimes 1 \otimes \sigma_1 + 1 \otimes D_2 \otimes \sigma_2, \quad \gamma = 1 \otimes 1 \otimes \sigma_3,$$

$$J = J_1 \otimes J_2 \otimes (BK),$$

where

$(\epsilon_1', \epsilon_2')$	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
B	σ_1	1	σ_3	$-i\sigma_2$.

This agrees, up to harmless phases, with Dąbrowski–Dossena, equation (25), Table 6.

Proof. In the odd-odd case,

$$B\sigma_1 = \epsilon_1'\sigma_1 B, \qquad B\sigma_2 = -\epsilon_2'\sigma_2 B.$$

The second minus sign compensates for complex conjugation of `sigma_2`. Consequently `JD=DJ`; computing `J^2` and `J gamma` gives the signs of the sum degree. The grading twists in the other two formulas follow by the same calculation. Order zero and first order pass to these products.

Thus the intended doubling has Hilbert space `H_0 ⊗ C^2`, with the displayed grading on the auxiliary factor. All subsequent statements use this corrected interpretation and the zero-Dirac point. ■

2. Theorem — Exact doubling criterion in KO-dimension six.

Let `(A,H,D,J,gamma)` be a finite KO-6 triple. For `s=±1`, the following are equivalent:

1. It is a product `T_n × P_k`, where

$$(s, n, k) = (+1, 7, 7) \quad \text{or} \quad (-1, 5, 1).$$

2. There is a complex-linear unitary `U` satisfying

$$U^2 = 1, \quad U\gamma = -\gamma U, \quad [U, a] = [U, a^o] = [U, D] = 0, \quad UJ = sJU,$$

   where `a^o=Ja^*J^{-1}`.

Write `H_L=H_{\gamma=+1}`, `H_R=H_{\gamma=-1}`, and

$$D = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix}.$$

Equivalently, there is an `A`-bimodule unitary `V:H_L→H_R` such that

$$V^*M = (V^*M)^*, \qquad (V^*C)^2 = s, \quad C = J|_{H_L} : H_L \longrightarrow H_R.$$

Here `C` and `V^*C` are antilinear.

Proof. Given `U`, identify

$$H_L \oplus H_L \longrightarrow H, \qquad (x, y) \longmapsto x + Uy.$$

Put

$$H_0 = H_L, \qquad d = (UD)|_{H_L}, \qquad j = (UJ)|_{H_L}.$$

The assumptions give

$$d = d^*, \qquad j^2 = s, \qquad jd = s\,dj.$$

Hence `(j^2,jd/dj)` is `(+,+)` for KO-7 and `(-,-)` for KO-5. In these coordinates,

$$a = \begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d \\ d & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$J(x,y) = (sjy, jx).$$

These are precisely the products in Result 1: `B=σ_1` for `(7,7)` and `B=-iσ_2` for `(5,1)`.

The descended opposite action is

$$ja_0^* j^{-1} = a^o|_{H_L}.$$

Since `[d,a_0]=U[D,a]|_{H_L}`, order zero and first order descend. Conversely, the product has `U=1⊗σ_1`, with all the required properties.

Finally,

$$U = \begin{pmatrix} 0 & V^* \\ V & 0 \end{pmatrix}.$$

Its commutation with `D` is equivalent to self-adjointness of `V^*M`, while `UJ=sJU` is equivalent to `(V^*C)^2=s`.

If one prefers an anticommuting convention, replacing `U` by `i gamma U` exchanges `[U,D]=0` with `{U,D}=0` and preserves `s`. ■

In particular, ordinary equality of left and right module multiplicities is only part of the criterion; compatibility with `D` and `J` must also be checked.

3. Counterexample — J does not select a unique reduction; electrodynamics admits both.

Consider the electrodynamics triple in the basis `(e_L,e_R,bar e_L,bar e_R)`. After a phase change compatible with `A,J,gamma`, its matrices are

$$\rho(a,b) = \mathrm{diag}(a,a,b,b), \qquad \gamma = \mathrm{diag}(1,-1,-1,1),$$

$$J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} K, \qquad D = m \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_1 \end{pmatrix}, \quad m \geq 0.$$

These are the input data of van den Dungen–van Suijlekom, §4.1, equations (4.1)–(4.2).

Define

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1).$$

Proof. Both involutions reverse the grading and commute with the algebra, opposite algebra, and `D`. Direct multiplication gives

$$U_+ J = JU_+, \qquad U_- J = -JU_-.$$

Thus the same triple, with the same `J`, admits both factorizations.

On `H_0=span(e_L,bar e_R)`, the two descended triples have

$$d_+ = mI_2, \qquad j_+ = \sigma_1 K$$

and

$$d_- = \mathrm{diag}(m,-m), \qquad j_- = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} K.$$

They have KO-degrees seven and five, respectively. This also holds at `m=0`. ■

4. Lemma — Exact neutral-sector criterion, including Majorana masses.

Suppose a KO-6 sector has

$$H = H_+ \oplus H_-, \qquad \dim H_+ = \dim H_- = r,$$

and both algebra actions are the same scalar character throughout this sector. Then:

- a KO-7 reduction always exists;
- a KO-5 reduction exists exactly when **every singular value of `M`, including zero, has even multiplicity**.

Proof. Choose an orthonormal basis of `H_+` and its `J`-image as the basis of `H_-`. Then

$$J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} K.$$

The equation `JD=DJ` says `M^T=M`. Takagi factorization, implemented by a basis change preserving this form of `J`, reduces `M` to

$$\Sigma = \mathrm{diag}(\mu_1, \dots, \mu_r), \qquad \mu_i \geq 0.$$

Taking `V=I` satisfies Result 2 with `s=+1`.

For `s=-1`, the descended antiunitary `j` squares to `-1` and commutes with `d^2=D^2|_{H_+}`. Every eigenspace of `d^2` must therefore have even complex dimension. These dimensions are exactly the singular-value multiplicities.

Conversely, pair equal entries of `Sigma` and choose `V=σ_2` on each pair. Then `V` is a skew-symmetric Hermitian unitary commuting with `Sigma`; it satisfies the criterion with `s=-1`. ■

A single neutral Weyl state and its conjugate therefore **are** a KO-7 doubling, even without a Dirac partner. They cannot be a KO-5 doubling. The same statement holds for a single Majorana pair, whether its mass vanishes or not.

5. Proposition — The strict one-generation heavy limit is a KO-7 doubling, not a KO-5 doubling.

Here is an explicit convention for the Standard Model representation. Set

$$F = \mathbb{C} \oplus \mathbb{C}^3, \qquad H_p = (\mathbb{C}^2 \otimes F)_L \oplus (\mathbb{C}^2 \otimes F)_R, \qquad H = H_p \oplus H_{\bar{p}}.$$

For `a=(lambda,q,m)`, put

$$Q_\lambda = \mathrm{diag}(\lambda, \bar{\lambda}), \qquad B_{\lambda,m} = \mathrm{diag}(\lambda, m),$$
$$\rho_p(a) = (q \otimes I_4) \oplus (Q_\lambda \otimes I_4),$$
$$\rho_{\bar{p}}(a) = (I_2 \otimes B_{\lambda,m}) \oplus (I_2 \otimes B_{\lambda,m}).$$

Use

$$J(p, \bar{p}) = (K\bar{p}, Kp), \qquad \gamma = \mathrm{diag}(I_8, -I_8, -I_8, I_8).$$

On particles, the Dirac block is

$$S_p = \begin{pmatrix} 0 & Y^* \\ Y & 0 \end{pmatrix},$$

where `Y` has entries `y_nu,y_e,y_u I_3,y_d I_3` on the corresponding species. The antiparticle block is `bar S_p`; the additional Majorana block connects `nu_R` with `J nu_R`. This realizes the bimodule and mass-block structure in CCM, Proposition 2.4 and Theorem 2.21.

Assume precisely the strict projection described in canonical §3: remove `nu_R,J nu_R` and the complete up-type Dirac sector, and restrict to

$$B = \{(\lambda, Q_\lambda, m)\} \simeq \mathbb{C} \oplus M_3(\mathbb{C}).$$

The surviving space consists of `d_L,d_R,e_L,e_R,nu_L` and their conjugates, with `D_light=PSP`.

Proof. The surviving down-quark and electron masses pair identical `B`-bimodules. After making their masses nonnegative, each sector has both involutions from Result 3, with the identity on color.

On

$$N = \mathrm{span}(\nu_L, J\nu_L),$$

both algebra actions are the scalar character `lambda`, and `PSP|_N=0`. Thus

$$\gamma_N = \sigma_3, \qquad J_N = \sigma_1 K, \qquad U_N = \sigma_1$$

gives a KO-7 reduction. Combining this with `U_+` on the charged sectors gives a global KO-7 reduction.

A KO-5 reduction is impossible: the full space has dimension eighteen, so its descended space would have dimension nine, which cannot carry an antiunitary squaring to `-1`. More specifically, the neutral bimodule is inequivalent to the charged bimodules, so any permitted `U` preserves it; Result 4 already obstructs KO-5 there. ■

Thus the anticipated claim "the neutral sector prevents any doubling" is false under the definition in Question 1.

6. Proposition — Modifications and the distinction from physical vectorlikeness.

For the reduced algebra `B` above, the following conclusions hold.

Surviving neutral content	Odd reductions of the full light triple
No neutral sector	Both KO-5 and KO-7
`nu_L,nu_R` and conjugates, pure Dirac mass	Both KO-5 and KO-7
Only `nu_L,J nu_L`	KO-7 only
A single isolated Majorana pair	KO-7 only
General neutral Dirac–Majorana block	Always KO-7; KO-5 exactly under Result 4

Proof. The charged sectors admit both reductions by Result 3. Pure Dirac neutrinos have the same four-state construction, so adding or retaining `nu_R` restores both possibilities, including when its Dirac mass is zero.

For one generation with Dirac mass `m` and right Majorana mass `R`, the neutral symmetric matrix can be written

$$M_N = \begin{pmatrix} 0 & m \\ m & R \end{pmatrix}.$$

Phase changes make `m,R` nonnegative. Its singular values are

$$\mu_\pm = \frac{\sqrt{|R|^2 + 4|m|^2} \pm |R|}{2}.$$

They coincide exactly when `R=0`. Hence any nonzero right Majorana mass obstructs KO-5 in this one-generation neutral sector, but never KO-7. Treating that sector separately leaves a charged KO-5 reduction; it does not turn the complete direct sum into a KO-5 reduction. ■

There is an additional physically meaningful condition absent from Question 1. Let `Q` be `+1` on particles and `-1` on antiparticles.

No-go theorem under the extra hypothesis `[U,Q]=0`. An unpaired `nu_L` prevents doubling.

Proof. Such a `U` must map particle states of positive grading to particle states of negative grading within the same bimodule. No particle `nu_R` exists. The neutral involution used in Result 5 instead maps `nu_L` to its conjugate and violates `[U,Q]=0`. ■

This is why physical chirality and doubling of the complete real triple are different tests.

7. Proposition — Three generations and arbitrary flavor mixing.

Assume the reduced algebra acts trivially on generation indices.

If all three right-neutrino Majorana pairs are removed and only the top Dirac sector is made heavy, the light space has

$$96 - 6 - 12 = 78$$

complex dimensions. If all three up-type sectors are removed, it has dimension fifty-four. In either case, with three surviving left neutrinos and `D_light=PSP`, the light triple admits KO-7 and excludes KO-5.

Dropping the remaining neutral sector, or retaining three right neutrinos with purely Dirac neutrino masses, permits both reductions.

Proof. Each surviving charged species has equal left/right flavor multiplicities. Singular-value decomposition of its mass matrix uses bimodule unitaries; conjugate transformations on antiparticles preserve `J`. Consequently every charged sector reduces to copies of Result 3, irrespective of mass rank or flavor mixing.

The three unpaired neutral states give `r=3`, `M_N=0` in Result 4, excluding KO-5 and admitting KO-7. Equivalently, the proposed descended spaces have odd dimensions thirty-nine and twenty-seven.

With three right neutrinos retained, the general neutral matrix is

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & R \end{pmatrix}, \qquad R^T = R.$$

For `R=0`, every singular value of `Y_nu` appears twice, so both reductions exist. For general `R`, KO-7 always exists, while KO-5 exists exactly when every singular value of this six-by-six matrix has even multiplicity. Thus nonzero Majorana masses are not an unconditional obstruction in several generations; the exact obstruction is the stated multiplicity condition. ■

8. No-go theorem — The massless Standard Model triple is not a doubling.

Let the algebra remain the full

$$A = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$$

with the representation of Result 5, for `g≥1` generations. Then no involution satisfying Result 2 exists, for any admissible `D`. In particular this applies at `v=0`, when only the Majorana block remains.

Proof. Consider the central projection

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0).$$

Its represented range lies entirely in the particle left-handed sector:

$$\dim \rho(e_{\mathbb{H}})H_+ = 8g, \qquad \dim \rho(e_{\mathbb{H}})H_- = 0.$$

An algebra-linear grading-reversing unitary would identify these spaces, which is impossible.

In physical representation-ring notation,

$$V_L = (3,2)_{1/6} \oplus (1,2)_{-1/2},$$

$$V_R = (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1} \oplus (1,1)_0.$$

Thus `I_G=[V_L]-[V_R]≠0`. For the complete real graded Hilbert space, the relevant gauge character is

$$[H_+] - [H_-] = I_G - \overline{I_G},$$

not merely `I_G`. It too is nonzero: the coefficient of `(3,2)_{1/6}` is `g`. Either this full-character calculation or the central-projection argument proves the obstruction. ■

Consequence for the 10 -> 9 claim

9. Proposition — What dimension reduction is actually established.

The heavy projection itself preserves KO-degree six, as canonical §3 explicitly states. Hence its spacetime product still has degree `4+6=10 mod 8`.

A subsequent choice of a KO-5 reduction removes a degree-one Clifford factor and gives the formal representative `4+5=9`. Such a reduction exists for the charged truncation or the pure-Dirac completion, but **both also admit KO-7 reductions**. Their fixed `J` does not select nine.

For the strict light spectrum with one or three unpaired left neutrinos, **KO-5 is obstructed; only KO-7 is available among the proposed reductions**. This gives the formal representative eleven, not nine. If particle preservation is additionally required, that strict light spectrum admits neither reduction.

Proof. These are Results 2–7. For completeness, the spacetime real structures are

$$J_{4+5} = (J_M \gamma_M) \otimes j_5, \qquad J_{4+7} = J_M \otimes j_7.$$

Thus even the odd reductions require different product conventions. No finite-dimensional operation here changes heat dimension: a finite internal factor remains spectrally zero-dimensional. Consequently the claimed loss of one dimension is, at most, a chosen Clifford-factor removal—not a proved metric collapse, and not a consequence of vectorlike charged matter alone. ■ —

Independent numerical check (Claude, 2 Nov 2026)

We used the electrodynamics data of Result 3, with m = 0.7, J = [[0, I_2],[I_2, 0]] K and gamma = diag(1,-1,-1,1). The triple has J^2 = 1, JD = DJ and J gamma = -gamma J, so its KO-dimension is 6. Both U_+ = diag(sigma_1, sigma_1) and U_- = diag(sigma_1, -sigma_1) satisfy U^2 = 1 and U gamma = -gamma U. Both commute with random algebra elements, their opposites and D; and U_+ J = J U_+, U_- J = -J U_-. On H_0 = span(e_L, bar e_R), U_+ descends to j^2 = +1 with jd = dj (KO-7) and U_- to j^2 = -1 with jd = -dj (KO-5). For No-go 8 we also rechecked by hand the dimension count dim rho(e_H) H_+ = 8g, dim rho(e_H) H_- = 0 in the representation of Result 5.

Terminology. Here "doubling" means the factorization of an even triple as an odd triple times a Clifford point (the doubling of an odd triple in Dabrowski-Dossena's product construction). It is unrelated to the "fermion doubling" of the noncommutative Standard Model (Lizzi-Mangano-Miele-Sparano). Prior work on products and on J versus J gamma: ../LITERATURE.md, section D; the factorization criterion and the 9/11 results were not found in print.

Referee note

Terra 6.1 (3 Nov 2026; report) checked Result 3, Lemma 4, Proposition 5, Result 7, No-go theorem 8 and the no-go under [U, Q] = 0: all correct. One amendment to Result 7: "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen onto the specified top Dirac sector; mixing between inequivalent species is outside the displayed Dirac operator. Escapes: for KO-5, neutral-sector pairing or a symmetry enforcing even multiplicities; for No-go 8, a change of algebra or representation, the even description, and boundary or defect realizations of chirality; for the no-go under [U, Q] = 0, a matching particle bimodule of opposite grading.

Part 07

Does the chiral interval select 9 or 11?

Questions 1-3 of prompts/07. The Astra job stopped on the codex usage limit before producing output, so these are Claude Opus 5.5's proofs, checked by hand. Astra job 08 confirmed Lemmas 1-3 under the pointwise reality hypothesis of Lemma 1 and answered Question 4: see lemma 08 and the amendments below.

Let I_l = [0, l], H = L^2(I_l, C^2) with psi = (L, R), D = E d/dx with E = [[0, 1], [-1, 0]], Dom D = {psi in H^1 : R(0) = 0, L(l) = 0}, Gamma = diag(1, -1) (job 05). D and Gamma are real.

Lemma 1 (real structures on the chiral interval). Let J be antiunitary with J f J^{-1} = conj(f) for f in C^infty(I_l), J Dom D = Dom D, J^2 = eps, JD = eps' DJ, J Gamma = eps'' Gamma J. Then J = W K with W = diag(alpha, beta) constant, abs(alpha) = abs(beta) = 1, and

```
eps = +1,   eps'' = +1,   alpha = eps' beta,   eps' in {+1, -1}.
```

Both values of eps' occur, and they are exchanged by J -> J Gamma. The chiral boundary conditions exclude every real structure anticommuting with Gamma; the bare line allows such ones, e.g. W = E with signs (-1, +1, -1).

Proof. Antilinearity together with J f = conj(f) J gives J = W K for a unitary matrix function W(x). Since E is real, D J psi = E W' conj(psi) + E W conj(psi'), while J D psi = W E conj(psi'). Hence eps' D J = J D forces E W' = 0, so W is constant, and W E = eps' E W. Preservation of R(0) = 0 for arbitrary L(0) forces W_21 = 0, and preservation of L(l) = 0 forces W_12 = 0. So W is diagonal and unitary. Then W conj(W) = 1 gives eps = +1, and W commutes with Gamma, so eps'' = +1. Finally diag(alpha, beta) E = [[0, alpha], [-beta, 0]] and E diag(alpha, beta) = [[0, beta], [-alpha, 0]], so alpha = eps' beta. Replacing J by J Gamma keeps eps = eps'' = +1 and flips eps'. ■

With Connes' even convention (eps' = +1) the chiral interval therefore has KO-dimension 0 mod 8, although its metric dimension is 1.

Lemma 2 (compression). Every J of Lemma 1 commutes with P = 1_{[0,E]}(abs(D)). On the truncation span(u_L, u_R) of job 05 it induces J_2 = diag(alpha, beta) K, with D_2 = m_1 sigma_1, Gamma_2 = sigma_3 and the signs (+1, eps', +1).

Proof. JD = eps' DJ gives J D^2 = D^2 J, so J commutes with abs(D) and with its spectral projections. The basis functions u_L = sqrt(2/l)(cos(pi x/2l), 0) and u_R = sqrt(2/l)(0, sin(pi x/2l)) are real, so J u_L = alpha u_L and J u_R = beta u_R. ■

Lemma 3 (the reflection fixes s = eps'). Let (sigma psi)(x) = a (R(l - x), L(l - x)). Then sigma preserves Dom D, anticommutes with Gamma and commutes with constants; [sigma, D] = 0 holds exactly when the phases on the two components agree, and sigma^2 = 1 forces a = +1 or -1. On the truncation, sigma = a sigma_1. For every J of Lemma 1,

```
sigma J = s J sigma   with   s = beta/alpha = eps'.
```

Proof. The swap with x -> l - x exchanges the conditions R(0) = 0 and L(l) = 0. With general phases a, b on the two components, (D sigma psi)_L = -b L'(l - x) and (sigma D psi)_L = -a L'(l - x), so commutation forces a = b. One checks sigma u_L = u_R and sigma u_R = u_L. Then sigma J psi = a (beta conj R(l - x), alpha conj L(l - x)) and J sigma psi = a (alpha conj R(l - x), beta conj L(l - x)), which gives s = beta/alpha = eps'. ■

Consequence. As the interval collapses, only constants keep bounded commutators, and the reflection becomes a canonical L <-> R involution. Its sign against J is s = eps', so the interval fixes s exactly as far as the real structure on it is fixed, and no further. The 9 versus 11 ambiguity of job 06 is the same as the standard ambiguity J versus J gamma for even real spectral triples (Dabrowski-Dossena). This matches job 06, Proposition 9: the two odd reductions use J_M gamma_M (for 4 + 5) and J_M (for 4 + 7) on spacetime. Under Connes' convention eps' = +1 the reflection gives s = +1, i.e. KO-7, formally 11.

Conjecture (Question 4; false as stated, see amendment 4). For F = F_0 x (collapsing chiral interval) with the product conventions of job 06, Lemma 1, the induced U = 1 (x) sigma satisfies UJ = sJU with s = eps' (interval) times a sign determined by the product grading twist. In particular, which of 9 and 11 is selected is fixed by choosing which spacetime real structure (J_M or J_M gamma_5) represents physical charge conjugation. A KO-0 interval does not change the KO-degree of a product, so the "Clifford point" of job 06 must come from the odd x odd product structure rather than from the interval alone; the open point is how the two descriptions match. Attempted by: Claude Opus 5.5 (Questions 1-3 above). The gpt-6-astra job was cut off by the usage limit before attempting it.

Amendments from job 08

1. Lemma 1 needs its hypothesis J f J^-1 = conj(f). Without it, J_ref = sigma_1 R_l K (R_l the reflection x -> l - x) preserves the chiral boundary conditions and has the KO-6 signs (Counterexample 08.4), so the chiral boundary conditions alone do not force KO-dimension 0.
2. Lemma 2 holds for the compressed operators. With the full compressed diagonal algebra and the inherited J_2, the two-point data fail the first-order condition: [[D_2, e], e°] = D_2 for e = diag(1, 0) (Counterexample 08.4).
3. The sign s = eps' in the Consequence above is the sign of the reflection against the interval's own J. In the standard internal-first KO-6 product the required grading twist changes it: the bare reflection

1 (x) sigma always has s = -1, formally 4 + 5 = 9, for either eps' (Theorem 08.5).
4. Question 4 (Conjecture 07.4) is false as stated: a fixed spacetime J_M and particle-number conservation admit both signs (Proposition 08.9, Counterexample 08.11, Theorem 08.12).

Part 08

Product real structures and the choice between 9 and 11

Prompt: prompts/08 (Task A: check Lemmas 07.1-07.3; Task B: Conjecture 07.4). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5: by hand (results 1-3, 4 second part, 8, 12) and numerically in Python (results 5, 6, 7, 11 and 8; section at the end).

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all 12 numbered statements are correct. Two points of scope: the case t = -1 in Result 1 uses the extended convention 0_-, since the standard KO-0 signs have t = +1; and Theorem 7 concerns the scalar-algebra truncation, since the full compressed two-sheet algebra with its inherited real structure fails first order (Counterexample 4).

Outcome. Lemmas 07.1-07.3 hold under the pointwise reality hypothesis. Conjecture 07.4 is false as stated: Counterexample 11 fixes the physical D_F, J_F, Gamma_F and particle number Q and still admits admissible involutions of both signs. What holds instead: in the standard internal-first KO-6 product, the bare interval reflection always has s = -1 (Theorem 5, formally 4+5 = 9); the product is exactly T_5(m) x P_1 with d_5 = D_0 + m Gamma_0 (Theorem 7); particle preservation pairs the two signs whenever [Q, D] = 0 (Theorem 12); and the literal collapse of the massless interval leaves no light sector (No-go theorem 8).

We use the sign convention

$$J^2 = \epsilon, \qquad JD = \epsilon' DJ, \qquad J\Gamma = \epsilon'' \Gamma J.$$

Unless stated otherwise, KO-6 means the standard signs $(+,+,-)$. We require the order-zero and first-order conditions when calling data a real spectral triple; we impose no orientability or Poincare-duality assumption.

1. Lemma (07.1: classification under pointwise reality).

Let

$$H_I = L^2([0,l],\mathbb{C}^2), \quad D_I = E\partial_x, \quad E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \Gamma_I = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with

$$\operatorname{Dom} D_I = \{(L,R) \in H^1 : R(0) = 0,\ L(l) = 0\}.$$

Suppose J_I is antiunitary, preserves this domain, and satisfies

$$J_I f J_I^{-1} = \overline{f}, \qquad J_I D_I = t D_I J_I, \quad t \in \{+1,-1\}.$$

Then

$$J_I = \operatorname{diag}(\alpha,\beta)K, \qquad |\alpha| = |\beta| = 1, \qquad \alpha = t\beta.$$

Consequently,

$$(\epsilon_I, \epsilon_I', \epsilon_I'') = (+,t,+).$$

Proof. The linear unitary $W = J_I K$ commutes with every scalar multiplication operator. Hence W is multiplication by a measurable unitary matrix $W(x)$.

Domain preservation, applied to compactly supported smooth sections, gives $W \in H^1_{\mathrm{loc}}$. Comparing the coefficients of derivatives in

$$WE\overline{\psi'} = tE(W'\overline{\psi} + W\overline{\psi'})$$

gives

$$WE = tEW, \qquad EW' = 0.$$

Thus W is constant. Preservation of the boundary conditions gives $W_{21} = W_{12} = 0$, because the remaining endpoint traces are arbitrary.

Writing $W = \mathrm{diag}(\alpha, \beta)$, the equation $WE = tEW$ becomes $\alpha = t\beta$. Finally,

$$W\overline{W} = 1, \qquad W\Gamma_I = \Gamma_I W.$$

These imply the asserted signs. Conversely, every displayed J_I satisfies all the hypotheses. □

Thus Lemma 07.1 is correct under its stated pointwise reality hypothesis. Up to phase, the two possibilities are K and $\Gamma_I K$. They are 0_+ and 0_- in the extended convention; Connes' even convention selects $t = +1$. See Dąbrowski–Dossena, arXiv v1, Table 1, p. 5.

2. Lemma (07.2: spectral compression).

Every J_I in Result 1 commutes with the spectral projections of $|D_I|$. On the minimal nonzero grading-preserving spectral subspace,

$$D_2 = m\sigma_1, \qquad \Gamma_2 = \sigma_3, \qquad J_2 = \mathrm{diag}(\alpha, \beta)K, \qquad m = \frac{\pi}{2l}.$$

Proof. The relation $J_I D_I = tD_I J_I$ implies that J_I preserves every eigenspace of D_I^2, hence every spectral subspace of $|D_I|$.

Solving $D_I\psi = \lambda\psi$ with the boundary conditions gives $\lambda = \pm(r - \frac{1}{2})\pi/l$. The lowest pair has the real orthonormal chiral basis

$$u_L = \sqrt{\frac{2}{l}}\begin{pmatrix}\cos(mx)\\ 0\end{pmatrix}, \qquad u_R = \sqrt{\frac{2}{l}}\begin{pmatrix}0\\ \sin(mx)\end{pmatrix}.$$

Direct differentiation gives $D_I u_L = mu_R$, $D_I u_R = mu_L$. Also $J_I u_L = \alpha u_L$, $J_I u_R = \beta u_R$. □

This proves Lemma 07.2 as a statement about the compressed operators. Result 4 below addresses the first-order condition.

3. Lemma (07.3: reflection sign).

Let $R_l\psi(x) = \psi(l - x)$. A component-exchanging reflection

$$\sigma = \begin{pmatrix}0 & a\\ b & 0\end{pmatrix} R_l, \qquad |a| = |b| = 1,$$

commutes with D_I exactly when $a = b$. Requiring additionally $\sigma^2 = 1$ gives $a = b = \pm 1$. For these reflections,

$$\sigma\Gamma_I = -\Gamma_I\sigma, \qquad \sigma J_I = tJ_I\sigma.$$

On the minimal truncation, $\sigma = \pm\sigma_1$.

Proof. Reflection exchanges the two endpoint conditions. Since $\partial_x R_l = -R_l\partial_x$, commutation with D_I requires

$$-E\begin{pmatrix}0 & a\\ b & 0\end{pmatrix} = \begin{pmatrix}0 & a\\ b & 0\end{pmatrix} E,$$

which is equivalent to $a = b$. Its square is then a^2, so $a = \pm 1$.

For real a, multiplication of the matrices representing σ and J_I gives

$$\sigma J_I = tJ_I\sigma$$

using $\alpha = t\beta$. The identities $\cos(m(l - x)) = \sin(mx)$ and $\sin(m(l - x)) = \cos(mx)$ give the compressed reflection. □

Thus all three lemmas agree with the direct calculation. Notice, however, that

$$\sigma f\sigma^{-1} = f(l - \cdot).$$

The reflection commutes with constants, but generally does not commute with the interval algebra.

4. Counterexample (two stronger interpretations of the interval lemmas fail).

First, chiral boundary conditions alone do not force KO-dimension zero. Without pointwise reality, the antiunitary

$$J_{\mathrm{ref}} = \sigma K, \qquad \sigma = \sigma_1 R_l,$$

has the standard KO-6 signs.

Second, the minimal two-point compression with its full diagonal algebra and inherited J_2 is not a real spectral triple satisfying the first-order condition when $m > 0$.

Proof. For the first assertion,

$$J_{\mathrm{ref}}^2 = 1, \qquad J_{\mathrm{ref}} D_I = D_I J_{\mathrm{ref}}, \qquad J_{\mathrm{ref}} \Gamma_I = -\Gamma_I J_{\mathrm{ref}}.$$

It implements

$$J_{\mathrm{ref}} f J_{\mathrm{ref}}^{-1} = \overline{f}(l - \cdot).$$

The opposite action is still scalar multiplication, so order zero and first order hold. This example violates precisely the pointwise reality hypothesis of Result 1.

For the second assertion, the compressed algebra is

$$A_2 = \{\mathrm{diag}(a, b)\}.$$

Indeed, compression sends 1 to 1 and $\cos(\pi x/l)$ to $\frac{1}{2}\sigma_3$. Let $e = \mathrm{diag}(1, 0)$. The inherited diagonal J_2 gives $e^o = e$, whereas

$$[[D_2, e], e^o] = D_2 \neq 0.$$

Moreover, $\sigma_1 e \sigma_1 = 1 - e$, so the reflection is not an A_2-module map. □

For the finite products below, we therefore explicitly restrict the interval algebra to scalar constants. This restriction makes the compressed interval a real triple and makes its reflection algebra-linear. It is an additional assumption about the proposed collapsed model.

5. Theorem (B1–B2: the internal-first KO-6 product).

Let

$$F_0 = (A_0, H_0, D_0, J_0, \Gamma_0)$$

be a finite real KO-6 triple:

$$J_0^2 = 1, \qquad J_0 D_0 = D_0 J_0, \qquad J_0 \Gamma_0 = -\Gamma_0 J_0.$$

Take either the interval or its scalar-algebra truncation, and use the internal-first product

$$D_F = D_0 \otimes 1 + \Gamma_0 \otimes D_I, \qquad \Gamma_F = \Gamma_0 \otimes \Gamma_I.$$

Among the grading-twisted tensor real structures

$$J_{a,b} = (J_0 \Gamma_0^a) \otimes (J_I \Gamma_I^b), \qquad a, b \in \{0, 1\},$$

the standard KO-6 conditions force

$$a = 0, \qquad t(-1)^b = -1.$$

For the induced reflection $U = 1 \otimes \sigma$,

$$\boxed{U J_F = -J_F U.}$$

Proof. Direct calculation gives

$$J_{a,b}^2 = (-1)^a, \qquad J_{a,b} \Gamma_F = -\Gamma_F J_{a,b}.$$

The two Dirac summands satisfy

$$J_{a,b}(D_0 \otimes 1) = (-1)^a (D_0 \otimes 1) J_{a,b},$$

$$J_{a,b}(\Gamma_0 \otimes D_I) = -t(-1)^b (\Gamma_0 \otimes D_I) J_{a,b}.$$

Thus $J_F^2 = 1$ requires $a = 0$, and $J_F D_F = D_F J_F$ requires $t(-1)^b = -1$.

Since $\sigma J_I = t J_I \sigma$ and $\sigma \Gamma_I = -\Gamma_I \sigma$,

$$U J_{a,b} = t(-1)^b J_{a,b} U.$$

The required product sign is therefore -1. □

Explicitly:

Interval $t = \epsilon'_I$	Required b	Product reflection sign s
$+1$	1	-1
-1	0	-1

For $t = +1$, this is exactly job 06's formula

$$J_F = J_0 \otimes (J_I \Gamma_I).$$

Changing J_I to $J_I \Gamma_I$ while adjusting the required twist leaves the resulting J_F unchanged, up to phase. The general formula $s = t(-1)^b$ is proved, but t and b are not independent choices when the product must have standard KO-6 signs. The relevant product constructions are Dąbrowski–Dossena, §4.1, equation (18) and Tables 2–3.

A KO-0 factor leaves the KO-degree unchanged. Hence F_0 must already have degree six in this ordinary even-even construction. An odd internal triple multiplied by an even interval remains odd under the standard product rules.

6. Proposition (B1–B2: the grading and the lift of reflection).

The preceding construction has three necessary distinctions.

1. The interval grading alone cannot be the total KO-6 grading for a tensor real structure of the displayed form.
2. Twisting the reflection by the internal grading can change s, but generally destroys commutation with D_F.
3. Transporting all data between equivalent product conventions preserves s.

Proof. If one sets $\Gamma_F = 1 \otimes \Gamma_I$, then every $J_{a,b}$ commutes with Γ_F, contradicting the required KO-6 relation. The product grading must be $\Gamma_0 \otimes \Gamma_I$.

For

$$U_c = \Gamma_0^c \otimes \sigma, \qquad c \in \{0, 1\},$$

we obtain

$$U_c J_F = (-1)^c \, t(-1)^b J_F U_c,$$

and

$$[U_c, D_F] = [\Gamma_0^c, D_0] \otimes \sigma.$$

Thus U_0 always commutes with D_F. For $c = 1$,

$$[U_1, D_F] = 0 \quad \Longleftrightarrow \quad D_0 = 0,$$

because $\{\Gamma_0, D_0\} = 0$. When $D_0 = 0$, both lifts are admissible and have opposite signs.

To compare product conventions, let

$$W = P^0_+ \otimes 1 + P^0_- \otimes \Gamma_I, \qquad P^0_\pm = \frac{1 \pm \Gamma_0}{2}.$$

Then

$$W D_F W = D_0 \otimes \Gamma_I + 1 \otimes D_I,$$

while

$$W(1 \otimes \sigma)W = \Gamma_0 \otimes \sigma.$$

Unitary transport preserves the relation $UJ = sJU$. Keeping the bare symbol $1 \otimes \sigma$ after changing convention can therefore mean choosing a different involution. □

The interval can still carry the physical L/R label. For example, if $\Gamma_0 = Q_0$ distinguishes particles from antiparticles, then

$$\Gamma_F = Q_0 \otimes \Gamma_I$$

has the usual reversal of finite grading between a particle and its conjugate.

Multiplying the total real structure by Γ_F also flips s, but changes its signs to $(-,-,-)$. It therefore leaves the standard KO-6 convention required by job 06.

7. Theorem (B1: exact reconciliation with the odd Clifford point).

For the scalar-algebra minimal truncation, choose the harmless overall phase so that

$$D_F = D_0 \otimes 1 + m\Gamma_0 \otimes \sigma_1, \quad \Gamma_F = \Gamma_0 \otimes \sigma_3, \quad J_F = J_0 \otimes (\sigma_3 K).$$

With $U = 1 \otimes \sigma_1$, this triple is exactly the odd-odd product

$$\boxed{F \cong T_5(m) \times P_1,}$$

where

$$T_5(m) = (A_0, H_0,\ d_5 = D_0 + m\Gamma_0,\ j_5 = J_0\Gamma_0),$$

and

$$P_1 = (\mathbb{C}, \mathbb{C}, 0, K).$$

For real algebras, use the scalar point algebra $\mathbb{R}$.

Proof. The operator d_5 is self-adjoint, and

$$j_5^2 = -1, \qquad j_5 d_5 = -d_5 j_5.$$

Its opposite action is unchanged, and $[d_5, a] = [D_0, a]$, so order zero and first order descend.

Let $e_\pm$ be the eigenvectors of σ_3, and define

$$R : H_0 \longrightarrow H_{F,+}, \qquad Rx = P^0_+ x \otimes e_+ + P^0_- x \otimes e_-.$$

This is unitary onto the positive total-grading subspace. Direct calculation gives

$$R^* U D_F R = D_0 + m\Gamma_0, \qquad R^* U J_F R = J_0 \Gamma_0.$$

Under the unitary identification

$$(x, y) \longmapsto Rx + URy,$$

the operators become

$$D_F = \begin{pmatrix} 0 & d_5 \\ d_5 & 0 \end{pmatrix}, \quad \Gamma_F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_F(x, y) = (-j_5 y, j_5 x).$$

These are precisely the odd-odd product matrices with auxiliary matrix $B = -i\sigma_2$. □

More generally, any admissible involution in a standard KO-6 triple gives

$$d = (UD)|_{H_+}, \qquad j = (UJ)|_{H_+},$$

with

$$j^2 = s, \qquad jd = s\,dj.$$

Thus

$$s = -1: \quad F \cong T_5 \times P_1, \qquad s = +1: \quad F \cong T_7 \times P_7.$$

The auxiliary matrices are respectively $-i\sigma_2$ and σ_1, in agreement with Dąbrowski–Dossena, equation (25), Table 6, p. 12.

The even KO-0 interval is therefore not identified with an odd point. Its two-dimensional space becomes the auxiliary doubling space after reorganizing the complete product, while its mass enters d_5. The point itself has one-dimensional Hilbert space and zero Dirac operator. Because its Dirac operator is zero, the distinction between P_1 and P_7 includes a choice of KO label.

8. No-go theorem (literal collapse produces no light finite sector).

Assume the heavy-limit model uses exactly the product Dirac operator of Result 5, with $m = \pi/(2l)$. At every fixed energy cutoff Λ, its light subspace is zero for sufficiently small l.

Proof. Anticommutation of D_0 and Γ_0 gives

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2.$$

The chiral interval and its minimal truncation satisfy

$$D_I^2 \geq \frac{\pi^2}{4l^2}.$$

Hence

$$|D_F| \geq \frac{\pi}{2l}, \qquad \mathbf{1}_{[0,\Lambda]}(|D_F|) = 0 \quad \text{if } l < \frac{\pi}{2\Lambda}.$$

The same conclusion holds for the alternative product convention by unitary equivalence. □

Consequently, the exact product supplies an algebraic model at finite l, but cannot yield a nonempty light vectorlike sector by literal collapse at fixed energy. Retaining such a sector requires additional dynamics or a modified limiting prescription.

9. Proposition (B3: spacetime grading twists do not select the finite involution).

Let the four-dimensional Euclidean spacetime triple satisfy

$$J_M^2 = -1, \qquad J_M D_M = D_M J_M, \qquad J_M \gamma_M = \gamma_M J_M.$$

For an odd reduction (d, j_s) with

$$j_s^2 = s, \qquad j_s d = s\,d j_s,$$

the job-06 even-odd convention requires

$$J_{M\times T_s} = (J_M \gamma_M^{a_s}) \otimes j_s, \qquad (-1)^{a_s} = s.$$

Therefore

$$J_{4+5} = (J_M \gamma_M) \otimes j_5, \qquad J_{4+7} = J_M \otimes j_7.$$

These formulas do not imply that choosing physical spacetime charge conjugation selects s.

Proof. For

$$D_{M\times T_s} = D_M \otimes 1 + \gamma_M \otimes d,$$

the proposed real structure has commutation signs $(-1)^{a_s}$ and s against the two summands. They must agree. Its square is $-s$, giving KO-1 for $s = -1$ and KO-3 for $s = +1$.

We can also retain the removed point and compare the complete products. Write

$$F = T_s \times P_k, \qquad J_F = j_s \otimes (B_s K),$$

where $B_- = -i\sigma_2$, $B_+ = \sigma_1$. The unitary

$$V = P_+^M \otimes 1 \otimes 1 + P_-^M \otimes 1 \otimes \sigma_1$$

identifies $(M \times T_s) \times P_k$ with $M \times F$. In particular,

$$V((J_M \gamma_M^{a_s}) \otimes j_s \otimes B_s K)V^* = J_M \otimes J_F.$$

Here the extra factor $\gamma_M^{a_s}$ cancels because $\sigma_1 B_s = s B_s \sigma_1$. □

Thus both reconstructions can use the same original J_M. The different spacetime factors in the reduced products are required by the product rules.

10. Proposition (B3: Euclidean and Lorentzian charge conjugation).

In the conventions used by Chamseddine–Connes–Marcolli and van den Dungen–van Suijlekom, the Euclidean spacetime charge-conjugation operator is J_M, with

$$(J_M^2, \epsilon'_M, \epsilon''_M) = (-, +, +).$$

The operator $J_M\gamma_5$ is its other even real-structure convention. Lorentzian charge conjugation requires changing signature and is not obtained merely by choosing between these two Euclidean operators.

Proof and source identification. CCM state, "The operator J is the charge conjugation operator," in §3.1, and use $J = J_M \otimes J_F$ in §3.2. Their calculation in §3.5.2 explicitly uses $J_M D_M = D_M J_M$. See CCM, arXiv v1, §§3.1–3.2, p. 20, and §3.5.2, p. 27.

Van den Dungen–van Suijlekom likewise identify J_M as "the charge conjugation operator on the spinors" in Example 2.1 and use $J = J_M \otimes J_F$ in Definition 2.3. Their fermionic action uses this J_M. See the published paper, pp. 435–436 and Proposition 4.3, p. 452.

For an explicit signature comparison, take Lorentzian signature $(+---)$, chiral gamma matrices, and

$$\gamma_E^0 = \gamma_L^0, \qquad \gamma_E^j = -i\gamma_L^j.$$

Up to overall phases, standard conjugations are

$$C_E = i\gamma_E^0\gamma_E^2 K, \qquad C_L = -i\gamma_L^2 K.$$

These are the conventions in D'Andrea–Kurkov–Lizzi, arXiv v2, equations (3.4), (3.13)–(3.15).

The Clifford relations give

$$C_E^2 = -1, \quad [C_E, \gamma_5] = 0, \quad [C_E, D_E] = 0, \qquad D_E = -i\gamma_E^\mu \nabla_\mu,$$

whereas

$$C_L^2 = 1, \qquad C_L\gamma_5 = -\gamma_5 C_L.$$

In the displayed common matrix basis,

$$C_L = -i\gamma_E^0 C_E.$$

Both C_E and $C_E\gamma_5$ square to -1 and preserve Euclidean chirality; neither is already C_L.

Finally, for real m, Lorentzian charge conjugation carries

$$(i\gamma_L^\mu(\partial_\mu + ieA_\mu) - m)\psi = 0$$

to the same equation with charge $-e$. Composing C_L with γ_5 instead reverses the scalar mass sign. Thus the usual massive Dirac-field identification distinguishes charge conjugation once the signature and Dirac-operator conventions are fixed. □

This identification determines the starting spacetime convention. By Result 9, it does not determine the finite involution.

11. Counterexample (fixed physical real structure permits both 9 and 11).

There is a single scalar-interval product with fixed D_F, J_F, Γ_F, Q admitting particle-preserving involutions of both signs.

Proof. Let τ_i act on an internal particle/antiparticle space and σ_i on the interval truncation. Set

$$\rho(a,b) = \mathrm{diag}(a,b) \otimes 1, \qquad D_F = m\tau_3 \otimes \sigma_1,$$

$$\Gamma_F = \tau_3 \otimes \sigma_3, \qquad J_F = (\tau_1 \otimes \sigma_3)K, \qquad Q = \tau_3 \otimes 1, \quad m > 0.$$

This is Result 5 with

$$D_0 = 0, \quad \Gamma_0 = \tau_3, \quad J_0 = \tau_1 K.$$

It has standard KO-6 signs. Its Dirac operator commutes with the algebra, so the first-order condition holds.

Define

$$U_- = 1 \otimes \sigma_1, \qquad U_+ = \tau_3 \otimes \sigma_1 = QU_-.$$

Both are grading-reversing unitary involutions commuting with D_F, the two algebra actions, and Q. Nevertheless,

$$U_- J_F = -J_F U_-, \qquad U_+ J_F = J_F U_+.$$

The unitary

$$W = \mathrm{diag}(1, \sigma_3)$$

on the particle/antiparticle blocks transforms the data into

$$D_F' = m\, 1 \otimes \sigma_1, \qquad J_F' = (\tau_1 \otimes 1)K,$$

while preserving Γ_F, Q, ρ. The transformed involutions are

$$U_-' = \mathrm{diag}(\sigma_1, -\sigma_1), \qquad U_+' = \mathrm{diag}(\sigma_1, \sigma_1).$$

Thus this is also the electrodynamics counterexample in job 06, expressed as an interval product. Tensoring with a fixed spacetime J_M leaves both possibilities available. □

This disproves the assertion in Conjecture 07.4 that physical spacetime charge conjugation alone selects the reduction. Selecting specifically the bare reflection $1 \otimes \sigma$ is an additional restriction and gives $s = -1$ in the internal-first convention.

12. Theorem (B4: particle preservation and sign pairing).

Let Q be a self-adjoint involution equal to $+1$ on particles and -1 on antiparticles.

For an interval product with $Q = Q_0 \otimes 1$, the bare reflection automatically satisfies

$$[1 \otimes \sigma, Q] = 0.$$

More generally, assume a standard KO-6 triple satisfies

$$[Q, D] = [Q, \Gamma] = [Q, a] = [Q, a^o] = 0, \qquad QJ = -JQ.$$

If an admissible doubling involution U preserves particle number, then QU is another particle-preserving doubling involution with the opposite sign against J.

Proof. The tensor-product assertion is immediate. For the general assertion, $[Q, U] = 0$ gives

$$(QU)^2 = 1.$$

The remaining commutation and grading properties follow from those of Q and U. If $UJ = sJU$, then

$$(QU)J = sQJU = -sJ(QU).$$

Also $[QU, Q] = 0$. □

Thus particle preservation does not select a sign in a particle-number-conserving sector. The additional assumption $[Q, D] = 0$ is essential to the sign-pairing argument; Majorana terms can violate it.

Conversely, particle-preserving doubling requires the particle spaces of opposite grading to be isomorphic as bimodules: U restricts to such an isomorphism. An unpaired left-handed neutrino with no matching right-handed particle in its bimodule therefore obstructs particle-preserving doubling. The formal interval product builds in the missing partner and cannot represent that strict light spectrum without changing its particle content.

Consequence for 9 versus 11

Lemmas 07.1–07.3 hold under pointwise reality, but their proposed physical consequence does not follow. For the internal-first standard KO-6 product, the required grading twist gives the bare interval reflection $s = -1$, hence a KO-5 reduction and formal $4+5 = 9$. Changing the interval's even real-structure convention consistently does not change this result. A fixed spacetime charge conjugation J_M, and even particle-number preservation, can coexist with both reductions; choosing the lift of L/R reflection remains additional data. Finally, literal collapse of this chiral interval leaves no light fermions, so neither 9 nor 11 has been established as the dimension of a surviving heavy-W/Z limit.

Independent numerical check (Claude, 3 Nov 2026)

Python (numpy), antiunitaries represented as J = M K, with J X J^-1 = M conj(X) M^-1, J^2 = M conj(M), and U J = s J U iff U M = s M conj(U). F_0: H_0 = C^2 (+) C^2 with Gamma_0 = diag(g, -g), D_0 = diag(d, conj(d)) (d off-diagonal, random complex entry), J_0 = block swap times K; KO-6 signs verified. Interval truncation D_I = m sigma_1 (m = 0.7), Gamma_I = sigma_3, J_I = diag(t, 1) K.

- **Theorem 5.** Of the eight choices (t, a, b), exactly (t, a, b) = (+1, 0, 1) and (-1, 0, 0) give the KO-6 signs (+, +, -); both give s = -1 for U = 1 (x) sigma_1, and U commutes with D_F.
- **Theorem 7.** With R x = P_+ x (x) e_+ + P_- x (x) e_-: R* U D_F R = D_0 + m Gamma_0 and R* U J_F R = J_0 Gamma_0, with j_5^2 = -1 and j_5 d_5 = -d_5 j_5 (KO-5 signs).
- **Proposition 6.** Gamma_0 (x) sigma_1 commutes with D_F for D_0 = 0 and fails to commute for D_0 != 0.
- **Counterexample 11.** KO-6 signs, order zero and first order hold for random diagonal algebra elements; U_- and U_+ are both admissible (involutive, grading-reversing, commuting with D_F, Q, the algebra and its opposite), with s = -1 and s = +1.
- **No-go theorem 8.** D_F^2 = D_0^2 (x) 1 + 1 (x) D_I^2 holds exactly.
- **Counterexample 4 (first order), by hand.** For e = diag(1, 0) and the inherited diagonal J_2, e° = e and [[m sigma_1, e], e] = m sigma_1.

Terminology. Here "doubling" means the factorization of an even triple as an odd triple times a Clifford point (the doubling of an odd triple in Dabrowski-Dossena's product construction). It is unrelated to the "fermion doubling" of the noncommutative Standard Model (Lizzi-Mangano-Miele-Sparano). Prior work on products and on J versus J gamma: ../LITERATURE.md, section D; the factorization criterion and the 9/11 results were not found in print.

Part 09

The chiral interval with a bulk mass: chirality, light pairs and both limits

Theorem T3 of ../STRATEGY.md, core part. Proved by Claude Opus 5.5 without Astra or Fable; checked numerically against an independent first-order discretization. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): of 11 statements, 4 correct, 3 needed amendments, 2 had gaps (closed by the referee) and

2 were wrong as first stated (Corollary 9.10 and Lemma 9.11). All corrections are incorporated below; the referee report is appended.

Setting

Let I = [0, l], H = L^2(I; C^2) with psi = (L, R), and Gamma = diag(1, -1). For M in R let A = -d/dx + M, so that A^dagger = d/dx + M, and

```
D_M psi = (A^dagger R, A L),   i.e.   (D psi)_L = R' + M R,   (D psi)_R = -L' + M L.
```

At M = 0 this is the operator of job 05. We consider three boundary conditions:

- (a) R(0) = R(l) = 0;
- (b) L(0) = L(l) = 0;
- (c) R(0) = 0 and L(l) = 0 (the chiral interval of job 05).

Lemma 9.1. For each of (a), (b) and (c), D_M is self-adjoint on {psi in H^1 : boundary condition} and has compact resolvent.

Proof. For psi, phi in H^1 the mass terms cancel (M is real) and integration by parts gives <D psi, phi> - <psi, D phi> = [conj(R) phi_L - conj(L) phi_R]_0^l. Each condition is a Lagrangian subspace of the four-dimensional space of boundary values: two independent conditions on which the form vanishes. Distributional differentiation shows Dom D_M^* is contained in H^1, so the boundary form forces the same conditions and D_M is self-adjoint. The estimate norm(psi') <= norm(D_M psi) + abs(M) norm(psi) makes the graph norm control the H^1 norm, so the resolvent maps L^2 boundedly into H^1, which embeds compactly in L^2. ■

Chirality lives in the boundary conditions

Theorem 9.2. For every M in R and l > 0:

- (a) ker D = C (e^{Mx}, 0), so ind = dim ker_{Gamma=+1} - dim ker_{Gamma=-1} = +1;
- (b) ker D = C (0, e^{-Mx}), so ind = -1;
- (c) ker D = 0, so ind = 0.

Proof. D psi = 0 iff A L = 0 and A^dagger R = 0, i.e. L = alpha e^{Mx} and R = beta e^{-Mx}. Exponentials never vanish, so a condition at either end forces the corresponding coefficient to 0. ■

Corollary 9.3 (equivariant index). Let a compact group G act on H_G = (+)_rho V_rho (x) L^2(I; C^2), a sum of finitely many finite-dimensional representations V_rho, with boundary type tau(rho) in {a, b, c} and mass M_rho on each summand. (With infinitely many summands the operator need not be Fredholm: the referee report gives a counterexample.) Then the G-index of D in R(G) is

```
sum over tau(rho) = a of [V_rho]   minus   sum over tau(rho) = b of [V_rho],
```

independent of every M_rho and of l. Assigning (a) to the summands of V_L and (b) to those of V_R realizes I_G = [V_L] - [V_R] of research_note 7. Vectorlike pairs of type (c) contribute 0. ■

Spectrum of the mixed condition with a bulk mass

Theorem 9.4. Take condition (c). The spectrum is symmetric under lambda -> -lambda, all eigenvalues are simple, and 0 is not an eigenvalue. Write c = mu l.

1. M = -mu with mu > 0 and c > 1: there is exactly one positive eigenvalue below mu,

   ```
   lambda_0 = mu sech(kappa_0 l),   where kappa_0 in (0, mu) is the unique positive root of
    ↪  tanh(kappa l) = kappa/mu,
   ```

   and the remaining positive eigenvalues are lambda_n = sqrt(mu^2 + k_n^2), n >= 1, with k_n the unique root of tan(k l) = k/mu in (n pi/l, (n + 1/2) pi/l).

2. M = -mu and c < 1: no eigenvalue in (0, mu]. The positive eigenvalues are sqrt(mu^2 + k_n^2), n >= 0, with k_0 in (0, pi/(2l)) and k_n in (n pi/l, (n + 1/2) pi/l) roots of tan(k l) = k/mu. At c = 1 the lowest eigenvalue is exactly mu.

3. M = +mu > 0: every positive eigenvalue is sqrt(mu^2 + k_n^2) with k_n in ((n - 1/2) pi/l, n pi/l), n >= 1, a root of tan(k l) = -k/mu, so all exceed sqrt(mu^2 + (pi/2l)^2).

For M = 0 the spectrum is +/-(n + 1/2) pi/l, n >= 0 (job 05).

Proof. If D psi = lambda psi with lambda > 0, then R = A L / lambda and A^dagger A L = lambda^2 L, where A^dagger A = -$d^{2/dx}$2 + M^2. The conditions become L(l) = 0 and L'(0) = M L(0). Conversely, every solution of this Sturm-Liouville problem with lambda^2 > 0 yields the eigenvector (L, A L/lambda), whose R-component vanishes at 0. The Sturm-Liouville problem has simple eigenvalues, and the condition L(l) = 0 forces each eigenfunction into one of three forms:

- E = M^2 + k^2: L = sin(k(l - x)), giving tan(k l) = -k/M;
- E = M^2 - kappa^2 with 0 < kappa < abs(M): L = sinh(kappa(l - x)), giving tanh(kappa l) = -kappa/M;
- E = M^2: L = l - x, giving M l = -1.

Counting roots for M = -mu, with t = k l or t = kappa l:

- g(t) = tanh t - t/c is concave on t > 0, with g(0) = 0, g'(0) = 1 - 1/c and g -> -infinity. It has a positive root iff c > 1, and that root is unique.
- h(t) = tan t - t/c is negative on (n pi - pi/2, n pi). On (n pi, n pi + pi/2) we have h' = sec^2 t - 1/c, which changes sign at most once from - to +. Since h(n pi) <= 0 and h tends to +infinity, there is exactly one root there for n >= 1. For n = 0 there is one iff c < 1.

For M = +mu, the function tan t + t/c is strictly increasing on (n pi - pi/2, n pi), from -infinity to n pi/c > 0, so it has exactly one root there. The hyperbolic equation and M l = -1 have no solutions. Finally, s = kappa_0 l satisfies s = c tanh s, which gives lambda_0^2 = mu^2 - kappa_0^2 = mu^2 sech^2 s. ■

Lemma 9.5 (size and shape of the light pair). For c > 1,

`mu sech(mu l) <= lambda_0 <= 2 mu e^{-kappa_0 l}, so lambda_0 = 2 mu e^{-mu l} (1 + o(1)) as mu l -> infinity.`

The pair is spanned by L_0 = sinh(kappa_0 (l - x)), concentrated at x = 0, and R_0 = A L_0 / lambda_0 = (kappa_0 cosh(kappa_0(l - x)) - mu sinh(kappa_0(l - x)))/lambda_0, which vanishes at 0 and is concentrated at x = l. Moreover ||R_0|| = ||L_0||.

Proof. The lower bound holds because sech is decreasing and kappa_0 < mu. For the upper bound, lambda_0^2 = (mu - kappa_0)(mu + kappa_0) <= 2 mu (mu - kappa_0) = 2 mu^2 (1 - tanh(kappa_0 l)) <= 4 mu^2 e^{-2 kappa_0 l}. The equality of norms follows from ||A L_0||^2 = <A^dagger A L_0, L_0> = lambda_0^2 ||L_0||^2. For the asymptotic, put s = kappa_0 l, so that c = s/tanh s and 0 < c - s = c(1 - tanh s) <= 2c e^{-2s} -> 0; hence lambda_0 = mu sech s = 2 mu e^{-c} e^{c-s}/(1 + e^{-2s}) = 2 mu e^{-c}(1 + o(1)). Finally, mu = kappa_0 coth(kappa_0 l) and lambda_0 = kappa_0/sinh(kappa_0 l) give the exact identity R_0(x) = sinh(kappa_0 x) = L_0(l - x): the two profiles are mirror images, decreasing away from their walls, with localization on the scale 1/kappa_0 as s -> infinity (and approaching multiples of l - x and x as c -> 1). ■

Theorem 9.6 (the light pair is Connes-Lott with m = lambda_0). Let P_0 project onto the eigenvalues +/- lambda_0 (c > 1), and use the basis u_L = (L_0, 0)/||L_0|| and u_R = (0, R_0)/||R_0||. Then:

- P_0 D P_0 = lambda_0 sigma_1 and P_0 Gamma P_0 = sigma_3;
- the compression of C(I) is Q(f) = diag(int f abs(u_L)^2, int f abs(u_R)^2), whose range is C + C;
- the truncated end states are at Connes distance 1/lambda_0.

Proof. We have D u_L = A L_0/||L_0|| = lambda_0 u_R. Pointwise orthogonality of the L and R components makes Q(f) diagonal. Q(1) = I and Q(x) = diag(_L, _R) with _L < _R, so the range is the

full diagonal algebra. Finally, ||[lambda_0 sigma_1, diag(a, b)]|| = lambda_0 abs(a - b). ■

The two-sheet distance is therefore 1/(fermion mass), as in the Standard Model finite geometry; it depends on l only through lambda_0. At M = 0 (job 05) it was 2l/pi.

The two limits

Lemma 9.7. For c > 1 let s(c) be the positive root of tanh s = s/c, and set F(c) = (lambda_0 l)^2 = c^2 sech^2 s(c). Then F is strictly decreasing on (1, infinity), F(1+) = 1 and F(infinity) = 0.

Proof. Differentiating tanh s = s/c gives s'(1/c - sech^2 s) = s/c^2. Here 1/c - sech^2 s > 0, since 1/c = tanh(s)/s and sinh 2s > 2s. Using c^2 - s^2 = c^2 sech^2 s, the condition F'(c) = 2c - 2 s s' < 0 is equivalent to c^2 sech^2 s < c^3 sech^2 s, i.e. to c > 1. The limits follow from s -> 0 as c -> 1 and s/c -> 1 as c -> infinity. ■

Theorem 9.8 (decompactification, l -> infinity at fixed mu, M = -mu). We have lambda_0 <= 2 mu e^{-kappa_0 l} -> 0 with kappa_0 -> mu. The normalized profiles of L_0 near x = 0 and of R_0 near x = l converge, uniformly on compact sets of y and in L^2([0, infinity)) after extension by zero, to sqrt(2 mu) e^{-mu y}, where y is the distance to the respective wall: these are exact chiral zero modes of the two half-line operators (proof: norm(L_0)^2 = sinh(2s)/(4 kappa_0) - l/2 ~ e^{2s}/(8 kappa_0), kappa_0 -> mu, L_0(y) = R_0(l - y), and the same formula gives a uniform exponential bound). Since k_n lies in (n pi/l, (n + 1/2) pi/l), the rest of the spectrum obeys

```
#{n : lambda_n <= E} = (l/pi) sqrt(E^2 - mu^2) + O(1)   for E > mu,
```

a one-dimensional Weyl law (for the positive spectrum; both signs double the leading term). One dimension emerges above the bulk gap mu, and the chirality sits on the two walls. ■

Theorem 9.9 (collapse at fixed light mass). Fix m_f > 0. For each l in (0, 1/m_f) there is a unique mu(l) > 1/l with lambda_0 = m_f. As l -> 0:

- c(l) = mu(l) l -> infinity, with c = log(2c/(m_f l)) + o(1); equivalently c = -W_{-1}(-m_f l/2) + o(1) (Lambert W, lower branch), so c = L + log L + o(1) with L = log(2/(m_f l)): the leading scaling log(1/(m_f l))/l carries a log log correction, and it is specific to this localization model;
- the next eigenvalue exceeds sqrt(mu(l)^2 + (pi/l)^2) -> infinity;
- the light pair stays Connes-Lott with mass m_f (Theorem 9.6), while the interval collapses to a point in Gromov-Hausdorff distance. This is a statement about the compression at each l. As whole triples in the spectral propinquity, the family converges to (C, C^2, m_f sigma_1), a point carrying a massive two-dimensional Dirac block, at distance at least min(sqrt(2)/2, (1/m_f - l)/2) from the Connes-Lott triple (Corollary 15.5 of lemma 15). Both statements hold: the metric compares the whole algebra, and the compression is a different object.

Proof. Existence and uniqueness follow from Lemma 9.7 applied to F(c) = (m_f l)^2 in (0, 1). As l -> 0 we have F -> 0, so c -> infinity, and Lemma 9.5 gives m_f = 2 (c/l) e^{-c}(1 + o(1)). The bound on the next eigenvalue is Theorem 9.4.1. ■

Corollary 9.10 (collapse at fixed bulk mass). If instead mu is fixed and l -> 0, then eventually c < 1, every eigenvalue has absolute value greater than mu, and the smallest absolute value is (pi/(2l))(1 + o(1)) (the lowest root t_0 = k_0 l of tan t/t = 1/(mu l) tends to pi/2). No light fermion survives. This extends Proposition 05-4 to M != 0.

Gauge fields on the interval (scale lemma for T4)

Lemma 9.11. For $-d^{2/dx}2$ on [0, l]:

- Neumann-Neumann conditions give eigenvalues (n pi/l)^2, n >= 0, including a massless mode;
- Neumann-Dirichlet (in either order) gives ((n - 1/2) pi/l)^2, n >= 1;
- Dirichlet-Dirichlet gives (n pi/l)^2, n >= 1.

A gauge field with a Dirichlet condition at one end and a Neumann condition at the other therefore has lightest mass pi/(2l), which equals the Connes-Lott mass of the massless chiral interval (job 05); with Dirichlet conditions at both ends the lightest mass is pi/l. ■

Remark: fermion mass as internal momentum

In this family, masses are internal momenta, with an explicit rule for their quantization. At M = 0 the Connes-Lott mass is the lowest allowed momentum (r - 1/2) pi/l, quantized by the chiral boundary conditions. With a bulk mass, lambda_n^2 = mu^2 + k_n^2 adds the bulk mass and the quantized momentum in quadrature. The light pair's mass, mu sech(kappa_0 l), is exponentially sensitive to mu l. Order-one differences of mu_i l between generations therefore give hierarchical masses of about 2 mu_i e^{-mu_i l}. This is the localization mechanism for Yukawa hierarchies (Arkani-Hamed-Schmaltz 2000; Gherghetta-Pomarol 2000; Grossman-Neubert 2000). In noncommutative-geometry terms, the finite operator D_F of the two-sheet space is the towerless remnant of internal momentum, and its texture becomes localization and overlap data.

Numerical check (Claude)

The secular equations of Theorem 9.4 were compared with the eigenvalues of an independent first-order discretization: the chain of job 05 with mass term (M/2)(R_{i+1} + R_i), N = 4000.

mu (l = 1)	secular	chain
1e-4	1.57073, 4.71237, 7.85397	1.57054, 4.71178, 7.85299
0.5	1.26828, 4.63129, 7.80591	1.26809, 4.63070, 7.80493
3	0.30255, 5.06274, 8.05194	0.30244, 5.06227, 8.05103
6	0.02975, 7.04564, 9.33873	0.02972, 7.04536, 9.33799

At mu = 6, 2 mu e^{-mu l} = 0.029745, against lambda_0 = 0.029747. Lemma 9.7 was confirmed on (1, 40] using the cancellation-free form c^2 sech^2 s. For the collapse at m_f = 1:

l	mu l	mu	lambda_0	next eigenvalue
0.5	2.234	4.47	1.000000	9.56
0.1	4.501	45.0	1.000000	59.2
0.01	7.284	728	1.000000	812
0.001	9.893	9893	1.000000	10487

Referee report (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_09.md. Reproduced as received; the corrections above implement it. Claude checked the two errors, the identity R_0(x) = L_0(l - x) and the asymptotic of 9.5 by hand.

We take $\ell > 0$ throughout. All checks below are analytic; we used no numerical computation.

Lemma 9.1 Verdict: Correct with amendment. The compact-resolvent argument needs the graph-norm estimate

$$\|\psi'\|_{L^2} \le \|D_M\psi\|_{L^2} + |M|\,\|\psi\|_{L^2}.$$

The boundary form in the proof is correct. Each proposed boundary subspace is maximal isotropic. To apply this to the adjoint, distributional differentiation shows that every vector in $\mathrm{Dom}\, D_M^*$ lies in H^1; the boundary form then forces it to satisfy the same boundary conditions. Thus D_M is self-adjoint.

The displayed estimate makes the graph norm control the H^1-norm. Consequently, the resolvent maps L^2 boundedly into H^1, and compactness of $H^1 \hookrightarrow L^2$ gives compact resolvent.

Theorem 9.2 Verdict: Correct.

The zero-mode equations give

$$L(x) = \alpha e^{Mx}, \qquad R(x) = \beta e^{-Mx}.$$

The boundary conditions force $\beta = 0$ in case (a), $\alpha = 0$ in case (b), and both coefficients to vanish in case (c). The grading preserves each domain and anticommutes with D_M, so the stated chiral indices follow.

Corollary 9.3 Verdict: Correct with amendment. Assume that the displayed direct sum has finitely many finite-dimensional G-representations. Under that hypothesis, Theorem 9.2 gives the stated element of $R(G)$.

Without such a hypothesis, the direct-sum operator need not be Fredholm, even if every summand has boundary type (c). For a counterexample, take $\ell = 1$, $G = S^1$, the distinct one-dimensional representations $V_j(z) = z^j$, $j \geq 2$, and $M_j = -j$. Choose a smooth χ equal to 1 on $[0, 1/2]$ and equal to 0 at 1, and put

$$f_j(x) = \chi(x) e^{-jx}.$$

Then $(f_j, 0)$ satisfies condition (c), while

$$A_j f_j = -\chi'(x) e^{-jx}, \qquad \frac{\|A_j f_j\|}{\|f_j\|} = O\big(\sqrt{j}\, e^{-j/2}\big) \longrightarrow 0.$$

After normalization, these vectors lie in distinct orthogonal representation summands. The direct sum has no kernel, but it is not bounded below and hence is not Fredholm. Its ordinary $R(G)$-valued Fredholm index is therefore undefined.

Theorem 9.4 Verdict: Correct with amendment. Specify $\mu > 0$ in the negative-mass cases and $n \geq 1$ in case 3. For an exhaustive list, add $M = 0$, where

$$\operatorname{spec} D_0 = \left\{ \pm \frac{(n + \frac{1}{2})\pi}{\ell} : n \geq 0 \right\}.$$

At $c = 1$, the threshold eigenvalue is μ, and the remaining positive roots have $n \geq 1$ in the intervals stated in case 1.

For $\lambda \neq 0$, eliminating R gives

$$-L'' + M^2 L = \lambda^2 L, \qquad L(\ell) = 0, \quad L'(0) = M L(0).$$

Conversely, every solution gives $R = AL/\lambda$. The squared operator is nonnegative, and Theorem 9.2 excludes $\lambda = 0$. Separated boundary conditions make each squared eigenvalue simple; $\Gamma D_M = -D_M \Gamma$ gives the symmetric spectrum.

For $M = -\mu$, put $c = \mu\ell$. In the subthreshold case, the equation is

$$\frac{\tanh t}{t} = \frac{1}{c}, \qquad t = \kappa\ell > 0.$$

The left side decreases strictly from 1 to 0, so there is exactly one positive root if $c > 1$, and none if $c \leq 1$. It satisfies $\kappa < \mu$.

For the oscillatory roots, let $h(t) = \tan t - t/c$. It is negative on each positive interval $(n\pi - \pi/2, n\pi)$. On $(n\pi, n\pi + \pi/2)$, its derivative increases strictly. For $n \geq 1$, it starts negative in value and tends to $+\infty$, so it has exactly one root. On $(0, \pi/2)$, it has exactly one positive root when $c < 1$, and none when $c \geq 1$. At $c = 1$, the separate linear solution $L = \ell - x$ gives $\lambda = \mu$.

For $M = +\mu$, $\tan t + t/c$ increases strictly from $-\infty$ to a positive value on each $((n - \frac{1}{2})\pi, n\pi)$, $n \geq 1$, giving exactly one root there. It has no positive root on the other half-periods, and there is no subthreshold or threshold solution. These counts establish all three stated cases.

Lemma 9.5 Verdict: Gap, closed here. The displayed proof does not justify the relative asymptotic or make "concentrated" precise.

Write $s = \kappa_0 \ell$ and $c = \mu \ell$. The root equation gives $c = s/\tanh s$. As $c \to \infty$, $s \to \infty$ and $s/c \to 1$; moreover,

$$0 < c - s = c(1 - \tanh s) \leq 2ce^{-2s} \longrightarrow 0.$$

Thus

$$\lambda_0 = \mu \operatorname{sech} s = 2\mu e^{-c} \frac{e^{c-s}}{1 + e^{-2s}} = 2\mu e^{-c}(1 + o(1)).$$

The two stated inequalities also follow directly from $s < c$ and $\operatorname{sech} s \leq 2e^{-s}$.

The root equation simplifies the right profile exactly:

$$R_0(x) = \sinh(\kappa_0 x) = L_0(\ell - x).$$

This proves the equality of norms and shows that the two profiles decrease away from their respective walls. Exponential localization on the scale $1/\kappa_0$ holds when $s \to \infty$. It is not uniform throughout $c > 1$: at fixed ℓ, as $c \downarrow 1$, the normalized profiles approach multiples of $\ell - x$ and x.

Theorem 9.6 Verdict: Correct.

The two chiral basis vectors span the $\pm\lambda_0$ eigenspaces together, and the first two matrix identities follow from

$$D_M u_L = \lambda_0 u_R, \qquad D_M u_R = \lambda_0 u_L.$$

By the reflection identity in Lemma 9.5, the normalized left density exceeds the right density on the left half of the interval, with the reverse inequality on the right half. Hence

$$\langle x \rangle_L < \ell/2 < \langle x \rangle_R.$$

Thus $Q(1)$ and $Q(x)$ span the diagonal algebra, and the commutator norm is $\lambda_0 |a - b|$. The distance between its two pure states is $1/\lambda_0$.

The pullbacks of these states through Q are the two smeared probability measures, not literal endpoint evaluations. Also, the unnumbered sentence immediately after the proof needs correction: the distance is **not determined by ℓ alone**, but it does depend on ℓ. In fact, Lemma 9.7 gives $1/\lambda_0 > \ell$ for $c > 1$.

Lemma 9.7 Verdict: Correct.

Set $q = \operatorname{sech}^2 s$. Since

$$\frac{1}{c} = \frac{\tanh s}{s} > q,$$

implicit differentiation is valid and gives

$$s' = \frac{s}{c(1 - cq)}, \qquad F'(c) = \frac{2cq(1 - c)}{1 - cq} < 0.$$

As $c \downarrow 1$, the inverse relation $c = s/\tanh s$ gives $s \to 0$, so $F \to 1$. As $c \to \infty$, $s \to \infty$ and $c/s \to 1$, so $c^2 \operatorname{sech}^2 s \to 0$.

Theorem 9.8 Verdict: Gap, closed here. The statement asserts convergence of the normalized profiles without proving it or specifying a mode of convergence. The convergence holds uniformly on compact sets of wall distance and in $L^2([0, \infty))$, after extension by zero.

Let $B_\ell = \|L_0\| = \|R_0\|$. With $s = \kappa_0 \ell$,

$$B_\ell^2 = \frac{\sinh(2s)}{4\kappa_0} - \frac{\ell}{2} \sim \frac{e^{2s}}{8\kappa_0}.$$

At fixed μ, $s/c \to 1$, so $\kappa_0 \to \mu$. For fixed $y \geq 0$,

$$\frac{L_0(y)}{B_\ell} = \frac{R_0(\ell - y)}{B_\ell} \longrightarrow \sqrt{2\mu}\, e^{-\mu y}.$$

The same formula supplies a uniform exponential bound, proving the stated L^2 convergence. The limits are exact zero modes of the two separately oriented half-line operators, with left and right chirality respectively. They are not zero modes at finite ℓ.

For $E > \mu$, set $K = \sqrt{E^2 - \mu^2}$. The root bounds

$$n\pi < k_n \ell < (n + \tfrac{1}{2})\pi$$

give

$$\#\{n \geq 1 : k_n \leq K\} = \frac{\ell K}{\pi} + O(1).$$

This is the displayed one-sided counting law. Counting both signs of the Dirac spectrum doubles its leading term.

Theorem 9.9 Verdict: Correct.

Lemma 9.7 makes F a continuous strictly decreasing map from $(1, \infty)$ onto $(0, 1)$. Since $m_f \ell \in (0, 1)$, it gives the unique $c = \mu\ell > 1$. As $\ell \to 0$, $c \to \infty$, and Lemma 9.5 yields

$$m_f \ell = 2ce^{-c}(1 + o(1)), \qquad c = \log \frac{2c}{m_f \ell} + o(1).$$

The first root above the light pair has $k_1 > \pi/\ell$, proving the stated lower bound and divergence.

The original interval metric is $|x - y|$: the mass term commutes with scalar functions, so it does not change the Dirac commutator seminorm. Therefore

$$d_{\mathrm{GH}}([0, \ell], \{\text{point}\}) = \ell/2 \longrightarrow 0.$$

The compressed two-state geometry uses a different seminorm and retains distance $1/m_f$.

Corollary 9.10 Verdict: Error. The assertion that "every eigenvalue exceeds μ" is false: the spectrum is symmetric and contains negative eigenvalues.

The corrected statement is that, for fixed bulk mass and sufficiently small ℓ, **every eigenvalue has absolute value greater than** μ, and the smallest absolute eigenvalue is

$$\min_{\lambda \in \operatorname{spec} D_M} |\lambda| = \frac{\pi}{2\ell}(1 + o(1)).$$

For $M = -\mu < 0$, the lowest root $t_0 = k_0 \ell \in (0, \pi/2)$ satisfies

$$\frac{\tan t_0}{t_0} = \frac{1}{\mu\ell} \longrightarrow \infty,$$

so $t_0 \to \pi/2$, proving the formula. For $M = 0$ it is exact; the positive-mass case follows similarly from the root just above $\pi/2$. Thus the conclusion that no light fermion survives is correct.

Lemma 9.11 Verdict: Error. The three eigenvalue lists are correct, but the gauge-field conclusion lacks a boundary condition at the other end.

For a Dirichlet condition at **both** ends, which in particular includes a Dirichlet condition at one end, the lightest mass is π/ℓ, not $\pi/(2\ell)$. The conclusion is correct for a Dirichlet condition at one end **and a Neumann condition at the other**, for the stated operator $-d^2/dx^2$.

Even if "at one end" is read as exactly one Dirichlet condition, another boundary condition need not be Neumann: $u(0) = 0$, $u'(\ell) = u(\ell)/\ell$ admits the zero mode $u(x) = x$.

Amendments and errors

- **9.1:** Add the graph-norm estimate for compact resolvent.
- **9.3:** Require a finite direct sum, or impose other hypotheses ensuring an equivariant Fredholm operator with index in $R(G)$.
- **9.4:** Specify the root indices and $\mu > 0$, and include $M = 0$ for completeness.
- **9.5:** Supply the relative-asymptotic argument and state localization precisely; the gap is closed above.
- **9.6:** Correct the adjacent unnumbered claim that the distance is unrelated to ℓ.
- **9.8:** Prove and specify the profile convergence; the gap is closed above.
- **9.10:** Replace "every eigenvalue exceeds μ" by the corresponding statement about absolute values.
- **9.11:** Require Neumann at the opposite endpoint for the claimed lightest mass.

Overall: 11 statements checked; 4 correct, 3 correct with amendment, 2 gaps closed above, and 2 errors.

Literature (../LITERATURE.md, section B). Opposite-chirality boundary modes with exponentially small mixing are in Jackiw-Rebbi 1976, Kaplan 1992 and Shamir 1993 (section 3); collapse of spectral triples to a point is treated by Farsi-Latremoliere (J. London Math. Soc., 2026). The tuned collapse mu(l) of Theorem 9.9 was not found in print.

Part 10

Toeplitz compression of the twistor coordinates on CP^3

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): the three identities are correct; the displayed fourth moment had an index error (pairing two barred indices), corrected below. This answers Part A of parked job 04 (research_note 4 and canonical 4, which labelled the result "Checked calculation", although no script computes it). Proved by Claude Opus 5.5 without Astra or Fable; Monte Carlo check with 2 x 10^6 samples.

Let Gamma_1, ..., Gamma_5 be Hermitian generators of Cl(5) on C^4, so that Gamma_a Gamma_b + Gamma_b Gamma_a = 2 delta_ab. For [w] in CP^3 put x_a([w]) = w* Gamma_a w / w* w, and define the level-one Berezin-Toeplitz compression

```
Q(f) = 4 int_{CP^3} f([w]) P_w dmu([w]),
```

where P_w is the rank-one projector onto w and mu is the Fubini-Study probability measure, normalized so that Q(1) = I.

Proposition 10.1.

```
Q(x_a) = Gamma_a / 5,   Q(x_a)^2 = I/25,   Q(x_a^2) = I/5.
```

In particular Q is not multiplicative.

Proof. Lift to w uniform on S^7 in C^4. For n = 4, writing w = g/abs(g) with g a standard complex Gaussian (abs(g) and g/abs(g) are independent), Wick's theorem gives

```
E[w_j w_k conj(w_i) conj(w_l)] = (delta_ij delta_kl + delta_jl delta_ki) / (n(n+1)),
```

and the sixth moment is the sum over the six matchings of unbarred with barred indices, divided by n(n+1)(n+2) = 120.

First, Q(x_a)_kl = 4 sum_ij (Gamma_a)_ij E[conj(w_i) w_j w_k conj(w_l)] = (4/20)(tr(Gamma_a) delta_kl + (Gamma_a)_kl) = Gamma_a/5, because tr Gamma_a = 0.

Second, Q(x_a^2)_kl = (4/120) sum over matchings of {j, q, k} with {i, p, l} of the corresponding contraction of (Gamma_a)_ij (Gamma_a)_pq. Matchings that contain tr Gamma_a vanish. The remaining three give tr(Gamma_a^2) delta_kl = 4 delta_kl, (Gamma_a^2)_kl = delta_kl and (Gamma_a^2)_kl = delta_kl. The total is 6 delta_kl, so Q(x_a^2) = (24/120) I = I/5.

The middle identity follows from Gamma_a^2 = I. ■

Numerical check. With Gamma_a built as Pauli tensor products (Clifford relations verified), the Monte Carlo deviations are max abs(Q(x_a) - Gamma_a/5) <= 0.0009 and max abs(Q(x_a^2) - I/5) <= 0.0006, at a sampling error of about 10^-3.

Part 11

Heavy W in the spectral-action family forces a heavy fermion

Theorem T5 of ../STRATEGY.md. Proved by Claude Opus 5.5 without Astra or Fable. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): Proposition 11.1 correct; Corollaries 11.2 and 11.3 and the regime amendment overstated their conclusions as first written; corrected below, report appended. The result is conditional on the Chamseddine-Connes-Marcolli normalization at the matching (unification) scale.

Hypothesis (CCM relation). In the spectral action of the almost-commutative Standard Model, the gauge normalization g^2 f_0 / (2 pi^2) = 1/4 and the Higgs normalization by a = Tr(Y_nu* Y_nu + Y_e* Y_e + 3(Y_u* Y_u + Y_d* Y_d)) give, at the matching scale,

```
8 M_W^2 = sum over generations of (m_nu^2 + m_e^2 + 3 m_u^2 + 3 m_d^2),
```

where the m are Dirac masses (Chamseddine-Connes-Marcolli, Adv. Theor. Math. Phys. 11 (2007), section 5). In outline (H the unrescaled Higgs field of CCM; the normalization is the three-generation one): the Dirac masses satisfy sum N_f m_f^2 = a abs()^2, while M_W^2 = (g^2/4)(a f_0/pi^2) abs()^2 with f_0 = pi^2/(2 g^2). Majorana masses do not enter a.

Proposition 11.1. Assume the CCM relation at fixed gauge coupling, with N_g generations, and let m_max be the largest Dirac mass. Then

```
m_max / (2 sqrt 2) <= M_W <= sqrt(N_g) m_max.
```

In particular M_W -> infinity if and only if m_max -> infinity.

Proof. Every term has weight at least 1, so 8 M_W^2 >= m_max^2. Each generation has weights 1 + 1 + 3 + 3 = 8, so 8 M_W^2 <= 8 N_g m_max^2. ■

Corollary 11.2 (no-go for the fixed-mass prescription). The heavy-W prescription of canonical 14.1 and research_state 9 (v -> infinity with y_f v fixed for every fermion and the gauge couplings fixed, so that the Yukawa couplings tend to zero) is incompatible with the CCM relation: the right-hand side stays bounded while M_W -> infinity. In the spectral-action family a heavy-W/Z limit forces m_max -> infinity, with m_max >= M_W/sqrt(N_g); some species is then unbounded along a subsequence, though no single fixed species need diverge along the whole family (the referee report gives a family with m_e^2 + m_nu^2 = 8 t^2 in which each mass returns to zero infinitely often). ■

Corollary 11.3 (only the top heavy). Assume that the up-type Dirac mass of one generation (three colors) diverges and every other Dirac mass stays bounded. Then

```
m_t^2 = (8/3) M_W^2 + O(1),   so   m_t / M_W -> sqrt(8/3), approximately 1.633.
```

Top dominance is a hypothesis here: the relation alone also allows, for example, y_e = 2g with every other Yukawa zero, which gives m_e = 2 sqrt(2) M_W and m_t = 0. Under the hypothesis, the surviving SU(3) x U(1)_em theory is vectorlike and anomaly-free, and job 03 gives the residual kinetic weights. ■

Remark (interval comparison). In the massless chiral interval of job 05, the fermion truncation mass and the Neumann-Dirichlet gauge mass both equal pi/(2l) (Lemma 9.11). Under top dominance (Corollary 11.3), the CCM relation instead fixes m_t / M_W = sqrt(8/3) at matching. The interval model and the spectral action therefore assign the top and the W to the same scale with different order-one ratios. Matching them is part of T4.

Amendment (regime of validity; review of 3 Nov 2026; corrected after the referee report). The CCM relation is a relation in the truncated asymptotic (heat-kernel) action at the matching scale Lambda; small Dirac masses relative to Lambda alone do not make it exact for the full finite-cutoff action, which also

needs control of momenta and of the Majorana sector. With fixed Yukawa couplings, M_W and every mass with nonzero Yukawa diverge together as v -> infinity, with ratios fixed by all the Yukawas through the matching constraint (for example y_t = y_nu = g gives m_t/M_W = sqrt 2); Corollary 11.3's ratio needs top dominance. At fixed Lambda a heavy-W limit leaves every fixed small-mass hierarchy (m_max <= eps Lambda implies M_W <= sqrt(N_g) eps Lambda), so it needs Lambda -> infinity jointly or a calculation of the finite-cutoff action. The weights f($m^{2/\mathrm{Lambda}}2$) of lemma 03, Part A, hold under its commuting finite-operator hypothesis and do not apply directly to the broken gauge and Higgs kinetic terms (Part C's w_L(x, 0) = 1 - 3x/4 + O(x^2) differs from f); near the cutoff the coefficients need their own calculation (STRATEGY T5b).

Literature (../LITERATURE.md, section E). The mass relation and the top ratio are in CCM 2007: equation (1.5), Lemma 4.6 equation (4.16) and section 5.4 equation (5.27). Corollary 11.3 restates that result; Proposition 11.1 and Corollary 11.2 apply it to the heavy-W prescription of the corpus.

Referee report on lemmas 10 and 11 (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_10_11.md. Reproduced as received; the corrections above (and in lemma 10) implement it. Claude checked the counterexamples (the rotating e/nu family; y_e = 2g; y_t = y_nu = g) and the corrected fourth moment by hand.

We count Proposition 10.1, Proposition 11.1, the two corollaries, and the regime-of-validity amendment as five statements. The checks below are analytic; we did not rerun the Monte Carlo calculation.

Proposition 10.1 **Verdict: Correct with amendment.** All three identities and the normalization of Q are correct. The displayed fourth-moment formula in the proof has an index error, and its subsequent contraction is consequently incorrect as written.

For w uniformly distributed on $S^7 \subset \mathbb{C}^4$, the moments are

$$\mathbb{E}\left[\prod_{r=1}^{p} w_{j_r}\overline{w_{i_r}}\right] = \frac{\displaystyle\sum_{\sigma\in S_p}\prod_{r=1}^{p}\delta_{j_r i_{\sigma(r)}}}{4\cdot 5\cdots(3+p)}.$$

Thus the fourth moment required in the proof is

$$\mathbb{E}[w_j w_k \overline{w_i}\,\overline{w_l}] = \frac{\delta_{ij}\delta_{kl}+\delta_{jl}\delta_{ki}}{20}.$$

The file instead has $\delta_{il}\delta_{kj}$ as its second term. For example, with $i = l = 1$ and $j = k = 2$, its formula gives $1/20$, whereas the expectation is zero by phase invariance. Contracting the erroneous formula with A_{ij} gives A_{lk}, not the claimed A_{kl}.

The sixth-moment statement is correct: its denominator is $4\cdot 5\cdot 6 = 120$, and its numerator is the sum over all six matchings. Also,

$$Q(1) = 4\mathbb{E}[ww^*] = I.$$

For any 4×4 matrix A, the corrected moments give

$$Q(w^*Aw) = \frac{\mathrm{tr}(A)I + A}{5}$$

and

$$Q((w^*Aw)^2) = \frac{((\operatorname{tr} A)^2 + \operatorname{tr}(A^2))I + 2\operatorname{tr}(A)A + 2A^2}{30}.$$

For $A = \Gamma_a$, conjugation by a different Clifford generator shows $\operatorname{tr}\Gamma_a = 0$, while $\Gamma_a^2 = I$ and $\operatorname{tr}(\Gamma_a^2) = 4$. Consequently,

$$Q(x_a) = \frac{\Gamma_a}{5}, \qquad Q(x_a)^2 = \frac{I}{25}, \qquad Q(x_a^2) = \frac{I}{5}.$$

Proposition 11.1 **Verdict: Correct.** For a fixed, finite N_g, the proof follows from the assumed relation. If S denotes its weighted mass sum, then

$$m_{\max}^2 \le S \le 8N_g m_{\max}^2.$$

Substituting $S = 8M_W^2$ proves both bounds and the equivalence of the two divergence statements.

The CCM relation is quoted correctly for the three-generation model. In Chamseddine–Connes–Marcolli, arXiv:hep-th/0610241v1, equations (1.5), (4.10), (4.15)–(4.17), and (5.25)–(5.26), the weights are $1, 1, 3, 3$; g is the weak coupling at matching; and the neutrino terms are **Dirac Yukawa mass parameters**, not the eigenmasses of the full seesaw matrix. The Majorana matrix does not enter the trace a, although it enters other terms in the action. CCM states that the mass relation is valid "at unification scale."

The outline preceding the proposition should identify its $\langle H \rangle$ as the **unrescaled** spectral Higgs field. CCM's canonically normalized field is $\mathbf{H} = (\sqrt{af_0}/\pi)\varphi$. Equivalently, using the standard normalized Yukawas and v,

$$m_f = \frac{y_f v}{\sqrt{2}}, \qquad M_W = \frac{gv}{2}, \qquad \sum_f N_f y_f^2 = 4g^2,$$

which gives the quoted mass relation.

There is also a scope distinction for $N_g \neq 3$. Proposition 11.1 remains true under its explicitly assumed relation, but CCM derives the coefficient 8 for three generations. If we replicate the same gauge representations N_g times and normalize their gauge kinetic term in the same way, the gauge trace scales by $N_g/3$, giving instead

$$\sum_f N_f m_f^2 = \frac{8N_g}{3} M_W^2.$$

Thus the N_g-generation version with coefficient 8 is an additional assumption, not the automatic extension of the cited normalization.

Corollary 11.2 **Verdict: Error.** The fixed-mass no-go and the displayed bound are correct. The subsequent assertion that a particular Dirac mass must *tend to infinity* is stronger than the relation proves for a general family.

If all finitely many Dirac masses are bounded, Proposition 11.1 bounds M_W. Moreover, it gives exactly

$$m_{\max} \ge \frac{M_W}{\sqrt{N_g}}.$$

Thus $M_W \to \infty$ forces $m_{\max} \to \infty$, and at least one fixed species is unbounded along a subsequence. It need not force one fixed species to tend to infinity along the whole family.

For a counterexample with fixed gauge coupling, let $t \to \infty$, put $M_W = t$, and take all Dirac masses except one charged-lepton mass and one neutrino Dirac mass to be zero. Set

$$\theta(t) = \frac{\pi}{4}(1 + \sin t), \quad m_e = 2\sqrt{2}\, t \cos\theta(t), \quad m_\nu = 2\sqrt{2}\, t \sin\theta(t).$$

Then $m_e^2 + m_\nu^2 = 8t^2$, so the assumed CCM relation holds. Nevertheless, each of m_e and m_ν vanishes on an unbounded sequence. The corresponding Yukawa couplings vary; the gauge coupling can remain fixed.

The description of the fixed-mass prescription should also say **gauge couplings fixed**: nonzero fixed fermion masses with $v \to \infty$ require their Yukawa couplings to tend to zero.

Corollary 11.3 **Verdict: Error.** The displayed asymptotic formula is correct under the stated "only the top heavy" hypothesis. The claim that a heavy top follows from the spectral action, without that choice or another hypothesis selecting it, is false.

Let R be the weighted sum of all the other Dirac masses squared. Under the hypothesis, $R = O(1)$, and

$$8M_W^2 = 3m_t^2 + R \quad \Longrightarrow \quad m_t^2 = \frac{8}{3}M_W^2 - \frac{1}{3}R.$$

This proves the stated $O(1)$ formula and $m_t/M_W \to \sqrt{8/3}$.

For a counterexample to the unconditional heavy-top claim, take $y_e = 2g$ and every other normalized Yukawa coupling, including y_t, equal to zero. The matching constraint is satisfied, $m_e = 2\sqrt{2}\, M_W$, and $m_t = 0$. The relation therefore permits a heavy charged lepton with no heavy top. CCM's equation (5.27) likewise obtains the top ratio by neglecting the other Yukawa couplings; its equation (5.29) considers a comparable neutrino Yukawa instead. The interval-comparison remark needs the same top-dominance qualification.

The assertion about the surviving $SU(3) \times U(1)_{\mathrm{em}}$ representation is correct for the stated Standard Model field content: each surviving charged Dirac fermion has identical left- and right-handed representations of the unbroken group. Their anomaly contributions cancel pairwise, and neutral Majorana states do not contribute to these gauge anomalies.

Amendment on the regime of validity **Verdict: Error.** The warning about a fixed cutoff is justified, but the amendment overstates both the fixed-Yukawa conclusion and the form of the threshold weights.

First, fixed Yukawa couplings do not by themselves imply a heavy top or the ratio in Corollary 11.3. A fixed nonzero y_t gives

$$\frac{m_t}{M_W} = \frac{\sqrt{2}\, y_t}{g},$$

but its value depends on the other Yukawas through the matching constraint. For example, $y_t = y_\nu = g$, with all others zero, satisfies that constraint and gives $m_t/M_W = \sqrt{2}$; both masses diverge as $v \to \infty$. If $y_t = 0$, the top does not diverge. Corollary 11.3 applies only when the other Dirac masses remain bounded.

Second, the displayed CCM relation is a relation in the **truncated asymptotic action**. Small Dirac masses relative to Λ alone do not establish it as an exact relation for the full finite-cutoff action: the expansion also needs control of the other relevant scales, including momenta and the Majorana sector, or a suitable uniform remainder estimate. CCM's own discussion places a Majorana scale of order Λ in one scenario; see its equations (5.16)–(5.21).

At fixed Λ, the amendment correctly observes that a heavy-W limit cannot remain within a fixed small-mass hierarchy. Indeed, if $m_{\max} \le \varepsilon\Lambda$, Proposition 11.1 gives $M_W \le \sqrt{N_g}\varepsilon\Lambda$. A limit within that hierarchy requires a joint family with $\Lambda \to \infty$, together with the necessary control of the remaining scales.

Finally, the simple weights $f(m^2/\Lambda^2)$ in lemma 03, Part A have a commuting finite-operator hypothesis. They cannot be assigned generally to broken gauge and Higgs kinetic terms. In particular, that same file's Part C gives, for a broken weak generator with one heavy and one massless partner and the squared-operator cutoff $f(x) = e^{-x}$,

$$w_L(x, 0) = 3 \int_0^1 u^2 e^{-xu}\, du = 1 - \frac{3}{4}x + O(x^2),$$

which differs from $f(x) = 1 - x + O(x^2)$. Near the cutoff, the appropriate coefficients and any replacement for the CCM relation require a calculation under specified hypotheses.

Amendments and errors found

- Correct the fourth-moment indices in Proposition 10.1's proof; the three proposition identities are unchanged.
- Specify the unrescaled Higgs field in the CCM outline and retain the three-generation scope of the cited normalization.
- In Corollary 11.2, assert divergence of $m_{\max}$, or add a hypothesis ensuring that one fixed species tends to infinity.
- In Corollary 11.3 and the interval remark, make top dominance a hypothesis; it is not forced by the mass relation.
- In the regime amendment, qualify fixed-Yukawa claims, distinguish the truncated action from the finite-cutoff action, and restrict the simple threshold weights to their applicable commuting setting.

Overall: 5 statements checked; 1 correct, 1 amended, 0 gaps, 3 errors.

Part 12

Boundary Yukawa matrices on the interval: from chiral to vectorlike boundary conditions, and towerless textures

Theorems T3 (remaining part) and T8 of ../STRATEGY.md. Proved by Claude Opus 5.5 without Astra or Fable. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): of 9 statements, 4 correct, 2 needed amendments and 3 were wrong as first stated (Theorem 12.4 item 2, Theorem 12.7, Corollary 12.9); all corrections are incorporated below and the referee report is appended. The secular equation was checked against direct numerical integration of the ODE; the bounds of Theorem 12.6 and Lemma 12.8 were checked on random samples (section "Numerical check").

Setting

Let I = [0, l]. A doublet-type field Q in L^2(I; C^2) (x) C^{N_Q} and a singlet-type field U in L^2(I; C^2) (x) C^{N_U} have components (L, R), with grading Gamma = diag(1, -1) on each. As in lemma 09, the bulk operator is (D psi)_L = R' + M R, (D psi)_R = -L' + M L, with diagonal bulk masses M_Q = diag(m_i^Q) and M_U = diag(m_j^U). For a complex N_Q x N_U matrix X, let D_X be this operator on the domain of H^1 functions with

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Q_R(0) = 0,   U_L(0) = 0,   Q_R(l) = X U_R(l),   U_L(l) = -X^dagger Q_L(l).
```

At X = 0, Q has boundary type (a) and U type (b) of lemma 09. The matrix X is a boundary Yukawa coupling: it couples the two fields only at the wall x = l. We write c_i^Q = -m_i^Q l and c_j^U = m_j^U l, so that c > 0 means that the zero mode of the field at X = 0 is localized near x = 0, away from the Yukawa wall.

Lemma 12.1. D_X is self-adjoint with compact resolvent, and Gamma D_X = -D_X Gamma.

Proof. The boundary form of lemma 9.1, summed over both fields, is [conj(Q_R) phi_{Q,L} - conj(Q_L) phi_{Q,R} + conj(U_R) phi_{U,L} - conj(U_L) phi_{U,R}] evaluated between 0 and l. At x = l the conditions leave Q_L and U_R free, which is half of the boundary values, and on them the form becomes conj(U_R) X^dagger phi_{Q,L} - conj(Q_L) X phi_{U,R} - conj(U_R) X^dagger phi_{Q,L} + conj(Q_L) X phi_{U,R} = 0. At x = 0 the conditions are those of types (a) and (b). Hence the conditions define a Lagrangian subspace, and the argument of lemma 9.1 applies. Gamma flips the signs of Q_R and U_R together, so it preserves each condition. ■

Kernel, index and symmetry breaking

Theorem 12.2. For every X,

```
ker D_X = {(e^{M_Q x} a, 0; 0, 0) : X^dagger e^{M_Q l} a = 0}  (+)  {(0, 0; 0, e^{-M_U x} b) : X
↪  e^{-M_U l} b = 0},
```

so dim ker_{Gamma=+1} = N_Q - rank X, dim ker_{Gamma=-1} = N_U - rank X, and ind D_X = N_Q - N_U.

Proof. As in Theorem 9.2, D psi = 0 forces L = e^{Mx} a and R = e^{-Mx} b on each field. The conditions at 0 remove Q_R and U_L, and the conditions at l reduce to X U_R(l) = 0 and X^dagger Q_L(l) = 0. ■

Let a compact group G act unitarily on C^{N_Q} and C^{N_U} by rho_Q and rho_U, commuting with the bulk masses. The domain of D_X is G-invariant exactly for g in the stabilizer H_X = {g : rho_Q(g) X = X rho_U(g)}.

Corollary 12.3 (equivariant index and a no-go). In R(H_X),

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ind_{H_X} D_X = [ker X^dagger] - [ker X] = [rho_Q] - [rho_U]   (restricted to H_X),
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because X maps (ker X)^perp isomorphically and H_X-equivariantly onto ran X = (ker X^{dagger})perp. At X = 0 we have H_X = G, and the index is the chiral class [V_L] - [V_R] of Corollary 9.3. D_X has no zero modes iff X is invertible. This requires rho_Q and rho_U to be isomorphic as H_X-representations, i.e. the residual theory must be vectorlike. For the Standard Model with H = SU(3) x U(1)_em, the quark sector satisfies this. A lepton sector without nu_R does not: a massless left-handed neutrino is forced for every X preserving electromagnetism (H contained in H_X), in agreement with the unpaired nu_L of job 06. ■

Exact secular equation

For a field of mass m and lambda > 0 let omega = (m^2 - $\mathrm{lambda}^{2)}\{1/2\}$, and read omega coth(omega l) as k cot(k l) when omega = ik and as 1/l when omega = 0. Define

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alpha_i(lambda) = omega coth(omega l) + m_i^Q,     beta_j(lambda) = omega coth(omega l) - m_j^U,
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A = diag(alpha_i) and B = diag(beta_j). Let W be the set of lambda > 0 with lambda^2 < m^2 + $\mathrm{pi}^{2/l}2$ for every bulk mass m of the system.

Theorem 12.4. For lambda in W, lambda is an eigenvalue of D_X iff det(lambda^2 - A(lambda) X B(lambda) X^dagger) = 0, and its multiplicity is dim ker(lambda^2 - A X B X^dagger). Moreover:

1. alpha_i(0) = 2m/(1 - e^{-2ml}) with m = m_i^Q, which equals f_{Q_i}^2, the value at x = l of the normalized zero mode e^{mx}; likewise beta_j(0) = 2m/(e^{2ml} - 1) = f_{U_j}^2 for e^{-mx}. In terms of c, f^2 = phi(c)^2/l with phi(c)^2 = 2c/(e^{2c} - 1) and phi(0) = 1.
2. On its own window 0 < lambda^2 < m^2 + $\mathrm{pi}^{2/l}2$, each alpha_i (and each beta_j) is a strictly decreasing, concave function of lambda^2, positive at 0 and tending to -infinity at the edge of that window, so it vanishes at exactly one point: the lowest positive eigenvalue Lambda_i^Q of Q_i alone with R(0) = 0 and L(l) = 0, respectively Lambda_j^U of U_j alone with L(0) = 0 and R(l) = 0 (the type-(c) problems of lemma 09). On the common window W it is still decreasing and concave but may stay positive (e.g. l = 1, M_Q = diag(0, 4): alpha_2 > 0 on all of W = (0, pi)).

Proof. With R(0) = 0, the Q-field solution is R = a sinh(omega x)/omega and L = a(cosh(omega x) + m sinh(omega x)/omega)/lambda. With L(0) = 0, the U-field solution is L = b sinh(omega x)/omega and R

= b(m sinh(omega x)/omega - cosh(omega x))/lambda. On W, every S = sinh(omega l)/omega is nonzero. Inserting these into the conditions at l and eliminating b gives lambda^2 c = X B X^dagger A c for c = diag(S_Q) a. The map c -> eigenfunction is linear and injective. Finally, lambda^2 - X B X^dagger A and lambda^2 - A X B X^dagger have kernels of equal dimension for lambda != 0. For item 2, the expansion x coth x = 1 + sum_{n >= 1} $2x^{2/(x}2$ + n^2 pi^2) gives

```
z^{1/2} coth(z^{1/2} l) = (1/l) [1 + sum_{n >= 1} 2 z l^2 / (z l^2 + n^2 pi^2)],
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which is strictly increasing and concave in z > $-pi^{2/l}2$. Substituting z = m^2 - lambda^2 gives the claim. The function tends to -infinity at the edge of its own window, and alpha_i = 0 means Q_L(l) = 0. ■

Example 12.5 (one flavor, no bulk mass: the interpolation). For N_Q = N_U = 1 and m = 0 we get alpha = beta = lambda cot(lambda l), and the equation reduces to tan^2(lambda l) = abs(X)^2. With theta = arctan abs(X) in [0, pi/2], the positive spectrum is

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theta/l,   (n pi - theta)/l,   (n pi + theta)/l   (n >= 1).
```

At X = 0 the light pair is the pair of chiral zero modes of types (a) and (b): zero then has multiplicity two and the positive eigenvalues are n pi/l, each of multiplicity two; theta = pi/2 is reached only as a limit. As abs(X) -> infinity the conditions tend to Q_L(l) = 0 and U_R(l) = 0, so each field becomes type (c), and the spectrum tends to two copies of (r - 1/2) pi/l. The light pair theta/l therefore interpolates continuously between the chiral zero modes and the Connes-Lott mass pi/(2l) of job 05. Its eigenvectors span a Gamma-graded two-dimensional space on which D acts as (theta/l) sigma_1, with the Gamma = +1 vector carried by (Q_L, U_L) and the Gamma = -1 vector by (Q_R, U_R). ■

Light spectrum and towerless limit

Let F_Q = diag(f_{Q_i}), F_U = diag(f_{U_j}) and Y_0 = F_Q X F_U. Let Lambda_Q = min_i Lambda_i^Q, Lambda_U = min_j Lambda_j^U and lambda_* = min(Lambda_Q, Lambda_U). For 0 < lambda < lambda_* the matrices A and B are positive, and we put Y(lambda) = A(lambda)^{1/2} X B(lambda)^{1/2}. Singular values sigma_k are listed in decreasing order.

Theorem 12.6.

1. lambda in (0, lambda_*) is an eigenvalue of multiplicity m iff exactly m indices k satisfy sigma_k(Y(lambda)) = lambda.
2. For each k, the equation lambda = sigma_k(Y(lambda)) has at most one solution lambda_k in (0, lambda_). *It has one whenever 0 < sigma_k(Y_0) < lambda_.*
3. Every such solution satisfies

   ```
   sigma_k(Y_0) (1 - lambda_k^2/Lambda_Q^2 - lambda_k^2/Lambda_U^2) <= lambda_k <= sigma_k(Y_0).
   ```

Proof. Each Lambda lies in the window of its own field, so (0, lambda_) *is contained in W. (1) Conjugating by A^{1/2} turns A X B X^dagger into Y Y^dagger, and dim ker(lambda^2 - Y Y^dagger) = #{k : sigma_k(Y) = lambda}. Then apply Theorem 12.4. (2) A and B decrease entrywise in lambda, and sigma_k(P Z Q) <= norm(P) norm(Q) sigma_k(Z). Hence sigma_k(Y(lambda)) is continuous and nonincreasing, so lambda - sigma_k(Y(lambda)) is strictly increasing. It is negative near 0 and at least lambda_* - sigma_k(Y_0) > 0* near lambda_*. (3) The upper bound is (2). For the lower bound, concavity places alpha_i above its chord: alpha_i(lambda) >= alpha_i(0)(1 - $\text{lambda}^{2/(\text{Lambda_i}}$Q)^2), and likewise for beta_j. Therefore Y(lambda) = D_1 Y_0 D_2 with diagonal D_1, D_2 whose entries are at least (1 - t_Q)^{1/2} and (1 - t_U)^{1/2}, where t_Q = $\text{lambda}^{2/\text{Lambda_Q}}$2 and t_U = $\text{lambda}^{2/\text{Lambda_U}}$2. Hence sigma_k(Y) >= ((1 - t_Q)(1 - t_U))^{1/2} sigma_k(Y_0) >= (1 - t_Q - t_U) sigma_k(Y_0). The last step uses a^2 + ab + b^2 <= a + b when a + b <= 1; for a + b > 1 the bound is trivial. ■

Theorem 12.7 (towerless limit of the texture). Fix the localization parameters c_i^Q and c_j^U and the matrix Y_hat = X/l, and let l -> 0. Put Phi_Q = diag(phi(c_i^Q)), Phi_U = diag(phi(c_j^U)) and M_F = Phi_Q Y_hat Phi_U. For every E > 0 and all small l, the spectrum of D_X in [-E, E] consists of:

- 0 with multiplicity N_Q + N_U - 2 rank Y_hat;

- plus/minus lambda_k(l) for those k <= rank Y_hat with sigma_k(M_F) <= E (since lambda_k(l) < sigma_k(M_F) and lambda_k(l) -> sigma_k(M_F)), where

  ```
  abs(lambda_k(l) - sigma_k(M_F)) <= sigma_k(M_F)^3 l^2 (1/L_Q^2 + 1/L_U^2),
  ```

  and L_Q = l Lambda_Q and L_U = l Lambda_U depend only on the c's.

All eigenvalues outside zero and the light set {+/- lambda_k} have absolute value greater than lambda_* = min(L_Q, L_U)/l, which tends to infinity. For E > norm(M_F) the light space has dimension N_Q + N_U and, after grading-preserving unitary identifications, its compressed Dirac matrices converge to D_F below. Calling the limit a spectral triple requires specifying an algebra and its representation; with that understood, the towerless remnant is the finite even triple with H_F = C^{N_Q} (+) C^{N_U}, Gamma_F = diag(1, -1) and

```
D_F = [[0, M_F], [M_F^dagger, 0]],   M_F = Phi_Q Y_hat Phi_U,
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up to unitaries on each chirality, which is the Yukawa block of the Chamseddine-Connes finite Dirac operator.

Proof. By item 1 of Theorem 12.4, Y_0 = (1/l) Phi_Q X Phi_U = M_F. Rescaling x -> x/l shows that l Lambda_i depends on c_i alone. Apply Theorem 12.6 with sigma_k(Y_0) fixed and lambda_* -> infinity, and use lambda_k <= sigma_k(M_F) in the error term. Theorem 12.2 counts the zero modes. ■

At fixed fermion masses, the towerless limit therefore requires X = l Y_hat -> 0: the boundary condition tends to the chiral type (a) (+) (b), while the normalized profiles at the wall grow like l^{-1/2}.

Hierarchies from localization

Lemma 12.8 (graded singular values). Let M = F X G with F = diag(f_1 <= … <= f_N) and G = diag(g_1 <= … <= g_N) positive, and X in M_N(C). For p = 1, …, N let T_p = {N-p+1, …, N}, Delta_p = abs(det X_{T_p T_p}), D_p = max over p-subsets I, J of abs(det X_{IJ}), eps_p^F = f_{N-p}/f_{N-p+1}, eps_p^G = g_{N-p}/g_{N-p+1} (both 0 for p = N), and K_p = binom(N, p) - 1. Then

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f_{T_p} g_{T_p} Delta_p  <=  sigma_1 ... sigma_p  <=  f_{T_p} g_{T_p} [Delta_p + D_p ((1 + K_p
↪  eps_p^F)(1 + K_p eps_p^G) - 1)],
```

where f_I is the product of f_i over i in I.

Proof. By Cauchy-Binet, the p-th compound is Lambda^p M = Lambda^p F Lambda^p X Lambda^p G, with entries f_I g_J det X_{IJ}, and sigma_1 … sigma_p = norm(Lambda^p M). The lower bound is the (T_p, T_p) entry. For the upper bound, norm(A) <= sum abs(A_{IJ}). Any p-subset I != T_p trades at least one index of T_p for an index <= N - p, so f_I/f_{T_p} <= eps_p^F. Hence sum_I f_I <= f_{T_p}(1 + K_p eps_p^F), and similarly for g. ■

Corollary 12.9 (masses from localization). Assume every trailing minor Delta_p is nonzero. Put E_0 = E_N = 0 and E_p = (D_p/Delta_p)((1 + K_p eps_p^F)(1 + K_p eps_p^G) - 1) for 1 <= p < N. Then

```
1/(1 + E_{p-1}) <= sigma_p / (f_{N-p+1} g_{N-p+1} Delta_p/Delta_{p-1}) <= 1 + E_p,
```

by dividing the bounds of Lemma 12.8 for p and p - 1. Indexing generations by k = N - p + 1, m_k = f_k g_k (Delta_{N-k+1}/Delta_{N-k})(1 + O(eps)) as eps = max_p (eps_p^F + eps_p^G) -> 0, for fixed X and N (uniform statements need uniform bounds on D_p/Delta_p). Without the minor condition the formula fails: F = G = diag(t, 1) and X = sigma_1 give Delta_1 = 0 while both singular values equal t. With M_F = Phi_Q Y_hat Phi_U and Y_hat anarchic (all minors of order one), the masses follow phi(c_k^Q) phi(c_k^U). Since phi(c) = (2c)^{1/2} e^{-c} (1 + O(e^{-2c})) for large c, order-one differences of the c's between generations give exponential hierarchies (the ratio also carries the factor sqrt(c_k^Q $c_k^{U/(c_{k+1}Q\ c_{k+1}^U)}$), bounded when adjacent c's differ by bounded amounts):

```
m_k / m_{k+1} ~ e^{-(c_k^Q - c_{k+1}^Q) - (c_k^U - c_{k+1}^U)}   up to factors of order one. ■
```

Remark 12.10 (mixing; numerical, not proved here). For an up sector and a down sector sharing Phi_Q, we drew 500 anarchic complex samples with c^Q = (3, 2, 0.5). The medians of abs(V_ij)/(phi(c_i^Q)/phi(c_jQ)) were 1.02, 0.92 and 1.09 for (ij) = (12), (23) and (13). So the left mixing follows the doublet localization alone, abs(V_ij) ~ phi(c_i^Q)/phi(c_jQ), as in the split-fermion and warped-flavor literature (Huber-Shafi 2001; Agashe-Perez-Soni 2005; Casagrande et al. 2008). An upper bound with explicit constants, |V_ij| <= 3 K_u K_d f_i/f_j for i < j, is proved in the referee report appended below, which also notes that there is no universal lower bound; Astra 6 wrote out the cofactor step that the report names (validation, item 10).

Consequences for STRATEGY

- **T3 (remaining part).** The boundary Yukawa X deforms the chiral types (a) (+) (b) (X = 0) continuously into two copies of the vectorlike type (c) (abs(X) -> infinity). The index N_Q - N_U is constant along the way. What changes is the symmetry: X reduces G to H_X, and the zero modes lift exactly as far as the H_X-index allows (Corollary 12.3).
- **T8.** In the towerless limit, the Yukawa block of D_F is Phi_Q Y_hat Phi_U. The texture of the finite geometry is localization data (the c's) times an order-one boundary matrix, the masses are internal momenta and overlaps, and the hierarchies are exponential in the c's (Corollary 12.9).

Numerical check (Claude)

Method: roots of the secular equation are bracketed on fine grids and refined by Brent's method; the shooting check integrates the ODE from x = 0 with scipy `solve_ivp` (rtol 1e-12) and finds the zeros of the boundary-condition determinant at x = l.

- **Secular equation against ODE shooting** (scipy `solve_ivp`, rtol 1e-12, real X). For one massless flavor and X = 0.3, 1 and 5, the shooting roots agree with (n pi +/- arctan X)/l to 6 digits. For three flavors with masses (-3, -1.5, 0.2) and (2.5, 1, -0.3), l = 1 and random X, both methods give 0.024689, 0.048653, 0.534569 and 2.03374.
- **Theorem 12.6** (c^Q = (2, 1, 0), c^U = (1.5, 0.5, -0.5), X = 0.05 x Gaussian, lambda_* = 0.577). The eigenvalues in the window are 0.000576, 0.035602 and 0.061096, against sigma_k(Y_0) = 0.000576, 0.035632 and 0.061188. All three two-sided bounds hold.
- **Theorem 12.7** (c^Q = (4, 2, 0.3), c^U = (3, 1, -0.2), Gaussian Y_hat, sigma(M_F) = 0.000735, 0.072042, 0.786929). The relative errors of the three light eigenvalues are:

l	relative errors	lambda_*
1	1.5e-5, 1.1e-2, 2.5e-1	0.147
0.1	below 1e-6, 1.1e-4, 4.2e-3	1.47
0.03	below 1e-6, 1.0e-5, 3.9e-4	4.90

  The errors decrease as l^2.
- **Lemma 12.8:** no violation in 2000 random complex 3 x 3 cases with log-uniform gradings over 8 e-folds (6000 inequalities).

Referee report (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_12.md. Reproduced as received; the corrections above implement it. Claude checked the three counterexamples (M_Q = diag(0, 4); arctan(l)/l -> 1; F = G = diag(t, 1), X = sigma_1) by hand.

We take $\ell > 0$ and the bulk masses to be real, as in Lemma 09. All arguments below are analytic; we used no numerical computation.

Lemma 12.1 Verdict: Correct.

At $x = \ell$, the free boundary values are $Q_L(\ell)$ and $U_R(\ell)$. Substitution of the boundary conditions into the boundary form gives zero, and the allowed boundary values have half the total boundary dimension. The conditions at $x = 0$ are likewise maximal isotropic. The adjoint boundary conditions therefore coincide with the stated ones, so D_X is self-adjoint.

The graph norm controls the H^1-norm, and the embedding $H^1([0,\ell]) \hookrightarrow L^2([0,\ell])$ is compact. Thus the resolvent is compact. Finally, Γ preserves the boundary conditions and anticommutes with the differential expression.

Theorem 12.2 Verdict: Correct.

The zero-mode equations give

$$L(x) = e^{Mx}a, \qquad R(x) = e^{-Mx}b.$$

The conditions at 0 remove Q_R and U_L, and those at ℓ impose exactly the two displayed kernel conditions.

The endpoint exponentials are invertible, so the two chiral kernel dimensions are $N_Q - \operatorname{rank} X$ and $N_U - \operatorname{rank} X$. Their difference is $N_Q - N_U$.

Corollary 12.3 Verdict: Correct with amendment. The neutrino conclusion holds for every X that preserves the stated electromagnetic symmetry, meaning $H \subseteq H_X$. The phrase "for every X" needs that qualification.

The endpoint exponentials commute with the group actions. They identify the actual chiral kernels equivariantly with $\ker X^\dagger$ and $\ker X$. The orthogonal complements are equivariantly isomorphic through X, proving the displayed index identity. Both kernels vanish exactly when X is invertible, in which case X is an equivariant isomorphism.

If electromagnetic symmetry is not required, the neutrino assertion can fail. For one lepton generation with zero bulk masses, take

$$Q_L = (\nu_L, e_L), \qquad U_R = e_R, \qquad X = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The left kernel is then spanned by e_L, while ν_L is coupled to e_R. This X breaks electromagnetic charge. If X intertwines the electromagnetic representations and there is no neutral right-handed lepton, the neutrino subspace lies in $\ker X^\dagger$, as claimed.

Theorem 12.4 Verdict: Error. The secular equation, multiplicity formula, endpoint values, monotonicity and concavity are correct. The assertion that *each* function has a zero **on the common window** W is false.

For example, take $\ell = 1$, $M_Q = \operatorname{diag}(0, 4)$, and $M_U = (0)$. Then $W = (0, \pi)$, but throughout W,

$$\alpha_2(\lambda) = \sqrt{16 - \lambda^2}\,\coth\sqrt{16 - \lambda^2} + 4 > 0.$$

Its individual lowest mixed-boundary eigenvalue is greater than 4, hence lies outside W. The proof's assertion that every function tends to $-\infty$ at the edge of W fails for the same reason.

The corrected zero statement uses each mass's **individual** first-pole window

$$0 < \lambda < \sqrt{m^2 + \pi^2/\ell^2}.$$

On that interval the corresponding α or β has exactly one zero, at the lowest positive eigenvalue of its mixed-boundary problem. The U problem is equivalent to the type-(c) problem after exchanging components and changing the mass sign.

For the monotonicity and concavity, write $s = \lambda^2$ and $a_n = (n\pi/\ell)^2$. Differentiating the displayed partial-fraction expansion on the individual window gives

$$\frac{d}{ds}(\omega \coth(\omega\ell)) = -\frac{2}{\ell}\sum_{n\geq 1}\frac{a_n}{(m^2 - s + a_n)^2} < 0,$$

$$\frac{d^2}{ds^2}(\omega \coth(\omega\ell)) = -\frac{4}{\ell}\sum_{n\geq 1}\frac{a_n}{(m^2 - s + a_n)^3} < 0.$$

The function is positive at $s = 0$ after the appropriate $\pm m$ is added, and tends to $-\infty$ at its own first pole. This proves the corrected zero claim.

For the secular equation, the proof's boundary variables c, d satisfy

$$\lambda c = -XBd, \qquad \lambda d = -X^\dagger Ac.$$

Eliminating d gives $\lambda^2 c = XBX^\dagger Ac$. For $\lambda \neq 0$, the kernels of $\lambda^2 I - CA$ and $\lambda^2 I - AC$, with $C = XBX^\dagger$, are isomorphic via A, with inverse C/λ^2. This also proves the stated multiplicity.

Example 12.5 Verdict: Correct with amendment. At $X = 0$, omit $\theta/\ell = 0$ from the **positive** spectrum; instead, zero has multiplicity two. The positive eigenvalues are then $n\pi/\ell$, each with multiplicity two. For finite X, $\theta < \pi/2$; the endpoint $\theta = \pi/2$ is a limit.

For all positive λ, direct substitution of the massless solutions into the boundary conditions gives

$$\sin^2(\lambda\ell) - |X|^2\cos^2(\lambda\ell) = 0.$$

For $X \neq 0$, its roots are exactly the listed values. This derivation also includes the roots at $n\pi/\ell$ when $X = 0$, even though the secular formula of Theorem 12.4 was derived only before the first pole. The negative spectrum follows by grading symmetry. The stated two-dimensional graded block and the limiting mixed-boundary spectrum follow from these roots and their eigenvectors.

Theorem 12.6 Verdict: Correct.

Although Theorem 12.4 requires the correction above, it still supplies what is needed here. The smallest individual first pole belongs to some field; that field's first zero lies before its pole. Consequently, λ_* lies before the smallest first pole, and $(0, \lambda_*) \subset W$.

On this interval A, B are positive. Conjugation reduces the secular equation to the eigenvalue equation for $Y(\lambda)Y(\lambda)^\dagger$, proving the multiplicity claim. If $\lambda_2 > \lambda_1$, then

$$Y(\lambda_2) = P\,Y(\lambda_1)\,Q$$

for diagonal contractions P, Q. Thus every singular value is continuous and nonincreasing, so it meets the strictly increasing function λ at most once. The stated condition at $\lambda = 0$ gives a crossing before λ_*.

Concavity gives, with $t_Q = \lambda^2/\Lambda_Q^2$ and $t_U = \lambda^2/\Lambda_U^2$,

$$\sigma_k(Y(\lambda)) \geq \sqrt{(1 - t_Q)(1 - t_U)}\,\sigma_k(Y_0) \geq (1 - t_Q - t_U)\sigma_k(Y_0).$$

Together with $\sigma_k(Y(\lambda)) \leq \sigma_k(Y_0)$, this proves both bounds.

Theorem 12.7 Verdict: Error. The fixed-energy spectrum cannot contain every listed light eigenvalue for every $E > 0$.

Take one massless flavor, $\widehat{Y} = 1$, and $E = 1/2$. Then $X = \ell$, and Example 12.5 gives the positive light eigenvalue

$$\lambda(\ell) = \frac{\arctan\ell}{\ell} \longrightarrow 1.$$

For sufficiently small ℓ, it is outside $[-1/2, 1/2]$, contrary to the stated list.

The corrected fixed-energy statement includes only those $\pm\lambda_k(\ell)$ with $\lambda_k(\ell) \le E$. Equivalently, for each fixed $E > 0$ and all sufficiently small ℓ, include exactly those indices for which

$$\sigma_k(M_F) \le E.$$

The equality case is included because the strict decrease of A and B gives $\lambda_k(\ell) < \sigma_k(M_F)$ for every positive light eigenvalue.

The displayed error estimate follows from Theorem 12.6:

$$0 < \sigma_k(M_F) - \lambda_k(\ell) \le \sigma_k(M_F)^3 \ell^2 \left(L_Q^{-2} + L_U^{-2}\right).$$

Zero has the stated multiplicity. Apart from zero and the **entire** light set $\{\pm\lambda_k\}$, the eigenvalues have **absolute value** greater than λ_* for small ℓ. At λ_*, the limiting matrices A, B are nonnegative and the relevant singular values are at most $\|M_F\|$, whereas $\lambda_* \to \infty$; hence equality at the boundary is excluded for small ℓ.

For $E > \|M_F\|$, the full low-energy space has dimension $N_Q + N_U$. After grading-preserving unitary identifications, its compressed Dirac matrices converge to the displayed D_F. Calling the limit a spectral triple additionally requires an algebra and its representation to be specified.

Lemma 12.8 Verdict: Correct.

The entries of $\bigwedge^p M$ are $f_I g_J \det X_{IJ}$, and its operator norm is $\sigma_1 \cdots \sigma_p$. Its (T_p, T_p) entry gives the lower bound.

For every $I \neq T_p$, at least one factor from the top p indices is replaced by one no larger than $N - p$, so $f_I / f_{T_p} \le \epsilon_p^F$; the analogous estimate holds for g. Bounding the operator norm by the sum of absolute entries, while keeping the (T_p, T_p) entry separate, gives the upper bound. For $p = N$, both bounds reduce to the determinant identity.

Corollary 12.9 Verdict: Error. The displayed mass formula is not valid under the hypotheses of Lemma 12.8: its denominator can vanish, and its leading term can be zero when the singular value is positive.

For $0 < t < 1$, take

$$F = G = \operatorname{diag}(t, 1), \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then FXG has both singular values equal to t, but $\Delta_1 = 0$ and $\Delta_2 = 1$. The formula for m_2 gives zero, and the formula for m_1 divides by zero.

A sufficient correction is to require every trailing minor Δ_p to be nonzero and to define the small parameter. More precisely, put $E_0 = E_N = 0$ and, for $1 \le p < N$,

$$E_p = \frac{D_p}{\Delta_p} \left[(1 + K_p \epsilon_p^F)(1 + K_p \epsilon_p^G) - 1\right].$$

Lemma 12.8 then gives the explicit bounds

$$\frac{1}{1 + E_{p-1}} \le \frac{\sigma_p}{f_{N-p+1} g_{N-p+1} \Delta_p / \Delta_{p-1}} \le 1 + E_p.$$

Indeed, apply the lemma to the products of the first p and first $p-1$ singular values and divide their bounds. For fixed X, N, if

$$\epsilon = \max_{p<N} (\epsilon_p^F + \epsilon_p^G) \longrightarrow 0,$$

these inequalities give the stated $1 + O(\epsilon)$ formula. Uniform estimates require uniform bounds on D_p/Δ_p.

The localization conclusion follows under quantitative order-one upper and lower bounds on the relevant minors. For large positive c, the mass ratio also contains the factor

$$\sqrt{\frac{c_k^Q c_k^U}{c_{k+1}^Q c_{k+1}^U}}.$$

It is bounded when the adjacent large c's differ by bounded amounts; without such control it cannot be suppressed uniformly as an order-one factor.

Remark 12.10 Bound. We can prove an upper bound independent of the right-hand localization profiles. It does not prove the reported sample medians or a universal lower bound.

For each sector $s = u, d$, let

$$M_s = F X_s G_s, \qquad F = \mathrm{diag}(f_1 \le f_2 \le f_3), \qquad G_s = \mathrm{diag}(g_1^{(s)} \le g_2^{(s)} \le g_3^{(s)}),$$

with all diagonal entries positive. Assume X_s is invertible and its trailing minors $\Delta_1^{(s)}, \Delta_2^{(s)}$ are nonzero. With $D_p^{(s)}$ as in Lemma 12.8, define

$$K_s = \left(1 + \frac{\sqrt{3} D_1^{(s)}}{\Delta_1^{(s)}}\right)\left(1 + \frac{\sqrt{3} D_2^{(s)}}{\Delta_2^{(s)}}\right).$$

Choose the columns of the left singular-vector matrices U_s in increasing order of singular value, and put $V = U_u^\dagger U_d$. Then, for $i < j$,

$$\boxed{|V_{ij}| \le 3 K_u K_d \frac{f_i}{f_j}.}$$

To prove this, set $H_s = M_s M_s^\dagger$, and temporarily suppress s. Write

$$a = \frac{\sqrt{3} D_1}{\Delta_1}, \qquad b = \frac{\sqrt{3} D_2}{\Delta_2}.$$

For the largest eigenvalue λ_3 of H,

$$\lambda_3 \ge f_3^2 g_3^2 \Delta_1^2, \qquad H_{ii} \le 3 D_1^2 f_i^2 g_3^2.$$

Cauchy–Schwarz therefore gives $|U_{i3}| \le a f_i / f_3$. Applying the same argument to H^{-1}, and using $(X^{-1})_{11} = \det X_{\{2,3\},\{2,3\}} / \det X$, gives $|U_{i1}| \le b f_1 / f_i$. Cofactor identities for the unitary U bound the two remaining off-diagonal entries. Together,

$$|U_{ij}| \le (1 + a)(1 + b) \min\left(\frac{f_i}{f_j}, \frac{f_j}{f_i}\right).$$

Summing the three terms in $V_{ij} = \sum_k \overline{(U_u)_{ki}} (U_d)_{kj}$ proves the boxed inequality.

There is no corresponding positive universal lower bound: identical up and down matrices have $V = I$, including when their minors are nonzero. The numerical medians also require the sampling distribution and sampled matrices, neither of which is specified in the remark.

Amendments and errors

- **12.3:** Restrict the neutrino assertion to couplings preserving electromagnetic symmetry.
- **12.4:** State the unique-zero claim on each field's individual first-pole window; it can fail on the common window.
- **12.5:** Treat zero separately at $X = 0$, and treat $\theta = \pi/2$ as a limit.
- **12.7:** Filter the light eigenvalues by E; state the gap in absolute value; specify an algebra and representation for a spectral-triple claim.

- **12.9:** Require nonzero trailing minors and define the separation parameter and its dependence. Retain the square-root localization factors unless they are uniformly bounded.

Overall: 9 statements checked: 4 correct, 2 correct with amendment, 0 gaps, and 3 errors.

Literature (../LITERATURE.md, section C). The one-flavour spectrum of Example 12.5 is equation (5.6) of Csaki-Grojean-Hubisz-Shirman-Terning, Fermions on an interval (2004); the warped multi-flavour analogue of the light spectrum and of the localization textures is in Casagrande et al. 2008 (equations 65, 87, 90, 94-97). No NCG paper deriving D_F from localization was found.

Part 13

T4: gauge fluctuations of the chiral interval

Prompt: prompts/13 (STRATEGY T4). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 by hand: Results 2, 3, 4, 5 and 6 (the compression integrals, the counterexample B = -2a/(3 pi) with trivial Wilson line, the Neumann-Dirichlet mass, the 1 : 3 ratio of the a_4 gauge and scalar terms with the canonical normalization, and the eigenvalues 0 and 6m^2); Results 7 and 8 reproduce the standard CCM relations (8 M_W^2 = sum N_f m_f^2, M_W/M_Z = sqrt(5/8) with Q_2 = 12, Q_Y = 20).

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 8 statements, 7 are correct and Proposition 8 needs one amendment, recorded there.

We prove an exact compression formula, but it does not identify the continuum gauge theory with the full Connes–Lott gauge theory. The fermion boundary conditions impose no boundary conditions on the gauge potentials, and fixed spectral compression produces only links of the form `phi = m I + i B`, with `B` Hermitian. The proposed ratio `M_W/m = sqrt(3/8)` is correct under the three-generation Standard Model CCM normalization, with one massive colour triplet and all other Dirac masses zero. It is not a consequence of the interval alone: the minimal two-sheet spectral action has a different gauge representation and gives a massive relative gauge field of mass `sqrt(6) m`.

1. Theorem (inner fluctuations and boundary conditions).

Let M be a closed Riemannian spin four-manifold. We use Hermitian Euclidean gamma matrices, $D_M = -i\gamma^\mu \nabla^S_\mu$, and

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad D = D_M \otimes 1 + \gamma_5 \otimes E\partial_x.$$

Let

$$\mathcal{H} = L^2(S_M) \otimes L^2([0,l], \mathbb{C}^2) \otimes \mathbb{C}^n,$$

with the algebra acting by the same matrix multiplication on L and R. The interval boundary conditions are

$$R(y,0) = 0, \qquad L(y,l) = 0.$$

We consider complex inner fluctuations $D + \omega$, where $\omega = \sum_j a_j[D, b_j]$ and $\omega^* = \omega$. No real structure or additional bimodule is assumed.

Then the fluctuated operators are exactly

$$\boxed{D_A = -i\gamma^\mu(\nabla^S_\mu + iA_\mu) + \gamma_5 E(\partial_x + iA_5),}$$

where $A_\mu(y,x)$ and $A_5(y,x)$ are arbitrary smooth Hermitian $n \times n$ matrices. They are connections on the trivial $U(n)$ bundle over $M \times I_l$. Restriction to traceless potentials gives the $SU(n)$ case.

The fermion conditions induce **no Dirichlet, Neumann, or endpoint-value conditions on A_μ or A_5.**

Proof. For a matrix-valued function b,

$$[D, b] = -i\gamma^\mu \partial_\mu b + \gamma_5 E \partial_x b.$$

Consequently, represented one-forms have precisely these five Clifford components. Self-adjointness makes their connection coefficients anti-Hermitian, hence iA_α with A_α Hermitian. Conversely, smooth matrix-valued one-forms are finite sums $a\,db$, using local coordinates and a partition of unity.

Every such fluctuation is a bounded self-adjoint perturbation of D, so D_A is self-adjoint on the original domain. A bounded perturbation need not map that domain into itself; requiring this would impose an additional, unjustified condition.

Every smooth $g : M \times I_l \to U(n)$ preserves the fermion boundary conditions:

$$(gR)(y, 0) = 0, \qquad (gL)(y, l) = 0.$$

The usual transformation

$$A_\alpha \longmapsto g A_\alpha g^* + i(\partial_\alpha g) g^*$$

therefore allows arbitrary endpoint values of g.

For smooth sections in $\mathrm{Dom}(D_A^2)$, the additional conditions are

$$(\partial_x + iA_5)L(y, 0) = 0, \qquad (\partial_x + iA_5)R(y, l) = 0.$$

These are conditions on the fermions, not on A_5. They follow by applying the original boundary conditions to $D_A\psi$.

An orbifold extension would add different hypotheses. Requiring A_μ to be even and A_5 odd under endpoint reflections gives $\partial_x A_\mu = 0$ and $A_5 = 0$ there. Those parity requirements do not follow from the stated fermion domain. □

2. Theorem (exact fixed-projection compression and its image).

Let

$$m = \frac{\pi}{2l}, \qquad u_L = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(mx) \\ 0 \end{pmatrix}, \qquad u_R = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(mx) \end{pmatrix},$$

and let P project onto their span, tensored with the spacetime and gauge factors. This is the unfluctuated lowest-pair projection of Lemma 05, Part 3.

Define

$$w_L(x) = \frac{2}{l}\cos^2(mx), \qquad w_R(x) = \frac{2}{l}\sin^2(mx), \qquad k(x) = \frac{1}{l}\sin\frac{\pi x}{l}.$$

Then, under the identification $P\mathcal{H} \simeq L^2(S_M) \otimes (\mathbb{C}^n_L \oplus \mathbb{C}^n_R)$,

$$\boxed{PD_AP = -i\gamma^\mu \left(\nabla^S_\mu + i \begin{pmatrix} A_{L,\mu} & 0 \\ 0 & A_{R,\mu} \end{pmatrix} \right) + \gamma_5 \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix},}$$

where

$$\boxed{A_{L,\mu} = \int_0^l w_L A_\mu \, dx, \qquad A_{R,\mu} = \int_0^l w_R A_\mu \, dx,}$$

and

$$\boxed{\phi = mI_n + iB, \qquad B = \int_0^l k(x) A_5(y, x)\, dx = B^*.}$$

Here ϕ denotes the total link; its additive fluctuation is $\phi - mI_n$.

The exact image consists of arbitrary Hermitian sheet connections together with links satisfying

$$\boxed{\phi + \phi^* = 2mI_n.}$$

In particular, not every Connes–Lott fluctuation arises by this compression.

Proof. The spacetime term is diagonal in interval chirality, giving the two weighted averages. The unfluctuated interval operator satisfies

$$D_I u_R = m u_L, \qquad D_I u_L = m u_R.$$

The upper-right matrix element of iEA_5 is

$$i\frac{2}{l}\int_0^l \cos(mx)\sin(mx)A_5\,dx = iB;$$

the lower-left element is $-iB$. This proves the operator identity without approximation.

The elementary integrals

$$\int_0^l w_L\,dx = \int_0^l w_R\,dx = 1, \quad \int_0^l w_L \cos\frac{\pi x}{l}\,dx = \frac{1}{2}, \quad \int_0^l w_R \cos\frac{\pi x}{l}\,dx = -\frac{1}{2}$$

show that any prescribed A_L, A_R have the lift

$$A_\mu(y,x) = \frac{A_{L,\mu} + A_{R,\mu}}{2} + (A_{L,\mu} - A_{R,\mu})\cos\frac{\pi x}{l}.$$

Since $\int_0^l k\,dx = 2/\pi$, any Hermitian B has the lift

$$A_5(y,x) = \frac{\pi}{2}B(y).$$

Thus the stated image is exact.

By contrast, the finite algebra

$$M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$$

with $D_F = m\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$ admits an arbitrary complex matrix as its total link. Indeed, writing p_L, p_R for its central projections, $p_L[D_F, p_R]$ supplies an arbitrary upper-right block after multiplication by the left algebra; adding its adjoint supplies the lower-left block. The allowed finite fluctuation $\phi = 0$, for example, has no continuum lift under the displayed compression.

For $SU(2)$, B is additionally traceless. The continuum A_5 is adjoint-valued; it is not automatically an electroweak Higgs doublet.

If an additional finite factor is included in the internal-first convention, and the gauge algebra commutes with D_0, Γ_0, its compressed internal operator is

$$D_0 \otimes 1 + \Gamma_0 \otimes \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix}.$$

There is no extra overlap factor. If $[D_0, a] \neq 0$, additional finite scalar fluctuations occur. Moreover, the cited Theorem 5 explicitly restricts the interval algebra to constants for its real-triple assertion; it does not supply a Standard Model real structure for the full compressed algebra. □

3. No-go theorem (fixed compression is not a gauge-covariant Wilson-link construction).

Let P be the fixed projection of Result 2. Its link $\phi[A_5]$ is not a functional of the Wilson line alone, and fixed compression does not descend to continuum gauge-equivalence classes.

Proof. It suffices to consider $U(1)$ potentials independent of y. Define parallel transport by

$$W'(x) = -iA_5(x)W(x), \qquad W(0) = 1.$$

Then

$$D_{I,A} = W D_I W^*.$$

Multiplication by W preserves both fermion boundary conditions, so the complete interval spectrum is unchanged.

Take

$$A_5(x) = a \cos \frac{2\pi x}{l}.$$

Its Wilson line is $W(l) = 1$, but

$$B = \frac{a}{\pi} \int_0^\pi \sin t \cos(2t)\, dt = -\frac{2a}{3\pi}.$$

Thus

$$\phi = m - \frac{2ia}{3\pi}, \qquad |\phi| = \sqrt{m^2 + \frac{4a^2}{9\pi^2}}.$$

The zero connection and this connection are gauge equivalent even through a transformation equal to 1 at both endpoints, yet their fixed compressed operators have different internal eigenvalues.

The obstruction is that multiplication by an x-dependent gauge transformation generally fails to preserve $P\mathcal{H}$. Compression of multiplication is not multiplicative, and PgP generally fails to be unitary.

The normalization difference is already visible for constant $A_5 = a$:

$$\frac{\phi}{m} = 1 + i\frac{4l}{\pi^2}a, \qquad W(l)^* = e^{ial}.$$

Neither equality with the Wilson link nor a universal correction to the mass follows from these expressions.

A different, gauge-adapted construction is possible. Use the transported interval frames

$$u_L(x)W(x), \qquad u_R(x)W(x)W(l)^*.$$

Their vertical Dirac matrix is exactly

$$\begin{pmatrix} 0 & mW(l)^* \\ mW(l) & 0 \end{pmatrix}.$$

Its link obeys $\phi^*\phi = m^2 I$, so it also omits general radial Higgs fluctuations. If W depends on y, derivatives of these frames additionally enter the sheet connections.

Finally, compression differs from eliminating the omitted fermions. With $Q = 1 - P$, block elimination adds

$$-PD_AQ\,[Q(D_A - z)Q]^{-1}QD_AP$$

whenever the inverse exists. This follows by solving the Q-component of $(D_A - z)\psi = 0$ and substituting into the P-component. It is the distinction relevant to the effective-operator discussion in research state §6.10. □

4. Lemma (the continuum vector mass requires specified gauge boundary conditions).

Let the quadratic transverse-vector operator in the interval direction be $-\partial_x^2$, with

$$f'(0) = 0, \qquad f(l) = 0.$$

Assume no boundary kinetic terms or other modifications of this operator. Then

$$\boxed{M_{\mathrm{KK}} = \frac{\pi}{2l} = m.}$$

Proof. The Neumann condition selects $f(x) = C\cos(kx)$. The Dirichlet condition gives

$$kl = \left(r + \frac{1}{2}\right)\pi, \qquad r = 0, 1, \ldots.$$

The lowest eigenvalue is therefore m^2.

A Dirichlet condition at one endpoint alone does not determine this answer. Dirichlet conditions at both endpoints give lowest mass π/l, whereas two Neumann conditions give a zero mode. Thus the comparison uses specifically the mixed boundary problem listed in Lemma 9.11, and Result 1 does not derive that problem from the fermion boundary conditions. □

5. Proposition (general spectral-action vector-mass formula).

Let H_F be a specified finite fermion space with Hermitian represented gauge generators T_a. Let

$$\mathcal{D} = -i\gamma^\mu(\nabla^S_\mu + iA_\mu) + \gamma_5 V,$$

where $V = V^*$ is the total finite scalar matrix. Use the ordinary spectral trace, an even cutoff function with $f_0 = f(0) > 0$, and the terms through order Λ^0. Assume no additional kinetic terms.

Let V_0 be the chosen constant vacuum value. Define

$$Q_{ab} = \mathrm{Re}\,\mathrm{Tr}_{H_F}(T_a T_b), \qquad K_{ab} = \mathrm{Re}\,\mathrm{Tr}_{H_F}([T_a, V_0]^*[T_b, V_0]).$$

On the faithfully represented gauge algebra, the canonically normalized vector mass-squared matrix is

$$\boxed{\mathcal{M}^2 = \frac{3}{2}\, Q^{-1/2} K Q^{-1/2}.}$$

Consequently, relative to the mixed-boundary continuum mass,

$$\boxed{\frac{M_j^2}{M_{\mathrm{KK}}^2} = \frac{3}{2m^2}\,\lambda_j\!\left(Q^{-1/2} K Q^{-1/2}\right).}$$

Proof. We use the heat coefficient

$$a_4 \supset \frac{1}{16\pi^2}\,\mathrm{Tr}\,\left(\frac{1}{2}\mathcal{E}^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\right),$$

as recorded in CCM, Theorem 6.1, equation (6.10). Here $\mathcal{D}^2 = \nabla^*\nabla - \mathcal{E}$.

Write

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], \qquad D_\mu V = \partial_\mu V + i[A_\mu, V].$$

The relevant endomorphism terms are

$$\mathcal{E}_F = \frac{i}{2}\gamma^\mu\gamma^\nu F_{\mu\nu}, \qquad \mathcal{E}_V = i\gamma^\mu\gamma_5 D_\mu V.$$

The Clifford traces give

$$\mathrm{tr}_S(\mathcal{E}_F^2) = 2F_{\mu\nu}F^{\mu\nu}, \quad \mathrm{tr}_S(\mathcal{E}_V^2) = 4D_\mu V D^\mu V, \quad \mathrm{tr}_S(\Omega_F^2) = -4F_{\mu\nu}F^{\mu\nu}.$$

Hence the relevant Lagrangian is

$$\boxed{\mathcal{L} = c\ \mathrm{Tr}_{H_F}(F_{\mu\nu}F^{\mu\nu}) + 3c\ \mathrm{Tr}_{H_F}(D_\mu V D^\mu V), \qquad c = \frac{f_0}{24\pi^2}.}$$

At $V = V_0$, the quadratic gauge terms are

$$cQ_{ab}F^a_{\mu\nu}F^{b,\mu\nu} + 3cK_{ab}A^a_\mu A^{b,\mu}.$$

The change of variables $A_{\mathrm{can}} = 2\sqrt{c}\,Q^{1/2}A$ gives kinetic normalization $F^2_{\mathrm{can}}/4$ and mass normalization $A_{\mathrm{can}}\mathcal{M}^2 A_{\mathrm{can}}/2$, proving the formula.

Thus the necessary data are the represented gauge generators, their full trace multiplicities, and the vacuum matrix V_0. The common coefficient f_0 cancels. However, the bare finite Dirac matrix need not itself be the vacuum: determining V_0 requires the scalar potential, including its cutoff moments and curvature dependence. All masses below are conditional on the stated vacuum and refer to this leading spectral-action approximation. □

6. Counterexample (the minimal two-sheet action does not give the CCM triplet ratio).

Let

$$H_F = (\mathbb{C}_L^n \oplus \mathbb{C}_R^n) \otimes \mathbb{C}^N, \qquad V_0 = m \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix} \otimes I_N.$$

Gauge both sheet groups, $U(1)_L \times U(1)_R$ for $n = 1$, or $SU(2)_L \times SU(2)_R$ for $n = 2$. Use the action of Result 5.

Then the diagonal gauge combination is massless and the relative combination has

$$\boxed{M_{\text{relative}} = \sqrt{6}\, m.}$$

This remains true for colour multiplicity $N = 3$.

Proof. For one normalized generator t, set

$$T_L = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}, \qquad T_R = \begin{pmatrix} 0 & 0 \\ 0 & t \end{pmatrix}, \qquad \kappa = N \operatorname{tr}(t^2).$$

Direct multiplication gives

$$Q = \kappa I_2, \qquad K = 2\kappa m^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Therefore

$$\mathcal{M}^2 = 3m^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

whose eigenvalues are 0 and $6m^2$. The factor N cancels between scalar and gauge normalization. For $SU(2)$, this calculation applies to each $t = \sigma_a/2$.

If only the diagonal group is gauged, every generator commutes with V_0, and all these vector masses vanish. Neither choice realizes the assumed continuum mixed-boundary breaking automatically.

The dependence on the full fermion trace can also be exhibited explicitly. In the $U(1)$ example, add S otherwise identical charged pairs with zero finite Dirac operator. They contribute to Q but not K, giving

$$M^2_{\text{relative}} = 6\frac{N}{N+S}m^2.$$

The mass and colour multiplicity of the original pair have not changed.

There is another distinction if one restricts the continuum field to its lowest mixed-boundary profile before evaluating the finite action:

$$A_\mu(y, x) = a_\mu(y) \cos(mx).$$

Result 2 then gives

$$A_{L,\mu} = \frac{8}{3\pi} a_\mu, \qquad A_{R,\mu} = \frac{4}{3\pi} a_\mu.$$

For a restriction $A_L = c_L a,\ A_R = c_R a$, Result 5 gives the one-field quadratic mass

$$M^2_{\text{restricted}} = 3m^2 \frac{(c_L - c_R)^2}{c_L^2 + c_R^2}.$$

Thus this particular restriction gives

$$M^2_{\text{restricted}} = \frac{3}{5} m^2.$$

It is an additional restriction of the action, not a mass eigenvector of the unrestricted two-sheet theory. It demonstrates that retaining a continuum gauge profile and fluctuating the full compressed algebra are different prescriptions. □

7. Corollary (the precise conditional CCM comparison).

Let the finite model instead have the full three-generation Standard Model representation and its usual Higgs doublet. Retain all charged fermion states in the spectral trace, including those whose Yukawa masses vanish. Identify one colour-triplet Dirac mass at the vacuum with

$$m = \frac{\pi}{2l},$$

and suppose every other Dirac mass is zero.

Then, at the matching scale,

$$\boxed{M_W^2 = \frac{3}{8}m^2, \qquad \frac{M_W}{M_{\mathrm{KK}}} = \sqrt{\frac{3}{8}}.}$$

No additional factor from the interval wavefunctions or from Wilson-line normalization multiplies this ratio.

Proof. We can recover the normalization directly from Result 5. For $T_1 = \sigma_1/2$, one Standard Model generation of particles contributes

$$\mathrm{Tr}(T_1^2) = \frac{1}{2} + 3\frac{1}{2} = 2.$$

Antiparticles double this, and three generations give

$$Q_{11} = 12.$$

For a left doublet and its right singlets with mass matrix $Y = \mathrm{diag}(m_u, m_d)$, the particle contribution to K_{11} is

$$2\,\mathrm{tr}(Y^* T_1^2 Y) = \frac{1}{2}(m_u^2 + m_d^2).$$

Including antiparticles and all multiplicities gives

$$K_{11} = \sum_f N_f m_f^2.$$

Therefore

$$M_W^2 = \frac{3}{2}\frac{K_{11}}{12} = \frac{1}{8}\sum_f N_f m_f^2.$$

This is the relation in CCM, Lemma 4.6, equation (4.16), assumed in Lemma 11. Substitution of $K_{11} = 3m^2$ proves the claimed ratio.

The normalization identity is equivalently

$$\frac{f_0 g_2^2}{2\pi^2} = \frac{1}{4}.$$

Its coefficient comes from $Q_{11} = 12$. Applying that coefficient after deleting the other charged states would no longer canonically normalize the gauge kinetic term.

The interval basis already satisfies $\|u_L\| = \|u_R\| = 1$, and $\langle u_L, D_I u_R\rangle = m$. Thus its unfluctuated fermion mass is exactly m. The factor $2/\pi$ belongs to the coupling of a constant A_5 to the fixed compressed link; it does not rescale the background mass. A change of scalar coordinates likewise cannot change a mass ratio when both kinetic and Yukawa terms are transformed consistently.

If other Dirac masses are nonzero, the exact conditional comparison becomes

$$\frac{M_W^2}{M_{\mathrm{KK}}^2} = \frac{1}{8}\left(3 + \sum_{\text{other } f} N_f \frac{m_f^2}{m^2}\right).$$

If "a coloured doublet" means two massive coloured Dirac species, both of mass m, their contribution is $6m^2$, giving $M_W/m = \sqrt{3}/2$ when all other masses vanish. □

8. Proposition (electroweak doublet and the neutral-vector ratio).

Amendment (referee). In the last paragraph of this result, the law phi -> g_L phi g_R^* holds for the link of the independently gauged finite two-sheet completion. The fixed compression of the continuum field does not obey it in general: Result 3 gives an endpoint-fixed gauge transformation that changes phi. The decomposition 1 + 3 under the diagonal SU(2) stays correct.

Let an additional electroweak assignment supply one complex Higgs doublet h, with

$$T_a = \frac{\sigma_a}{2}, \qquad Y_h = \frac{1}{2}, \qquad \langle h \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix},$$

where $Q_{\mathrm{em}} = T_3 + Y$. Let the gauge fields have canonical kinetic terms and couplings g_2, g_Y.

Then

$$\boxed{M_W^2 = \frac{g_2^2 v^2}{4}, \qquad M_Z^2 = \frac{(g_2^2 + g_Y^2) v^2}{4},}$$

and consequently

$$\boxed{\frac{M_W}{M_Z} = \cos\theta_W, \qquad \rho = \frac{M_W^2}{M_Z^2 \cos^2\theta_W} = 1, \quad \tan\theta_W = \frac{g_Y}{g_2}.}$$

Proof. At the vacuum,

$$|D_\mu h|^2 = \frac{v^2}{8} \left[g_2^2 ((W_\mu^1)^2 + (W_\mu^2)^2) + (g_2 W_\mu^3 - g_Y B_\mu)^2 \right].$$

Diagonalizing the neutral quadratic form gives one massless photon and the displayed Z mass.

Under a common spectral-action coefficient, canonical normalization additionally gives

$$g_2^2 Q_2 = g_Y^2 Q_Y.$$

For the full three-generation Standard Model trace, including antiparticles,

$$Q_2 = 12, \qquad Q_Y = 20,$$

so

$$\boxed{\frac{M_W}{M_Z} = \sqrt{\frac{5}{8}}}$$

at that matching scale. These trace weights agree with the per-generation calculation in research state §6.10.

The original interval multiplication representation acts identically on both chiralities. Its matrix link transforms as

$$\phi \longmapsto g_L \phi g_R^*,$$

and under the diagonal $SU(2)$ it decomposes as $1 \oplus 3$. It therefore does not specify the electroweak doublet assignment used above. That assignment, its hypercharges, and the charged fermion trace must be supplied separately. Likewise, the continuum value of ρ requires the neutral-sector boundary conditions and kinetic normalization; the single mixed-boundary eigenvalue in Result 4 cannot determine it. □

Consequence for T4 and hypothesis H

The minimal fermion compression supports the geometric identification $m = \pi/(2l)$, but it does not establish an equivalence of gauge fluctuations, gauge boundary conditions, or bosonic actions. The ratio $\sqrt{3/8}$ is correct for the specified CCM Standard Model completion and receives no interval-overlap correction. Hypothesis H therefore requires an additional gauge-covariant reduction and a matching of representations, vacuum data, and kinetic forms before it can identify the finite vector spectrum with the interval Kaluza–Klein spectrum.

Part 14

T4b: gauge-covariant projection and conditional bosonic matching

Prompt: prompts/14 (STRATEGY T4b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 by hand: Theorem 5 (both mass matrices as quadratic forms), Counterexample 6 (the ratio 3/5 agrees with the restricted computation of lemma 13, Result 6: c_L = 8/(3 pi), c_R = 4/(3 pi)), Proposition 7 (t = 3 pi/4: M = 3 pi/(4l), lambda^2/m2 = 9/2, normalization l(1/2 + 1/(3 pi)), ratio 27/26, then the intermediate value theorem) and No-go theorem 8 (M_spec^2 <= 6 norm(V_0)^2). Reading: the spectral-action vector mass matches the Neumann-Dirichlet tower only through a fermion whose mass is at the tower scale (the matching bulk mass M_* > 0 leaves no light pair, Theorem 9.4 case 3), and with bounded light fermions no matching is possible as l -> 0.

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 9 statements, 5 are correct, Theorems 1 and 5 and Proposition 4 need amendments, and the global form of Theorem 3 is wrong as first written. Each is recorded at its statement.

We prove that a gapped spectral projection defines a gauge-covariant light-fermion bundle with its induced Grassmannian connection. For the mixed interval with a gauge-invariant bulk mass, its finite link has fixed singular values; it does not supply the full scalar fluctuation space of a two-sheet finite triple. At the leading four-dimensional spectral-action order, equality with a specified Kaluza–Klein vector mass is equivalent to an explicit trace-and-overlap sum rule. Covariance does not enforce that rule: the massless-bulk example gives $M_{\mathrm{spec}}^2 = \frac{3}{5} M_{\mathrm{KK}}^2$, whereas a suitable positive bulk mass gives exact equality. We also prove that bounded retained fermion masses cannot match a Neumann–Dirichlet vector mass diverging as $1/l$. Matching one vector mass does not, by itself, determine relative gauge couplings or all scalar kinetic coefficients.

1. Theorem (covariant spectral projection and the data controlling its spectrum).

Amendment (referee). The double-coset statement needs the endpoint transformations to act independently through H_0 and H_l, with extendability understood, and it is a local pointwise classification unless the global topology is specified. Periodic conditions require g(l) = g(0), and then the holonomy is classified by conjugation.

Let $I = [0, l]$, and let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Let the vertical operator be

$$D_{A_5,\mathsf{M},\mathcal{B}} = E(\partial_x + iA_5) + \begin{pmatrix} 0 & \mathsf{M} \\ \mathsf{M} & 0 \end{pmatrix},$$

where A_5 and M are Hermitian matrix-valued functions, and $\mathcal{B}$ specifies a self-adjoint boundary domain. Gauge transformations act in the same representation on both interval chiralities.

Let the operators form a smooth family in y, in the norm-resolvent sense after local identification of their domains. Assume that the selected light cluster is separated from the rest of the spectrum and contains complete eigenspaces, including any gauge degeneracy. Let $P_A(y)$ be its spectral projection.

Then

$$\boxed{P_{A^g} = G_g P_A G_g^\dagger,}$$

provided the transformation includes

$$A_5^g = gA_5g^\dagger + i(\partial_x g)g^\dagger, \qquad \mathsf{M}^g = g\mathsf{M}g^\dagger, \qquad \mathcal{B}^g = G_g\mathcal{B}.$$

For fixed $\mathsf{M}, \mathcal{B}$, this assertion applies exactly to the transformations preserving those data.

Define parallel transport by

$$\partial_x W = -iA_5 W, \qquad W(0) = 1, \qquad W_l = W(l).$$

The vertical spectrum is determined by the axial-gauge data

$$\boxed{\widetilde{\mathsf{M}}(x) = W(x)^\dagger \mathsf{M}(x) W(x), \qquad \widetilde{\mathcal{B}} = W^\dagger \mathcal{B}.}$$

Consequently:

- For a gauge-invariant bulk mass and the mixed conditions

$$R(0) = 0, \qquad L(l) = 0,$$

  the entire spectrum is independent of A_5.
- With nontrivial endpoint data and gauge-invariant bulk coefficients, the relevant holonomy information is the Wilson line relative to those endpoint data. If the endpoint groups are H_0, H_l, its gauge class is represented by

$$H_l \backslash G / H_0.$$

- A bulk matrix not commuting with parallel transport can introduce dependence on the whole transported profile $\widetilde{\mathsf{M}}(x)$, beyond W_l.

An open Wilson line is not generally gauge invariant:

$$W_l^g = g(l) W_l g(0)^\dagger.$$

It is invariant under transformations fixed at both endpoints; it transforms by conjugation when the endpoint transformations are identified. With independently unrestricted endpoints, its double-coset class is trivial.

Proof. Direct differentiation, including the boundary domain, gives

$$D_{A^g, \mathsf{M}^g, \mathcal{B}^g} = G_g D_{A, \mathsf{M}, \mathcal{B}} G_g^\dagger.$$

For a contour enclosing the selected cluster,

$$P_A = \frac{1}{2\pi i} \oint (z - D_A)^{-1} \, dz.$$

Conjugation of the resolvent proves covariance. Smoothness follows from the same formula and the assumed gap.

Multiplication by W removes the connection:

$$W^\dagger D_{A_5, \mathsf{M}, \mathcal{B}} W = D_{0, \widetilde{\mathsf{M}}, \widetilde{\mathcal{B}}}.$$

This proves the assertion about spectral data. For a commuting bulk mass and the mixed boundary conditions, both transformed data equal their unfluctuated values.

Finally, the transport equation gives $W^g(x) = g(x) W(x) g(0)^\dagger$. The endpoint transformation law and the stated quotients follow. If the gap closes, no fixed-rank smooth light bundle is asserted. □

2. Theorem (the induced connection and finite operator).

Let M_4 be a closed Riemannian spin four-manifold. Use Hermitian Euclidean gamma matrices and

$$\mathcal{D}_A = -i\gamma^\mu (\nabla_\mu^S + \partial_\mu + i A_\mu) + \gamma_5 D_{I, A_5}.$$

Assume the hypotheses of Result 1, and let

$$\mathcal{E}_y = \operatorname{Ran} P_A(y).$$

Then the induced connection and vertical endomorphism are

$$\boxed{\nabla_\mu^{\mathcal{E}} = P_A (\partial_\mu + i A_\mu) P_A, \qquad V = P_A D_{I, A_5} P_A \big|_{\mathcal{E}}.}$$

The compressed first-order operator is exactly

$$\boxed{P_A \mathcal{D}_A P_A\big|_{\mathcal{E}} = -i\gamma^\mu(\nabla^S_\mu + \nabla^{\mathcal{E}}_\mu) + \gamma_5 V.}$$

In a local orthonormal frame $U(y) : \mathbb{C}^r \to \mathcal{E}_y$,

$$\boxed{\mathcal{A}_\mu = U^\dagger A_\mu U - iU^\dagger \partial_\mu U, \qquad V_F = U^\dagger D_{I,A_5} U.}$$

The inner products here include integration over the interval. Its curvature is

$$[\nabla^{\mathcal{E}}_\mu, \nabla^{\mathcal{E}}_\nu] = P_A(iF_{\mu\nu})P_A + P_A[\nabla_\mu P_A, \nabla_\nu P_A]P_A,$$

where $\nabla_\mu P_A = \partial_\mu P_A + i[A_\mu, P_A]$.

Proof. Write a light section as $U\chi$. Applying $\partial_\mu + iA_\mu$ and multiplying by $U^\dagger$ gives the displayed connection. The spectral projection commutes with the vertical operator, giving the displayed finite endomorphism. Expanding the commutator of the projected covariant derivatives and using $P_A^2 = P_A$ gives the curvature formula.

If

$$U^g = G_g U h^\dagger,$$

then

$$\mathcal{A}^g_\mu = h\mathcal{A}_\mu h^\dagger + i(\partial_\mu h)h^\dagger, \qquad V^g_F = hV_F h^\dagger.$$

Thus the construction is gauge covariant.

This is an exact compression. It is not generally the operator obtained by eliminating the other fermions. With $P^\perp = 1 - P_A$, elimination additionally produces

$$-P_A \mathcal{D}_A P^\perp \left[P^\perp(\mathcal{D}_A - z)P^\perp\right]^{-1} P^\perp \mathcal{D}_A P_A,$$

when the inverse exists. Horizontal derivatives and $A_\mu(x)$ generally make these off-diagonal blocks nonzero. □

3. Theorem (the finite algebra and fluctuation image for the mixed interval).

Correction (referee; the global statement as first written is false). The statement that the continuum link image consists of all links with Phi^* Phi = lambda^2 I holds locally in spacetime. Globally the unitary link must admit a path from the identity in the chosen trivialization. For M_4 = S^1 x S^3, n = 1 and Phi = lambda exp(i theta), every globally smooth real A_5 gives Phi = lambda exp(i int A_5 dx), with zero winding around S^1, so this Phi is not attained. The displayed formulas, the compression range and the reducing-subalgebra criterion are correct.

Let

$$D_{I,A_5} = \begin{pmatrix} 0 & \partial_x + iA_5 + M \\ -\partial_x - iA_5 + M & 0 \end{pmatrix}, \qquad M \in \mathbb{R},$$

with $R(0) = 0$, $L(l) = 0$, acting on $\mathbb{C}^n$-valued fields. Let the parent algebra be $C^\infty(I, M_n(\mathbb{C}))$, acting identically on both interval chiralities.

Let p_L, p_R be real normalized profiles for the lowest pair:

$$(-\partial_x + M)p_L = \lambda p_R, \qquad (\partial_x + M)p_R = \lambda p_L, \qquad \|p_L\| = \|p_R\| = 1.$$

Use the endpoint-adapted light frame

$$U_L = p_L W, \qquad U_R = p_R W W_l^\dagger.$$

Then

$$\boxed{V_F = \begin{pmatrix} 0 & \lambda W_l^\dagger \\ \lambda W_l & 0 \end{pmatrix}.}$$

Writing $w_L = p_L^2,\ w_R = p_R^2$ and

$$B_\mu = W^\dagger A_\mu W - iW^\dagger \partial_\mu W,$$

the induced connections are

$$\boxed{\begin{aligned} \mathcal{A}_{L,\mu} &= \int_0^l w_L B_\mu \, dx, \\ \mathcal{A}_{R,\mu} &= W_l \left(\int_0^l w_R B_\mu \, dx \right) W_l^\dagger - iW_l \partial_\mu W_l^\dagger. \end{aligned}}$$

The range of algebra compression is

$$A_F = M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$$

on

$$H_F = \mathbb{C}_L^n \oplus \mathbb{C}_R^n, \qquad \Gamma_F = \mathrm{diag}(1, -1).$$

Taking this range with its matrix product gives a complex even finite triple. Its unitary group is $U(n)_L \times U(n)_R$, and its unrestricted inner fluctuations have an arbitrary complex matrix as their total link.

The continuum construction, however, produces exactly

$$\boxed{\Phi^\dagger \Phi = \Phi \Phi^\dagger = \lambda^2 I_n}$$

for the link. With unrestricted A_μ, it produces arbitrary pairs of Hermitian sheet connections locally in spacetime. Thus its field space is the fixed-singular-value restriction of that finite triple's fluctuation space.

The inherited local gauge action is

$$h = \mathrm{diag}(g(0), g(l)).$$

Additional gauge boundary conditions restrict this endpoint group. Globally, only endpoint transformations extending across $M_4 \times I$ are inherited.

Proof. Gauge removal and the two profile equations give the formula for V_F. Applying Result 2 to U_L, U_R gives the connection formulas. Under a continuum transformation,

$$U_L^g = gU_L g(0)^\dagger, \qquad U_R^g = gU_R g(l)^\dagger,$$

which proves the endpoint action.

The two functions w_L, w_R are linearly independent: one vanishes at $x = 0$, and the other does not. Therefore the map

$$b(x) \longmapsto \left(\int w_L b, \int w_R b \right)$$

is onto $M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$. This proves both the algebra-range assertion and the connection-surjectivity assertion. Every unitary W_l has a local smooth path lift, proving the stated link image.

For the finite triple with

$$D_F = \lambda \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix},$$

multiplication of $p_L[D_F, p_R]$ by the left algebra generates arbitrary upper-right matrices; adjoining the adjoint generates arbitrary self-adjoint odd fluctuations. The full finite fluctuation space therefore includes radial changes absent from the continuum image.

There is a further distinction concerning algebra multiplication. For any compression $C(a) = PaP|_{PH}$,

$$C(a^\dagger a) - C(a)^\dagger C(a) = Pa^\dagger(1-P)aP.$$

Hence C is a representation on a unital $*$-subalgebra precisely when that subalgebra reduces P. In the present model, multiplication by $a(x)$ reduces the light pair precisely when

$$a(x) = W(x)bW(x)^\dagger$$

for a constant matrix b. Its restriction is a single diagonal $M_n(\mathbb{C})$, commuting with V_F, and its finite scalar one-forms vanish.

Thus the two-sheet algebra is obtained by equipping the compression range with its matrix product; it is not a multiplicative quotient of the interval multiplication algebra. No real structure, real gauge-group prescription, or Standard Model representation follows from this construction. □

4. Proposition (boundary Yukawa data determine a finite operator, but not a unique finite triple).

Amendment (referee). To obtain the entire limiting block, retain the full zero-mode-descended cluster of dimension N_Q + N_U: for two massless flavours and X = l diag(1, 2), the cluster containing only the lowest pair is gapped and has a 1 x 1 limiting block. The nonzero singular values of Y are the retained positive eigenvalues; zero singular values and unmatched chiral zero modes are included separately.

Let D_X have the boundary conditions of Lemma 12:

$$Q_R(0) = 0, \quad U_L(0) = 0, \qquad Q_R(l) = XU_R(l), \quad U_L(l) = -X^\dagger Q_L(l).$$

Let the bulk masses be M_Q, M_U, and let the gauge representations commute with these masses.

Gauge covariance requires

$$X^g = g_Q(l)Xg_U(l)^\dagger.$$

If X is fixed, the allowed endpoint transformations satisfy

$$g_Q(l)X = Xg_U(l).$$

After axial gauge fixing,

$$\widetilde{X} = W_Q(l)^\dagger X W_U(l).$$

Let P_A retain a gapped grading-invariant light cluster. Then

$$H_F = H_{F,+} \oplus H_{F,-}, \qquad V_F = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \end{pmatrix},$$

where the singular values of Y are the retained positive vertical eigenvalues. These eigenvalues alone do not determine the represented gauge generators or their trace data.

In the towerless scaling of Theorem 12.7,

$$X = l\widehat{X}, \qquad M_Q l,\ M_U l \text{ fixed},$$

the limiting block is, up to changes of chiral orthonormal frames,

$$\boxed{Y = \Phi_Q \widehat{X} \Phi_U,} \qquad \phi(c)^2 = \frac{2c}{e^{2c}-1}, \quad \phi(0) = 1.$$

Every rectangular complex matrix Y is attainable as an operator block by choosing

$$\widehat{X} = \Phi_Q^{-1} Y \Phi_U^{-1}.$$

This does not establish that every finite algebra, real structure, gauge representation, or inner-fluctuation space is attained.

Proof. Transforming the boundary equations gives the transformation law for X. Applying $W_Q^\dagger, W_U^\dagger$ gives $\widetilde{X}$. Since the domain is grading invariant, its symmetric spectral projection commutes with the grading; the restricted operator is therefore odd and has the displayed block form.

For an explicit spectral determination, define

$$\alpha_i(\lambda) = \omega_i \coth(\omega_i l) + m_i^Q, \qquad \beta_j(\lambda) = \omega_j \coth(\omega_j l) - m_j^U, \qquad \omega^2 = m^2 - \lambda^2.$$

In the low window where the transfer denominators are nonzero, the boundary equations give

$$\det\left[\lambda^2 I - \operatorname{diag}(\alpha_i)\widetilde{X}\operatorname{diag}(\beta_j)\widetilde{X}^\dagger\right] = 0.$$

This follows by solving the constant-coefficient first-order equations and imposing the two conditions at l, as in Theorem 12.4.

At zero spectral parameter, the normalized wall values give

$$F_Q = l^{-1/2}\Phi_Q, \qquad F_U = l^{-1/2}\Phi_U.$$

Consequently

$$F_Q X F_U = \Phi_Q \widehat{X} \Phi_U.$$

The bounds of Theorem 12.6 give convergence of the retained positive eigenvalues to its singular values, with error $O(l^2)$; the remaining rescaled eigenvalues stay separated from zero. Positivity of the diagonal entries of Φ_Q, Φ_U proves attainability of every operator block.

To obtain a finite triple, one must additionally specify the algebra to be represented on the retained space. A multiplicative restriction must satisfy the reducing-subalgebra criterion in Result 3; taking the algebra generated by compressed operators is a different prescription. Once an algebra A_F is specified, its complex gauge group is the represented $U(A_F)$, and its scalar one-forms are

$$\Omega^1_{V_F}(A_F) = \operatorname{span}\{a[V_F, b] : a, b \in A_F\}.$$

Neither this algebra nor this space is determined by the light eigenvalues. □

5. Theorem (the exact quadratic matching condition at leading spectral-action order).

Amendment (referee). Both quadratic forms must be compared on one common profile space on which G^(5) and Q are positive, because a Kaluza-Klein profile can be invisible to compression. For the massless mixed pair, f = cos(2 pi x/l) satisfies Neumann conditions at both ends and has positive norm, while int w_L f = int w_R f = 0, so Q = K = 0 for this mode.

Let a five-dimensional Yang–Mills theory have no boundary kinetic or mass terms. Let $\langle\ ,\ \rangle_5$ be its positive invariant gauge form, including its coupling constants. Expand the retained transverse vectors about $A_5 = 0$ as

$$A_\mu(y, x) = \sum_a a_\mu^a(y) F_a(x),$$

where $F_a(x)$ are fixed Hermitian Lie-algebra-valued profiles satisfying the chosen gauge boundary conditions. Define

$$G^{(5)}_{ab} = \int_0^l \langle F_a, F_b\rangle_5 \, dx, \qquad H^{(5)}_{ab} = \int_0^l \langle F_a', F_b'\rangle_5 \, dx.$$

Then

$$\mathcal{L}^{(2)}_{\mathrm{KK}} = \frac{1}{4} G^{(5)}_{ab} f^a_{\mu\nu} f^{b,\mu\nu} + \frac{1}{2} H^{(5)}_{ab} a^a_\mu a^{b,\mu},$$

and

$$\boxed{\mathcal{M}^2_{\mathrm{KK}} = (G^{(5)})^{-1/2} H^{(5)} (G^{(5)})^{-1/2}.}$$

Let U_0 be the light frame at this background, and set

$$T_a = U_0^\dagger F_a U_0, \qquad V_0 = U_0^\dagger D_I U_0.$$

For the literal spectral trace, retain the gauge and scalar derivative terms through a_4, assume $f_0 > 0$, and add no independent kinetic terms. Define

$$Q_{ab} = \operatorname{Re}\operatorname{Tr}(T_a T_b), \qquad K_{ab} = \operatorname{Re}\operatorname{Tr}([T_a, V_0]^\dagger [T_b, V_0]).$$

On the subspace where $Q > 0$,

$$\boxed{\mathcal{M}^2_{\text{spec}} = \frac{3}{2} Q^{-1/2} K Q^{-1/2}.}$$

Thus equality of the vector mass spectra is equivalent to equality of the generalized spectra

$$\boxed{\operatorname{spec}\left(\tfrac{3}{2}K, Q\right) = \operatorname{spec}\left(H^{(5)}, G^{(5)}\right).}$$

For a retained sector whose KK mass matrix is $\kappa^2 I$, this is equivalent to the sum rule

$$\boxed{K = \frac{2}{3}\kappa^2 Q.}$$

Equality of the quadratic actions in the same field coordinates is stronger:

$$\boxed{G^{(5)} = 4\eta Q, \qquad H^{(5)} = 6\eta K, \qquad \eta = \frac{f_0}{24\pi^2}.}$$

For a chiral finite block

$$V_0 = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \end{pmatrix}, \qquad T_a = \operatorname{diag}(t_{L,a}, t_{R,a}),$$

put

$$C_a = t_{L,a} Y - Y t_{R,a}.$$

The trace data are explicitly

$$\boxed{Q_{ab} = \operatorname{Re}\operatorname{Tr}(t_{L,a} t_{L,b} + t_{R,a} t_{R,b}), \qquad K_{ab} = 2\operatorname{Re}\operatorname{Tr}(C_a^\dagger C_b).}$$

Here Y and the $t_{L/R,a}$ must both be computed from the interval eigenfunctions and the retained gauge profiles.

Proof. Expanding the five-dimensional action gives the two integral forms $G^{(5)}, H^{(5)}$. Canonical normalization by $(G^{(5)})^{1/2}$ gives its mass matrix.

For

$$\mathcal{D} = -i\gamma^\mu(\nabla^S_\mu + i\mathcal{A}_\mu) + \gamma_5 V,$$

the relevant heat-coefficient traces are

$$\operatorname{tr}_S(\mathcal{E}_F^2) = 2F_{\mu\nu}F^{\mu\nu}, \quad \operatorname{tr}_S(\mathcal{E}_V^2) = 4D_\mu V D^\mu V, \quad \operatorname{tr}_S(\Omega_F^2) = -4F_{\mu\nu}F^{\mu\nu}.$$

Their coefficients $1/2, 1/2, 1/12$ give

$$\mathcal{L}_{\text{spec}} = \eta \operatorname{Tr} F_{\mu\nu}F^{\mu\nu} + 3\eta \operatorname{Tr} D_\mu V D^\mu V.$$

At V_0, this becomes

$$\eta Q_{ab} f^a_{\mu\nu} f^{b,\mu\nu} + 3\eta K_{ab} a^a_\mu a^{b,\mu}.$$

The normalization $a_{\text{can}} = 2\sqrt{\eta}\, Q^{1/2} a$ proves the coefficient $3/2$, agreeing with Proposition 13.5.

Equality with a scalar matrix $\kappa^2 I$ is equivalent to

$$Q^{-1/2} K Q^{-1/2} = \frac{2}{3}\kappa^2 I.$$

The block formulas follow by multiplying

$$[T_a, V_0] = \begin{pmatrix} 0 & C_a \\ -C_a^\dagger & 0 \end{pmatrix}.$$

This proves all the matching conditions.

For mixed fermion pairs labelled by r, let their masses be λ_r, and let a single gauge profile f have overlaps

$$c_{L,r} = \int |p_{L,r}|^2 f, \qquad c_{R,r} = \int |p_{R,r}|^2 f.$$

If $\tau_r = \mathrm{tr}_r(t^2)$ includes every multiplicity, the one-vector condition becomes

$$\boxed{3\sum_r \tau_r \lambda_r^2 (c_{L,r} - c_{R,r})^2 = \kappa^2 \sum_r \tau_r (c_{L,r}^2 + c_{R,r}^2).}$$

This is the trace-and-overlap sum rule for that gauge reduction.

If instead both sheet connections are fluctuated independently, with the same representation on each pair, their relative mass is

$$M^2_{\text{relative}} = 6\frac{\sum_r \tau_r \lambda_r^2}{\sum_r \tau_r}.$$

Equating this single eigenvalue to κ^2 requires

$$6\sum_r \tau_r \lambda_r^2 = \kappa^2 \sum_r \tau_r.$$

That comparison has an additional massless diagonal vector and is not the same restriction as retaining one interval gauge profile.

All these spectral-action equalities concern the stated a_4 approximation and chosen vacuum. For a full finite scalar fluctuation space, identifying V_0 with the compressed bare operator additionally requires the scalar potential to select that vacuum. At masses comparable to the cutoff, Lemma 03, Part C shows why weighting Q alone does not give the broken-generator answer.

Finally, a nonabelian truncation to nonconstant lowest profiles need not be closed under gauge transformations: $[ft_a, ft_b] = f^2[t_a, t_b]$ generally leaves the retained span. The theorem compares quadratic transverse actions; it does not assert equality of full interacting theories. □

6. Counterexample (the covariant lowest-mode reduction still gives 3/5).

Let the gauge group be $U(1)$, and impose

$$\partial_x A_\mu(0) = 0, \qquad A_\mu(l) = 0,$$

with complementary conditions

$$A_5(0) = 0, \qquad \partial_x A_5(l) = 0.$$

Retain

$$A_\mu = a_\mu(y)\cos(\kappa x), \qquad A_5 = b(y)\sin(\kappa x), \qquad \kappa = \frac{\pi}{2l}.$$

Let one charge-one fermion have the mixed boundary conditions and scalar bulk mass M. Let λ be its lowest positive vertical eigenvalue, and define

$$c_L = \int_0^l p_L^2 \cos(\kappa x)\, dx, \qquad c_R = \int_0^l p_R^2 \cos(\kappa x)\, dx.$$

The gauge-covariant light projection gives

$$\boxed{M^2_{\text{spec}} = 3\lambda^2 \frac{(c_L - c_R)^2}{c_L^2 + c_R^2}, \qquad M^2_{\text{KK}} = \kappa^2.}$$

For $M = 0$,

$$\boxed{M^2_{\text{spec}} = \frac{3}{5} M^2_{\text{KK}}.}$$

Proof. The vector boundary problem selects $f(x) = \cos(\kappa x)$. Direct integration gives

$$\frac{\int (f')^2}{\int f^2} = \kappa^2,$$

which is the Neumann–Dirichlet result of Lemma 9.11.

The retained abelian gauge transformations have parameter

$$\epsilon(y, x) = \alpha(y) \cos(\kappa x).$$

With the conventions of Result 1,

$$a_\mu \mapsto a_\mu - \partial_\mu \alpha, \qquad b \mapsto b + \kappa\alpha.$$

Thus this vector-scalar restriction is gauge closed.

Parallel transport is

$$W(x) = \exp\left[-\frac{ib}{\kappa}(1 - \cos \kappa x)\right].$$

Result 3 gives

$$\Phi = \lambda e^{ib/\kappa},$$

and

$$\mathcal{A}_L = c_L a - \frac{1 - c_L}{\kappa}\, db, \qquad \mathcal{A}_R = c_R a + \frac{c_R}{\kappa}\, db.$$

Consequently,

$$D\Phi = i\Phi(c_L - c_R)\left(a + \frac{db}{\kappa}\right), \qquad F_L = c_L\, da, \quad F_R = c_R\, da.$$

The quadratic spectral action is therefore

$$\eta(c_L^2 + c_R^2)(da)^2 + 6\eta\lambda^2(c_L - c_R)^2 \left(a + \frac{db}{\kappa}\right)^2.$$

Canonical vector normalization proves the mass formula. In particular, the Berry terms have been included; the calculation does not use a fixed projection.

At $M = 0$,

$$\lambda = \kappa, \qquad p_L = \sqrt{\frac{2}{l}} \cos(\kappa x), \qquad p_R = \sqrt{\frac{2}{l}} \sin(\kappa x).$$

Elementary integration gives

$$c_L = \frac{8}{3\pi}, \qquad c_R = \frac{4}{3\pi}.$$

Hence

$$\frac{M^2_{\text{spec}}}{M^2_{\text{KK}}} = 3\frac{(8-4)^2}{8^2 + 4^2} = \frac{3}{5}.$$

The matrices $\mathrm{diag}(c_L, c_R)$ describe the coefficient of the retained vector in the induced connection; they are not the endpoint $U(1)$ charge assignment. The explicit transformation formulas above establish covariance. □

7. Proposition (exact matching is possible after imposing an additional bulk-mass condition).

Keep the gauge and fermion boundary conditions of Result 6. At fixed l, let the fermion bulk mass vary over

$$0 \le M \le \frac{3\pi}{4l}.$$

Then at least one value

$$\boxed{0 < M_* < \frac{3\pi}{4l}}$$

satisfies

$$M_{\mathrm{spec}}(M_*) = M_{\mathrm{KK}}.$$

An exact defining condition is

$$\boxed{3\lambda(M_*)^2(c_L(M_*) - c_R(M_*))^2 = \frac{\pi^2}{4l^2}(c_L(M_*)^2 + c_R(M_*)^2).}$$

No uniqueness is asserted.

Proof. For $M \ge 0$, write

$$t = kl \in [\pi/2, \pi), \qquad M = -k\cot t, \qquad \lambda = \frac{k}{\sin t}.$$

The lowest chiral profiles are proportional to

$$p_L(x) \propto \sin(k(l-x)), \qquad p_R(x) \propto \sin(kx).$$

The relation $M = -k\cot(kl)$ follows by imposing $p_L'(0) = Mp_L(0)$; substitution in the first-order equation gives the displayed p_R and λ.

At

$$t = \frac{3\pi}{4}, \qquad M = \frac{3\pi}{4l},$$

we have

$$\lambda^2 = \frac{9\pi^2}{8l^2}.$$

The common squared normalization integral is

$$\int_0^l \sin^2(kx)\,dx = l\left(\frac{1}{2} + \frac{1}{3\pi}\right).$$

Integrating against $\cos(\pi x/(2l))$ gives

$$c_L = \frac{9}{3\pi + 2}, \qquad c_R = \frac{6}{3\pi + 2}.$$

Therefore

$$\frac{M_{\mathrm{spec}}^2}{M_{\mathrm{KK}}^2} = 3\frac{9}{2}\frac{(9-6)^2}{9^2 + 6^2} = \frac{27}{26} > 1.$$

At $M = 0$, Result 6 gives $3/5 < 1$. The lowest eigenvalue is simple, and the displayed root and normalized profiles depend continuously on M. The intermediate value theorem proves the existence of M_*. Thus universal impossibility of mass matching is false, while automatic matching is also false. □

8. No-go theorem (bounded light fermions cannot match the collapsing Neumann–Dirichlet tower).

Let a family of induced finite operators satisfy

$$\|V_0(l)\| \le C < \infty.$$

Use the leading spectral action of Result 5, with no additional vector-mass or kinetic terms. The retained gauge couplings may depend on l, and any finite multiplicities may be included.

Then every canonically normalized spectral-action vector mass satisfies

$$\boxed{M^2_{\mathrm{spec}} \le 6C^2.}$$

Consequently it cannot equal

$$M^2_{\mathrm{KK}} = \frac{\pi^2}{4l^2}$$

for arbitrarily small l.

This applies to the fixed-light-mass collapse of Theorem 9.9 and to the bounded-texture limit of Result 4.

Proof. For every Hermitian retained coupling matrix T,

$$\|[T, V_0]\|_{\mathrm{HS}} \le \|TV_0\|_{\mathrm{HS}} + \|V_0T\|_{\mathrm{HS}} \le 2C\|T\|_{\mathrm{HS}}.$$

Hence

$$K \le 4C^2Q$$

as quadratic forms. Result 5 gives

$$\mathcal{M}^2_{\mathrm{spec}} \le 6C^2 I.$$

The same inequality holds after restricting to any gauge-profile subspace. Multiplicities cannot evade it because they enter both Hilbert–Schmidt norms.

The Neumann–Dirichlet mass diverges as l^{-1}, proving the contradiction. More quantitatively, matching any such vector requires

$$\boxed{\|V_0(l)\| \ge \frac{\pi}{2\sqrt{6}\,l}.}$$

Thus some retained fermion scale must grow at least as $1/l$, unless the bosonic action or the stated approximation is changed. □

9. Proposition (gauge-factor and scalar normalizations require additional conditions).

Let the finite gauge kinetic form be diagonal on two specified factors, with fixed generator normalizations,

$$Q|_{\mathfrak{g}_s} = q_s I, \qquad s = 1, 2.$$

Let their five-dimensional kinetic terms have independent couplings $g_{5,s}$, and let

$$J_s = \int_0^l f_s(x)^2\, dx.$$

Then the leading finite spectral action predicts

$$\frac{1}{g^2_{4,s}} = 4\eta q_s, \qquad \frac{g^2_{4,1}}{g^2_{4,2}} = \frac{q_2}{q_1},$$

whereas the KK action gives

$$\frac{1}{g^2_{4,s}} = \frac{J_s}{g^2_{5,s}}.$$

Vector-mass matching does not imply equality of these relative normalizations.

For scalar coordinates φ^i, let the induced connection at vanishing ambient A_μ be

$$\mathcal{A}_\mu = \mathcal{B}_i(\varphi)\partial_\mu\varphi^i,$$

and define

$$H_i = \partial_i V + i[\mathcal{B}_i, V].$$

The spectral-action scalar metric, in the convention $\frac{1}{2}Z_{ij}\partial_\mu\varphi^i\partial^\mu\varphi^j$, is

$$\boxed{Z_{ij}^{\mathrm{spec}} = 6\eta\ \mathrm{Re}\,\mathrm{Tr}(H_iH_j).}$$

Mass matching constrains this metric on the corresponding gauge-orbit directions. It does not generally determine the metric on additional physical scalar directions.

Proof. Canonical normalization of $\eta q_s F_s^2$ gives the finite coupling formula. Integrating the five-dimensional kinetic term gives the KK formula. Rescaling $g_{5,s}^{-2}$ multiplies both $G_s^{(5)}$ and $H_s^{(5)}$ by the same factor and leaves every KK mass unchanged. Therefore mass matching cannot determine the independent ratio $g_{5,1}/g_{5,2}$. A common simple parent group with a specified invariant form could remove this freedom, but that is additional geometric input.

For the scalar metric,

$$D_\mu V = H_i\partial_\mu\varphi^i.$$

Substitution into $3\eta\,\mathrm{Tr}(D_\mu V D^\mu V)$ proves the formula.

The eaten scalar in Result 6 illustrates the constrained directions precisely. Put

$$G = \frac{l}{2g_5^2}, \qquad q = c_L^2 + c_R^2, \qquad d = c_L - c_R.$$

Its KK action is

$$\frac{G}{4}(da)^2 + \frac{G}{2}(db + \kappa a)^2.$$

If both the gauge kinetic coefficient and the vector mass match,

$$4\eta q = G, \qquad 3\lambda^2 d^2 = \kappa^2 q,$$

then the spectral scalar coefficient automatically satisfies

$$Z_b^{\mathrm{spec}} = \frac{12\eta\lambda^2 d^2}{\kappa^2} = G.$$

Thus no separate tuning is needed for this eaten scalar once those two equalities hold.

A scalar direction transverse to the gauge orbit is not tested by that mass equation. In the commuting-bulk mixed model, A_5 supplies only the unitary link of Result 3, so there is no radial Higgs kinetic term to match. If boundary Yukawa matrices are promoted to independent scalar fields, their continuum dynamics, including any boundary kinetic form, must be specified separately. □

Consequence for T4b and hypothesis H

Gauge-covariant projection repairs the covariance defect and supplies an exact light-bundle connection, but it does not establish equality with the full finite-triple fluctuation space or with the interacting KK theory. At fixed length, vector-mass equality is possible precisely when the appropriate trace-and-overlap condition holds; covariance supplies no selection principle for it or for the remaining kinetic normalizations. Hypothesis H therefore requires additional geometric relations, and its collapse regime cannot reproduce a Neumann–Dirichlet vector of mass $\pi/(2l)$ from a leading finite spectral action whose retained fermion masses all remain bounded.

Part 15

Collapse in the spectral propinquity: a gap criterion and limits of commutative spaces

Lemmas R2 and R3 of the Farsi-Latrémolière assessment, written out and extended. Proved by Claude Opus 5.5 without Astra or Fable, from the definitions in the LaTeX source of arXiv:2404.00240v3, Section 2. The sketches came from the reading subagent; the lower bound in the sketch of R2 omitted the cap sqrt(2)/2 of the metric and is corrected here. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): Theorem 15.2 item 2 had a gap, closed by the referee; Remark 15.4 omitted the cap; Proposition 15.9 needed amendments; the T0 section first contained two invalid arguments (on unbroken symmetry and on Morita equivalence). All corrections are incorporated below. Proposition 15.10 and Counterexample 15.11 are the referee's.

Scope. Everything here concerns one metric, the spectral propinquity, applied to the whole algebra of the parent triple. The section "What this says about T0" states what follows for the founding problem and what does not; lemma 16 treats truncations, where other answers hold.

Conventions

These are the definitions of Farsi-Latrémolière, Section 2.

- A *quantum compact metric space* (A, L) is a unital C*-algebra A with a seminorm L on a dense subspace of sa(A) such that ker L = R 1; the metric mk_L(phi, psi) = sup{abs(phi(a) - psi(a)) : L(a) <= 1} metrizes the weak* topology of the state space S(A); L satisfies the Leibniz inequality for the Jordan product a o b = (ab + ba)/2 and the Lie product {a, b} = (ab - ba)/(2i); and {L <= 1} is closed.
- A *quantum isometry* is a *-epimorphism pi with L_B(b) = inf{L_A(a) : pi(a) = b}. Its dual map is an isometry of state spaces.
- A *tunnel* is a quantum compact metric space (E, L_E) with quantum isometries pi_1, pi_2 onto the two spaces. Its *extent* chi is the larger of the Hausdorff distances, in mk_{L_E}, from S(E) to pi_1^*S(A_1) and to pi_2^*S(A_2). The *propinquity* Lambda is the infimum of extents.
- A *metric spectral triple* T = (A, H, D) is a spectral triple for which L_T(a) = norm([D, a]) is an L-seminorm. In particular ker L_T = R 1. We write DN_T(xi) = norm(xi) + norm(D xi) and diam T for the diameter of S(A) in mk_{L_T}.
- A *metrical tunnel* between T_1 and T_2 consists of a C*-correspondence P from E to a unital C*-algebra B, with L_E and L_B as above and a norm DN_P on a dense subspace of P such that: the unit ball of DN_P is compact in P; DN_P dominates the module norm; DN_P(e Theta) <= (norm(e) + L_E(e)) DN_P(Theta) (*modular Leibniz*); L_B applied to the real and to the imaginary part of <Theta, Theta'>*B is at most 2 DN_P(Theta) DN_P(Theta')* (inner Leibniz*; this is the condition of Latrémolière, arXiv:1811.10843, in the definition of a metrical C*-correspondence, whereas the display in Farsi-Latrémolière applies L to the non-self-adjoint inner product and names L_A). It carries maps (Pi_j, pi_j, theta_j) onto (H_j, A_j, C), compatible with the actions and inner products, with DN*{T_j}(xi) = inf{DN_P(Theta) : Pi_j Theta = xi}. Its extent is the larger of the extents of (E, pi_1, pi_2) and (B, theta_1, theta_2).
- The *pseudo-states* of T are the functionals xi -> <omega, xi> with DN_T(omega) <= 1. The *separation* of two operator families (a_t), (b_t) is the Hausdorff distance between {(phi a_t Pi_1)_t} and {(psi b_t Pi_2)_t}, phi and psi ranging over pseudo-states, in the metric sup_t sup{abs(F_t(Theta) - G_t(Theta)) : DN_P(Theta) <= 1}. The *dispersion* is the larger of extent and separation.
- The *spectral propinquity* Lambda_spec(T_1, T_2) is the infimum of sqrt(2)/2 and of those eps > 0 for which some metrical tunnel has dispersion < eps for the families exp(i t D_1), exp(i t D_2), 0 <= t <= 1/eps.

Three elementary consequences.

(a) If tau is a tunnel of extent chi, then abs(diam_1 - diam_2) <= 2 chi. Take phi, phi' realizing diam_1; their pullbacks lie within chi of pullbacks of states of A_2, and the dual maps are isometries.

(b) Lambda_spec(T_1, T_2) >= min(sqrt(2)/2, Lambda(A_1, A_2)), because the dispersion dominates the extent of the top tunnel. So convergence in Lambda_spec implies convergence in Lambda.

(c) Combining: Lambda_spec(T_1, T_2) >= min(sqrt(2)/2, abs(diam T_1 - diam T_2)/2).

The cap is dimensionful: Lambda_spec has the units of a length and is cut off at sqrt(2)/2 in the chosen unit.

(d) If a = a* in (A, L), and z is the midpoint of the convex hull of Sp(a), then norm(a - z) <= (diam/2) L(a). The endpoints of the hull are values of states on a.

Comparison along a tunnel

Lemma 15.1. Let tau = (E, L_E, pi_1, pi_2) be a tunnel of extent chi, and let e = e* with L_E(e) finite. For {j, k} = {1, 2},

```
norm(pi_j e) <= norm(pi_k e) + chi L_E(e),     norm(e) <= norm(pi_j e) + chi L_E(e).
```

For a real polynomial p = sum p_m x^m, the element p(e) has finite L_E, with L_E(p(e)) <= L_E(e) sum_{m >= 1} abs(p_m) m norm(e)^{m-1}.

Proof. The norm of a self-adjoint element is the supremum of abs(state) on it. A state phi of A_j pulls back to S(E), and lies within chi of the pullback of a state psi of A_k, so abs(phi(pi_j e)) <= abs(psi(pi_k e)) + chi L_E(e). The second inequality is the same argument starting from a state of E. For powers, e^m = e o e^{m-1}, and induction with the Jordan-Leibniz inequality gives L_E(e^m) <= m norm(e)^{m-1} L_E(e); L_E(1) = 0. ■

Limits of commutative spaces (Lemma R3)

Theorem 15.2. Let (A_n, L_n) -> (A, L) in the propinquity.

1. If every A_n is commutative, then A is commutative.
2. If every A_n = C(X_n) with X_n having at most k connected components, then the spectrum of A has at most k connected components. In particular, a finite-dimensional limit is C^j with j <= k, and for k = 1 it is C.

Both statements hold for limits in Lambda_spec, by (b).

Proof. Let tau_n be tunnels of extent chi_n -> 0.

(1) Let a_1, a_2 in sa(A) have finite L. Choose lifts d_i in E_n with pi_A(d_i) = a_i and L_{E_n}(d_i) <= L(a_i) + 1. Lemma 15.1 bounds norm(d_i) uniformly. The Lie product e = {d_1, d_2} satisfies pi_n(e) = 0 and pi_A(e) = {a_1, a_2}, and L_{E_n}(e) <= norm(d_1) L(d_2) + norm(d_2) L(d_1) is uniformly bounded. So norm({a_1, a_2}) <= chi_n L_{E_n}(e) -> 0. Elements of finite L are dense in sa(A), so A is commutative.

(2) By (1), A = C(X). Suppose X has at least k + 1 components. Since components and quasicomponents agree in a compact Hausdorff space, X splits into nonempty clopen sets X_0, ..., X_k. Let a in sa(A) have finite L and norm(a - sum_j j 1_{X_j}) < 1/8, so that Sp(a) meets each [j - 1/8, j + 1/8] and lies in their union. Take lifts d_n as above, and b_n = pi_n(d_n), a real function on X_n. All spectra lie in a fixed compact interval I for large n, by Lemma 15.1.

Let g : I -> [0, 1] be continuous, 0 on the union of the [j - 1/8, j + 1/8] and 1 outside the union of the open intervals (j - 3/8, j + 3/8), and let p be a polynomial with abs(p - g) < 1/8 on I. Then norm(p(a)) <= 1/8, so Lemma 15.1 gives norm(p(b_n)) <= 1/4 for large n, because L_{E_n}(p(d_n)) is uniformly bounded. Hence the range of b_n lies in the union of the disjoint intervals (j - 3/8, j + 3/8), and each component of X_n maps into one of them. So b_n reaches at most k of them, and along a subsequence it misses one fixed j_*. Let h : I -> [0, 1] be continuous, 1 on [j_* - 1/8, j_* + 1/8] and 0 outside radius 3/8 around j_*, and let q be a polynomial with abs(q - h) < 1/8 on I. Then norm(q(a)) >= 7/8 and norm(q(b_n)) <= 1/8, contradicting Lemma 15.1 for large n. ■

The theorem is one-directional: noncommutative spaces may converge to commutative ones, as matrix algebras do to the sphere (Rieffel; Latrémolière) and spectral truncations of tori do to the tori in the Gromov-Hausdorff sense on state spaces (Leimbach-van Suijlekom). We did not find Theorem 15.2 stated in the sources we searched (arXiv:1302.4058, 1311.0104, 2404.00240, 2603.19128); its proof is elementary and it may be folklore.

The gap criterion (Lemma R2)

Theorem 15.3. Let T = (A, H, D) be a metric spectral triple with delta = diam T. Let Sigma be a finite subset of R, P = 1_Sigma(D) of rank N >= 1 and Q = 1 - P, and let g > 0 be such that abs(lambda) >= g for every lambda in Sp(D) outside Sigma. Let D_F be a Hermitian N x N matrix, W : PH -> C^N a unitary, extended by 0 on QH, and Delta = norm(W D W* - D_F). For T_F = (C, C^N, D_F),

```
min(sqrt(2)/2, delta/2) <= Lambda_spec(T, T_F) <= 1/g^2 + max(delta/2, Delta) + Delta +
↪ sqrt(Delta).
```

For Delta = 0 the upper bound reads delta/2 + 1/g^2. No anticommutation hypothesis enters.

Proof. The lower bound is (c), since diam T_F = 0. For the upper bound fix rho > 0 with rho >= max(delta/2, Delta), and put s = rho + 1/(2 g^2), c = 1/(1 + Delta), D_B = W D W*.

Top tunnel. On E = A (+) C let L_E(a, z) = max{L_T(a), norm(a - z)/rho}. Then ker L_E = R (1, 1). The Leibniz inequality for the Jordan product follows from a o b - z w = (a - z) o b + z (b - w), and for the Lie product from {a, b} = {a - z, b}. The set {L_E <= 1, mu(a) = 0}, for a fixed state mu of A, is contained in a product of compact sets, since abs(z) <= norm(a) + rho and norm(a) <= delta there, so L_E is an L-seminorm by Rieffel's criterion, with the closed ball following from lower semicontinuity. The projection onto A is a quantum isometry by (d), as rho >= delta/2; the projection onto C is one because L_E(z 1, z) = 0. A state of E has the form (a, z) -> t mu(a) + (1 - t) z. On L_E(a, z) <= 1 it differs from (a, z) -> mu(a) by (1 - t) abs(z - mu(a)) <= norm(a - z) <= rho, and from (a, z) -> z by t abs(mu(a) - z) <= rho. So the extent is at most rho.

Base tunnel. On B = C (+) C let L_B(x, y) = abs(x - y)/s. The two projections are quantum isometries onto (C, 0), and the extent is at most s.

Module. Let P_mod = H (+) C^N, a right Hilbert B-module with <(xi, eta), (xi', eta')> = (<xi, xi'>, <eta, eta'>) and left action (a, z)(xi, eta) = (a xi, z eta). On Dom D (+) C^N set

```
DN_P(xi, eta) = max{DN_T(xi), DN_F(eta), norm(W xi - eta)/rho}.
```

The unit ball is closed in a product of compact sets, because D has compact resolvent. The norm is dominated.

Modular Leibniz. DN_T(a xi) <= (norm(a) + L_T(a)) DN_T(xi) and DN_F(z eta) = abs(z) DN_F(eta). Further

```
norm(W a xi - z eta) <= norm(W (a - z) xi) + abs(z) norm(W xi - eta) <= norm(a - z) norm(xi) +
↪ abs(z) rho DN_P(xi, eta),
```

and dividing by rho gives at most (L_E(a, z) + norm(a, z)) DN_P(xi, eta). No commutation of P with A is needed, because the limit algebra acts by scalars.

Inner Leibniz. On Q H the spectrum has abs(lambda) >= g, so norm(Q xi) <= norm(D xi)/g <= DN_T(xi)/g. Then <xi, xi'> = <W xi, W xi'> + <Q xi, Q xi'> and

```
abs(<W xi, W xi'> - <eta, eta'>) <= norm(W xi - eta) norm(W xi') + norm(eta) norm(W xi' - eta') <=
↪ 2 rho DN_P DN_P',
```

so abs(<xi, xi'> - <eta, eta'>) <= (2 rho + 1/g^2) DN_P DN_P' = 2 s DN_P DN_P'. This complex estimate implies the condition on real and imaginary parts.

Quotient conditions. Given xi, take eta = c W xi. Since P reduces D and D_F W = W D - (D_B - D_F) W on Dom D, we get DN_F(c W xi) <= c (norm(W xi) + norm(W D xi) + Delta norm(W xi)) <= DN_T(xi), and norm(W xi - c W xi) <= Delta norm(xi) <= rho DN_T(xi). So DN_P(xi, c W xi) = DN_T(xi). Given

eta, take xi = c W* eta. Then DN_T(xi) = c (norm(eta) + norm(D_B eta)) <= DN_F(eta) and norm(W xi - eta) <= Delta norm(eta), so DN_P(c W* eta, eta) = DN_F(eta).

Separation. Let DN_T(omega) <= 1 and put omega' = c W omega, so that DN_F(omega') <= 1. For DN_P(xi, eta) <= 1 and real t, since P commutes with exp(i t D),

```
<omega, e^{itD} xi> - <omega', e^{itD_F} eta> = <Q omega, e^{itD} Q xi> + <W omega, (e^{itD_B} -
↪  e^{itD_F}) W xi> + <W omega, e^{itD_F} (W xi - eta)> + (1 - c) <W omega, e^{itD_F} eta>,
```

and the four terms are bounded by 1/g^2, abs(t) Delta, rho and Delta. Conversely, let DN_F(omega') <= 1 and put omega = c W* omega', so that DN_T(omega) <= 1 and Q omega = 0; the same decomposition without the first term gives at most abs(t) Delta + rho + Delta.

Conclusion. Put K = 1/g^2 + rho + Delta. The extent is at most s <= K and the separation on [0, 1/eps] is at most K + Delta/eps. If eps > K + sqrt(Delta), then Delta/eps < sqrt(Delta) (or Delta = 0), so the dispersion is below eps. Hence Lambda_spec <= K + sqrt(Delta), and we let rho decrease to max(delta/2, Delta). ■

Remark 15.4 (rescaling). For T_eps = (A, H, D/eps) with 0 in Sp(D), Sigma = {0} and D_F = 0, the theorem gives min(sqrt(2)/2, eps delta_1/2) <= Lambda_spec <= eps delta_1/2 + $\text{eps}^{2/g_1}2$, where delta_1 = diam T_1 and g_1 is the gap of D. This is Corollary 4.1 of Farsi-Latrémolière with a sharp leading rate for small eps. Counterexample R1 of the assessment violates the gap hypothesis, since the eigenvalue x stays bounded outside Sigma.

Corollary 15.5 (the tuned interval, Theorem 9.9). Let T_l = (C([0, l]), L^2([0, l]; C^2), D_{-mu(l)}) with condition (c) and mu(l) as in Theorem 9.9, c = mu(l) l, and let P project onto +/- m_f. For l small,

```
l/2 <= Lambda_spec(T_l, (C, C^2, m_f sigma_1)) <= l/2 + l^2/(pi^2 + c^2).
```

For the Connes-Lott triple CL(m_f) = (C + C, C^2, m_f sigma_1) of diameter 1/m_f,

```
Lambda_spec(T_l, CL(m_f)) >= min(sqrt(2)/2, (1/m_f - l)/2).
```

Proof. Multiplication by a Lipschitz function preserves condition (c), and the mass term commutes with functions, so L(f) = norm(f')_infinity and diam T_l = l. Theorem 9.6 gives W with W D W* = m_f sigma_1, so Delta = 0, and Theorem 9.4 gives g >= sqrt(mu^2 + (pi/l)^2). The second bound is (c). ■

So the whole triple converges to a point carrying a two-dimensional massive Dirac block, while the compression at each l is the Connes-Lott triple (Theorem 9.6). The two statements concern different objects, and both hold; see "What this says about T0".

Corollary 15.6 (towerless Yukawa block, Theorem 12.7). Fix the c's and Y_hat of Theorem 12.7, with T_{X,l} = (C([0, l]), L^2 (x) (C^{N_Q} (+) C^{N_U}), D_X). Then T_{X,l} is a metric spectral triple of diameter l, and

```
Lambda_spec(T_{X,l}, (C, C^{N_Q + N_U}, D_F)) = O(l),     D_F = [[0, M_F], [M_F^dagger, 0]],   M_F
↪  = Phi_Q Y_hat Phi_U.
```

Proof. Multiplication by f preserves the conditions at l because f takes one value there, and norm([D_X, f]) = norm(f')_infinity. Theorem 12.7 gives g >= min(L_Q, L_U)/l. Mapping eigenvectors of +/- lambda_k to the singular vectors of M_F gives W with Delta = max_k abs(lambda_k - sigma_k(M_F)) = O(l^2). Theorem 15.3 applies, with sqrt(Delta) = O(l). ■

This answers the remark in Theorem 12.7 that a limit triple needs a choice of algebra: in this metric the algebra of the limit is C.

The converse

Theorem 15.7. Let T_n = (A_n, H_n, D_n) be metric spectral triples with Lambda_spec(T_n, T_F) -> 0, T_F = (C, C^N, D_F). Then diam T_n -> 0; for every R > norm(D_F), D_n has exactly N eigenvalues in [-R, R] for large n, so that the gap g_n to the remaining spectrum tends to infinity (with g_n infinite if

no spectrum remains); and there are unitaries W_n from the light space onto C^N with norm(W_n D_n W_n* - D_F) -> 0. With Theorem 15.3,

```
Lambda_spec(T_n, T_F) -> 0   if and only if   diam T_n -> 0, g_n -> infinity, Delta_n -> 0.
```

Proof. The diameter statement is (c). Let tau_n have dispersion < eps_n on [0, 1/eps_n], with eps_n -> 0.

Inner estimate. The characters theta_1, theta_2 are states of the base algebra B_n, and S(C) is one point, so mk_{L_B}(theta_1, theta_2) < eps_n. Applying this to the real and imaginary parts of <Theta, Theta'> and using inner Leibniz,

```
abs(<Pi_1 Theta, Pi_1 Theta'> - <Pi_2 Theta, Pi_2 Theta'>) <= 4 eps_n DN_P(Theta) DN_P(Theta').
```

At most N eigenvalues. Suppose D_n has N + 1 orthonormal eigenvectors xi_i with eigenvalues in [-R, R], so DN(xi_i) <= 1 + R. Lift them to Theta_i with DN_P(Theta_i) <= 2(1 + R), and put eta_i = Pi_2 Theta_i in C^N. The Gram matrix G of the eta_i has rank at most N, so norm(1 - G) >= 1, while the inner estimate gives norm(1 - G) <= 16 (N + 1)(1 + R)^2 eps_n. This fails for large n.

Concentration. Let e be a unit eigenvector of D_F with eigenvalue lambda, C = 1 + norm(D_F), and lift e to Theta with DN_P(Theta) <= 2C and xi = Pi_1 Theta. Apply the separation to the pseudo-state omega = xi/(2C): there is omega' with DN_F(omega') <= 1 and, for t in [0, 1/eps_n], abs(<omega, e^{itD_n} xi> - e^{it lambda} <omega', e>) <= 2C eps_n. Comparing with t = 0 and multiplying by 2C,

```
abs(<xi, e^{itD_n} xi> - e^{it lambda} norm(xi)^2) <= 8 C^2 eps_n.
```

Let nu be the spectral measure of D_n at xi. Taking real parts and averaging over t in [0, 1/eps_n] gives the integral of 1 - sinc((x - lambda)/eps_n) d nu(x) at most 8 C^2 eps_n. Since 1 - sinc(u) >= 1/2 for abs(u) >= 2, nu{abs(x - lambda) >= 2 eps_n} <= 16 C^2 eps_n.

Multiplicities. Apply this to an orthonormal eigenbasis e_i of D_F, with lifts xi_i. The inner estimate makes the Gram matrix of the xi_i tend to the identity, and concentration keeps it so after projecting each xi_i to the spectral window of radius 2 eps_n around its eigenvalue. So each window around an eigenvalue of multiplicity m contains at least m eigenvalues of D_n, for large n. The total is at most N, by the previous step, so equality holds. Mapping eigenvectors to the e_i gives W_n with norm(W_n D_n W_n* - D_F) <= 2 eps_n. ■

Corollary 15.8 (finite limits of connected commutative triples). Let T_n be metric spectral triples with A_n = C(X_n), X_n connected, converging in Lambda_spec to a triple with finite-dimensional Hilbert space. Then the limit is (C, C^N, D_F), and the convergence is characterized by Theorem 15.7. Every Hermitian D_F arises: on the circle of length l, the operator -i d/dx + diag(lambda_i) on $L^{2(S^1_l; C^N)}$ has the block diag(lambda_i) on constants and gap at least 2 pi/l - norm(D_F). This construction concerns ungraded triples; a grading or real structure imposes its own compatibility conditions.

Proof. A metric triple is faithful: a self-adjoint a acting as 0 has [D, a] = 0, so a = t 1 by ker L = R 1, and then t = 0. So the limit algebra is finite-dimensional, and Theorem 15.2 gives C. ■

Corollary 9.10 fits the converse: at fixed mu every eigenvalue diverges, the spectra have no bounded sequences, and no limit exists. Theorem 9.8 lies outside the metric, because the diameter diverges.

Grading and real structure

The metric ignores gamma and J. They pass to the block exactly.

Proposition 15.9. In Theorem 15.3 let Sigma = -Sigma, and let D anticommute with a grading gamma and satisfy J D = eps' D J for an antiunitary J with J^2 = eps and J gamma = eps'' gamma J. Then P commutes with gamma and J, and gamma_F = W gamma W*, J_F = W J W* satisfy the same relations exactly with D_B = W D W*, and with D_F up to defects bounded by 2 Delta. The tunnel of Theorem 15.3 controls them: for DN_T(omega) <= 1 and DN_P(xi, eta) <= 1,

```
abs(<omega, gamma xi> - <W omega, gamma_F eta>) <= 1/g^2 + rho,     abs(<omega, J xi> - <W omega,
↪  J_F eta>) <= 1/g^2 + rho.
```

For Delta > 0, W omega need not be a pseudo-state of T_F; the normalized c W omega is one, and adds at most Delta to these bounds.

Proof. Spectral projections of D for a symmetric set commute with gamma and J, because gamma and J map the lambda-eigenspace to the eigenspaces of -lambda and eps' lambda. Decompose along P and Q as in Theorem 15.3, using additivity of J_F. ■

So gamma added to the families converges along the same tunnels, and the KO-signs of the parent reach the block unchanged when Delta = 0. For the interval, P Gamma P = sigma_3 (Theorem 9.6). Composition with the antilinear J gives antilinear functionals, which the definition of the separation does not cover; a statement of convergence for J needs an extension of the pseudo-state formalism, which we have not developed.

Unbroken symmetry and the metric class

Proposition 15.10 (referee). Let (A, H, D, J) be a finite-dimensional real triple with order zero, order one and J D = eps' D J, and let u be a unitary in A. If D commutes with u J u J^-1, then [D, u] = 0. If moreover the triple is metric, u is scalar, and the inner gauge transformation is trivial.

Proof. Put v = J u J^-1. From [D, u v] = 0, we get [D, u] v = - u [D, v], so a := u* [D, u] = - [D, v] v*. Now [D, v] = eps' J [D, u] J^-1 commutes with A by order one, and v* commutes with A by order zero. So a commutes with u, and [D, u^n] = n a u^n by induction. Then u^n D u^-n = D - n a. D is bounded, so a = 0. For a metric triple, the self-adjoint parts of u lie in ker L = R 1. ■

Counterexample 15.11 (referee; order one is needed). Let H = M_2(C) with the Hilbert-Schmidt product, A = M_2(C) acting by left multiplication, J x = x*, and D x = x - (1/2) tr(x) I. Applying [D, L_a] to I gives a - (1/2) tr(a) I, so [D, L_a] = 0 forces a scalar, and the triple is metric. Yet every conjugation x -> u x u* = L_u J L_u J^-1 commutes with D. This triple fails order one.

What this says about T0

Corollary 15.8 describes one notion of convergence: the state space of the whole algebra C(K_l) must converge in the Monge-Kantorovich metric. Under it, a collapsing connected parent yields a point, and the finite data survive as (C, C^N, D_F, gamma_F, J_F). Three limits on its reach:

1. **The truncation is a different object.** Along Theorem 9.9, the compression at each l is exactly Connes-Lott, with distance 1/m_f, while the parent has diameter l. The compression is not a quantum isometry, so this metric cannot see it. This is the towerless phenomenon itself: the two-sheet distance is unrelated to the size of the internal space.
2. **The class excludes a kind of target.** A metric spectral triple has ker L_T = R 1, so no nonscalar element of A commutes with D. The finite triple of the Standard Model lies outside the class: D_F commutes with the nonscalar subalgebra C_F = {(lambda, lambda, 0)}, which keeps the photon massless, and colour is unbroken (Chamseddine-Connes-Marcolli 2007, section 2.6). Its almost-commutative product lies outside as well, through the constant copy of such an element. A_F is moreover a real algebra, so applying the complex framework also needs a complexification. Among finite real first-order triples, the metric class has no nontrivial unbroken inner gauge transformation (Proposition 15.10); without order one this fails (Counterexample 15.11).
3. **The theorem concerns the scalar algebra.** Theorem 15.2 restricts limits of the whole scalar algebra C(K). Morita equivalence does not remove that restriction: C(K) and C(K, M_n) are Morita equivalent but not isomorphic, and their limits differ, and a bundle E can carry matrix-valued connections while C(K) still acts by scalars. Replacing the represented algebra by End(E) is a change of data, and a possible enlargement of T0. In that enlargement, Farsi-Latrémolière Theorem 4.6 makes an even metric finite (B, H_F, D_F) with noncommutative B the limit of B (x) C(K_l) with K_l collapsing, with nonzero multiplicity space ker D_K; there B is put in by hand.

Theorem 15.2 is a real constraint within its scope: this metric, on scalar function algebras. T0 also allows truncations, where it does not apply; lemma 16 treats these.

Part 16

Noncommutative finite real triples from commutative data: truncating the coarse correspondence

Proved by Claude Opus 5.5 without Astra or Fable, in response to the challenge that the T0 note overread lemma 15 and the Farsi-Latrémolière limits. Signs and the level-two algebra were checked with numpy on random data (section "Numerical check"). The term "coarse correspondence" is that of Connes, *Noncommutative Geometry* (1994), chapter V, for the bimodule of Hilbert-Schmidt operators, and the form D = Delta + J Delta J^-1 is in Krajewski (arXiv:hep-th/9701081) and Paschke-Sitarz (arXiv:q-alg/9612029); all checked against the sources. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): the examples and Corollary 16.4 are correct; amendments to Proposition 16.1, Theorem 16.2, Corollary 16.3 and the caveats are incorporated below; Remark 16.5 and Example 16.6 come from the referee's assessment of what the lemma achieves. The two-sheet section was refereed separately (Terra 6.1, 3 Nov 2026; report), with one error in Corollary 16.8 and one in Example 16.9, corrected below.

Outcome. Compressing functions on a connected commutative space to a fixed spectral window produces a noncommutative finite algebra with no limit taken. With the standard reality J a J^-1 = a*, the inherited structure fails order zero exactly when that algebra is noncommutative (Proposition 16.1). With the parent Hilbert space doubled to H_0 (x) conj(H_0), order zero and order one hold for every such algebra (Theorem 16.2, Corollary 16.3); these two conditions are formal consequences of the tensor square, available for any finite even representation (Remark 16.5). The KO-6 signs and the count dim H_F = 2 k^2 are reached by a product with a Clifford point, which is an input and fails finite orientability (Corollary 16.4), and more faithfully by a parent with two sheets, which gives the form of the Chamseddine-Connes classification (Theorem 16.7, Corollary 16.8, Example 16.9). This selects no particular algebra and gives no dynamics.

Setting

Let (A_0, H_0, D_0, gamma_0) be an even spectral triple, A_0 commuting with gamma_0, with D_0 of compact resolvent. Let Sigma = -Sigma be a finite subset of R and P_0 = 1_Sigma(D_0), of rank k, so that P_0 commutes with gamma_0. Put V = P_0 H_0, graded by gamma_V, with d = D_0 restricted to V, and

```
Q(a) = P_0 a P_0 on V,     O_F = Q(A_0),     A_F = C*(O_F) in B(V).
```

O_F is an operator system, in general not an algebra, and A_F consists of even operators. Compression is standard: for functions it is the Toeplitz compression, and its use in truncating spectral triples is due to D'Andrea-Lizzi-Martinetti and Connes-van Suijlekom. The latter note that the C*-algebra generated by PAP is in many examples a noninformative full matrix algebra. Since Q(1) = 1, A_F is the bicommutant of O_F, so its structure is set by the operators commuting with all compressed functions, such as the grading in example (b) and j in Example 16.9. That is where the information lies, and the fact that A_F is noncommutative is modest.

Examples of A_F.

(a) *Chiral interval, lowest level.* For condition (c) with M = 0 and Sigma = {+/- pi/(2l)}, O_F = C + C = A_F (lemma 05).

(b) *Chiral interval, next level.* For Sigma = {+/- pi/(2l), +/- 3 pi/(2l)}, V has basis (c_1, 0), (c_2, 0), (0, s_1), (0, s_2), with c_r = (2/l)^{1/2} cos((r - 1/2) pi x/l) and s_r = (2/l)^{1/2} sin((r - 1/2) pi x/l). With C_n = (1/l) int f cos(n pi x/l),

Q(f) = (Q_L, Q_R), Q_L = [[C_0 + C_1, C_1 + C_2], [C_1 + C_2, C_0 + C_3]], Q_R = [[C_0 - C_1, C_1 - C_2], [C_1 - C_2, C_0 - C_3]].

The values C_0, C_1, C_2, C_3 are independent and determine Q(f), so O_F has dimension 4. Each of Q_L and Q_R alone ranges over all symmetric 2 x 2 matrices, which generate M_2(C), so A_F projects onto each summand. For f_1 = 1/2 + cos(pi x/l) + cos(2 pi x/l) + cos(3 pi x/l) and f_2 = 1/2 - cos(pi x/l) + cos(2 pi x/l) - cos(3 pi x/l),

```
Q(f_1) = ([[1, 1], [1, 1]], 0),     Q(f_2) = (0, [[1, -1], [-1, 1]]).
```

M_2(C) is simple, so multiplying Q(f_1) and Q(f_2) by elements of A_F gives M_2(C) + 0 and 0 + M_2(C). Hence A_F = M_2(C) + M_2(C), of dimension 8. The finite algebra of the interval itself is noncommutative at the second level.

(c) *Twistor space.* For the level-one Toeplitz compression on CP^3, Q(x_a) = Gamma_a/5 (Proposition 10.1), and these generate A_F = M_4(C) on V = C^4. Here V is the kernel of the Dolbeault-Dirac operator twisted by O(1), so d = 0 on V.

The standard correspondence

Proposition 16.1 (order zero fails with inherited reality). Let J_0 be a real structure on H_0 (antiunitary, J_0^2 = +/-1, with the grading sign), with J_0 a J_0^-1 = a* for all a in A_0 and J_0 D_0 = eps' D_0 J_0, and let J_F = J_0 restricted to V. Then J_F Q(a)* J_F^-1 = Q(a), and the data (A_F, V, d, J_F) satisfy order zero if and only if A_F is commutative. The same holds for a product with a fixed finite factor that itself satisfies order zero, with the compatible product real structure.

Proof. J_0 preserves V because Sigma = -Sigma. Then J_0 P_0 a* P_0 J_0^-1 = P_0 a P_0. Order zero would make Q(a) commute with Q(b)° = Q(b) for all a, b, so O_F and hence A_F would be commutative; the converse is immediate, since J_F A_F J_F^-1 = A_F. For the product, take the unit in the finite factor: [1 (x) Q(a), 1 (x) Q(b)] = 1 (x) [Q(a), Q(b)]. ■

Together with Amendment 2 of lemma 07 (first order fails for C + C with inherited J_2), this is the correct form of the obstruction. It concerns the standard correspondence L^2(A_0) with the pointwise real structure.

The coarse correspondence

Let conj(H_0) be the conjugate Hilbert space, with canonical antiunitary xi -> xi^c, and with gamma^c, D^c the conjugate operators, gamma^c xi^c = (gamma_0 xi)^c and D^c xi^c = (D_0 xi)^c. On H = H_0 (x) conj(H_0), which is the coarse correspondence of A_0, equivalently the Hilbert-Schmidt operators on H_0, define for homogeneous xi, eta

```
pi(a) = a (x) 1,
DD = D_0 (x) 1 + gamma_0 (x) D^c,
Gamma = gamma_0 (x) gamma^c,
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c.
```

Theorem 16.2. (A_0, H, DD, J, Gamma) is a real even spectral triple with signs J^2 = 1, J DD = DD J, J Gamma = Gamma J (KO-dimension 0). For even a, b with bounded commutators,

```
a° := J pi(a)\* J^-1 = 1 (x) (a\*)^c,     [pi(a), b°] = 0,     [[DD, pi(a)], b°] = 0,
```

and norm([DD, pi(a)]) = norm([D_0, a]). So if the original triple is metric, the new triple is metric with the same Connes distance on A_0. If the Weyl exponent p = lim log N_0(R)/log R of D_0 exists, that of DD is 2 p, since N_0(R/sqrt(2))^2 <= N_DD(R) <= N_0(R)^2.

Proof. J is antilinear, and J^2 = 1 because the sign is applied twice with the degrees exchanged. Gamma has eigenvalue (-1)^{|xi| + |eta|}, symmetric under the exchange. For the Dirac operator,

```
J DD (xi (x) eta^c) = (-1)^{(|xi| + 1) |eta|} eta (x) (D_0 xi)^c + (-1)^{|xi|} (-1)^{|xi| (|eta| +
↪  1)} D_0 eta (x) xi^c
                    = (-1)^{|xi| |eta|} [D_0 eta (x) xi^c + (-1)^{|eta|} eta (x) (D_0 xi)^c] = DD J
                     ↪  (xi (x) eta^c).
```

For even a, a* preserves degree, so J pi(a)* J^-1 (eta (x) xi^c) = eta (x) (a* xi)^c, which gives the formula for a° and order zero. Since a is even, [DD, pi(a)] = [D_0, a] (x) 1, which commutes with 1 (x) (b*)^c. Finally DD^2 = D_0^2 (x) 1 + 1 (x) ($D^{c)}$2, because gamma_0 anticommutes with D_0. An eigenbasis of D_0 decomposes DD into finite-dimensional invariant blocks, so DD is essentially self-adjoint on the algebraic tensor product, with compact resolvent, and the spectrum is +/- (lambda_i^2 + lambda_$j^{2)}${1/2}. ■

Geometrically, when H_0 = L^2(S_X) on an even-dimensional spin manifold X, a charge conjugation commuting with D_X identifies conj(H_0) with H_0, H is L^2(S_{X x X}) and DD is the ordinary product Dirac operator. (The grading may acquire an overall sign under this identification.) So the data are a product-manifold Dirac operator on X x X with a restricted observable algebra C(X) (x) 1, and a J covering the flip of the two factors, which differs from the pointwise spin real structure. They count as commutative Kaluza-Klein data only with that restriction and that J: they are neither the ordinary spin triple of X x X with its full function algebra nor a finite-rank twist over X. The restriction is essential: on all of C(X x X), the compressions generate A_F (x) conj(A_F), and order zero generally fails again.

Corollary 16.3 (truncation). Let P = P_0 (x) conj(P_0). It commutes with DD, Gamma and J. It is a joint cutoff in the two factors, and need not be a spectral projection of DD: for absolute eigenvalues 1, 5, 7, sqrt(73) of D_0 with P_0 selecting the first three, the selected pair (5, 7) and the excluded pair (1, sqrt(73)) both give DD^2 = 74. The compression of pi(A_0) is O_F (x) 1; passing to the generated algebra gives the finite real even triple

```
T_F = (A_F, V (x) conj(V), D_F, J_F, Gamma_F),     D_F = d (x) 1 + gamma_V (x) d^c = Delta + J_F
↪  Delta J_F^-1,     Delta = d (x) 1,
```

with A_F acting on the left. It has KO-dimension 0, and satisfies order zero and order one for every A_F, noncommutative or not, with dim H_F = k^2. The family is exact at each truncation; no limit is taken.

Proof. The proof of Theorem 16.2 applies verbatim on V with arbitrary even a, b in B(V), since it uses only the grading and the tensor structure. For J_F Delta J_F^-1, apply J to (d (x) 1) J (xi (x) eta^c): the signs combine to (-1)^{|xi|}, giving gamma_V (x) d^c. ■

So H_F has the bimodule form of Connes' finite geometries, with A_F on the left and its opposite on the right, and D_F has the first-order form of Krajewski and Paschke-Sitarz.

Example (a) gives (C + C, C^2 (x) C^2, m (sigma_1 (x) 1 + sigma_3 (x) sigma_1)). There [[D_F, e], e°] = 0 for e = diag(1, 0), which repairs the first-order failure of Counterexample 08.4 at the price of doubling. Example (b) gives (M_2(C) + M_2(C), C^16, D_F).

Corollary 16.4 (KO-6 by a Clifford point). The product of T_F with (C, C^2, 0, sigma_1 K, sigma_3) is

```
(A_F, V (x) conj(V) (x) C^2, D_F (x) 1, J_F (x) sigma_1 K, Gamma_F (x) sigma_3),
```

with signs (+, +, -), order zero, order one and dim H = 2 k^2. It fails the usual zero-dimensional orientability axiom: every zero-cycle built from the left and opposite actions acts as T (x) 1_2, while the grading is Gamma_F (x) sigma_3.

Proof. (sigma_1 K)^2 = 1 and sigma_1 K sigma_3 = -sigma_3 sigma_1 K, while the D term is untouched. a° is a° of T_F tensored with 1. ■

Remark 16.5 (the order conditions are formal). For every finite even representation (A, V, d, gamma), the map V -> V (x) conj(V), d -> d (x) 1 + gamma (x) d^c gives a real bimodule with order zero and order one, using no prior finite real structure. So these two conditions follow from choosing the tensor square, and say nothing specific about A_F. The construction does not recover an arbitrary prescribed finite (H, D, J). What is specific to T0 is that A_F is generated by compressed functions, that the tensor square is itself the truncation of a commutative parent in the restricted sense above, and families such as the following.

Example 16.6 (a tower-decoupling family, from the referee). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with V = ker D_0 = C^4, and let DD_l = DD/l on the doubled parent. Its kernel is V (x) conj(V), and its gap grows like 1/l. Here P is a spectral projection. Compression followed by algebra generation gives the constant finite data A_F = M_4(C), dim H_F = 16, D_F = 0, and the Clifford point gives dim H = 32 with the KO-6 signs. Order one is automatic, since D_F = 0. The whole triple converges to (C, C^16, 0) by Theorem 15.3, consistently with lemma 15: the two statements concern the whole algebra and the truncation.

Two sheets: the form of the classification

Chamseddine and Connes (*Why the Standard Model*, arXiv:0706.3688, sections 2-3) classify irreducible pairs (A, J) with order zero and a separating vector: the centre of the complexified algebra is C, with A_C = M_k(C) on itself and J x = x*, or C + C, with A_C = M_k(C) + M_k(C) on itself and J(x, y) = (y*, x*). With centre C, J gamma = -gamma J is impossible. The pair of Corollary 16.4 is therefore reducible: the projection 1 (x) (1 + sigma_1)/2 commutes with A_F and J. The irreducible form comes from a parent with two sheets. At the finite level the bimodule (W (x) conj(V)) (+) (V (x) conj(W)) with its adjoint-flip J is already stated in Remark 2.9 of that paper; what we add is its realization from parents and compressed functions.

Theorem 16.7 (two-sheet coarse parent). Let (A_X, H_X, D_X, gamma_X) and (A_Y, H_Y, D_Y, gamma_Y) be even spectral triples. On H = (H_X (x) conj(H_Y)) (+) (H_Y (x) conj(H_X)), for even a in A_X and b in A_Y, put

```
pi(a, b) = (a (x) 1) (+) (b (x) 1),
DD = (D_X (x) 1 + gamma_X (x) D_Y^c) (+) (D_Y (x) 1 + gamma_Y (x) D_X^c),
Gamma = (gamma_X (x) gamma_Y^c) (+) (-gamma_Y (x) gamma_X^c),
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c,
```

where J maps each summand to the other. Then J^2 = 1, J DD = DD J and J Gamma = -Gamma J, the KO-6 signs. Moreover

```
(a, b)° = (1 (x) (b\*)^c) (+) (1 (x) (a\*)^c),
```

so the first summand is an (A_X, A_Y)-bimodule and the second an (A_Y, A_X)-bimodule, and order zero and order one hold.

Proof. The computation of Theorem 16.2 applies with D_Y in place of the second factor: on xi (x) eta^c in the first summand, J DD and DD J both give (-1)^{|xi| |eta|} [D_Y eta (x) xi^c + (-1)^{|eta|} eta (x) (D_X xi)^c]. The sign in Gamma on the second summand gives J Gamma = -Gamma J. For a°, apply J, then pi(a, b)*, then J, as in Theorem 16.2: an element of the first summand goes to the second, where b* acts on the left, and returns with b* on the right factor. Since a and b are even, [DD, pi(a, b)] = ([D_X, a] (x) 1) (+) ([D_Y, b] (x) 1), which commutes with the opposite actions. ■

Corollary 16.8 (truncation; the form of the classification). For symmetric windows with V_X, V_Y of dimensions k_X, k_Y, and even subalgebras A_{F,X} in B(V_X), A_{F,Y} in B(V_Y), the compression

```
T_F = (A_{F,X} + A_{F,Y}, (V_X (x) conj(V_Y)) (+) (V_Y (x) conj(V_X)), D_F, J_F, Gamma_F)
```

has the KO-6 signs, order zero and order one, and dim H_F = 2 k_X k_Y. If gamma_{V_X} lies in A_{F,X} and gamma_{V_Y} in A_{F,Y}, the real orientability relation holds: with c = (gamma_{V_X}, gamma_{V_Y}) and z = (1, -1), Gamma_F = pi(z c) c°, a zero-cycle with coefficients in the algebra and its opposite. This is weaker than Gamma_F in pi(A_F). If V_X and V_Y are ungraded with k_X = k_Y = k and the generated algebras are full, then (A_F, H_F, J_F) is exactly the irreducible pair with centre C + C: identify both summands with M_k(C), so that J_F(x, y) = (y*, x*); an invertible matrix in each summand is a separating vector, and a projection commuting with both full matrix actions is scalar on each summand, with J equating the scalars. In all cases D_F commutes with the two sheet projections, so the sheets are not connected. It need not commute with the whole centre: for V_X = C^2, gamma_X = sigma_3, d_X = sigma_1, A_{F,X} = C + C acting diagonally and V_Y = C, the central element (diag(1, 0), 0) does not commute with D_F.

Proof. Theorem 16.7 applies on the truncation with arbitrary even operators. For orientability, pi(c) c° = (gamma_{V_X} (x) gamma_{V_Y}^c) (+) (gamma_{V_Y} (x) gamma_{V_X}^c), and z flips the sign of the second summand. The last statements are direct. ■

Example 16.9 (twistor level one, and M_2(H) + M_4(C)). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with kernel $H^{0(CP}3, O(1)) = C^4$, and let j be the antiunitary with j^2 = -1 commuting with the Gamma_a of Proposition 10.1, which makes C^4 a quaternionic space H^2. Use this

space twice, as W (with the functions of S^4 pulled back along the twistor projection) and as V (with the functions of CP^3).

1. A real function f on S^4 pulls back to a function invariant under w -> j w. So j Q(f) j^-1 = Q(f), and Q(f) is a Hermitian element of End_H(W). Since Q(x_a) = Gamma_a/5, the real algebra generated is End_H(W) = M_2(H), of real dimension 16.
2. For real functions on CP^3, Q(w* B w / w* w) = (B + tr(B) 1)/5 for Hermitian B, so Q is onto the Hermitian matrices, and the real algebra generated is End_C(V) = M_4(C), of real dimension 32.

Hence Corollary 16.8, with the parent gradings, which are trivial on the kernels, gives the real algebra A = M_2(H) + M_4(C) on H_F = Hom(V, W) (+) Hom(W, V) = E (-) E* of dimension 32, with J(x, y) = (y*, x*). These are the algebra, bimodule and real structure of Chamseddine and Connes. The generation is of real algebras from real functions: complex generation on the S^4 side would give M_4(C).

The nontrivial grading of W in that construction is not obtained from this parent. An even parent has [P gamma P, P a P] = 0 for every compressed function, while Gamma_5 = 5 Q(x_5) anticommutes with Q(x_1). So the grading has to be added after truncation, as abstract finite data, fixing an axis. The full algebra is then a graded prealgebra rather than the algebra of an even triple, with even part H + H + M_4(C), and Gamma_F(xi, eta) = (Gamma_5 xi, -eta Gamma_5), which is Proposition 3.2 of that paper; the real orientability relation holds on the even part. Similarly d = mu n^a Gamma_a with n orthogonal to the axis is the compression of a bounded real multiplication operator, but adding that operator to the parent Dirac operator keeps neither the kernel as a spectral projection nor the parent grading as an odd perturbation, so d too is abstract data. It commutes with the two sheet projections and with the centre R + C of the full prealgebra, and not with the centre of the even algebra.

This bears on the step left open in canonical section 4, where M_2(H) in M_4(C) is a nested pair on one four-space, with "the independent E, F module roles" to be derived. The example closes that abstract finite-module problem after adding a second sheet: the nested actions on one four-space become independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the Dirac operator connecting the two pieces that the selection of A_F = C + H + M_3(C) requires. The fibre-rank problem remains open.

The inputs are the second sheet, the restriction of one algebra to base functions, real algebra generation, the four-dimensional window, the grading axis and d. The content beyond the formal bimodule is the concrete compression calculation in items 1 and 2, already in the corpus, and its explicit match with the classified data.

Caveats

These results should be read as narrowly as the note they correct.

1. **Metric.** T_F is usually not a metric spectral triple. For A_F = B(V_+) + B(V_-) with V_+ of dimension at least 2 and d odd, the commutant of d in A_F contains the pairs (x, B^-1 x B) with x commuting with B B*, when B is the invertible block of d. In example (b), (diag(1, 0), diag(1, 0)) has L = 0, so d leaves a residual U(1) x U(1) of U(2) x U(2) unbroken. The Standard Model finite triple has the same feature, through M_3(C).
2. **Spectrum.** The eigenvalues of D_F are +/- (lambda_i^2 + lambda_j^2)^{1/2}. In example (a) they are +/- sqrt(2) m with Connes distance 1/m, so the doubled spectrum differs from the masses in d, and the mass relations of Theorem 9.6 and lemma 11 would have to be redone.
3. **Parent.** The doubled parent is a choice. Its algebra sees one factor of X x X, its J covers the flip, which is outside the spin real structure of X x X, and generating A_F from O_F is an additional operation. We have no dynamical reason for it. For an ordinary compact spin X of dimension n, its Weyl exponent is 2 n with metric dimension n, which bears on the counting of 9, 10 and 11 and has not been analyzed.
4. **KO-6.** The Clifford point is an input, as F_0 is in Theorem D, and it fails finite orientability. In Theorem 16.7 the sign comes from the block structure of two sheets, which is itself an input. The

documents checked here establish neither a construction of two sheets from one connected parent nor a no-go theorem. The grading-reversing structure of lemma 08 is a candidate we have not tested.

5. **Count.** For k = 4, dim H = 32, the value in the classification of Chamseddine-Connes, and k = 4 occurs in examples (b) and (c). For k = 4 in examples (b) and (c) alone this is a count and nothing more: the classification's algebra is M_2(H) + M_4(C), acting differently, and those examples do not produce it. Example 16.9 does, as abstract data with the listed inputs.

Consequence for T0 and T0b

T0 asks for finite real triples from commutative Kaluza-Klein data under specified notions of convergence. For one specified truncation procedure (compression, then generation, on the doubled parent with restricted algebra and flip J), this lemma gives a noncommutative finite algebra with an explicit bimodule, and a family with the tower decoupled (Example 16.6). It does not establish the target under every meaning of ordinary Kaluza-Klein data. For T0b, it gives examples with the requested signs and first-order relations; with two sheets it reaches the form of the classification, and in Example 16.9 the real algebra M_2(H) + M_4(C), bimodule and real structure of Chamseddine and Connes, with the grading added as abstract data. It leaves open an operator connecting the two pieces and hence the selection of A_F, a parent that yields the grading and nonzero mass data, and the beyond-first-order and twisted frameworks. See the background note.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random odd d and even a, b for dim V_+ = dim V_- = 2 with J = M K: J^2 = 1, J DD = DD J, J Gamma = Gamma J, DD Gamma = -Gamma DD, order zero and order one all hold to machine precision, with norm([a, b]) = 7.67; the KO-6 signs of Corollary 16.4 hold. For Theorem 16.7 and Example 16.9, with k = 4, W graded by Gamma_5 and V ungraded: J^2 = 1, J Gamma = -Gamma J, J D = D J, D Gamma = -Gamma D, D = Delta + J Delta J^-1, order zero and order one for random a, b, and Gamma_F = pi(z c) c° all hold to machine precision; j^2 = -1 commutes with the Gamma_a, and x_a(j w) = x_a(w); the real algebras generated by Gamma_a, by i Gamma_a and by all Hermitian matrices have real dimensions 16, 32 and 32; and sum_a x_a Gamma_a = 2 (P_w + P_{jw}) - 1. For example (b), by quadrature with 23 test functions: dim O_F = 4 and the generated algebra has dimension 8; the two displayed values Q(f_1) and Q(f_2) were reproduced by quadrature.

Part 17

Explicit masses in the reduced triple, the chiral limit and the set of reductions

Routine part of STRATEGY T10. Proved by Claude Opus 5.5 without Astra or Fable. The identities of Theorems 17.1 and 17.6 were checked with numpy on random finite models (section "Numerical check"). Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 10 items, 1 correct, 4 needed amendments, and 5 were wrong as first written (Corollary 17.4, Remark 17.7, Proposition 17.8, Proposition 17.9 and the consequence for T10, each by overreach). All corrections are incorporated below. Proposition 17.10 is the referee's.

Outcome. The fermion action of a product M x F, restricted to its physical subspace, is exactly the action of the odd triple M x T on an unconstrained spinor, with the Yukawa block becoming the constant matrix d in D_M + gamma_5 d (Theorem 17.1). A paired Dirac sector gives one Dirac action for either sign s, up to a change of Grassmann variables (Corollary 17.2); other sectors, such as a neutral one, have Majorana-type actions. On a subalgebra commuting with D_F there are no scalar fluctuations, so the mass is explicit (Proposition 17.3). At M = 0 the symmetry of the quark sector is U(N_f) x U(N_f) (Proposition 17.5). The admissible involutions of the quark sector are the Sigma in U(N_f) with Sigma* M Hermitian, which are exactly the stationary points of the mass term Re tr(M Sigma*); at M = 0 they form a homogeneous space of the chiral group (Theorem 17.6). All of this is kinematic. A pion kinetic term does not come from the fixed spectral action (Proposition 17.10). The condensate requires a measure on gauge fields, infinite volume and an order of limits (Proposition 17.8).

"D = 9" below is the formal KO-label 4 + 5 of Theorem D. The metric dimension of M x T is 4. The label exists only where an involution of sign s = -1 exists: charged paired sectors have one, and the strict light spectrum with one or three unpaired neutrinos has none, since its H_0 has odd dimension (lemma 06, Result 7).

Setting

Let M be a compact Riemannian spin 4-manifold with (D_M, J_M, gamma_5), where J_M^2 = -1, J_M D_M = D_M J_M, J_M gamma_5 = gamma_5 J_M and gamma_5 D_M = -D_M gamma_5. Let F = (A_F, H_F, D_F, J_F, gamma_F) be a finite real triple with the KO-6 signs. On L^2(S) (x) H_F put

```
D = D_M (x) 1 + gamma_5 (x) D_F,     J = J_M (x) J_F,     Gamma = gamma_5 (x) gamma_F,
```

with J^2 = -1, J D = D J and J Gamma = -Gamma J. Let H^+ be the Gamma = +1 subspace. The fermion action is (1/2) <J xi, D xi> on Grassmann variables xi in H^+ (Chamseddine-Connes-Marcolli 2007).

Let U be an admissible involution in the sense of Theorem D: U^2 = 1, U unitary, U gamma_F = -gamma_F U, [U, a] = [U, a°] = [U, D_F] = 0 and U J_F = s J_F U. On H_0 = {gamma_F = +1} put d = U D_F and j = U J_F, restricted to H_0, so that j^2 = s and j d = s d j.

The reduced action

Theorem 17.1. Define R : L^2(S) (x) H_0 -> H^+ by R(phi (x) x) = phi_+ (x) x + phi_- (x) U x, with phi_+- the chiral parts of phi. Then R is unitary, and

```
<J R psi, D R psi'> = <jj psi, DD psi'>,     DD = D_M (x) 1 + gamma_5 (x) d,
```

where jj = J_M (x) j for s = +1 and jj = gamma_5 J_M (x) j for s = -1, with jj^2 = -s and jj DD = s DD jj. Hence the action is (1/2) <jj psi, DD psi> on an unconstrained Grassmann spinor psi. Moreover DD^2 = D_M^2 (x) 1 + 1 (x) d^2, and if f($D^{2/\Lambda}$2) is trace class, Tr f($D^{2/\Lambda}$2) = 2 Tr f($DD^{2/\Lambda}$2).

Proof. R is unitary because U exchanges the two gamma_F-eigenspaces. Let UU = 1 (x) U, which commutes with D and is an involution, so that <J xi, D xi'> = <UU J xi, UU D xi'>.

Put r x = (x + U x)/sqrt(2). Then D_F r = r d, since D_F r x = (D_F x + U D_F x)/sqrt(2) with U D_F x = d x and D_F x = U d x. Also (1 + UU) R/sqrt(2) = 1 (x) r. Since UU commutes with D, this gives UU D R = R DD.

For the real structure, J_F x = U j x and U J_F U = s J_F. As J_M preserves chirality,

```
UU J R(phi (x) x) = J_M phi_+ (x) j x + s J_M phi_- (x) U j x = R((J_M phi_+ + s J_M phi_-) (x) j
↪  x) = R jj (phi (x) x).
```

Therefore <J R psi, D R psi'> = <UU J R psi, UU D R psi'> = <R jj psi, R DD psi'> = <jj psi, DD psi'>. The square follows from gamma_5 D_M = -D_M gamma_5. Finally, D is the direct sum of its restrictions to the eigenspaces of UU, on which it is equivalent to D_M + gamma_5 d and D_M - gamma_5 d, and conjugation by gamma_5 sends the first to minus the second, so their squares are unitarily equivalent. ■

So the doubling is removed by the chirality projection, and the physical space is exactly the Hilbert space of M x T.

Corollary 17.2 (a paired sector gives one Dirac fermion, for both signs). Let H_0 = span(e_L, conj(e_R)) with m > 0 and psi = (psi_1, psi_2). For (d_+, j_+) = (m I, sigma_1 K) and (d_-, j_-) = (diag(m, -m), [[0, 1], [-1, 0]] K) of lemma 06, the action is

```
S_+ = <J_M psi_2, (D_M + gamma_5 m) psi_1>     and     S_- = <gamma_5 J_M psi_2, (D_M + gamma_5 m)
↪  psi_1>
```

respectively, with S_-(psi_1, psi_2) = S_+(psi_1, gamma_5 psi_2). After the charge-conjugation matrix is absorbed, the left argument is an independent Euclidean variable psi-bar, in the Grassmann-bilinear sense of Chamseddine-Connes-Marcolli, section 4. With R_5 = exp(i pi gamma_5/4), R_5 (D_M + gamma_5

m) R_5 = D_M + i m. So the action is that of one Dirac fermion of explicit mass m, whichever sign s is used, up to the displayed invertible change of Grassmann variables.

Proof. The total form is antisymmetric, so on Grassmann variables the two cross terms are equal, which cancels the factor 1/2. The rotation uses R_5 D_M R_5 = D_M. ■

This agrees with Theorem D, item 5: at the level of the classical fermion action, s labels two coordinate descriptions of one bilinear form, and no anomaly or theta shift follows from changing s alone. The rotation R_5 is a formal chiral change of both independent variables; in a quantized theory its Jacobian must be included.

An admissible reduction need not have this paired form. For gamma_F = sigma_3, J_F = sigma_1 K, D_F = m sigma_1 and U = sigma_1 we have s = +1, H_0 is one-dimensional, and the action is the Majorana-type (1/2) <J_M psi, (D_M + gamma_5 m) psi>. The neutral sector of the strict light spectrum has this form with m = 0.

Proposition 17.3 (no Higgs on a commuting subalgebra). Let B in A_F be a unital subalgebra with [D_F, B] = 0, and let U commute with B and B°. The inner fluctuation of D by C^infinity(M) (x) B contains only Clifford one-forms, D_A commutes with UU, and D_A reduces to

```
DD_A = D_M(A) + gamma_5 (x) d,
```

with d constant on M. If [D_F, B] != 0, the same reduction holds, and the fluctuation permits a finite scalar field d_A(x) in place of d.

Proof. For a = f (x) alpha and b = g (x) beta, we have a [D, b] = f [D_M, g] (x) alpha beta + f g gamma_5 (x) alpha [D_F, beta], whose second term vanishes. The term J A J^-1 is a Clifford one-form whose finite coefficients lie in B°. Since U commutes with B and with B°, Theorem 17.1 applies pointwise with D_M(A) in place of D_M. In the other case, U commutes with [D_F, beta], so the fluctuation preserves the reduction. ■

Corollary 17.4 (Yukawa couplings to explicit masses). In the representation of lemma 06, Result 5, with B = {(lambda, Q_lambda, m)} = C + M_3(C), u_L and u_R carry the same left character lambda and the same opposite colour action, d_L and d_R carry the left character conj(lambda) and the same opposite colour action, and e_L and e_R pair as well. So the kept mass blocks commute with B, in the strict projection of lemma 06 and also when only a vectorlike top pair is removed. Hence DD_A = D_M(A) + gamma_5 d, with d the direct sum of the kept mass matrices, and no Higgs field.

Two qualifications apply. First, U(B) contains U(1) x U(3), so before unimodularity or an inherited restriction from the Standard Model gauge group, two independent abelian gauge fields act on the surviving charged fermions; the reading "photon and gluons" needs that restriction. Second, a canonically normalized radial field couples to a fermion of fixed mass m_f with strength m_f/v, which tends to 0 if v tends to infinity at fixed gauge coupling. Under the CCM relation (lemma 11), keeping all Dirac masses fixed while M_W tends to infinity is impossible: some Dirac mass must become unbounded. It need not be the top; top dominance is a further hypothesis. The relation holds in the asymptotic matching regime, and the regime near the cutoff is T5b.

The chiral limit

Consider N_f quark flavours, with q_L, q_R in C^{N_f} (x) C^3, and q-bar_L, q-bar_R = J q_L, J q_R. Let D_F act on (q_L, q_R) by [[0, M], [M*, 0]] (x) 1_3, and by the complex conjugate block on the antiparticles. Let M_3(C) act on the antiparticles on the left and on the particles through its opposite action.

Proposition 17.5 (the symmetry restored at M = 0). Let G(M) be the group of unitaries on this sector commuting with both actions of M_3(C), gamma_F, J_F and D_F. Then

```
G(M) = {(V_L, V_R) in U(N_f) x U(N_f) : V_L M = M V_R},
```

acting by V_L, V_R on q_L, q_R and by the complex conjugates on q-bar_L, q-bar_R. So G(0) = U(N_f) x U(N_f), G(m I) = U(N_f)_V for m != 0, and G(M) = U(1)^{N_f}_V for M positive diagonal with distinct entries. With m_u = m_d = 0 and m_s != 0 in one block, G contains U(2) x U(2). If V is also required to commute with the electric charge, through lambda in B, then each V_L, V_R is diagonal between u and d, and G(0) = U(1)^4 for two flavours.

Proof. The colour identity separates particles from antiparticles, since its left action is zero on particles and the identity on antiparticles. The commutant of the colour matrix algebra is scalar on colour, so V = V' (x) 1_3 on each. The grading separates L and R, and J_F fixes the antiparticle part. Commutation with D_F gives V_L M = M V_R, whose adjoint with unitarity gives the other block equation. ■

The nonabelian flavour transformations are not colour gauge transformations, and at M = 0 the axial part is an ungauged symmetry of the reduced triple. If the central U(1) of U(3) is retained in the gauge group, it contains a common vector phase, which an imposed unimodularity condition can remove. The Lagrangian symmetry U(1)_A is anomalous in the quantum theory, which the classical data do not show.

Theorem 17.6 (the set of reductions). Let B contain only the colour algebra. The admissible involutions of sign s on this sector are exactly the operators, tensored with 1_3,

```
U_{Sigma,s} = [[0, Sigma], [Sigma\*, 0]] on (q_L, q_R),     s [[0, conj(Sigma)], [Sigma^T, 0]] on
↪  (q-bar_L, q-bar_R),
```

with Sigma in U(N_f) and Sigma* M Hermitian. This condition holds if and only if Sigma is a stationary point of V_M(Sigma) = -Re tr(M Sigma*) on U(N_f). The reduced mass operator on H_0 = (q_L, q-bar_R) is

```
d_{Sigma,s} = (Sigma M\* (x) 1_3) (+) (s conj(Sigma\* M) (x) 1_3).
```

The group G(0) acts by Sigma -> V_L Sigma V_R*. Hence:

1. At M = 0 the set is all of U(N_f), for each s, and the action is transitive with stabilizer of Sigma = 1 the diagonal U(N_f): the reductions form the homogeneous space (U(N_f) x U(N_f))/U(N_f)_V.
2. At M = m I with m > 0 it is {Sigma : Sigma^2 = 1, Sigma* = Sigma}, the disjoint union of the Grassmannians of C^{N_f}.
3. For M positive diagonal with distinct entries, it is the 2^{N_f} matrices diag(+-1, …, +-1).
4. For M invertible, the unique Sigma with Sigma* M positive is the polar part of M. It is the unique global minimum of V_M, and it is the unique Sigma for which the first block of d_{Sigma,s} is positive.

With the electric charge in B, the admissible set is the intersection of this set with the commutant of the charge. For two differently charged flavours this keeps diagonal phases at M = 0, and only signs +-1 for positive nonzero diagonal masses.

Proof. An admissible U commutes with both colour actions, so it acts on each of q, q-bar as U' (x) 1_3; it reverses the grading, so it is off-diagonal between L and R; U^2 = 1 with U unitary gives the form [[0, Sigma], [Sigma*, 0]] with Sigma unitary. Then U J_F = s J_F U fixes the antiparticle block. Commutation with D_F is Sigma M* = M Sigma* and Sigma* M = M* Sigma. If H = Sigma* M is Hermitian, then Sigma M* = Sigma H Sigma* is Hermitian, so the two are equivalent. Along Sigma exp(i t X), X Hermitian, the derivative of Re tr(M Sigma*) at t = 0 is Im tr(H X), which vanishes for all Hermitian X exactly when H is Hermitian. The formula for d is direct, using q-bar_R = J q_R. Item 1 is immediate. For item 2 the condition is Sigma* = Sigma. For item 3, H^2 = M* M forces H to be diagonal with entries +- m_i. Item 4 is von Neumann's trace inequality, Re tr(M Sigma*) <= tr |M|, with equality exactly at the polar part, and the first block is Sigma H Sigma*. ■

Remark 17.7 (scope). Item 1 is an equivariant statement: the reductions form a homogeneous space of the chiral group, with the same stabilizer as in the symmetry breaking of QCD. The mass condition has an elementary origin, since it says that H = Sigma* M is Hermitian, and for invertible M the positive choice is the polar decomposition. Its equality with the stationarity equation of -Re tr(M Sigma*) is exact, and it captures all stationary points, including saddles and maxima. Vacuum alignment (Dashen) asks which stationary point minimizes a physical energy, and the data here contain no such energy; the stationarity calculation is our derivation, not an attribution of this reduction theorem to Dashen.

So the theorem shows the shape of the set of reductions. Identifying a reduction with a condensate direction would need an explicit map to a quantum order parameter, which we do not have. In QCD, the anomalous singlet axial direction also means that the full classical U(N_f) cannot be identified without qualification with the pion manifold.

For invertible M, every stationary point other than the polar part has at least one negative eigenvalue of Sigma* M; for singular M there may be a continuum of minima, as for M = diag(1, 0) and Sigma = diag(1, exp(i theta)). A negative entry in the reduced mass block comes from choosing another reduction of one fixed even triple, and is not a physical mass-sign change at fixed fermion measure. A physical sign change is different: for one Dirac flavour in a background with n_0 zero modes, the determinants at m and -m differ by the factor (-1)^{n_0}, the factor associated with a shift of theta by pi.

Spacetime-dependent Sigma

Proposition 17.10 (referee; no kinetic term from the fixed action). Let U(x) be a pointwise admissible involution on M, with c^mu = [D_M, x^mu] and B_mu = (U d_mu U) restricted to H_0. With R defined pointwise,

```
R^-1 UU D R = D_M + c^mu P_- B_mu + gamma_5 d(x),
```

where P_- = (1 - gamma_5)/2. This operator is generally not self-adjoint. So:

1. If Sigma(x) only changes coordinates for the fixed operator D, the spectral action of D does not depend on Sigma, and conjugating D by a smooth unitary preserves its spectrum. Neither produces a two-derivative term.
2. A kinetic term needs an additional choice of operator. For D_new = D_M + gamma_5 d_Sigma(x) on flat M, with M = m I and Sigma* = Sigma, Sigma^2 = 1, the heat expansion of Tr f(D_new$^{2/\mathrm{Lambda}}$2) contains

   ```
   (N_c f(0) m^2/(4 pi^2)) int tr(d_mu Sigma\* d^mu Sigma),
   ```

   tracing over both reduced blocks. The sign is positive when f(0) > 0. At M = 0 this operator has no dependence on Sigma.

Proof. The first display is direct differentiation in Theorem 17.1. Since B_mu is anti-Hermitian and P_- c^mu = c^mu P_+, the derivative term has adjoint c^mu P_+ B_mu. For the coefficient, D_new^2 = -(nabla^2 + E) with E containing -i gamma^mu gamma_5 d_mu d, and tr over spinors of the square of that term is 4 tr(d_mu d d^mu d); then the a_4 coefficient (1/16 pi^2) tr(E^2/2) gives (1/(8 pi^2)) tr_{H_0}(d_mu d d^mu d), and the two blocks of d give 2 N_c m^2 tr(d_mu Sigma d^mu Sigma). ■

So the idea of a pion as a spacetime-dependent reduction, in the form we first proposed, does not produce a kinetic term: the kinetic term needs a nonzero m and a new operator, or a quantum effective action. On the even triple, D_M + gamma_5 S(x) with S built from Phi = m Sigma(x) is self-adjoint for every unitary Sigma. Its kinetic term is the Higgs-type term of an inner fluctuation, available when A_F contains flavour algebras acting on L and R, and then flavour gauge fields accompany it. The coefficient is proportional to N_c m^2, as in constituent-quark estimates of f_pi^2 (quoted from memory); here it is a classical spectral-action coefficient, and m is not derived.

What the spectral data say about the condensate

Proposition 17.8. Let D_M(A) have eigenvalues lambda_n, with n_0 zero modes, on M of volume V, and for m > 0 put

```
c_V(m; A) = (1/V) sum_n m/(lambda_n^2 + m^2) = (1/V) d/dm log |det (D_M(A) + gamma_5 m)|,
```

which is formal on a compact four-manifold and needs ultraviolet regularization, with subtraction of divergent terms including m Lambda^2, m^3 log Lambda and possibly curvature terms.

1. With a subtraction compatible with the pairing +-lambda whose finite remainder is O(m), for example odd and analytic in m, at fixed A on compact M, c_V(m; A) = n_0/(V m) + O(m) as m -> 0. A

subtraction merely regular at m = 0 may leave a constant. So at finite volume and fixed background, no nonzero limit exists beyond the zero-mode term.

2. Let rho_m(lambda) = lim_{V -> infinity} <(1/V) sum_n delta(lambda - lambda_n)>*m for a measure* <.>m on gauge fields. Assume that the family is equicontinuous at 0 with rho_m(0) -> rho_0(0), that the tails and the ultraviolet subtraction are controlled uniformly in m, and that the infinite-volume limit may be taken inside the integral. Then lim*{m -> 0} lim*{V -> infinity} = pi rho_0(0). This has the form of the Banks-Casher relation as stated by Giusti and Lüscher (the original, Nucl. Phys. B 169, eqs. (49), (50), has neither the factor pi nor an order of limits with the volume); the sign and flavour normalization of a condensate depend on convention, and the hypotheses here are ours.

Equicontinuity alone is insufficient: rho_m(lambda) = (1 + (1/2) sin(1/m)) exp(-lambda^2) gives no limit.

Proof. Item 1: for a cutoff preserving the pairs +-lambda, the determinant has absolute value m^{n_0} times the product of (lambda^2 + m^2) over lambda > 0, and the nonzero eigenvalues are bounded away from 0. Item 2: m/(lambda^2 + m^2) is pi times the Poisson kernel at scale m. ■

So the spectral triple supplies the operator D_M + gamma_5 m, the density rho_A of each fixed operator, and the continuity of m -> 0 at fixed background, since DD^2 = D_M^2 + m^2. The gauge-averaged density, the regulated determinant or Pfaffian measure, the sequence of volumes and the order of limits need further input. A spectral action may contribute to a proposed gauge weight, but a single compact spectral triple does not specify the functional integral or the thermodynamic limit.

A proof of existence of QCD with fermions would supply a measure. It would not by itself give rho_0(0) > 0. The points to check when it appears are those in STRATEGY T10: whether m_q = 0 is included, and whether infinite volume is reached. With the anti-self-adjoint operator slashed-D = -i D_M and a compatible regulator, det(slashed-D + m) = m^{n_0} times the product over lambda > 0 of (lambda^2 + m^2), which is positive for m > 0; this positivity is part of the method of Vafa and Witten.

The chiral limit on the interval

Proposition 17.9. In the scalar-algebra realization F_0 x I_l of Theorem 08.7, DD = D_M (x) 1 + gamma_5 (x) (D_0 + m Gamma_0), so the interval mass is an explicit mass term with the internal grading. The limit m -> 0 has two distinct routes:

1. *Localization (lemma 09).* With bulk mass -mu and mu l > 1, m = lambda_0 = mu sech(kappa_0 l), of order 2 mu exp(-mu l), which is positive at finite parameters. At fixed mu > 0, l -> infinity gives m -> 0 as an exponentially small residue. The relevant control is the dimensionless suppression: an absolute limit needs a scaling hypothesis, since mu = 2/l gives m -> 0 at fixed mu l, and mu l -> infinity does not ensure m -> 0 if mu grows fast enough. The localization and overlap mechanism is that of domain-wall fermions (Kaplan; Shamir), though this is neither lattice operator.
2. *Boundary Yukawa (lemma 12).* At X = 0 the zero modes are exact (Theorem 12.2), with boundary types (a) and (b). For flavour-degenerate bulk masses, D_0 commutes with U(N_Q) x U(N_U) on the whole tower, and at any l the boundary coupling X preserves exactly the stabilizer {(V_Q, V_U) : V_Q X = X V_U}. So X acts as a spurion located on the wall x = l, and the symmetry is exact at finite l when X = 0. In the towerless limit X = l Y_hat, M_F is proportional to X, and for N_Q = N_U the stabilizer is G(M_F) of Proposition 17.5. At finite l, M_F is not proportional to X: for one flavour with zero bulk masses, the light eigenvalue is arctan(abs(X))/l.

The boundary-coupled operator of route 2 is a different construction from the tensor-product truncation of Theorem 08.7, and an identification of the two at finite l would need a proof.

Proof. Route 1 is Theorem 9.4 and Lemma 9.5, and the scaling examples are direct. Route 2 follows from the invariance of the domain of D_X (lemma 12), Theorem 12.7 and Example 12.5. ■

Caveats

1. **Classical data.** Everything here is at the level of the classical fermion operator. Anomalies, the running of masses and confinement do not appear.

2. **QED.** With the photon present, the axial group reduces as stated in Proposition 17.5, and the set of reductions reduces as stated in Theorem 17.6.
3. **Regime.** Corollary 17.4 inherits the regime of the CCM relation.
4. **Label.** A reduction is a rewriting of one even triple. It gives no parent nine-dimensional geometry, in agreement with Theorem D, and the KO-5 label exists only where an s = -1 involution exists.

Consequence for T10

The reduced action is (1/2) <jj psi, (D_M(A) + gamma_5 d) psi> on an unconstrained spinor, whenever the chosen admissible involution exists. A paired charged sector gives one Dirac action of explicit mass for either sign, up to a change of variables; a neutral unpaired sector gives a Majorana-type action and has only the KO-7 reduction, so the formal label 9 is available only for sectors with the required s = -1 involution. The symmetry restored at m_u = m_d = 0 is U(2) x U(2) on the colour-only algebra, and the reductions there form the homogeneous space U(2), with the mass condition picking out the stationary points of the mass term. The spectral data state the operator, the symmetry, this homogeneous space and the density of each fixed operator. The condensate and the pions as dynamical objects need the QCD measure and a quantum effective action.

Numerical check

Python (numpy, scipy), run on 3 Nov 2026 in a temporary directory, not kept. On a random finite model with KO-4 signs for M (dimension 8) and a quark sector with N_f = 3 and random complex M, for three sign patterns of Sigma and both s: U^2 = 1, U gamma_F = -gamma_F U, [U, D_F] = 0 and U J_F = s J_F U hold to machine precision; R is unitary; the form <J R psi, D R psi'> equals <jj psi, DD psi'> to about 10^-14 and is antisymmetric; j^2 = s and j d = s d j; the formula for d_{Sigma,s} holds, after a first version of it was found wrong by this check; finite differences confirm that V_M is stationary at these Sigma; a generic unitary Sigma gives norm([U, D_F]) of about 10. R_5 (D_M + gamma_5 m) R_5 = D_M + i m holds to machine precision. The spin trace in Proposition 17.10 was evaluated with explicit Dirac matrices (tr of the square equals 4 |a|^2).

Part 18

T5b: vector masses in the spectral action at fixed cutoff

Prompt: prompts/18 (STRATEGY T5b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 with Python (numpy, scipy), by computations independent of the kernel derived here: a direct heat trace with constant a reproduces equation (2) for five choices of (m_1, m_2), to about 0.03%; a direct heat trace on Fourier modes with a_2(x_1) = a cos(q x_1), including the F_{12} and [T, V] vertices, reproduces the kernel (37) at three values of (m, q), to 0.1-0.3%, the size of the discretization error. The closed forms (16)-(19), the bound (28), the limits (27), the neutral matrix (30) recomputed from Standard Model charges, and the identities behind (31) also check. The largest value of the Taylor mass is M_W^2 = 0.275 Lambda^2, at m_t^2 = 2.02 Lambda^2. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all ten numbered results are correct, with small amendments incorporated below. The referee added Propositions 11 to 14 and Remark 15, which widen the scope in both directions; the ratio in Proposition 14 was added by Claude.

Scope. Within the three-generation, zero-Majorana, top-only family at fixed cutoff, these results refute an infinitely heavy W, Z, t limit for the stated quadratic mass definitions. They do not rule out other finite data (Proposition 14), a different action or constraint, a joint limit with Lambda -> infinity, or a quantum effective theory. They do not show unbroken SU(2) x U(1) on the light space (Remark 15).

For the heat cutoff, on $\mathbb{R}^4$, we obtain the exact quadratic vector form without expanding in m/Λ. In the three-generation Standard Model finite representation with only the top Dirac mass nonzero, both the mass defined by the q^0/q^2 Taylor ratio and the unique positive timelike pole mass satisfy $M_W \sim \Lambda^2/m_t$ as $m_t/\Lambda \to \infty$ at fixed Λ. The broken vectors consequently do not decouple by becoming heavy, although the heavy fermion

contribution disappears from the gauge coefficients. These statements concern the classical bosonic quadratic action about a prescribed constant background; moreover, the positive Laplace class considered below does not itself select a finite nonzero stationary background when radial variations of that background are allowed.

1. Theorem (exact quadratic kernel).

Let

$$D_A = -i\gamma^\mu(\partial_\mu + iA_\mu) + \gamma_5 V, \qquad V = V_0 = V_0^\dagger,$$

on $\mathbb{R}^4$, with Hermitian Euclidean gamma matrices, $\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}$, $\gamma_5^2 = 1$, and four spin components. Let the finite trace be the literal trace on H_F, including every stated multiplicity, without a subsequent projection; this agrees with the bosonic trace of Chamseddine-Connes-Marcolli, which counts particles, antiparticles and four spin components. Take A smooth and compactly supported.

We use the vacuum-subtracted action, or equivalently its quadratic variation, since the constant-background trace on $\mathbb{R}^4$ has infinite volume. Write

$$A_\mu(q) = a_\mu(q)T, \qquad T = T^\dagger, \qquad q^\mu a_\mu(q) = 0, \qquad \int_q = \int \frac{d^4q}{(2\pi)^4}.$$

Diagonalize V_0:

$$V_0 e_i = m_i e_i, \qquad m_i \in \mathbb{R}.$$

The signed eigenvalues are needed; their squares alone do not determine the answer.

For $S_t = \operatorname{Tr} e^{-tD_A^2}$, define

$$b(u) = u(1-u), \quad d_{ij} = m_i - m_j,$$

$$I_t(a,b;q^2) = \int_0^1 e^{-t[(1-u)a+ub+u(1-u)q^2]}\, du.$$

Then

$$S_t^{(2)} = \int_q |a_T(q)|^2\, C_{t,T}(q^2),$$

where

$$\boxed{C_{t,T}(q^2) = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2\left[\left(\frac{2}{t} + d_{ij}^2 + q^2\right) I_t(m_i^2, m_j^2; q^2) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t}\right].} \tag{1}$$

For several generators, replace $|T_{ij}|^2$ by $\operatorname{Re}((T_a)_{ij}(T_b)_{ji})$.

In particular, writing $C(q^2) = C_0 + q^2 C_1 + O(q^4)$, we obtain

$$\boxed{C_{0,t} = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2\left[\left(\frac{2}{t} + d_{ij}^2\right) I_t(m_i^2, m_j^2; 0) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t}\right],} \tag{2}$$

and

$$\boxed{C_{1,t} = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2 \int_0^1 e^{-t[(1-u)m_i^2 + um_j^2]}\left[1 - 2b(u) - td_{ij}^2 b(u)\right] du.} \tag{3}$$

These formulas also apply, after integration over $t = s/\Lambda^2$, to

$$f(z) = \int_{(0,\infty)} e^{-sz}\, d\nu(s), \quad 0 < f(0) = \nu((0,\infty)) < \infty, \quad \int (s^{-2} + s^2)\, d\nu(s) < \infty, \tag{4}$$

where ν is positive. These hypotheses suffice for the Euclidean action density and the Taylor coefficients used below. They do not, by themselves, guarantee analytic continuation to every negative q^2.

Proof. With $\nabla_\mu = \partial_\mu + iA_\mu$,

$$D_A^2 = -\nabla^2 + V_0^2 - \frac{i}{2}\gamma^{\mu\nu}F_{\mu\nu} + \gamma^\mu\gamma_5[A_\mu, V_0].$$

Let $P_0 = -\partial^2 + V_0^2$ and write $D_A^2 = P_0 + P_1 + P_2$, according to degree in A. Duhamel expansion and cyclicity give

$$S_t^{(2)} = -t\,\mathrm{Tr}(e^{-tP_0}P_2) + \frac{t^2}{2}\int_0^1 \mathrm{Tr}\,\left(e^{-(1-u)tP_0}P_1 e^{-utP_0}P_1\right) du. \tag{5}$$

The linear vertex from j, p to $i, p+q$ is

$$T_{ij}\left[(2p+q)\cdot a + \gamma^{\mu\nu}q_\mu a_\nu + (m_j - m_i)\gamma^\mu\gamma_5 a_\mu\right].$$

Completing the momentum square produces

$$(4\pi t)^{-2}e^{-t[(1-u)m_i^2 + um_j^2 + b(u)q^2]}.$$

For transverse a, the averaged spin trace of the vertex product is

$$4\left(\frac{2}{t} + d_{ij}^2 + q^2\right)|a|^2.$$

The P_2 term contributes

$$-\frac{1}{4\pi^2 t}\sum_{ij}|T_{ij}|^2 e^{-tm_i^2}|a|^2.$$

Symmetrizing in i, j proves (1). Differentiation proves (2)–(3).

The divided-difference structure is explicit:

$$I_t(a,b;0) = \frac{e^{-ta} - e^{-tb}}{t(b-a)}, \qquad I_t(a,a;0) = e^{-ta}, \tag{6}$$

and

$$\int_0^1 u(1-u)e^{-t[(1-u)a+ub]}\,du = \frac{1}{t^2}\partial_a\partial_b I_t(a,b;0). \tag{7}$$

Thus (2) uses a first divided difference, while (3) also uses its mixed derivative. The full momentum dependence follows from the convergent series

$$I_t(a,b;q^2) = \sum_{n=0}^{\infty}\frac{(-q^2/t)^n}{n!}\partial_a^n\partial_b^n I_t(a,b;0). \tag{8}$$

Repeated-node values are obtained continuously.

For completeness, the mass coefficient for (4) can be written entirely in dimensionless variables. Put

$$r_i = m_i/\Lambda, \quad x_i = r_i^2, \quad z(u) = (1-u)x_i + ux_j, \quad \Phi_1(z) = \int s^{-1}e^{-sz}\,d\nu(s).$$

The bracket replacing that in (2) is

$$\Lambda^2\left[2\int_0^1 \Phi_1(z(u))\,du + (r_i - r_j)^2\int_0^1 f(z(u))\,du - \Phi_1(x_i) - \Phi_1(x_j)\right]. \tag{9}$$

The corresponding bracket in (3) is

$$\int_0^1 \left[(1-2b)f(z) + (r_i - r_j)^2 b f'(z)\right] du. \tag{10}$$

Each unweighted integral is a divided difference of an antiderivative. The moment assumptions justify the integrations and differentiations. □

On a finite flat torus, (5) remains exact, but the Gaussian momentum integral must be replaced by the spin-structure-dependent lattice sum. Explicitly, for a transverse polarization a, its two-vertex contribution per unit volume is

$$\frac{2t^2}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2\int_0^1 e^{-t[(1-u)(p^2+m_i^2)+u((p+q)^2+m_j^2)]}\left(|(2p+q)\cdot a|^2+(d_{ij}^2+q^2)|a|^2\right)du,$$

and the quadratic-vertex contribution is

$$-\frac{4t}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2e^{-t(p^2+m_i^2)}|a|^2.$$

Consequently, the continuum formulas below are statements on $\mathbb{R}^4$, or in the infinite-volume limit. A finite torus can have additional holonomy dependence, including for constant commuting connections.

2. Lemma (gauge invariance and the Goldstone direction).

Let scalar fluctuations include the gauge orbit of V_0. Then the nonzero vector coefficient C_0 in (2) is compatible with gauge invariance.

For one generator, parameterize the orbit by

$$V(x)=e^{i\pi(x)T}V_0e^{-i\pi(x)T}.$$

With our conventions,

$$a_\mu\longmapsto a_\mu-\partial_\mu\alpha,\qquad \pi\longmapsto\pi+\alpha.$$

The invariant combination is

$$B_\mu=a_\mu+\partial_\mu\pi. \tag{11}$$

Proof. Gauge covariance gives

$$D_{A^U,V^U}=UD_{A,V}U^{-1}.$$

Applying $U=e^{-i\pi T}$ transforms V to V_0 and a to B. Hence

$$S[a,e^{i\pi T}V_0e^{-i\pi T}]=S[B,V_0]. \tag{12}$$

This establishes the Goldstone completion directly, without a derivative expansion.

For comparison, the longitudinal vector kernel is obtained from (1) by replacing the q^2 in its prefactor by $(1-2u)^2q^2$ inside the integral. If $[T,V_0]=0$, its integral vanishes by

$$\int_0^1[2+tq^2(1-2u)^2]e^{-tq^2u(1-u)}\,du=2.$$

For a broken generator the longitudinal kernel need not vanish; it belongs to the invariant combination (11).

If V_0 is not stationary, a Cartesian scalar Hessian alone need not annihilate the linear gauge-orbit tangent. The quadratic term in $e^{i\pi T}V_0e^{-i\pi T}$, contracted with the scalar tadpole, must also be retained. Equation (12) includes that term. Transverse vectors do not mix quadratically with scalar fluctuations on this translation-invariant background. □

3. Corollary (the asymptotic normalization).

Let all finite eigenvalues be small relative to Λ, let f satisfy (4), and let Q below be positive definite on the chosen generator space (otherwise pass to the quotient by its null space). Define

$$Q_{ab}=\mathrm{Re}\,\mathrm{Tr}(T_aT_b),\qquad K_{ab}=\mathrm{Re}\,\mathrm{Tr}([T_a,V_0]^\dagger[T_b,V_0]).$$

At leading order,

$$C_0 = \frac{f(0)}{8\pi^2}K, \qquad C_1 = \frac{f(0)}{12\pi^2}Q,$$

and therefore

$$\boxed{\mathcal{M}^2_{\text{Taylor}} = \frac{3}{2}Q^{-1/2}KQ^{-1/2}.} \tag{13}$$

This reproduces Proposition 13.5.

For the full three-generation Standard Model representation,

$$Q_{11} = 12, \qquad K_{11} = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2),$$

so

$$\boxed{8M_W^2 = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2).} \tag{14}$$

Proof. In (2), the orbital expression

$$\frac{2}{t}I_t(a,b;0) - \frac{e^{-ta}+e^{-tb}}{t}$$

starts at order $t(a-b)^2$. The leading term is consequently d_{ij}^2. In (3), the leading integral is $\int_0^1(1-2u(1-u))du = 2/3$, proving (13).

For $T_1 = \sigma_1/2$ on each left doublet, one generation of particles contributes 2 to Q_{11}. Charge conjugates and three generations give 12. A doublet with Dirac masses m_u, m_d contributes $\frac{1}{2}(m_u^2+m_d^2)$ to the particle commutator trace, before color multiplicity. This proves (14).

Equivalently, the gauge coefficient is

$$\frac{f(0)g_2^2}{2\pi^2} = \frac{1}{4}.$$

For comparison with CCM's $\operatorname{Tr} F(D/\Lambda)$, take $F(v) = f(v^2)$; then $F(0) = f(0)$. The normalization and mass constraint are precisely equations (4.10) and (4.16), with the latter stated in Lemma 4.6 of the checked preprint. CCM describes the relation as "valid at unification scale." Chamseddine–Connes–Marcolli, pp. 37–39. □

The leading bound $M^2 \le 6\|V_0\|^2$ in Lemma 14 follows from (13) and $K \le 4\|V_0\|^2 Q$. Formula (13), rather than an exact finite-cutoff identity, is its applicable approximation.

4. Proposition (the exact charged-vector Taylor mass).

Let the finite data be the full three-generation Standard Model representation, including charge conjugates. Let the Majorana matrix and every Dirac mass except one up-type color triplet be zero. Write

$$m = \mu\Lambda, \qquad x = \mu^2, \qquad r = e^{-x}.$$

Let $f(z) = e^{-z}$. Define

$$H(y) = \int_0^1 e^{-yu(1-u)}\,du,$$

$$g(y) = (2+y)H(y) - 2,$$

$$h_x(y) = (2+x+y)\int_0^1 e^{-xu-yu(1-u)}\,du - 1 - e^{-x}. \tag{15}$$

Then the full charged-vector kernel is

$$\boxed{C_W(q^2) = \frac{\Lambda^2}{4\pi^2}\left[\frac{9}{2}g(y) + \frac{3}{2}h_x(y)\right], \qquad y = q^2/\Lambda^2.} \tag{16}$$

Define

$$A(x) = h_x(0) = \frac{2[1-(1+x)e^{-x}]}{x},$$

$$w(x) = 3\int_0^1 u^2 e^{-xu}\,du = \frac{3[2-(x^2+2x+2)e^{-x}]}{x^3}. \tag{17}$$

Continuous values are understood at $x = 0$. The particle-trace kinetic coefficient is

$$K_W(x) = \frac{9}{2} + \frac{3}{2}w(x) > 0.$$

The Taylor mass, defined by $M^2_{W,\mathrm{Taylor}} = C_W(0)/C'_W(0)$, is

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A(x)}{4K_W(x)} = \frac{3[1-(1+x)e^{-x}]}{x[3+w(x)]}.} \tag{18}$$

Its limiting behavior is

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{8}x - \frac{23}{128}x^2 + O(x^3), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{1}{x} \quad (x\to\infty). \tag{19}$$

Consequently,

$$\frac{m}{M_{W,\mathrm{Taylor}}} \longrightarrow \sqrt{\frac{8}{3}} \quad (\mu\to 0), \qquad \frac{m}{M_{W,\mathrm{Taylor}}} \sim \mu^2 \quad (\mu\to\infty). \tag{20}$$

Proof. On the particle space, $Q_{11} = 6$. The selected quark doublet contributes $3/2$, while the remaining doublets contribute $9/2$. Diagonalizing the massive left-right block produces eigenvalues $+m, -m$. The charged generator connects these to the massless down states; every such pair has $d^2_{ij} = m^2$. Formula (1) therefore gives h_x for the selected doublet and g for the remaining doublets. Charge conjugation supplies the factor two in (16).

Integration by parts gives

$$h_x(y) = 2\int_0^1 u\,[x + 2y(1-u)]e^{-xu-yu(1-u)}\,du, \tag{21}$$

and

$$\boxed{h'_x(y) = 2\int_0^1 u^2[1-y(1-u)]e^{-xu-yu(1-u)}\,du.} \tag{22}$$

The boundary term proving (22) is $\partial_u[u(1-u)e^{-xu-yu(1-u)}]$. Thus

$$h'_x(0) = \frac{2}{3}w(x), \qquad g'(0) = \frac{2}{3}.$$

This proves (18). Expanding the elementary expressions at zero and taking their large-x limits proves (19)–(20). □

The same Taylor result extends to every f in (4). Put

$$\ell = f(0), \qquad w_f(x) = \int w(sx)\,d\nu(s), \qquad A_f(x) = \int \frac{A(sx)}{s}\,d\nu(s).$$

Then

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A_f(x)}{4\left(\frac{9}{2}\ell + \frac{3}{2}w_f(x)\right)}.} \tag{23}$$

All denominators are positive, and dominated convergence gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{3}{8}m^2 \quad (x \to 0), \qquad \boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{\int s^{-2}\, d\nu(s)}{\ell x} \quad (x \to \infty).} \tag{24}$$

Indeed, $A(z)/z \to 1$, $w(z) \to 1$ at zero, while $zA(z) \to 2$, $w(z) \to 0$ at infinity.

If the other Dirac masses are merely small and nonzero, the answer depends on those additional parameters and is given by (1). In particular, the vanishing limit in (19) assumes they are zero: masses of surviving light doublets can leave a residual vector mass.

5. Theorem (the charged-vector pole).

Under the heat-cutoff hypotheses of Proposition 4, define a timelike pole by analytic continuation of the quadratic inverse propagator:

$$C_W(-M^2_{W,\mathrm{pole}}) = 0. \tag{25}$$

For every $x > 0$, this equation has exactly one solution with $M^2_{W,\mathrm{pole}} > 0$. It is simple and has positive residue in the convention that the Euclidean propagator near the pole is a positive multiple of $(q^2 + M^2_{W,\mathrm{pole}})^{-1}$.

Let $z(x) = M^2_{W,\mathrm{pole}}/\Lambda^2$. Its exact specification is the unique positive root

$$\boxed{\frac{9}{2}g(-z) + \frac{3}{2}h_x(-z) = 0.} \tag{26}$$

Furthermore,

$$z(x) \sim \frac{3}{8}x \quad (x \to 0), \qquad z(x) \sim \frac{1}{x} \quad (x \to \infty), \tag{27}$$

and both mass definitions obey

$$\boxed{0 < M^2_{W,\mathrm{Taylor}},\ M^2_{W,\mathrm{pole}} \le \frac{\Lambda^2}{e}.} \tag{28}$$

Proof. Let

$$F_x(y) = \frac{9}{2}g(y) + \frac{3}{2}h_x(y).$$

Equation (22) implies $h'_x(y) > 0$ for $y \le 0$, and also $g'(y) > 0$, since $g = h_0$. Hence F_x is strictly increasing on that half-axis. We have

$$F_x(0) = \frac{3}{2}A(x) > 0.$$

As $y \to -\infty$, the prefactors in (15) become negative and both integrals grow exponentially, so $F_x(y) \to -\infty$. This proves existence, uniqueness, simplicity, and the residue sign.

For $y \le 0$, equation (22) gives $g'(y) \ge 2/3$, and therefore

$$F'_x(y) \ge 3.$$

The mean-value theorem yields

$$z(x) \le \frac{1}{2}A(x).$$

The Taylor denominator gives the same bound for $M^2_{W,\mathrm{Taylor}}/\Lambda^2$. Finally,

$$A(x) = 2\int_0^1 xu\, e^{-xu}\, du \le \frac{2}{e},$$

proving (28).

As $x \to 0$, the bound implies $z(x) \to 0$, and $F'_x(y) \to 4$ near the root. As $x \to \infty$, $z(x) \to 0$, $h'_x(y) \to 0$ near the root, and $F'_x(y) \to 3$. Together with $F_x(0) \sim 3x/2$ and $F_x(0) \sim 3/x$, respectively, these prove (27). □

The pole mass governs the linearized dispersion relation of this analytically continued quadratic action. The Taylor mass governs its two-derivative truncation; they need not coincide near the cutoff. We assert no closed-form inversion of (26), no classification of complex poles, and no Lorentzian unitarity theorem.

The pole proof also extends to positive measures supported in a compact subset of $(0,\infty)$, using

$$F_{f,x}(y) = \int \frac{1}{s}\left[\frac{9}{2}g(sy) + \frac{3}{2}h_{sx}(sy)\right] d\nu(s).$$

It gives the analogues of (24) for the pole mass. For the larger class (4), only the Euclidean and Taylor conclusions have been established here; its moment assumptions alone do not justify that continuation.

6. Proposition (unbroken coefficients, neutral vectors, and weak angles).

Retain the heat-cutoff Standard Model family of Proposition 4. Normalize color generators by $\operatorname{tr}_{\mathbf{3}} t_a t_b = \frac{1}{2}\delta_{ab}$, and use $Q = T_3 + Y$. Define the full-trace coefficients by

$$S^{(2)}_{\text{unbroken}} = \int \left(\kappa_{\text{em}} F_{\text{em}}^2 + \kappa_s \sum_a G_a^2\right).$$

Then

$$\boxed{\kappa_{\text{em}} = \frac{1}{24\pi^2}\frac{16}{3}(5+r), \qquad \kappa_s = \frac{1}{24\pi^2}2(5+r), \qquad \frac{\kappa_{\text{em}}}{\kappa_s} = \frac{8}{3}.} \tag{29}$$

Both coefficients approach 5/6 of their all-massless values.

Let K denote the particle-trace neutral q^2 coefficient in coordinates (A_3, A_Y). Then

$$\boxed{\begin{aligned} K_{33} &= \frac{21}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{103}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}} \tag{30}$$

The matrix K, the charged coefficient K_W, and the unbroken coefficients are positive for every $x \geq 0$.

Define the neutral-current angle from the current of the massive direction after diagonalizing the two-derivative neutral kinetic form:

$$J_Z \;\propto\; T_3 - \sin^2\theta_{\text{NC}}\, Q.$$

Then

$$\boxed{\cos^2\theta_{\text{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}} = \frac{5}{8}.} \tag{31}$$

A different continuation, namely

$$\cos^2\theta_{\text{match}} := \frac{K_{YY}}{K_W + K_{YY}},$$

satisfies

$$\boxed{\cos^2\theta_{\text{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\text{match}}(\infty) = \frac{103}{157}.} \tag{32}$$

Thus these definitions cease to describe one common electroweak angle.

Proof. Photon and color generators commute with V_0. Their coefficients therefore follow from (3) with $d_{ij} = 0$ on each invariant mass block. The charge traces give (29).

A Dirac species of mass m, with neutral charges q_L^a, q_R^a, contributes to K

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right]. \tag{33}$$

Applying (33) to the three top colors, and retaining all other states, proves (30).

Write

$$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad B_\infty = \begin{pmatrix} 21/4 & -1/4 \\ -1/4 & 103/12 \end{pmatrix}, \quad U = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 17/12 \end{pmatrix}.$$

Then

$$K = B_\infty + rU - \frac{3xr}{8} vv^T.$$

We have $B_\infty \geq 5I$, $U \geq 0$, and $xr \leq 1/e$. Hence

$$K \geq \left(5 - \frac{3}{4e}\right) I > 0.$$

Positivity of K_W follows from (17).

The neutral mass direction is v, while the photon direction is $(1, 1)$. The kinetic-orthogonal massive field is proportional to $K^{-1}v$, giving the first expression in (31). Directly,

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

which proves (31). Taking limits proves (32). □

The neutral mass also tends to zero. More precisely,

$$\boxed{\frac{M^2_{Z,\mathrm{Taylor}}}{\Lambda^2} = \frac{9}{4} xr\, v^T K^{-1} v \sim \frac{2}{3} xe^{-x} \quad (x \to \infty).} \tag{34}$$

For $x > 0$ there is likewise one positive massive neutral pole, besides the photon pole. To prove this, put $B = B_\infty + rU$. The exact neutral kernel, apart from the common factor $\Lambda^2/(4\pi^2)$, is

$$B\, g(y) + \frac{3}{2} xr\, H(y) vv^T. \tag{35}$$

After congruence by $B^{-1/2}$, its massive eigenvalue is

$$g(y) + \beta H(y), \qquad \beta = \frac{3}{2} xr\, v^T B^{-1} v > 0.$$

Its timelike root obeys

$$z - 2 + \frac{2}{H(-z)} = \beta. \tag{36}$$

The left side increases from zero to infinity, because

$$\frac{d}{dz}\left(z - 2 + \frac{2}{H(-z)}\right) = 1 - \frac{2\,\mathbb{E}_z[b]}{H(-z)} \geq \frac{1}{2},$$

where $\mathbb{E}_z$ uses density proportional to $e^{zb(u)}$. The pole is therefore unique and has positive residue. Since $v^T B_\infty^{-1} v = 8/27$,

$$\frac{M^2_{Z,\mathrm{pole}}}{\Lambda^2} \sim \frac{2}{3} xe^{-x}.$$

In particular, M_W^2/M_Z^2 does not continue to equal $5/8$.

For every f in (4), equations (29)–(33) extend by replacing 1 by $\ell = f(0)$, r by $f(x)$, and xr by $-xf'(x)$. Positivity follows by averaging the positive heat-cutoff matrices. The ratio $8/3$, the neutral-current value $5/8$, and the matching limit $103/157$ persist. At the endpoint both neutral masses vanish, so the distinguished massive neutral direction—and hence its angle—requires continuation from finite x.

7. Counterexample (positive cutoff does not ensure positive broken kinetic coefficients).

Let $H_F = \mathbb{C}^2$,

$$V_0 = m\sigma_3, \qquad T = \sigma_1, \qquad x = m^2/\Lambda^2,$$

with the same spacetime and trace conventions. For the heat cutoff,

$$C_T(q^2) = \frac{\mathrm{Tr}(T^2)\Lambda^2}{8\pi^2} e^{-x} \left[(2 + 4x + y)H(y) - 2\right]. \tag{37}$$

Its q^2 Taylor coefficient is proportional to

$$e^{-x}(1 - x).$$

It vanishes at $x = 1$ and becomes negative for $x > 1$. Thus the two-derivative truncation has a wrong-sign vector kinetic term in this example.

Nevertheless, the full quadratic form has a unique positive real timelike pole with positive residue, satisfying

$$\boxed{4x < \frac{M^2_{\mathrm{pole}}}{\Lambda^2} < 4x + 2.} \tag{38}$$

Consequently $M_{\mathrm{pole}} \sim 2m$ diverges at fixed cutoff.

Proof. The generator connects the eigenvalues $m, -m$, so $d^2_{ij} = 4m^2$ while both squared eigenvalues equal m^2. Substitution in (1) proves (37). Its Taylor ratio is

$$M^2_{\mathrm{Taylor}} = \frac{6m^2}{1 - x},$$

where this is a positive mass only for $x < 1$.

The pole equation is (36) with $\beta = 4x$. The monotonicity proof there gives uniqueness and the residue sign. Evaluating (37) at $y = -4x$ gives $2H(-4x) - 2 > 0$, whereas at $y = -4x - 2$ it gives -2; this proves (38).

For general f satisfying (4), the Taylor kinetic coefficient in this example is proportional to

$$f(x) + xf'(x) = \frac{d}{dx}[xf(x)]. \tag{39}$$

It must be negative somewhere: $xf(x) > 0$ for $x > 0$, but $xf(x) \to 0$ at infinity under (4). A nonnegative derivative everywhere would contradict these facts. □

This counterexample concerns arbitrary finite data, not the top-only Standard Model family. It also distinguishes a wrong-sign **Taylor truncation** from a negative-residue pole of the **full** form factor. We have not proved absence of complex-pole or other nonlocal problems in either model.

8. Counterexample (heavy fermions do not dynamically select the centralizer).

Let the data be those of Proposition 4, and let

$$\mathfrak{g}_{V_0} = \{X \in \mathfrak{g}_{\mathrm{SM}} : [X, V_0] = 0\}.$$

For nonzero top mass this is the represented unbroken gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$. The limit $x \to \infty$ at fixed Λ does **not** produce a theory containing only these vector directions by quadratic mass decoupling.

Proof. For fixed y, equation (21) implies

$$h_x(y) \longrightarrow 0.$$

Consequently

$$\boxed{C_W(q^2) \longrightarrow \frac{\Lambda^2}{4\pi^2}\frac{9}{2} g(q^2/\Lambda^2).} \tag{40}$$

For positive y,

$$g(y) = 4y \int_0^1 u(1 - u)e^{-yu(1-u)}\, du > 0.$$

Thus a charged generator outside $\mathfrak{g}_{V_0}$ retains a finite nonzero quadratic kinetic form and a nonvanishing propagator at every fixed positive momentum. Its mass approaches zero. Equation (35) similarly leaves the positive matrix $B_\infty g(y)$, including the additional neutral direction. These facts contradict decoupling of the broken vectors through diverging masses or vanishing propagators. □

A narrower decoupling statement is true. If we **restrict the connection from the outset** to the centralizer, the finite mass blocks are invariant and the heat trace factorizes. On the heavy block its contribution carries e^{-sm^2/Λ^2}, which tends to zero. The restricted action therefore converges to the light-block spectral action. This proves decoupling of a finite fermion block under that restriction; it does not prove dynamical elimination of the other gauge fields.

The amendment to Lemma 03 Part C is therefore:

- Its transverse kinetic weights, including $w_L(x,0) = w(x)$, and its two weak-angle formulas are correct under their stated conventions.
- They must be accompanied by the mass coefficient (2), including the quadratic-vertex subtraction.
- A centralizer-only endpoint does not follow from those weights. In the present exact quadratic family, the broken-vector masses vanish while their remaining kinetic coefficients are positive.

9. No-go theorem (finite radial vacuum for the positive Laplace class).

Let f satisfy (4), and suppose the allowed scalar configurations include

$$V = cV_*, \qquad c > 0, \qquad V_* \neq 0,$$

with $A = 0$. Let the action be the literal spectral action, without an added scalar potential or a constraint fixing c. Then no finite $c > 0$ is stationary along this radial direction.

Proof. On $\mathbb{R}^4$, the constant-background action density is

$$\mathcal{U}(c) = \frac{\Lambda^4}{4\pi^2} \int s^{-2} \operatorname{Tr}_{H_F} e^{-sc^2V_*^2/\Lambda^2}\, d\nu(s).$$

Differentiating gives

$$\boxed{\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \int s^{-1} \operatorname{Tr}_{H_F} \left(V_*^2 e^{-sc^2V_*^2/\Lambda^2}\right) d\nu(s) < 0.} \tag{41}$$

The inequality is strict because $V_* \neq 0$ and ν is nonzero and positive. The same argument applies on a finite flat torus before performing its positive momentum sum. □

Thus the masses above are conditional quadratic-background quantities. In the top-only family, the ordinary Higgs radial variation supplies the direction used in this theorem. A finite symmetry-breaking minimum of the polynomial asymptotic scalar potential cannot automatically be regarded as a minimum of this full positive-Laplace spectral action.

10. Corollary (comparison of heavy limits).

Let the background family and mass definitions be those above.

At fixed Λ, with $\mu \to \infty$:

quantity, heat cutoff	behavior
m_t	$\mu\Lambda \to \infty$
$M_{W,\mathrm{Taylor}}$, $M_{W,\mathrm{pole}}$	$\sim \Lambda/\mu \to 0$
$M_{Z,\mathrm{Taylor}}$, $M_{Z,\mathrm{pole}}$	$\sim \sqrt{2/3}\,\Lambda\mu e^{-\mu^2/2} \to 0$
m_t/M_W	$\sim \mu^2 \to \infty$

(42)

At fixed $0 < \mu < \infty$, with $\Lambda \to \infty$, all four vector masses in (42) instead grow linearly with Λ, as does m_t. A controlled asymptotic family can also satisfy

$$\Lambda \to \infty, \qquad \mu \to 0, \qquad \mu\Lambda \to \infty;$$

then $M_W, m_t \to \infty$ while

$$m_t/M_W \to \sqrt{8/3}.$$

Proof. The first assertions follow from (19), (27), and (34)–(36). At fixed positive μ, their dimensionless mass functions are fixed and positive, so multiplication by Λ proves the joint-limit assertions. The final assertion follows from the small-μ limits. □

Accordingly, the obstruction has three distinct scopes. The CCM identity cannot be extrapolated to $\mu \to \infty$ at fixed cutoff because it belongs to the asymptotic truncation. The exact top-only quadratic action obstructs a fixed-cutoff heavy-W limit for the heat cutoff under both mass definitions, and for class (4) under the Taylor definition. A family of actual nonzero stationary vacua of the unmodified positive-Laplace action encounters the separate obstruction (41). Counterexample 7 excludes extending the first of these exact mass conclusions to arbitrary finite representations.

11. Proposition (the radial derivative for general f; referee). On flat $\mathbb{R}^4$, for a sufficiently integrable real cutoff f, with λ_j^2 the eigenvalues of V_*^2,

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{43}$$

For nonnegative f, a finite $c > 0$ is stationary exactly when every tail associated with a nonzero λ_j vanishes. So if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray; if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$. Sign-changing cutoffs can have cancellations. A nonconstant cutoff flat near zero lies outside (4), since a nonzero positive measure gives $f'(0) < 0$; (43) covers the nonnegative ones. A radial path with an additional fixed noncommuting mass matrix need not have the form cV_*. □

12. Proposition (the top-only Taylor mass for general f; referee). For nonnegative smooth compactly supported f with $f(0) > 0$, the formulas of Proposition 4 hold with

$$A_f(x) = \frac{2}{x}\int_0^x z f(z)\, dz, \qquad w_f(x) = \frac{3}{x^3}\int_0^x z^2 f(z)\, dz,$$

by approximating f with its derivatives by real combinations of e^{-nz}. Hence

$$\frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} \sim \frac{\int_0^\infty z f(z)\, dz}{f(0)\, x}.$$

This includes cutoffs flat near zero. For a nonanalytic cutoff there need be no analytic continuation defining the pole. □

13. Proposition (several heavy doublets; referee). For the heat cutoff, let P be the total particle-trace weight of complete massless doublets, and let heavy doublet i, of weight $d_i > 0$, have one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\Big(Pg(y) + \sum_i d_i h_{x_i}(y)\Big), \qquad \frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}.$$

For finitely many doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, then $M_{W,\mathrm{Taylor}}^2 \sim m^4/(2\Lambda^2)$, the form factor has a unique positive pole with $M_{W,\mathrm{pole}}^2 \sim m^2/2$, and the whole unnormalized kernel tends to zero at every fixed Euclidean momentum. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory. □

14. Proposition (the fully massive family; referee; a complete proof by Astra 6 is in the validation, item 11). Give both members of every Standard Model doublet the same Dirac mass m, with no Majorana masses, and let z_W, z_Z be the charged and massive neutral poles in units of Λ^2. The charged form factor is proportional to $e^{-x}[g(y) + 2xH(y)]$, and

$$2x < z_W < 2x + 2, \qquad \tfrac{16}{5}x < z_Z < \tfrac{16}{5}x + 2.$$

So both pole masses diverge with m at fixed cutoff, and

$$\frac{M^2_{W,\text{pole}}}{M^2_{Z,\text{pole}}} \longrightarrow \frac{5}{8} \qquad (x \to \infty),$$

as also for $x \to 0$, where $z_W \sim 3x$ and $z_Z \sim \frac{24}{5}x$. In between the ratio differs, reaching about 0.66 near $x = 3$ (Python, solving the pole equations). The kernel tends to zero at every fixed Euclidean momentum, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1 - x/2)$, negative for $x > 2$.

Proof. The species with equal mass have eigenvalues $\pm m$. The charged generator connects them with $d^2 = 0$ and $d^2 = 4m^2$ in equal weights, which gives the charged form. For the neutral generators the diagonal charge has $d = 0$ and the off-diagonal charge $(q_L - q_R)/2 = T_3 v/2$ has $d^2 = 4m^2$; on the particle trace the kernel is $e^{-x}[\text{diag}(6, 10)g + 12xHvv^T]$, so $\beta = 12x\, v^T \,\text{diag}(6, 10)^{-1} v = \frac{16}{5}x$, and for the charged direction $\beta = 2x$. Equation (36) gives the bounds, since $H(-z) > 1$. □

This is a counterexample to any general no-go for diverging pole locations at fixed cutoff. It has no light fermions, and the physical quality of that limit is open.

15. Remark (the light space does not carry unbroken $SU(2) \times U(1)$; referee). In the top-only limit the broken masses vanish, but this does not restore the electroweak group on the whole light space. The light space retains b_L after removing t_L. If P_{light} projects off the top mass block, then on b_L we have $P_{\text{light}} T_1 P_{\text{light}} = P_{\text{light}} T_2 P_{\text{light}} = 0$ and $P_{\text{light}} T_3 P_{\text{light}} = -\frac{1}{2}$, which do not obey the $SU(2)$ relations. Correspondingly $K_W = 9/2$, $K_{33} = 21/4$, $K_{3Y} = -1/4$ in the limit, while a linearly realized unbroken action would have equal weak coefficients and no mixing. The remaining complete massless doublets carry the ordinary representation, and their contribution is gauge invariant. If both members of every removed doublet are heavy, the retained light space can be invariant under the full group; a full limiting theory still needs control of the remaining fields, normalizations, interactions and vacuum. "The heavy fermion leaves the loop" is shorthand for the suppression in this classical trace.

The light-space statement should be read together with the separate role of the top (lemma 21): among single charged species only an up-type quark can be removed while keeping cos^2 theta_NC = 5/8 and the ratio 8/3, and removing it leaves the five other quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R.

16. Remark (the CCM relation as the small-mass form of a weighted sum rule; Claude). For the heat cutoff and doublets with one massive member, Proposition 13 reads

$$8\kappa M^2_{W,\text{Taylor}} = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{A(x)}{x} = \frac{2[1 - (1 + x)e^{-x}]}{x^2},$$

with $N_f = 2d_f$ and $\kappa = (P + \sum_i d_i w(x_i))/6$. As $x \to 0$, $\rho, \kappa \to 1$, which is the CCM relation (14). The relation is therefore the leading term of an expansion, and its range is $m \ll \Lambda$. For the top alone, $\kappa = (3 + w)/4$; m/M_W departs from $\sqrt{8/3}$ by 5% at $m \approx 0.45\Lambda$, and M_W peaks at about 0.52Λ at $m \approx 1.42\Lambda$ (Python). The maximum is unique, at $x = 2.024$ with $M_W/\Lambda = 0.5242$ (referee, by a sign argument and Python). A massive diagonal contribution to the trace is exponentially suppressed above the cutoff, while the mixed loop with a massless partner has the algebraic $\rho \sim 2/x^2$ at large x; a vacuum with no other mass leaves the W mass with no source. Whether the classical action is meaningful for $m > \Lambda$ is a separate question, on which these formulas do not decide. In the top-only heat-cutoff family, the ratio differs from $\sqrt{8/3}$ for every $x > 0$, since $3 + w - 4\rho$ vanishes at zero and is increasing, so the limit with M_W and m_t diverging together and $m_t/M_W \to \sqrt{8/3}$ needs $\Lambda \to \infty$ faster than m_t (Corollary 10). This necessity is for that family: with one complete massless lepton doublet and one member of each other doublet of common mass m, the weighted sum equals the unweighted CCM sum at some $x_* > 0$ (referee).

As algebraic checks, we used **Python 3 with SymPy** to verify the integration-by-parts identities, the displayed small-x Taylor series, and $v^T B_\infty^{-1} v = 8/27$. The arguments above provide the proofs; these computations introduce no fitted constants.

Consequence for T5b. At fixed cutoff, the specified top-only Standard Model spectral action does not realize an infinitely heavy W, Z, t family: the top grows while both vector masses tend to zero, for the heat

cutoff and the stated quadratic mass definitions. A joint limit with $\Lambda \to \infty$ permits all three background mass scales to diverge, but neither centralizer-only decoupling nor a stationary Higgs vacuum follows from that scaling. These conclusions concern the classical bosonic action and contain no quantum corrections or claims about the physical Standard Model at those scales. Other finite data behave differently: with every doublet heavy and no light fermions, the pole masses diverge at fixed cutoff with M_W^2/M_Z^2 -> 5/8 (Proposition 14). With a massless complete doublet retained, the Taylor mass tends to zero (Proposition 13). So in the families treated here, a heavy W and Z at fixed cutoff is found only without a massless complete doublet. Light nonzero masses are not treated: they leave a residual vector mass that depends on them.

Part 19

Inner fluctuations of the two-point data without order one

Step 1 of the next steps in the background note, for STRATEGY T0b. Proved by Claude Opus 5.5 without Astra or Fable; every identity checked with numpy on random data (section "Numerical check"). Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): Theorem 19.1 and the comparison with arXiv:1304.7583 are correct; Corollary 19.2 item 3 and the Wilson phase after the table were wrong as first written, and these and the amendments are incorporated below.

Outcome. The truncated two-point data satisfy order zero and fail order one, for each natural real structure. So the inner fluctuations of Chamseddine-Connes-van Suijlekom (arXiv:1304.7583), with their quadratic term, are the applicable ones. The result factorizes. With the inherited pointwise real structure, the fluctuated operator is m |Phi|^2 sigma_1: a real gauge-neutral radial field, with trivial gauge group. With the reflection real structure, which has the KO-6 signs, the link is m Phi^2: all of C is reached, and the gauge group U(1) acts with charge 2. Along a gauge orbit the quadratic term keeps the modulus of the link fixed. This parallels the fixed singular value of the covariant projection of lemma 14, which the fixed compression of lemma 13 lacks. So the first-order failure of Counterexample 08.4 points to the framework of arXiv:1304.7583 for these truncated data.

Setting

Let A = C + C act diagonally on H = C^2, with D = m sigma_1, m > 0, and gamma = sigma_3. For a real structure J = C K put x-hat = J x J^-1. We use two choices.

- **J_0 = K** (or sigma_3 K, the other sign of lemma 08), inherited from pointwise reality on the interval. Then x-hat = conj(x) on A, and J_0 commutes with gamma.
- **J_r = sigma_1 K**, inherited from the reflection real structure sigma_1 R_l K of Counterexample 08.4. Then diag(x_1, x_2)-hat = diag(conj(x_2), conj(x_1)), J_r^2 = 1, J_r D = D J_r and J_r gamma = -gamma J_r: the KO-6 signs.

Both satisfy order zero, since A is commutative and x-hat lies in A. Both fail order one: for e = diag(1, 0), [[D, e], e-hat] = m sigma_1 for J_0 and -m sigma_1 for J_r.

For a universal one-form A = sum_j a_j delta(b_j), the fluctuated operator of arXiv:1304.7583 is

```
D' = D + A_(1) + A~_(1) + A_(2),
A_(1) = sum a_j [D, b_j],   A~_(1) = sum a_j-hat [D, b_j-hat] = eps J A_(1) J^-1,   A_(2) = sum
↪  a_j-hat [A_(1), b_j-hat],
```

with J D J^-1 = eps D. Gauge transformations are U = u J u J^-1 for unitary u in A, and U D(A) U* = D(gamma_u(A)), where gamma_u(A) = u delta(u*) + u A u*.

Put

```
phi_1 = sum a_{j,1} (b_{j,2} - b_{j,1}),     phi_2 = sum a_{j,2} (b_{j,1} - b_{j,2}),
```

so that A_(1) = m [[0, phi_1], [phi_2, 0]], which is self-adjoint exactly when phi_2 = conj(phi_1). The pair (phi_1, phi_2) takes every value in C^2, as a = diag(x, y) and b = diag(0, 1) show.

The fluctuated operator

Theorem 19.1. For every universal one-form,

```
J_0:   D' = m [[0, |1 + phi_1|^2], [|1 + phi_2|^2, 0]],
J_r:   D' = m [[0, (1 + phi_1)(1 + conj(phi_2))], [(1 + phi_2)(1 + conj(phi_1)), 0]].
```

For a self-adjoint form, with Phi = 1 + phi_1,

```
J_0:   D' = m |Phi|^2 sigma_1,          J_r:   D' = m [[0, Phi^2], [conj(Phi)^2, 0]].
```

Proof. For diagonal b, [X, b] for off-diagonal X = [[0, x], [y, 0]] is [[0, x (b_2 - b_1)], [y (b_1 - b_2), 0]].

For J_0: a-hat and b-hat are the conjugates. Then A~*(1) = m [[0, conj(phi_1)], [conj(phi_2), 0]], and the first row of* $A(2)$ is m phi_1 sum conj(a_{j,1}) (conj(b_{j,2}) - conj(b_{j,1})) = m phi_1 conj(phi_1); the second row is m phi_2 conj(phi_2). So the (1,2) entry of D' is m (1 + phi_1 + conj(phi_1) + |phi_1|^2) = m |1 + phi_1|^2, and likewise (2,1).

For J_r: a-hat = diag(conj(a_2), conj(a_1)) and b-hat = diag(conj(b_2), conj(b_1)). Then A~*(1) = sigma_1 conj(*$A(1)$) sigma_1 = m [[0, conj(phi_2)], [conj(phi_1), 0]], and the first row of A_(2) is m phi_1 sum conj(a_{j,2}) (conj(b_{j,1}) - conj(b_{j,2})) = m phi_1 conj(phi_2), and the second is m phi_2 conj(phi_1). This gives the stated factorization. ■

Corollary 19.2 (gauge action and image). Let u = diag(e^{i alpha}, e^{i beta}) and psi = alpha - beta. Under gamma_u, Phi -> e^{i psi} Phi.

1. For J_0: U = 1. The image of the self-adjoint forms is {t sigma_1 : t >= 0}, so the link is real, and t is a gauge-neutral field. The value t = 0 is reached at Phi = 0.
2. For J_r: U = diag(e^{i psi}, e^{-i psi}), and the link m Phi^2 goes to e^{2 i psi} m Phi^2. So U(1) acts with charge 2 in the variable of the link, and the image is all of C, with Phi -> Phi^2 two-to-one away from 0. The image statement is pointwise: a globally prescribed smooth link need not have a global smooth square root.
3. The gauge orbit of the zero form has Phi = e^{i psi}. It gives D' = D for J_0, whose operator orbit is the single point D, and link m e^{2 i psi} for J_r. Without A_(2), the link is m (2 cos psi - 1) for J_0, of modulus m (1 - psi^2 + O(psi^4)), although U = 1; and m (2 e^{i psi} - 1) for J_r, of modulus m (5 - 4 cos psi)^{1/2} = m (1 + psi^2 + O(psi^4)), off the orbit of D.

Proof. gamma_u(0) has phi_1 = u_1 conj(u_2) - 1 = e^{i psi} - 1. The rest is direct. ■

Item 3 shows, in the smallest example, why the quadratic term is needed: it restores gauge covariance.

Comparison with the parent

Proposition 19.3. Consider the chiral interval at M = 0, with m = pi/(2l), and the links of lemmas 13 and 14.

1. *With no real structure*, the finite fluctuation is m (1 + phi_1), all of C, while the fixed compression of a gauge field gives only the line m + i B (Proposition 13.2).
2. *With pointwise reality*, the parent has no scalar fluctuation at all: the algebra C(I) is commutative with J f J^-1 = conj(f), it satisfies order one, and A + J A J^-1 = 0 for a scalar one-form. The truncated data do have a fluctuation, the neutral radial field t. So the finite fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation. This does not establish a dynamical scalar field or its action, and it says nothing about parent fields in other models. Compression is not multiplicative, so there is no canonical identification of finite potentials with parent potentials.
3. *With the reflection real structure*, the parent satisfies order one, since J_r f J_r^-1 = conj(f) composed with the reflection commutes with [D, g]. One finds J_r D_{A_5} J_r^-1 = D_{A_5 o R}, with R the reflection, so that the symmetrized field A_5 + A_5 o R survives, and the fixed compression gives m + i B^s with B^s = 2 B, because the weight sin(pi x/l) is even about l/2. The modulus along a pure-gauge field behaves as follows.

object	modulus of the link along a gauge orbit
covariant projection (lemma 14, Theorem 3)	constant: lambda W_l^* has fixed singular value
finite data with A_(2) (Corollary 19.2)	constant: m
fixed compression (Proposition 13.2), with B the integral of the symmetrized coefficient for J_r	(m^2 + B^2)^{1/2}, growing at second order along paths where B has a linear term
finite data with J_r, linear terms only	m (5 - 4 cos psi)^{1/2}, growing at second order

Proof. Item 1 is lemma 13. For item 2, a represented self-adjoint parent one-form is E h(x) with h = i a and a real, and K (i E a) K^-1 = -i E a, so the real fluctuation cancels. An individual commutator need not change sign, since K [D, x] K^-1 = [D, x] = E: self-adjointness is the step that matters. For item 3, K changes the sign of i A_5, the reflection changes the sign of the derivative and composes A_5 with R, and sigma_1 E sigma_1 = -E for E = [[0, 1], [-1, 0]]. The table collects Theorem 3 of lemma 14, Corollary 19.2 and Proposition 13.2. ■

The table states a parallel and no identification. The variables differ: for constant unsymmetrized A_5 = a, the transported link is m e^{i a l} = m + i (pi a/2) + O(a^2), with Wilson phase a l, against m + i (2 a/pi) in the compression. For the reflection-real fluctuation from the same a, these become m e^{2 i a l} and m + i (4 a/pi). The phase of a link depends on the frame, while its modulus does not. In the parent, the covariance defect is repaired by moving the projection, and in the finite data by the quadratic term.

Reading

Spectral compression can break order one, and it can also break order zero: on the circle with J = K, compression to the zero mode preserves both, while compression to the modes -1, 0, 1 sends e^{ix} to a truncated shift S, with [S, S*] != 0, so that order zero fails. Where a represented algebra, a preserved real structure and order zero hold, as here, the framework of arXiv:1304.7583 applies to the truncated data as they stand. So the first-order failure of Counterexample 08.4 has an escape, and this calculation is a first use of it. Two points were open. Whether the parallel in the table extends to an exact statement for a fluctuated parent and its truncation: this holds on gauge orbits, by Corollary 22.6 of lemma 22, and remains open for the nonunitary part. And whether the neutral radial parameter t is related to the singlet of the grand-symmetry model (Devastato-Lizzi-Martinetti, where sigma comes from a Majorana-operator fluctuation in a larger algebra). This is an analogy only: in arXiv:1304.7583 the quadratic term does not generate new scalar fields, and here too it depends on the same two parameters. An identification would need a map of algebras, representations, fields and gauge actions.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random complex a_j, b_j, with m = 1.3: the three real structures have the stated signs; the formulas of Theorem 19.1 hold to about 10^-15, without assuming self-adjointness; D(gamma_u(0)) = U D U* holds to 10^-16 for both real structures, and fails by 4.5 for the sample when A_(2) is dropped; U D(A) U* = D(gamma_u(A)) holds for a general form; the values of [[D, e], e-hat] are as stated.

Part 20

Sheet-connecting operators and boundary gluing

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/20_connecting_operator.md. Reproduced as delivered, with links made relative and one link to a temporary copy of a source written as plain text. The sections “Numerical check” and “Readings” at the end are by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): Results 1 to 4 and 6 to 11 correct, with one amendment to Result 7 given in the note after its proof; the readings are corrected below.

A rank-one operator can connect the two finite summands with the required real signs and grading, and the Chamseddine–Connes maximal algebra follows from its intertwining algebra. A boundary coupling supplies such an operator in an explicit interval model. However, agreement of scalar functions at a wall generally does not survive compression as the finite order-one restriction: an exact one-flavour counterexample exhibits this failure. The interval construction realizes the classified rank-one block when its matrix algebra, grading and coupled channels are supplied as inputs; it does not derive those inputs from Example 16.9. The three requested obstruction checks hold in the scopes specified below.

1. Theorem — Finite connecting blocks and their admissible algebras

Hypotheses and conventions. Let

$$E = \mathrm{Hom}_{\mathbb{C}}(V, W), \qquad E^* = \mathrm{Hom}_{\mathbb{C}}(W, V),$$

with Hilbert–Schmidt inner products, conjugate-linear in the first argument. Here E^* denotes the second Hom space; operator adjoints also carry a superscript $*$.

Let $c : E \to E^*$ be the antiunitary occurring in the real structure:

$$J(x, y) = (c^{-1}y, cx), \qquad \Gamma = \begin{pmatrix} g & 0 \\ 0 & h \end{pmatrix}, \qquad h = -cgc^{-1}.$$

In Example 16.9, after choosing the grading of W,

$$cx = x^*, \qquad gx = \gamma_W x, \qquad hy = -y\gamma_W.$$

For the more general graded flip of Corollary 16.8, c includes its stated grading sign.

Let $\mathcal{A}^{\mathrm{ev}}$ act diagonally through π_1, π_2. Write $\rho_i(b)$ for its opposite action on the two summands. Assume that the sheet-diagonal D_F already satisfies self-adjointness, oddness, $JD_F = D_F J$, and order one on $\mathcal{A}^{\mathrm{ev}}$, as in Corollary 16.8.

For a complex-linear map $T : E \to E^*$, put

$$S_T = \begin{pmatrix} 0 & T^* \\ T & 0 \end{pmatrix}, \qquad D = D_F + S_T, \qquad \tau(T) = cT^*c.$$

Then D has the required signs and satisfies order one on a unital involutive subalgebra $B \subseteq \mathcal{A}^{\mathrm{ev}}$ precisely when

$$\boxed{T = \tau(T), \qquad hT + Tg = 0,} \tag{1}$$

and

$$\boxed{(T\pi_1(a) - \pi_2(a)T)\rho_1(b) = \rho_2(b)(T\pi_1(a) - \pi_2(a)T) \quad (a, b \in B).} \tag{2}$$

Self-adjointness is automatic from the definition of S_T. In bases related by c, the first equation in (1) says that the matrix of T is complex symmetric.

In the ordinary adjoint-flip convention, for $a = (a_W, a_V)$,

$$\pi_1(a)x = a_W x, \quad \pi_2(a)y = a_V y, \quad \rho_1(a)x = xa_V, \quad \rho_2(a)y = ya_W.$$

Thus (2) is an entirely explicit equation on Hom spaces.

For an operator $R : E \to E^*$, define exactly the commutation algebra used in section 4 of Chamseddine–Connes:

$$\mathcal{A}(R) = \left\{ a \in \mathcal{A}^{\mathrm{ev}} : \begin{array}{l} \pi_2(a)R = R\pi_1(a), \\ \pi_2(a^*)R = R\pi_1(a^*) \end{array} \right\}. \tag{3}$$

For fixed T satisfying (1), its maximal admissible algebras are the inclusion-maximal members of

$$\boxed{\{\mathcal{A}(R) : \ hR + Rg = 0, \quad T = R + \tau(R)\}.} \tag{4}$$

Consequently, maximal dimension for fixed T is obtained by maximizing the dimension of the algebras in (4). The distinction between T and R is necessary.

Proof. Block multiplication gives (1). The lower off-diagonal block of the order-one equation is exactly (2); adjoints give the other block.

To prove (4), suppose (2) holds. Average over the compact unitary group of the finite-dimensional real C^*-algebra B:

$$R_0 = \int_{U(B)} \pi_2(u) T \pi_1(u)^* \, du.$$

Then R_0 intertwines the left actions. Equation (2) implies that

$$T - \pi_2(u) T \pi_1(u)^*$$

intertwines the opposite actions. Therefore $Q_0 = T - R_0$ intertwines the opposite actions.

The involution τ exchanges these two intertwining conditions. Since $\tau(T) = T$,

$$R = \frac{1}{2}(R_0 + \tau(Q_0))$$

intertwines the left actions and satisfies $T = R + \tau(R)$. Averaging preserves oddness because B is even; τ also preserves oddness. Hence $B \subseteq \mathcal{A}(R)$ for an R occurring in (4).

Conversely, on $\mathcal{A}(R)$, the term R intertwines the left actions and $\tau(R)$ intertwines the opposite actions. Their sum satisfies (2). This proves the characterization and the maximality statement.

For an explicit reason why $\mathcal{A}(T)$ alone is insufficient, take ungraded $W = V = \mathbb{C}^2$, $g = 1$, $h = -1$, and

$$B = \{(\mathrm{diag}(\alpha, \beta), \mathrm{diag}(\alpha, \beta))\}.$$

Let $e_{ij} \in E$, $f_{ij} = ce_{ij}$, and define

$$Te_{11} = f_{12}, \qquad Te_{12} = f_{11},$$

with all other images zero. This T is symmetric and odd. The first arrow preserves the right B-character; the second preserves the left character. Each therefore satisfies order one. But the first arrow fails to intertwine the left actions when $\alpha \neq \beta$. ■

2. Theorem — The classified algebra is realized by a rank-one block

Hypotheses. Let

$$W = W_+ \oplus W_-, \qquad \dim_{\mathbb{H}} W_+ = \dim_{\mathbb{H}} W_- = 1, \qquad V = \mathbb{C}^4,$$

and use the adjoint-flip convention of Result 1. Thus

$$\mathcal{A}^{\mathrm{ev}} = \mathbb{H}_+ \oplus \mathbb{H}_- \oplus M_4(\mathbb{C}).$$

Choose unit vectors $w \in W_+$, $v \in V$, and $\kappa \neq 0$. Put

$$e = wv^* \in E, \qquad f = vw^* = ce,$$

and

$$\boxed{T_\kappa x = \kappa \, (w^* x v) \, v w^* = \kappa \, |f\rangle\langle e| x.} \tag{5}$$

Then T_κ has complex rank one, satisfies (1), and has intertwining algebra

$$\boxed{\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m) : \lambda \in \mathbb{C},\ q \in \mathbb{H},\ m \in M_3(\mathbb{C})\}.} \tag{6}$$

Here the first λ acts on the quaternionic line W_+ as

$$\lambda \longmapsto \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix}$$

in a basis starting with w, and the second acts on $\mathbb{C}v \subset V$.

The algebra (6) has real dimension 24. Up to an automorphism of $\mathcal{A}^{\mathrm{ev}}$, it is the unique maximal-dimensional involutive subalgebra admitting a nonzero sheet-connecting Dirac block with order one. The off-diagonal self-adjoint operator S_{T_κ} has complex rank two.

Proof. Since $ge = e$ and $hf = -f$, T_κ is odd. Its matrix has the single symmetric entry κ in c-paired bases, so $\tau(T_\kappa) = T_\kappa$.

The two intertwining equations in (3) require that $\mathbb{C}w$ and $\mathbb{C}v$ be reducing subspaces for the respective actions, with the same scalar action on these lines. A quaternion preserving $\mathbb{C}w$ belongs to the displayed copy of $\mathbb{C}$. On V, the orthogonal complement $v^\perp$ carries an arbitrary $M_3(\mathbb{C})$-action. The other quaternionic summand is unrestricted. This proves (6), and $2 + 4 + 18 = 24$.

We reproduce the dimension argument in Chamseddine–Connes, section 4, whose local source defines (3) at line 875. A nonzero connecting block implies, by Result 1, a nonzero intertwiner R for B. Compress R on either side by projections in the commutants of the full even algebra. Some nonzero compression remains, and its support and range can be placed in irreducible representation spaces

$$K : \mathbb{C}^2 \longrightarrow \mathbb{C}^4.$$

These compressions enlarge the intertwining algebra.

If K has rank two, its image is a reducing two-plane in $\mathbb{C}^4$. The matrix action on that plane is determined by the quaternion action. The remaining two-plane contributes at most $M_2(\mathbb{C})$. Including the unused quaternionic summand gives

$$\dim_{\mathbb{R}} B \leq 4 + 4 + 8 = 16.$$

If K has rank one, its support and range are complex lines. The intertwining algebra is (6), of dimension 24. Unitary changes of these lines, and exchange of the quaternionic summands, give the stated uniqueness.

Finally, T_κ and T_κ^* have orthogonal source and target summands, giving $\operatorname{rank} S_{T_\kappa} = 2$. ■

The connecting condition here is relative to the sheet projections of the original prealgebra: $[S_{T_\kappa}, e_1] \neq 0$. On the selected algebra (6), this particular Majorana block commutes with the algebra. Additional sheet-diagonal Dirac terms are needed for the usual nontrivial finite one-forms and Yukawa data.

3. Lemma — Boundary preservation is an intertwining condition

Hypotheses. Use the domain of Lemma 12:

$$Q_R(0) = 0, \quad U_L(0) = 0, \quad Q_R(l) = XU_R(l), \quad U_L(l) = -X^* Q_L(l). \tag{7}$$

Let multiplication act through smooth matrix functions F_Q, F_U, with the same matrix on both chiral components of each field.

Multiplication preserves this domain precisely when

$$\boxed{F_Q(l)X = XF_U(l), \qquad F_U(l)X^* = X^* F_Q(l).} \tag{8}$$

These equations define a unital involutive algebra. For scalar functions and $X \neq 0$, they reduce to

$$f_Q(l) = f_U(l). \tag{9}$$

For $X = 0$, no equality between the boundary values is required.

Proof. At $x = l$, the traces $Q_L(l)$ and $U_R(l)$ can be chosen independently. Applying multiplication to the first coupled condition in (7) gives

$$F_Q(l)XU_R(l) = XF_U(l)U_R(l)$$

for every $U_R(l)$. This is the first equation in (8). The other condition gives the second equation. Conversely, these equations preserve both boundary conditions; the conditions at zero are automatically preserved.

Taking the adjoint of the second equation gives

$$F_Q(l)^* X = X F_U(l)^*.$$

Thus (8) has precisely the two-equation form (3). Closure under products and adjoints follows directly. ■

For matrix-valued functions, "agreement at the wall" therefore means agreement through X on the coupled boundary channels. It need not mean equality of the entire boundary matrices.

4. Counterexample — Scalar gluing does not imply finite order one

Hypotheses. Let $N_Q = N_U = 1$, both bulk masses vanish, and $X = t > 0$. Set

$$\theta = \arctan t \in (0, \pi/2), \qquad m = \theta/l.$$

Take the full domain-preserving scalar algebra

$$\mathcal{A}_{\text{glue}} = \{(f_Q, f_U) \in C^\infty([0,l])^2 : f_Q(l) = f_U(l)\}. \tag{10}$$

The bulk operator is

$$D(Q_L, Q_R; U_L, U_R) = (Q_R', -Q_L'; U_R', -U_L')$$

on (7). Use physical chirality

$$\Gamma(Q_L, Q_R; U_L, U_R) = (Q_L, -Q_R; U_L, -U_R)$$

and the antiunitary

$$\mathcal{J}(Q_L, Q_R; U_L, U_R) = (\overline{U_R}, -\overline{U_L}; -\overline{Q_R}, \overline{Q_L}). \tag{11}$$

This parent satisfies the signs $(+,+,-)$, order zero and order one. Its lowest symmetric spectral window gives

$$A_{\text{fin}} = \mathbb{C} \oplus \mathbb{C}, \qquad D_{\text{fin}} = m\sigma_1, \qquad J_{\text{fin}} = \sigma_1 K, \qquad \Gamma_{\text{fin}} = \sigma_3, \tag{12}$$

and these finite data fail order one.

Proof. The boundary form proves self-adjointness, as in Lemma 12.1. Equation (11) preserves (7), squares to one, commutes with D, and reverses Γ. The opposite algebra acts by the two scalar functions exchanged. Hence it commutes both with the multiplication algebra and with its differential commutators.

The lowest window is spanned by the orthonormal chiral vectors

$$\begin{aligned} u_+(x) &= l^{-1/2}(\cos mx, 0; -\sin mx, 0), \\ u_-(x) &= l^{-1/2}(0, \sin mx; 0, \cos mx). \end{aligned} \tag{13}$$

Direct differentiation gives $Du_+ = mu_-$, $Du_- = mu_+$. The boundary conditions hold because $\tan(ml) = t$. Solving the same elementary ODE gives the next absolute eigenvalue $(\pi - \theta)/l$, so any symmetric cutoff between m and this value selects exactly (13).

Compression of (10) is

$$Q(f_Q, f_U) = \begin{pmatrix} \dfrac{1}{l}\displaystyle\int_0^l (f_Q \cos^2 mx + f_U \sin^2 mx)\, dx & 0 \\ 0 & \dfrac{1}{l}\displaystyle\int_0^l (f_Q \sin^2 mx + f_U \cos^2 mx)\, dx \end{pmatrix}. \tag{14}$$

Let f_Q be a nonnegative bump supported sufficiently near zero, and let $f_U = 0$. The wall equality holds, while the difference of the diagonal entries is

$$\frac{1}{l}\int_0^l f_Q(x)\cos(2mx)\, dx > 0.$$

Together with the identity, this gives the full diagonal algebra $\mathbb{C}^2$.

On (13), (11) gives $J_{\text{fin}} = \sigma_1 K$. For $p = \text{diag}(1, 0)$,

$$p^\circ = 1 - p, \qquad [[D_{\text{fin}}, p], p^\circ] = -D_{\text{fin}} \neq 0.$$

Thus finite order one fails.

There is also a precise distinction between the original field sheets and the two finite chiral summands. The projection onto the Q-field compresses to

$$\begin{pmatrix} \frac{1}{2} + \frac{\sin 2\theta}{4\theta} & 0 \\ 0 & \frac{1}{2} - \frac{\sin 2\theta}{4\theta} \end{pmatrix}, \tag{15}$$

which is not a projection. The coupled spectral window does not preserve the original field decomposition. ■

This proves failure of statement 2(c) for the full scalar gluing algebra, even when the parent has all the required real signs and order conditions. It also specifies the qualification needed in 2(b): the finite summands require an identification, such as continuation from the uncoupled zero modes.

5. Theorem — An exact interval parent supplies the rank-one connector

Hypotheses. Use the Hom representation, c, g, h of Result 1. Set

$$k = cgc^{-1}, \qquad h = -k.$$

Choose a unit vector $e \in E$ with $ge = e$, let $f = ce$, and put

$$S = |f\rangle\langle e|, \qquad X = tS^* : E^* \to E, \qquad t > 0. \tag{16}$$

Take two massless interval fields, Q valued in E and U valued in E^*, with boundary conditions (7).

As candidate multipliers, take **constant** elements of $\mathcal{A}^{\text{ev}}$, acting by π_1 on Q and π_2 on U. Their domain-preserving algebra is

$$B = \mathcal{A}(S). \tag{17}$$

Define

$$\hat{\Gamma}(Q_L, Q_R; U_L, U_R) = (gQ_L, -gQ_R; kU_L, -kU_R), \tag{18}$$

$$\widehat{J}(Q_L, Q_R; U_L, U_R) = (c^{-1}U_R, -c^{-1}U_L; -cQ_R, cQ_L). \tag{19}$$

Let $n = \dim_{\mathbb{C}} E$, $\theta = \arctan t$, $m = \theta/l$, and choose

$$m < \Lambda_0 < (\pi - \theta)/l.$$

Then the spectral projection

$$P = 1_{[-\Lambda_0, \Lambda_0]}(D_X)$$

has rank $2n$. Under an explicit identification $P\mathcal{H} \simeq E \oplus E^*$, its finite data are

$$\boxed{\left(B,\ E \oplus E^*,\ \begin{pmatrix} 0 & mS^* \\ mS & 0 \end{pmatrix},\ J,\ \begin{pmatrix} g & 0 \\ 0 & -k \end{pmatrix}\right).} \tag{20}$$

They satisfy self-adjointness, oddness, order zero, order one, and the signs $(+, +, -)$. The connecting block is $T = mS$, of complex rank one.

Proof. Result 3, with $X = tS^*$, gives (17). The rank-one condition makes $\mathbb{C}e$ and $\mathbb{C}f$ reducing subspaces for B, with the same action on the two lines.

The coupled channels have exactly the eigenvectors (13), with 1 replaced by e in the Q-components and by f in the U-components. Every orthogonal Q-channel has boundary type (a), and every orthogonal U-channel

has type (b). They contribute $2n-2$ constant chiral zero modes; their next absolute eigenvalue is π/l. The coupled channels contribute the two eigenvalues $\pm m$. This proves the window and dimension assertions.

The identification in (20) sends e and f to these two coupled chiral vectors. It sends $e^{\perp} \subset E$ to constant Q_L-profiles and $f^{\perp} \subset E^*$ to constant U_R-profiles. Since B acts identically on the coupled lines, this identification intertwines its representations exactly. Direct application of (18) and (19) gives the displayed finite grading and real structure.

On the parent, (19) squares to one and commutes with the real differential expression. The relations $kf = f$ and $ge = e$ ensure grading preservation of the boundary conditions. The identity

$$cX = X^* c^{-1}$$

ensures their preservation by $\widehat{J}$. These establish the real signs on the operator domain.

The opposite representation is the original right Hom action, so order zero holds. Every admissible constant multiplier commutes with D_X, including preservation of its domain. Order one therefore holds on the parent and on (20). ■

The wall coupling can also be converted explicitly into a bounded bulk term. On the coupled field channels, let

$$F = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad E_{\text{spin}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad U(x) = \exp(mxF).$$

Apply the same field rotation to both spin components. It maps the uncoupled boundary domain to the coupled domain and gives

$$U^* D_X U = D_0 + mF \otimes E_{\text{spin}}. \tag{21}$$

Compression of the added bounded term to the uncoupled zero modes is exactly $m(S+S^*)$. Thus the parent supplies the connector through an explicit change of domain and an explicit unitary transport.

Two specializations specify the extent of this construction.

- **A commutative example.** Let $E = E^* = \mathbb{C}$, $g = 1$, $c = K$, and start with constant pairs $\mathbb{C}^2$. Boundary preservation selects their common scalar:

  $$B = \{(\lambda, \lambda)\} \simeq \mathbb{C}.$$

  The finite triple is

  $$(\mathbb{C}, \mathbb{C}^2, m\sigma_1, \sigma_1 K, \sigma_3).$$

  This is the simplest successful instance of 2(a)–(c), with the candidate observable algebra restricted to constant functions on each sheet.

- **The classified block.** Take $E = \mathrm{Hom}(\mathbb{C}^4, \mathbb{H}^2)$, the chosen quaternionic grading, and $e = wv^*$ from Result 2. Equations (17) and (20) give

  $$B = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \qquad \dim_{\mathbb{C}} H_{\text{fin}} = 32,$$

  and the rank-one connecting block (5) with $\kappa = m$. This parent contains the Hom fibres and constant matrix algebra as inputs. It is not a construction from scalar functions on the twistor parent.

An exact tower-decoupling family is obtained by fixing $m > 0$ and setting

$$t(l) = \tan(ml), \qquad 0 < ml < \pi/2.$$

The finite mass remains exactly m, while the next absolute eigenvalue is $\pi/l - m \to \infty$.

The usual zero-dimensional orientability axiom fails in these examples: the linked vectors e, f have identical left and opposite characters, whereas their grading eigenvalues are opposite. Every represented zero-cycle acts identically on these two lines. The claims above concern real even spectral triples with the stated order conditions and signs.

6. Proposition — Point evaluation bounds rank by the boundary-channel dimension

Hypotheses. Let K_i be finite-dimensional windows of smooth sections, let

$$\mathrm{ev}_i : K_i \to F_i$$

be evaluation at the proposed wall, and suppose a finite coupling is defined by

$$T = \mathrm{ev}_2^* \; X \; \mathrm{ev}_1 \,. \tag{22}$$

Then

$$\operatorname{rank} T \le \min\{\operatorname{rank} X, \dim F_1, \dim F_2\}. \tag{23}$$

A single scalar evaluation channel gives rank at most one. A point with several fibre components can give larger rank.

Proof. Equation (23) follows by factoring (22) through the range of X.

For a counterexample to an unrestricted rank-one claim, take Lemma 12 with

$$N_Q = N_U = N, \qquad M_Q = M_U = 0, \qquad X = tI_N.$$

It is the orthogonal sum of N copies of Result 4. Its lowest chiral block is

$$T = \frac{\arctan t}{l} I_N,$$

of rank N, although the coupling occurs at the single point $x = l$. ■

For an exact boundary-domain perturbation, formula (22) is an additional hypothesis. The operators D_X and D_0 have different domains, so their difference is not automatically a point-evaluation perturbation. The exact interval mass $\arctan(t)/l$ also distinguishes the finite operator from its small-t approximation t/l. An extended wall carries a space of boundary sections and supplies no rank-one conclusion by itself.

7. Proposition — Point gluing does not select the algebra in the unchanged twistor window

Hypotheses. Let $Q_i(f) = P_i M_f P_i$, where P_i project onto finite-dimensional spaces of smooth sections on compact manifolds. Let Z_i be closed subsets with empty interior. Impose any scalar gluing condition which allows independent functions vanishing near Z_1 and Z_2, including equality at two marked points.

Then the image of this glued algebra under the **unchanged separate compressions** $Q_1 \oplus Q_2$ equals the image of the unrestricted pair algebra.

Proof. We first show

$$Q_i(C_c^\infty(M_i \setminus Z_i)) = Q_i(C^\infty(M_i)). \tag{24}$$

If the left image were a proper subspace of the right image, finite-dimensional linear algebra would give a linear functional separating them. Applied to $Q_i(f)$, that functional has the form

$$f \longmapsto \int_{M_i} f(x)\, r(x)\, dx$$

for a smooth density r, because matrix entries of $Q_i(f)$ are integrals against products of smooth sections.

Vanishing for every test function supported outside Z_i forces $r = 0$ there. Since that complement is dense and r is continuous, $r = 0$ everywhere, a contradiction. The argument also applies over the real numbers.

By (24), arbitrary compressed values on either sheet can be obtained using functions vanishing near its wall. Such pairs satisfy the gluing condition independently, proving the assertion. ■

In Example 16.9, both analytic section spaces come from $\mathbb{CP}^3$; one observable algebra is restricted to pullbacks from S^4. The corresponding matrix-entry densities on S^4 are smooth after integration along the twistor fibres, so the same proof applies.

Neither closed space comes with the boundary required by Lemma 12. One may choose a base point $p \in S^4$, a point $q \in \mathbb{CP}^3$, or a twistor fibre, and impose scalar agreement there. In the original level-one windows, the generated real algebra remains

$$M_2(\mathbb{H}) \oplus M_4(\mathbb{C}).$$

Thus this operation derives no further algebra reduction.

A marked base point does provide a quaternionic line through

$$\Pi_p = \frac{1}{2}(1 + p_a \Gamma_a).$$

Choosing a complex line inside that quaternionic line, a complex line in the other four-space, and the rank-one map between the corresponding Hom vectors gives the data of Result 2. The grading $2\Pi_p - 1$ and the rule selecting this connector are then additional finite data. The stabilizer calculation derives $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ from those choices.

To obtain an analytic parent for Example 16.9, further work must specify an interface or interval collar, its boundary trace spaces, self-adjoint coupling conditions, a compatible real structure and grading, and the spectral window of the coupled operator. Result 7 concerns the unchanged windows. A new coupled window, boundary degrees of freedom, or a different observable representation lies outside its hypotheses.

Referee amendment. For a general subset upstairs in the twistor space, add the hypothesis that its image in S^4 has empty interior, so that independent base test functions exist.

8. Proposition — Scope of the Hořava–Witten comparison

Hypotheses. Compare chiralities using one common ten-dimensional chirality convention, and retain the unbroken boundary gauge group $E_8 \times E_8$.

The same-chirality interval analogue is type **(a) at both ends**, or type **(b) at both ends** after reversing the chirality convention. To model two separately localized multiplets, take two fields of that same type, with opposite signs of their localization masses. Lemma 12 at $X = 0$ contains one type-(a) and one type-(b) field, so its displayed pair has opposite chiralities.

Proof and scope. Hořava–Witten impose $\Gamma_{11}\chi = \chi$ on the boundary gaugino; the boundary vector multiplets are intrinsic boundary fields. Their chirality projector is stated in section 2, immediately before the boundary Yang–Mills Lagrangian. Labels based on outward normals change sign between the endpoints; this does not change the comparison in a fixed ten-dimensional convention.

Type (a) in Lemma 09 retains the L-zero mode at both ends, while type (b) retains R. Different mass signs localize two such fields near opposite ends. Type (c) instead produces opposite-chirality wall profiles.

On a space containing only the two same-chirality modes, an operator odd for that chirality grading has zero matrix elements between them. Moreover,

$$\mathrm{Hom}_{E_8 \times E_8}((248, 1), (1, 248)) = 0,$$

because the first factor acts trivially on the target and has no invariant vector in its adjoint representation. Thus these boundary multiplets do not supply a gauge-invariant bilinear connector of the interval mass-block type. ■

Bulk propagation can define effective boundary-to-boundary interactions. Identifying one with the finite T requires a specified field, representation and grading, potentially after compactification or symmetry breaking. The Hořava–Witten comparison alone supplies no such identification.

9. No-go theorem — Corollary 11.2 under its matching-scale hypothesis

Verdict and hypotheses. The corrected Corollary 11.2 is correct. Assume a fixed finite set of Dirac mass parameters, N_g generations, and the matching-scale relation

$$8M_W^2 = \sum_{g=1}^{N_g}(m_{\nu,g}^2 + m_{e,g}^2 + 3m_{u,g}^2 + 3m_{d,g}^2). \tag{25}$$

For $N_g \neq 3$, the coefficient 8 is an explicitly assumed normalization; it does not follow automatically by replicating the three-generation gauge trace.

Then $M_W \to \infty$ is incompatible with keeping every Dirac mass bounded. More precisely,

$$\frac{m_{\max}}{2\sqrt{2}} \leq M_W \leq \sqrt{N_g}\, m_{\max}. \tag{26}$$

Proof. The weighted sum in (25) lies between $m_{\max}^2$ and $8N_g m_{\max}^2$, proving (26). Thus a fixed nonzero gauge coupling and $v \to \infty$, with all $y_f v$ fixed, contradict (25).

The conclusion concerns $m_{\max}$. One fixed species need only be unbounded along a subsequence. For example, with $M_W = t$, put

$$\alpha(t) = \frac{\pi}{4}(1 + \sin t), \quad m_e = 2\sqrt{2}\, t \cos\alpha(t), \quad m_\nu = 2\sqrt{2}\, t \sin\alpha(t).$$

Their squared sum is $8t^2$, while each mass vanishes on an unbounded sequence. ■

Scope and escapes. Equation (25) belongs to the stated normalization of the truncated asymptotic spectral action at matching. It concerns Dirac Yukawa masses, including the neutrino Dirac parameters, rather than physical seesaw eigenmasses. Changing the action or normalization, controlling the full finite-cutoff action, varying the particle content, or allowing Dirac masses to grow changes the problem. Extrapolation to a heavy-mass limit at fixed cutoff requires a separate calculation; the theorem directs attention to that calculation.

10. Theorem — Proposition 16.1 holds for inherited pointwise reality

Verdict and hypotheses. The corrected Proposition 16.1 is correct. Let

$$J_0 a J_0^{-1} = a^*, \qquad [P, J_0] = 0, \qquad Q(a) = PaP|_{P\mathcal{H}},$$

and let $J_F = J_0|_{P\mathcal{H}}$. Let A_F be the involutive algebra generated by the compressed functions. Then order zero holds for this inherited structure exactly when A_F is commutative.

Proof.

$$J_F Q(b)^* J_F^{-1} = P J_0 b^* J_0^{-1} P = Q(b).$$

Order zero therefore requires

$$[Q(a), Q(b)] = 0$$

for every pair of generators. Their generated algebra is then commutative.

Conversely, the opposite operation reverses products and fixes each compressed generator. On a commutative generated algebra its image lies in the same algebra, proving order zero. ■

For a product with a fixed finite factor, the unit of that factor gives the same necessity. Sufficiency also requires that the finite factor satisfy order zero and that the product real structure be compatible. The commutativity conclusion concerns the compressed function algebra; the separate finite factor may be non-commutative.

Scope and escapes. The theorem concerns compression followed by algebra generation with this inherited pointwise reality and a J_0-invariant window. The coarse correspondence supplies a separate opposite action and escapes the hypothesis, as demonstrated positively by Theorem 16.2. Other representations or real structures require their own order-zero calculation.

11. Proposition — The stated inherited grading cannot equal Γ_5

Verdict and hypotheses. The grading obstruction in Example 16.9 is correct under the following simultaneous requirements:

1. Every parent observable under consideration is even: $[\gamma, a] = 0$.
2. The finite grading is inherited as $g = P\gamma P|_{P\mathcal{H}}$.
3. This compression is an actual grading: $g^2 = 1$.
4. The retained observables include $Q(x_a) = \Gamma_a/5$.

Under these hypotheses, $g = \Gamma_5$ is impossible.

Proof. Even without initially assuming $[P, \gamma] = 0$, the grading identity implies

$$0 = P - (P\gamma P)^2 = P\gamma(1-P)\gamma P = \big((1-P)\gamma P\big)^*\big((1-P)\gamma P\big).$$

Hence $(1-P)\gamma P = 0$, and self-adjointness gives $[P, \gamma] = 0$. Therefore

$$[g, Q(a)] = P[\gamma, a]P = 0$$

for every retained even observable.

But

$$[\Gamma_5, Q(x_1)] = \frac{1}{5}[\Gamma_5, \Gamma_1] = \frac{2}{5}\Gamma_5\Gamma_1 \neq 0.$$

Thus the proposed grading and all the stated coordinate compressions cannot simultaneously be inherited in this manner. ■

Scope and escapes. The conclusion does not exclude Γ_5 as additional finite data or as a grading of a prealgebra. Possible changes include compressing a graded prealgebra with odd observables, restricting to the algebra commuting with the proposed grading, or constructing an additional grading. Changing the window alone does not remove the contradiction while retaining all four hypotheses. In the specified Dolbeault kernel, the inherited grading is already scalar.

The three checks above agree with the corrected repository statements. The recorded author is Claude Opus 5.5, and the existing independent Terra checks appear in the Lemma 11 report, the Proposition 16.1 report, and the Example 16.9 report. The present response adds a sequential Codex check; no subagents or additional model consultations were used.

Consequence for T0b and T11

For T0b, an exact interval parent supplies a rank-one connector with KO-6 signs, and a parent with chosen Hom fibres and constant matrix multipliers realizes the classified connecting block and its algebra. Deriving that algebra, grading and connector from the commutative twistor parent remains open: scalar wall agreement in the original window does not perform the selection. For T11, same-chirality boundary multiplets require the type-(a) or type-(b) comparison, and the unbroken Hořava–Witten boundary representations supply no identified finite mass connector.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. On W = V = C^4 with random elements:

- T_kappa x = kappa K x K with K = v w*, the rank-one map (5): J D = D J, J Gamma = -Gamma J, D Gamma = -Gamma D; complex rank 1; real dimension of A(T_kappa) inside the 40-dimensional H + H + M_4(C) equal to 24, by linear algebra.
- K of rank two on W_+: dimension 16, as in the proof of Result 2, and x -> K x K has rank 4.
- On C + H + M_3(C): order zero, and order one with defect about 10^-15, and [D, a] = 0 with defect about 10^-15. For a random element off this algebra the order-one defect is about 40.

- The example at the end of Result 1: order-one defect 0, and a nonzero defect of left intertwining.
- Result 4: the values in (15), with a quadrature. By hand: the eigenvectors (13), their boundary conditions, the action of (11), and the value of the commutator.
- Equation (21): by hand, including the compression to m (S + S*).
- Pi_p is a projection of rank 2 and 2 Pi_p - 1 = p_a Gamma_a; the bound (26) on a random sample.

Against the source: (3) is equation (comrel) of arXiv:0706.3688, read in the LaTeX source. The proof there also takes an intertwiner distinct from the Dirac block, in line with the distinction between T and R in Result 1. The other statements are checked by the referee.

Readings

By Claude Opus 5.5, as proposals, corrected after the referee report.

1. The connector is the Majorana block. In the Standard Model dictionary (Chamseddine-Connes-Marcolli 2007, section 2, quoted from memory), W_+ holds (nu_R, e_R) and V = C^4 holds the lepton line and the three colours, so that with w along nu_R and v along the lepton line, the vector e = w v* is nu_R, and the off-diagonal block of D_F acts on nu_R alone. After this identification, and an overall grading convention if needed, T = m S in Result 5 has the form of the one-generation Majorana block (equation (2.34) there, as the referee checked), with m = arctan(t)/l. In this window m < pi/(2l), so the first level of the tower bounds it, and for t of order one m is of the order of that level. As a pattern, this agrees with the placement of the Majorana mass near the cutoff in that model. Three identical copies give three equal masses, and a general three-generation Majorana matrix needs additional flavour data.

2. The connector as a marked pair of points and a coupling. Since T_kappa x = kappa K x K with K = v w*, the data of the connector are a complex line w in W = C^4, a complex line v in V = C^4, that is, a point of CP^3 on each sheet, and the coupling kappa. In Example 16.9, the point w lies over a point p of S^4 in the twistor fibration, and Pi_p = (1 + p_a Gamma_a)/2 projects onto its quaternionic line. Since Q(x_a) = Gamma_a/5, the grading is

```
2 Pi_p - 1 = p_a Gamma_a = 5 Q(p . x),
```

the compression of the height function at p. So one marked point fixes the quaternionic line and the complex line w, and a second one fixes v. Sp(2) acts transitively on the unit vectors of H^2, and U(4) on those of C^4, so at fixed nonzero |kappa| all line choices are conjugate, in agreement with the uniqueness in Result 2. The modulus |kappa| remains, rephasing w and v changes K x K by a phase, and other finite data can introduce relative moduli. The inputs are then: mark a point on each sheet, with a coupling and a prescription for evaluation and normalization. This has the form (22) with one channel per sheet.

The height function 5 Q(p . x) fails to commute with the other coordinate compressions, so it can serve as additional finite grading data only after restricting to its even algebra. This evades the inherited-grading hypothesis of Result 11, and it does not derive the grading from the even parent. Result 7 shows that scalar agreement at the points does not do this in the unchanged windows, so a coupled window with the data listed after Result 7 is needed. Whether a parent with an interface at those points yields this connector is open.

3. The wall coupling as a Wilson line. Equation (21) says that the wall coupling X = tan(m l) is unitarily equivalent to the uncoupled boundary conditions with a constant matrix connection m F, of relative endpoint rotation angle theta = m l, and the mass is that angle over the length, k = theta/l. On an interval, open-path transport depends on endpoint frames, and the physical quantity is the transport together with the boundary conditions. The comparison with gauge-Higgs unification (Y. Hosotani, Phys. Lett. B 126 (1983) 309; Hosotani, Noda and Takenaga, arXiv:hep-ph/0403106) is an analogy at this stage: the calculation supplies a fixed matrix connection, with no Higgs fluctuation or effective potential. It bears on the question of mass as internal momentum in the steer for the next paper, and compares with the link of lemma 19 and the covariant projection of lemma 14.

4. Next calculation. The connector commutes with B, so it gives no finite one-forms and no Higgs field. The next test is one interval parent carrying both the Dirac Yukawa block of lemma 12, odd for g and

diagonal in the sheets, and this connector, for instance at opposite walls: whether order one survives on B, and what the joint form of (21) is.

Part 21

The separate role of the top: removal identities and counting

Prompted by the observation that the top has a separate role, which matters more than the loss of an electroweak partner for b_L. Proposition 21.1 and the counts below are exact calculations by Claude Opus 5.5, in Python with exact fractions. The readings in the last section are proposals, and nothing there is derived. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): one error, in Proposition 21.2, and several amendments, incorporated below.

Outcome. The spectral action detects the top: in the exact quadratic action of lemma 18, among single charged species only an up-type quark can be removed while keeping both the neutral-current value cos^2 theta_NC = 5/8 and the ratio 8/3 of the photon and gluon coefficients. Removing it leaves 84 of the 96 states of the finite Hilbert space. Those 84 are the seed of an SO(30), inside SO(32), whose adjoint contains the charged scalar counts of three generations, and 84 is also the number of on-shell components of the three-form of eleven-dimensional supergravity.

Removal identities

For a Dirac species with charges q_L, q_R in the coordinates (T_3, Y), equation (33) of lemma 18 gives a contribution to the neutral kinetic matrix K with weight e^{-x}. The off-diagonal charge is (q_L - q_R)/2 = (T_3/2)(1, -1), which is orthogonal to the photon direction (1, 1). So the combinations K_33 + K_3Y and K_YY + K_3Y depend on x only through the factor e^{-x} of each massive species, and cos^2 theta_NC = (K_YY + K_3Y)/(K_33 + 2 K_3Y + K_YY) is determined by the sums of T_3 Q and Y Q.

Proposition 21.1. Per species, with colour multiplicity, on the particle trace for the first two columns and with charge conjugates and both chiralities for the last two:

species	sum T_3 Q	sum Y Q	Q^2 coefficient	colour coefficient
up-type quark	1	5/3	16/3	2
down-type quark	1/2	1/6	4/3	2
charged lepton	1/2	3/2	4	0
neutrino	0	0	0	0
three generations	6	10	32	12

Hence the ratios of the whole, 5/3 and 8/3, are those of an up-type quark alone. As one species is removed, that is, as its mass tends to infinity at fixed cutoff in the setting of lemma 18:

removed	cos^2 theta_NC	photon/gluon coefficient
up-type quark	5/8	8/3
down-type quark	59/92	46/15
charged lepton	17/28	7/3

So among single charged species only an up-type quark preserves both values. For an up-type quark they hold at every x, as in Proposition 6 of lemma 18. Within one generation, the down-type quark and charged lepton together contribute exactly what the up-type quark contributes: (1, 5/3; 16/3, 2).

Proof. Arithmetic with the Standard Model charges, Y = Q - T_3. The last statement is direct addition. ■

The scope is that of lemma 18: classical bosonic trace, with these definitions of the coefficients and of the angle, three generations, and the other species with the mass assignments of that calculation. Other definitions of the weak angle do not stay at 5/8 (Proposition 6 there). At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished, so its angle is read as the limit from finite mass, as lemma 18 states.

The state count in the action

Proposition 21.2. With three generations and nu_R, dim H_F = 96 (Chamseddine-Connes-Marcolli). With only an up-type Dirac mass m and no Majorana mass, the constant-background density of lemma 18 has finite trace Tr exp(-V_$0^{2/\mathrm{Lambda}}$2) = 84 + 12 e^{-x} for the heat cutoff. So the coefficient of the vacuum density moves from 96 to 84 along the radial direction, with no finite nonzero stationary point on this free radial ray (No-go theorem 9 and Proposition 11 of lemma 18). This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints and paths with an additional fixed noncommuting mass lie outside it. At m = 0 the derivative is zero.

Proof. The top block has 3 colours, 2 chiralities and charge conjugates, so 12 states, each with V_0^2 = m^2. ■

This also gives the top-only family a place in the radial problem, as the endpoint of a descending direction. A radial direction giving other species masses descends further, so this is the endpoint for that direction only.

SO(30)

Let the five quarks other than the top, in three colours, with their conjugates, form the vector 30 = (5, 3) + (5-bar, 3-bar) of SU(5) x SU(3) inside SO(30). The adjoint, of dimension 435, is

```
Lambda^2(30) = (24, 8) + (24, 1) + (1, 8) + (1, 1) + (10, 6) + (15, 3-bar) + conjugates of the last
↪  two.
```

With Q = 2/3 for u, c and Q = -1/3 for d, s, b, that is, Q = diag(3/5, 3/5, -2/5, -2/5, -2/5) + (1/15) I_5, which includes the U(1) commuting with SU(5), the charges are:

representation	charge: number
(15, 3-bar)	-2/3: 6; +1/3: 6; +4/3: 3
(10, 6)	-2/3: 3; +1/3: 6; +4/3: 1
(24, 1) + (1, 1)	+1: 6; -1: 6; 0: 13

Three generations of scalar partners in a supersymmetric Standard Model with nu_R need, for antisquarks in the 3-bar, 6 of charge -2/3 and 6 of charge +1/3, and for sleptons 6 of charge +1, 6 of charge -1 and 12 neutral real states. So (15, 3-bar) and (24, 1) + (1, 1) contain exactly the charged counts. What is left over in them is three triplets of charge +4/3 and one neutral state. The counting of 15 diquarks and 25 mesons, with the charge +4/3 states, and the remark that the top is unable to form them, go back to an early preprint (hep-ph/0512065), which finds neither SO(30) nor a role for SO(32). What SO(30) adds is one group whose adjoint contains both families. It also sits in SO(32), with 496 -> (435, 1) + (1, 1) + (30, 2) under SO(30) x SO(2), which hints at a role for SO(32); we have not explored this. These are representation and charge counts: identifying the entries with physical scalar partners needs a field construction, spin and chirality assignments, interactions and dynamics.

The two families of 84

Proposition 21.3. Count four states per Dirac fermion and colour, and take Dirac neutrinos. Then

```
96 = 84 + 12:   five quarks and all leptons (60 + 12 + 12), and the top (12);
96 = 84 + 12:   all electrically charged fermions (72 + 12), and the neutrinos (12).
```

The two families of 84 overlap in 72 states, and each is the other with the top and the neutrinos exchanged. Without nu_R, the first family has 78 states and the second stays at 84. In eleven dimensions, the three-form C_3 has C(9, 3) = 84 on-shell components, and its dual C_6 has C(9, 6) = 84. The 84 in H_F is a complex finite-Hilbert-space dimension, and the three-form count is a real bosonic polarization count, so an identification needs a map of representations and degrees of freedom.

Readings, as proposals

These are proposals to test, with no derivation behind them.

1. **Seiberg duality.** For SU(3) with five flavours, N_f = 5 lies in the conformal window 3 N_c/2 < N_f < 3 N_c, and the magnetic dual is SU(N_f - N_c) = SU(2) with five flavours and a 5 x 5 gauge-singlet meson chiral superfield coupled by W = M q q-tilde (N. Seiberg, Nucl. Phys. B 435 (1995) 129, arXiv:hep-th/9411149, checked by the referee). The removal of the top is what gives N_f = 5, and SU(2) is the magnetic group there. To become a statement this needs an N = 1 supersymmetric theory with the required massless matter and dynamics, for example an emergent one, with an operator and anomaly map. Five-flavour ordinary QCD alone does not supply the duality, and the magnetic SU(2) needs a separate identification with any electroweak group.
2. **Branes.** The first family of 84 could relate to C_3 or to an M2 or M5 brane, and the second family to the dual object. The exchange of the top and the neutrinos would then be the content of the duality on the Standard Model side. The counts match, and equality of the number is insufficient: a representation and dynamics must relate the finite fermionic states to the bosonic form or brane degrees of freedom. The pairing of item 1 is one candidate. The first family also needs Dirac neutrinos for its 84, and the second does not.
3. **Relation to the odd labels.** The removal of the top leaves the five quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R. So the loss of the electroweak partner of b_L has no role at the label 9, where SU(2) is absent. A statement here needs the required involution on this finite sector, including its action on D, J, the algebra and its opposite; the vectorlike charges alone do not establish it, and the neutral states still need treatment.

Numerical check

Python with exact fractions, run on 3 Nov 2026 in a temporary directory, not kept: all tables above, and the dimension 435 = 224 + 1 + 105 + 105 with 105 = 60 + 45 and 224 = 192 + 24 + 8.

Part 22

The finite Dirac operator as a boundary condition on an interval

For STRATEGY T0, T0b, T10 and T13, following lemma 12 and lemma 20. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and one gap as first written, and amendments; all are incorporated below.

Outcome. Let (H_F, D_F, J_F, Gamma_F) be finite data of signs (+, +, -), and let l ||D_F|| < pi/2. Then D_F, with J_F and Gamma_F, is exactly the lowest spectral window of the pure derivative on an interval of length l, with no bulk mass and no Yukawa term, on the domain fixed by the boundary coupling tan(l D_F). So all the eigenvalues of D_F, Yukawa and Majorana blocks alike, are quantized internal momenta k with tan(k l) equal to a singular value of the coupling. The input is the finite triple: the content is that a consistent parent exists, with these properties:

- Preservation of the domain forces the observables at the wall into C_F, the commutant of D_F in A_F, while A_F acts freely in the bulk.
- The parent satisfies order one for any finite data with order zero.
- The compression is the exact average Q_l(a) = (2l)^-1 int_{-l}^{l} e^{ixD_F} a e^{-ixD_F} dx, so it tends to the finite representation as l -> 0, with defects of order l^2.

- The scalar inner fluctuations of the parent are pure gauge under its allowed endpoint gauge group. On the moving window, they act by the gauge orbit of D_F through the transport along the interval. This reproduces the quadratic term of Chamseddine-Connes-van Suijlekom on the orbit, and it leaves the singular values fixed, so the radial mode of the Higgs field is absent from this parent.

With the family X(l) = tan(l M) and l -> 0, the couplings of the light fermions vanish as m l at fixed mass, the tower decouples, and the broken gauge directions go to pi/(2l). The parent algebra is matrix-valued, so T0 for a commutative parent remains open.

Setting

Let H_F = H_+ (+) H_- be finite-dimensional, Gamma_F = diag(1, -1), let c : H_+ -> H_- be antiunitary, and put J_F(a, b) = (c^-1 b, c a), so that J_F^2 = 1 and J_F Gamma_F = -Gamma_F J_F. Let D_F = [[0, M*], [M, 0]] with M : H_+ -> H_-. Then J_F D_F = D_F J_F is equivalent to M = c M* c. By the Takagi factorization, in bases related by c, we may write M = V |M| with a unitary V satisfying V = c V* c.

Let l > 0 with l ||D_F|| < pi/2, and define X : H_- -> H_+ by

```
tan(l D_F) = [[0, X], [X\*, 0]],   that is,   X = tan(l |M|) V\*.
```

Since J_F commutes with real functions of D_F, c X = X* c^-1. Conversely, every X with c X = X* c^-1 arises from exactly one such D_F, namely D_F = l^-1 arctan of the block operator. In the limit, X = l M* + O(l^3).

The parent. On H_l = L^2([0, l]; C^2) (x) (H_+ (+) H_-), write psi = (Q_L, Q_R; U_L, U_R), with Q valued in H_+ and U in H_-. Let (D psi)_L = psi_R' and (D psi)_R = -psi_L' on both fields, the operator of lemma 12 with no bulk mass, on the domain

```
Q_R(0) = 0,   U_L(0) = 0,   Q_R(l) = X U_R(l),   U_L(l) = -X\* Q_L(l).
```

Call it D_X. Put Gamma^ psi = (Q_L, -Q_R; U_L, -U_R) and J^ psi = (c^-1 U_R, -c^-1 U_L; -c Q_R, c Q_L). Let (E psi)_L = psi_R and (E psi)_R = -psi_L, so that D_X = E d/dx on its domain.

In the Standard Model form of the classification, with E = Hom(V, W) and W = W_+ (+) W_-, one takes H_+ = E_+ (+) E_-* and H_- = E_- (+) E_+*, with c the adjoint map. At X = 0, the Q-fields have boundary type (a) and the U-fields type (b), in the terms of lemma 09; for X != 0 the modes continue from those zero modes, which generally lift. A single matrix X encodes the Yukawa blocks and the Majorana block together. Its entries need not keep the original block pattern, since the tangent brings in higher powers of D_F: for M = [[a, b], [b, 0]] real symmetric, X_22 = (l^3/3) a b^2 + O(l^5).

The window

Theorem 22.1. D_X is self-adjoint with compact resolvent, and J^^2 = 1, J^ D_X = D_X J^, J^ Gamma^ = -Gamma^ J^, Gamma^ D_X = -D_X Gamma^. Let R(x) = exp(x Gamma_F D_F), and define W : H_F -> H_l so that for phi = (a, b), the L-components (Q_L, U_L) of W phi at x are l^{-1/2} R(x)(a, 0), and the R-components (Q_R, U_R) are l^{-1/2} R(x)(0, b). Then W is an isometry and

```
D_X W = W D_F,    J^ W = W J_F,    Gamma^ W = W Gamma_F.
```

With sigma_k the singular values of M and theta_k = l sigma_k, the signed spectrum of D_X is (n pi +- theta_k)/l, n in Z, with multiplicities added when values coincide; a zero singular value gives a zero mode of multiplicity two. So the spectrum of |D_X| consists of the sigma_k and of (n pi +- theta_k)/l for n >= 1, and ran W is the spectral subspace of |D_X| <= ||D_F||, and the next level is at least (pi - l ||D_F||)/l.

Proof. Self-adjointness is Lemma 12.1. Since Gamma_F D_F is anti-Hermitian, R(x) is unitary, and with C(x) = cos(x |M|), C'(x) = cos(x |M*|), S(x) = V sin(x |M|),

```
R(x)(a, 0) = (C(x) a, -S(x) a),    R(x)(0, b) = (S(x)\* b, C'(x) b).
```

Isometry follows from the unitarity of R(x) and the orthogonality of the L- and R-components. The conditions at 0 hold since S(0) = 0. At l, using V* cos(l |M*|) = cos(l |M|) V*, we get X C'(l) = S(l)* and X* C(l) = S(l), which are the conditions at the wall. For the derivative, the L-part of D W phi is d/dx R(x)(0, b) = R(x) Gamma_F D_F (0, b) = R(x)(M* b, 0), and the R-part is -d/dx R(x)(a, 0) = R(x)(0, M a); together these are W D_F phi. The relations for Gamma^ are direct. For J^, use c |M| c^-1 = |M*| and V = c V* c, which give c^-1 C' = C c^-1 and c S(x)* = S(x) c^-1. The signs of the parent follow in the same way, using c X = X* c^-1 for the domain. For the spectrum, constant unitaries on Q and on U commute with E d/dx and bring X to diag(tan theta_k), which reduces to Example 12.5 in each channel. ■

Corollary 22.2 (conjugated form). D_X = R (D_0 + E Gamma_F D_F) R*, with R acting pointwise on the pair (Q, U) and D_0 the uncoupled operator, X = 0. So after this unitary change of domain, D_F is a constant matrix connection, odd for the grading, with relative endpoint rotation R(l). This is equation (21) of lemma 20, for general D_F.

Proof. D(R phi) = R(D phi + E R* R' phi), and R* R' = Gamma_F D_F. For phi in the uncoupled domain, the computation at the wall in the proof above shows that R phi satisfies the coupled conditions. ■

So in this family a fermion mass is a momentum: D_X has no mass term, and the light eigenvalue k = theta/l is fixed by tan(k l) = tan(theta), a singular value of X. In the finite description, the condition relating k to l is lost and the value appears as free data in D_F.

The parent as a real spectral triple

Let (A_F, H_F, D_F, J_F, Gamma_F) be finite data as above, with an even representation pi = pi_+ (+) pi_- of a unital finite-dimensional real or complex *-algebra A_F, faithful, unital and satisfying order zero, with pi°(b) = J_F pi(b)* J_F^-1. Order one is not assumed. Let

```
C_F = {a in A_F : [pi(a), D_F] = 0},    A_l = {f in C^infinity([0, l], A_F) : f(l) in C_F},
```

with f acting pointwise by pi_+(f(x)) on Q and pi_-(f(x)) on U.

Theorem 22.3.

1. A function f in C^infinity([0, l], A_F) preserves Dom D_X if and only if f(l) lies in C_F.
2. (A_l, H_l, D_X, J^, Gamma^) is a real even spectral triple, with [D_X, pi(f)] = E pi(f') and J^ pi(g)* J^^-1 = pi°(g) pointwise.
3. It satisfies order zero and order one, with signs (+, +, -).

So A_F acts freely in the bulk, and the boundary condition restricts the observables at the wall to the commutant of D_F. In the Standard Model data with nonzero Yukawa blocks, this is the algebra {(lambda, lambda, m)} of the photon and gluons; with vanishing Yukawa blocks the commutant is larger. Order one holds in this parent whatever the finite data, for its pointwise algebra: the finite condition appears only on the truncation, and arbitrary compressions need not preserve either order condition.

Proof. Item 1: at l, the traces Q_L(l) and U_R(l) are free, and applying f to the conditions gives pi_+(f(l)) X = X pi_-(f(l)) and pi_-(f(l)) X* = X* pi_+(f(l)), as in Result 3 of lemma 20. That is [pi(f(l)), tan(l D_F)] = 0, equivalent to [pi(f(l)), D_F] = 0 because l ||D_F|| < pi/2 makes D_F a function of tan(l D_F). At 0 the conditions are preserved since pi is even. Item 2: the commutator is the Leibniz rule. Conjugation by J^ acts on (Q, U) as J_F does, and pi(g) acts identically on the two spin components, which gives the second formula. This preserves the domain since J_F commutes with D_F. Item 3: pi(f(x)) and pi°(g(x)) commute at each x by finite order zero, and E commutes with both; so [E pi(f'), pi°(g)] = E [pi(f'), pi°(g)] = 0. ■

This is consistent with Counterexample 4 of lemma 20, whose finite data A_F = C + C, D_F = m sigma_1 fail order one while the parent satisfies it.

The compression

Let Q_l(T) = W* T W for a bounded operator T on H_l. For T(x) a bounded function with values in the even operators on H_F, acting pointwise and identically on both spin components, we write T for the

multiplication operator.

Theorem 22.4.

```
Q_l(T) = l^-1 int_0^l cos(x ad_{D_F})(T(x)) dx,      Q_l(E T) = -l^-1 int_0^l sin(x ad_{D_F})(T(x))
↪ dx.
```

For constant a in A_F,

```
Q_l(a) = sinc(l ad_{D_F})(pi(a)) = (2l)^-1 int_{-l}^{l} e^{ix D_F} pi(a) e^{-ix D_F} dx,
```

so in an eigenbasis of D_F with eigenvalues mu_i, the matrix elements are pi(a)_{ij} sin(l (mu_i - mu_j))/(l (mu_i - mu_j)). Moreover:

1. Q_l(A_l) = Q_l(C^infinity([0, l], A_F)), so the image of the parent algebra contains Q_l(a) for every a in A_F.
2. ||Q_l(a) - pi(a)|| <= (l^2/6) |||[D_F, [D_F, pi(a)]]|| <= (2/3) l^2 ||D_F||^2 ||a||.
3. Q_l([D_X, f]) = [D_F, Q_l(f)] for f in A_l.
4. With a' = Q_l(a) and b' = J_F Q_l(b)* J_F^-1, |||[a', b']|| <= (8/3) l^2 ||D_F||^2 ||a|| ||b||, and |||[[D_F, a'], b'] - [[D_F, pi(a)], pi°(b)]|| <= (16/3) l^2 ||D_F||^3 ||a|| ||b||.

So as l -> 0 with D_F fixed, D_F, J_F and Gamma_F stay exact, and the compressed observables converge to pi(A_F) with rate l^2. The finite order conditions hold in the limit exactly when they hold for the finite data.

Proof. From the definition of W, the expectation of T splits into the L- and R-parts, so Q_l(T) is l^-1 int of the Gamma_F-even part of R(x)* T(x) R(x). Put K = Gamma_F D_F. For even Z, ad_K Z = Gamma_F [D_F, Z], which is odd, and ad_K (Gamma_F [D_F, Z]) = -[D_F, [D_F, Z]]. So the even part of e^{-xK} T e^{xK} is cos(x ad_{D_F})(T). Likewise, E exchanges the L- and R-parts with a sign, so Q_l(E T) is l^-1 int of Gamma_F times the odd part of R* T R, and that odd part is -Gamma_F sin(x ad_{D_F})(T). The formula for constant a follows from cos(x ad_D)(Z) = (e^{ixD} Z e^{-ixD} + e^{-ixD} Z e^{ixD})/2.

Item 1: the map f -> Lambda(Q_l(f)), for a linear functional Lambda, is integration against a smooth density. If it vanishes on A_l, which contains all f of compact support in [0, l), the density vanishes on [0, l), hence everywhere by continuity. Item 2: by Taylor's formula with integral remainder, ||cos(x ad_D)(Z) - Z|| <= (x^2/2) ||ad_D^2 Z||, since cos(x ad_D) has norm at most 1 by the average formula; averaging over x gives l^2/6. Item 3: P commutes with D_X. It also follows from the two formulas by integration by parts, where the boundary term at l vanishes because f(l) is in C_F. Item 4: J_F commutes with D_F, so b' is the same average of pi°(b). Expand both commutators with the bound of item 2 and ||[D_F, Z]|| <= 2 ||D_F|| ||Z||. ■

The convergence is a statement about the maps Q_l, on the operator systems, in the setting of Connes and van Suijlekom (arXiv:2004.14115, sections 1 and 3). The generated algebra is not continuous at l = 0. An analytic example, from the referee: let M = [[0, 0, 1], [0, 0, 1], [1, 1, 0]], D = [[0, M], [M, 0]], with A_F = C^3 acting diagonally and equally on the two blocks. With P_1, P_2 the first two channel projections, [Q_l(P_1), Q_l(P_2)]_{12} = -l^2/3 + O(l^4), so the compressions of commuting projections generate a noncommutative algebra for small nonzero l. On the one-generation Standard Model data, a numerical closure with tolerance 10^-7 gives complex dimension 15 for the algebra generated by pi(B), with B = C + H + M_3(C), and 30 for Q_l(B) at l = 0.3; these are numerical values with no certified rank.

Scalar fluctuations

Theorem 22.5. Let h be smooth with values in the anti-Hermitian elements of A_F. The self-adjoint one-forms of the parent are exactly the E pi(h). Let u' = -h u, u(0) = 1, and let w = u(l), a unitary of A_F. Then

```
D_X + E pi(h) = pi(u) D_{X_w} pi(u)\*,      X_w = pi_+(w)\* X pi_-(w),
```

and the window of D_{X_w} carries pi(w)* D_F pi(w). With the real structure, for A = E pi(h) we have J^ A J^^-1 = E J_F pi(h) J_F^-1, and D_X + A + J^ A J^^-1 has the window U_w* D_F U_w, with

U_w = pi(w) J_F pi(w) J_F^-1, and this commutes with J_F. Moreover, with u~(x) = u(x) w*, which lies in A_l since u~(l) = 1, and U~ = pi(u~) J_F pi(u~) J_F^-1,

```
D_X + A + J^ A J^^-1 = U~ D_X U~\*.
```

So every scalar connection in this model is gauge equivalent to the original parent under the allowed endpoint gauge group (referee).

Proof. A one-form is sum f_1 [D_X, f_2] = E pi(sum f_1 f_2'), and E pi(h) is self-adjoint exactly when h is anti-Hermitian, since E is anti-Hermitian and commutes with pi(h) and pi is faithful. Every smooth such h arises: with b(x) = int_l^x h, we have b(l) = 0, so b is in A_l, and [D_X, b] = E pi(h). Next, pi(u)* D pi(u) = D + E pi(u* u'), and pi(u)* E pi(h) pi(u) = E pi(u* h u), so the sum is D with no connection term. At 0 the conditions are unchanged. At l, the substitution psi = pi(u) phi gives the conditions with X_w. Then tan(l pi(w)* D_F pi(w)) has the block X_w. For general w, this operator need not commute with J_F, so we use the spectrum and the isometry of Theorem 22.1 without its real-structure part. For the real fluctuation, J^ commutes with E, u(x) is replaced by pi(u) J_F pi(u) J_F^-1, using finite order zero, and U_w commutes with J_F, so the real window follows. The last formula follows by differentiating, with the endpoint condition. ■

Corollary 22.6 (the quadratic term as transport). By Chamseddine-Connes-van Suijlekom (arXiv:1304.7583, section III), for unitary v in A_F, with U = v J v J^-1, one has U D U* = D + A_(1) + A~*(1)* + *A*(2), for the universal form v delta(v*). Taking v = w*, on the gauge orbit the fluctuation of the parent, which satisfies order one and has no quadratic term, gives on the moving window exactly the fluctuated operator of that framework (equations (10)-(11) of section III.2 there), quadratic term included, for finite data that fail order one. This answers the first open point of lemma 19 on these gauge orbits and with the moving window. For the two-point data with J = sigma_1 K and w = diag(e^{i alpha}, e^{i beta}), it gives the upper link m e^{-2 i (alpha - beta)}, which is m e^{2 i psi} of Corollary 19.2 with psi defined from v = w*. Theorem 22.5 and the covariant projection of lemma 14 share a transport principle. The fixed compression, as in lemma 13, is a different object: by Theorem 22.4, it gives Q_l(E pi(h)) = -[D_F, l^-1 int_0^l x pi(h) dx] + O(l^3 ||D_F||^3 ||h||) for h fixed, so that for constant h it is -(l/2)[D_F, h], while the first derivative of the transported operator is -l [D_F, h]. The difference is the contribution of the moving spectral projection.

No-go theorem 22.7 (the radial mode). *Hypotheses:* the parent above, on one interval, with fixed X, no bulk masses, and scalar inner fluctuations from the even algebra A_l, with the moving spectral window. *Conclusion:* the fluctuations leave all singular values of the finite Dirac operator fixed. In the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through the unit quaternions, a three-dimensional gauge orbit, with no radial mode. Degenerate data can have a smaller orbit. A fixed compression can show changes in singular values at second order, so the window has to be named.

Proof. Theorem 22.5: D_F and pi(w)* D_F pi(w) are unitarily equivalent. The same holds with U_w. ■

Known escapes.

These lie outside the hypotheses, and each needs its own construction.

- The coupling X as a field on the wall, with its own kinetic term and potential.
- Connections odd for the grading, as in Corollary 22.2 and lemma 23: the fluctuation K -> K + delta K changes the singular values.
- Bulk masses that do not commute with the gauge action.
- Two or more internal dimensions, where a connection has curvature, although curvature alone does not prove a radial mode.
- An additional composite or other degree of freedom; a quantum effective action on the same configuration space does not by itself add a radial coordinate.
- A fluctuation of the finite data alone, as with the radial parameter of lemma 19, which is not the moving window.

Direction. On the interval, the even connection is pure gauge up to its boundary effect, so the finite orbit is

all it can reach. The physical Higgs modulus must then be sought at the wall, in odd directions, or in more dimensions. At the 10 -> 9 endpoint this agrees with T10, where the Higgs is frozen.

The gauge sector and the 10 -> 9 family

Proposition 22.8. Consider the product of M^4 with this parent, with algebra of functions f(y, x) valued in A_F and f(y, l) in C_F. Then the gauge one-forms sum f_1 d_mu f_2 take values in C_F at x = l. Assume, in addition, a flat product, a positive five-dimensional Yang-Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Then the directions in the Lie algebra of C_F have masses n pi/l, n >= 0, and the orthogonal directions have masses (n - 1/2) pi/l, n >= 1. The physical gauge algebra comes from the skew-adjoint elements through the real gauge representation, with its kernel and any unimodularity condition.

Proof. The first statement is direct evaluation at x = l. The rest is the interval spectrum in the gauge A_5 = 0, with Neumann conditions at 0, and with Neumann or Dirichlet conditions at l. ■

In the Standard Model data with nonzero Yukawas, M_W = M_Z = pi/(2l). The Lie algebra of C_F has dimension 10 before unimodularity; with the unimodularity condition of Chamseddine-Connes-Marcolli (arXiv:hep-th/0610241, section 2.5), the massless modes are the photon and the gluons. These are the boundary conditions of Higgsless models (Csaki, Grojean, Murayama, Pilo and Terning, Phys. Rev. D 69 (2004) 055006), and the fermion formula tan(m L) = M_D L is equation (5.6) of Csaki, Grojean, Hubisz, Shirman and Terning (Phys. Rev. D 70 (2004) 015012); LITERATURE.md, section C. Boundary spectral geometry for the Standard Model has been studied by Chamseddine and Connes (arXiv:1008.3980, section 5). What is added here, as far as our searches went, is the combination: the real structure, the finite triple as an exact window, the wall algebra C_F, the compression formula and the fluctuations. A wider claim of novelty would need a fuller comparison.

Corollary 22.9 (the 10 -> 9 family). Fix a block M_0 for the light fermions and a block Theta/l on an orthogonal channel, compatible with J_F and the grading, with 0 < Theta < pi/2, for the top and if wanted for the Majorana block. Let D_F(l) have M(l) = M_0 (+) Theta/l, and let l -> 0. Then:

1. X(l) has the light block tan(l |M_0|) V_0* = l M_0* + O(l^3) and a heavy block with singular values tan(Theta): the light couplings vanish as m l at fixed mass m.
2. The heavy channel has mass Theta/l, the broken gauge directions have mass pi/(2l), and the other levels diverge.
3. The massless gauge sector is C_F. Let P_L be the projection reducing D_F(l) to the fixed light block D_L. For constant a, P_L Q_l(a) P_L = sinc(l ad_{D_L})(P_L pi(a) P_L), with ||P_L Q_l(a) P_L - P_L pi(a) P_L|| <= (2/3) l^2 ||D_L||^2 ||pi(a)||, also when pi(a) mixes light and heavy channels. The limiting light compression need not be a representation, since products can pass through the heavy subspace. The full estimate of Theorem 22.4 does not apply, because l ||D_F(l)|| tends to Theta. For example, with D_F(l) = sigma_1 (x) diag(m, Theta/l) and pi(a) = 1 (x) sigma_1, one has Q_l(a) = sinc(Theta - l m) pi(a), which tends to sinc(Theta) pi(a).
4. With l = pi/(2 M_W), we get X_q ~ (pi/2) m_q/M_W, which has the form of y_q = sqrt(2) m_q/v -> 0 at fixed mass, as the coupling vanishes while the mass stays. Identifying X_q with the four-dimensional Yukawa coupling is a hypothesis.
5. The ratio is m_t/M_W = 2 Theta/pi < 1, and M_W/M_Z = 1.

So the 10 -> 9 proposal is realized at this level as a limit l -> 0, with W, Z and top at the tower scale and the light fermions keeping explicit masses. Item 5 does not reach m_t/M_W = sqrt(8/3), which belongs to the interior. This is a statement in the stated family: lowest level, no bulk mass, strict wall algebra. Bulk masses for the top, and a changed gauge wall condition, are directions beyond it.

A numerical comparison for the interior (Python). A separate model, with a finite wall scalar and Neumann conditions at 0, lies outside Proposition 22.8 and No-go theorem 22.7. In it, the mass of a broken direction satisfies t tan t = s, with t = M l and s = g^2 v^2 l^2/4 (for the Z, s/cos^2 theta_0), which has the form of M_k tan(M_k pi R) = -V in Csaki et al. (arXiv:hep-ph/0305237, section on mixed boundary conditions, read in the LaTeX source), with the sign convention of V there. If $g'^{2/g}2 = 3/5$ is taken at all

scales, M_W^{2/M_Z}2 goes from 5/8 as s -> 0 to 1 as s -> infinity. It equals the measured 0.777 at M_W l = 1.03, with 1/l = 78 GeV and a next W level at 280 GeV; whether collider searches allow this would need the couplings, branching fractions and an applicable limit, which we have not analyzed. The top needs M_W l < 0.96 for m_t/M_W = sqrt(8/3), or M_W l < 0.73 for the measured ratio, in the lowest level with no bulk mass. So in the flat tree-level toy without bulk masses or running, the top and gauge requirements are incompatible. It does show the interior interpolating between the spectral-action value 5/8 and the endpoint value 1. The directions are bulk masses, warping and running.

What is input and what is derived

The finite data (A_F, H_F, D_F, J_F, Gamma_F) are all inputs, so the existence of the window is close to a reparametrization: X = tan(l D_F). The derived statements are the wall algebra C_F for that input, the order conditions of the parent, the compression formula with its rate, the form of the fluctuations, and the gauge spectrum under the stated assumption. The window identifies D_F, J_F and Gamma_F exactly, while the represented algebra is obtained through compression and is not an exact representation of A_F at l > 0. In particular, this does not select A_F. T0 asks for a commutative parent: a natural next step is a two-stage parent, compressing functions on S^4 and CP^3 at level one to get A_F-valued functions on the interval, and then taking this window, where the difficulty is the same interface problem as after Result 7 of lemma 20.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^5, l ||M|| = 1.3: D u_+(a) = u_-(M a) and D u_-(b) = u_+(M* b) by finite differences, to about 10^-9; boundary conditions and J^ to about 10^-15; norm of W to 10^-13.
- Q_l(a) = sinc(l ad)(a), by quadrature, to 2 x 10^-8. The formula for Q_l(E T), with a nonconstant T, to 10^-14.
- One-generation Standard Model data with Yukawas and the rank-one Majorana block, with signs, order zero and order one exact on B = C + H + M_3(C), and commutant of D in B of real dimension 20. At l = 0.1 and 0.03, ||Q_l(a) - a + (l^2/6)[D, [D, a]]|| is 10^-5 and 4 x 10^-8, and the order-one defect of Q_l is 1.5 x 10^-4 and 1.2 x 10^-5, which scales as l^2.
- The dimensions of the generated algebras quoted above, 15 and 30, from a numerical closure with tolerance 10^-7, not retained and not certified; at smaller l the numerical rank depends on the tolerance.
- The roots of t tan t = s, and the values quoted in the comparison for the interior.
- On the two-point data with J = sigma_1 K: D + A_(1) + A~*(1)* + *A*(2) = U D U* to 4 x 10^-16, and the link of U_w* D U_w equal to m e^{-2 i psi}.
- The bounds of Theorem 22.4, items 2 and 4, on 30 random samples of the one-generation data: largest ratios to the bound 0.33, 10^-4 and 10^-3.

Part 23

Two gradings on an interval: masses as principal angles, the Higgs modulus, and exact W and Z masses

For STRATEGY T0b, T5b, T10 and T13, following lemma 22. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): four errors and one gap as first written, and amendments, all incorporated below. The simpler form of the mass formulas, and the proof of item 3 of Theorem 23.6, are by the referee.

Outcome. A self-adjoint boundary condition for the derivative on an interval is a grading at each wall. The whole finite Dirac operator is then the relative position of two gradings: its singular values are the principal angles between them, divided by l, and

```
D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0).
```

With a bulk prealgebra, this gives four results.

1. Aligned walls compress a prealgebra stable under the grading to its even part, with the grading inherited, which is the step that Example 16.9 had to add by hand.
2. Odd scalar fluctuations move the second grading, so they change the singular values and supply the radial mode that No-go theorem 22.7 excludes for even fluctuations. For M_2(H) the field space is S^4, and the Higgs modulus is half the geodesic distance between the points chosen by the two walls.
3. With a wall connector on one channel and a five-dimensional Yang-Mills action with common normalization, the masses are exact: sin^2(M_W l) = sin^2(2 theta)/2 and sin^2(M_Z l) = (4/5) sin^2(2 theta), for universal fermions of mass theta/l. At small theta this gives M_W^2 = 2 m^2 and M_Z^2 = (16/5) m^2, the constants in the lower bounds of the fully massive family of lemma 18, with $M_W^{2/M_Z}2$ tending to 5/8 at both ends.
4. The heat-cutoff spectral action on the interval is a Jacobi theta function of theta, decreasing on [0, pi/2]. This family lies outside the hypotheses of the radial no-go of lemma 18: the radial direction is compact, with a stationary point at m = pi/(2l), the Connes-Lott value, where M_W = M_Z = 0.

Everything finite is still an input, and the parent algebra is matrix-valued. For a commutative parent, lemma 24 finds the window exactly, with spectral walls, while direct compression of scalar functions gives only R + R on the S^4 factor. The formulas for gauge masses depend on the assumed action. The relation to measured values fails in the flat toy, as stated at the end.

Two gradings

Let H be a finite-dimensional Hilbert space, and let Gamma_0, Gamma_1 be self-adjoint involutions on H. On L^2([0, l]; C^2) (x) H, with psi = (psi_L, psi_R), let D = E d/dx, where (E psi)_L = psi_R and (E psi)_R = -psi_L, on the domain

```
Gamma_0 psi_L(0) = psi_L(0),   Gamma_0 psi_R(0) = -psi_R(0),   Gamma_1 psi_L(l) = psi_L(l),
↪  Gamma_1 psi_R(l) = -psi_R(l).
```

Call it D(Gamma_0, Gamma_1), and let Gamma^ act as +1 on psi_L and -1 on psi_R.

Theorem 23.1.

1. D(Gamma_0, Gamma_1) is self-adjoint with compact resolvent, and it anticommutes with Gamma^.
2. Suppose Gamma_0 and Gamma_1 have eigenspaces of equal dimensions, and that -1 is not an eigenvalue of Gamma_1 Gamma_0. Let theta_k in [0, pi/2) be the principal angles between the +1 eigenspaces. Then K = (1/2l) log(Gamma_1 Gamma_0), with the principal branch, is anti-Hermitian, Gamma_0 K Gamma_0 = -K, and D_F = Gamma_0 K is self-adjoint, odd for Gamma_0, with singular values theta_k/l on the +1 eigenspace. Moreover Gamma_1 = e^{lK} Gamma_0 e^{-lK}.
3. With H_+, H_- the eigenspaces of Gamma_0, and X the H_- -> H_+ block of tan(l D_F), as in lemma 22, D(Gamma_0, Gamma_1) = D_X, and the spectral part of Theorem 22.1 gives the spectrum and the window, with no real structure needed. Conversely, every D_F odd for Gamma_0 with l ||D_F|| < pi/2 arises from exactly one such Gamma_1.
4. Let an antiunitary J_1 satisfy J_1^2 = -1, J_1 Gamma_i J_1^-1 = -Gamma_i for i = 0, 1. Then (J^ psi)_L = J_1 psi_R, (J^ psi)_R = -J_1 psi_L defines a real structure of signs (+, +, -) for D(Gamma_0, Gamma_1). With J_F = Gamma_0 J_1 up to sign, J_F^2 = 1, J_F Gamma_0 = -Gamma_0 J_F and J_F D_F = D_F J_F, which are the signs of lemma 22.

So J_F itself does not anticommute with Gamma_1 in general: J_F Gamma_1 J_F^-1 = -Gamma_0 Gamma_1 Gamma_0, the reflection of Gamma_1 through Gamma_0.

Proof. Item 1: the boundary form is [<psi_R, phi_L> - <psi_L, phi_R>] evaluated between 0 and l, which vanishes on the domain because L-values and R-values lie in orthogonal eigenspaces at each wall, and these are half of the boundary data.

Item 2: U = Gamma_1 Gamma_0 is unitary and Gamma_0 U Gamma_0 = Gamma_0 Gamma_1 = U^-1, so K is anti-Hermitian and anticommutes with Gamma_0. The eigenvalues of U are e^{+-2 i theta_k},

by the standard two-projection decomposition, and K then has eigenvalues +-i theta_k/l. Since e^{lK} Gamma_0 e^{-lK} = e^{2lK} Gamma_0 = U Gamma_0 = Gamma_1, the last formula follows.

Item 3: in the notation of Theorem 22.1, R(l) = e^{lK} carries H_+ to the graph {(C a, -S a)} = graph(-X*), and H_- to graph(X). These are the +1 and -1 eigenspaces of Gamma_1, which is the condition at the wall. The converse uses the same formula, with uniqueness from injectivity of the exponential on the stated range.

Item 4: J^ commutes with E by direct computation, and J^^2 = -J_1^2 = 1. The domain is preserved since J_1 exchanges the eigenspaces of each Gamma_i. For the last statement, J_1 U J_1^-1 = U, so as J_1 is antilinear, J_1 K J_1^-1 = K in the sense of the principal branch, and J_F = Gamma_0 J_1 then satisfies J_F K J_F^-1 = -K and commutes with D_F = Gamma_0 K. ■

When theta_k = pi/2 the channel has type (c) of lemma 09, with no zero mode and lowest level pi/(2l), the Connes-Lott mass of Theorem A in the paper. There the principal logarithm and the isolated window cease to be unique.

The bulk prealgebra

Let P be a *-subalgebra of End(H), which need not commute with Gamma_0. Let

```
A(Gamma_0, Gamma_1) = {f in C^infinity([0, l], P) : [f(0), Gamma_0] = 0, [f(l), Gamma_1] = 0},
```

acting pointwise and identically on both spin components.

Theorem 23.2.

1. A function f in C^infinity([0, l], P) preserves the domain exactly when it lies in A(Gamma_0, Gamma_1). The commutators are [D, f] = E f', and Gamma^ commutes with every f, including those with values odd for Gamma_0.
2. Under the hypotheses of Theorem 23.1, item 2, with W the isometry of Theorem 22.1, Q(T) = l^-1 int_0^l E_0(R(x)* T(x) R(x)) dx, where E_0(Z) = (Z + Gamma_0 Z Gamma_0)/2 is the conditional expectation onto the even part.
3. *Aligned walls.* If Gamma_1 = Gamma_0, then D_F = 0, the window consists of constant L-values in H_+ and constant R-values in H_-, and Gamma^ restricts on it to Gamma_0. For an ambient constant multiplier a in P, Q(a) = E_0(a), and the image of the boundary algebra A(Gamma_0, Gamma_0) is E_0(P), by compactly supported profiles. If Gamma_0 P Gamma_0 = P, then E_0(P) is the even part of P, so the compressions of the prealgebra are exactly its even part, with the grading inherited from the even parent operator Gamma^. Without that hypothesis this can fail: for H = C^2, Gamma_0 = sigma_z and P generated by sigma_x + sigma_z, E_0(P) = span{1, sigma_z}, while the even part of P is C.
4. *Order conditions.* Let J_1 be as in Theorem 23.1 and let P° = {J_1 b* J_1^-1 : b in P} commute with P. Then the parent has order zero and order one for A(Gamma_0, Gamma_1).

Proof. Item 1: as in Theorem 22.3, using that the conditions at a wall say that L- and R-values lie in the eigenspaces of Gamma_i. Item 2: as in the proof of Theorem 22.4. Item 3 is the case R = 1. Item 4: [D, f] = E f', and pointwise commutation gives both conditions. ■

Item 3 evades the inherited-grading obstruction of Result 11 of lemma 20, because hypothesis 4 there fails: for odd a, here Q(a) = 0, while there the coordinates are retained as Q(x_a) = Gamma_a/5. So at the level of a matrix-valued parent this addresses the question left in step 3 of the framework note: the prealgebra is the bulk algebra, the even algebra is what the window sees, and the grading is a boundary condition. The commutative-parent and retained-coordinate questions remain open, and Theorem 4 of lemma 24 shows a real difference: scalar functions compressed directly give R + R on the S^4 factor, and H + H needs the matrix algebra generated at the intermediate stage.

Remark 23.3 (the twist). On H = E (+) E*, with J(x, y) = (y*, x*), the operator J_1 = Gamma_0 J is the one that anticommutes with both gradings. For an odd anti-Hermitian h in P, J_1 h J_1^-1 is y -> +y h on E*, where J alone gives y -> -y h. So the opposite representation b° = J_1 b* J_1^-1 incorporates on the W-factor the grading automorphism sigma(b) = gamma_W b gamma_W. In this parent the twist comes

from the requirement that J_1 preserve both boundary conditions. Twisted spectral triples for the grand symmetry (Devastato, Lizzi and Martinetti, JHEP 01 (2014) 042, arXiv:1304.0415; Devastato and Martinetti, Math. Phys. Anal. Geom. 20 (2017) 2, arXiv:1411.1320, Eqs. (1.8)-(1.11)) use a different automorphism: their rho exchanges left and right spinorial indices and algebra summands, while conjugation by gamma_W is inner on M_2(H) and fixes its centre. So this is a structural analogy between involutive automorphisms, and an identification needs a comparison of the representations, opposites and first-order conditions.

Odd fluctuations and the radial mode

Theorem 23.4. Let h be smooth with values in the anti-Hermitian elements of P, and let u' = -h u, u(0) = 1, w = u(l). Then

```
D(Gamma_0, Gamma_1) + E h = u D(Gamma_0, w\* Gamma_1 w) u\*.
```

So scalar fluctuations move the second grading along its orbit under the unitary group of P. If h has an odd part, the principal angles between Gamma_0 and w* Gamma_1 w change in general.

Proof. u* D u = D + E u* u', and u* E h u = E u* h u, so the connection cancels. Since u(0) = 1, the wall at 0 is unchanged. At l, psi = u phi gives the stated grading for phi. ■

This lies outside No-go theorem 22.7, whose hypotheses include the even algebra: an even connection gives a w commuting with Gamma_0, and the angles are kept, while an odd connection changes them. For a real fluctuation, add the conjugate one-form: D + E h + J^(E h)J^^-1, with generator of the form (L_F, R_F) in Example 23.5. The invariant is a double coset. With H_i = Stab(Gamma_i) in the connected unitary group G of P, a gauge transformation removing h has the form u(x) c, and it preserves both endpoints exactly when c is in H_0 and w c in H_1, that is, when w is in H_1 H_0. So the scalar moduli of the parent are the double cosets H_1 G / H_0, the relative position of the two gradings. This is with the moving window, as in lemma 22.

Example 23.5 (M_2(H) and S^4). Let W = H^2 = C^4, with the Clifford generators Gamma_a of lemma 16, and for a unit vector p in R^5 put Gamma_p = p_a Gamma_a. Each grading of W with one-dimensional quaternionic +1 and -1 eigenspaces is such a Gamma_p, so these gradings form S^4 = Sp(2)/(Sp(1) x Sp(1)), the quaternionic projective line. For two points, Gamma_q Gamma_p = p . q + (p wedge q), with eigenvalues e^{+-i d}, where d is the geodesic distance; so all the principal angles equal theta = d/2.

Take H = Hom(V, W) (+) Hom(W, V), J(x, y) = (y*, x*), Gamma_0(x, y) = (gamma x, -y gamma) with gamma = Gamma_{p_0}, and K(x, y) = (theta/l)(F x, y F), with F an odd anti-Hermitian element of M_2(H), F^2 = -1. Then:

- D_F = Gamma_0 K has all singular values equal to theta/l: a universal Dirac mass, for all colours.
- The second grading is Gamma_1(x, y) = (Gamma_p x, -y Gamma_{p'}), where p is at distance 2 theta from p_0 and p' is its mirror image through p_0. J_1 = Gamma_0 J satisfies Theorem 23.1, item 4, while J alone does not.
- The coordinates (theta, q) on S^4, with F determined by the unit quaternion q, are the polar coordinates of a Higgs quaternion: theta is the radial coordinate, bounded by pi/2, and q gives three angular directions, undefined at the poles. At theta = pi/2 the finite logarithm and the lowest window are degenerate. Quotienting by the stabilizer leaves the radial interval. A canonically normalized field needs the scalar kinetic term, which we have not computed.

With bulk masses, by Theorem 12.6, the light masses are about (tan(theta)/l) Phi_Q Phi_U when small compared with the tower thresholds, with a universal matrix before flavour data; general finite-angle eigenvalues need the secular equation. This field space is the coset of the minimal composite Higgs model (Agashe, Contino and Pomarol, Nucl. Phys. B 719 (2005) 165, sections 2-3, with their h/f_pi corresponding to 2 theta). Their construction uses a warped interval and additional fermion and boundary data, so the shared coset does not identify the actions or potentials. The appearance here of the S^4 of the twistor window is an observation, and no identification between the two roles of S^4 is made.

The weak bosons

Take P = M_2(H) (+) M_4(C), acting on E = Hom(V, W) on the left and on the right. Assume the following.

- A five-dimensional Yang-Mills field valued in sp(2) (+) su(4), with the trace form Tr_W + Tr_V, so with one coupling (the normalization of the spectral action, g_3 = g_2 = sqrt(5/3) g_Y).
- At x = 0: Neumann for the stabilizer of p_0, Dirichlet for its complement. At x = l: Neumann for the stabilizer of p_l and of a connector, that is, the a with [a_W, Gamma_p] = 0, a_W w = lambda w and a_V v = lambda v, as in Results 2 and 5 of lemma 20, for a line w in the +1 eigenspace of Gamma_p and a line v in V; Dirichlet for the complement.
- The connector is prescribed gauge-breaking wall data here. If it is instead a term in the fermion boundary operator, as for a unit vector e in the +1 eigenspace of Gamma_1, f = J_1 e, K_c = |f><e| - |e><f| and Gamma_{1,c} = e^{eta K_c} Gamma_1 e^{-eta K_c}, then J_1 still anticommutes with Gamma_{1,c}, the parent order conditions are kept, and the affected fermion tower changes, so the universal fermion formula must be recalculated for that channel.

Theorem 23.6. With theta the angle above, the modes that are massless for all theta are exactly nine, the photon and the eight gluons, with three additional electroweak zero modes at theta = 0 and theta = pi/2. The lowest charged and neutral massive modes satisfy

```
sin^2(M_W l) = sin^2(2 theta)/2,
sin^2(M_Z l) = (4/5) sin^2(2 theta),
```

equivalently tan^2(M_W l) = sin^2(2 theta)/(1 + cos^2(2 theta)) and tan^2(M_Z l) = 4 sin^2(2 theta)/(1 + 4 cos^2(2 theta)).

The photon has 1/e^2 = (8/3)/g^2. Moreover:

1. As theta -> 0, with m = theta/l: M_W^2 -> 2 m^2 and M_Z^2 -> (16/5) m^2, so $M_W^{2/M_Z}2$ -> 5/8.
2. As theta -> pi/2: M_W, M_Z -> 0, with $M_W^{2/M_Z}2$ -> 5/8, while m -> pi/(2l).
3. At theta = pi/4: M_W l = pi/4, tan(M_Z l) = 2, and $M_W^{2/M_Z}2$ = ((pi/4)/arctan 2)^2 = 0.5032..., the global minimum. So $M_W^{2/M_Z}2$ stays in [0.5032, 0.625), with 5/8 as a limit at the ends, where the ratio itself is undefined.
4. m/M_W goes from 1/sqrt(2) at theta -> 0 to infinity at theta -> pi/2. It equals sqrt(8/3) at theta = 1.0589.

Proof. In the gauge A_5 = 0 the field is free, with cos(k x) on the Neumann directions at 0 and sin(k x) on the Dirichlet ones, and the condition at l holds in the frame rotated by Ad exp(theta F). The generators split according to the frequency of Ad F. Write L, R for su(2) on W_- and W_+, S = (L + R)/sqrt(2), A = (L - R)/sqrt(2), and O for the odd generators; Ad F has frequency 0 on S, and frequency 2 on the plane (A, O). So S = s cos kx, A = alpha cos kx, O = beta sin kx.

At l, in the rotated frame, A' = cos(2 theta) A + sin(2 theta) O and O' = -sin(2 theta) A + cos(2 theta) O, with O' = 0 at l, which gives beta = alpha tan(2 theta) cot(k l), and so A'(l) = alpha cos(k l)/cos(2 theta).

Charged directions. The connector forces R^+- to vanish at l, with L^+- Neumann. So S(l) = A'(l), and d/dx (S + A') = 0 at l. Substituting gives sin^2(kl) (1 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Neutral directions. Add B = b cos kx for the generator T_15 = diag(1, 1, 1, -3)/(2 sqrt 6) of su(4). With Y = sqrt(3/5) R + sqrt(2/5) B and Z' = -sqrt(2/5) R + sqrt(3/5) B, the connector forces Z' to vanish at l, with L and Y Neumann. The Neumann conditions give b = -sqrt(3) s, and Z' = 0 gives s = alpha/(4 cos(2 theta)). Substituting in the condition for L gives sin^2(kl) (1/4 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Massless modes and limits. At k = 0, a zero mode is a constant in the intersection of the two stabilizers. This consists of a_W = diag(q, q) on W_+ and W_-, identified by F, with q preserving the line of w, together with a_V = lambda (+) m, and with the trace condition this has dimension 1 + 8. In it, L = R and B/L = sqrt(2/3), which gives 1/e^2 = (1 + 1 + 2/3)/g^2. The limits follow by expanding. A second

derivation, from the referee, uses the principal angles between the Neumann subspaces at the two walls: for each charged direction the squared sine of the angle is sin^2(2 theta)/2, and in the neutral sector the single nonzero value is (4/5) sin^2(2 theta); it also covers cos(2 theta) = 0, where the divisions above need a limit. For item 3, with a = sin(2 theta), the ratio is [arcsin(a/sqrt 2)/arcsin(2a/sqrt 5)]^2, and this decreases with a on (0, 1], since x/(sqrt(1 - x^2) arcsin x) increases. ■

The constants in item 1 are the values $|||[D_F, a]|||^{2/||a||}2$ in the Hilbert-Schmidt norm, for universal mass m, for a = L^+- and for the direction orthogonal to the photon and Z', which has A-component of squared length 4/5. These are the coefficients of the lower bounds for the poles of the fully massive family in Proposition 14 of lemma 18 (2 x < z_W, (16/5) x < z_Z); the small-x pole coefficients there are 3 and 24/5. The limit 5/8 is common to both, with different behaviour: the finite-action poles grow in the heavy limit of lemma 18, while the interval masses here tend to zero at the antipodal point. The spectral-action Taylor mass for universal m is M_W^2 = 3 m^2, which exceeds the value here by the factor 3/2 of Proposition F in the paper: there the mass matrix carries the relative heat coefficients of the action, and here we have assumed a five-dimensional Yang-Mills action, which gives the commutator norm alone. A direct matching claim needs the full parent spectral action, with its tower and boundary conditions.

Item 2 gives a geometric form of the behaviour that lemma 18 found: heavy fermions with light vectors. At antipodal points Gamma_p = -Gamma_{p_0}, so both walls have the same stabilizer in sp(2) again, and the fermion channels have type (c) with mass pi/(2l).

The spectral action on the interval

Proposition 23.7. For the heat cutoff and universal theta, with s = l Lambda, the trace over a channel is

```
Tr_channel exp(-D^2/Lambda^2) = 2 sum_{n in Z} exp(-(n pi + theta)^2/s^2).
```

This is proportional to the Jacobi function theta_3(theta, exp(-s^2)). It is strictly decreasing on [0, pi/2], with a maximum at theta = 0 and a minimum at theta = pi/2, where it is stationary, and it is symmetric under theta -> pi - theta. In the towerless regime s -> 0 at fixed x = theta$^{2/s}$2, it tends to 2 e^{-x}, the form of the vacuum density in lemma 18 and Proposition 21.2.

Proof. The eigenvalues in a channel are +-(n pi + theta)/l, by Theorem 22.1, and the sum is over Z after combining the signs. By Poisson summation the sum equals (s/sqrt(pi)) sum_{k in Z} exp(-k^2 s^2) e^{2 i k theta}, and the Jacobi triple product writes this as a constant times the product over n >= 1 of (1 + 2 q^{2n-1} cos(2 theta) + q^{4n-2}), with q = exp(-s^2). Each factor is positive and strictly decreasing in theta on [0, pi/2]. ■

In the form of the Jacobi function, the trace is (2 s/sqrt(pi)) theta_3(theta, exp(-s^2)). This family lies outside the hypotheses of No-go theorem 9 of lemma 18, which concerns finite internal data and a free ray V = c V_*: here the operator family is different, the radial direction is a compact orbit of relative gradings, and the tower, which the finite description omits, supplies a second branch with a stationary point at theta = pi/2. In the present family the literal action has its minimum at antipodal gradings, m = pi/(2l), with M_W = M_Z = 0 by Theorem 23.6, where the lowest eigenspace is degenerate. This is a vacuum with massless vectors, so it is not the physical point. The sign of the term agrees with the fermion loop in the Hosotani mechanism, where gauge-boson loops enter with the opposite sign. So an interior minimum could arise from that competition, which the classical spectral action does not contain. We have not computed it.

The one-loop potential in the same family

From the exact spectra, the products over the towers are, relative to theta = 0 and in Euclidean momentum x = p l,

```
fermion channel:  1 + sin^2(theta)/sinh^2(x),
charged vectors:  1 + sin^2(2 theta)/(2 sinh^2(x)),
neutral vector:   1 + (4/5) sin^2(2 theta)/sinh^2(x).
```

So in the gauge-boson and fermion-loop form of the Hosotani mechanism, assuming the usual one-loop formula with three polarizations per vector, and with n_f = 2 x (number of physical Dirac species),

```
8 pi^2 l^4 V(theta) = int_0^infinity x^3 dx [ 3 log(1 + s^2/(2 sinh^2 x)) + (3/2) log(1 + (4/5)
↪  s^2/sinh^2 x) - n_f log(1 + sin^2(theta)/sinh^2 x) ],
```

with s = sin(2 theta). A Dirac top with three colours has n_f = 6, which needs a separately specified fermion sector, since the universal connection gives masses to all: one generation has n_f = 16 and three have n_f = 48. With I = int $x^{3/\sinh}2$ x dx = (3/2) zeta(3) and I' = int $x^{3/\cosh}2$ x dx = (9/8) zeta(3), near theta = 0 the bracket is (10.8 - n_f) I theta^2 + o(theta^2), and with epsilon = pi/2 - theta, near theta = pi/2 it is V(pi/2) + (10.8 I + n_f I') epsilon^2 + o(epsilon^2), where we have written V for the left side.

Proposition 23.8 (referee). In the massless flat family with this content, for every n_f > 0, theta = pi/2 is the unique global minimum on [0, pi/2], since the gauge logarithms are nonnegative and vanish there, and the negative fermion logarithm is minimized at sin^2(theta) = 1. For n_f >= 10.8, V is strictly decreasing on the interior. Indeed, with t = sin^2(theta) and J(z) = int_0^infinity x^3 dx/(sinh^2 x + z), the derivative with respect to t is, up to a positive factor,

```
(1 - 2t) [6 J(2t(1 - t)) + (24/5) J((16/5) t (1 - t))] - n_f J(t).
```

For t <= 1/2, both arguments in the bracket are at least t, and J decreases; for t > 1/2 both contributions have the required sign.

For 0 < n_f < 10.8, on grids of theta (Python), for n_f in {1, 2, 3, 4, 6, 8, 10}, V has one interior maximum, at about 0.33 for n_f = 8, and no interior minimum. Theorem 9 of lemma 24, by Astra 6, proves this for all n_f in that range: exactly one interior stationary point, a strict maximum, using a log-concavity property of the integral.

These are negative results in an exact scope: one interval, no bulk masses, fermions of frequency 1 under F, this gauge content, one loop, this condition at the connector wall. As a formal leading approximation, not uniform at x = 0, V has the form B sin^2(2 theta) - A sin^2(theta) with A, B > 0, whose interior stationary point cos(2 theta) = A/(4B) is a maximum. *Bulk masses.* With bulk masses, the fermion product becomes cos^2(theta) + sin^2(theta) G(x), with G = (w_Q coth w_Q - c_Q)(w_U coth w_U - c_U)/x^2 and w = $(c^2 + x^2)\{1/2\}$ in the secular form of lemma 12. For a = c_Q + c_U != 0, G = 1 - a/x + O(x^-2), so the potential has angle-dependent ultraviolet divergences, and a finite value depends on a regulator and subtraction: as the referee showed, an earlier scan of general pairs here did not establish a potential, and it is withdrawn. The exceptional case c_U = -c_Q = c has a common bulk mass on the two mixed components and G = 1 + $w^{2/(x}2$ sinh^2 w), which decays exponentially. Keeping the flat gauge products as a separate tree-level assumption, a scan for c in {0, 1/2, 1, 3/2, 2, 3, 5}, n_f in {1, 3, 6, 8, 10, 16, 48}, on a grid of 721 values of theta, finds no interior minimum (Python). General bulk masses need a gauge-invariant completion, a regulator and wall counterterms.

So an interior minimum needs another change: a negative coefficient of sin^2(2 theta), as from fermions of frequency 2 under F, which the bimodule E = Hom(V, W) does not contain; general bulk masses or other gauge form factors, as with warping; a wall scalar with its own potential; or tree-level terms at the walls. With h/f_pi = 2 theta, the leading flat form has the angular form alpha cos(h/f_pi) - beta sin^2(h/f_pi) of Agashe, Contino and Pomarol (section 2.1, Eqs. (19)-(23)) with spinorial SO(5) fermions, with alpha = A/2 and beta = -B, up to a constant. Their form factors can give the sign and competition needed for an interior minimum, so the flat results here do not exclude that construction. This points away from a pure flat Wilson line, toward those changed form factors or a wall scalar with its own potential, for the physical interior.

Comparison with measured values in the flat toy

These are numerical comparisons (Python), with no running and no bulk masses, in lowest levels, using the Particle Data Group 2026 values quoted by the referee, m_t = 172.60 GeV as a direct-measurement mass, M_W = 80.3625 GeV and M_Z = 91.1879 GeV, so m_t/M_W = 2.1478 and M_$W^{2/M_Z}2$ = 0.7767. The ratio m_t/M_W reaches sqrt(8/3) at theta = 1.0589 and the measured value at theta = 1.1626, where M_W l = 0.541, so 1/l is about 148.5 GeV. There, M_$W^{2/M_Z}2$ = 0.581. The wall-scalar model of lemma 22 covers [0.625, 1], and this family covers [0.5032, 0.625); in the wall-scalar model, fitting the measured vector ratio gives M_W l = 1.028, where the lowest no-bulk-mass fermion channel has m/M_W < 1.528. So

neither flat family fits both ratios on its own, and the flat one-loop potential above does not select an interior point. Whether a tower at 1/l of this size is allowed would need the couplings and an applicable experimental limit, which we have not analyzed. The directions are bulk masses for the light fermions, warping, running of the couplings, and families combining a Wilson line with a wall scalar. These negative statements hold in the flat tree-level toy only.

Open questions

1. The interior vacuum: the effective potential with a wall scalar, or with warping, or with general bulk masses, with a gauge-invariant completion, a regulator and wall counterterms. The flat cases above have none on the grids.
2. The relation between the two roles of S^4: the base of the twistor window, and the field space of the Higgs quaternion.
3. A commutative parent: functions on S^4 x I and CP^3 x I, with the gradings as local boundary conditions.
4. The factor 3/2 between the five-dimensional normalization and the spectral action: whether the spectral action on the parent itself, with its tower, reproduces the formula of Theorem 23.6.
5. The comparison with twisted spectral triples, Remark 23.3.
6. The contribution of a real connector to the fermion tower, the degenerate antipodal window, and the scalar kinetic normalization needed for a canonically normalized Higgs field.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^4 with l ||M|| = 1.25: Gamma_1 is an involution, Gamma_1 = exp(2 l K) Gamma_0; D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0) to 3 x 10^-15; the principal angles over l equal the singular values of M to five digits.
- Sp(2) x SU(4), of dimension 25, with the Neumann and Dirichlet subspaces built from the stated stabilizers: the two-point boundary problem was solved as a linear system on a grid in k l. For theta = 0.2, pi/4, 1.06 and 1.3, the two lowest nonzero roots are (0.279, 0.356), (0.785, 1.107), (0.647, 0.868), (0.373, 0.479), equal to the formulas of Theorem 23.6 to three digits, with other roots at pi/2 and above. The intersection of the two stabilizers has dimension 9 at each theta tested.
- Small theta: $M_W^{2/m}2 = 2.000$ and $M_Z^{2/m}2 = 3.200$.
- The universal connection on E (+) E*: J D = D J, Gamma_0 D = -D Gamma_0, all |eigenvalues| equal to theta/l, Gamma_1 = (left Gamma_p, right mirror), J_1^2 = -1, J_1 Gamma_1 = -Gamma_1 J_1, while J alone fails; J_1 h J_1^-1 is right multiplication by +h on E*, for odd h.
- The theta function: monotone on a grid for s = 0.3, 1, 2, with the towerless value at s = 0.3 equal to e^{-x} to five digits; Poisson summation and the triple product to 10^-16.
- The ratio $M_W^{2/M_Z}2$ on a grid: range [0.5032, 0.6250], symmetric under theta -> pi/2 - theta to 10^-15, minimum at pi/4. At theta = 1.1625, M_W l = 0.541, and at theta = 1.0589, M_W l = 0.648.
- The one-loop integrals by adaptive quadrature (scipy), and by Gauss-Legendre quadrature on 400 nodes for the scan with c_U = -c_Q. The referee recomputed the derivative with scipy, found one sign change for each n_f in {1, 2, 3, 4, 6, 8}, with the stationary angle about 0.3382 at n_f = 8, and checked I = (3/2) zeta(3).

Part 24

Commutative cylinders, spectral walls, and the two-stage compression

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/24_commutative_parent.md. Reproduced as delivered, with links made relative. The section "Numerical check" at the end is by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): no error in the displayed equations, and one gap in the general statement on compression maps. The record above the section "Referee amendments" stays as delivered, and those amendments apply to it.

A commutative cylinder admits an exact realization of the **operator window** of lemmas 22 and 23 when the two finite gradings are prescribed through nonlocal spectral boundary conditions. The higher K-modes then decouple as the radius of K tends to zero. Two qualifications remain essential: these boundary conditions restrict scalar functions at the walls, and compression of scalar functions through both stages differs from generating a matrix algebra between the stages. In particular, aligned walls give $\mathbb{R} \oplus \mathbb{R}$ on the quaternionic factor under direct scalar compression, whereas the intervening matrix-algebra construction gives $\mathbb{H} \oplus \mathbb{H}$. Local boundary gradings preserve all scalar functions, but their reduction to the unchanged four-section window has a precise restriction proved below. The radial no-go of lemma 22 survives rechecking in its stated scope. For the massless one-loop potential of lemma 23, an analytic argument strengthens the numerical statement to all $n_f > 0$.

1. Theorem — An exact spectral boundary construction.

Hypotheses and conventions. Let

$$K = \mathbb{CP}^3, \qquad \mathcal{H}_K = L^2(K, \Lambda^{0,*}T^*K \otimes \mathcal{O}(1)),$$

and let

$$A = \sqrt{2}(\bar\partial + \bar\partial^*), \qquad \gamma_K = (-1)^{\deg}.$$

Use the Fubini–Study metric at a fixed reference scale. Write

$$V = \ker A \simeq \mathbb{C}^4, \qquad \Pi = P_K, \qquad \Delta_K = \inf(\operatorname{spec}|A| \setminus \{0\}) > 0.$$

Here $\gamma_K|_V = 1$. The radius parameter $r > 0$ replaces A by A/r, after the usual unitary identification of the L^2-spaces.

These are the analytic data of Example 16.9. Its S^4 option uses functions pulled back to this same twistor space. An actual operator on S^4 with the required kernel and compression map would need to be specified separately.

Let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

On $L^2(I_\ell; \mathbb{C}^2 \otimes \mathcal{H}_K)$, set

$$\mathcal{D}_r = E\partial_x + Z\,A/r, \qquad \hat\gamma = Z\gamma_K.$$

All unspecified tensor factors are identities.

Choose self-adjoint involutions Γ_0, Γ_1 on V, each with two-dimensional positive and negative eigenspaces, and assume

$$-1 \notin \operatorname{spec}(\Gamma_1\Gamma_0).$$

Extend Γ_i by zero on $V^\perp$, writing $C_i = \Pi\Gamma_i\Pi$. Define $\operatorname{sgn} A = 0$ on V, and impose

$$\chi_0\psi(0) = \psi(0), \qquad \chi_\ell\psi(\ell) = \psi(\ell),$$

where

$$\chi_0 = ZC_0 - S\operatorname{sgn} A, \qquad \chi_\ell = ZC_1 + S\operatorname{sgn} A. \tag{1}$$

Then:

1. The H^1 realization of $\mathcal{D}_r$ is self-adjoint with compact resolvent and anticommutes with $\hat\gamma$.
2. $\Pi \otimes 1$ reduces the operator, including its domain.
3. On this reducing subspace, the operator is exactly $D(\Gamma_0, \Gamma_1)$ of lemma 23.
4. Every eigenvalue on the complementary subspace satisfies

$$|\lambda| \geq \Delta_K/r. \tag{2}$$

Proof. Put $B = SA/r$. Since $ES = Z$,

$$\mathcal{D}_r = E(\partial_x + B).$$

The operators χ_i are self-adjoint involutions and anticommute with E. Their positive eigenspaces are therefore maximal isotropic for the boundary form

$$[\langle \psi, E\phi \rangle]_0^\ell.$$

On $V^\perp$, condition (1) selects the negative spectral subspace of B at 0 and its positive spectral subspace at ℓ. These are APS conditions with the inward-normal sign reversed between the two walls. On $\ker B = \mathbb{C}^2 \otimes V$, condition (1) supplies a Lagrangian subspace:

$$\Gamma_i \psi_L = \psi_L, \qquad \Gamma_i \psi_R = -\psi_R.$$

For q in the complementary domain, integration by parts gives

$$\|\mathcal{D}_r q\|^2 = \|q'\|^2 + \|Aq/r\|^2 + [\langle q, Bq \rangle]_0^\ell \geq \|q'\|^2 + \frac{\Delta_K^2}{r^2}\|q\|^2. \tag{3}$$

The boundary term is nonnegative by the spectral choices.

An eigenbasis of A decomposes the complementary problem into finite-dimensional interval problems with maximal isotropic endpoint conditions. Each is self-adjoint. Estimate (3) identifies the domain of their self-adjoint direct sum with the stated H^1 domain and gives compactness of its resolvent. The finite-dimensional tangential kernel contributes an ordinary self-adjoint interval problem. This proves self-adjointness, compactness, and (2).

Every boundary projector commutes with Π, proving reduction. Finally, $\{\gamma_K, A\} = 0$ implies $\{\hat{\gamma}, \mathcal{D}_r\} = 0$, and $\hat{\gamma}$ preserves both boundary subspaces. ■

Define

$$K_F = \frac{1}{2\ell}\log(\Gamma_1 \Gamma_0), \qquad D_F = \Gamma_0 K_F, \qquad R(x) = e^{xK_F},$$

with the principal logarithm. Let $P_\pm = (1 \pm \Gamma_0)/2$. The finite-window embedding is

$$W_\ell v(x) = \ell^{-1/2} \begin{pmatrix} R(x)P_+ v \\ R(x)P_- v \end{pmatrix}. \tag{4}$$

Direct differentiation gives

$$\mathcal{D}_r W_\ell = W_\ell D_F, \qquad \hat{\gamma} W_\ell = W_\ell \Gamma_0.$$

Thus the prescribed finite operator and grading are exact, provided $\Delta_K/r > \|D_F\|$. The next eigenvalue in absolute value is at least

$$\min\left\{\frac{\Delta_K}{r}, \frac{\pi - \ell\|D_F\|}{\ell}\right\}. \tag{5}$$

---

2. Proposition — Local walls and spectral walls have different scalar domain algebras.

Hypotheses. Use the product differential expression of Result 1.

For the local option, let $G_i(y)$ be smooth self-adjoint bundle involutions and impose

$$G_i \psi_L = \psi_L, \qquad G_i \psi_R = -\psi_R \quad \text{at } x = i.$$

Assume

$$[G_i, \gamma_K] = 0, \qquad [G_i, \sigma_A(y, \xi)] = 0 \quad (\xi \in T_y^* K), \tag{6}$$

where σ_A is the tangential principal symbol.

Then these local conditions give a self-adjoint elliptic realization with compact resolvent, and **every** smooth scalar function on $K \times I_\ell$ preserves its domain.

For the spectral option (1), the maximal smooth scalar $*$-algebra preserving the domain is

$$\mathcal{A}_{\mathrm{APS}} = \{f \in C^\infty(K \times I_\ell) : f(\,\cdot\,,0), f(\,\cdot\,,\ell) \text{ are constant on } K\}\,. \tag{7}$$

The same conclusion holds for the subalgebra pulled back from $S^4 \times I_\ell$.

Proof. For the local option, the boundary involution is ZG_i. It anticommutes with the normal matrix E. Condition (6) makes it anticommute with the principal symbol of the adapted tangential operator SA/r. Hence its positive eigenspace is an elliptic Lagrangian boundary condition. The corresponding local ellipticity and adjoint-domain criteria are described in Bär–Ballmann, Corollary 4.18 and Example 4.20. Primary reference

Scalar multiplication commutes pointwise with G_i, so it preserves the endpoint conditions and the H^1 domain. Moreover,

$$[\mathcal{D}_r, f] = E\,\partial_x f + Z[A, f]/r$$

is bounded for each $r > 0$.

For spectral walls, preservation by a scalar $*$-algebra is equivalent to

$$[M_{f_i}, \chi_i] = 0.$$

Independence of the spin matrices Z, S gives

$$[M_{f_i}, C_i] = 0, \qquad [M_{f_i}, \operatorname{sgn} A] = 0.$$

Since $C_i^2 = \Pi$, the first equation implies

$$[M_{f_i}, \Pi] = 0. \tag{8}$$

Consequently multiplication by f_i preserves $H^0(K, \mathcal{O}(1))$. If s is such a section, then

$$0 = \bar\partial(f_i s) = (\bar\partial f_i)s.$$

The sections of $\mathcal{O}(1)$ have no common zero, so $\bar\partial f_i = 0$ everywhere. A holomorphic function on compact connected $\mathbb{CP}^3$ is constant. Constants plainly preserve (1), proving (7). ■

The hypotheses in (6) are a sufficient local ellipticity condition. An arbitrary bundle grading supplies a pointwise isotropic condition, but self-adjointness on an H^1 domain and compact resolvent still require an ellipticity check. Likewise, prescribing a grading only on V leaves the boundary problem incomplete until a condition on $V^\perp$ is supplied.

---

3. Proposition — Compression of a local grading and the unchanged level-one window.

Hypotheses. Let G be a self-adjoint bundle involution acting by multiplication, and let

$$g = \Pi G \Pi|_V, \qquad Q_K(f) = \Pi M_f \Pi|_V.$$

Keep the level-one space and scalar representation of Example 16.9 unchanged.

Then

$$1_V - g^2 = ((1-\Pi)G\Pi)^*((1-\Pi)G\Pi). \tag{9}$$

In particular, g is a grading exactly when $[G, \Pi] = 0$. In that event,

$$[g, Q_K(f)] = 0 \quad \text{for every scalar } f. \tag{10}$$

For either the full $\mathbb{CP}^3$ scalar algebra or the S^4 pullback algebra, this forces

$$g = 1_V \quad \text{or} \quad g = -1_V. \tag{11}$$

Proof. Expand $\Pi G^2 \Pi = \Pi$ using $1 = \Pi + (1 - \Pi)$; this gives (9). Vanishing of its positive right-hand side implies $(1 - \Pi)G\Pi = 0$, and self-adjointness gives the other off-diagonal block.

Since scalar multiplication commutes with G, reduction then gives (10). The compressed $\mathbb{CP}^3$ functions generate $M_4(\mathbb{C})$. The compressed real S^4 functions generate $M_2(\mathbb{H})$, whose complex-linear commutant on $\mathbb{C}^4$ is also $\mathbb{C}1$. Thus a self-adjoint involution satisfying (10) is ± 1. ■

For **aligned local walls** with $[G, A] = 0$, the finite zero-mode window is explicitly

$$v \longmapsto \ell^{-1/2} \begin{pmatrix} P_+(g)v \\ P_-(g)v \end{pmatrix}.$$

The induced grading is g, and scalar compression is

$$P_+(g)Q_K(f)P_+(g) + P_-(g)Q_K(f)P_-(g) = Q_K(f).$$

By (11), the unchanged four-section window has trivial grading. Its generated algebra remains $M_2(\mathbb{H})$ or $M_4(\mathbb{C})$, according to the scalar algebra chosen. The globally constant choices $G = \pm 1$ give concrete self-adjoint examples with this window and a separated higher spectrum.

There is also an explicit local field illustrating failure of reduction. Add a trivial auxiliary $\mathbb{C}^4$ bundle and put

$$G(y) = \sum_{a=1}^{5} y_a \Gamma_a^{\text{aux}},$$

where $y \in S^4$ is the twistor projection and the Γ_a satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$. This is a local involution satisfying (6), but now the tangential kernel has dimension 16, and

$$g = \frac{1}{5} \sum_{a=1}^{5} \Gamma_a^{\text{section}} \otimes \Gamma_a^{\text{aux}}.$$

Using $\operatorname{tr}(\Gamma_a \Gamma_b) = 4\delta_{ab}$,

$$\operatorname{tr}(g^2) = \frac{16}{5} \neq 16.$$

Thus g is not a grading. Equation (9) even gives

$$\|(1 - \Pi)G\Pi\| \geq \frac{2}{\sqrt{5}},$$

independently of r. This is a bound on boundary-trace mixing; it does not identify the actual low spectral window of the coupled local problem.

A local $G(y)$ specifies a **field of gradings**. It does not select a marked point p. In the tautological quaternionic line bundle,

$$\Pi_y = \frac{1 + y_a \Gamma_a}{2},$$

the natural field $2\Pi_y - 1$ acts as $+1$ on the fibre $\operatorname{ran} \Pi_y$. By contrast, the finite grading

$$\Gamma_p = p_a \Gamma_a = 5Q_K(p \cdot y)$$

uses a separately chosen point p. Its action on global section coefficients is nonlocal.

The restriction proved here concerns the unchanged level-one space, a local multiplication grading, and its compression as an actual involution. A different coupled window, additional fibres, or spectral boundary conditions remain available directions.

4. Theorem — The composed compression map and the aligned-wall algebra.

Hypotheses. Use the exact spectral construction of Result 1. For a scalar multiplier $f(y,x)$, let

$$T_f(x) = Q_K(f(\,\cdot\,,x)), \qquad \mathcal{E}_0(T) = \frac{1}{2}(T + \Gamma_0 T \Gamma_0).$$

Use real-valued functions and real algebra generation when discussing $M_2(\mathbb{H})$.

Then the compression directly from the scalar cylinder to the finite window is

$$\boxed{\Phi_\ell(f) = \frac{1}{\ell}\int_0^\ell \mathcal{E}_0\big(R(x)^* T_f(x) R(x)\big)\, dx.} \tag{12}$$

It is independent of r.

For aligned walls $\Gamma_1 = \Gamma_0 = \Gamma_p$, direct scalar compression on the S^4 factor generates

$$\mathbb{R} \oplus \mathbb{R}. \tag{13}$$

If instead one first generates $M_2(\mathbb{H})$ and then applies the interval compression, the result is

$$\mathbb{H} \oplus \mathbb{H}. \tag{14}$$

Proof. Insert (4) on both sides of the multiplication operator. Orthogonality of the two interval spin components gives exactly (12).

For real S^4 functions, Example 16.9 implies

$$Q_K(C^\infty(S^4, \mathbb{R})) = \mathrm{span}_{\mathbb{R}}\{1, \Gamma_1, \dots, \Gamma_5\}. \tag{15}$$

Indeed all compressed functions are quaternionic Hermitian matrices; that space has real dimension six, and the displayed six matrices belong to the image and are independent.

For aligned walls $R = 1$. Clifford multiplication gives

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p.$$

Therefore the image of direct scalar compression is

$$\mathrm{span}_{\mathbb{R}}\{1, \Gamma_p\}.$$

This space is already the algebra $\mathbb{R} \oplus \mathbb{R}$, proving (13).

On the other hand,

$$\mathcal{E}_p(M_2(\mathbb{H})) = \{a \in M_2(\mathbb{H}) : [a, \Gamma_p] = 0\} = \mathbb{H} \oplus \mathbb{H},$$

proving (14). The noncommuting products needed to generate the quaternionic diagonal blocks have been inserted between the two compressions in this second procedure. ■

On the single $\mathbb{CP}^3$ factor, real scalar compression is onto the Hermitian 4×4 matrices. A chosen $2+2$ aligned grading therefore gives all Hermitian block-diagonal matrices and generates

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

For the **chosen two-sheet coarse correspondence** of Example 16.9, with

$$H_0 = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V), \qquad \Gamma_0(\xi, \eta) = (\Gamma_p \xi, -\eta \Gamma_p),$$

the same calculation gives

$$\boxed{\mathrm{alg}_{\mathbb{R}}(\text{direct aligned scalar compression}) = \mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).} \tag{16}$$

The V-side matrices act on the left of η and commute with its right grading, so that factor survives unchanged.

Endpoint restrictions do not change these finite-dimensional images. If a linear functional vanishes on compressions of functions supported in $0 < x < \ell$, its smooth matrix-entry density vanishes in the interior and hence everywhere. Finite-dimensional separation then gives equality with the unrestricted image, exactly as in Result 7 of lemma 20.

At the intermediate stage, however, spectral walls still give scalar endpoint values. Even after algebra generation, those values remain scalar. Enlarging them to the full stabilizers of Γ_0, Γ_1, as in lemma 23, is another change of the parent algebra. Equality of finite compression images does not identify the parent gauge groups or one-forms.

---

5. Theorem — Radius collapse, interval collapse, and controlled mixing.

Hypotheses. First use the exact construction of Result 1. Identify the varying Hilbert spaces by metric rescaling. For resolvent formulas, fix units in which the nonreal spectral parameter is i.

Then, at fixed ℓ,

$$\left\| (\mathcal{D}_r - i)^{-1} - \iota_\Pi (D(\Gamma_0, \Gamma_1) - i)^{-1} \iota_\Pi^* \right\| \le \frac{r}{\Delta_K}. \tag{17}$$

Thus radius collapse gives the entire matrix interval operator in generalized norm-resolvent sense.

If D_F is fixed, $\ell \to 0$, and

$$\Gamma_1(\ell) = e^{\ell \Gamma_0 D_F} \Gamma_0 e^{-\ell \Gamma_0 D_F},$$

then $r \to 0$ and $\ell \to 0$ decouple both towers and leave the finite operator D_F. No relation between the two rates is required in this exactly reducing construction.

Proof. The resolvent is block diagonal with respect to Π. Estimate (2) bounds its complementary block by r/Δ_K, proving (17). For the finite-window limit, (5) bounds the complementary resolvent by

$$\max\left\{ \frac{r}{\Delta_K}, \frac{\ell}{\pi - \ell \|D_F\|} \right\},$$

which tends to zero. ■

A useful mixing statement requires additional hypotheses. Suppose a transformed problem has a common domain and the form

$$\widetilde{\mathcal{D}}_r = \mathcal{D}_r + V_r, \qquad V_r = V_r^*, \qquad \|V_r\| \le b_r.$$

Write

$$v_r = \|(1 - \Pi) V_r \Pi\|.$$

For an eigenvector $\psi = p + q$, with $p = \Pi\psi$, $q = (1 - \Pi)\psi$, and $|\lambda| \le L$, assume

$$g_r = \Delta_K / r - b_r - L > 0.$$

Then

$$\boxed{\|q\| \le \frac{v_r}{g_r} \|p\|, \qquad \|\text{Schur correction at } \lambda\| \le \frac{v_r^2}{g_r}.} \tag{18}$$

Proof. The complementary eigenvalue equation gives

$$q = -(\mathcal{D}_{r,Q} + V_{QQ} - \lambda)^{-1} V_{QP} p.$$

The inverse has norm at most g_r^{-1}. Substituting this formula into the p-equation gives the second bound. ■

For bounded b_r, v_r, both errors are $O(r/\Delta_K)$. If a common-domain construction instead produces terms of order $1/\ell$, sufficient conditions include

$$\frac{rv_r}{\Delta_K} \to 0, \qquad \frac{rv_r^2}{\Delta_K} \to 0,$$

together with convergence of the Π-diagonal perturbation.

An arbitrary local $G_i(y)$ generally fails $[G_i, \Pi] = 0$. Although the differential expression commutes with Π, its domain then does not. Such a boundary change is not automatically a bounded perturbation satisfying (18). The explicit local field in Result 3 has nonvanishing boundary mixing at every radius. No convergence to the prescribed two-gradings operator is established here for that nonreducing local problem.

At the level of observables, radius collapse leaves (12) unchanged. For an x-independent $T = Q_K(f)$,

$$\|\Phi_\ell(f) - \mathcal{E}_0(T)\| \le \ell \|D_F\| \, \|T\|. \tag{19}$$

If T is even for Γ_0, the stronger formula of Theorem 22.4 applies:

$$\Phi_\ell(f) = \operatorname{sinc}(\ell \operatorname{ad}_{D_F})(T),$$

$$\|\Phi_\ell(f) - T\| \le \frac{\ell^2}{6} \|[D_F, [D_F, T]]\|. \tag{20}$$

Equation (19) follows by integrating $\|R^* T R - T\| \le 2x \|D_F\| \|T\|$; equation (20) follows by taking the even part of the conjugation series.

Consequently there is convergence of the **compression maps on their scalar operator systems**. The limit is $\mathcal{E}_0 Q_K$. It does not turn that image into every element of $\mathcal{E}_0(P)$; Result 4 computes the difference explicitly. Algebra generation remains a separate operation.

---

6. Proposition — The second sheet, reality, even algebra, and connector remain prescribed data.

Hypotheses. Consider the scalar cylinder construction above and, separately, its extension using the two-sheet coarse correspondence of Example 16.9.

The resulting distinctions are:

Structure	What follows under the stated choices
Second sheet and the two Hom spaces	They are inputs in the coarse correspondence. A single four-section cylinder window does not supply them.
$J_1^2 = -1$, $J_1 \Gamma_i = -\Gamma_i J_1$	These are additional compatibility conditions on the prescribed gradings and antiunitary.
$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$	This is the derived even part after choosing the quaternionic real form, the grading, and intermediate algebra generation. Direct aligned scalar compression gives (16).
Connector and $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	The connector's lines and coupling are chosen. Their intertwining algebra is then derived.

Proof. The one-factor construction starts with $V = \mathbb{C}^4$, and (4) has domain V. The two-sheet construction instead starts with

$$(W \otimes \overline{V}) \oplus (V \otimes \overline{W}).$$

Geometrically, its analytic realization uses product factors and a restricted scalar algebra, as Theorem 16.7 specifies. It is additional correspondence data over the scalar spaces.

For reality, suppose an antiunitary $\mathcal{J}_1$ on the tangential Hilbert space satisfies

$$\mathcal{J}_1^2 = -1, \quad \mathcal{J}_1 A = -A\mathcal{J}_1, \quad [\mathcal{J}_1, \gamma_K] = 0,$$

and its restriction J_1 to V anticommutes with both Γ_i. Then

$$\widehat{J} = E\mathcal{J}_1$$

preserves (1) and obeys

$$\widehat{J}^2 = 1, \qquad [\widehat{J}, \mathcal{D}_r] = 0, \qquad \widehat{J}\hat{\gamma} = -\hat{\gamma}\widehat{J}.$$

On the window,

$$J_F = \Gamma_0 J_1.$$

These identities follow directly from the stated commutation relations. They specify sufficient conditions; the natural $\mathcal{O}(1)$ cylinder has not supplied such an extension or its order-zero property here.

In particular, the quaternionic antiunitary j used to recognize $M_2(\mathbb{H})$ commutes with the Γ_a. It does not itself satisfy the required anticommutation with Γ_p.

The even-algebra assertion is the centralizer calculation in Result 4 after intermediate generation.

Finally, choose

$$w \in W_+, \quad v \in V, \quad \|w\| = \|v\| = 1, \quad \kappa \neq 0,$$

and prescribe

$$T_\kappa \xi = \kappa(w^* \xi v)\, v w^*.$$

Its intertwining conditions require the lines $\mathbb{C}w, \mathbb{C}v$ to be reducing, with the same scalar action. A quaternion preserving $\mathbb{C}w$ lies in its chosen copy of $\mathbb{C}$; $v^\perp$ carries arbitrary $M_3(\mathbb{C})$, and W_- carries arbitrary $\mathbb{H}$. Hence

$$\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m)\} \simeq \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$$

as in Result 2 of lemma 20. ■

A marked $p \in S^4$ fixes a quaternionic line, but choosing w also requires a complex line within it. The second line v, the coupling, and an interface realizing the connector remain necessary. Scalar agreement at marked points has unchanged compression image by Result 7 of lemma 20, so it supplies none of this selection by itself.

---

7. No-go theorem — Recheck of the radial statement in lemma 22.

Hypotheses. Let the parent be exactly the one-interval model of lemma 22: fixed wall coupling X, zero bulk mass, the even algebra $\mathcal{A}_\ell$, scalar inner fluctuations Eh with $h^* = -h$, and the **moving spectral window**. For real fluctuations, include the conjugate one-form and assume the stated order-zero representation.

Then every fluctuated finite operator has the same singular values as D_F. This is an exact finite-amplitude spectral statement; it uses no action or loop expansion.

Proof. Solve

$$u' = -hu, \qquad u(0) = 1, \qquad w = u(\ell),$$

and set $v(x) = u(x)w^*$. Then

$$v(\ell) = 1, \qquad v(0) = w^*.$$

Since h is even, w preserves the boundary splitting at 0; the identity at ℓ preserves the fixed coupled wall. Thus v preserves the original domain, and

$$v D_X v^* = D_X + Eh.$$

For the real fluctuation, the analogous implementing unitary is

$$vJ_F vJ_F^{-1};$$

order zero separates the two factors in its derivative.

The full operators are unitarily equivalent, so their moving spectral windows are unitarily equivalent. In the finite identifications of lemma 22, the operators are $w^* D_F w$, or $U_w^* D_F U_w$ in the real case. Their singular values coincide. ■

The moving-window hypothesis matters. For example, with $D_F = m\sigma_1$ and $h = i\sigma_3$, fixed compression gives

$$Q_\ell(Eh) = -\frac{1-\cos(2m\ell)}{2m\ell}\sigma_2.$$

Adding a multiple of this operator to the fixed $m\sigma_1$ changes its singular values at second order.

The verdict agrees with the author's amended theorem and Terra's independent endpoint-preserving proof. In the usual nondegenerate Standard Model representation, the normalized shifted Higgs quaternion lies on the unit-quaternion orbit; its norm is fixed.

The known escapes are a variable wall coupling, odd connection directions, suitable bulk masses, additional internal directions with nontrivial connection data, or additional elementary or composite fields. A quantum effective action on the same configuration space does not add a radial coordinate. The proposed $K \times I_\ell$ parent has tangential directions and tangential one-forms, so this theorem does not decide its full fluctuation space.

---

8. Lemma — A logarithmic concavity property of the one-loop integral.

Hypotheses. Let

$$\mathcal{J}(z) = \int_0^\infty \frac{x^3}{\sinh^2 x + z}\, dx, \qquad z > 0,$$

and define

$$e(z) = -\frac{z\mathcal{J}'(z)}{\mathcal{J}(z)}.$$

Then $0 < e(z) < 1$, and e is nondecreasing.

Proof. Positivity and

$$0 < \frac{z}{\sinh^2 x + z} < 1$$

show that $e(z)$, expressed as a weighted average of this fraction, lies strictly between zero and one.

Put $x = e^v$, $z = e^u$. Then

$$\mathcal{J}(e^u) = \int_{\mathbb{R}} \exp\left(4v - \log(\sinh^2(e^v) + e^u)\right) dv. \tag{21}$$

The function

$$b(v) = 2\log\sinh(e^v)$$

is convex, because, with $x = e^v$,

$$b''(v) = 2x(\coth x - x\ \operatorname{csch}^2 x) > 0.$$

The last inequality is equivalent to $\sinh x \cosh x > x$.

The function $(u, v) \mapsto \log(e^{b(v)} + e^u)$ is therefore convex. The integrand in (21) is jointly log-concave. Marginal integration preserves log-concavity by Prékopa's theorem, so $u \mapsto \log \mathcal{J}(e^u)$ is concave. We use precisely Theorem 6, p. 342, of Prékopa's original paper.

Finally,

$$\frac{d}{du}\log \mathcal{J}(e^u) = -e(e^u).$$

Concavity makes this derivative nonincreasing, proving the assertion. ■

---

9. Theorem — The massless one-loop potential has no interior local minimum.

Hypotheses. Let the action be exactly the one-loop determinant displayed in lemma 23: a flat interval, zero bulk masses, frequency-one fermions, the stated charged and neutral gauge towers and multiplicities, and the prescribed connector wall condition. Assume that no fermion connector modifies the displayed fermion tower, and add no wall potential or other angle-dependent term.

Let $n_f > 0$, set $t = \sin^2\theta$, and write

$$\mathcal{V}(t) = 8\pi^2\ell^4 V(\theta).$$

Then:

1. $t = 1$, equivalently $\theta = \pi/2$, is the unique global minimum.
2. If $n_f \geq 54/5$, the potential is strictly decreasing on the interior.
3. If $0 < n_f < 54/5$, there is exactly one interior stationary point, and it is a strict maximum.

Proof. Define

$$F(z) = \int_0^\infty x^3 \log\left(1 + \frac{z}{\sinh^2 x}\right) dx, \qquad F'(z) = \mathcal{J}(z).$$

The exact potential is

$$\mathcal{V}(t) = 3F(2t(1-t)) + \frac{3}{2}F\Big(\frac{16}{5}t(1-t)\Big) - n_f F(t). \tag{22}$$

Its gauge terms are nonnegative and vanish at $t = 1$. Since F is strictly increasing, the fermion term is uniquely minimized there. This proves item 1.

Differentiate:

$$\mathcal{V}'(t) = (1-2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\Big(\frac{16}{5}t(1-t)\Big)\right] - n_f\mathcal{J}(t). \tag{23}$$

For $t \geq 1/2$, this is strictly negative.

For $0 < t < 1/2$ and $c \geq 2$, put

$$R_c(t) = (1-2t)\frac{\mathcal{J}(ct(1-t))}{\mathcal{J}(t)}.$$

Let $a = ct(1-t)$. Then $a \geq t$, and Result 8 gives $e(a) \geq e(t)$. Consequently,

$$\begin{aligned}\frac{d}{dt}\log R_c(t) &= -\frac{2}{1-2t} - \frac{1-2t}{t(1-t)}e(a) + \frac{e(t)}{t} \\ &\leq -\frac{2}{1-2t} + \frac{e(t)}{1-t} \\ &< -\frac{2}{1-2t} + \frac{1}{1-t} < 0.\end{aligned}$$

Thus every R_c is strictly decreasing. Since $\mathcal{J}(0)$ is finite and positive,

$$R_c(0+) = 1, \qquad R_c(1/2-) = 0.$$

Therefore

$$H(t) = 6R_2(t) + \frac{24}{5}R_{16/5}(t)$$

decreases strictly from 54/5 to zero. Equation (23) becomes

$$\mathcal{V}'(t) = \mathcal{J}(t)\,[H(t) - n_f] \quad (0 < t < 1/2).$$

For $n_f \geq 54/5$, its sign is negative. For $0 < n_f < 54/5$, it changes sign exactly once, from positive to negative. Since $t = \sin^2\theta$ is strictly increasing on the interior, the same stationary-point classification holds in θ. ■

This proves the massless statement for the entire parameter range, including the previously numerical range. No numerical computation is used in this proof.

The recorded author and Terra referee calculations agree on the sampled massless cases. The logarithmic-concavity proof above is an additional single-agent derivation; it has not received a fresh independent model review in this turn. The recorded agreement therefore should not be described as a multi-model review of this new proof.

The escapes remain changes to the stated determinant: frequency-two fermions, modified form factors through warping or other geometry, suitable bulk-mass sectors, wall potentials, and wall kinetic or interaction terms. The full effective action of the proposed $K \times I_\ell$ parent, including its K-tower, has not been computed here.

---

10. Proposition — Scope of the bulk-mass scans.

Hypotheses. Retain the flat gauge terms of (22) as an independent assumption. Replace the fermion factor by the expression in lemma 23:

$$1 + t(G(x) - 1), \qquad G(x) = \frac{(w_Q \coth w_Q - c_Q)(w_U \coth w_U - c_U)}{x^2}, \quad w_i = \sqrt{x^2 + c_i^2}.$$

If $c_U = -c_Q = c$, then the potential is finite and $t = 1$ is its unique global minimum for every $n_f > 0$. The reported absence of additional interior **local** minima at nonzero c remains a numerical statement here.

If $a = c_Q + c_U \neq 0$, the displayed unrenormalized integral has angle-dependent ultraviolet divergences.

Proof. For opposite masses,

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w}, \qquad w = \sqrt{x^2 + c^2}.$$

The added fraction is positive and decays exponentially at infinity. The negative fermion logarithm is strictly decreasing in t. The gauge logarithms are nonnegative and vanish at $t = 1$, proving the global-minimum assertion.

For general masses,

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

and hence

$$\log(1 + t(G - 1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by the integration measure $x^3 dx$ produces angle-dependent divergences. A finite potential therefore requires a specified completion, regulator, and counterterms. ■

The opposite-mass scan in lemma 23 samples finitely many masses, multiplicities, and angles. It cannot exclude unsampled local minima. General bulk masses and altered gauge form factors remain directions for constructing an interior vacuum; these conclusions leave that question open.

Consequence for T0 and T0b

For T0, a commutative scalar representation with prescribed spectral walls gives an exact interval operator window and controlled tower decoupling, while local walls preserve the full scalar algebra under the ellipticity hypotheses above. The derivation of a nontrivial local grading and the desired algebra from one coupled scalar window remains open, and intermediate algebra generation must be stated explicitly. For T0b, the

second sheet, compatible reality, quaternionic grading choice, and connector remain inputs; their algebraic consequences can be derived once those choices are supplied. Neither the radial theorem nor the specified one-loop potential closes the commutative-parent problem.

Referee amendments

From Terra 6.1. These apply to the text above.

- **Result 2.** Write G_0 at x = 0 and G_l at x = l. Equation (6) supplies the principal-symbol condition of Bar-Ballmann, Corollary 4.18 and Theorem 4.11; their Example 4.20 assumes more. On the unaugmented Dolbeault bundle, the Clifford representation is irreducible, so a fibre endomorphism commuting with every sigma_A is scalar, and a smooth involution of this form is +1 or -1. Additional fibres and other local elliptic projectors lie outside this sufficient criterion.
- **Result 3.** The statement on marked points concerns the natural field displayed there. A local field can encode a chosen point: on the auxiliary bundle, G_p(y) = p_a Gamma_a^aux is a local constant field whose coefficients contain p.
- **Result 4.** Equation (16) has the two Hom spaces and the two scalar algebras of Example 16.9 as inputs. The endpoint argument proves equality of the finite-dimensional linear images, and does not identify the parent algebras, norms, gauge groups or one-forms.
- **Result 5, gap in the final assertion.** Convergence of compression maps needs a common domain and a collapse map. With aligned walls, D_F = 0 and f_l(y, x) = 6 (x/l)(1 - x/l), both endpoint values vanish while Phi_l(f_l) = 1. A correct form: for F(y, s) fixed on K x [0, 1] and f_l(y, x) = F(y, x/l), Phi_l(f_l) tends to E_0 Q_K(int_0^1 F(., s) ds). For x-independent ambient multipliers the limit is E_0 Q_K, by (19). The rate discussion after (18) also needs control of the denominator: a sufficient addition is r (b_r + L) <= (1 - epsilon) Delta_K. The case b_r = O(1/l) follows from the stronger estimate |D_{r,Q}| >= (Delta_K$^{2/r}$2 + pi^2/(4 l$^{2)}$){1/2}, from the one-ended Poincare inequality on the positive and negative B-spectral components.
- **Result 6.** At the finite level, if the gradings are Gamma_p, Gamma_q, a unit n orthogonal to p and q gives J_1 = Gamma_n j, with J_1^2 = -1 and the required anticommutation. Extending this choice to the parent remains a separate task.
- **Lemma 8 and Theorem 9.** An independent check of the key monotonicity, without log-concavity or Prekopa's theorem: with rho(s) = (arsinh sqrt(s))^3/(2 sqrt(s (1 + s))), J(z) = int_0^infinity rho(s)/(s + z) ds; with L(v) = log rho(e^v), L'(v) = 3 tanh(x)/(2x) - (1 + tanh^2 x)/2 for v = 2 log sinh x, which is strictly decreasing. For z = e^u, with the probability density in y proportional to e^y rho(e^{u+y})/(1 + e^y), e(e^u) is the mean of 1/(1 + e^y), and its derivative in u is Cov(1/(1 + e^y), L'(u + y)) > 0, as both are strictly decreasing. Theorem 9 is for theta in [0, pi/2]. At theta = pi/2 the determinant has its continuous value, while the principal-logarithm window is degenerate. Its hypotheses are the flat massless one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no other angle-dependent term.
- **Result 10.** The divergence is stated for nonzero fermion multiplicity, with no cancellation by additional sectors. The opposite-mass scan in lemma 23 is a numerical result on its sampled parameters.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept: E_p(Gamma_a) = p_a Gamma_p to 10^-15, which gives equation (13); tr(g^2) = 16/5 for the auxiliary field of Result 3; on grids, e(z) lies in (0, 1) and is nondecreasing for z in [10^-3, 50]; H(t) is strictly decreasing on (0, 1/2), with H near 54/5 at small t and near 0 at t = 1/2; for n_f in {1, 3, 6, 8, 10}, H - n_f has one sign change, with the stationary angle 0.34 at n_f = 8, which agrees with the referee of lemma 23. Results 1, 2, 5 and 6 were read and not separately computed.

Part 25

The weak-boson sum rule and the effective dimension of the interval

For STRATEGY T5b, T13 and T1, following lemma 18, lemma 22 and lemma 23. Proved by Claude Opus 5.5, with checks in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and two gaps as first written, and amendments, all incorporated below.

Outcome. Unfolded, the interval parent is a circle of length 2l, and the parent operator is -i d/dx (x) 1 + 1 (x) D_F: the tower is D_F shifted by the momenta k_n = n pi/l. So the exact quadratic kernel of lemma 18 applies termwise over the tower. For small masses, at leading order, for the heat cutoff, the Taylor mass of a vector with constant profile in the transported frame is

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2 / ||T||^2,      d_eff(s) = d log S / d log s,      S(s) =
↪  sum_{n in Z} exp(-n^2 pi^2/s^2),   s = l Lambda,
```

in Hilbert-Schmidt norms, where d_eff is the spectral dimension at the scale of the cutoff of the unfolded circle, equivalently of the paired Neumann and Dirichlet interval: 0 for l Lambda « 1 and 1 for l Lambda » 1. For the Standard Model representation, at the same order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,       m_t/M_W = (8/(3 - d_eff))^{1/2} for the top alone,
```

with $M_W^{2/M_Z}2 = 5/8$ for all s. So the relation 8 M_W^2 = sum N_f m_f^2 of the asymptotic spectral action is the zero-dimensional value, with m_t/M_W = sqrt(8/3). A resolved interval gives 12 M_W^2 = sum N_f m_f^2, with m_t = 2 M_W, as five-dimensional covariance requires for this operator on the circle. The endpoint m_t = 2 M_W is known in flat gauge-Higgs models with suitable representations (references below); in the inspected records the referee did not find the coefficient (3 - d_eff)/24 or its crossover, and priority remains open pending a wider search. The factor 3/2 in Proposition F of the paper, the finite-action coefficient relative to the Yang-Mills commutator quotient, is the ratio of these two endpoints. The crossover is at l Lambda about 2.

Unfolding

Let D_X = E d/dx be the parent of lemma 22, and let D_0 + E Gamma_F D_F be its form after the transport of Corollary 22.2, where D_0 is the operator with X = 0.

Proposition 25.1. On the circle x in [-l, l] with periodic conditions, for psi in the domain of D_0, put

```
Psi_Q(x) = Q_L(|x|) + i sgn(x) Q_R(|x|),      Psi_U(x) = U_R(|x|) - i sgn(x) U_L(|x|).
```

After multiplying by 1/sqrt(2), this is a unitary from the parent Hilbert space onto L^2(circle) (x) (H_+ (+) H_-), which maps the H^1 domain of D_0 onto the periodic H^1 domain, and it carries D_0 + E Gamma_F D_F to

```
-i d/dx (x) 1 + 1 (x) D_F.
```

So the signed spectrum is k_n + mu_i, with k_n = n pi/l for n in Z and mu_i the eigenvalues of D_F. An even operator T = T_Q (+) T_U on H_F, constant in x in this transported frame and acting identically on both spin components, is carried to 1 (x) T.

Proof. Since Q_R(0) = Q_R(l) = 0 and U_L(0) = U_L(l) = 0, the functions Psi are continuous and periodic, and every smooth function on the circle arises from its even and odd parts. For x > 0, -i d/dx Psi_Q = Q_R' - i Q_L' = (D psi)_L + i (D psi)_R on Q, and -i d/dx Psi_U = -i U_R' - U_L' = (D psi)_R - i (D psi)_L on U, which are the transforms of D_0 psi; x < 0 is the same with the odd parts. For the connection term, with M* the block of D_F, the term E Gamma_F D_F sends Psi_Q to M* Psi_U and Psi_U to M Psi_Q. ■

In this form D_F is a constant connection on the circle, with holonomy exp(2 i l D_F) up to the convention for parallel transport, and the two walls are the fixed points of the reflection that carries D_F to -D_F. With U = exp(-i pi Gamma_0/4), the identity of lemma 23 reads Gamma_1 Gamma_0 = exp(2 l Gamma_0

D_F) = U exp(2 i l D_F) U*. This form of the interval as a circle with a reflection is the usual orbifold description. It also gives a one-line proof of the spectrum in Theorem 22.1.

The frame. By Corollary 22.2, D_X = R (D_0 + E Gamma_F D_F) R*, with R(x) = exp(x Gamma_F D_F). A generator T constant after transport is R(x) T R(x)* in the original frame of D_X, which generally varies with x and has an odd bulk component. Conversely, a constant generator in the original frame need not preserve the coupled wall, if it is broken. So the field used here requires an admissible enlarged gauge algebra, as in the prealgebra P of lemma 23, and it is generally not a field of the even algebra of lemma 22.

The kernel over the tower

Let A_mu(y) = a_mu(y) T on M^4 = R^4, with T even and constant along x, and consider D = -i gamma^mu (d_mu + i A_mu) + gamma_5 V with V = -i d/dx (x) 1 + 1 (x) D_F. Theorem 1 of lemma 18 has this form, with V_0 = V, whose eigenvalues are k_n + mu_i, and T_{n i, n' j} = delta_{n n'} T_{ij}.

Theorem 25.2. With d_ij = mu_i - mu_j, for the heat cutoff t = 1/Lambda^2, the kernel is

```
C_t(q^2) = (1/(8 pi^2)) sum_{n in Z} sum_{ij} |T_ij|^2 [ (2/t + d_ij^2 + q^2) I_t((k_n + mu_i)^2,
↪  (k_n + mu_j)^2; q^2) - (e^{-t (k_n + mu_i)^2} + e^{-t (k_n + mu_j)^2})/t ],
```

with I_t as in lemma 18. The sum converges, and it holds without expansion in D_F. Expanding in D_F, with C_t = C_0 + q^2 C_1 + …, and with t = 1, so with k_n = n pi/(l Lambda) dimensionless,

```
C_0 = (1/(8 pi^2)) ||[D_F, T]||^2 sum_n e^{-k_n^2} (1 - (2/3) k_n^2) + O(||D_F||^4),
C_1 = (1/(8 pi^2)) (2/3) ||T||^2 sum_n e^{-k_n^2} + O(||D_F||^2).
```

So, with <k^2> = sum k_n^2 e^{-k_n^2} / sum e^{-k_n^2}, at fixed s and for fixed nonzero T,

```
M^2 = C_0/C_1 = (3/2 - <k^2>) ||[D_F, T]||^2/||T||^2 + O(||D_F||^4 / Lambda^2).
```

Proof. V and T split into finite-dimensional blocks indexed by n, so equation (1) of lemma 18 applies to symmetric finite sets of blocks, and the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify the Taylor expansion under the sum. Terms odd in k cancel between n and -n: exactly, the bracket at q = 0 is d_ij^2 int_0^1 E(u) [1 - u(1-u)(2k + mu_i + mu_j)^2] du, with E(u) = exp(-(1-u)(k + mu_i)^2 - u (k + mu_j)^2), whose next term is d_ij^2 e^{-k^2} ((2k^3 - 5k)/3)(mu_i + mu_j), and the bracket at order q^2 begins e^{-k^2} (2/3 - (2k/3)(mu_i + mu_j)). For the leading term of the expansion, put a_i = (k + mu_i)^2 = k^2 + p mu_i + mu_i^2 with p = 2k. To second order in mu,

```
I(a_i, a_j) = e^{-k^2} [1 - (p/2)(mu_i + mu_j) - (mu_i^2 + mu_j^2)/2 + (p^2/6)(mu_i^2 + mu_j^2 +
↪  mu_i mu_j)],
```

and e^{-a_i} = e^{-k^2} [1 - p mu_i - mu_i^2 + p^2 mu_i^2/2]. Therefore

```
2 I(a_i, a_j) - e^{-a_i} - e^{-a_j} = -(p^2/6) d_ij^2 e^{-k^2},
```

and adding d_ij^2 I gives d_ij^2 e^{-k^2} (1 - p^2/6) = d_ij^2 e^{-k^2} (1 - (2/3) k^2). Finally, sum_ij |T_ij|^2 d_ij^2 = ||[D_F, T]||^2. For C_1, at D_F = 0 the bracket in equation (3) of lemma 18 has d_ij = 0 and integrates to 2/3. ■

Differentiating directly, <k^2> = (1/2) d log S/d log s = (s/2) d log S/ds, with S(s) = sum_n exp(-n^2 pi^{2/s}2). This S is the heat trace of the Dirac operator of the unshifted circle of length 2l at t = 1/Lambda^2, equivalently the paired Neumann and Dirichlet trace on the interval, with S_N = (S + 1)/2 and S_D = (S - 1)/2 separately. So d_eff = d log S/d log s = -2 d log Tr e^{-t D^2}/d log t is its spectral dimension at that scale. For nonzero D_F, the parent trace has e^{-t (k_n + mu_i)^2}, and S is the unperturbed trace used in the leading expansion. Hence, at leading order,

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2/||T||^2.
```

The two limits. We first take the coefficient at small D_F/Lambda at fixed s, and then the limit in s. Explicitly, d_eff ~ (4 pi$^{2/s}$2) exp(-pi$^{2/s}$2) as s -> 0, and d_eff = 1 - 4 s^2 exp(-s^2) + O(s^2 exp(-2 s^2)) as s -> infinity.

1. For s « 1, only n = 0 contributes. Then d_eff = 0, the factor is 3/2, and this is the towerless regime of lemma 18 and Proposition F of the paper.
2. For s » 1, the sum is a Gaussian integral, with <k^2> = 1/2. Then d_eff = 1, and the factor is 1. This is forced by covariance on the unfolded circle: V is the fifth component of a five-dimensional Dirac operator with constant A_5 = D_F, and the local five-dimensional heat coefficient contains F_{MN} F^{MN}, with F_{5 mu} = i [D_F, A_mu] up to convention and with the same coefficient as F_{mu nu}^2. So the mass is the Hilbert-Schmidt commutator norm alone, as at small angle for the Yang-Mills parent of Theorem 23.6. On a general space with boundary, the bulk coefficient alone does not exclude boundary contributions; here the operator with these transported fields is unitarily equivalent to the circle problem, which is the reason they are absent.

In table form, at small masses:

s = l Lambda	0.3	1	1.5	2	2.5	3	5
(3 - d_eff)/2	1.5000	1.4990	1.3935	1.1414	1.0240	1.0022	1.0000

Corollary 25.3 (the sum rule). For the three-generation Standard Model representation, with the literal particle and antiparticle trace, with Dirac masses small compared with Lambda and with 1/l, and with T a charged generator of SU(2)_L, at leading order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,
```

which at d_eff = 0 is 8 M_W^2 = sum N_f m_f^2, and at d_eff = 1 is 12 M_W^2 = sum N_f m_f^2. For the top alone,

```
m_t/M_W = (8/(3 - d_eff))^{1/2},
```

from sqrt(8/3) = 1.633 at d_eff = 0 to 2 at d_eff = 1. The neutral direction scales with the same factor, so M_W$^{2/M_Z}$2 = 5/8 at all s to this order, and the massless directions keep zero mass coefficient, while their kinetic coefficients can change. These are leading-order statements: already in the towerless top-only family, M_W^2 = (3/8) m_t^2 - (23/128) m_t$^{4/\mathrm{Lambda}}$2 + … (equation (19) of lemma 18).

Proof. The trace gives ||T||^2 proportional to Q_11 = 12 and ||[D_F, T]||^2 proportional to K_11 = sum N_f m_f^2, which gives the coefficient. At d_eff = 0 this is the CCM relation (hep-th/0610241, Eq. (4.16)), as recovered in Proposition 4 and Remark 16 of lemma 18 at small x. The factor is common to all generators in Theorem 25.2. ■

In terms of Yukawa couplings, with m_t = y_t v/sqrt(2) and M_W = g v/2, this reads y_t^2 = 4 g^2/(3 - d_eff), so y_t = 2 g/sqrt(3) at d_eff = 0, the top-dominance value of CCM (Eq. (5.27) there), and y_t = sqrt(2) g at d_eff = 1.

Remark (positive Laplace cutoffs). For f(z) = int e^{-tz} dnu(t) as in equation (4) of lemma 18, with t measured in units of 1/Lambda^2, the same computation gives at leading order

```
M^2 = ((3 - d_nu)/2) ||[D_F, T]||^2/||T||^2,     S_nu(s) = int S(s/sqrt(t)) dnu(t),     d_nu = d
↪ log S_nu/d log s = 2 <t k^2>_nu,
```

where the average has weights dnu(t) e^{-t k_n^2}. Under the moment hypotheses there, dominated convergence gives the same endpoints, and the crossover and its corrections depend on nu. A general smooth cutoff, such as one flat near zero, need not belong to this class.

Scope

- This is the Taylor quotient of a prescribed constant-profile vector, at leading order in D_F, about a prescribed background and not a demonstrated vacuum. Taylor and pole masses differ in general. Calling it the leading term of a lowest-mode mass also needs an admissible gauge field, a positive kinetic form and an isolated lowest branch; on the circle, translation invariance separates the zero Fourier sector, within which the generator matrix must still be diagonalized. For finite angles in the Yang-Mills parent, the exact masses are in Theorem 23.6, which expands as M_W^2 = 2 m^2 - (4/3) m^4 l^2 + … and M_Z^2 = (16/5) m^2 - (64/75) m^4 l^2 + …, in agreement with d_eff = 1 at leading order. That theorem is for its own Yang-Mills boundary problem, and it does not establish the full finite-cutoff gauge spectrum of the spectral action.
- T must be even and constant in the transported frame, so it preserves both walls there, and it needs the enlarged gauge algebra noted above. Vectors that are Dirichlet at a strict wall, as in Proposition 22.8, are a different boundary problem.
- All fields have flat profiles on one interval, with no bulk masses. In particular, for the top-only statement, the kinetic term counts all twelve doublets of three generations on the same interval, with the same S. With bulk masses to localize light fields, the spectra and overlaps change, and the kinetic coefficient must be recomputed from all eigenstates and overlaps.
- The spectral action is taken as a function of the fermion operator at a matching scale, with no running.
- For a boundary with reflection, the walls contribute heat-kernel boundary terms in general. Here the unfolding removes them for this operator and these generators.
- Other priority: the endpoint m_t = 2 M_W appears at tree level in Cacciapaglia, Csaki and Park, Fully radiative electroweak symmetry breaking, arXiv:hep-ph/0510366, section 5, for a suitable rank-four representation, and a limit approaching 2 M_W in Adachi and Maru, arXiv:1804.06012, Eq. (3.23). These are representation-dependent mechanisms, and they do not give the spectral-action crossover. Universal propagation alone does not impose it in ordinary universal extra dimensions (Cordero-Cid et al., arXiv:1108.2926, section 2).

Reading

This connects three items of the project.

1. *The emergence of a dimension.* The heat dimension of the interval, which distinguishes the labels 10 and 11, enters the mass relation directly. As the cutoff resolves the interval, the coefficient moves from the finite-space value to the five-dimensional value.
2. *Proposition F and lemma 13.* The factor 3/2 is the finite-action coefficient relative to the Yang-Mills commutator quotient, and in this constant-profile calculation the two are the two ends of one function. Proposition F also contains a bounded-fermion against tower-scale obstruction and overlap conditions, which keep their separate hypotheses.
3. *The measured ratio.* With the Particle Data Group 2026 listings, m_t = 172.60 +- 0.27 GeV as a direct-measurement Monte Carlo mass, and M_W = 80.3625 +- 0.0077 GeV, the ratio is 2.148, which would formally correspond to d_eff = 1.27, outside the range. With a running mass of about 163 GeV (Yoon and Peskin, arXiv:1810.12352, section 3.3 and Eq. (23), for a pole mass of 173 GeV, with about 147 GeV at 2 TeV), it is 2.03, which corresponds to d_eff = 1.06. Dividing a running mass by an on-shell W mass is only a formal orientation and not a matching test at a common scale, and no running from the matching scale has been computed, so we draw no conclusion from this comparison. The known spectral-action value y_t = 2 g/sqrt(3) is the d_eff = 0 boundary condition. A tower near the weak scale, as in the separate unrefereed estimate of the steer note, item 9, would use the d_eff = 1 boundary condition y_t = sqrt(2) g, at a matching scale of that order, in a specified effective theory with thresholds and a renormalization prescription. In a one-loop orientation with universal bulk fields (the steer note, item 11, no referee), the running approaches an infrared quasi-fixed point near 211 to 232 GeV, nearly independently of the boundary value, so that family does not support the closeness of 2 M_W to the running top mass.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. The kernel of lemma 18, with quadrature in u, was summed over the tower, with explicit 4 x 4 level blocks for the operator on the interval, independently of the unfolding.

- Random complex K_1 and Hermitian T_Q, T_U on C^3, with ||K|| = 10^-3 min(s, 1): the ratio of C_0/C_1 to the Hilbert-Schmidt quotient agrees with 3/2 - <k^2> to six digits, for s in {0.3, 1, 1.5, 2, 2.5, 3, 5, 10}.
- Three generations, top only, with T = sigma_x/2 on the U-fields: m_t/M_W agrees with (8/(3 - d_eff))^{1/2} to six digits on the same grid, from 1.63299 at s = 0.3 to 2.00000 at s = 5.
- The neutral direction, with weights for lepton and colour: M_Z$^{2/m}$2 goes from 4.801 to 3.200 for universal mass, and M_W$^{2/M_Z}$2 stays at 0.625 to four digits.
- Finite mass in the top-only family: M$^{2/m}$2 at s = 5 is 0.2500, 0.2498, 0.2477, 0.2409, 0.2241, 0.1478 for m/Lambda = 0.02, 0.1, 0.3, 0.6, 1, 2, and at s = 0.5 it is 0.3749, 0.3732, 0.3591, 0.3145, 0.2277, 0.0555, the latter matching lemma 18.

Part 26

The interior vacuum and the matching scale in the flat parent

For STRATEGY T13, T5b and T12, following lemma 23, lemma 24, lemma 25 and the steer note, items 9 and 11. Proved by Claude Opus 5.5; the numerical statements are in Python and each is labelled. Refereed by Terra 6.1 (effort max, 3 Nov 2026; first report): three errors and two gaps as first written, and amendments. Proposition 26.3c and Result 26.8 were added after that report and answer its gap on the wall mass, and a second report corrected one error in the summary. The textual corrections of both are incorporated below, and the calculations they call for are listed under "Outstanding".

Outcome.

1. Pure frequency-2 fermion channels give the flat family an interior global minimum, and we classify it for n_2 >= n_c (Theorem 26.2). With frequency-2 channels alone it sits exactly at theta = pi/4, where M_KK = 4 M_W.
2. In the displayed channels of the multiplets containing Q_L and U_R, a pure channel carries an exact massless coloured exotic, and a wall mass of any finite positive size leaves a coloured state of charge 2/3 below M_W (Proposition 26.3c). So a top at 2 M_W in such a channel has a lighter partner, whose couplings are not analyzed. In E there is no such partner: m_1 = 2 M_W at theta = 1.1377, with M_KK = 444 GeV, and the vacuum then needs a separate sector.
3. In the step-threshold model of the parent, without frequency-2 multiplets, the left-right difference of the couplings runs only logarithmically (Proposition 26.5). The couplings then meet at mu_* = 2.2 x 10^14 GeV, with M_KK = 3.1 x 10^13 GeV (Result 26.6, one loop, conditional on the spectrum).
4. At a low tower, with the hypercharge coupling fitted by a wall kinetic term, the finite interval shifts M_W$^{2/M_Z}$2 below its four-dimensional value by 0.05 at the top-compatible point, and by less than 0.006 only above about 1.2 TeV (Result 26.8, tree level).
5. These give two regimes of one family, stated at the end with hypotheses and escapes.

Setting

We keep the flat interval of length l and the gauge content, couplings and connector wall of Theorem 23.6. A fermion channel is *pure of frequency q* when its two wall gradings differ by a rotation through q theta in the channel. Its spectrum is +-(n pi + q theta)/l, n in Z, so that its product over the tower, relative to theta = 0, is 1 + sin^2(q theta)/sinh^2(x), with x = p l.

The bimodule E = Hom(V, W) is pure with q = 1 (Proposition 23.7). Let n_q >= 0 be twice the number of physical Dirac species in such channels, each colour counted once. With t = sin^2(theta), u = t (1 - t), and F, J as in Results 8 and 9 of lemma 24, the one-loop potential of the form displayed in lemma 23 is

```
8 pi^2 l^4 V = 3 F(2u) + (3/2) F((16/5) u) - n_1 F(t) - n_2 F(4u),        (1)

dV/dt = (1 - 2t) G(u) - n_1 J(t),   G(u) = 6 J(2u) + (24/5) J((16/5) u) - 4 n_2 J(4u),        (2)
```

up to the factor 8 pi^2 l^4. Equation (1) assumes precisely these towers and multiplicities; a full representation can contain others. We use J(0) = (3/2) zeta(3) and J(1) = (9/8) zeta(3), so J(0)/J(1) = 4/3.

Lemma 26.1. Let R(u) = [6 J(2u) + (24/5) J((16/5) u)] / (4 J(4u)) on (0, 1/4]. Then R is strictly increasing, with R(0+) = 27/10 and R(1/4) = n_c, where

```
n_c = [6 J(1/2) + (24/5) J(4/5)] / (4 J(1)) = 2.8855033... (numerical quadrature).
```

Proof. For c < 4, d/du log[J(c u)/J(4u)] = [e(4u) - e(c u)]/u, with e(z) = -z J'(z)/J(z). By Result 8 of lemma 24 and the covariance argument in the referee amendment there, which is strictly positive, e is strictly increasing. So both ratios J(c u)/J(4u), for c = 2 and c = 16/5, strictly increase. ■

The classification

Theorem 26.2. Let n_1, n_2 >= 0 in the family (1), with theta in [0, pi/2].

1. *n_1 = 0.* The potential is a function of u.
   - If n_2 <= 27/10, it increases with u: theta = 0 and theta = pi/2 are the two global minima, degenerate at n_2 = 27/10, where the leading term is (99/25) theta^4 log(1/theta^2), and theta = pi/4 is the maximum.
   - If 27/10 < n_2 < n_c, there are exactly two global minima, at theta_* and pi/2 - theta_*, *with theta_* in (0, pi/4), and theta = pi/4 is a local maximum.
   - If n_2 >= n_c, theta = pi/4 is the unique global minimum, with a quartic leading displacement at n_2 = n_c.
2. *n_1 > 0 and n_2 >= n_c.* Let n_* = (4/3)(4 n_2 - 54/5).
   - If n_1 < n_*, there is exactly one stationary point in the open interval. It is the strict global minimum, and it lies in (pi/4, pi/2).
   - If n_1 >= n_*, *V decreases strictly on (0, pi/2), with unique global minimum at theta = pi/2, degenerate at n_1 = n_*.

The endpoints are stationary in theta in every case, since dt/d(theta) vanishes there. The range n_1 > 0, n_2 < n_c is not classified here.

Proof. Item 1: dV/du = 4 J(4u) (R(u) - n_2), so Lemma 26.1 gives the three cases. Item 2: since n_2 >= n_c >= R(u), G(u) = -4 J(4u) (n_2 - R(u)) <= 0 on (0, 1/4], with equality only at u = 1/4 and n_2 = n_c. For t <= 1/2, (2) gives dV/dt <= -n_1 J(t) < 0. For t > 1/2 write dV/dt = J(t) [Psi(t) - n_1], with Psi(t) = (2t - 1) |G(u)|/J(t). Here 2t - 1 increases, 1/J(t) increases, and |G(u)| increases with t, since u decreases with t and both factors J(4u) and n_2 - R(u) decrease in u. So Psi increases strictly from Psi(1/2) = 0 to Psi(1) = (4 n_2 - 54/5) J(0)/J(1) = n_*. ■

No numerical computation enters the proof, apart from the value of n_c. A Python check on a grid of 11600 pairs, n_2 from 3 to 60 and n_1 from 0 to 199, with theta on 20001 points, finds the stated stationary points in each case.

For the remaining integer counts: n_2 = 0 is Theorem 9 of lemma 24. For n_1 = 0 and n_2 = 1, 2, Theorem 26.2 gives both endpoints as global minima. For n_1 > 0 and n_2 = 1, 2, a scan of n_1 up to 200 (Python) finds no interior local minimum, which we have not proved. So at integer counts every interior minimum found has theta >= pi/4. The window 27/10 < n_2 < n_c, with its minima at small theta, contains no integer.

Proposition 26.2b (referee; no global minimum at small angle). For n_1 > 0 and 0 < t < 1/2, V(1 - t) - V(t) = -n_1 [F(1 - t) - F(t)] < 0, so no point of (0, pi/4) is a global minimum. With item 1 of Theorem 26.2 for n_1 = 0, this covers all integer counts. It does not exclude metastable minima at small angle in the cases left to scans. ■

Corollary 26.3 (the spectrum at the minimum). Write g(y) = arcsin(sin(y)/sqrt(2)), so that M_W l = g(2 theta) by Theorem 23.6, for theta in (0, pi/2), where M_W > 0. On [pi/4, pi/2), m_1 l = theta and m_2 l = pi - 2 theta for pure channels.

1. At theta = pi/4, the minimum for n_1 = 0 and n_2 >= n_c: M_KK := pi/l = 4 M_W, m_1 = M_W, m_2 = 2 M_W, tan(M_Z l) = 2 and M_W$^{2/M_Z}$2 = 0.5032…
2. As theta increases from pi/4 to pi/2, m_1/M_W increases from 1 to infinity and m_2/M_W decreases strictly from 2 to sqrt(2).
3. M_KK >= 4 M_W for all theta in (0, pi/2), and M_KK/M_W grows without bound as theta -> pi/2 and as theta -> 0.

Proof. Item 1 follows from Theorem 23.6. For item 2, g'(y) = cos(y)/(1 + cos^2 y)^(1/2) is positive and decreasing on (0, pi/2), so g is concave with g(0) = 0, and y/g(y) increases. With y = pi - 2 theta, this gives m_2/M_W = y/g(y). Item 3 follows from sin^2(M_W l) <= 1/2. ■

Evaluating the minimum (Python, on the root given by the proof, recomputed by the referee), with M_W = 80.36 GeV:

n_1	n_2	theta_*	M_KK (GeV)	m_1/M_W	m_2/M_W	M_W$^{2/M_Z}$2
0	>= 3	0.7854	321	1	2	0.503
6	8	0.941	342	1.27	1.71	0.525
14	8	1.138	444	2.00	1.52	0.576
16	8	1.186	491	2.31	1.50	0.586
16	6	1.446	1437	8.23	1.42	0.621
48	12	1.502	2597	15.5	1.42	0.624

A frequency-1 top with m_1/M_W = 2 needs theta = 1.13774, on the line n_1 = 2.71387 n_2 - 7.71725, with M_KK = 444 GeV. A larger tower needs n_1 close to n_* from below. With even counts this reaches 29 TeV at (n_1, n_2) = (604, 116), and the frequency-1 fermions then sit near M_KK/2, as in item 2 of Theorem 23.6.

Remark (the quadratic coefficient). The coefficient of theta^2 in (1) is (54/5 - n_1 - 4 n_2) (3/2) zeta(3), which does not vanish for integer n_1, n_2. So in (1) the quadratic coefficient has no exact cancellation. This concerns that coefficient only: F(z) = J(0) z + (1/4) z^2 log z + O(z^2), so a theta^4 log(theta^2) term remains at a tuned small angle. For added bosons the statement needs a stated normalization: a real scalar determinant carries 1/2 and a real vector 3/2, and mixed boundary projections can give other prefactors.

Where frequency 2 lives

With multiplicities for the eigenvalues of F: the 4 has +-1, each twice; the 5 has 0 three times and +-2 once each; the 10 has 0 four times and +-2 three times each. So E = (4, 4bar) has no frequency-2 channel. For a top with Q_L in (2, 2) and U_R in the same multiplet, the charges Y = R_3 + (B - L)/2 require colour triplets with B - L = 4/3 for the 5, smallest in the 15, so (5, 15), of 75 components; or B - L = -2/3 for the 10, smallest in the 6, so (10, 6), of 60 components, with Q_L at R_3 = +1/2 and U_R at R_3 = +1. Also Lambda^2 E = (10, 6) (+) ((1 (+) 5), 10bar). The (5, 15) with light fermions localized is the pattern of Contino, Da Rold and Pomarol.

Proposition 26.3b (the frequency-2 plane does not separate the top). Take the (10, 6), with F the generator J_45 of so(5). The frequency-2 planes are (A^a, O^a), with A^a = (L^a - R^a)/sqrt(2) and O^a in (2, 2). The component O^+ is the left-handed top, with (T_3L, R_3) = (+1/2, +1/2), and its partner A^+ contains the right-handed top R^+ with weight 1/sqrt(2), together with the triplet component L^+. Let phi = 2 theta, A' = cos(phi) A + sin(phi) O, O' = -sin(phi) A + cos(phi) O, and S = (L + R)/sqrt(2). Consider the self-adjoint problem, by the referee: at x = 0, psi_R is Neumann on S, A and Dirichlet on O, and psi_L has the complementary conditions.

1. If at x = l, psi_R is Neumann on S and A' and Dirichlet on O', then (O^+, A^+) is a pure frequency-2 channel, and S_R is an exact massless coloured mode of charge 2/3.
2. If at x = l, psi_R is Dirichlet on O' and on S + A', and Neumann on S - A', the determinant is proportional to cos(a) (2 sin^2 a - sin^2 phi), with a = m l. So the angle-dependent tower obeys sin^2(m l) = sin^2(2 theta)/2, the equation of the charged vectors: m_t = M_W at every theta. There is also an angle-independent half-integer tower.
3. In the (5, 15) the same holds with the chiralities exchanged: the frequency-2 plane contains U_R and (t_L + X)/sqrt(2), with X the exotic of Y = 7/6.

Proof. These are the commutation relations of so(5) and the boundary problem, solved below as the limits of Proposition 26.3c. ■

These statements concern the displayed channel. A gauge-invariant completion needs conditions on complete wall multiplets, and the other modes of the multiplet also enter the potential.

Proposition 26.3c (a wall mass leaves a light state). In the problem above, normalize u = (S + A')/sqrt(2), v = (S - A')/sqrt(2), and let a wall Weyl field chi couple to g_u at x = l with parameter w: the self-adjoint operator is H(f, g, chi) = (-g', f', w g_u(l)), with domain f_u(l) = w chi, g_{O'}(l) = 0, f_v(l) = 0 and the stated conditions at x = 0 (referee). Let omega = w^2 l and a = m l. Eliminating chi for m != 0 gives f_u(l) = (omega/a) g_u(l). With s = sin(2 theta), the nonzero eigenvalues solve

```
2 a Delta sin(a) = omega cos(a) (Delta + sin^2 a),   Delta = sin^2 a - s^2,        (3)
```

where omega = 0 gives Delta sin(a) = 0, item 1 above, with the decoupled wall Weyl zero mode counted separately, and omega -> infinity gives cos(a) (2 sin^2 a - s^2) = 0, item 2. For every theta in (0, pi/2) and every omega in (0, infinity), there is an eigenvalue with 0 < m < M_W.

At theta = pi/4 equation (3) factorizes as cos(a) (omega cos(2a) - a sin(2a)) = 0: the state m = pi/(2l) = 2 M_W does not depend on omega, and the other equation has a root in (0, pi/4) for every omega > 0, moving from 0 to M_W l = pi/4.

Proof. On x in [0, l], with f the left and g the right components and the massless bulk operator, f' = m g and g' = -m f. The conditions at 0 give f_O = alpha cos(m x), g_O = -alpha sin(m x), f_A = beta sin(m x), g_A = beta cos(m x), f_S = gamma sin(m x), g_S = gamma cos(m x). These enter three linear conditions at x = l, g_{O'}(l) = 0, f_v(l) = 0 and f_u(l) = (omega/a) g_u(l), and (3) is the vanishing of their determinant, which we use before any division, so that it covers cos(2 theta) = 0 as well. For the root, let u_W = M_W l = arcsin(s/sqrt(2)) <= pi/4. On 0 < a < u_W,

```
omega = 2 a tan(a) (s^2 - sin^2 a)/(s^2 - 2 sin^2 a)
```

is continuous and strictly increasing from 0 to +infinity (referee), and it takes every value in (0, infinity). ■

So in the displayed charge-2/3 channel a state lies below M_W for every finite positive wall mass, and in the infinite-mass limit the lowest root of this isolated channel is M_W. This does not exclude identifying a higher eigenstate with the top: equation (3) has the eigenvalue m = 2 M_W when omega = 4 u_W sin(2 u_W)/(1 + 2 cos(2 u_W)). For example, at theta = 1 this is omega = 2.04, with another state at 0.85 M_W (Python, referee). That higher state has nonzero O_L and R_R components; its physical identification and the couplings of the lighter state need analysis. This compares with the top-mass limits for low representations in flat gauge-Higgs models (Cacciapaglia, Csaki and Park; Adachi and Maru; massless exotics and their wall masses in Hosotani, Oda, Ohnuma and Sakamura). A complete gauge-invariant wall multiplet, other boundary mixings, bulk masses and the other towers need separate analysis, and they can add still lighter states.

The vacuum exists in case 2 as well. Let n_d be twice the number of Dirac species with the product 1 + sin^2(2 theta)/(2 sinh^2 x), in place of n_2. Then G(u) acquires -2 n_d J(2u), and for n_d >= 27/5 we have G <= 0 on (0, 1/4], since J(16u/5) <= J(2u). So with n_1 = 0, theta = pi/4 is the unique global minimum, by the argument of item 1 of Theorem 26.2. One Dirac colour triplet has n_d = 6. A scan (Python) gives minima at 0.8585 and 1.0011 for (n_d, n_1) = (12, 2), (12, 6), and no interior stationary point for n_d = 6 and n_1 = 2, 6. For E with n_1 and such a sector, m_1/M_W = 2 lies on the line n_d

= 5.292 + 0.6857 n_1; for example (n_1, n_d) = (4, 8) gives theta = 1.1423, with m_1/M_W = 2.026. Such a sector has its own state at M_W in that isolated channel.

The tower of the parent in the step-threshold model

Take the conditions of Theorem 23.6 at theta = 0, and let S = mu/M_KK with M_KK = pi/l. A massive vector, with its fifth component and ghosts, contributes -(7/2) tr Q^2 per level, the trace over real components.

generators	conditions (x = 0, x = l)	tower	SU(3)	SU(2)	U(1), GUT normalization
gluons	N, N	n	-21/2		
W_L	N, N	n		-7	
O = (2, 2), Y = +-1/2	D, D	n		-7/2	-21/10
R^+-, Y = +-1	N, D	n - 1/2			-21/5
X = 3 + 3bar, Y = +-2/3	N, D	n - 1/2	-7/2		-28/5
hypercharge combination	N, N	n			0
orthogonal neutral combination	N, D	n - 1/2			0

The totals -21/2 for SU(2) and -14 for SU(3) are -(7/2) C_2 of Sp(2) and SU(4). The Higgs is the zero mode of A_5 along O, counted in the Standard Model coefficients, and the nonzero modes of A_5 are the gauge partners counted in the vector towers. Three complete massless bulk generations of E add F = 8 to each coupling on the integer tower, in the limit theta -> 0; at nonzero angle, frequency-one masses are shifted to (n pi +- theta)/l. With the fermions on the walls, F = 0.

So above M_KK, with b = (41/10, -19/6, -7), we take as a model

```
alpha_i^-1(mu) = alpha_i^-1(M_Z) - (b_i/2 pi) log(mu/M_Z) - (1/2 pi) [ (B_i + F) I_n(S) + C_i
↪  I_h(S) ],        (4)

B = (-21/10, -21/2, -21/2),   C = (-49/5, 0, -7/2),
```

where I_n(S) is the sum of log(S/n) over n <= S, and I_h(S) the sum of log(S/(n - 1/2)) over n - 1/2 <= S. For large S, I_n ~ S - (1/2) log(2 pi S) and I_h ~ S - (1/2) log 2, so I_h - I_n ~ (1/2) log S + (1/2) log pi.

The model (4) omits the frequency-2 multiplets used for the vacuum above. A complete five-dimensional Dirac (10, 6) has level contributions (296/15, 24, 40/3), before its threshold splitting and zero-mode subtractions, and these are not a universal F. Boundary kinetic counterterms are also generated by loops and need a renormalization prescription (Carena, Tait and Wagner). So (4) is the running of this spectrum, and of the parent only as far as this spectrum is the parent's.

Proposition 26.4. In (4), the differences of the alpha_i^-1 are independent of F. ■

Proposition 26.5 (the left-right difference in Sp(2) is logarithmic). Define alpha_R^-1 = (5/3) alpha_1^-1 - (2/3) alpha_3^-1, the coupling of SU(2)_R in 1/g_Y^2 = 1/g_R^2 + (2/3)/g_4^2. Then, from (4),

```
alpha_R^-1(mu) - alpha_2^-1(mu) = D_0 - (44/3)/(2 pi) log(mu/M_Z) + (7/pi) [I_h(S) - I_n(S)],
↪  (5)
```

with D_0 = alpha_R^-1(M_Z) - alpha_2^-1(M_Z), and it has no power of S. With the inputs (59.0, 29.6, 8.47), D_0 = 63.09.

Proof. The zero modes give (5/3)(41/10) + (2/3)(7) + 19/6 = 44/3. The integer tower gives (5/3)(-21/10 + F) - (2/3)(-21/2 + F) - (-21/2 + F) = 14, and the half-integer tower gives (5/3)(-49/5) - (2/3)(-7/2) = -14. ■

The cancellation concerns the difference within Sp(2); differences involving colour keep a power of S. This is the differential running of orbifold models (Nomura, Smith and Weiner). Without the tower the difference vanishes at 5 x 10^13 GeV. With a tower, for S » 1, it vanishes at

```
log(mu_LR/M_Z) = (3/23) [ 2 pi D_0 + 7 log(pi) - 7 log(M_KK/M_Z) ].        (6)
```

So at M_KK = 321 GeV, the difference is 61.69 at S = 1 in (5), where I_h = log 2, against 60.15 from the Standard Model extrapolation alone, and by (6) it vanishes near 2 x 10^24 GeV. It lies above the Planck mass, 1.2 x 10^19 GeV, for M_KK below about 2 x 10^8 GeV.

Result 26.6 (the meeting point in (4); Python, one loop). The two conditions alpha_1 = alpha_2 = alpha_3 in (4) have the solution

```
M_KK = 3.15 x 10^13 GeV,   mu_* = 2.21 x 10^14 GeV,   S = 7.04,
```

with alpha^-1 = 46.0 for F = 8 and 52.6 for F = 0. It is the only solution with S >= 1 (referee): eliminating log(mu/M_Z) between the left-right equation and the equation for alpha_2 = alpha_3 leaves a continuous function of S whose derivative between thresholds is (315 N_h - 161 N_n)/(44 S) > 0, with N_n, N_h the numbers of levels below S. Shifts of 0.1 in alpha_1^-1 or alpha_2^-1 and of 0.15 in alpha_3^-1, one at a time, move M_KK within 2.9 to 3.4 x 10^13 GeV, and all three together within 2.66 to 3.70 x 10^13 GeV. Larger effects are possible: one additional light colour-triplet Weyl fermion changes alpha_3^-1 by about -1.4 up to that scale, the Standard Model gauge part of the two-loop terms by about (-0.20, -0.27, +0.39) (referee), and finite wall terms and split thresholds are not included. So in this model the relation g_3^2 = g_2^2 = (5/3) g_Y^2 holds at one scale with no added thresholds, and that scale fixes the tower. An interval Pati-Salam model with a compactification threshold of order 10^13 GeV, with a different intermediate group and matter, is in Abe, Goto and Kawamura.

The top against the matching scale

In a flat parent with the Higgs as A_5, the tree-level relation at small angle, for a pure channel of frequency q, is y_q = (q/2) g, since m_q = q theta/l and M_W = sqrt(2) theta/l. The same covariant derivative fixes the local bulk vertices. Physical four-dimensional gauge and Yukawa couplings still receive different finite corrections and thresholds (Lim and Maru). So the absence of separate running holds for the local bulk coupling, in a stated matching scheme. This differs from item 11 of the steer note, where the Higgs is an independent bulk field, y_t runs with its own power law, and the matching sits at the crossings. Item 11 stays correct for that content, and the two calculations have different actions.

At finite angle a mass ratio does not give the Higgs coupling. With v_ang = 2 sqrt(2) theta/(g l), the coupling is sqrt(2) dm/dv_ang. At theta = pi/4, the pure frequency-2 branches have slopes corresponding to |y| = g and are degenerate, so the coupling is a matrix on that subspace, while the tower with m = M_W has dm/d(theta) = 0 there (referee).

Result 26.7 (illustration; Python, one loop, Standard Model below M_KK). The running mass m_t solving y(m_t) v/sqrt(2) = m_t, from the boundary value y(M_KK) = r g_2(M_KK):

M_KK (GeV)	r = 1/2	r = 1/sqrt(2)	r = 1	r = 2/sqrt(3)	r = sqrt(2)
321	63	87	119	136	164
10^4	74	101	136	153	179
10^9	99	129	163	176	195
3.1 x 10^13	114	143	172	183	196

The measured value in the same convention is 163 GeV, and from it y_t/g_2 equals sqrt(2) at 265 GeV, 2/sqrt(3) at 2.8 x 10^5 GeV and 1 at 1.3 x 10^9 GeV. These are boundary-value calculations, and they are

predictions of no spectrum: at low M_KK a real parent has other thresholds below M_KK, so Standard Model running there is an illustration. Reading the entry 164 GeV as the pure channel at theta = pi/4 fails for three reasons: the Higgs coupling at that angle is not sqrt(2) g, as above; the channel has its massless exotic and a second fermion at M_KK/2; and Proposition 26.3c gives a lighter partner for any wall mass.

Result 26.8 (the neutral problem with a wall term; Python, tree level). Let the bulk couplings of Sp(2) and SU(4) be separate, and let a wall kinetic term r at x = l act on the hypercharge combination Y-hat = cos(phi) R + sin(phi) B, with cos^2(phi) = alpha_2^-1/(alpha_2-1 + (2/3) alpha_3^-1). We fix these, as an estimate, by Standard Model running to M_KK, with alpha_Y^-1 = (1 + r/l)(alpha_2^-1 + (2/3) alpha_3^-1). The charged sector is unchanged. At x = 0, S, A, B are Neumann and O is Dirichlet. At x = l, with L_l = (S + A')/sqrt(2), R_l = (S - A')/sqrt(2), p = cos(phi), q = sin(phi): O' = 0, -q R_l + p B = 0, d_x L_l = 0 and d_x(p R_l + q B) = r m^2 (p R_l + q B). The lowest neutral root a = M_Z l solves, with t = r/l (referee),

```
(1 + t a cot a) sin^2 a = (sin^2(2 theta)/2) (1 + p^2 + t a cot a),
```

whose r = 0, one-coupling limit is Theorem 23.6, and which our 4 x 4 determinant reproduces.

theta	M_KK (GeV)	m_1/M_W	M_W$^{2/M_Z}$2	four-dimensional value at M_KK	r/l
0.7854	321	1.00	0.629	0.762	1.63
1.1377	444	2.00	0.709	0.761	1.60
0.30	615	0.73	0.735	0.759	1.56
1.30	677	3.48	0.739	0.759	1.55
0.15	1199	0.71	0.750	0.756	1.48
0.08	2236	0.71	0.751	0.752	1.42
1.53	4378	26.5	0.749	0.749	1.35

So within this matching prescription the finite interval shifts the ratio by -0.13 at theta = pi/4 and by -0.05 at the theta where m_1/M_W = 2. In the rows of the table the shift falls below 0.006 from about 1.2 TeV, where m_1/M_W is either near 0.71 or above 8, as a numerical statement for this prescription. The four-dimensional number is a reference within this prescription. A physical precision comparison needs the coupling definitions, wavefunction overlaps, threshold matching and radiative corrections of the model, which can enter at the same order as a small shift. We have not done an electroweak fit, and the measured on-shell value, 0.7767, is not comparable with these columns without that.

Two regimes

We state the outcome in the exact scope of (1) and (4): one interval, flat profiles, this gauge content, one loop, no bulk masses. This is one independent review of a statement about one family, and broader consensus on its negative parts remains open. It leaves T13 open.

Low tower. With frequency-2 channels of sufficient count, the one-loop potential has a natural interior minimum, at theta >= pi/4 and M_KK from 321 GeV upward (Theorem 26.2). In E, m_1 = 2 M_W at theta = 1.1377, with M_KK = 444 GeV, and the frequency-2 sector that stabilizes it has its own light states in the displayed channels (Propositions 26.3b, 26.3c). A state at 2 M_W in a frequency-2 channel has a lighter partner, whose couplings are not analyzed. In (5), the left-right difference at the tower is about 60, so if only a hypercharge wall term and separate bulk couplings are admitted, that wall term is of that size in alpha^-1, comparable with twice the bulk value, as an estimate; general non-universal wall kinetic terms are another way to fit the couplings. The weak angle is then an input, and the finite interval shifts M_W$^{2/M_Z}$2 by the amounts of Result 26.8. Whether a tower at this scale is allowed needs its couplings and an applicable experimental limit, which we have not analyzed.

High tower. The relation of one coupling holds in the model (4) at mu_* = 2.2 x 10^14 GeV, with M_KK = 3.15 x 10^13 GeV, and the weak angle runs down from it. With the boundary values of Result 26.7, r = 1 gives a running mass near 172 GeV and r = 1/2 gives 114 GeV. The small-angle relation M_W l =

sqrt(2) theta then requires theta = 5.7 x 10^-12, and Proposition 26.2b shows that no such angle is a global minimum of (1) at integer counts. Metastable minima, in the cases left to scans, are not excluded by it.

Escapes. All of these are model-dependent: wall kinetic terms and separate bulk couplings, which give up the relation as a prediction; a wall scalar with its own potential; warping, whose spectra, counterterms and matching the flat sums do not determine; bulk masses; antiperiodic conditions; larger representations with brane mixing; boundary interactions.

Direction. Warping and larger representations are the standard directions for items 2 and 4 of the outcome. In this project the next computations are the same family in a warped interval, with the spectral action of lemma 25 at the level of its heat trace, and the items below.

Outstanding. The low-tower potential (1), the wall-mass spectrum (3), the wall kinetic term of Result 26.8 and the step-threshold running (4) use different simplified spectra, so their conclusions have not been combined into one model. A combined model needs: complete gauge-invariant wall multiplets, with their determinants in (1) and their thresholds in (4); the coupling matrix at finite angle; the couplings of the light partner in (3); and the matching of the neutral sector at one loop. These calculations are open.

Priority

The sources are listed in LITERATURE, section J. The referee's search covered orbifold differential running, Pati-Salam interval scales, flat large-representation tops, SO(5) vector and adjoint embeddings, exotics and Wilson-line minima, and found no prior occurrence of the classification with n_c or of the pair (M_KK, mu_*) from (4), with forward citation chains incomplete. So priority for those exact statements is open. Hosotani et al. find theta_H = pi/2, corresponding to 2 theta = pi/2, in a warped model with other matter.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept; the referee recomputed the items marked in the report.

- J(0) and J(1) by quadrature against the closed forms, to 15 digits. The function e(z) is nondecreasing and R(u) is increasing on grids, with R(0+) = 2.7001 at u = 10^-5 and R(1/4) = 2.88550.
- Theorem 26.2 on the grid described after its proof, with no failure. For n_1 = 0: n_2 = 2.75 has minima at 0.1527 and 1.4180 and a maximum at pi/4, and n_2 = 2.90 has the minimum at pi/4.
- Equation (3) against a direct 3 x 3 determinant of the boundary conditions, for theta = 0.3, pi/4, 1.0, 1.1377 and omega from 10^-3 to 10^4: the lowest roots agree to four digits, and the lowest root lies below M_W l in every case.
- Equations (4), (5): the difference computed directly from (4) agrees with (5) to 13 digits at S = 1000. For S above 10^8 the closed sums lose precision in floating point, so (6) is used there.
- Result 26.6 by nested root finding, over a scan of 1401 values of M_KK. The inputs are those of the steer note, with no thresholds at the top or W.
- Result 26.7: one-loop top Yukawa equation of the steer note with no tower, below M_KK. The asymptotics of I_n, I_h and their difference, against the closed sums at S = 10.3, 100.3, 1000.3, to about 10^-3 or better.
- Result 26.8: the 4 x 4 determinant, with root finding on 6401 points. Its limit r = 0, one coupling against Theorem 23.6 at theta = 0.2, pi/4, 1.06, 1.1377, 1.3.

Part 27

The axial anomaly on the reduced action, and the two-flavour discrete term

For STRATEGY T10 and T13, following lemma 17 and open question 6 of the steer note. Proved by Claude Opus 5.5, using the index theorem and Witten 1982 as references. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors of interpretation, in Propositions 27.4 and 27.5, and amendments, all incorporated below.

Outcome.

1. On the reduced operator D_M + gamma_5 d, with masses commuting with the gauge field, the regulated axial trace has an exact form at every cutoff: each flavour block contributes its index, weighted by f(m$^{2/\mathrm{Lambda}}$2) (Theorem 27.1). The regulated mass term equals it identically (Proposition 27.3). As Lambda -> infinity this is the anomaly with the coefficient 2 N_c tr(T^3 Q^2) = 1, on the action of Theorem 17.1 (Corollary 27.2).
2. The operator of Proposition 17.10 at M = 0, in independent quark and antiquark variables, is the massless Dirac operator with a pure-gauge connection on one chirality. So for two flavours and a reduction Sigma : S^4 -> SU(2) in the nontrivial class, the spectral action is unchanged, and the Weyl Pfaffian transported along connections from 0 to that pure gauge has the relative sign (-1)^{N_c} (Proposition 27.5). The same sign appears in Witten's anomaly, in the discrete term of the pion sigma model, and in the mod-two index on S^5 of the research state. The sigma field represents a class in the reduced spin bordism of SU(2) in dimension four, and the anomaly a class in Omega_5^Spin(BSU(2)); these agree through the anomaly comparison and are distinct groups. The corpus S^5 is an internal space, and the S^5 here is the compactified S^4 x R of an auxiliary flow.

Setting

Let M be a closed Riemannian spin 4-manifold. Following Corollary 17.2, a paired sector gives one Dirac fermion: we take the operator

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DD = D_M(A) + gamma_5 d,   d = diag(m_f) on E = C^{N_c} (x) C^{N_f},   m_f >= 0,
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on L^2(S) (x) E, with action psi-bar DD psi, and with A the gauge connection of the commuting subalgebra: colour and electromagnetism, so that A is diagonal in the flavour index, with charge Q_f on flavour f. For topologically nontrivial backgrounds, E is replaced by the colour and electromagnetic bundles E_f, with charges normalized so that the flavour representations are globally defined. Let T be a Hermitian matrix on C^{N_f}, diagonal in the same basis, with eigenvalues tau_f. Then d and T commute with A, and Theorem 17.1 gives DD^2 = D_M(A)^2 + d^2.

Let f be a cutoff with f(DD$^{2/\mathrm{Lambda}}$2) trace class, for example in the positive Laplace class of lemma 18. Choose gamma_5 = +1 on S^+, and let ind_f = n_{f,+} - n_{f,-} be the index of D_f^+ : S^+ (x) E_f -> S^- (x) E_f.

The rotation psi -> exp(i alpha gamma_5 T) psi, psi-bar -> psi-bar exp(i alpha gamma_5 T) sends DD to D_M + gamma_5 d exp(2 i alpha gamma_5 T), since gamma_5 anticommutes with D_M. So the classical action is invariant at d = 0. Its Jacobian, regulated by the spectrum of DD, is exp(-2 i alpha Tr(gamma_5 T f(DD$^{2/\mathrm{Lambda}}$2))), up to the orientation convention.

The axial trace

Theorem 27.1. For every Lambda > 0,

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Tr(gamma_5 T f(DD^2/Lambda^2)) = sum_f tau_f f(m_f^2/Lambda^2) ind_f.
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Proof. On block f, DD^2 = D_f^2 + m_f^2, with D_f = D_M(A) on that block. The operator D_f anticommutes with gamma_5, so it maps the eigenspace of D_f^2 with eigenvalue mu > 0 isomorphically between the two chiralities. So these eigenspaces contribute zero to the trace with gamma_5, and only ker D_f remains, with weight f(m_f$^{2/\mathrm{Lambda}}$2) and signed dimension ind_f. ■

No expansion in m/Lambda enters. By the index theorem,

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ind_f = (1/8 pi^2) int tr_f(F wedge F) - (N_c/24) p_1(M),
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with F Hermitian, and with F = Q_f e F_em + g G on block f, tr_f(F wedge F) = N_c Q_f^2 e^2 F_em wedge F_em + g^2 tr_c(G wedge G), since G is traceless. Reversing the chirality convention reverses the index.

Corollary 27.2 (the anomaly on the reduced action). Let f be continuous at 0, and let Lambda -> infinity at fixed masses. If tr T = 0, with the flavours in the same colour bundle, then

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2 Tr(gamma_5 T f(DD^2/Lambda^2)) -> f(0) 2 N_c tr(T Q^2) (e^2/8 pi^2) int F_em wedge F_em,
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with no gluonic and no gravitational term. For two flavours with T = T^3 and Q = diag(2/3, -1/3), the coefficient is 2 N_c tr(T^3 Q^2) = 1, so that the right side is f(0) (e^2/16 pi^2) int F_em,mu nu F-tilde_em^mu nu, with F-tilde^mu nu = (1/2) epsilon^mu nu rho sigma F_rho sigma. A singlet T = 1 on N_f flavours gives in addition the gluonic term f(0) 2 N_f (g^2/8 pi^2) int tr_c(G wedge G) and the gravitational term -f(0) (N_c N_f/12) p_1(M). ■

The normalization f(0) = 1 is the usual convention for a regulator, and it is the value that makes the Jacobian the anomaly. For a smooth cutoff class in which the heat expansion can be used, with E and Omega defined by the Laplace-type square, the local form, with a smooth function chi on M inserted, follows at Lambda -> infinity from the a_4 coefficient, (1/16 pi^2) tr(gamma_5 T (E^2/2 + Omega^2/12)), which gives the integrands above. The other a_4 terms and the lower coefficients have vanishing gamma_5 spin trace in this problem, and the factor exp(-t d^2) commutes with the heat kernel and changes only higher orders in t. So the algebraic coefficient quoted in the steer note, item 6, now has the regulated trace and measure that it lacked.

Proposition 27.3 (the mass term equals it at every cutoff). Let m_f > 0 for all f. Then

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Tr(f(DD^2/Lambda^2) d T DD^-1) = Tr(gamma_5 T f(DD^2/Lambda^2)) = sum_f tau_f f(m_f^2/Lambda^2)
↪ ind_f.
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Proof. DD^-1 = (D_M + gamma_5 d)(D_M^2 + $d^{2)}$-1. The term with D_M has zero trace, by conjugation with gamma_5, since all other factors commute with gamma_5. The remaining term is Tr(f gamma_5 T d^2 (D_M^2 + $d^{2)}$-1). On the nonzero spectrum of D_f it vanishes by the pairing of Theorem 27.1, and on ker D_f the factor $m_f^{2/m_f}2$ is 1. ■

Since {gamma_5 T, D_M + gamma_5 d} = 2 d T, the insertion is psi-bar d T psi in this mass convention, pseudoscalar in a conventional scalar-mass basis. So if the Jacobian and the composite mass insertion are given the same specified regularization, this is the integrated Ward identity in a fixed background at every Lambda, with a common factor $f(m_f^{2/\mathrm{Lambda}}2)$ on both sides. The trace identity alone constructs neither an independently regulated measure nor a local current, and the local identity has a current-divergence term as well. The surviving zero modes contribute the signed count n_{f,+} - n_{f,-}, whereas the scalar density of Proposition 17.8 sees the unsigned count n_0 = n_{f,+} + n_{f,-}.

Proposition 27.4 (a heavy flavour at fixed cutoff). With the weights of Theorem 27.1, for T^3 on a doublet (t, b) with m_b/Lambda -> 0, the electromagnetic coefficient is

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2 N_c (1/2) (Q_t^2 f(x) - Q_b^2 f(0)) / f(0),   x = m_t^2/Lambda^2.
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It equals 1 at x = 0. For the heat cutoff it is (4 exp(-x) - 1)/3, which tends to -N_c Q_b^2 = -1/3 as x -> infinity, with gluonic and gravitational regulated traces proportional to f(x) - f(0). ■

This is a trace in the fixed-cutoff family of lemma 18, where the top is heavy at fixed Lambda. An explicitly massive flavour breaks the axial symmetry, so this calculation alone does not establish a Wess-Zumino term in the low-energy theory. Such a term can be required once the retained symmetry, its background fields or spurions, and the anomaly-matching prescription are specified, as in D'Hoker and Farhi (Nucl. Phys. B 248 (1984) 59 and 77, named in the steer note, and not read here). The weights depend on the chosen regulator, and Proposition 27.3 carries them on both sides of the Ward identity. Making f part of a finite-scale spectral action, as in lemmas 18 and 25, does not by itself identify the regulator of the fermion measure with that action or make these weights observable.

The two-flavour discrete term

Take N_f = 2 and M = 0, with electromagnetism off, so that SU(2)_L x SU(2)_R is a symmetry of the reduced sector (Proposition 17.5). By Theorem 17.6, at M = 0 the admissible involutions are U_Sigma,

with Sigma in U(N_f), exchanging the two eigenspaces of gamma_F by Sigma. Let Sigma(x) be a smooth map, and take the pointwise reduction of Proposition 17.10. Its operator at d = 0 is

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D_M + c^mu P_- B_mu,   B_mu = Sigma d_mu Sigma^*,        (1)
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up to the convention for which side carries Sigma, in independent quark and antiquark variables. On the full H_0 = (q_L, q-bar_R) the two blocks of (U_Sigma d U_Sigma) are Sigma d Sigma^* and Sigma^T d conj(Sigma). On the chirality selected by P_- this is the massless Dirac operator with the pure-gauge connection B of SU(2), and on the other chirality it has no flavour connection. The operator is generally not self-adjoint. The colour connection is unaffected.

Let Sigma : S^4 -> SU(2) generate pi_4(SU(2)) = Z_2, on the round S^4, with colour background zero, so that D_M has no kernel. Such a Sigma cannot be joined to 1 through maps to SU(2). So let Pf_-(A_t) be the square root of Witten 1982 for one Weyl doublet, transported along the path of connections A_t = t B, with the endpoint fibres compared by the gauge transformation.

Proposition 27.5. For the operator (1), with N_c doublets of the chirality selected by P_-, the transport along A_t gives

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Pf_-(B)^{N_c} = (-1)^{N_c} Pf_-(0)^{N_c}.
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The spectral action of the fixed operator D does not depend on Sigma. The fixed classical form is unchanged as a form, while its expression in the fixed reduced variables acquires the connection in (1).

Proof. The display counts N_c Weyl doublets, on which we use the main result of Witten 1982: (det i D-slash)^(1/2) of one doublet has the opposite sign after a gauge transformation in the nontrivial class of pi_4(SU(2)) (Eq. (6), p. 325 there, read on 3 Nov 2026). The last sentences are item 1 of Proposition 17.10. ■

This is a relative sign for the stated Pfaffian transport and endpoint identification. The pointwise R_Sigma of Theorem 17.1 changes fermion coordinates, and pulling the original measure through that change preserves the value of the same integral, because the Jacobian of the measure accompanies the transformed action. So the sign is the comparison between that pulled-back measure and the Pfaffian continued along connections, and for N_c odd they differ between the two classes of reductions. Matching this to a pion field needs the sigma model and its anomaly-matching assumptions. The coupling that results has the sign of the discrete term in the N_f = 2 sigma model: Lee, Ohmori and Tachikawa (arXiv:2009.00033, section 2.5 and Eq. (2.29), read in the LaTeX source) define the coupling as (-1)^{N_c [sigma]}, with [sigma] the class in the reduced spin bordism group of SU(2) in dimension four, which is Z_2, on spin four-manifolds. The original discussion is Witten 1983 (*Global aspects of current algebra*), which we have not read. Proposition 17.10 constructs neither a pion action nor its measure: a kinetic term needs a new operator or a quantum effective action, and Proposition 27.5 supplies only the sign.

Remark 27.6 (the relation with the S^5 index). By Witten 1982, for the flow from A = 0 to A = B, the sign equals (-1) to the number of zero modes of the five-dimensional Dirac operator on S^4 x R. The two ends compactify, with the appropriate gauge trivializations, to the nontrivial SU(2) bundle on S^5, and footnote 2 there describes the invariant zero mode for a special connection. The general protected statement is the parity of the kernel. For isospin j the parity is that of 2 tr T^2 = 2 j (j + 1)(2j + 1)/3, which is odd exactly at j = 2k + 1/2 (LITERATURE, section G). So the discrete term, with N_c colours, has the parity of N_c times the mod-two index of the doublet on the nontrivial SU(2) bundle on S^5. After choices of identification, that bundle is the one in section 6.12 of the research state, and the operator D_{S^5,V} there has the same mod-two index, on the generator of Omega_5^Spin(BSU(2)) = Z_2.

The scope of this identification is the bundle and the index. In the research state the S^5 is an internal space, the base of the chain 7 -> 5. In Proposition 27.5 the S^5 is the compactification of spacetime and the auxiliary flow parameter, with the flavour group of the quarks as structure group. We have not proved a relation between those two spaces, or between their SU(2) groups, and open question 6 of the steer note stays open for that part.

Caveats

- The background is fixed in Theorem 27.1 and Proposition 27.3. A measure on gauge fields and the order of limits are separate, as in Proposition 17.8.
- T, d and A commute by hypothesis. So T^3 and the W are excluded: this is the commuting subalgebra of Proposition 17.3. Off-diagonal d is outside Theorem 27.1, where DD^2 has a term gamma_5 [d, D_M].
- Proposition 27.5 has colour background zero and M = 0. For a nonzero colour background the integral is complex in general, and the statement on the sign is on the class of Sigma through the flow, which we have not developed.
- The sign in Proposition 27.5 is a statement on the fermion measure at one configuration. A sum over Sigma needs a sigma model that is not constructed here.
- The regulated Jacobian by the spectrum of DD is one choice. It is gauge invariant under the commuting group, which is why the anomaly falls entirely on the axial current.

Consequence for T10

The chiral limit now has, on the reduced action: the anomaly with its coefficient, the Ward identity at every cutoff, the top against the cutoff, and the discrete term. Still to be supplied are a measure on gauge fields, the map to a quantum order parameter, and the kinetic term of a pion field, which need a quantum effective action.

Numerical check

By Claude Opus 5.5, in Python (fractions, numpy), 3 Nov 2026, in a temporary directory, not kept. These check the arithmetic and the pairing, and the proofs stand without them.

- 2 N_c tr(T^3 Q^2) = 1 and tr T^3 = 0, in exact fractions. The third-generation coefficient for the heat cutoff at x = 0, 0.1, 1, 5, 50 is 1.0000, 0.8731, 0.1572, -0.3243, -0.3333.
- On T^2 x T^2 with fluxes n_1, n_2 and Landau spectra, truncated to 2000 levels: Tr(gamma_5 exp(-t(D^2 + m^2))) = exp(-t m^2) n_1 n_2 to 15 digits, at (n_1, n_2, t, m) = (1, 1, 0.3, 0.7), (2, 3, 0.05, 1.2), (4, 1, 1, 0).
- tr T^2 half-integer at j = 2k + 1/2 and integer at j = 2k + 3/2 for k up to 20, in exact fractions.

Part 28

Two sheets as the two ends of one connected commutative parent

For STRATEGY T0 and T0b, following lemma 16 (Example 16.9), lemma 24 (Proposition 6) and lemma 09. Propositions 28.1 to 28.4 and 28.13 are by Claude Opus 5.5. Results 28.8 to 28.12 and 28.14 to 28.21 are by Astra 6, job 28 (report, with the full proofs; prompt), checked by Claude Opus 5.5 by hand and in Python. Proposition 28.13 and the numerics are refereed by Terra 6.1 (report), with its corrections applied. The window is a conjecture with numerical support; the numbers come from tools/lemma28_radial.py (Python, numpy and scipy).

Correction, 3 Nov 2026. The first version of Result 28.5 and of the numbers in Remark 28.7 (commits dd1b14c, 530fa06) used the wrong sign of the spin connection in the Python slice operator. That changed the singular values and the boundary fibre parts, so those numbers are withdrawn, including the next singular value 1.04, the value 1.35 on the representation 16, and the boundary fibre parts -0.011 to -0.853. The exponents at the origin and the multiplicities are unaffected. Astra 6 did not reproduce the old boundary values and its proofs do not use them. We found the error when that operator gave a core zero mode in the representation 20, against the exact count of Proposition 28.11. The singular values, fibre parts and densities below are recomputed, and the script tests the operator against exact statements: {A_0, A_1} = 0, A_0^2 equal to the Casimir formula on S^4, D^2 on (0, 0)-forms with values in O(1) at the Fubini-Study radius equal to j(j + 4), and A_2 equal to its curvature contraction (the last test proposed by Terra 6.1).

Update, 4 Nov 2026. Lemma 29 proves Conjecture 28.6 in case (b) for the metric class of its Setting 29.0, and in case (a) at the Fubini-Study radius, refereed by Terra 6.1. Item 3 below and the Status of Proposition 28.13 describe the position before that lemma.

Outcome.

1. *The mechanism is proved at the level of compression maps.* A smooth function on the disc bundle X has independent values on the zero section S^4 and on the boundary CP^3, and at the zero section it is a function on S^4 alone (Proposition 28.8). For prescribed windows over this X, real functions generate exactly M_2(H) + M_4(C) at every localization scale (Theorem 28.21). In Example 16.9 the second sheet and the restriction to S^4 were inputs, so this is a connected source for them.
2. *The two ends of an actual local operator are understood.* There is a self-adjoint operator with a local wall (Theorem 28.9). The core carries exactly four states in the representation 4 (Proposition 28.11). At the wall, large mass keeps a tangential mixing (Lemma 28.12). For the pull-back connection the boundary mode then has the fibre part -w(rho) in closed form, with w(rho) = 2(rho$^{2+2)}$2/(9+2(rho$^{2+2)}$2), which reaches the level-one value of Example 16.9 only as rho -> infinity (Proposition 28.13). This is a statement on the finite wall model, and its convergence from the full operator is numerical. A connection preserving the line summands in a collar, with the Fubini-Study metric at the wall, gives the level-one value in that model.
3. *The isolated window of the whole operator stays a conjecture* (Conjecture 28.6): what is missing is the normal reductions with their errors, the matching, and a coercive estimate uniform in the mass and in all representations of Sp(2) (Lemma 28.14).
4. *The algebra M_2(H) + M_4(C) is the algebra of the limit map.* Generated algebras need not converge when compression maps do (Counterexample 28.17). At the sample points resolved numerically, the core mode has a nonzero density breaking the quaternionic structure, empirically of order lambda^2; if it is nonzero in the continuum, Proposition 28.16 gives M_4(C) for the core block at those masses. This is numerical evidence and does not establish the exact algebra at any finite mass.

Setting

Let G = Sp(2), K = Sp(1)_a x Sp(1)_b and H = Sp(1)_a x U(1)_b, so that G/K = S^4, K/H = S^2 and G/H = CP^3. Let S be the bundle G x_K (1, 2), so that CP^3 = P(S). The normal bundle of the zero section is G x_K (1, 3), and away from it X = (0, 1] x CP^3, with radial coordinate x and S^2 fibres of radius rho(x), rho(x) ~ x at x = 0. Charges of U(1)*b are normalized so that the doublet of Sp(1)*b has +-1. The isotropy representation is (2)*{+-1} (+) (1)*{+-2}. The spinors of X restrict on a slice to Sigma_6 = Sigma^+ (+) Sigma^-, with c(d_x) = Z the chirality of Sigma_6, where

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Sigma^+ = (1)_2 (+) (1)_0 (+) (2)_{-1},   Sigma^- = (2)_1 (+) (1)_0 (+) (1)_{-2}.
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The representation 4 of G restricts to (2)_0 (+) (1)*1 (+) (1)*{-1}, that is (2, 1) (+) (1, 2) of K. Let Gamma_a be the Clifford generators on 4, with Gamma_5 commuting with K and equal to +-1 on the two summands, and let x_a(g) be defined by lambda(g) Gamma_5 lambda(g)^-1 = x_a(g) Gamma_a.

Proposition 28.1 (the twist).

1. For k != 0, O(k) on CP^3 does not extend to a line bundle on X, since H^2(X) = H$^{2(S}$4) = 0 while O(k) has degree k on the fibre S^2.
2. The pullback of S to X restricts on CP^3 to O(1) (+) O(-1), as the tautological splitting. So E = S^3(S), which is pulled back from S^4 and has trivial connection along the fibres, restricts to O(3) (+) O(1) (+) O(-1) (+) O(-3). The summand O(3) gives the Dolbeault operator on O(1) of Example 16.9, since Sigma (x) O(3) = Lambda^{0,*} (x) O(1).
3. On E the section Phi(v) = phi(|v|) (v-hat . T), with T the spin-3/2 generators and phi odd and smooth, is smooth and G-equivariant, with zero set exactly S^4. In the frame of the slice it is phi(x) tau, with tau diagonal in the summands, with eigenvalues +-3/2, +-1/2.

So we take the Callias-type operator T = D_X(E) + i Phi on spinors of X tensored with E, and the self-adjoint operator D-hat = [[0, T^*], [T, 0]], with the grading Gamma-hat = diag(1, -1). At x = l we take the local condition Gamma-hat Z = -1 on the doubled spinor, that is u_even(l) = 0 for T and v_odd(l) = 0 for

T^*. It is a chirality condition, elliptic and local, and it preserves the domain under all smooth functions on X.

Proposition 28.2 (multiplicities). On a slice, Hom_H(4, Sigma^+ (x) E) and Hom_H(4, Sigma^- (x) E) each have dimension 5. For the summand O(3), the dimensions are 1 and 0. So on 4, the operator on the summand O(3) with its U(1) connection vanishes for every G-invariant metric on the slice, and its kernel contains $H^{0(CP}3$, O(1)) = 4 for every fibre radius. This concerns the separate line connection. The pull-back connection of E does not preserve the summands, and its slice operator is invertible for small fibres (Proposition 28.10).

Proof. Tensoring the lists above with the charges of E gives the multiplicities. The operator is odd under Z and maps the multiplicity space of 4 into itself. ■

Proposition 28.3 (the zero section). The fibre operator c_N . nabla + i Phi on R^3 with values in E has zero modes (1, 3) (+) (1, 1) of K, by the Callias count for the positive weights 3/2 and 1/2, with one sign of Phi. On S^4 this gives the Dirac operator twisted by (1, 3) (+) (1, 1). Then Hom_K(4, S^- (x) (1, 3)) has dimension 1, through (1, 2), and Hom_K(4, S^+ (x) (1, 3)) = 0, so the twisted operator vanishes on that copy of 4, in agreement with its index 4. The untwisted part (1, 1) has eigenvalue 2 on the representation 4. Every map in Hom_K(4, S^- (x) (1, 3)) annihilates (2, 1) in 4. These are the twistor spinors of S^4, as sections of S.

Proof. The representation count is as in Proposition 28.2, and the Callias count is a reference, with the exact count in Proposition 28.11. For the last sentence, the first-order operator from (1, 2) to (2, 3) in the slice limit is the twistor operator, and its kernel is the 4. ■

On a representation V the twisted operator has D^2 = Cas_G(V) + 3/2 - Cas_K(tau), for the twist tau, with Cas_G(4) = 5/2, Cas_G(16) = 15/2, Cas_G(20) = 21/2, Cas_G(40) = 29/2, Cas_K(j_a, j_b) = 2 j_a(j_a+1) + 2 j_b(j_b+1). With the twist (1, 3) this gives 0, 5, 8, 12, and with the twist (1, 1) on the representation 4 it gives 4. The slice operator reproduces these (Result 28.5).

Proposition 28.4 (the compression). Let s_v(g) = F lambda(g)^-1 v be a section in the multiplicity space of 4 on a slice, with B = F^* F, which commutes with lambda(H). Write t-hat = 2 i t_3 of Sp(1)_b. Then, with normalization by the norm of s_v,

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Q(x_a) = (tr(B Gamma_5)/tr B) Gamma_a/5,
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and for the functions y_T(g) = tr(T Ad(g) t-hat) with T in the Lie algebra sp(2), Q(y_T) is a multiple of T, proportional to tr(B t-hat)/tr B.

Proof. End_H(4) is spanned by 1, Gamma_5 and t-hat. Since int lambda(g) Gamma_5 lambda(g)^-1 x_a = int x_a x_b Gamma_b = Gamma_a/5, the part with Gamma_5 gives the first formula. Equivariant maps from 5 to End(4) are multiples of Gamma_a, and tr(Gamma_a Ad(g) t-hat) = 0, so 1 and t-hat contribute nothing. The second formula is Schur's lemma on the representation 10. ■

Consequently, a mode with F supported on (1, 2) and K-symmetric has tr(B Gamma_5)/tr B = +-1 and tr(B t-hat) = 0. That is the zero section: Q(x_a) = +-Gamma_a/5, and the part of a function that depends on the fibre gives zero. On the boundary with the pure level-one mode, B is the projection on (1)_{-1}, so that tr(B t-hat)/tr B = +-1. Example 16.9 has these values.

The operator and its two ends

Proposition 28.8 (endpoint restrictions). Restriction C^infinity(X, R) -> $C^{infinity(S}4$, R) + $C^{infinity(CP}3$, R) is surjective, and in normal coordinates a(y, v) = a(y, 0) + O(|v|), uniformly on bounded subsets of C^1(X). (Report, Proposition 1.)

Theorem 28.9 (a self-adjoint realization). Take the connection metric g = dr^2 + h(r)^2 $pi^{* g_\{S}4\}$ + rho(r)^2 g_{S^2} with h smooth, even and positive at 0, rho odd and smooth with rho(r) = r + O(r^3), and a product near r = l. Let Phi_M = M b(|v|^2) v . J with b smooth and positive, T_M = D_X^E + i Phi_M, and D-hat_M the doubled operator. At r = l set Z = -i c(dr), chi = Gamma-hat Z, and impose chi psi = -psi. Then D-hat_M on this H^1 domain is self-adjoint, elliptic and odd, with compact resolvent,

and every smooth scalar function preserves the domain. No boundary condition enters at the zero section: after the outer condition is fixed, removing S^4 creates no further self-adjoint extension. (Report, Theorem 2, using Bär-Ballmann, Corollary 4.18.)

Proposition 28.10 (three slice operators). For the standard complex structure on CP^3: (1) the kernel of the Dolbeault-Dirac operator on O(1) is C^4 for every Hermitian metric; (2) the Levi-Civita Dirac operator twisted by the separate line bundle O(3) with its invariant connection vanishes on the 4 for every invariant squashing, which gives a kernel of dimension at least four, with any further kernel in other representations at a locally discrete set of squashings; (3) the Levi-Civita Dirac operator twisted by the full pull-back connection on E is invertible for small fibres, since $(D^{E_rho})2 >= 1/(2\ rho^2) - C_0 - C_1\ rho^2$. (Report, Proposition 3.) In Python, on the representation 4, the smallest |eigenvalue| of that operator is 3.235, 1.842, 1.200, 0.722, 0.148 at rho = 0.3, 0.5, 0.707, 1, 2. So the window comes from the competition between the normal operator and the mass.

Proposition 28.11 (normal model). Let Q = sigma . d_y + y . J on $L^{2(R}3$; C^2 (x) C^4), with J Hermitian of spin 3/2. Then ker Q = 0 and ker Q^* = 3 (+) 1 under the normal Sp(1), with a positive gap above the zero modes; changing the sign of the mass exchanges the two kernels. The proof uses Q^* Q = -Laplacian + (y . J)^2 - sigma . J >= 0, with equality excluded since sum_i J_i^2 = 15/4, and the Callias index for the positive eigenbundles of n . J, of degrees 3 and 1. The induced bundle over S^4 is S^2(S) (+) 1, the operator on Sigma (x) S^2(S) has exactly four zero modes, in one chirality, and on the parallel inclusion S -> S (x) S^2(S) the twisted equation is a multiple of the twistor equation. So the kernel consists of harmonic spinors of the twisted operator, which are twistor spinors as sections of S. Rescaling gives the localization scale r = O(M^{-1/2}) at the core and O(M^{-1}) at the wall. (Report, Proposition 4.) In Python, in an oscillator basis: Q^* Q starts at 1.02, QQ^* has four near-zero values, with total spin 0 and three values near spin 1, then 1.02.

Lemma 28.12 (tangential mixing). On a product collar, normalized stable profiles for mass eigenvalues M q_i overlap by 2 sqrt(q_i q_j)/(q_i + q_j), independently of M, which is sqrt(3)/2 for q = 1/2, 3/2. So large mass does not remove the coupling between the stable channels O(3) and O(1), which in the pull-back connection has order 1/rho. (Report, Lemma 5.)

Proposition 28.13 (the boundary operator on the representation 4). Take the sign of Phi for which the right singular vector of T is at the zero section. Its left singular vector v is at the wall, in the domain Z v = +v of Theorem 28.9 for T^. *A wall profile exp(q(x - l)), q > 0, for T^* = i Z d/dx + D_6 + i phi tau, with phi > 0, has q = -phi Z tau at leading order. So the stable channels of v are (Z, tau) = (+, -3/2) and (+, -1/2) twice, and they map to the channels (-, -1/2) twice. In unitary bases of these, the compression of D_6(rho) with the overlaps of Lemma 28.12 is

```
A(rho) = ( 3/(2 rho)   1/rho   rho + 1/rho )
         (     0       sqrt 2     -sqrt 2  )
```

with the matrix elements computed in Python, and reproduced by Terra 6.1 in exact symbolic arithmetic from the bracket and Clifford data. Its kernel is spanned by c = (-(2/3)(rho^2 + 2), 1, 1). The channel tau = -3/2 and one channel tau = -1/2 are supported on (1)*{-1} of the 4; the other channel tau = -1/2 is supported on (1)*{+1}. All three lie in Gamma_5 = -1, and their target spaces are mutually orthogonal, so the cross terms vanish in the traces of Proposition 28.4. Hence for the mode c

```
tr(B Gamma_5)/tr B = -1,    tr(B t-hat)/tr B = -w(rho),    w(rho) = 2(rho^2 + 2)^2 / (9 + 2(rho^2 +
↪ 2)^2),
```

for every rho, with horizontal scale 1; for horizontal scale h, replace rho^2 by $rho^{2/h}2$. So w = 8/17 as rho -> 0, w = 2/3 at the Fubini-Study radius rho = 1 (the nearly Kähler radius is 1/sqrt 2), and w -> 1 as rho -> infinity, the level-one value. The nonzero singular values of A are the next wall values on the 4: 1.863 and 2.789 at rho = 1, with the core value 2 in between. With the opposite sign of Phi, the right singular vector is at the wall, in the channels Z = -1, tau > 0, with fibre part +w(rho), and its left singular vector at the zero section.

For the connection preserving the line summands, the first row is replaced by (0, 1/rho, rho + 1/rho), the kernel is (1, 0, 0), so w = 1 for every rho, and the block on tau = -1/2 has singular values with sum of

squares rho^2 + 2/rho^2 + 6 and product sqrt 2 (rho^2 + 2)/rho, so it is invertible for every rho > 0, with D^2 = 3, 6 at rho = 1. This is the collar repair of the report: an invariant connection preserving the four line summands in a collar extends into X with a cutoff, with the pull-back connection near the zero section. At the Fubini-Study metric the stable boundary model is then a sum of line-twisted Dirac operators, with four zero modes on O(3) and none on O(1), since $H^{q(CP}$3, O(-1)) = 0 for all q.

Status. The kernel and the traces are exact for this finite block, which is the leading formal wall operator; the next correction is O(1/M) in the fixed-block expansion and does not change the limiting kernel, since the kernel is separated from the other singular values. That the boundary mode of T_M converges to it as M -> infinity is a numerical statement here (Result 28.5), since a uniform global reduction is unproved: the difference falls as phi_0 grows, from 0.037 to 0.009 at wall radius 0.5 and from 0.008 to 0.002 at wall radius near 1, between phi_0 = 16 and 32. The value 2/3 tests the factor sqrt 3/2, since without it w(1) would be 3/5.

Lemma 28.14 (a sufficient spectral criterion). Let U_M : C^8 -> Dom D-hat_M be an isometry with ||D-hat_M U_M|| <= eps_M -> 0 and ||D-hat_M psi|| >= delta ||psi|| for psi orthogonal to the range, with delta > 0 independent of M. Then for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), and the spectral space is within O(eps_M/delta) of the range. (Report, Lemma 6.) The missing steps for Conjecture 28.6 are the normal reduction with its errors, and the coercive estimate on every representation of Sp(2).

Conjecture 28.6 (the window). Take Theorem 28.9 with the sign of Proposition 28.13 and with (a) the pull-back connection and wall radius rho, or (b) the collar connection of Proposition 28.13 and the Fubini-Study metric at the wall. For large M the hypotheses of Lemma 28.14 hold, with the eight states being one copy of the 4 at the zero section in one Gamma-hat degree and one at the boundary in the other. For a family of metrics and mass profiles with a uniform positive decay rate, the off-diagonal block is exponentially small in l. The compression of a smooth function tends to (Q_B(a|*{S^4}), Q_Y(a*/{CP^3})), with Q_B and Q_Y(x_a) = -Gamma_a/5, and Q_Y(y_T) a multiple of T, equal to the level-one value of Example 16.9 times w(rho) in case (a), and equal to it in case (b). This concerns compression maps, in line with Counterexample 28.17.

Result 28.5 (Python, the radial system). We solve T on the multiplicity spaces, with Hermitian Clifford generators, so that c(dx) = i Z in the skew convention of Theorem 28.9. For horizontal scale h, after absorbing the half-density h^2 rho, T = i Z d/dx + A_0/h + A_1/rho + (rho/h^2) A_2 + i phi tau, with -s(x) A_E/rho added for a collar connection. We take h = 1, so that T = i Z d/dx + D_6(rho(x)) + i phi(x) tau, with D_6 built from the structure constants of sp(2), rho(x) = rho_inf tanh(x/rho_inf), phi(x) = phi_0 tanh(x/w_0), w_0 = 0.3, in units with S^4 of radius 1, by the quadratic form of T^* T with P1 elements on a graded grid. This rho is not exactly a product near the wall. Rayleigh quotients of the exact form bound the values from above, and with three-point quadrature they approximate those bounds. Small entries have a mesh floor that depends on the parameters: at (phi_0, l) = (6, 4) the lowest value is 0.00548, 0.00401, 0.00355 for 2500, 5000, 10000 elements, so such entries do not establish a zero continuum value. At x = 0 the exponents are the values for R^3: +-1, +-2 (three times), +-3 on the 4.

phi_0	l = 2	l = 3	l = 4	next singular value, l = 4
3	0.406	0.159	0.066	1.293
4	0.288	0.088	0.028	1.452
5	0.200	0.045	0.011	1.665
6	0.132	0.021	floor	1.818
8	0.049	floor	floor	1.852

(rho_inf = 1, representation 4; the next value at larger phi_0 is 1.86 to 1.87.) The lowest value falls with the exponent of the constant-coefficient system, the smallest |Re kappa| of i Z (D_6(rho) - i phi tau): measured between l = 3 and 4, 0.877, 1.155, 1.464 at phi_0 = 3, 4, 5, against 0.856, 1.151, 1.476, as numerical evidence for these profiles, without a general tunnelling theorem. At large phi, kappa = phi/2 - 2/rho + O(1/phi)

(phi = 64: 30.07 at rho = 1, 31.07 at rho = 2). The formula 2 mu exp(-mu l) of lemma 09 belongs to a separable interval and does not carry over.

On the two modes, at rho_inf = 1, l = 3, with the sign of Proposition 28.13, the boundary mode being the left singular vector; the core fibre parts at phi_0 = 16, 40 are below what the computation resolves:

phi_0	zero section: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B	boundary: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B, Z, tau
10	0.39, -1.0000, -3e-7	2.93, -1.0000, -0.645, 0.944, -1.117
16	0.28, -1.0000, 1e-11	2.96, -1.0000, -0.658, 0.976, -1.146
40	0.16, -1.0000, 1e-11	2.99, -1.0000, -0.663, 0.996, -1.162

Against Proposition 28.13, at l = 3:

rho_inf	wall radius	-w(rho)	phi_0 = 16	phi_0 = 32	next value, phi_0 = 16	second singular value of A
0.5	0.500	-0.529	-0.493	-0.521	1.992	1.992
0.7	0.700	-0.579	-0.562	-0.575	1.966	1.966
1.0	0.995	-0.665	-0.658	-0.663	1.864	1.865
1.5	1.446	-0.788	-0.785	-0.787	1.660	1.661
2.0	1.810	-0.861	-0.858	-0.860	1.557	1.556
3.0	2.285	-0.921	-0.918	-0.919	1.487	1.486

Collar connection. With rho(l) = 1 and s(x) = (1 + tanh((x - l + 1)/0.15))/2 multiplying the vertical connection term to be removed, at phi_0 = 8, 16, 32: the boundary mode has tr(B t-hat)/tr B = -1.0000 and Q(x_a)/(Gamma_a/5) = -1.0000, the next value is 1.733, that is sqrt 3, and the core mode is unchanged. The switch is nonzero, though tiny, at x = 0, whereas the smooth extension needs a cutoff vanishing near the zero section, so these rounded values support the limiting model and do not establish exact finite-mass equality.

Other representations. The multiplicity is zero on 1, 5, 10, 14, 35. At phi_0 = 16, rho_inf = 1, l = 3, the lowest singular values are 2.188 (wall) and 2.235 (core) on the 16, 1.920 (wall) and 2.828 (core) on the 20, and 3.381 (wall) and 3.463 (core) on the 40, the core values being sqrt 5, sqrt 8 and sqrt 12 up to the mass. The multiplicity zero follows from the central character. The finite wall operator A_V(rho) has no zero at 1500 sampled radii in [0.1, 20] on the 16, 20, 40, 64, a sampled statement; its smallest singular value tends to the core value as rho -> 0 and falls slowly at large rho on the 16 and the 40 (0.33 and 0.67 at rho = 20). Higher representations were not computed, and no estimate in the representation is proved.

Core block at finite mass. The density of Proposition 28.16 is nonzero at these points, stable in the mesh to five digits:

phi_0, l	lambda	tr(B t-hat)/tr B, integrated	max of abs tr(B t-hat) over max of tr B	sup of the pointwise ratio	integrated value over lambda^2
4, 3	0.0882	-5.71e-3	6.48e-3	0.849	-0.73
5, 3	0.0453	-1.84e-3	1.97e-3	0.856	-0.89
6, 3	0.0212	-4.55e-4	4.73e-4	0.858	-1.01

The pointwise ratio reaches its maximum at the outer wall, where the core density is small, so the breaking comes from the tail of the core mode at the wall. Over phi_0 = 3 to 6, l = 2 to 4, the integrated value

over lambda^2 lies between -0.4 and -1.0, an empirical observation with no O(lambda^2) asymptotic proved. These values support the hypothesis of Proposition 28.16 at the sample points, conditional on the density being nonzero in the continuum.

The compression and the algebra

Theorem 28.15 (compression, conditional on localization). Assume an eight-dimensional spectral window splitting under Gamma-hat as W_M (+) V_M with dimensions 4, 4. Then scalar compression is exactly block diagonal. If the isometries differ by at most eta_M from model embeddings with angular compressions Q_B, Q_Y and normal measures with first moments at most C M^{-1/2} at the core and C M^{-1} at the wall, then

```
|| Q_M(a) - (Q_B(a|_B) (+) Q_Y(a|_Y)) || <= C ||a||_{C^1} M^{-1/2} + 4 ||a||_infinity eta_M.
```

(Report, Theorem 7.) For a spectral wall as in lemma 24, the domain has constant boundary values, and if a|Y = *c then any boundary-localized family has* //Q{Y,M}(a) - c|| <= C M^{-1} ||d_r a||_infinity. So localization just inside a spectral wall does not recover arbitrary boundary functions for a fixed smooth a, and the limit needs a local wall or an explicitly different comparison algebra.

Proposition 28.16 (the exact symmetry test). For an equivariant copy of 4, B(r) = F(r)^* F(r) = diag(b_a(r) I_2, b_+(r), b_-(r)). All real scalar functions compress to operators commuting with the quaternionic antiunitary j if and only if b_+(r) = b_-(r) for almost every r. If in addition the 5 component is nonzero, the block generates M_2(H). If both the 5 and the 10 components are nonzero, it generates M_4(C). A vanishing integral of tr(B(r) t-hat) alone does not decide this. If a radial h has Q_M(h) = alpha I_W + beta I_V with alpha != beta, then the block projections lie in the generated algebra. (Report, Proposition 8.) At the limit, the real linear image is Herm_2(H) + Herm_4(C), of real dimension 6 + 16 = 22, and the generated real algebra has dimension 16 + 32 = 48.

Counterexample 28.17 (generation is discontinuous under convergence of compression maps). Let R_0 = Q_B E_fib, with E_fib the fibre average on CP^3, R_1 = Q_Y, and R_eps = (1 - eps) R_0 + eps R_1. These are unital completely positive, ||R_eps - R_0|| <= 2 eps, and real functions generate M_4(C) under R_eps for every eps > 0, and M_2(H) at eps = 0. So the correct limit statement concerns specified compression maps, and it gives no convergence of the generated algebras. (Report, Counterexample 9.)

Theorem 28.21 (prescribed windows on the same X). Let P_y = (1 + y_a Gamma_a)/2 and p_z <= P_{pi(z)} the tautological rank-one projector. With normalized radial profiles chi_0, chi_l supported in disjoint eps-neighbourhoods of the two ends, U_0 v = sqrt 2 chi_0 P_y v and U_l v = 2 chi_l p_z v are isometries into L^2(X; C^4) (+) L^2(X; C^4). Real functions compressed by them generate M_2(H) + M_4(C) at every eps, and the compression differs from Q_B (+) Q_Y by at most C eps ||a||_{C^1}. (Report, Theorem 13.) This proves the connected-space mechanism without Conjecture 28.6, with the window as input. Allowing nonlocal smoothing perturbations, (1 - P) D (1 - P) + P d P makes such a window reducing, which enlarges the operator class and keeps the window as input.

The finite data

Proposition 28.18 (charge conjugation and the wall). With Hermitian generators J_i, C = C_7 (x) j_E has C^2 = -1 and commutes with T_M, D-hat_M and Gamma-hat as a differential expression, and CZ = -ZC. So it exchanges the two local wall conditions and does not preserve the chosen domain. If a pointwise parent antiunitary preserves a window and commutes with real multiplication, every real compressed multiplier commutes with its restriction, which cannot hold on a generated full M_4(C). (Report, Proposition 10.) The complex boundary block therefore requires this loss of the inherited symmetry, or a different real correspondence.

Remark 28.7 (why one block is quaternionic and the other complex). Proposition 28.18 proves the signs argument of the first version, with one correction of wording: j_E commutes with the anti-Hermitian Lie algebra action and reverses the Hermitian generators, so that C commutes with i Phi. Numerically (Python, phi_0 = 16, l = 3, rho_inf = 1), exchanging the wall condition exchanges (Z, tau, tr(B t-hat)/tr

B) = (0.976, -1.146, -0.658) and (-0.976, 1.146, 0.658) on the boundary mode, with the same singular values, and leaves the zero-section mode unchanged. The formal parent symmetry does not by itself prove that the finite-mass core space is preserved, since C changes the wall domain, and the breaking computed in Result 28.5 shows the size of the effect at the sample points.

Proposition 28.19 (tunnelling and the grading). If the window consists of one copy of the 4 in each Gamma-hat degree, then d_M = [[0, conj(lambda_M) I_4], [lambda_M I_4, 0]] after equivariant unitary identifications, by Schur's lemma, and lambda_M = (1/4) Tr(U_{Y,M}^* T_M U_{B,M}). The inherited grading is I_W (+) (-I_V), the auxiliary grading of the doubling, which labels the two ends under localization. The radial Clifford involution Z does not extend across the zero section, so it does not serve as an interval chirality. This grading leaves all four core states in one degree, so W = H (+) H, for the even algebra H + H + M_4(C), still needs a chosen quaternionic grading, such as a marked Gamma_p. (Report, Proposition 11.)

Proposition 28.20 (the doubled bimodule). With H = W (+) V, Theorem 16.2 gives K = H (x) conj(H), of dimension 64, with DD = d (x) 1 + Gamma-hat (x) d^c. Its odd part is exactly K^- = Hom(V, W) (+) Hom(W, V), with the desired bimodule actions, and order zero and order one hold exactly on the full K, for its actual even algebra. If lambda != 0, then DD maps K^- into K^+, P_- DD P_- = 0 and DD^2 = 2 lambda^2. So K^- is not a reducing subspace. On K^- the inherited tensor grading is constant; the sheet-sign grading of Theorem 16.7 gives J(x, y) = (y^, x^), J^2 = 1, J Gamma_6 = -Gamma_6 J, with zero Dirac operator. The tunnelling map W -> V does not provide the map Hom(V, W) -> Hom(W, V) of lemma 20. (Report, Proposition 12.)

Balance against Proposition 6 of lemma 24

(Report, Proposition 14, with the numerical statements updated.)

Structure	What follows
Connected scalar parent	Supplied by the chosen disc bundle X.
Independent endpoint functions	Derived, by restriction and extension.
Base-function restriction at the collapsed end	Derived geometrically; quantitative for localized states.
Four core states	Proved for the linear normal model and the resulting twisted operator on S^4.
Four boundary states	Proved for the separated collar model; for the pull-back connection, the finite block of Proposition 28.13, with the convergence numerical.
Isolated 4 + 4 window	Conjecture 28.6.
Limiting quaternionic core compression	Derived once convergence to the core model is established.
Exact finite-mass quaternionic block	Requires b_+ = b_-; numerically not satisfied at the sample points resolved.
Exact level-one boundary compression	Requires rho -> infinity or the separated collar connection.
End grading	Inherited from the chosen self-adjoint doubling.
Quaternionic grading axis	Input.
The two Hom spaces	The odd part of the coarse correspondence; the selection is additional.
KO-6 reality and order conditions	Exact after the specified correspondence and grading choices.
Nonzero connector of lemma 20	Input.
Mass value and dependence on l	Radial matching; equivariance gives only lambda I_4; the exponent is computed numerically in Result 28.5.

Choices that stay inputs: X, E = S^3(S), the hedgehog, the wall condition, the collar connection in case (b), and the doubling of Theorem 16.2. A double with two identical collapsed zero sections removes the outer boundary condition, but the same local core construction gives two quaternionic endpoint maps, so obtaining a complex block needs different structure at one end. T0 stays open, and this is a candidate for its first half.

Direction. The next analytic step is the coercive estimate of Lemma 28.14, in case (b) first, where both end models have exact kernels. Since Phi commutes with the Clifford symbol, D + iM Phi is a concentrating pair in the sense of Maridakis (arXiv:1510.07004, Definition 2.1 and Lemma 2.2), with [D, Phi] of order zero; the symbol condition goes back to Prokhorenkov and Richardson (arXiv:math/0307251). Two things are missing there. The spectral separation theorem of arXiv:2110.11654 (section 2, Assumption 3) needs a leading endomorphism independent of the normal direction, and this fails here, since (n . J)^2 depends on n for spin 3/2; so a separate core analysis or an anisotropic extension is needed. And the wall adds a boundary layer: on a half-line ||(d_t + M) f||^2 = ||f'||^2 + M^2 ||f||^2 - M |f(0)|^2, and exp(-Mt) is a zero mode away from the zero set of the mass. (Read by Terra 6.1 in arXiv HTML; the LaTeX sources were not available to it.)

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026: tools/lemma28_radial.py runs the tests as assertions, and with the argument `tables` it prints the main tables. The slice operator on each multiplicity space is Hermitian, odd under Z, and closes on Hom_H. The structure constants reproduce [t_1, t_2] = t_3 in Sp(1)_a and Sp(1)_b. The decomposition D_6(rho) = A_0 + A_1/rho + A_2 rho holds. The form uses three-point Gauss quadrature on each element. The exact tests are {A_0, A_1} = 0, A_0^2 = 0, 4 on the 4, 5, 9 on the 16, 0, 8 on the 20, 12, 16 on the 40 (the value 0 on the 20 has the twist (1, 5) and gives no core zero mode), A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha, and at rho = 1 with the line connection the values D^2 = 0, 5, 5, 12 on the summand O(3) of the 4, 16, 20, 40, against j(j + 4) for (0, 0)-forms in O(1); the whole summand also has the value 10 on the 20. The closed form of Proposition 28.13 is tested to ten digits, and the core mode density in the mesh.

Part 29

The eight-state window on the disc bundle

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T0, without an Astra job. It continues lemma 28 and answers the prompt 29_window_coercive_estimate.md. Refereed by Terra 6.1 (report): case (b) stands with amendments, and the two errors it found in case (a), the definition of A_eff and the sign of the fibre part, are corrected below. The referee ran no computation. A fresh reading of the corrected case (a) in the referee report on lemma 30 found one more sign and one missing step in Proposition 29.10; these are corrected, and the corrections have not been read again.

Outcome

1. **Conjecture 28.6 holds in case (b)** for the class of metrics and mass profiles of Setting 29.0: for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), for every delta < sqrt 3 (Theorem 29.7).
2. **Conjecture 28.6 holds in case (a) at the Fubini-Study radius** rho = 1, with fibre part -2/3 (Theorem 29.11). At another wall radius, case (a) follows from one kernel condition K(rho), which is a finite check at each radius.
3. **The off-diagonal block is exponentially small**, |lambda_M| <= C exp(-(1 - eta)(M/2) int f dr), by an exact tunnelling identity and Agmon estimates (Proposition 29.8).
4. **The compression converges** to (Q_B(a|*{S^4}), Q_Y(a/*{CP^3})) with error O(M^{-1/2}) (Theorem 29.9). The statement concerns compression maps, in line with Counterexample 28.17.

The proof is short because Sp(2)-equivariance gives exact sums of squares at the core and at the wall.

Each estimate holds on all of L^2 of a fixed manifold, with the boundary operators elliptic on CP^3 or S^4, so uniformity over the representations of Sp(2) needs no separate argument. The spectral separation hypothesis of Maridakis, which fails for this mass (reviews/2026-11-03_terra_check_28.md, item 6), is never used. General forms of two identities are in the literature: the square with its zero-order cross term in the bulk (Maridakis, arXiv:1510.07004, Lemma 2.2), and a horizontal and vertical factorization with a curvature contraction (Kaad and van Suijlekom, arXiv:1710.03182, Proposition 18, under other hypotheses).

Scope. These are spectral statements for a chosen metric family, with flat fibres near the zero section. They give the finite window from the operator, for the sign of Proposition 28.13. They leave open: other metrics in the class of Theorem 28.9, including the Bryant-Salamon metric; the connector of Proposition 28.20; the quaternionic grading of Proposition 28.19; any dynamics selecting M, l or rho. Finite-mass generated algebras stay outside the statement (Counterexample 28.17).

Setting

Setting 29.0. Let X be the disc bundle of lemma 28, G = Sp(2), K = Sp(1)_a x Sp(1)_b, H = Sp(1)_a x U(1)_b, E = S^3(S), Phi_M = eps M f(r) n . J, and T_M = D_X^E + i Phi_M, with D-hat_M the doubled operator and the wall condition of Theorem 28.9: Z u = -u for T_M, and Z v = +v for T_M^*. Assume:

- (H1) g = dr^2 + pi^* g_{S^4} + rho(r)^2 g_{S^2}, with S^4 of radius 1, rho(r) = r on [0, 2e], and rho constant, equal to rho_l, on the collar [l - 2a, l]. Case (b) and Theorem 29.11 take rho_l = 1, the Fubini-Study metric.
- (H2) f(r) = r on [0, 2e], f = f_0 > 0 on the collar, and f > 0 on (0, l]. This is M b(|v|^2) v . J with b positive.
- (H3) The connection on E is the pull-back connection on r <= 2e. In case (b) it preserves the four line summands on the collar, with any invariant interpolation in between. In case (a) it is the pull-back connection throughout. We use radial parallel gauge: nabla_{d/dr} = d/dr. An invariant dr-component a(r) q could otherwise be added to a line-preserving connection; it is removed by radial parallel transport, which changes the displayed formulas by a phase on each line.
- (H4) eps = -gamma_b, where gamma_b is the chirality of Sigma_4 whose K-type is (1, 2). This is the sign of Proposition 28.13: its right singular vector is at the zero section. The opposite sign exchanges T_M and T_M^* throughout. In the convention of tools/lemma28_radial.py, gamma_b = +1.

On the collar put q = eps n . J, with eigenvalues +-3/2, +-1/2 on the line summands, and m = M f_0. In general q = -gamma_b n . J. In the notation of Proposition 28.13 and tools/lemma28_radial.py, q = -tau, and T_M = i Z d/dx + A + i m q on the collar, with x increasing towards the wall. The line q = +3/2 is the summand O(3), and q > 0 are the stable channels of Proposition 28.13.

Let C_R = ||A_2||, the norm on Sigma_6 (x) E of the curvature contraction A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha; it does not depend on V, and C_R = 1 in Python. Let mu_1 > 0 be the smallest positive eigenvalue of Q^* Q for Q = sigma . d_y + y . J of Proposition 28.11. It is positive since the resolvent is compact, because (y . J)^2 >= |y|^2 / 4, and ker Q = 0. Python gives 1.02.

The zero section

Proposition 29.1 (exact splitting). Identify sections over the tube r <= 2e with K-equivariant maps s : G -> F, F = Sigma_4 (x) L^2(R^3; C^2 (x) C^4), with c(e_a) = gamma_a (x) 1 and c(f_alpha) = gamma_5 (x) sigma_alpha. Let D_H = i sum gamma_a e_a, with e_a acting as left-invariant derivatives on G, and Q^ = gamma_5 sigma . d_y + eps M y . J. Then on this tube T_M = T_0 + R, with T_0 = D_H + i Q^ and R Hermitian of order zero with |R| <= C_R |y|, and

```
T_0^* T_0 = D_H^2 + Q^^* Q^,    T_0 T_0^* = D_H^2 + Q^ Q^^*,
```

with D_H^2 commuting with Q^^* Q and Q^ Q^^*.

Proof. The fibres are flat and totally geodesic by (H1), so the Levi-Civita connection of X differs from the connection associated with the principal K-bundle G -> S^4 only by the O'Neill tensor. Its component is

the curvature of N evaluated at v. This gives the zero-order term R, which equals r A_2 in the radial form D_6 = A_0 + A_1/rho + A_2 rho, of norm at most C_R |y|. For the pull-back connection, the remaining operator is D_H + i gamma_5 sigma . d_y + i Phi_M. The operator Q^ acts on the values of s, and it is K-equivariant, since sigma . d_y and y . J are invariant under the simultaneous rotation of y, of C^2 and of C^4. So Q^ maps equivariant maps to equivariant maps, and it commutes with each derivative e_a, because it does not depend on the point of G. Since gamma_a anticommutes with gamma_5 and commutes with y . J, D_H Q^ = Q^~* D_H, where Q^~* = -gamma_5 sigma . d_y + eps M y . J. Therefore

```
T_0^* T_0 = (D_H - i Q^~*)(D_H + i Q^) = D_H^2 + Q^~* Q^ + i (D_H Q^ - Q^~* D_H) = D_H^2 + Q^~* Q^,
```

and likewise for T_0 T_0^. *Finally D_H Q^{* Q} = Q^ Q^~* D_H*, so D_H^2 commutes with both. ■

In radial form. T_0 = i Z d/dx + A_0 + A_1/rho + i phi q, and the cross terms vanish by {A_0, Z} = 0, [A_0, tau] = 0 and {A_0, A_1} = 0. tools/lemma28_radial.py asserts these three identities on the representations 4, 16, 20, 40 (Python).

Proposition 29.2 (the model spectrum). On the total space of N, with f = r and T_0 as above, let Pi be the projection onto ker T_0. Then:

1. ker T_0 is one copy of the 4 of G; ker T_0^* = 0.
2. ||T_0 u||^2 >= min(4, mu_1 M) ||u||^2 for u orthogonal to ker T_0, and ||T_0^{* v}||2 >= min(4, mu_1 M) ||v||^2 for all v in the domain.

Proof. Write Q^_gamma for Q^ on chirality gamma. By Proposition 28.11, Q^_gamma has kernel exactly when gamma = -eps; this is the rank-four K-representation W_0 = (1, 3) + (1, 1), and on the orthogonal complement Q^{* Q} >= mu_1 M, after the rescaling y -> M^{-1/2} y. The same bound holds on the other chirality, and for Q^ Q^~* with the chiralities exchanged, since Q^* Q and Q Q^* have the same nonzero spectrum. The fibrewise projection P_W onto W_0, on both chiralities of Sigma_4, commutes with D_H, by Proposition 29.1. (The projection onto the fibre kernel of Q^ also selects gamma = -eps, so it does not commute with D_H, which is odd; it commutes with D_H^2.) On chirality -eps, Q^ vanishes on the range of P_W and is bounded below by sqrt(mu_1 M) on its complement; on chirality +eps, it is bounded below by sqrt(mu_1 M) throughout. On Sigma^gamma (x) W_0, D_H is the Dirac operator of S^4 twisted by the homogeneous bundle of W_0, so by Parthasarathy's formula D_H^2 = Cas_G(V) + 3/2 - Cas_K(tau), with Cas_K = 2 j (j + 1) on Sp(1)_b, that is 0 and 4 on W_0. Only V odd under -1 occur, with Cas_G(V) >= 5/2 and Cas_G(V) >= 15/2 for V not the 4. So D_H^2 >= 4 except for D_H^2 = 0 at V = 4, twist j = 1. This kernel needs Hom_K(4, Sigma^gamma (x) (1, 3)) != 0, which holds exactly for Sigma^gamma = (1, 2), where (1, 2) (x) (1, 3) contains (1, 2) once. By (H4), gamma_b = -eps, so T_0 sees this chirality and T_0^* the other. Decomposing u by chirality and by P_W, and using the sum of squares, gives both bounds. The computation holds on compactly supported smooth sections, and extends by closure, since the doubled model is essentially self-adjoint on that core; no boundary condition enters at infinity. ■

Lemma 29.3 (core estimate). Let u be in H^1 with support in r < 2e. Then

```
|| |y| u || <= (2/M) ||T_0 u|| + sqrt(10/M) ||u||,
```

and the same holds with T_0^. *So ||T_M u|| >= (1 - 2 C_R/M) ||T_0 u|| - C_R sqrt(10/M) ||u||, and likewise for T_M^* with T_0^**. In particular, for such u with Pi u = 0, and for all such v,

```
||T_M u|| >= (2 - o(1)) ||u||,    ||T_M^* v|| >= (2 - o(1)) ||v||,    M -> infinity.
```

Proof. Q^{* Q} = -Laplacian_y + M^2 (y . J)^2 - eps M gamma_5 sigma . J. The eigenvalues of sigma . J are 3/2 and -5/2, and (y . J)^2 >= |y|^2/4. Hence M^2 || |y| u ||^2 / 4 <= ||Q^ u||^2 + (5/2) M ||u||^2 <= ||T_0 u||^2 + (5/2) M ||u||^2, by Proposition 29.1. The rest is the triangle inequality with |R| <= C_R |y|, and Proposition 29.2. ■

The bulk

Lemma 29.4 (bulk estimate). Let w be in H^1 with support in a compact subset of 0 < r < l, and let f_1 be the minimum of f on that support. Then

```
||T_M w||^2, ||T_M^* w||^2 >= (M^2 f_1^2 / 4 - C_Phi M) ||w||^2,
```

with C_Phi the supremum of the operator norm of the Clifford contraction sum_i c(e_i) nabla_{e_i} (f n . J). With 0 < r_1 < r_2 < l and a radial cutoff psi equal to 1 on [r_1, r_2], with compact support in 0 < r < l, we have for u in Dom T_M,

```
||u||^2_{L^2(r_1 <= r <= r_2)} <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

Proof. ||(D + i Phi) w||^2 = ||D w||^2 + ||Phi w||^2 + <w, i [D, Phi] w>, with [D, Phi] of order zero and pointwise bounded independently of V, and Phi^2 >= M^2 f^2 / 4. For the second statement, T_M (psi u) = psi T_M u + [D, psi] u. ■

The interpolation of the connection in (H3) takes place in this region, where it enters only through C_Phi.

The wall

Lemma 29.5 (line-twisted operators at the Fubini-Study metric). Let D_k be the Dirac operator on CP^3 twisted by O(k) with its invariant connection, k odd, at rho = 1. Then D_k^2 = C_2(L) + 3/2 - (3/8) k^2 on the SU(4)-type L, with C_2 normalized by C_2(4) = 15/8. Consequently:

1. On O(3): ker D_3 is one copy of the 4 of Sp(2), in Z = +1; and D_3^2 >= 5 on the V-isotypic part for V not the 4.
2. On O(1): ker D_1 = 0; D_1^2 >= 3 on the 4 and D_1^2 >= 6 on the V-isotypic part for V not the 4.

So the smallest nonzero |eigenvalue| on O(3) + O(1), over all V, is kappa_w = sqrt 3, attained on the 4; the lowest wall value on the 16 is sqrt 5.

Proof. The form of D_k^2 is Parthasarathy's formula for SU(4)/U(3), since the twist is a character of U(3). The Sp(2)-invariant connection on a line summand is unique, because the isotropy representation has no trivial summand, so it is the Chern connection. The normalization: j(j + m + 3) is the eigenvalue of 2 box on (0,0)-forms in O(m) on the type [m + j, 0, j], with C_2([m + j, 0, j]) - C_2([m, 0, 0]) = j (j + m + 3), which fixes the coefficient of C_2; scalar curvature 12 gives the constant 3/2; and the term in k^2 is fixed by the kernel of D_3 on the 4. Two checks: D_0^2 = 4 on the type 6, the Kirchberg limit R/3 at R = 12 (Ann. Global Anal. Geom. 4 (1986) 291-325); and the values in the Python table below. The kernel statements are $H^{q(CP}3$, O(1)) = C^4 at q = 0 only, and $H^{q(CP}3$, O(-1)) = 0, with K^{1/2} = O(-2). The types of odd 4-ality with C_2 < 55/8 are the 4, its conjugate, the 20 and its conjugate (enumerated in Python). The 4 occurs on O(3) once, by the multiplicity on the slice, and this copy is the kernel; since each 20 restricts to Sp(2) as 4 + 16, neither 20 occurs on O(3). So V != 4 has C_2 >= 55/8 on O(3), giving D_3^2 >= 5. On O(1), the 4 and its conjugate restrict only to the 4 of Sp(2), so V != 4 has C_2 >= 39/8, giving D_1^2 >= 6; and the slice block on the 4 has D_1^2 = 3, 6. The chirality of ker D_3 is that of (0,0)-forms, and the slice block fixes it as Z = +1 for q = +3/2 (Python). ■

Lemma 29.6 (wall estimate in case (b)). Let w be in H^1 with support in the collar, vanishing near its inner end, and p(x) = sqrt(3 m) exp((3/2) m (x - l)), so that the kernel of T_M^* on the half-cylinder is p (x) ker D_3. Then:

1. If Z w = -w at r = l, then ||T_M w||^2 >= min(m^2/4, 3) ||w||^2.
2. If Z w = +w at r = l, and w is orthogonal to p (x) ker D_3, then $||T_M^{*\,w}||2$ >= min(m^2/4, 3) ||w||^2.

Proof. On the collar, A commutes with q and with d/dx, and anticommutes with Z. Writing w = w_+ + w_- by Z, integration by parts gives, for T_M,

```
||T_M w||^2 = ||(d + m q) w_+||^2 + ||A w_+||^2 + ||(d - m q) w_-||^2 + ||A w_-||^2,
```

where the cross terms are purely imaginary, using w_+(l) = 0. For T_M^* the same holds with q replaced by -q and with w_-(l) = 0. There are three one-dimensional cases on (-infinity, l]. (i) If g(l) = 0, then ||(d +- m q) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2. (ii) With g(l) free, ||(d + m |q|) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2 + m |q| |g(l)|^2. (iii) With g(l) free, ||(d - m |q|) g||^2 is the form of -d^2 + m^2 q^2 with a Robin

condition at l, whose spectrum consists of 0, on the profile sqrt(2 m |q|) exp(m |q| (x - l)), and of [m^2 q^2, infinity). Cases (i) and (ii) give the lower bound m^2/4. Case (iii) arises exactly in the channels (Z, q) = (-, q > 0) for T_M and (+, q > 0) for T_M^. *On those, expand in eigenvectors of A^2: the term //A w//^2 is at least 3 //w//^2, except on ker D_3 in Z = +1, which by Lemma 29.5 only arises for T_M^* at q = 3/2, where the Robin gap (9/4) m^2 applies. ■

The window

Theorem 29.7 (the window, case (b)). Under Setting 29.0 with rho_l = 1 and connection (b), let delta < sqrt 3. For M large, D-hat_M has exactly eight eigenvalues in (-delta, delta). The spectral subspace splits under Gamma-hat as W_M (+) V_M, each a copy of the 4 of Sp(2), with W_M within O(M^{-1/2}) of the cut-off core modes in the domain of T_M, and V_M within O(M^{-1/2}) of the cut-off wall modes p (x) ker D_3 in the domain of T_M^*.

Proof. We verify Lemma 28.14. Take radial functions chi_c, chi_b, chi_w with chi_c^2 + chi_b^2 + chi_w^2 = 1, chi_c = 1 on r <= e with support in r < 2e, chi_w = 1 on [l - a, l] with support in the collar. Let U_M span (chi_c u_i, 0) and (0, chi_w v_k), where u_i is a basis of ker T_0 and v_k of p (x) ker D_3, with Gram-Schmidt. Radial functions preserve the wall condition.

Trial states. T_M (chi_c u_i) = [D, chi_c] u_i + R chi_c u_i, of norm O(M^{-1/2}), since u_i has Gaussian decay at the scale M^{-1/2}; and T_M^* (chi_w v_k) = [D, chi_w] v_k, which is O(exp(-(3/2) m a)). So eps_M = O(M^{-1/2}).

Coercivity. Since [D, chi] = c(d chi) with c(d chi)^2 = |d chi|^2, and sum chi_i d chi_i = 0,

```
||T_M u||^2 = sum_i ||T_M chi_i u||^2 - sum_i || |d chi_i| u ||^2,
```

and likewise for T_M^*. The functions d chi_i vanish outside the bulk region e <= r <= l - a. So by Lemma 29.4,

```
(1 + C M^{-2}) ||T_M u||^2 >= ||T_M chi_c u||^2 + ||T_M chi_w u||^2 - C M^{-2} ||u||^2,    ||chi_b
↪  u||^2 <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

If u is orthogonal to every chi_c u_i, then Pi (chi_c u) = 0 exactly, since <u_i, chi_c u> = <chi_c u_i, u>, so Lemmas 29.3 and 29.6 (1) give ||T_M u||^2 >= (3 - o(1)) ||u||^2. If v is orthogonal to every chi_w v_k, then chi_w v is orthogonal to p (x) ker D_3, so Lemmas 29.3 and 29.6 (2) give the same for T_M^*. Since ||D-hat (u, v)||^2 = ||T_M u||^2 + ||T_M^* v||2, Lemma 28.14 gives the count and the distance to the range. The window is G-invariant and close to the range of U_M, so as representations W_M and V_M are copies of the 4. ■

Remark. The gap is set by the wall at sqrt 3, with the core next at 2. On the 16, 20 and 40 the lowest values are sqrt 5, sqrt 5, sqrt 12.

Proposition 29.8 (tunnelling). Let T_M u = lambda v and T_M^* v = lambda u be a singular pair of the window, with unit vectors. By Proposition 28.19, the four singular values coincide. For a radial chi equal to 1 near r = 0 and 0 near r = l,

```
lambda (<u, chi u> - <v, chi v>) = <v, c(d chi) u>.
```

For every eta in (0, 1) there are M_eta and C_eta with

```
|lambda_M| <= C_eta exp(-(1 - eta)(M/2) int_e^{l - a} f dr),    M >= M_eta.
```

Proof. The function chi u lies in Dom T_M, since chi vanishes near the wall. So <T_M^* v, chi u> = <v, T_M chi u>, and T_M chi u = chi lambda v + c(d chi) u. This gives the identity.

For the decay of u, let xi = psi exp(g) be radial, with psi = 0 on r <= e/2 and psi = 1 on r >= e, with g = 0 on r <= e, g' = (1 - eta)(M/2) f on [e, r_*], and g constant on r >= r_*. Since xi^2 u lies in Dom T_M,

```
||T_M (xi u)||^2 = lambda^2 ||xi u||^2 + || |d xi| u ||^2.
```

As in the proof of Theorem 29.7, with no core piece, the bulk bound of Lemma 29.4 and Lemma 29.6 (1), or Proposition 29.10 in case (a), bound the left side from below by the integral of (M^2 f^2/4 - C M) |xi u|^2

on the bulk, plus (3 - o(1)) ||xi u||^2 on the collar, minus C M^{-2} ||xi u||^2. On the bulk the excess over (g')^2 is at least (eta/4)(M^2 f^2 / 4) for M large, with f >= f_1 > 0, so ||exp(g) u||*{L^2(r >= e)}* <= *C_eta. For v we use the same argument from the wall inward, with g-tilde = 0 on [l - a, l], with -g-tilde' = (1 - eta)(M/2) f, and with Lemma 29.3 replacing Lemma 29.6 on the core, since T_M^* has no low states there. Take chi with |d chi| <= 2/s, supported on [r_, r_ + s] in the bulk. By Theorem 29.7,* <u, chi u> *-> 1 and* <v, chi v> *-> 0, so the identity gives |lambda| <= (4/s)* ||u||{[r_, r_ + s]} ||v||*{[r_, r_ + s]}. Run the weights with eta' = eta/2, and take s so small that (1 - eta')(I - int*{r_}^{r_ + s} f) >= (1 - eta) I, with I = int_e^{l - a} f. This gives the stated exponent. The absorption uses lambda -> 0, supplied by the trial states. The factor 1/2 holds on every channel, since (n . J)^2 >= 1/4. ■

For a family with l -> infinity, M >= M_eta, with uniform bounds on the end geometry, the cutoffs, the connection and the positive bulk mass, the constants are uniform, so lambda is exponentially small in l at the rate (1 - eta)(M/2) inf f. This is the uniform decay rate required by Conjecture 28.6, as amended by the referee.

Theorem 29.9 (compression). Under Theorem 29.7 or Theorem 29.11, for a in C^1(X),

```
|| Q_M(a) - (Q_B(a|_{S^4}) (+) Q_Y(a|_{CP^3})) || <= C ||a||_{C^1} M^{-1/2},
```

where Q_B(a) = int a(x) 2 P_x d mu(x), with P_x = (1 - x . Gamma)/2, and Q_Y(a) = int a(z) B(z) d mu(z). In case (b), B(z) = 4 p_z, the level-one map of Example 16.9. In case (a) at radius rho, B(z) = 2((1 + w) p_z + (1 - w) p_{jz}), with p_{jz} the quaternionic partner of p_z and w = w(rho); so Q_Y(x_a) = -Gamma_a/5, Q_Y(y_T) is w(rho) times the level-one value, and the signed fibre part is -w(rho). Restriction to the two ends is surjective (Proposition 28.8), so real functions on X generate M_2(H) + M_4(C) under the limit map, by lemma 10 and Example 16.9, for every w(rho) > 0.

Proof. Theorem 28.15, with eta_M = O(M^{-1/2}) from Theorems 29.7 and 29.11, after choosing equivariant unitary identifications of the window spaces with the model copies of the 4. The normal first moments are O(M^{-1/2}) at the core, by the Gaussian decay of the model, and O(M^{-1}) at the wall, by the profiles. At the core, F in Hom_K(4, (1, 2) (x) (1, 3)) kills (2, 1), so B = 2 P_x in the normalization Q_B(1) = I. The sign in P_x is the value tr(B Gamma_5)/tr B = -1 of Proposition 28.13. At the wall, the pointwise inner product of different line summands vanishes, so the density is a sum over channels with weights from Proposition 28.13. ■

Case (a)

On the collar in case (a), A = A_d + A_o, where A_d = D_line preserves the lines and A_o = A_E/rho has order zero, with [A_o, q] != 0. In Python, the block of A_E between q = 3/2 and q = 1/2 has norm sqrt 3 and ||A_E|| = 2, on the full fibre, hence for all V. Let P be the projection onto {sum_{q > 0} p_q g_q : Z g_q = -g_q}, with p_q = sqrt(2 m q) exp(m q (x - l)), and Y the same with Z = +1, with Pi_Y the projection onto the second space. Let F = Pi_Y (A_d + A_o) P, from the range of P to Y, and let A_eff be the operator on the sum of both ranges with blocks F and F^*. Identified with sections over CP^3 of the lines q > 0, it is, by Lemma 28.12,

```
A_eff = D_3(rho) (+) D_1(rho) + (sqrt 3 / 2) B(rho),
```

with B(rho) the block of A_o between the stable channels, of norm sqrt 3 / rho. This is self-adjoint and elliptic on the fixed manifold CP^3, so its nonzero spectrum has a gap kappa_a(rho) > 0, uniformly in V. On the 4, F^* is the matrix A(rho) of Proposition 28.13, from Y to the range of P.

Condition K(rho). The kernel of A_eff(rho) is exactly one copy of the 4, in Z = +1, so F is injective and ker F^* is that copy.

Proposition 29.10 (wall estimate in case (a)). Assume K(rho). For w as in Lemma 29.6, with the kernel p (x) ker D_3 replaced by the kernel of the half-cylinder operator T_M^*, the two statements of Lemma 29.6 hold with the constant 3 replaced by kappa_a(rho)^2 - O(1/m). The half-cylinder kernel is exactly one copy of the 4 for m large, with fibre part tending to -w(rho) at rate O(1/m).

Proof. T_d = i Z d/dx + A_d + i m q preserves the block structure: T_d P = Pi_Y T_d P, since T_d

(p_q g) = p_q A_d g. Also T_d maps the complement of P to the complement of Y, because for y in Y the function T_d^* y = A_d y lies in P. On the complement of P, ||T_d w_2|| >= (m/2) ||w_2||, by the sum of squares of Lemma 29.6, where the channels with Robin kernel are removed. So with w = w_1 + w_2 and w_1 = P w,

```
||T_M w|| >= ||(1 - Pi_Y) T_M w|| >= (m/2) ||w_2|| - ||A_o|| ||w||,
||T_M w|| >= ||Pi_Y T_M w|| >= ||F w_1|| - ||A_o|| ||w_2||.
```

Eliminating ||w_2||, one gets (1 + 2 ||A_o|| / m) ||T_M w|| >= ||F w_1|| - (2 ||A_o||^2 / m) ||w||. Under K(rho), F is bounded below by kappa_a(rho), and F^* is bounded below on the complement of its kernel. The same elimination holds for T_M^, *with P and Y and F and F^* exchanged. On the half-cylinder T_M is thus injective with closed range, so T_M^* is surjective; its kernel is the kernel of the Schur complement of the block T_d + (1 - Pi_Y) A_o (1 - P), whose inverse has norm at most (m/2 - ||A_o||)^-1. So this Schur complement differs from F^* by at most ||A_o||^2 / (m/2 - ||A_o||), and the graph of the Schur solution gives ||P_{ker T_M^} - *P_{ker F^}*|| = O(1/m). This projection estimate turns orthogonality to the exact half-cylinder kernel into orthogonality to ker F^* up to O(1/m), which gives the adjoint bound. Off the 4, this is invertible. On the 4 the block is a 2 x 3 matrix of rank 2, so its kernel has dimension exactly 1, within O(1/m) of (-(2/3)(rho^2 + 2), 1, 1). ■

Theorem 29.11 (the window in case (a)). Under Setting 29.0 with the pull-back connection, K(1) holds, and for delta < min(2, kappa_a(1)) and M large the conclusion of Theorem 29.7 holds at the Fubini-Study radius, with the wall modes replaced by the kernel of Proposition 29.10, and with fibre part tending to -w(1) = -2/3. Here kappa_a(1) >= sqrt 5 - 3/2 off the 4, and it equals 1.863 on the 4, the squared values there being (45 +- 3 sqrt 33)/8. At another wall radius the same conclusion holds under K(rho).

Proof. By Lemma 29.5, on V != 4 we have |D_3 (+) D_1| >= sqrt 5, while ||(sqrt 3/2) B(1)|| = 3/2. So A_eff is invertible there, with lower bound sqrt 5 - 3/2 > 0. On the 4, F^* is A(1), with kernel of dimension one in Z = +1, the vector (-2, 1, 1), and nonzero singular values 1.863 and 2.789, by Proposition 28.13. So K(1) holds. The proof of Theorem 29.7 then applies with Proposition 29.10 in place of Lemma 29.6. The trial wall states are cut-offs of the exact half-cylinder kernel, so their defect is exponentially small. ■

Other radii. For K(rho), only finitely many V can carry a kernel, those with |D_3(rho) (+) D_1(rho)| <= 3/(2 rho), by ellipticity. In Python, on the 16, 20, 40 and for rho in [0.2, 4], the compressed block has smallest singular value at least 1.30. A check of K(rho) on all relevant V at a given radius needs a lower bound for D_line(rho) on all V, which we have not written down away from rho = 1.

Remarks

Remark 29.12 (the index). The summands of E on CP^3 have indices 4, 0, 0, -4, of total zero, as equivariant cobordism invariance requires: E extends over X (Witten 1983, preprint pp. 43-44). The window has one copy of the 4 in each Gamma-hat degree, so its Sp(2)-index is 4 - 4 = 0. So index theory allows lambda != 0, and the eight states come from the estimates above. The sign in (H4) is consistent with this: with the opposite sign, the end-model modes at the core belong to T_M^* and those on O(-3) at the wall to T_M (Proposition 28.13), so the two ends never lie in the same Gamma-hat degree. At finite mass the full operator can have nonzero lambda, so these states need not lie in exact kernels. The relative sign of the two ends enters through the chirality of ker D_3. It follows from the slice weights, which contain the 4 once in Z = +1 and not in Z = -1, and the sign of the core is the chirality statement behind (H4). If the O(3) zero mode lay in Z = -1 with q > 0, it would be a wall zero mode of T_M, in conflict with Lemma 29.6 (1). As a check in Python, the mean position of the modes is 0.51 at the core for T_M and 2.96 at the wall for the adjoint problem, with l = 3.

Remark 29.13 (physical units). All statements are in shape units. Under the dilation g -> R^2 g, with D -> D/R and the potential divided by R, the operator is unitarily equivalent to D-hat_M / R. So the light mass is lambda_M / R, the gap delta / R, and their ratio is unchanged, while the compression maps stay the same; this observation is from the audit of 3 November 2026 by Astra 6, section 5.2, conditional there on the estimates proved here. A family with R -> infinity, with lambda_M / delta small, closes the physical gap and keeps the eight states separated. Whether a dynamics selects such a family is open. The limiting

geometry of such a family is a separate question: nothing here identifies it with a complete Bryant-Salamon manifold.

Remark 29.14 (other metrics). In the class of Theorem 28.9, T_M - T_0 also contains first-order terms with coefficients O(r^2), and M (b(r^2) - b(0)) y . J. One route shrinks the core piece to r < M^{-1/2 + s}, where Lemma 29.4 controls the complement since M^2 r^2 dominates. We have not carried this out, and the horizontal first-order terms need a bound in terms of D_H that takes account of the unbounded curvature terms.

Numerical check

Python (numpy, scipy): tools/lemma29_window.py. It asserts that [A, q] = 0 and {A, Z} = 0 at rho = 1 with the line-preserving connection, the values of Lemma 29.5 against the formula, and the norms used above; tools/lemma28_radial.py asserts the identities of Proposition 29.1. The direct solve is a test of the prediction: the limit values below were fixed by Proposition 29.2 and Lemma 29.5 before the run, and a failure would have refuted the argument. Setting: rho(x) = x for x <= 0.4, rho(x) = 1 for x >= 1.2, phi = M rho, l = 3, with the connection switched on [0.6, 1.2], and P1 elements at 2500 cells. Lowest singular values of T_M:

V	M = 8	M = 16	M = 32	predicted limit
4	small, 1.7321, 1.9861, 2.4495	small, 1.7321, 1.9963, 2.4495	small, 1.7323, 1.9991, 2.4497	0, sqrt 3, 2, sqrt 6
16	2.2245, 2.2361, 2.4495	2.2324, 2.2367, 2.4495	2.2351, 2.2409, 2.4497	sqrt 5 (core), sqrt 5 (wall), sqrt 6
20	2.2361, 2.8148, 2.8284	2.2367, 2.8248, 2.8284	2.2409, 2.8275, 2.8286	sqrt 5 (wall), sqrt 8 (core), sqrt 8 (wall)
40	3.4461, 3.4641, 3.4641	3.4585, 3.4641, 3.4645	3.4626, 3.4642, 3.4672	sqrt 12 three times

The wall values have a mesh error growing with M, of size 0.005 at M = 32 on the 16 and 20. The small value on the 4 is uninformative in this table: it halves when the mesh doubles, so it is a mesh floor, and Proposition 29.8 is proved without it. The wall mode of the adjoint problem has fibre part -1.00000 and Gamma_5 part -1.00000 at M = 8, 16, 32, as Theorem 29.9 says in case (b).

Part 30

The massless limit as a joint limit: a finite window and an opening tower

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T14. Theorem 30.1, Corollary 30.2, Proposition 30.3, Corollary 30.4 and Proposition 30.6 were proposed in an audit of commit b2ddf98 by Astra 6 (assessment in the review); the proofs below were redone against the source lemmas. Theorem 30.5 and Proposition 30.7 are new here and rest on lemma 29, whose referee is queued. Refereed by Terra 6.1, together with the note (report); its corrections are applied and have not been read again, since the Codex quota is spent.

Outcome

The claim that W and Z become massless as a dimension opens needs a family. Along a general path to M_W = M_Z = 0 nothing forces a tower to open: lemma 23 has theta -> 0 at fixed l, with the tower spacing fixed. The families below have the hierarchy

```
m_weak << E_eps << Delta_KK -> 0,
```

so a moving low-energy window keeps a fixed finite number of states. The hierarchy alone does not make the count at fixed energy diverge: the signed spectrum 0, +-eps^alpha, +-1, +-2, … has the hierarchy with a count bounded at fixed energy below 1. So the divergence at fixed positive energy is a separate property, proved below for each family.

1. On the exact interval this holds with an internal dimension of one (Theorem 30.1), also with localized end modes (Proposition 30.3).
2. On the cylinder of lemma 24 and on the disc bundle of lemma 29 it holds with seven internal dimensions. On the disc bundle the window comes from the operator, with one copy of the 4 at each end, and the count at fixed energy has seven-dimensional Weyl growth (Theorem 30.5).
3. On the disc bundle the core block of the window is quaternionic up to a defect that is O(M^{-1/2}) in general, and exponentially small when the connection preserves the quaternionic structure of E, for all bounded real observables (Propositions 30.6, 30.7).

Scope. These are statements about operators, compression maps and spectra along families given by hand. No dynamics selects them (Remark 30.8). The vector masses of Corollary 30.2 belong to the five-dimensional Yang-Mills toy of lemma 23. On the disc bundle only the fermion window is treated, so the relation between lambda and M_W is unproved there.

The interval

Theorem 30.1 (window and tower on the interval). Let (H_F, D_, *J_F, Gamma_F) satisfy the hypotheses of Theorem 22.1, with the algebra A_F represented by pi as in Theorem 22.4, r = dim H_F, and for 0 < alpha < 1, l_* > 0 and small eps > 0 let D_F(eps) = eps D_, *l_eps = l_* eps^{-alpha}, with D_eps the interval operator D_X of Theorem 22.1 for these data. Let Delta_eps be the smallest element of spec |D_eps| outside spec |eps D_*|. Then for eps small:

1. W_eps^* D_eps W_eps = eps D_*, with J_F and Gamma_F exact.
2. Delta_eps = (pi / l_*) *eps^alpha - eps ||D_||*.
3. ||Q_eps(a) - pi(a)|| <= (2/3) l_^2 *||D_||*^2 eps^{2(1 - alpha)} ||a||.
4. For alpha < beta < 1, E_* > 0 and E_eps = E_* eps^beta, the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals r.
5. For fixed E > 0, N_eps(E) ~ (2 r l_* E / pi) eps^{-alpha}.

Proof. The condition l_eps ||D_F(eps)|| = l_* ||D_|| *eps^{1 - alpha} < pi/2 holds for small eps. Theorem 22.1 gives item 1, and the next level is (pi - theta_max)/l = pi/l - ||D_F||, which is item 2. Theorem 22.4 (2) gives item 3. For item 4, eps ||D_||* « eps^beta « eps^alpha. For item 5, the signed spectrum is (n pi +- theta_k)/l_eps, which is r arithmetic progressions of spacing pi / l_eps, counted with multiplicity also when singular values coincide or vanish, each with (2 E l_eps / pi)(1 + o(1)) points in [-E, E]. ■

Theorem 22.1 starts from supplied finite data and a matrix-valued parent algebra, and the wall data change with eps by hand. A fixed Majorana block does not scale as eps D_*, so applying the theorem to an electroweak Dirac block needs a reducing sector or a treatment of the heavy neutral sector.

Corollary 30.2 (vector poles in the toy of lemma 23). For the universal channel of lemma 23, with m_eps = eps m_* and theta_eps = l_eps m_eps -> 0,

```
M_W = arcsin(sin(2 theta_eps)/sqrt 2) / l_eps,    M_Z = arcsin(2 sin(2 theta_eps)/sqrt 5) / l_eps,
```

so M_W^2 = 2 m_eps^2 (1 + O(theta_eps^2)) and M_Z^2 = (16/5) m_eps^2 (1 + O(theta_eps^2)). The vector poles follow the hierarchy. The constants 2 and 16/5 belong to this five-dimensional action. The small-mass pole coefficients of the fully massive finite spectral action are 3 and 24/5 (lemma 18, Proposition 14, and lemma 23), for the family with both members of every doublet at the same Dirac mass and no Majorana mass. Lemma 25 connects the two pairs at the level of the leading coefficient: its M^2 = ((3 - d_eff(s))/2) ||[D_F, T]||^2 / ||T||^2, with s = l Lambda, is the leading Taylor quotient of the heat-cutoff action for a generator constant in the transported frame, so the coefficient is 3 and 24/5 at d_eff = 0, and 2 and 16/5 at d_eff = 1. This does not identify the spectral-action pole problem with the Yang-Mills toy, and a joint-limit use needs a remainder uniform in s, since lemma 25 expands at fixed s. Along Lambda =

E_eps we have s = l_* E_* eps^{beta - alpha} -> 0, the finite value, and at fixed Lambda > 0 we have s -> infinity, the interval value. Varying the cutoff of the action is an extra prescription: an observational cutoff alone does not change the action.

Proposition 30.3 (a localized family). Take M = -mu and the mixed conditions of lemma 9, with mu_* > 0, 0 < alpha < 1, mu_eps = mu_* eps^alpha and the length l_eps^loc = c_eps / mu_eps, where c_eps = (1 - alpha) log(1/eps) + c_0. This length differs from the l_eps of Theorem 30.1. Let s = kappa l be the root of s = c tanh s, and m = mu sech s. Then

```
m_eps ~ 2 mu_* exp(-c_0) eps,    Delta_eps ~ mu_* eps^alpha,    (1/kappa) / l_eps = 1/s -> 0.
```

With the left profile L(x) proportional to exp(-kappa x) - r exp(kappa x), r = exp(-2 kappa l), the following identities are exact:

```
r = (m / (mu + kappa))^2,    mu - kappa = m^2 / (mu + kappa).
```

Proof. Lemma 9.5 gives m = 2 mu exp(-c)(1 + o(1)) and kappa/mu -> 1. The first oscillatory root has pi < k_1 l < 3 pi/2 and lambda_1^2 = mu^2 + k_1^2, so Delta/mu -> 1. From mu = kappa coth s and m = kappa / sinh s we get mu + kappa = kappa exp(s) / sinh s, hence m / (mu + kappa) = exp(-s). The second identity is m^2 = mu^2 - kappa^2. ■

The localization width and the length both grow, with a shrinking ratio. The finite two-point distance 1/m and the length l are both divergent, with (1/m)/l ~ exp(c)/(2c), so a metric convergence statement would need its own comparison maps.

Corollary 30.4 (the cylinder of lemma 24). Use the operator D_r = E d_x + Z A / r on the interval times CP^3, with A as in lemma 24, finite data on V = ker A = C^4 with gradings of type (2, 2) as in lemma 24, so that r = 4, the scaling of Theorem 30.1, and r_eps = r_* eps^{-gamma}, 0 < gamma < 1. Other values of r need an enlarged tangential bundle, which we do not construct. Then for max(alpha, gamma) < beta < 1, the cutoff E_* eps^beta isolates the finite window, since the next level is at least min((Delta_K / r_) *eps^gamma, (pi / l_)* eps^alpha - eps ||D_*||), by lemma 24, (5). At fixed E > 0 the count is proportional to l_eps r_eps^6 E^7, the seven-dimensional Weyl growth, by Dirichlet and Neumann bracketing of the squared product operator as l_eps and r_eps grow. For gamma = alpha the geometry expands homothetically in all seven directions.

The window here is prescribed by the spectral walls of lemma 24.

The disc bundle

Theorem 30.5 (joint limit on the disc bundle). Fix a shape as in Setting 29.0, with connection (b), or with connection (a) at rho = 1. Let delta > 0 be as in Theorem 29.7 or 29.11, G_0 = int_e^{l - a} f dr, 0 < alpha < beta < 1, E_* > 0, eta in (0, 1), and

```
M_eps = (2 (1 - alpha) / ((1 - eta) G_0)) log(1/eps),    R_eps = eps^{-alpha},    D_eps =
↪ D-hat_{M_eps} / R_eps.
```

By the dilation of Remark 29.13, which is exact under the canonical spinor identification and the L^2 unitary factor R^{-7/2}, D_eps is the operator of the metric R_eps^2 g with potential (M_eps / R_eps) f(s / R_eps) n . J in the physical radial coordinate s = R_eps r, of size M_eps / R_eps -> 0. For eps small:

1. D_eps has exactly eight eigenvalues in (-delta eps^alpha, delta eps^alpha), and they are +-m_eps with multiplicity 4 (or zero with multiplicity 8), where m_eps <= C eps. The next level is at most C log(1/eps) eps^alpha, so it tends to zero.
2. For E_eps = E_* eps^beta the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals 8.
3. The compression converges to the endpoint maps, with error C ||a||_{C^1} (log 1/eps)^{-1/2} in shape coordinates.
4. For fixed E > 0, N_eps(E) ~ C_X E^7 eps^{-7 alpha}, with C_X = 64 omega_7 Vol_g(X) / (2 pi)^7 the Weyl constant of the elliptic operator D-hat_0 on X, where omega_7 is the volume of the unit ball and 64 = 8 x 4 x 2 counts the spinors, E and the doubling.

Proof. Item 1: Theorem 29.7 or 29.11 applies since M_eps -> infinity, Proposition 28.19 gives the multiplicity, and Proposition 29.8 gives lambda <= C exp(-(1 - eta)(M_eps/2) G_0) = C eps^{1 - alpha}, hence m_eps = lambda / R_eps <= C eps. The bound on the next level: the shape-unit value is at most s_9(D-hat_0) + c M_eps, with the comparison below. Item 2: eps « eps^beta « eps^alpha. Item 3: Theorem 29.9, with M_eps^{-1/2} of the stated order. Item 4: D-hat_M = D-hat_0 + M V on the same domain, with V bounded by c = (3/2) sup f, so by min-max N(D-hat_0, Lambda - c M) <= N(D-hat_M, Lambda) <= N(D-hat_0, Lambda + c M). Take Lambda = E R_eps, so that M_eps / Lambda -> 0, and use the Weyl law for the fixed self-adjoint elliptic operator D-hat_0. ■

So the eight finite states stay separated at the moving scale, and at fixed physical energy the spectrum has the growth of a seven-dimensional internal space, for eleven dimensions in M^4 x X. Item 1 is an upper bound on m_eps, since a lower bound on lambda is unproved. A family with l_eps growing at fixed M has the same form, with the rate of Proposition 29.8. The constants M_eta, C_eta of Proposition 29.8 are fixed once the shape and eta are fixed, so no uniformity in a varying shape is used. The limit at fixed energy resolves a neighbourhood whose shape is fixed and whose size grows, so the pointed limit is flat at a fixed interior point and a half-space at a fixed boundary point. Nothing here identifies it with a complete Bryant-Salamon manifold, or with a solution of field equations.

The quaternionic defect

For the 4-copy of the window at the core, let B(r) = F(r)^* F(r) = diag(b_a I_2, b_+, b_-) as in Proposition 28.16, let j be the quaternionic antiunitary commuting with Sp(2), and P_H(A) = (A + j A j^-1)/2. Put

```
eta_t = int |b_+(r) - b_-(r)| dr / int tr B(r) dr.
```

Proposition 30.6 (all bounded real observables). For the normalized compression Q of this mode, ||Q(f) - P_H Q(f)|| <= 2 eta_t for every real f with ||f||_infinity <= 1. If X_1, …, X_k are such compressions, then ||X_1 … X_k - P_H(X_1 … X_k)|| <= 2 k eta_t.

Proof. With angular Haar measure of total mass 1, Schur's lemma gives the angular average (tr B / 4) I, so Q(f) = (4 / int tr B) int int f(r, g) lambda(g)^-1 B(r) lambda(g) dg dr. Since j commutes with lambda(g), Q(f) - P_H Q(f) is the same integral with B - P_H B. Here j exchanges b_+ and b_- and fixes the block b_a, so ||B - P_H B|| = |b_+ - b_-|/2. The triangle inequality gives the first bound. For the second, write X-tilde = j X j^-1, so that X - P_H X = (X - X-tilde)/2 and ||X_i - X-tilde_i|| <= 4 eta_t. Telescoping, X_1 … X_k - X-tilde_1 … X-tilde_k is a sum of k terms, each a product of contractions with one factor X_i - X-tilde_i, so its norm is at most 4 k eta_t; dividing by 2 gives 2 k eta_t. ■

So bounded expressions become quaternionic when eta_t -> 0. An unrestricted generated algebra may use coefficients of size 1/eta_t and retain the other directions, as in Counterexample 28.17.

Proposition 30.7 (the size of the defect). Under Theorem 29.7 or 29.11, eta_t = O(M^{-1/2}). Assume moreover that the quaternionic structure j_E of E is parallel on all of X. This holds for the pull-back connection and for the interpolation (1 - s(r)) nabla^pull + s(r) nabla^line with real s, and it can fail for other invariant interpolations allowed by (H3). Then for every eta in (0, 1) there are M_eta and C_eta with

```
eta_t <= C_eta exp(-(1 - eta) M int_{2e}^{l - a} f dr),    M >= M_eta.
```

This is the exponent of the bound on lambda in Proposition 29.8 doubled, with the integral starting at 2e. A bound in terms of lambda itself would need a lower bound on lambda, which is open.

Proof. For the first bound, fix a normalized equivariant model F_0 at the core with C F_0 = F_0 for the local real structure below, so that j B_0 j^-1 = B_0. Theorem 29.7 gives ||u - chi_c F_0|| = O(M^{-1/2}) in L^2, and |b_+ - b_-| <= (|F| + |C F|) |F - C F| gives the bound, by the Cauchy-Schwarz inequality.

For the second, j_E is parallel and j_E (n . J) j_E^-1 = -n . J, so C = C_7 (x) j_E commutes with D_X^E and with i Phi_M, hence with T_M as a differential expression, with C^2 = -1, and it exchanges the wall conditions (Proposition 28.18). For the line-preserving connection, j_E P_q j_E^-1 = P_{-q} gives j_E nabla^line j_E^-1 = nabla^line, and real affine interpolation keeps this. On Hom_G(4, -) put (C F) =

C F j^-1, which is antilinear with square +1. It commutes with D_H, with the full fibre term i Q^ and with R separately, since each part of the Clifford structure transforms in the same way, so it acts on the one-dimensional space Hom_G(4, ker T_0) as a real structure. Fix F_0 there with C F_0 = F_0, so that B_0 is j-invariant. Let u = F be the mode with T_M u = lambda v, T_M^* v = lambda u, with the phase for which <chi_c F_0, u> is real, with chi_c as in Theorem 29.7. Put zeta = u - C u and nu = v - C v. In the interior,

```
T_M zeta = lambda nu,    Pi (chi_c zeta) = 0,
```

where the second holds since <chi_c F_0, C u> is the conjugate of <chi_c F_0, u>. Take the weights of the proof of Proposition 29.8 starting at r = 2e and meeting at r_* = l - a: g = 0 on r <= 2e, g' = (1 - eta)(M/2) f on [2e, l - a], g constant beyond; and g-tilde = G - g on r <= l - a, g-tilde = 0 beyond, with G = g(l - a). So g-tilde is constant on r <= 2e. That proof gives ||exp(g) u|| <= C_eta on X, and ||exp(g-tilde) v|| <= C_eta on r <= l - a/2, and lambda <= C_eta exp(-G). The same bounds hold for C u and C v, since C is pointwise isometric.

Let xi = psi-tilde exp(g-tilde), with psi-tilde = 1 on r <= l - a and 0 on r >= l - a/2. Then ||T_M (xi zeta)|| <= lambda ||xi nu|| + || |d xi| zeta ||. Since xi is constant on the support of chi_c, Pi (chi_c xi zeta) = 0, so Lemmas 29.3 and 29.4 and the partition of Theorem 29.7 bound ||T_M (xi zeta)|| from below by the bulk term with (M/2) f and by (2 - o(1)) on the core. On the bulk |d xi| = (1 - eta)(M/2) f xi, which is absorbed with the factor eta to spare, leaving

```
||xi zeta|| <= C (lambda + ||u||_{L^2(l - a <= r <= l - a/2)}) <= C exp(-G).
```

Finally B - j B j^-1 = F^* F - (C F)^* (C F), so |b_+ - b_-| <= (|F| + |C F|) |zeta| pointwise in r. On r <= l - a, the Cauchy-Schwarz inequality with the weight exp(g) gives

```
int (|F| + |C F|) |zeta| <= 2 ||exp(g) u|| ||exp(-g) zeta|| = 2 ||exp(g) u|| exp(-G) ||xi zeta|| <=
↪  C exp(-2 G),
```

since exp(-g) = exp(g-tilde - G). On r >= l - a, both |F|^2 and |C F|^2 integrate to at most C exp(-2 G). ■

The samples in section 6.1 of the audit (Python, in the pull-back connection, rows reproduced by Claude Sonnet 5.5) have eta_t / lambda^2 between 0.5 and 0.9. They vary the length at fixed mass, with profiles outside Setting 29.0, so they are a qualitative comparison and do not test this statement as M -> infinity. Under the added hypothesis the defect comes from the wall, which breaks C: for an operator with the same interior and no wall, on a complete manifold, Hom_G(4, ker T) carries a real structure, and if it is one-dimensional the core block is exactly quaternionic.

Remark 30.8 (what is missing, and the direction). The next steps are, in order:

1. The vector sector on the family of Theorem 30.5: the kinetic matrix and mass Hessian of the vector fluctuations, H_eps u = M^2 K_eps u, and then the full momentum-dependent kernel. The aim is M_W, M_Z ~ c lambda / R with finite nonzero couplings. The tunnelling eigenvalue alone does not establish this.
2. The gauge normalization. For a bulk Yang-Mills field with fixed microscopic coupling and constant profile, g_4^-2 = Vol(X) / g_11^2, so plain dilation gives g_4 proportional to R^{-7/2}, which must be separated from restoration at finite coupling. Gauge fields from isometries have Killing overlap integrals instead.
3. The action in the radius and the electroweak order parameter. A toy V = a r^-p + b eps^2 r^q with a, b, p, q > 0 has its unique minimum at r proportional to eps^{-2/(p+q)}, so alpha = 2/(p+q) lies in (0, 1) if p + q > 2. The repository has not derived such a potential, and adding it by hand would derive nothing.
4. The connector of Proposition 28.20, and the origin of the quaternionic grading of Proposition 28.19.

Numerical check

No statement here needs a computation. The audit tabulates the localized family in Python (mu_* = 1, alpha = 1/2, c_0 = 2), and Claude Sonnet 5.5 recomputed the rows; for eps = 10^-2, c = 4.3026 and m = 2.710 x 10^-3, against the asymptotic 2.707 x 10^-3.

Part 11D (README, referee pending)

Eleven dimensions from massless W and Z

Branch `11D`, opened 3 Nov 2026. This folder holds one line of research: how the limit M_W, M_Z -> 0 of hypothesis H can give an eleven-dimensional theory, related on one side to Witten's family of seven-manifolds with the isometry group of the Standard Model, and on the other to the M2 and M5 details of lemma 21 and STRATEGY T12. All files are new; the only edits outside the folder are pointer lines in README.md, INDEX.md, STRATEGY.md and one include line in ledger.pl, so the branch can be fast-forwarded into master while master has not moved.

Attempted by Claude (Anthropic) on 3 Nov 2026. The Astra 6 audit of the same day (reviews/2026-11-03_astra_audit_d11.md, of snapshot b2ddf98, before this branch) is used in the second version of the folder: its joint-limit family defines the path along which v -> 0 opens a dimension, and its points corrected Proposition 1.4 and Proposal 1.3; see the section on the audit below. Every proposition of this folder is checked only by its author so far, and the ledger marks it as referee pending; prompts/terra_check_11d.md is the referee request. The conversation in which the folder was written is recorded at https://claude.ai/code/cshk_01BpscANdX3GG8D45SQ1BxsaHviSe7ufxxCLNiNaDv4fWT.

The picture

H reads the two sheets of the finite geometry as the minimal truncation of an interval with opposite chiral walls, and v -> 0 as the interval growing. The labels 9, 10 and 11 of H are KO labels; the heat dimension at the 11 endpoint is 5. To make 11 a metric count, the finite KO-6 factor is resolved into a six-dimensional base, B_6 = CP^2 x S^2, which has isometry SU(3) x SU(2) and is the base of Witten's spaces:

```
11 = 4 (spacetime) + 1 (the interval of H) + 6 (B_6).
```

There are two ways to complete 4 + 6 by one dimension, and they correspond to the two branes of M-theory:

	electric (M2) completion	magnetic (M5) completion
eleven-dimensional vacuum	AdS4 x Y_{3,2}, Y_{3,2} = M^{111}, a circle bundle over B_6	AdS5 x CP^2 x S^2
the interval of H	the fibre circle, antiperiodic, two walls at the fixed points of a reflection	the radial direction of AdS5, infrared end at z_IR ~ 1/v
SU(3) x SU(2)	isometries	isometries
hypercharge	the vector of C_3 on the harmonic two-form (b_2 = 1)	one combination of the two C_3 vectors (b_2 = 2)
v -> 0	the M2-brane moves to the apex; AdS4 x Y_{3,2} with the Standard Model group unbroken	the infrared end moves to infinity

Results

#	result	status	file
1.1	The chiral interval of length l is unitarily the antiperiodic circle of radius R = l/pi; the grading is the reflection, the walls its fixed points, m = 1/(2R)	Proposition, proved	01

#	result	status	file
audit §3, §4	Joint family: D_F = epsilon D_*, l ~ epsilon^{-alpha}, alpha < 1, gives m_weak « E_epsilon « 1/R -> 0, a finite window at the weak scale and an opening tower at fixed energy; a localized variant with a bulk mass	Proposition (Astra 6), rechecked here in circle language, constant of the Weyl count confirmed, localized table reproduced	01
1.3	The circle of H as the M-theory circle: v -> 0 along the joint family is strong coupling with the Wilson-line angles going to zero, the 9 endpoint has no bulk fermions as in type 0 strings	Proposal	01
1.4	The angle-dependent one-loop term of lemma 24 is -93 zeta(5) n_f/(512 pi^2 l^4) at its minimum, relative to theta = 0, where M_W = M_Z = 0 at fixed l (first version misread it as the whole wall energy; corrected)	Proposition, proved (scope of lemma 24)	01
1.5	A scale-free potential l^{-4} G(theta) runs away in l, or with theta = l m fixes l m: it selects only l ~ 1/M_W (alpha = 1); alpha < 1 needs a second scale	Proposition, proved	01
2.1	M^{111} is the hypercharge quotient of the 5-bar: (S^9 ∩ mu_Y^{-1}(0))/ U(1)_Y, Calabi-Yau cone because tr Y = 0, isometries (U(3) x U(2))/U(1)_Y	Theorem, proved	02
2.2	Only the 5 and 5-bar of SU(5) give a seven-dimensional quotient; among the circle quotients of S^5 x S^3 only (3, 2) is Calabi-Yau	Proposition, proved	02
2.3	Y_{a,b}: H^2 = Z (+ torsion), H^4 of order a^2 (Z/9 for M^{111}), spin iff a odd and b even; H^2(B_6) -> H^2(Y) is (x, y) -> 2x - 3y	Proposition, proved	02
2.4	The Z_6 of the Standard Model is the kernel of SU(3) x SU(2) -> Isom(Y_{3,2}); no bosonic Kaluza-Klein mode is a triplet or doublet	Proposition, proved	02
2.5	No generation from abelian fluxes on CP^2 x CP^1 along Y and B - L: u^c and d^c need one line bundle, and 2j + 1 must be linear in the fluxed charges	Negative result in scope, pending check	02
2.6	One generation exactly, from fluxes along a universal charge, the grading chi and B - L; u^c, d^c share O(1, 0), so the flux sees SU(2)_R doublets and leaves T_3R free	Proposition, proved	02
3.1	Hypercharge is the Betti vector of Y_{3,2}; the N = 1 fivebrane baryons are d^c and L	Proposal	03

#	result	status	file
3.2, 3.3	One generation is Λ^odd(5-bar), with 6Y the homology class of the wrapped fivebranes; conjecture: a Fock space of wrapped fivebranes	Proposition (standard), Conjecture	03
3.4	The Higgs is the M2 position on the doublet line of the resolved cone; v -> 0 is the apex and gives AdS4 x Y_{3,2}; at fixed N this is the fixed-tower path, and opening needs N -> infinity, against N = 1 for the matter	Proposal (stabilizer proved), obstruction stated	03
3.5	AdS5 x CP^2 x S^2 with G_4 = f_1 vol(CP^2) + f_2 J vol(S^2) solves eleven-dimensional supergravity iff abs(f_1) < 2 abs(f_2)	Proposition, proved	03
3.6, 3.7	Two completions of one base; the M2/M5 exchange as the electric-magnetic duality of hypercharge, which does not permute the 96 states	Proposals	03
3.8, 3.9	The 84 of C_3 is not colour-equivariantly either family of 84, neither through the little group nor as Kaluza-Klein modes on Y_{3,2}	Negative result in scope, pending check; Corollary	03
4.1	The two 12-state blocks of lemma 21 are exactly the ratio-preserving blocks of whole species: one up-type quark, or the three neutrinos	Proposition, proved	04
4.2, 4.3	The colour contents of F1 and F2 differ, and for every SU(3) -> SU(9) the space Lambda^3 C^9 is never either family (only 3 + 6 (1) gives triplets only)	Negative results in scope, pending check	04
4.A, 4.4	Two commuting order-3 gradings of e8, by SU(9) (brane level mod 3: gravity, C_3, C_6) and by SU(3) x E6 (families); the 10 and H_u lie in the C_3 sector, (d^c, L) and nu^c in the C_6 and gravity sectors according to a sign t	Theorem, proved	04
4.5	Up-type Yukawa couplings lie inside the C_3 sector, the others have one field per sector; the top mass is the wedge Lambda^3 -> Lambda^6	Proposition, proved	04
4.6	The M2/M5 exchange is the Chevalley involution, acting as the real structure J; it preserves both families and does not exchange the top with the neutrinos	Negative result in scope, pending check	04
4 §7, §8	nu^c as the gravitational state selects t = +1/2; a discrete Z_3 from the M-grading	Proposal; Remark	04
5.1	As configurations of E8 roots the two families of 84 have equal pair counts but are not isometric (the neutrino of the top's own family carries the obstruction), so no element of W(E8) exchanges them	Negative result in scope, pending check	05

#	result	status	file
5.2	With four families, M = -P in W(E8) transposes Pati-Salam colour with family and maps the up-type quark of family 0 onto the neutrinos of families 1, 2, 3	Proposition, proved	05
5.3	In a four-family finite geometry the colour-family flip commutes with gamma and J and conjugates the heavy-top limit into the heavy Dirac-neutrino limit of the other three families; the Pati-Salam and Standard Model algebras are not flip-invariant	Proposition, proved	05
5.4	No assignment of the four types to three of four family slots, aligned or misaligned as at the 9 label, admits the exchange as an isometry of the E8 roots (192 configurations)	Computer-assisted negative result in scope, pending check	05
5.5	Families as zero modes of an SU(4) bundle: the four slots are the fibre and three families an index	Proposal	05
6.1	The top channel of lemma 22 is an SO(32) Wilson-line channel with m_t = a/R	Proposition, proved	06
6 §2, §3, §5	The heterotic web 9, 10, 11 as the realization of the target; dictionary with D_F as a Wilson line, SO(30) x SO(2) as fifteen D8-branes and one displaced, composite scalars as open strings	Proposals	06
7.1	The angle of a channel of lemma 22 lives on the interval [0, pi/2], whose ends are the periodic and the antiperiodic spectra	Proposition, proved	07
7.2	The top channel is the SO(32) Wilson line 1_30 + R(2 pi a); the one-loop minimum of lemma 24 is the far end a = 1/2, the top's D8-brane on the other wall	Proposition, proved	07
7.3	A torus with flux p, doubled, has KO-6 signs, kernel C^p + conj(C^p) and window M_p(C), the colour form for p = 3	Proposition, proved	07
7 §2, §4, §5, §6	Readings of the dualities 1 to 4 for the finite geometry: walls and D8 positions, Landau levels as towers, stacks of D0-branes as the algebra	Proposals; known remark	07
8.1	The descended antiunitary of lemma 06 with j^2 = s gives O(n) or USp(n) on every eigenspace of d^2	Proposition, proved	08
8.2	Wall channels of the parent come with doubled multiplicity relative to bulk channels	Proposition, proved	08
8.3, 8 readings	O^- for label 11 and O^+ for label 9; E8 walls from 15 + 1, with the far wall enhancing to SU(2) x U(1) at infinite coupling	Remark conditional on recalled rules; proposals	08

#	result	status	file
9.1	The type I' function z (with e^phi = z^(-5/6)) is z(0) + c Tr g_x(l abs(D_F)) over sixteen channels; infinite coupling at the far wall iff z(0) = (2cL/ pi)(4 pi - l Sigma m_k)	Proposition, proved from the recalled slope rule	09
9.2	The diquark counts of lemma 21 (-2/3: 6, +1/3: 6, +4/3: 3) are the strings between the fifteen-brane stack and its image on an O8^- plane; O8^+ gives (3, 6, 1); a single brane has no string to its image on O8^-	Proposition, proved from representation theory and the recalled projections; reading as proposal	09
9.3	The diquark block between V and Vbar is symmetric iff eps eps' = +1: labels 9, 10, 11 give S^2 V with counts (3, 6, 1); the counts (6, 6, 3) of lemma 21 come from Lambda^2 V, as for a connection component along a circle or interval	Proposition, proved (linear algebra and the KO sign table); scope and escapes stated	09
9 §1	Comparison with A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397): the chain SO(32) > SO(2) x SO(30) > SO(2) x SU(5) x SU(3) x U(1) coincides with lemma 21 and the dictionary of 06; the type I' realization was not found there	Comparison, read through a summarizing fetch	09
9 §4	E-type walls as local del Pezzo surfaces (dP_2 for the top's wall); the colour-family transposition of 05 as a nine-dimensional T-duality, which by itself is a gauge relabelling	Proposals	09
5 Readings	Three families as the obstruction; M exchanges the two heavy directions of the algebra chain; 128 = 44 + 84 under Spin(9)	Proposals	05

The checks are in checks_11d.py (Python 3, exact fractions; every check asserts; about 20 s).

Added on master, 4 Nov 2026, by Claude Opus 5.5, unrefereed: 10 tests the dilaton as second scale (negative in scope; computes a_0; corrects the dictionary of 06 to 09: nonlinear angle map, T-dual lengths, Romans mass) and 11 computes the SO(30) fluctuations (the antisymmetric block needs M_15(C) at both walls, and the window sees its commutator with the masses).

How the massless limit gives eleven dimensions here

Vanishing W and Z masses open a dimension only along a specified path (audit §2.2): at fixed l, both theta -> 0 and theta -> pi/2 make them massless with the tower fixed. The criterion used here is the joint family of the audit, m_weak « E_epsilon « 1/R -> 0, with finite gauge couplings.

1. In the circle language of Proposition 1.1, the joint family is R -> infinity for an antiperiodic circle with two fixed points, with the Wilson-line angles of D_F going to zero. Read as the M-theory circle (Proposal 1.3), it is the strong-coupling limit, with the weak scale below the momentum tower of the eleventh dimension.
2. What selects the path: a potential with l and the angles as its only scales selects alpha = 1, the original l ~ 1/M_W of H, on which the finite window does not separate (Proposition 1.5). The joint family

needs a second scale; the localized family of the audit (bulk mass mu, mu l ~ (1 - alpha) log(1/epsilon)) is the first candidate, and the four-dimensional couplings must stay finite, which on the circle needs g_5^2 ~ R.

3. In the brane language of Proposal 3.4, v -> 0 moves the M2-brane to the apex of the cone over the Witten space with Killing spinors, and the near-horizon geometry is AdS4 x Y_{3,2}: 4 + 7 = 11 with SU(3) x SU(2) from isometries and U(1)_Y from C_3 (Proposal 3.1). This is the family of manifolds with the Standard Model group, singled out by the 5-bar (Theorem 2.1, Proposition 2.2). At fixed N it is the fixed-tower path; a joint family there needs N -> infinity, in tension with the N = 1 matter content.
4. On the magnetic side, AdS5 x CP^2 x S^2 carries the same base, and v -> 0 sends the infrared end of AdS5 to infinity (Proposal 3.6), which is the interval of H growing.

The angle-dependent one-loop term (Proposition 1.4) is minimized at the massless point at fixed l; the direction in which l moves depends on the angle-independent Casimir coefficient, which is not yet computed.

Next computations, ranked

0a. **The main point** (06, PLAN task B): one family realizing the three regimes; first steps listed in 06 §6 (type I' interval with the D8 configuration 15 + 1 as the parent of the lemma-22 window; Chan-Paton algebras against C + H + M_3(C); inner fluctuations against Λ^2(30)). 0b. **A nine-dimensional spectral triple on the type I' interval** (07 §7): sixteen D8 positions as the eigenvalues of D_F, and its strong-coupling limit with E8 walls; the dilaton as a spectral function and the diquark strings are in 09 (Propositions 9.1, 9.2), and whether the type I' coupling supplies the second scale of Proposition 1.5 is the next question there. 0. **A potential with a second scale that selects alpha < 1** (Proposition 1.5 and audit §7.1): the one-loop potential of the localized family, G(l m, l mu), from the exact secular equations of lemma 09, and the angle-independent Casimir coefficient a_0 of the flat family; then the vector kinetic matrix and mass Hessian along the selected path (audit eq. (17)), with g_4 finite. 1. **The referee of this folder** (prompts/terra_check_11d.md): Propositions 1.1, 1.4, 2.1 to 2.6, 3.2, 3.5, 3.8 and Corollary 3.9, with the negative results 2.5 and 3.8 needing the independent check before they are relied on. 2. **Spinors on Y_{3,2}** (prompts/astra_11d_01.md): the lift of (U(3) x U(2))/U(1) *Y to the spin bundle, whether fermionic Kaluza-Klein modes can carry triality or isospin 1/2, and which spin structures give the antiperiodic fibre momenta (k + 1/2)/R of Proposition 1.1. 3.* ***W and Z masses from the resolution****: in the warped resolved cone over Y*{3,2} with the M2-brane on the doublet line, compute the masses of the SU(2) x U(1)_Y vectors broken by the brane position, as functions of zeta and N, and compare M_W/M_Z with the cos^2 theta = 5/8 of the spectral action. 4. **The 9D reduction of Proposition 2.6**: with SU(2)_R broken by wall data, derive the mass basis of u and d against the charge basis, and the vectorlike limit at the 9 label, where the left-handed interactions are off. 5. **Fermionic zero modes of wrapped fivebranes** (Conjecture 3.3) at N = 1, and three generations. 6. **The magnetic completion**: flux quantization, stability and the two C_3 vectors of AdS5 x CP^2 x S^2 against the fluxed pair (chi, B - L) of Proposition 2.6. 7. **A four-family completion with the fourth family decoupled** (PLAN, steps 4-5): a finite geometry or Hořava-Witten configuration in which the colour-family transposition M of Proposition 5.2 is a symmetry, so that the three-family statement of lemma 21 follows as a limit; Proposition 5.1 shows that with three families the exchange is not an isometry.

The Astra 6 audit of 3 Nov 2026

The audit (reviews/2026-11-03_astra_audit_d11.md) reviews the master snapshot b2ddf98 and proposes a route from the massless electroweak limit to eleven dimensions with X = D(Λ^2_- T^*S^4) as principal candidate and M^{111} as comparison. Its use here:

audit item	use in this folder
§2.2, the massless limit must specify a path	adopted as the criterion; Proposal 1.3 revised; Proposition 1.5 shows what a scale-free action selects

audit item	use in this folder
§3, joint family	restated in circle language in 01, checked against Theorems 22.1 and 22.4; the constant 2 r l E/pi of its Weyl count confirmed (Python, section H)
§3.1, vector poles of the lemma-23 toy	quoted in 01: M_W$^{2/M_Z}$2 -> 5/8 along the family
§4, localized family and tail identity	table of roots and the identity (8) reproduced (Python, section H, bisection), independently of the audit's script
§7.1, modulus target	Proposition 1.5: in the flat family the electroweak term is m$^{2/1}$2 (q = -2), so the ansatz with q > 0 needs a second scale
§7.2, gauge normalization	test 2 of Proposal 1.3, and the large-N couplings in 03
Appendix A, M^{111}	agrees with Theorem 2.1 and Proposition 2.3 for (3, 2), and with the placement of hypercharge in the charged module (03)
§2.1, §5, §6, X and lemma 28	outside this folder; the 7-dimensional parent X is a third possible completion, M-theory on R^4 x X with the Bryant-Salamon comparison and the circle of solidification §7 as a candidate M-circle, listed as a direction only
§2.4, docs and distribution	outside this branch: refreshing the PDFs, adding lemmas 17 and 28 to paper/build_supplement.sh and repairing the supplement layout belong to master's build, and LaTeX is not available in this session

References used here

Read in this session (3 Nov 2026), from the arXiv abstract pages or the ar5iv text through a summarizing fetch: D. Fabbri, P. Fré, L. Gualtieri and P. Termonia, *M-theory on AdS4 x M^{111}: the complete Osp(2|4) x SU(3) x SU(2) spectrum from harmonic analysis*, Nucl. Phys. B 560 (1999) 617, hep-th/9903036 (coset, Killing spinors only for p = q, R-symmetry, Betti multiplet, Kaluza-Klein states neutral under it); D. Fabbri, P. Fré, L. Gualtieri, C. Reina, A. Tomasiello, A. Zaffaroni and A. Zampa, *3D superconformal theories from Sasakian seven-manifolds*, hep-th/9907219 (D-term 2|U|^2 = 3|V|^2, U(1) fibration over P^1 x P^2, baryons det U and det V as wrapped M5-branes, E_0 = 4N/9 and N/3, b_2 = b_5 = 1). M. Kreck and S. Stolz, *A diffeomorphism classification of 7-dimensional homogeneous Einstein manifolds with SU(3) x SU(2) x U(1)-symmetry*, Ann. of Math. 127 (1988) 373 (title and scope only).

Recalled and not reread, to be checked before any of them carries an argument: E. Witten, *String theory dynamics in various dimensions*, hep-th/9503124 (D0 states as momentum modes); O. Bergman and M. Gaberdiel, *Dualities of type 0 strings*, hep-th/9906055; M. Fabinger and P. Hořava, *Casimir forces between end-of-the-world branes*, hep-th/0002073; E. Witten, *SL(2, Z) action on three-dimensional conformal field theories with Abelian symmetry*, hep-th/0307041; C. N. Pope and P. van Nieuwenhuizen, *Compactifications of d = 11 supergravity on Kähler manifolds*, Commun. Math. Phys. 122 (1989) 281; J. Gauntlett, D. Martelli, J. Sparks and D. Waldram, *Supersymmetric AdS5 solutions of M-theory*, hep-th/0402153.

Already in the corpus: Witten 1981 and Castellani-D'Auria-Fré (LITERATURE.md §F); Hořava-Witten, Atiyah-Witten and Acharya-Witten-type cones (solidification §6).

Prior work for the observation of Theorem 2.1, that the weights of M^{111} are the hypercharges of the 5-bar, was not searched; this is an open item for LITERATURE.md.

Files

- 01_interval_and_eleventh_dimension.md: the "1".
- 02_witten_spaces_hypercharge_quotient.md: the "6 + 1" of Witten's spaces, the Z_6, chirality from fluxes.
- 03_branes_hypercharge_and_the_84.md: C_3, M2, M5 and the 84.
- 04_two_families_m2_m5.md: the two 12-state blocks, the two gradings of E8, Yukawa couplings as brackets, the M2/M5 exchange as J.
- 05_colour_family_exchange.md: the two families as E8 root configurations, the colour-family transposition with four families.
- 06_main_point_heterotic_route.md: the main point: target, the heterotic web 9/10/11, dictionary, Proposition 6.1.
- 07_dualities_9_11.md: dualities between 9 and 11 adapted to the spectral triple: the top angle as a D8 position between two orientifold walls, the one-loop vacuum as the far wall, a torus with flux as the colour form.
- 08_orientifold_sign.md: the sign of lemma 06 as an orientifold type (O^- for label 11, O^+ for label 9); E8 walls from 15 + 1: the far wall with the top's brane enhances to SU(2) x U(1) at infinite coupling.
- 09_dilaton_spectral_and_so32.md: the type I' dilaton as a trace of a function of D_F; the SO(32) of the EPJC paper compared with lemma 21; diquarks as strings between the stack and its image on an O8^- plane; del Pezzo walls and the transposition as a T-duality (proposals).
- PLAN.md: the handover plan of the line.
- checks_11d.py: exact checks.
- ledger_11d.pl: ledger facts, included by ../ledger.pl.
- prompts/: the referee request and the first Astra job.
- ../reviews/2026-11-03_astra_audit_d11.md: the Astra 6 audit used in the second version.

Part 11D (01_interval_and_eleventh_dimension, referee pending)

The interval of H as an eleventh dimension

Part of the 11D line of research (README). Propositions 1.1, 1.4 and 1.5 are proved here; Proposal 1.3 is a dictionary to be tested. The joint-limit family of the section of that name is from the Astra 6 audit of 3 Nov 2026 (reviews/2026-11-03_astra_audit_d11.md, §3 and §4), restated here in circle language and checked. Attempted by Claude (Anthropic), 3 Nov 2026; checks in Python, checks_11d.py, sections F, F2 and H. Referee pending.

Correction (3 Nov 2026). There was an error in the first version of Proposition 1.4: it read the potential of lemma 24 as the whole energy of the walls, but that potential is taken relative to theta = 0 and omits the theta-independent Casimir energy. The corrected statement is below; the reading "the walls attract" is withdrawn. The first version of Proposal 1.3 also placed the W at the bottom of the tower, M_W ~ 1/(2R), along every path to v = 0; after the audit (its §2.2), that holds only on the scale-free path of Proposition 1.5.

Setting

Hypothesis H of STRATEGY.md reads the two sheets of the finite geometry as the minimal truncation of an interval I_l with opposite chiral boundary conditions, with l ~ 1/M_W. Its three regimes carry the labels 9, 10 and 11. These are KO labels. At the 11 endpoint (v -> 0, l -> infinity) the heat dimension is 4 + 1 = 5, since the finite factor F keeps heat dimension 0 for every v (canonical §2.1). To read 11 as a metric dimension, the KO-6 factor must itself be resolved into a six-dimensional space, which gives the count

```
11 = 4 (spacetime) + 1 (the interval) + 6 (a space resolving F).
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The files of this folder take the six-dimensional space to be B_6 = CP^2 x S^2, the base of Witten's spaces (02), and study two ways of completing 4 + 6 to eleven dimensions (03). This file treats the "1", and the

path along which v -> 0 opens it.

Proposition 1.1 (the chiral interval is the antiperiodic circle)

Statement. Let D_l act on H_l = L^2([0, l], C^2) as in lemma 05, eq. (8): D_l(L, R) = (R', -L'), with R(0) = 0 and L(l) = 0, grading Gamma = diag(1, -1) and the scalar functions C^infinity([0, l]). Let S_R be the circle of length 2l, that is of radius R = l/pi, with coordinate x in [-l, l] and antiperiodic functions u(x + 2l) = -u(x). The map

```
(W(L, R))(x) = (L(|x|) + i sgn(x) R(|x|)) / sqrt 2
```

is a unitary H_l -> L^2_antiperiodic(S_R) with

1. W D_l W^* = -i d/dx on its antiperiodic domain;
2. W Gamma W^* = P, the reflection (Pu)(x) = u(-x);
3. W f W^* = multiplication by f(|x|), so the algebra goes to the reflection-even functions;
4. the walls x = 0 and x = l go to the two fixed points x = 0 and x = l ~ -l of the reflection.

Hence spec D_l = {(k + 1/2)/R : k in Z}, each eigenvalue simple, which is (9) of lemma 05, and the Connes-Lott mass of the minimal truncation is m = pi/(2l) = 1/(2R).

Proof. The inverse map is L(x) = (u(x) + u(-x))/sqrt 2, R(x) = (u(x) - u(-x))/(i sqrt 2) for x in [0, l], so W is a bijection, and |u(x)|^2 + |u(-x)|^2 = |L(x)|^2 + |R(x)|^2 gives unitarity. For H^1 data, u is continuous at 0 exactly when R(0) = 0, and u(l) + u(-l) = sqrt 2 L(l), so the antiperiodic matching u(-l) = -u(l) is the condition L(l) = 0. On x > 0, -i (L + iR)' = R' + i(-L'), the image of (R', -L') = D_l(L, R). On x < 0, u(x) = (L(-x) - i R(-x))/sqrt 2 and -i u'(x) = (R'(-x) - i(-L'(-x)))/sqrt 2, which is again the image of (R', -L'). Gamma(L, R) = (L, -R) maps to L(|x|) - i sgn(x) R(|x|) = u(-x). Items 3 and 4 are immediate. The antiperiodic spectrum of -i d/dx on a circle of length 2l is pi(2k + 1)/(2l) = (k + 1/2)/R. ■

So the two sheets of H are the two fixed points of a reflection on a circle whose spinors are antiperiodic, and the grading of the finite geometry is that reflection. With finite data D_F supplied as in lemma 22, the parent unfolds to -i d/dx + D_F on this circle (lemma 25): D_F is a constant connection, a Wilson line, and the wall angles theta_k = l mu_k (mu_k the eigenvalues of D_F) are its holonomy phases.

The joint family: weak masses below a descending tower

At fixed l, the vector masses of lemma 23 vanish as theta -> 0 and also as theta -> pi/2, while the tower spacing pi/l stays fixed; vanishing masses alone open no dimension (audit §2.2). The audit gives a family in which they do.

Proposition (Astra 6, audit §3; restated). Let D_F(epsilon) = epsilon D_* and l_epsilon = l_* epsilon^{-alpha}, 0 < alpha < 1, with wall data X = tan(l D_F) as in lemma 22. Then the exact window is W^* D W = epsilon D_, *the rest of the spectrum lies at |lambda| >= pi/l_epsilon - epsilon ||D_||*, the compression error is at most (2/3) l_^2 ||D_||^2 epsilon^{2(1-alpha)} ||a|| (Theorem 22.4, item 2), and for alpha < beta < 1 a cutoff E_epsilon = E_* epsilon^beta contains exactly the r = dim H_F finite states, while at every fixed E > 0 the number of states N_epsilon(E) grows like l_epsilon E, which is proportional to epsilon^{-alpha}.

In the circle language of Proposition 1.1 this reads: the radius R_epsilon = l_epsilon/pi goes to infinity, and the Wilson-line angles theta_epsilon = l_epsilon epsilon mu_k go to zero like epsilon^{1-alpha}. The hierarchy is

```
m_weak ~ epsilon  <<  E_epsilon ~ epsilon^beta  <<  1/R_epsilon ~ epsilon^alpha  ->  0.
```

In the five-dimensional Yang-Mills toy of lemma 23 the vector poles along this family are M_W^2 = 2 m^2 (1 + O(theta^2)) and M_Z^2 = (16/5) m^2 (1 + O(theta^2)), so $M_W^{2/M_Z}2$ -> 5/8 (audit §3.1, from Theorem 23.6).

Check (Claude). The parent space of lemma 22 is L^2([0, l]; C^2) ⊗ H_F, of fibre dimension 2r. By Theorem 22.1 its signed spectrum is (n pi +- theta_k)/l, n in Z, two progressions of spacing pi/l for each of the r/2

singular values, so N(E) = 2 r l E/pi + O(r), the constant of the audit (on the circle of Proposition 1.1, fibre C^2, the same count gives 2 E l/pi). The window, gap and compression statements follow from Theorems 22.1 and 22.4 as stated. (Python, section H: exact spectra for random singular values.)

The audit also gives a localized family (its §4): a bulk mass mu_epsilon = mu_* epsilon^alpha and mu l = (1 - alpha) log(1/epsilon) + c_0, with light mass m ~ 2 mu_* e^{-c_0} epsilon, gap ~ mu_* epsilon^alpha, and the exact tail identity e^{-2 kappa l} = (m/(mu + kappa))^2. Its table of roots is reproduced in section H (Python, bisection on tanh s = s/c and tan y = y/c): the four light masses and next masses agree with the audit to the digits printed there.

Proposal 1.3 (the circle of H as the M-theory circle)

Take the circle of Proposition 1.1 to be the eleventh direction of M-theory over a ten-dimensional type IIA background, so R = g_s l_s. Then, along the joint family:

H	circle of Proposition 1.1	M-theory / string side (recalled, to check)
v -> 0 along the joint family, label 11	R -> infinity, Wilson-line angles -> 0, m_weak R -> 0	strong coupling: D0-brane bound states of mass n/(g_s l_s) become light and are the momentum modes of the eleventh dimension (Witten 1995); the weak masses lie below this tower
v finite, label 10, two sheets	R finite, fixed points of a reflection	M-theory on an interval with two walls (Hořava-Witten); here with walls of opposite chirality
v -> infinity, label 9, sheets merge	R -> 0, antiperiodic spinors	a circle with antiperiodic fermions shrinking: bulk fermions get masses >= 1/(2R); compare type 0 strings, which have no bulk spacetime fermions (Bergman-Gaberdiel 1999), and the walls of opposite chirality of Fabinger-Hořava 2000

Each row matches a statement already in the corpus: the 11 endpoint needs the tower (canonical §2.1, lemma 05 Part 4), and at the 9 endpoint the bare interval keeps no light fermion, so light fermions must come from another sector (STRATEGY, T1 and T3). On the string side the same happens: the light fermions of a type 0 theory come from open strings on branes. The proposal therefore places the light fermions of the 9 endpoint on branes or walls, and 03 develops a brane origin for them.

Tests.

1. Along the joint family m_weak/M_KK ~ epsilon^{1 - alpha} -> 0, so the weak scale is far below 1/(g_s l_s); the scale-free path of Proposition 1.5, with M_W R fixed, is the other possibility and keeps no finite window.
2. Gauge couplings. A five-dimensional gauge field with constant profile on the interval has g_4^{-2} = l/g_5^2 = pi R/g_5^2 (audit §7.2 for the general volume formula), so at fixed g_5 the four-dimensional couplings vanish as R -> infinity, and M_W = g v/2 could vanish through g. The limit sought keeps g finite, which needs g_5^2 ~ R, and this must be shown for the vectors in question before the masses are read as restoration of the symmetry.
3. The path must be selected by an action: Proposition 1.5 shows what the flat interval alone selects.

Proposition 1.4 (the angle-dependent one-loop term at the vacuum angle)

Statement. Under the hypotheses of Theorem 9 of lemma 24 (flat interval, zero bulk masses, frequency-one fermions, the stated gauge towers and multiplicities, no wall potential), the angle-dependent part of the one-loop potential, taken relative to theta = 0 as in lemma 23, has its minimum at theta = pi/2 with value

```
V(pi/2) - V(0) = - 93 zeta(5) n_f / (512 pi^2 l^4),
```

and near theta = 0 it is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) theta$^{2/l}$4 + o(theta^2). At theta = pi/2 the exact vector poles of lemma 23 give M_W = M_Z = 0, with the fermion at the tower scale m = pi/(2l).

Proof. In the notation of lemma 24, V = /(8 pi^2 l^4), with F(0) = 0, and (1) = -n_f F(1) since the gauge terms vanish at t = 1. With 1 + 1/sinh^2 x = coth^2 x and log coth x = 2 sum over odd k of e^{-2kx}/k,

```
F(1) = 2 int_0^infinity x^3 log coth x dx = 4 sum_{k odd} (1/k) 3!/(2k)^4 = (3/2) sum_{k odd}
 ↪  k^{-5} = (3/2)(31/32) zeta(5) = 93 zeta(5)/64.
```

Near t = 0, '(0) = (6 + 24/5 - n_f) J(0) with J(0) = int x$^{3/\sinh}$2 x dx = (3/2) zeta(3), the coefficient already given in lemma 23. Python: F(1) = 1.506786, J(0) = 1.803085, series against quadrature to 10^-6. ■

Reading. The -independent Casimir energy of the towers, a_0/l^4, is not part of this potential and is not computed in lemmas 23 or 24; the direction in which l moves depends on the sign of a_0 - 93 zeta(5) n_f/(512 pi^2). The minimum over the angle is the point with massless W and Z at fixed l, which opens no dimension (lemma 23 already notes that it is not the physical point).

Proposition 1.5 (a scale-free potential fixes the path with M_W l constant)

Statement. Let V(l, theta) = l^{-4} G(theta) with G of class C^1, the form of every one-loop potential of a flat interval whose only scales are l and the wall angles (lemmas 23 and 24, with a_0 included in G).

1. If the angle is free, V is monotone in l at every fixed theta with G(theta) != 0. At the minimizing angle theta_min the length runs away: to l -> infinity if G(theta_min) > 0, to l -> 0 if G(theta_min) < 0. Stationary points exist only on the set G = G' = 0.
2. If the angle is tied to a fixed mass, theta = l m (as for X = tan(l M) in lemma 22), the stationary lengths satisfy theta G'(theta) = 4 G(theta) at theta = l m. So l = theta_*/m for a root theta_* that does not depend on m, and M_W l, M_Z l (functions of theta alone by Theorem 23.6) are the same at every m.

Hence a scale-free action realizes only the path l ~ 1/M_W of H's original statement, which is alpha = 1 in the scaling of the joint family, and on it the finite window never separates from the tower. The joint family with alpha < 1 needs a second scale in the action: a bulk mass (the localized family, G(l m, l mu)), the cutoff of the spectral action (lemma 25, G(theta, l Lambda)), curvature, flux or a wall term.

Proof. d/dl [l^{-4} G(l m)] = l^{-5} (theta G'(theta) - 4 G(theta)) at theta = l m. For item 1, d/dl V = -4 l^{-5} G(theta) and d/dtheta V = l^{-4} G'(theta). ■

Consequence for the modulus target of the audit (§7.1). The audit proposes V = V_*(a r^{-p} + b epsilon^2 r^q) with a, b, p, q > 0, whose minimum r ~ epsilon^{-2/(p+q)} gives alpha = 2/(p + q) < 1 when p + q > 2. In the flat one-loop family the epsilon-dependent term near theta = 0 is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) m$^{2/l}$2, so q = -2, and with the Casimir term p = 4 the stationary point has l ~ 1/m, alpha = 1, in agreement with item 2. A term growing with l at fixed v, as the ansatz requires, is absent from the flat family and must come from one of the second scales listed above.

What remains open here

- Whether the identification of Proposal 1.3 survives the gauge sector: lemmas 13, 14 and 25 tie M_W to the tower with model-dependent coefficients, which here become a relation between M_W, g_s and l_s, and test 2 above requires g_5^2 ~ R.
- The theta-independent Casimir coefficient a_0 of the flat family, which decides the runaway direction of Proposition 1.5, item 1.

- A potential with a second scale that selects alpha < 1: the localized family of the audit is the first candidate, with G(l m, l mu) computable from the exact secular equations of lemma 09.
- A fermion with both a fibre momentum and a base twist: on a nontrivial circle bundle (Witten spaces, file 02) the antiperiodic shift corresponds to a spin structure whose fibre circle does not lift the circle action. Which Witten spaces carry such a structure is an open item for the next job.

Part 11D (02_witten_spaces_hypercharge_quotient, referee pending)

Witten spaces as the hypercharge quotient of the five-dimensional representation

Part of the 11D line of research (README). Theorem 2.1 and Propositions 2.2 to 2.4 and 2.6 are proved here; Proposition 2.5 is a negative result in a stated scope, pending an independent check. Attempted by Claude (Anthropic), 3 Nov 2026, with a remark from the steer of the same day on the mixing of u and d. Checks: checks_11d.py, sections A to D, Python with exact fractions. Referee pending.

Setting

Witten's spaces M^{pqr} = G/H, with G = SU(3) x SU(2) x U(1) and H = SU(2) x U(1) x U(1), are the seven-dimensional spaces of least dimension with the isometry group of the Standard Model, so 4 + 7 = 11 (Witten 1981; LITERATURE.md §F). Castellani, D'Auria and Fré gave them Einstein metrics; Killing spinors exist only for p = q, where the AdS4 vacuum has N = 2 and the space is M^{111} (Fabbri, Fré, Gualtieri, Termonia, hep-th/9903036). Fabbri et al. (hep-th/9907219, §4) write the cone over M^{111} as the Kähler quotient of C^5 by a circle acting with weights 2 on a triplet U and -3 on a doublet V, with D-term 2|U|^2 = 3|V|^2, and M^{111} as a circle bundle over P^2 x P^1. In later notation it is M^{3,2}.

These weights are six times the hypercharges of the 5-bar of SU(5). The rest of the file develops that observation.

Theorem 2.1 (M^{111} is the hypercharge quotient of the 5-bar)

Statement. Let 5bar = C^3 + C^2 carry the Standard Model action on the left-handed fields d^c in (3bar, 1)*{1/3} and L in (1, 2)*{-1/2}; write U for d^c and V for L, so 6Y = 2 on U and -3 on V. Let mu_Y = (2|U|^2 - 3|V|^2)/6 be the moment map of hypercharge, and

```
Y_{3,2} = (mu_Y^{-1}(0) ∩ S^9) / U(1)_Y.
```

1. Y_{3,2} is a smooth, simply connected seven-manifold, the circle bundle of O(3, 2) over CP^2 x CP^1 (up to orientation), and its cone is the Kähler quotient C^5 // U(1)_Y of Fabbri et al.
2. The cone is Calabi-Yau (Gorenstein, sum of weights 0) because tr Y = 0 on the 5-bar, that is, because U(1)_Y lies in SU(5).
3. The group acting effectively on Y_{3,2} through the quotient is (U(3) x U(2))/U(1)_Y. Its subgroup S(U(3) x U(2))/U(1)_Y is PSU(3) x SO(3), and the overall phase of U(5) supplies one further circle, the Reeb (R-symmetry) direction.
4. The functions on Y_{3,2} that are polynomial in U, V and their conjugates have U(1)_Y-charge zero; the holomorphic ones are spanned by the monomials of degree (3k, 2k) in (U, V), in Sym^{3k}(3bar) x Sym^{2k}(2): for k = 1 this is the (10, 3) of Fabbri et al.

Proof. On mu_Y^{-1}(0) ∩ S^9 we have |U|^2 = 3/5 and |V|^2 = 2/5, a product of round spheres S^5 x S^3. The circle acts by (lambda^2 U, lambda^{-3} V); a fixed point would need lambda^2 = lambda^3 = 1, so lambda = 1 and the action is free. The quotient of S^5 x S^3 by the full torus T^2 of phases of U and V is CP^2 x CP^1, and Y_{3,2} -> CP^2 x CP^1 is a principal bundle for T^2/U(1)_Y. A character lambda_U^m lambda_V^n of T^2 is trivial on U(1)_Y exactly when 2m - 3n = 0, so the characters of T^2/U(1)*Y are generated by (m, n) = (3, 2), whose line bundle is O(3) O(2) up to the sign convention; this gives item 1 and item 4. Simple connectivity follows from the exact homotopy sequence of U(1) -> S^5 x S^3 -> Y*{3,2}. Item 2: the sum of weights is 3 * 2 + 2 * (-3) = 0. Item 3: an element (g, h) of U(3) x

U(2) acts trivially on the quotient exactly when (g, h) = (lambda^2, lambda^{-3}), because mu_Y^{-1}(0) spans C^5. The intersection of SU(3) x SU(2) with U(1)_Y consists of lambda^6 = 1, generated by (e^{2 pi i/3} I_3, -I_2), the centre Z_3 x Z_2. ■

So the isometry group of the Witten space with Killing spinors, SU(3) x SU(2) x U(1) (Fabbri et al.), is the centralizer of hypercharge in U(5), divided by hypercharge. The SU(3) x SU(2) part is the Standard Model one; the remaining circle is the R-symmetry, and hypercharge itself has been quotiented. Witten assigned the U(1) of G to hypercharge; in this realization hypercharge is a different U(1) (03, Proposal 3.1).

Proposition 2.2 (uniqueness of the count 7)

Statement. (a) Among the quotients Y_{a,b} = (S^5 x S^3)/U(1), with weights b on C^3 and -a on C^2 and gcd(a, b) = 1, a, b > 0, the cone is Calabi-Yau only for (a, b) = (3, 2). (b) For a complex representation of dimension n and a free circle action, (S^{2n-1} ∩ mu^{-1}(0))/U(1) has dimension 2n - 3; this is 7 only for n = 5. Among representations of SU(5), with hypercharge as the circle, only the 5 and the 5-bar give a seven-dimensional quotient.

Proof. (a) The Calabi-Yau condition is 3b - 2a = 0. (b) Dimension count: 2n - 1 for the sphere, minus one for the moment map, minus one for the circle. The irreducible representations of SU(5) of dimension 5 are the 5 and the 5-bar. ■

Witten's count reads: 11 = 4 + (2 * 5 - 3). The seven internal dimensions come from the smallest representation of SU(5), with hypercharge as the circle. For the 10 the quotient has dimension 17, and for the 5 + 10 it has 27.

Proposition 2.3 (topology of Y_{a,b})

Statement. Let Y_{a,b} -> B = CP^2 x CP^1 be the circle bundle with Euler class a h_1 + b h_2, a != 0. Then H^2(Y) = Z + Z/gcd(a, b), b_3(Y) = 0, H^4(Y) is a group of order a^2, cyclic when gcd(a, b) = 1, and Y is spin exactly when a is odd and b is even. For (a, b) = (3, 2): H^2 = Z, H^4 = Z/9, spin. The pullback H^2(B) = Z^2 -> H^2(Y) = Z is (x, y) -> 2x - 3y up to sign: the two divisor classes h_1 = [{l(U) = 0}] and h_2 = [{m(V) = 0}] go to 2 and -3, the values of 6Y on d^c and L.

Proof. The Gysin sequence H^{k-2}(B) -> H^k(B) -> H^k(Y) -> H^{k-1}(B) -> H^{k+1}(B), with the first map the cup product with c = a h_1 + b h_2 and H^*(B) = Z[h_1, h_2]/(h_1^3, h_2^2). In degree 2 the cokernel of 1 -> (a, b) is Z^2/(a, b). In degree 4 the map H^2(B) -> H^4(B) has matrix [[a, 0], [b, a]] in the bases (h_1, h_2), (h_1^2, h_1 h_2), with determinant a^2 and Smith invariants (gcd(a, b), a^2/gcd(a, b)); since H^3(B) = 0 this gives H^4(Y), and the same matrix is injective, so H^3(Y) = 0. For spin: w_2(B) = h_1 mod 2, and the kernel of H^2(B; Z/2) -> H^2(Y; Z/2) is spanned by c mod 2; so w_2(Y) = 0 exactly when (1, 0) = (a, b) mod 2. For (3, 2), Z^2/(3, 2) -> Z by (x, y) -> 2x - 3y is an isomorphism. ■ (Python: Smith invariants and the spin condition, section C.)

The product member CP^2 x S^2 x S^1 of Witten's family (a = b = 0) is not spin.

Proposition 2.4 (the Z_6 of the Standard Model)

Statement. (a) Every monomial in U and V, and every state of the exterior algebra Λ^*(5bar), satisfies 2t + 3d + 6Y = 0 mod 6, with t the triality (1 for 3, -1 for 3bar) and d = 2j mod 2. (b) The U(1)*Y-neutral monomials have t = 0 and d = 0. (c) Every tensor field on Y*{3,2} carries a representation of (U(3) x U(2))/U(1)*Y, on which SU(3) x SU(2) acts through PSU(3) x SO(3). So no bosonic Kaluza-Klein mode on Y*{3,2} is a colour triplet or a weak doublet.

Proof. (a) holds for U (t = -1, d = 0, 6Y = 2) and V (t = 0, d = 1, 6Y = -3) and is additive. (b) U^alpha V^beta is neutral when 2 alpha = 3 beta, so 3 | alpha and 2 | beta. (c) The effective group of Theorem 2.1 acts on the frame bundle and on every tensor bundle, and its intersection with SU(3) x SU(2) is the centre Z_6 (Theorem 2.1, item 3). ■

(a) is the statement that the Standard Model group is (SU(3) x SU(2) x U(1))/Z_6 = S(U(3) x U(2)). In the quotient realization the Z_6 is the kernel of SU(3) x SU(2) -> Isom(Y_{3,2}). States with fractional hypercharge are those charged under the quotient circle, and they cannot be functions or tensors on Y_{3,2}. For spinors the isometry group may lift to a double cover; whether fermionic Kaluza-Klein modes on Y_{3,2} can carry half-integer weak isospin or triality is open (next job, item 2 of the README).

Chiral zero modes on the base with abelian fluxes

Witten's argument makes Kaluza-Klein fermions on an odd-dimensional space vectorlike, for gauge fields acting by isometries. The even base B_6 = CP^2 x CP^1 escapes it: abelian fluxes on B_6 give chiral four-dimensional fermions through the index of the twisted Dirac operator. This section asks which fluxes give one generation.

Setting. Ten-dimensional Weyl fermions on M^4 x CP^2 x CP^1, each with definite charges under some U(1) factors with fluxes on B_6. Since CP^2 is not spin, a fermion is a spin^c spinor, $\Lambda^{0,\bullet} \otimes K^{1/2} \otimes N$ *with N the flux line bundle of its charges, and the four-dimensional chiral content is the virtual representation* $H^\bullet(O(\text{alpha}, \text{beta}))$ with (alpha, beta) = (-3/2, -1) + pairing of the charges with the fluxes. The cohomology is H^0(O(alpha)) = Sym^alpha(3bar) for alpha >= 0, H^2(O(alpha)) = Sym^{-alpha-3}(3) for alpha <= -3, and zero otherwise on CP^2; on CP^1, H^0(O(beta)) = Sym^beta(2) for beta >= 0 and H^1(O(beta)) = Sym^{-beta-2}(2) for beta <= -2. Each Standard Model Weyl field must come from one ten-dimensional field, from the whole cohomology of one line bundle.

Lemma (required bidegrees). Writing P_s = (alpha + 3/2, beta + 1) for the pairing of the charges of field s (up to the common sign of charge and chirality):

field	P_s	note
Q (3, 2)	(-5/2, 2)	H^2(O(-4)) x H^0(O(1))
u^c, d^c (3bar, 1)	(5/2, 1)	H^0(O(1)) x H^0(O(0))
L (1, 2)	(+-3/2, 2)	
e^c, nu^c (1, 1)	(+-3/2, 1)	

The second entry is 2j + 1 and the first is +-5/2 on colour triplets and +-3/2 on singlets. (Python, section D, by enumeration of the line bundles with these cohomologies and the Riemann-Roch check.)

Proposition 2.5 (no generation from fluxes along hypercharge and B - L) Statement. In the setting above, with the canonical spin^c root and fluxes only in U(1) directions spanned by Y and B - L (any flux quanta), no choice reproduces one generation, with or without nu^c. More generally, the pairing must give u^c and d^c the same image, and 2j + 1 = (2, 1, 1, 2, 1, 1) on (Q, u^c, d^c, L, e^c, nu^c) must be a linear function of the fluxed charges.

Proof. The pairing M is linear in the charges, and M c_s = P_s with c_s the charges of the left-handed field s. With only Y and B - L fluxed, c_u - c_d = (-6, 0) in the coordinates (6Y, 3(B - L)), so M vanishes on the hypercharge direction and P_s = 3(B - L)_s m for one vector m. The second entries 2 on Q and 1 on u^c would then need 3(B - L) to take values in the ratio 2 : 1 on (Q, u^c), against 1 : -1. In general, 2j + 1 is not in the span of 6Y = (1, -4, 2, -3, 6, 0) and 3(B - L) = (1, -1, -1, -3, 3, 3): from Q and u^c the coefficients are -1 and 3, and then d^c gives -5 against 1. ■ An exhaustive search adding a third integer charge z in [-3, 3]^6 to Y and B - L finds no solution either (Python, section D).

Scope and escapes. Abelian fluxes on CP^2 x CP^1 with the Fubini-Study complex structure; one ten-dimensional Weyl field per Standard Model field; canonical spin^c root; Dolbeault index as the chiral content. Known escapes: nonabelian bundles (an SU(2)_R doublet bundle), several ten-dimensional fields per four-dimensional field, a changed common twist, orbifolds and walls, and non-Kaluza-Klein (brane) states. The result points to the next proposition.

Proposition 2.6 (one generation from a universal charge, the grading and B - L) Statement. Let the fluxed charges be F (one on every field: a change of the common spin^c twist), chi (+1 on Q and L, -1 on u^c, d^c, e^c, nu^c) and B - L. Then one generation with nu^c is the exact index content for the pairing

```
P^{(1)} = c_0 chi + c_1 (B - L),   P^{(2)} = (3F + chi)/2,
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with (c_0, c_1) = (-9/4, -3/4) or (-3/2, -3). The bidegrees are Q: (-4, 1); u^c, d^c: (1, 0); L: (-3, 1) or (0, 1); e^c, nu^c: (0, 0) or (-3, 0). Without chi, or without B - L, there is no solution.

Proof. Substitution, and the two failures by the span test (Python, section D2). ■

Reading.

1. u^c and d^c see the same line bundle O(1, 0), and so do e^c and nu^c. The flux treats them as SU(2)_R doublets with a common internal profile; hypercharge Y = T_3R + (B - L)/2 enters through T_3R, which carries no flux. This is the left-right pattern of Pati-Salam, produced here by the index.
2. chi is the finite grading gamma_F on the particle basis. The flux therefore couples to the grading, an axial abelian connection on B_6; in the language of the finite geometry this is a fluctuation proportional to gamma_F, the structure of the two gradings of lemma 23.
3. **Mass and charge bases.** Because u^c and d^c share one zero-mode profile, overlap integrals on B_6 cannot separate them; their masses come from data that break SU(2)_R (Higgs or wall couplings), and the basis that diagonalizes the masses is set by those data, independently of the basis in which the charges are diagonal. The steer of 3 Nov 2026 remarks that u and d should still mix, since mass eigenvalues are not charge eigenvalues, and that at the 9 label, with the left-handed interactions off, charge eigenvalues can coincide. Proposition 2.6 puts the coincidence in the flux charges: equal flux charges for u^c and d^c, unequal hypercharges through T_3R. At the 9 label the electroweak part of the algebra is removed (lemma 21, STRATEGY T2), and what distinguishes u^c from d^c in the light theory is then the vectorlike charge alone. Making this precise needs the SU(2)_R-breaking wall data and the 9D reduction of this index; listed as item 4 of the README.

What this file establishes, and the direction

The Witten space with Killing spinors is the hypercharge quotient of the 5-bar (Theorem 2.1), the count 7 is fixed by SU(5) (Proposition 2.2), and the Z_6 of the Standard Model is the kernel of the isometry action (Proposition 2.4). The price is that hypercharge is no longer an isometry, and no bosonic Kaluza-Klein mode carries triality, weak isospin 1/2 or fractional hypercharge. Kaluza-Klein chirality on the base needs fluxes that see the grading and leave T_3R free (Propositions 2.5 and 2.6). Both results point to states outside the Kaluza-Klein spectrum of Y_{3,2} for the quarks and leptons: 03 takes them to be wrapped branes.

Part 11D (03_branes_hypercharge_and_the_84, referee pending)

Hypercharge from the three-form, matter from branes, and the two families of 84

Part of the 11D line of research (README). Propositions 3.2 and 3.5 and Corollary 3.9 are proved here, and Proposition 3.8 is a negative result in a stated scope, pending an independent check; the numbered proposals are conjectures to test. Attempted by Claude (Anthropic), 3 Nov 2026. Checks: checks_11d.py, sections B, E and G, Python with exact fractions. Referee pending.

Setting

02 realizes the Witten space with Killing spinors as Y_{3,2} = (5bar)//U(1)*Y. Its isometries give SU(3) x SU(2) x U(1)*R, and no bosonic Kaluza-Klein mode on it carries triality, weak isospin 1/2 or fractional hypercharge (Proposition 2.4). The M-theory vacuum AdS4 x Y*{3,2} is the near-horizon geometry of N*

M2-branes at the apex of the cone C(Y{3,2}), with N units of flux of *G_4 through Y_{3,2}. Lemma 21 proposed relating its two families of 84 states to the three-form C_3, to the M2 or M5 brane, and to their duality. This file connects the two.

Proposal 3.1 (hypercharge is the Betti vector of Y_{3,2})

In AdS4 x M^{111} the massless vectors are those of the isometries, SU(3) x SU(2) x U(1)_R, and one more from C_3 on the harmonic two-form (b_2 = 1), the Betti multiplet; the Kaluza-Klein states are neutral under it, and only non-perturbative states carry its charge (Fabbri et al., hep-th/9903036 and hep-th/9907219). By Proposition 2.3 the Betti class pairs with the divisors {l(U) = 0} and {m(V) = 0} as 2 and -3, the values of 6Y on d^c and L.

Proposal. Identify the Betti U(1) of Y_{3,2} with hypercharge. The gauge group of the vacuum is then

```
SU(3) x SU(2) (isometries) x U(1)_Y (from C_3) x U(1)_R (isometry),
```

thirteen vectors, the Standard Model group and one R-symmetry. With massless W and Z this is the eleven-dimensional vacuum of the v -> 0 limit; the identification is consistent with three facts already established:

1. Kaluza-Klein modes have Y = 0, as they must by Proposition 2.4, which leaves them only PSU(3) x SO(3) representations.
2. The states with Y != 0 are non-perturbative: M5-branes wrapped on five-cycles and M2-branes wrapped on the two-cycle.
3. The M5 baryons of Fabbri et al. are det U on one five-cycle and det V on the other, with baryonic charges 2N and -3N and representations Sym^N(3bar) and Sym^N(2). For N = 1 they are U in (3bar, 1)*{1/3} and V in (1, 2)*{-1/2}: d^c and L, the 5-bar.

In this reading hypercharge is the homology class of a wrapped fivebrane, 6Y = [Sigma_5] in H_5(Y_{3,2}; Z) = Z, and Witten's 1981 chirality argument, which assumes gauge fields acting by coordinate transformations, no longer covers U(1)_Y. It still covers SU(3) x SU(2) for Kaluza-Klein fermions, so chiral doublets must also be brane states.

Proposition 3.2 (one generation is the odd part of the exterior algebra of the 5-bar)

Statement. As SU(3) x SU(2) x U(1)_Y representations,

```
Λ^1(5bar) + Λ^3(5bar) + Λ^5(5bar) = (d^c + L) + (Q + u^c + e^c) + nu^c = 16,
Λ^0 + Λ^2 + Λ^4 = the conjugates,
```

and the charge of SO(10)/SU(5) is X = 5 - 2k on Λ^k. The 32 states are the finite Hilbert space of one generation with nu^c (dim H_F = 96 for three).

Proof. Enumeration with U, V of charges 2, -3 (Python, section B); the identification of the 16 of SO(10) with the odd forms on C^5 is standard. ■

The new element is the reading of the generators: U and V are the coordinates whose zero loci are the baryonic five-cycles of Y_{3,2}, so a state of Λ^k carries the total class of k such cycles, which is its 6Y. For example Q = U^1 U^2 V^1 has class 2 + 2 - 3 = 1.

Conjecture 3.3 (a generation as a Fock space of wrapped fivebranes). At N = 1, the states of one generation and its conjugate are the states of an exterior algebra generated by five fermionic zero modes, one for each of the cycles {U^i = 0} and {V^A = 0}, with hypercharge the total homology class and X the number of fivebranes. Needed: the fermionic statistics of the wrapped states (from fermionic zero modes of the M5 worldvolume theory on Sigma_5), the regime N = 1, where the supergravity description fails and the M2-brane theory is the abelian quotient of 02, and three generations.

Proposal 3.4 (the Higgs is the position of the M2-brane, and the massless limit is the apex)

Resolve the cone with a Fayet-Iliopoulos parameter: 2|U|^2 - 3|V|^2 = zeta. For zeta < 0 the locus U = 0 is the projective line of the doublet, P(V) = CP^1; for zeta > 0 the locus V = 0 is CP^2.

Statement (proved part). On the brane coordinates (U, V), on which the Betti U(1)_Y of Proposal 3.1 acts with charges (2, -3), the value V = (v, 0), U = 0 on the doublet line is fixed by SU(3) and by T_3 + Y, and moved by T_3 - Y. (In the quotient geometry, where U(1)_Y is divided out, the same point is fixed by T_3, which is T_3 + Y up to the quotient.) Q = T_3 + Y gives U the charge +1/3 of d^c and V the charges (0, -1) of (nu, e). (Python, section B2.)

Proposal. The scalar that breaks SU(2) x U(1)_Y to U(1)*em is the position of the M2-brane on the doublet line, with the quantum numbers of V, that is of H^* (or of a sneutrino). As zeta -> 0 and the brane moves to the apex, the full group is restored and the near-horizon geometry is AdS4 x Y*{3,2}. So the eleven-dimensional theory reached at v -> 0 is the Freund-Rubin vacuum on the Witten space with Killing spinors, with SU(3) x SU(2) from its isometries and U(1)_Y from C_3. The relation between M_W and zeta, or the brane position, is the next computation (README, item 3).

Which path this is. In AdS4 x Y_{3,2} the Kaluza-Klein scale is 1/L with L ~ N^{1/6} l_p, independent of zeta. At fixed N, the limit zeta -> 0 therefore restores the group at a fixed tower: it is the path of the Astra audit (§2.2) on which masslessness opens no dimension, and the eleven dimensions are present above 1/L at every v. A joint family in the sense of 01 (weak masses vanishing faster than the tower) needs L/l_p -> infinity, so N -> infinity, with the isometry couplings, g^{-2} proportional to the Killing-vector norms times the volume of Y in eleven-dimensional units, going to zero unless compensated (audit §7.2). Large N conflicts with item 3 of Proposal 3.1, which needs N = 1 for the baryons to be d^c and L rather than Sym^N(3bar) and Sym^N(2). This tension, between the regime in which the matter content is right and the regime in which the dimension opens, is the main obstruction of the M2 side in its present form, stated for the Freund-Rubin vacuum with N units of flux; escapes are N = 1 matter on a separate stack, a varying flux along the family, or a different vacuum.

Agreement with the audit (its Appendix A). The audit states independently that the circle bundle of M^{111} has c_1 = 3h + 2u, that the total space is spin, that its geometric U(1) is an R-symmetry, and that hypercharge belongs to the charged module and the gauge lift, since a fixed circle-charge sector acts by a scalar phase. This agrees with Theorem 2.1, Proposition 2.3 for (3, 2), and the placement of hypercharge in Proposal 3.1. The audit also notes that Toeplitz compression on the first holomorphic sector H^0(O(3, 2)), the (10, 3) of Theorem 2.1, item 4, generates the full M_30(C), a noninformative algebra in the sense of STRATEGY T0.

Proposition 3.5 (a magnetic completion: AdS5 x CP^2 x S^2)

Statement. On AdS5 x CP^2 x S^2 with product metric and four-form

```
G_4 = f_1 vol(CP^2) + f_2 J  vol(S^2),
```

with J the Kähler form of CP^2 and orthonormal-frame components f_1, f_2, the equations of eleven-dimensional supergravity hold with Ricci components

```
R_AdS = -(f_1^2 + 2 f_2^2)/6,   R_CP2 = f_1^2/3 + f_2^2/6,   R_S2 = 2 f_2^2/3 - f_1^2/6,
```

so a solution exists with round factors exactly when |f_1| < 2|f_2|.

Proof. dG_4 = 0, and *G_4 = vol(AdS5) (f_1 vol(S^2) + f_2 J), which is closed. G_4 G_4 is an eight-form on the six-dimensional factor, so it vanishes and d*G_4 + G_4 G_4/2 = 0. The Einstein equation R_MN = (G_MPQR G_N^PQR - g_MN G^2/12)/12 gives the displayed components and zero off the diagonal (Python, section G, by summing all frame components); each factor is Einstein, and the radii follow from the three values. ■

Supersymmetry, stability and flux quantization are not checked. The product with flux on KE_4 x S^2 appears in Pope and van Nieuwenhuizen (1989), and supersymmetric AdS5 solutions fibre the S^2 over the Kähler-Einstein base (Gauntlett, Martelli, Sparks and Waldram, hep-th/0402153); both recalled, not reread here.

Proposal 3.6 (two completions of one base)

The base B_6 = CP^2 x S^2 has isometry SU(3) x SU(2), and its dimension is the KO-dimension of the finite factor. It has two eleven-dimensional completions, both of the form 4 + 1 + 6:

	electric completion	magnetic completion
geometry	AdS4 x Y_{3,2}, Y a circle bundle over B_6	AdS5 x B_6
flux	*G_4 through Y: N M2-branes	G_4 through CP^2 and CP^1 x S^2: M5 charges (N_1, N_2)
the "1"	the fibre circle (Proposition 1.1 of 01)	the radial direction of AdS5
vectors from C_3	b_2(Y) = 1: U(1)_Y	b_2(B_6) = 2: U(1)*{h_1}* x *U(1)*{h_2}
further vectors	SU(3) x SU(2) x U(1)_R	SU(3) x SU(2)
v -> 0	the M2-brane at the apex	the infrared end of AdS5 moves to infinity

Under reduction along the fibre of class (3, 2), the combination of U(1)*{h_1}* x *U(1)*{h_2} along (3, 2) is replaced by the fibre isometry, and the quotient Z^2/(3, 2) = Z, (x, y) -> 2x - 3y, is hypercharge. The magnetic side therefore carries two abelian factors, a candidate for the fluxed pair needed by Proposition 2.6 together with the universal twist. The magnetic completion realizes the counting of H directly: four-dimensional spacetime on the M5 worldvolume, the interval as the conformal radial coordinate z in (z_UV, z_IR) with M_KK ~ 1/z_IR, as in Higgsless models on an interval, and v -> 0 as z_IR -> infinity.

Proposal 3.7 (the M2/M5 exchange acts on the Betti vector)

In AdS4, M2-branes on the two-cycle and M5-branes on five-cycles are electric and magnetic for the Betti vector (C_3 = A omega_2 and C_6 = Ã omega_5), with Dirac pairing given by the intersection of H_2 and H_5. Which of the two is electric depends on the boundary condition of the bulk vector, and the S operation of Witten's SL(2, Z) action on three-dimensional theories with a U(1) symmetry exchanges them (hep-th/0307041, recalled). With Proposal 3.1 this exchange acts on hypercharge as electric-magnetic duality.

Test against lemma 21. S maps charges to monopole charges; it does not permute the 96 states of H_F. The exchange of the top and the neutrinos in lemma 21 is a permutation, so it does not come from this S. In the brane reading, the 84 states of the second family (Q_em != 0) are the states whose fivebrane class 6Y is not cancelled by T_3, after the breaking of Proposal 3.4; the first family (all states except the top) has no reading here. This points away from the M2/M5 duality as the origin of the exchange, and toward a property of the top (lemma 21, Proposition 21.1) that the present construction does not see.

Proposition 3.8 (no colour-equivariant identification of the 84 with the three-form)

Statement. For every homomorphism SU(3) -> SO(9), Λ^3 R^9 ⊗ C is not isomorphic, as an SU(3) representation, to the colour content of either family of 84 in lemma 21: 10 (3 + 3bar) + 24 (1) for the five quarks and all leptons, 12 (3 + 3bar) + 12 (1) for the charged fermions. The same holds for Λ^6 R^9, the polarizations of C_6.

Proof. A real nine-dimensional representation has a self-conjugate complexification of dimension 9, which is 9 (1), 3 + 3bar + 3 (1) or 8 + 1 (the sextets and two triplet pairs already exceed 9). In the first case Λ^3 has only singlets. In the second, Λ^2(3) ⊗ 3bar = 3bar ⊗ 3bar contains the 6bar. In the third, Λ^3(8) = 1 + 8 + 10 + 10bar + 27. Both families contain only 1, 3 and 3bar. Λ^6 R^9 is isomorphic to Λ^3 R^9 under SO(9). (Python, section E, by weights.) ■

Scope and escapes. A linear action of colour on the nine transverse directions of the eleven-dimensional little group. Escapes: colour acting on the internal space instead of the little group, a nonlinear or brane

realization, or a map that is not colour-equivariant, in which case the equality of the two 84s is an equality of numbers only.

Corollary 3.9 (the coloured states of the 84 are not Kaluza-Klein modes of C_3 on Y_{3,2}) Under the quotient realization, every Kaluza-Klein mode of C_3 (or of the metric) on Y_{3,2} carries a representation of PSU(3) x SO(3) (Proposition 2.4), so none is a colour triplet. The 60 coloured states of the first family, and the 72 of the second, are therefore not modes of C_3 on Y_{3,2}. Together with Proposition 3.8 this closes the two direct routes, the little group and the Kaluza-Klein tower, for a colour-equivariant identification, in their stated scopes. The route left open is the brane one: wrapped M5 and M2 states, which carry triality through their hypercharge class (Proposition 2.4 (a)).

Summary of the direction

The M2 and M5 details of lemma 21 enter in a specific place. Hypercharge is the C_3 vector of the Witten space with Killing spinors, the 5-bar consists of the N = 1 fivebrane baryons, the Higgs is the M2 position, and the M2/M5 exchange is the electric-magnetic duality of hypercharge. The count 84 of C_3 does not survive as a colour-equivariant identification. The next computations are in the README.

Part 11D (04_two_families_m2_m5, referee pending)

The two families of 84 and the M2 and M5 sides

Part of the 11D line of research (README, PLAN). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 4.1 to 4.3 and Theorem 4.4 and Propositions 4.5, 4.6 are proved here by hand (branching rules and counting); the negative results 4.3 and 4.6 await an independent check; readings are labelled as proposals. Referee pending.

The question

Lemma 21, Proposition 21.3: with three generations and Dirac neutrinos, dim H_F = 96 = 84 + 12 in two ways,

```
F1 = all states except the top          (the top: 3 colours x 4 = 12 states),
F2 = all states except the neutrinos    (the neutrinos: 3 generations x 4 = 12 states),
```

where 4 counts chirality and particle/antiparticle. The two families overlap in the 72 states of the charged fermions other than the top. Lemma 21 proposed relating one family to C_3 (or the M2-brane) and the other to C_6 (or the M5-brane), with the exchange of the top and the neutrinos as the content of the duality on the Standard Model side, and recorded that equal numbers are insufficient. The files 02 and 03 showed that colour-equivariant identifications through the little group or through Kaluza-Klein modes on the Witten space fail (Proposition 3.8, Corollary 3.9). This file asks the question in the Hořava-Witten wall group E8 and in the finite geometry.

1. What singles out the two blocks

Proposition 4.1. Consider blocks of 12 states made of whole Dirac species (all colours, both chiralities, particles and antiparticles; 12 states for a quark species, 4 for a lepton species). The blocks whose removal keeps both spectral-action values of lemma 21, cos^2 theta_NC = 5/8 and kappa_em/kappa_s = 8/3, are exactly: one up-type quark (u, c or t), and the three neutrinos together. So F1 (with the top, the heavy up-type quark) and F2 are the two kinds of solutions: a charged one and a neutral one.

Proof. A 12-state block of whole species is one quark species or three lepton species. By Proposition 21.1, with the totals (sum T_3 Q, sum Y Q; Q^2, colour) = (6, 10; 32, 12) and cos^2 theta_NC = (sum Y Q)/(sum T_3 Q + sum Y Q), removing a quark keeps both values only for an up-type quark, (1, 5/3; 16/3, 2). For three lepton species with k charged leptons, each contributing (1/2, 3/2; 4, 0), the angle becomes (10 - 3k/2)/(16 - 2k), which is 5/8 only for k = 0; neutrinos contribute zero to all four sums. ■

Reading at the 9 label. At the 9 label the light gauge algebra is colour plus electromagnetism (canonical §3, lemma 21). H_F then splits into three parts: the 72 charged light states (F1 ∩ F2), the 12 neutrino states, which carry no charge of that algebra and are free, and the 12 top states, which are charged and infinitely heavy. F1 = charged light + free light; F2 = charged light + charged heavy. The exchange of lemma 21 exchanges the block that decouples through zero charge with the block that decouples through infinite mass.

2. Colour obstructions

Proposition 4.2. (a) As representations of colour SU(3), F1 = 10 (3 + 3bar) + 24 (1) and F2 = 12 (3 + 3bar) + 12 (1); they are not isomorphic. Hence no unitary of H_F commuting with colour maps F1 onto F2, and no automorphism of the finite spectral triple of the Standard Model does. (b) Any bijection of the top block onto the neutrino block that is linear in each index exchanges the colour index of the top (r, g, b) with the generation index of the neutrinos (e, mu, tau).

Proof. (a) Counting: F1 has the five quarks other than the top (5 x 2 triplet pairs, counting chirality) and 24 leptonic states; F2 has the six quarks and 12 charged-lepton states. Colour acts through A_F. (b) The top block is C^3_colour ⊗ e_3 ⊗ C^4 (third generation, 4 for chirality and conjugation) and the neutrino block is e_lepton ⊗ C^3_generation ⊗ C^4 in the Pati-Salam form C^4_colour ⊗ C^3_generation of the up-type sector; a map of the first onto the second that respects the tensor factors sends C^3_colour to C^3_generation. ■

So the exchange cannot be a symmetry of any structure in which colour acts and generations do not; it needs a structure in which colour and generation can be exchanged.

Proposition 4.3 (complex case of Proposition 3.8). For every homomorphism SU(3) -> SU(9), the colour content of Λ^3 C^9 is not that of F1 or F2. Λ^3 C^9 contains only 1, 3 and 3bar exactly when C^9 = 3 + 6 (1) or its conjugate, and then Λ^3 C^9 = 15 (3) + 6 (3bar) + 21 (1).

Proof. C^9 is a sum of irreducibles of dimensions 1, 3, 6, 8 with total 9. With a sextet: 6 + 3 (1) has the 6 in Λ^1(6) ⊗ Λ^2(3 (1)); 6 + 3 has Λ^3(6) = 10 + 10bar. With an octet: 8 + 1 has Λ^3(8) = 1 + 8 + 10 + 10bar + 27. With two triplets T, T' (each 3 or 3bar) and three singlets: Λ^1(T) ⊗ Λ^1(T') ⊗ Λ^1(1) contains T ⊗ T', which is 6 + 3bar, 6bar + 3 or 8 + 1. With three triplets: T ⊗ T' ⊗ T'' contains a 10, a 10bar, a 15 or a 15bar. With nine singlets there is no triplet. With one triplet and six singlets, Λ^3(3 + 6 (1)) = Λ^3(3) + 6 Λ^2(3) + 15 (3) + 20 (1) = 15 (3) + 6 (3bar) + 21 (1). Neither family has unequal numbers of 3 and 3bar. ■

Together with Proposition 3.8 (real nine-dimensional representations) this closes the colour-equivariant route through Λ^3 of a linear nine-dimensional space, in that scope. The positive construction below places the two blocks instead of identifying the two families.

3. Two commuting gradings of E8

E8 is the gauge group of each wall in Hořava-Witten theory, and it also models the C-field topologically (Witten 1996, "On flux quantization in M-theory"; Diaconescu, Moore and Witten 2000; recalled). Over C, e8 has an inner automorphism of order three with fixed algebra sl(9):

```
e8 = sl(9) ⊕ Λ^3 C^9 ⊕ Λ^6 C^9          (80 + 84 + 84),
```

with [Λ^3, Λ^3] Λ^6 (wedge product), [Λ^3, Λ^6] sl(9) (contraction) and [Λ^6, Λ^6] Λ^3. In the U-duality algebra e8(8) of eleven-dimensional supergravity on T^8, the decomposition under gl(8) by level (West's level, recalled) places the C_3 shifts at level 1 (Λ^3 of 8, 56 states), the C_6 shifts at level 2 (Λ^6 of 8, 28), the dual graviton at level 3 (8) and gl(8) at level 0; then Λ^3 C^9 = 56 (level 1) + 28 (level -2) and the grading above is the level mod 3. So M-grade b = 1 is one unit of M2 charge (or minus an M5), b = 2 is one unit of M5 charge (or minus an M2), and b = 0 holds the metric and the dual graviton (levels 0 and ±3). The bracket Λ^3 × Λ^3 -> Λ^6 is the algebraic form of the term C_3 G_4 in G_7 = dC_6 + (1/2) C_3 G_4, equivalent by duality to the Chern-Simons coupling; Λ^3 × Λ^6 -> sl(9) adds levels 1 +

2 = 3, the dual graviton. Under SO(9) SL(9), 248 = 36 + 44 + 84 + 84, the SO(9) content of so(9), the graviton, C_3 and C_6. We call the three pieces the gravity, C_3 and C_6 sectors.

A second element of order three, the centre of SU(3) in E8 (SU(3) × E6)/Z_3, gives the family grading e8 = (su(3) ⊕ e6) ⊕ (3, 27) ⊕ (3bar, 27bar), with family grade a = 0, 1, 2. With this SU(3) as family group, (3, 27) holds three families of E6, each containing a generation 16 27, and (3bar, 27bar) holds their conjugates.

Lemma 4.A (the two gradings commute). Take SU(9) SU(3) × SU(6) × U(1) with C^9 = (3, 1)*{-2} + (1, 6)*{1}, and let s_f be the centre of this SU(3). Then s_f commutes with the SU(9) grading, its centralizer is (SU(3) × E6)/Z_3, the U(1) is the Cartan of the SU(2) in E6 SU(6) × SU(2) with charge q = 6 T_3, and the joint pieces g_{a,b} are

	b = 0 (gravity)	b = 1 (C_3)	b = 2 (C_6)
a = 0	su(3) + su(6) + u(1): 44	(1,1)*{-6} + (1,20)*{3}: 21	(1,1)*{6} + (1,20)*{-3}: 21
a = 1	(3, 6bar)_{-3}: 18	(3, 15)_{0}: 45	(3, 6bar)_{3}: 18
a = 2	(3bar, 6)_{3}: 18	(3bar, 6)_{-3}: 18	(3bar, 15bar)_{0}: 45

so that b a + q/3 = a + 2 T_3 (mod 3).

Proof. Λ^3 of (3,1)*{-2} + (1,6)*{1} is (1,1)*{-6} + (3bar,6)*{-3} + (3,15)*{0} + (1,20)*{3}; the adjoint is (8,1) + (1,35) + (1,1) + (3,6bar)*{-3} + (3bar,6)*{3}; Λ^6 is the conjugate of Λ^3. The column a = 0 sums to 86 = 8 + 78 and the row a = 1 to 81 = 3 x 27, which matches E8 SU(3) × E6 with E6 SU(6) × SU(2): 27 = (15, 1) + (6bar, 2) and 78 = (35, 1) + (1, 3) + (20, 2), the SU(2) triplet being (1,1)*{-6}, the u(1) and (1,1)*{6}. ■

4. Placement of the generation

Let SU(5) SU(6) E6 be the Georgi-Glashow group, with 15 = 10 + 5 and 6bar = 5bar + 1. The commutant of SU(5) in E6 is SU(2) × U(1), and its SU(2) is the SU(2) of the lemma; we call it SU(2)*I. With SO(10) × U(1)*psi and SU(5) × U(1)*chi, 27 = 16_1 + 10*{-2} + 1_4, 16 = 10*{-1} + 5bar_3 + 1*{-5} and 10 = 5_2 + 5bar_{-2}. The Cartan T_3 of SU(2)*I vanishes on the singlets 10 and 5, so it is proportional to psi + chi: T_3 = t on 5bar_16 and on N = 1*{E6}, and T_3 = -t on 5bar_10 and on nu^c = 1_16, with t = ±1/2. The SU(2)_I doublets are (d^c, D^c), (L, H_d) and (N, nu^c).

Theorem 4.4 (the generation in the three sectors). For the generation 16 27 of family grade a = 1:

1. the 10 = (Q, u^c, e^c) and the 5 = (D, H_u) lie in g_{1,1}, the C_3 sector;
2. (d^c, L) and N lie in g_{1,1+2t}, and (D^c, H_d) and nu^c lie in g_{1,1-2t}: one pair in the C_6 sector and the other in the gravity sector, according to the sign t; the two choices differ by the Weyl reflection of SU(2)_I;
3. the antiparticles lie in the conjugate pieces g_{2,-b}.

So every top state of a particle (t_L in Q, t^c in u^c), and the up-type Higgs, lie in the C_3 sector, and the anti-top in the C_6 sector. The particle neutrinos lie outside the C_3 sector, nu_L and nu^c in different sectors (C_6 and gravity). With three families and antiparticles, the 96 states of H_F split as 45 (C_3) + 45 (C_6) + 6 (gravity: nu^c and its conjugate) for t = +1/2, and 33 + 33 + 30 for t = -1/2.

Proof. b a + 2 T_3, with T_3 = 0 on the 10 and the 5, and T_3 = ±t on the doublet members; for antiparticles a = 2 and T_3 changes sign. The counts follow per family: for t = +1/2, C_3 holds the 10 and the 5 of antiparticles (15), C_6 their conjugates, gravity nu^c and its conjugate; for t = -1/2, C_3 holds the 10 and the conjugate of nu^c (11), C_6 their conjugates, gravity the 5bar and the 5 (10). ■

5. Yukawa couplings as brackets

Proposition 4.5. Suppose the cubic couplings of the matter come from the structure constants of e8, as in heterotic and Hořava-Witten compactifications, and both gradings are symmetries, so that a coupling needs

grades summing to (0, 0) mod 3. Take H_u in the 5 g_{1,1} and H_d in 5bar_10 g_{1,1-2t}. Then, for either sign t:

1. the up-type Yukawa couplings Q u^c H_u have all three fields in the C_3 sector;
2. the down-type, charged-lepton and Dirac-neutrino couplings Q d^c H_d, L e^c H_d and L nu^c H_u have one field in each of the three sectors;
3. in Dirac form, the top mass is ad() : g_{1,1} -> g_{2,2}, the wedge Λ^3 -> Λ^6 with the Higgs trivector, while the Dirac-neutrino mass is a bracket that passes through the gravity sector: Λ^6 -> sl(9) (levels 1 + 2 = 3) for t = +1/2, sl(9) -> Λ^3 for t = -1/2.

Proof. Grades from Theorem 4.4: (1, 1) three times for the up type; for the others the b-grades are {1, 1 + 2t, 1 - 2t} = {1, 2, 0}. The Dirac masses pair the left-handed field with the conjugate of the right-handed one, through the Killing form between g_{a,b} and g_{-a,-b}. ■

The family grading alone allows only epsilon_{ijk} in exact family SU(3); in compactifications the family label is a cohomology label and the couplings are intersection numbers. The M-grading survives when its element commutes with the structure group of the bundle: for the standard embedding, s_M = s_f exp(2 pi i (2/3) T_3), a product of the centre of the holonomy SU(3) and an element of SU(2)_I E6, which does.

Reading (proposal). The up-type Yukawa couplings are the only Standard Model Yukawa couplings inside one sector, and that sector's bracket is the C_3 C_3 -> C_6 term, metric-independent; the other couplings use one field of each sector, including the gravity sector. The top, the only fermion with a Yukawa coupling of order one and the one singled out by the spectral action in lemma 21, takes its mass from the C_3 bracket.

6. In this realization the M2/M5 exchange is the real structure J

Proposition 4.6.

1. No automorphism of e8 that preserves the family grading maps the C_3 sector of the matter, g_{1,1} (dimension 45), onto its C_6 sector, g_{1,2} (dimension 18).
2. The Chevalley involution, the element -1 of the Weyl group of E8, maps g_{a,b} to g_{-a,-b}. It normalizes SU(9), acting as complex conjugation (Λ^3 <-> Λ^6), and on the placed H_F it exchanges every state with its conjugate: it is the real structure J of the finite geometry. The family grade a is the four-dimensional chirality in the left-handed basis (a = 1 for the left-handed fields, a = 2 for their conjugates), and J inverts it, as KO-dimension six requires (J gamma = -gamma J).
3. J preserves F1 and F2, which are unions of species with their conjugates.
4. The Weyl reflection of SU(2)_I fixes s_f and replaces b by 2a - b: on the matter (a = 1) it fixes the C_3 sector and exchanges the C_6 and gravity sectors, together with (d^c, L) <-> (D^c, H_d) and nu^c <-> N.

Proof. (1) Dimensions. (2) -1 acts on a Cartan subalgebra by -1, so it inverts every element of the maximal torus containing s_f and s_M, and maps the joint eigenspace for (omega^a, omega^b) to that for (omega^{-a}, omega^{-b}); it maps the root system A_8 of SU(9) to itself. (3) Immediate. (4) The reflection inverts T_3 and commutes with SU(3) × SU(6). ■

Negative result (scope: e8 with these two gradings, automorphisms preserving the pair up to inversion). The M2/M5 exchange does not exchange the top with the neutrinos; it exchanges each block with its conjugate. The exchange of lemma 21 must therefore come from an automorphism that does not preserve the family grading. The colour-family exchanges inside E8 SU(3)^4, where the four SU(3) factors are permuted by the normalizer, are the remaining candidates (PLAN, step 4). What the gradings do give is a placement: the top on the M2 side, the neutrinos on the M5 and gravity sides.

7. Agreement with files 02 and 03, and the finite geometry

- **nu^c as the gravitational state.** nu^c is the only Standard Model state with Y = 0, so on the Witten space it is the only one that can be a Kaluza-Klein mode (Proposition 2.4 with Proposal 3.1), while every hypercharged state is a brane state. In e8 with t = +1/2 it is the only Standard Model state in the gravity sector, and every hypercharged state lies in the C_3 or C_6 sector. The two

placements agree for t = +1/2; for t = -1/2 the gravity sector holds (d^c, L) instead. We take the agreement as selecting t = +1/2 (proposal).

- **The finite geometry.** On H_F, the family grading is the chirality, the Chevalley involution is J, and the Yukawa blocks of D_F are brackets with the Higgs: the top block maps the C_3 sector to the C_6 sector, the Majorana block acts inside the gravity sector (nu^c with nu^c, t = +1/2). A finite triple built on the graded e8, with D_F = ad(Higgs), is the construction to attempt next (PLAN, step 5).
- **The two families.** F1 removes a block of the C_3 sector (with its conjugate in C_6): the states joined by the cubic C_3 coupling. F2 removes the block that meets the gravity sector. In this sense the top belongs to the M2 side and the neutrinos to the M5 and gravity sides, which is the correspondence lemma 21 asked for, at the level of the removed blocks. The two families of 84 are not themselves the 84 of C_3 and the 84 of C_6 (Propositions 3.8, 4.3, 4.6).

8. Remark: a discrete symmetry from the M-grading

For t = +1/2, s_M acts on the 27 by omega^b: omega on the 10 and on the 5 (H_u, D), omega^2 on (d^c, L) and N, and 1 on (D^c, H_d) and nu^c. It allows Q u^c H_u, Q d^c H_d, L e^c H_d, L nu^c H_u, N H_u H_d, the Majorana mass of nu^c and the Weinberg operator L L H_u H_u, and forbids the mu term H_u H_d, the dimension-five operators Q Q Q L and u^c u^c d^c e^c, and the trilinear couplings L L e^c, L Q d^c, u^c d^c d^c; the bilinear L H_u is allowed. This is the pattern of E6-inspired models in which the singlet N supplies the mu term. It is a symmetry of the low-energy theory only if the breaking of E6 commutes with s_M; for t = -1/2 the charges change. Stated as a remark, not used above.

9. What remains open

1. Colour-family exchange in E8 SU(3)^4 acting on the placed states: does an element map the top block onto the neutrino block and F1 onto F2? (PLAN, step 4.)
2. A finite real spectral triple on the graded e8 with D_F = ad(Higgs): order zero and order one, and comparison with C + H + M_3(C) (PLAN, step 5).
3. Hořava-Witten dynamics: whether an M5-brane moving into a wall (small-instanton transition) can remove one of the 12-state blocks of Proposition 4.1 (PLAN, step 6).
4. The E11 level decomposition beyond E8, where C_3, C_6 and the dual graviton sit at levels 1, 2, 3: test the assignment 10 -> level 1, (d^c, L) -> level 2, nu^c -> level 0 or 3 (PLAN, step 7).

References for this file

Recalled, not reread in this session: E. Witten, "On flux quantization in M-theory and the effective action", hep-th/9609122; D.-E. Diaconescu, G. Moore and E. Witten, "E8 gauge theory, and a derivation of K-theory from M-theory", hep-th/0005090; P. West, "E11 and M theory", hep-th/0104081, for the level decomposition and the commutator of the level-one generators; E. Cremmer and B. Julia (1979) for E8(8) in three dimensions; E. B. Vinberg and A. G. Elashvili (1978) for the Z_3-graded e8 and trivectors of C^9; the E6 subgroup chains are standard (R. Slansky, Phys. Rep. 79 (1981) 1).

Part 11D (05_colour_family_exchange, referee pending)

The exchange of the top and the neutrinos as a colour-family transposition in E8

Part of the 11D line of research (README, PLAN, step 4). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 5.1 (a negative result in its scope, pending an independent check) and Proposition 5.2 are proved by hand; an exploratory search in Python with exact fractions, in a scratch directory and not kept, found the same results (no isometry for three families, signed coordinate permutations for four). Readings are labelled as proposals. Referee pending.

Setting

Proposition 4.6 showed that the M2/M5 exchange of the graded e8 acts as J and cannot exchange the top with the neutrinos, and Proposition 4.2 that any such exchange must trade the colour index of the top for the generation index of the neutrinos. This file tests the colour-family exchanges inside E8.

Use E8 SO(16) SO(10) × SO(6), with the roots of E8 in the coordinates of R^8: ±e_i ± e_j, and (1/2)(±1, …, ±1) with an even number of minus signs. The 128 half-integer roots form the chiral spinor of SO(16), which under SO(10) × SU(4) is (16, 4) + (16bar, 4bar): four families of the Standard Model with ^c, with their conjugates. Write a root as (| w |), with the signs of the three colour coordinates, w those of the two weak coordinates and those of the three family coordinates; a particle has an even number of minus signs among , w and an even number among . The four family patterns are = (+++) (family 0) and the three patterns with two minus signs (families 1, 2, 3); the colour patterns with one minus sign are the quark colours, the pattern − is the lepton. Then, for a colour pattern c with one minus sign and a family f,

state	root (times 2)
u_L, d_L	(c | −+ | f), (c | +− | f)
u^c, d^c	(−c | ++ | f), (−c | −− | f)
_L, e_L	(− | −+ | f), (− | +− | f)
^c, e^c	(| ++ | f), (| −− | f)

and antiparticles are the negatives. Three families give the 96 roots of H_F, closed under negation; the four family slots are permuted by the Weyl group of SU(4), so the choice of three is immaterial. The same 96 roots are (16, 3) + (16bar, 3bar) in E8 E6 × SU(3) SO(10) × U(1) × SU(3), the embedding of file 04.

Proposition 5.1 (the two families of 84 are not isometric)

Statement. Let H_F be the 96 roots of three families, the top block T the 12 roots of the up-type quark of one family f_t with its conjugate, and the neutrino block N the 12 roots of the neutrinos of the three families with their conjugates; F1 = H_F − T and F2 = H_F − N. Then F1 and F2 have the same numbers of pairs at each inner product (42 at −2, 816 at −1, 1812 at 0, 816 at +1), but no isometry of R^8 maps F1 onto F2. In particular no element of the Weyl group of E8 does, and no isometry of H_F maps T onto N.

Proof. For a root v of a set S let d_S(v) be the number of roots of S at inner product +1 with v; it is preserved by isometries. Two half-integer roots have inner product +1 exactly when their signs agree in six of the eight places.

1. d_{H_F} = 22 for every root. H_F is a single orbit of the group generated by the Weyl group of SO(10) (transitive on the 16, a minuscule representation), the permutations of the three families and the negation. For ^c of family 0, (| ++ |), the roots of H_F with exactly two minus signs are: two in the first five places with family (10), two among the family places (2, the patterns of families 1 and 2 in H_F), and antiparticles with one minus sign in each block (5 x 2). The total is 22.
2. Removing N. For u_L = (c | −+ | f): the inner product with _L(g) is (1 + f · g)/4 and with ^c(g) is (1 + f · g)/4, where f · g = 3 for g = f and −1 otherwise; so u_L has exactly 2 roots of N at +1 (its own family's _L and ^c), and the same holds for u^c and the conjugates. For d_L = (c | +− | f): the inner product with _L(g) is (−3 + f · g)/4, which is −1 for g ≠ f, so the two roots − _L(g), g ≠ f, are at +1, and ^c(f) is at +1: 3 in all; the same count holds for d^c, e_L, e^c and the conjugates. So in F2, d = 20 on the 36 up-type quark roots and d = 19 on the 48 others.
3. Removing T. For a neutrino of the top's family, _L(f_t) = (− | −+ | f_t): its inner products with t_L(c) = (c | −+ | f_t) and with t^c = (−c | ++ | f_t) are (−1 + 2 + 3)/4 = 1 and (1 + 0 + 3)/4 = 1, six roots at +1; for g ≠ f_t, _L(g) has inner product (−1 + 2 − 1)/4 = 0 with t_L and (1 + 0 − 1)/4 = 0 with t^c. The same holds for ^c. So in F1, d = 16 on the 4 roots of the neutrinos of the top's family and d = 22 on the 8 roots of the other neutrinos (the up-type quarks of the other families

have 2 roots of T at +1 and the down-type quarks and charged leptons 3, by inner products of the same kind as in item 2, so d = 20 and d = 19).

F1 has roots with d = 16 and d = 22, and F2 has none. The sums of d agree, $4 \cdot 16 + 48 \cdot 19 + 24 \cdot 20 + 8 \cdot 22 = 48 \cdot 19 + 36 \cdot 20 = 1632 = 2$ x 816, which is the equality of the pair counts at +1. Both sets are closed under negation, so the pairs at −1 are as many as those at +1 and the pairs at −2 are the 42 antipodal pairs; the 1812 pairs at 0 are the remainder of the 3486 pairs. ■

Scope. The roots of H_F in E8, through the embedding above, which is canonical up to the Weyl group of E8; isometries of R^8 acting on the roots. Escapes: maps that are not linear on the roots (bijections of states with no geometric meaning), other realizations of H_F in a larger group, and a fourth family (Proposition 5.2). The tau neutrino carries the obstruction: in F1 the neutrino of the top's own family is the only state whose neighbourhood changes by six.

Proposition 5.2 (four families: the transposition of colour and family)

Statement. Let P be the permutation of coordinates that exchanges the three colour coordinates with the three family coordinates and the two weak coordinates with each other, and M = −P. Then M is an involution in the Weyl group of E8 that preserves the 128 roots of the four families and SU(2)_L, and acts on the up-type and down-type states as the transposition of the four Pati-Salam colours (r, g, b, lepton) with the four families (1, 2, 3, 0), with the lepton paired with family 0 and the colour c paired with the family −c. In particular:

1. the up-type quark of family 0 (t_L and t^c, three colours, with conjugates) goes onto the neutrinos of families 1, 2, 3 (_L and the conjugate of ^c), and back;
2. the down-type quark of family 0 goes onto the charged leptons of families 1, 2, 3;
3. the leptons of family 0 are fixed, and the quarks of families 1, 2, 3 go to quarks of families 1, 2, 3 with colour and family transposed.

So the four-family sets 128 − T(0) and 128 − N(1, 2, 3), each of 116 roots, are exchanged.

Proof. P is an even permutation of coordinates and −1 is in the Weyl group of E8, so M is in it; M^2 = P^2 = 1. Both preserve the set of half-integer vectors with an even number of minus signs. P maps (| w_1 w_2 |) to (| w_2 w_1 |), so M maps it to (− | −w_2 −w_1 | −). The weak part (−+) goes to (−+) and (++) to (−−), so T_3 of SU(2)_L, (w_2 − w_1)/2, is preserved and T_3 of SU(2)_R is reversed. Then t_L(c) = (c | −+ |) goes to (− | −+ | −c) = _L(−c); t^c = (−c | ++ |) goes to (− | −− | c) = − ^c(−c); b_L(c) = (c | +− |) goes to (− | +− | −c) = e_L(−c); _L(0) = (− | −+ |) and e_L(0) are fixed; u_L(c, f) = (c | −+ | f) goes to u_L(−f, −c), with −f a quark colour and −c a family pattern. ■

The search found further signed permutations with the same effect on the two blocks; P itself maps T(0) onto N(1, 2, 3) up to conjugation, and M = J P, with J = −1 the real structure of Proposition 4.6.

Proposition 5.3 (the flip in a four-family finite geometry)

Statement. Let H_f = (C^2_L ⊕ C^2_R) ⊗ C^4_c ⊗ C^4_f, H = H_f ⊕ conj(H_f), with J the antilinear exchange of the two summands and equal to +1 on C^2_L and −1 on C^2_R in H_f and to its negative on conj(H_f), the signs of KO-dimension six. Fix an isomorphism : C^4_c -> C^4_f sending the lepton line to the family line e_0 and the colour space C^3 onto the span of e_1, e_2, e_3, and let F be the flip x ⊗ y -> ^{-1}(y) ⊗ (x) on C^4_c ⊗ C^4_f, extended by the identity on C^2_L ⊕ C^2_R and to conj(H_f) by J F J^{-1}. Then:

1. F is unitary, F^2 = 1, and F commutes with and J.
2. For the up-type Dirac blocks T_top = Y ⊗ P_q ⊗ P_0 and T_ = Y ⊗ P_ ⊗ (1 − P_0), where Y maps the up state of C^2_L to the up state of C^2_R (and Y^* back), P_ and P_q = 1 − P_ are the lepton and quark projectors of C^4_c and P_0 the projector on e_0, one has F T_top F^{-1} = T_ , and likewise for the down-type blocks of family 0 and the charged leptons of families 1, 2, 3.

3. So the heavy limits D + t T_top and D + t T_ , for D commuting with F, are conjugate by F: their light spaces, of 116 states each, correspond. The light algebras A ∩ {T}' correspond when the algebra is invariant under F; the Pati-Salam and Standard Model algebras act on C^4_c alone and are not.

Proof. F acts on the factor C^4_c ⊗ C^4_f only, where it conjugates a ⊗ b to ^{-1} b ⊗ a ^{-1}, so P_q ⊗ P_0 goes to P_ ⊗ (1 − P_0) by the choice of . and Y act on the other factor, and J F J^{-1} on conj(H_f) is the conjugate flip. ■

This is the finite-geometry form of Proposition 5.2. It also makes reading 2 below precise: F exchanges the projector P_ , which the Majorana direction selects in colour space, with P_0, which the top Yukawa coupling selects in family space; as operators, it maps the top Yukawa block to the Dirac blocks of the neutrinos of families 1, 2, 3, and it does not map the Majorana block to a Dirac block, since the Majorana block joins H_f to conj(H_f).

Proposition 5.4 (no three-slot assignment admits the exchange)

Statement. Place the top in slot 0, and let the up-type quarks, the down-type quarks, the charged leptons and the neutrinos each occupy three of the four family slots, independently (the up-type quarks in slots containing 0). This includes the aligned families of Proposition 5.1 and the configurations of the 9 label, where SU(2)_L is off and does not tie the slots of up-type and down-type quarks. For none of the 192 configurations S is there an isometry of R^8 mapping S − T onto S − N. In every case S − N has d = 19 and d = 20 only, while S − T has d = 16 on the neutrino roots of one slot and d = 22 on those of the two others.

Proof. Exhaustive computation of the neighbour counts d (Python, exact fractions, scratch script not kept); the counts of S − T and S − N differ for every configuration, so no isometry exists. ■ This is a computer-assisted statement; Proposition 5.1 is its hand-proved aligned case.

Scope. It assumes that each family occupies one slot, so that every state is a single root. Four slots then cannot be reduced to three while keeping the exchange as an isometry of the roots.

Proposal 5.5 (four slots, three zero modes)

In heterotic compactifications on a Calabi-Yau threefold with E8 SO(10) × SU(4), 248 = (45, 1) + (1, 15) + (16, 4) + (16bar, 4bar) + (10, 6), the SU(4) is the structure group of a rank-four bundle V, and the families are the zero modes, the 16s coming from H^1(X, V); their number is the index |c_3(V)|/2, which is 3 in the heterotic Standard Models built on SU(4) bundles (Braun, He, Ovrut and Pantev, hep-th/0501070; recalled, not reread). In that reading the four slots of this file are the fibre of V, and the three families are not three of the slots: a zero mode is a section of V and in general involves all four. Propositions 5.1 and 5.4 do not apply to it, since they assume one slot per family.

The transposition M of Proposition 5.2 then exchanges the Pati-Salam SU(4), a gauge group, with the structure group SU(4) of V: colour and holonomy trade places, and SO(10) goes to the commutant SO(10)' of the other SU(4). The top-neutrino exchange becomes a statement about the fibre, which an index map must carry to the zero modes. The test is an explicit bundle. For a split bundle V = L_0 + L_1 + L_2 + L_3 (line-bundle models), slot i carries n_i = − (L_i) families with n_0 + … + n_3 = 3; such models have aligned families and fall under Proposition 5.1. A non-split V, whose zero modes are spread over the slots, is the case to compute.

Readings (proposals)

1. **Three families are the obstruction.** The exchange of lemma 21 is a symmetry of the E8 geometry exactly in the four-family completion, where the colour-family square is 4 x 4 and the top belongs to the family paired with the lepton colour. With three families the square is 4 x 3 and the exchange is not an isometry (Proposition 5.1). Lemma 21's 96 = 84 + 12 in two ways is the three-family shadow of 128 = 116 + 12 in two ways, exchanged by M.

2. **The two heavy directions of the algebra chain.** Canonical §2.2 and §3 obtain the chain Pati-Salam -> Standard Model -> colour plus electromagnetism from two heavy operators: the Majorana direction, which selects the lepton among the four Pati-Salam colours and _R among the SU(2)_R states, and the top Yukawa coupling, which selects the heavy family and the up-type state. M exchanges the lepton colour with family 0 and keeps the up-type direction: it exchanges the direction selected by the first heavy operator with the direction selected by the second. If the 11 label is the grand algebra M_2(H) + M_4(C) (lemma 28) and the 9 label colour plus electromagnetism, M exchanges the two steps of 11 -> 10 -> 9. This needs a finite geometry in which the generation space is C^4 with an algebra exchangeable with M_4(C), and a Dirac operator compatible with the exchange.
3. **Four families and eleven dimensions.** The 128 of the four families is the chiral spinor of SO(16) E8; under Spin(9) SO(16), embedded through its spinor, it decomposes as 44 + 84, the graviton and the three-form of eleven-dimensional supergravity (Pengpan and Ramond, hep-th/9808190; Kostant's cubic Dirac operator on F4/Spin(9); recalled). The two families of 116 are not Spin(9) subrepresentations; whether the exchange M has a Spin(9) meaning is open.
4. **NCG.** In the finite geometry with three generations the exchange is not an automorphism (Proposition 4.2). A Pati-Salam finite geometry with C^4_colour ⊗ C^4_family, where the flip of the two factors exchanges the σ field with the top Yukawa direction, is the candidate in which it is (PLAN, step 5).

What remains open

- A finite spectral triple, or a Hořava-Witten configuration, with four families in which M is a symmetry and the fourth family is decoupled in a controlled way, so that the three-family statement of lemma 21 follows as a limit.
- Whether the fourth family can be made heavy by the same mechanism as the top (lemma 21 counts only one heavy up-type quark).

Part 11D (06_main_point_heterotic_route, referee pending)

9 and 11 as limits of the modern spectral triple: the target, and a route through the heterotic strings

Part of the 11D line of research (README, PLAN, task B). Attempted by Claude (Anthropic), 4 Nov 2026, prompted by two remarks in the steer of the same day: that SO(32) against E8 × E8 resembles non-chiral against chiral, and that the SO(32) of lemma 21 was mainly scalars, a way in which supersymmetry could be hidden in the finite geometry as scalar composites. Proposition 6.1 is proved; everything else is a proposal, and the string-theory facts used are recalled, not reread (marked). Referee pending.

1. The target

The modern spectral triple is the real even finite triple of KO-dimension six with order zero and order one, fermion doubling, dim H_F = 96 (three generations with _R), A_F = C + H + M_3(C), or its grand algebra M_2(H) + M_4(C), and D_F with Yukawa and Majorana blocks. The main point of the project asks for one family of spectral triples with towers, T_s, and three regimes, each described by the three records of canonical §1 (geometry, fermions, physics):

regime	geometry	fermions	physics
finite v (label 10)	towerless window at the weak scale	the modern finite triple, with J, gamma, D_F exact on the window	the Standard Model
v -> 0 (label 11)	eleven resolved dimensions along a joint family (01)	chirality on walls or defects	massless W, Z with finite couplings

regime	geometry	fermions	physics
v -> infinity (label 9)	the interval collapses	vectorlike; KO-5 or KO-7 by the sign s of lemma 06	colour and electromagnetism

What exists: the exact window for D_F, J and gamma on an interval parent (lemma 22) and its circle form (lemma 25, Proposition 1.1); the joint family that opens the tower while keeping the window (Astra audit, 01); the grand algebra from the seven-dimensional parent X, conditionally (lemma 28); the KO parity of the odd endpoint (lemma 06, T2c); eleven-dimensional vacua with the Standard Model isometry (02, 03); placements of H_F in E8 (04, 05). What is missing: one family doing all three; the six internal dimensions resolved in the same family; and an action selecting the path (Proposition 1.5).

2. The heterotic web has the shape 9, 10, 11

Recalled facts (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209 and hep-th/9603142; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; to be checked before use):

- **9:** the heterotic strings on a circle. SO(32) and E8 × E8 are T-dual there, joined through the moduli space of Wilson lines; the theory is non-chiral, nine being odd. Its type I' form is an interval with two orientifold walls (O8) and sixteen D8-branes, whose positions are the Wilson-line eigenvalues.
- **10:** the two heterotic strings, both chiral, distinct.
- **11:** M-theory on an interval (Hořava-Witten), with an E8 multiplet on each wall, chiral at the walls; the length of the interval grows with the string coupling.

This is the pattern of hypothesis H: vectorlike at 9, chiral and KO-6 at 10, chiral walls at 11, with an interval whose two ends are the two sheets. The steer's remark places SO(32) at the non-chiral end and E8 × E8 at the chiral end; in this web the two meet at 9.

3. Dictionary (proposal)

corpus	heterotic web
the two sheets of the finite geometry	the two walls of the interval (O8 planes in type I', E8 walls in Hořava-Witten)
D_F as a constant connection on the circle (lemma 25)	a Wilson line on the heterotic circle, T-dual to D8 positions on the interval
the chiral interval as the antiperiodic circle (Proposition 1.1)	the interval as the orbifold S^1/Z_2 of that circle
lemma 21: removing the top keeps 5/8 and 8/3 and leaves SO(30) x SO(2) SO(32)	fifteen D8-branes at one wall and the sixteenth displaced: the gauge group on the D8-branes is SO(30) x U(1)
the 30 = (5, 3) + (5bar, 3bar): five light quarks in three colours	the Chan-Paton labels of the fifteen-brane stack, with images
the SO(30) adjoint, Λ^2(30), which contains the counts of squarks and sleptons (lemma 21)	open strings with both ends on the stack: bilinear in the Chan-Paton labels, so composite in exactly the steer's sense, and in the adjoint scalars of a theory with sixteen supercharges
the (30, 2) of SO(32) SO(30) x SO(2), states that the top cannot form	strings stretched between the stack and the displaced brane, with masses set by the displacement
v -> 0 opens the eleventh dimension (Proposal 1.3)	strong coupling: the interval of Hořava-Witten grows

4. Proposition 6.1 (the top channel as an SO(32) Wilson line)

Statement. On a circle of radius R, take the SO(32) Wilson line W = 1_30 ⊕ R(2 pi a) (rotation by the angle 2 pi a in the last two directions). The adjoint 496 decomposes under SO(30) x SO(2) as (435, 1)*0 + (1, 1)*0 + (30, 2), where (30, 2) = (30)*{+1} + (30)*{-1}; the Kaluza-Klein masses are n/R on the first two pieces and |n ± a|/R on (30)_{±1}, n in Z. With l = pi R and theta = pi a this is the spectrum (n pi ± theta)/l of a channel of lemma 22 (Theorem 22.1) and of the circle form of lemma 25, with mass parameter m = theta/l = a/R; so the top channel of lemma 22 and the SO(32)-charged states of this Wilson line have the same free spectrum when m_t = a/R.

Proof. The Wilson line commutes with SO(30) x SO(2) and acts on the vector 2 of SO(2) by e^{±2 pi i a} in the complex basis; a field of charge q on the circle with holonomy e^{2 pi i q a} has momenta (n + q a)/R. Substitution of l = pi R and theta = pi a. ■

Prior work (search of 4 Nov 2026). The content of Proposition 6.1 is known. A field of charge q on a circle with holonomy e^{2 pi i q a} has momenta (n + q a)/R: the Hosotani mechanism (Y. Hosotani, Phys. Lett. B 126 (1983) 309), which also gives the fermion masses of gauge-Higgs unification, including top-driven models (for example arXiv:0806.0480). An SO(32) Wilson line with one nonzero eigenvalue breaks SO(32) to SO(30) x U(1) and is T-dual to fifteen D8-branes at one point and the sixteenth displaced (Bergman, Gaberdiel and Lifschytz, hep-th/9711098; Kaplunovsky, Sonnenschein, Theisen and Yankielowicz, hep-th/0108135; seen at abstract level). The equality with the channel spectrum of lemma 22 is the circle form of lemma 25. Not found: the assignment of the top to the displaced D8-brane and of the five light quarks in three colours to the Chan-Paton labels of the fifteen-brane stack. That assignment is the proposal of section 3, and it has a gap: in the heterotic string the (30, ±1) are gauge bosons and gauginos, while the top is a quark, so the two spectra belong to fields of different spin unless the composite reading of section 5 supplies the map.

This identifies spectra, which is all it does. The physical identification needs the fermion content (in the heterotic string the (30, 2) are vectors and gauginos), the chirality at 10, and the map from the Yukawa block to the Wilson line beyond the top channel.

5. Supersymmetry hidden as composite scalars (the steer's second remark)

In the type I' form the adjoint scalars of the D8 stack are open strings with two Chan-Paton ends, so they are bilinear in the labels that the dictionary assigns to the five light quarks in three colours: the composites of lemma 21 (the 15 diquarks and 25 mesons of five flavours recorded there, and the SO(30) adjoint containing the squark and slepton counts). The theory on the D8-branes has sixteen supercharges, which pair these adjoint scalars with gauge fields and gauginos. A finite spectral triple in which the scalars of the inner fluctuations are such bilinears would carry the supersymmetry in hidden form. The test is to compute the inner fluctuations of a finite triple whose algebra is the Chan-Paton algebra of the stack, and to compare them with Λ^2(30).

6. What would make this a theorem, and the first steps

1. **One family, three regimes.** Realize T_s as the type I' interval with D8-branes in the configuration 15 + 1, with the corpus parent (lemma 22) as its low-energy window: the window must reproduce the modern D_F, J and gamma (lemma 22 does this for any finite data), the tower must open along a joint family as the interval grows (strong coupling, 11), and the collapse must give the non-chiral nine-dimensional theory (9).
2. **The algebra.** The Chan-Paton algebras of orientifold stacks are of the types O(n), Sp(n), U(n); the Connes algebra C + H + M_3(C) has the form of stacks of one brane, two branes with a symplectic projection, and three branes. Decide whether the modern algebra is the Chan-Paton algebra of a stack configuration on the interval (for the colour factor, lemma 21 puts colour inside SO(30), on the same stack as the light flavours).
3. **The six dimensions and chirality.** Ten-dimensional chirality and the KO-6 class must come from the internal six dimensions (02: CP^2 x S^2 with fluxes, or a Calabi-Yau with an SU(4) bundle as in Proposal 5.5).

4. **The path.** Proposition 1.5 requires a second scale; in the heterotic web the dilaton gradient of type I' (whose slope depends on the D8 positions) is a candidate; Proposition 9.1 of 09 writes it as a trace of a function of |D_F|.

References for this file

Recalled, not reread: J. Polchinski and E. Witten, "Evidence for heterotic-type I string duality", hep-th/9510169; P. Hořava and E. Witten, hep-th/9510209 and hep-th/9603142; O. Bergman, M. Gaberdiel and G. Lifschytz, "String creation and heterotic-type I' duality", hep-th/9711098; L. Ibáñez, F. Marchesano and R. Rabadán, "Getting just the Standard Model at intersecting branes", hep-th/0105155, for stacks with symplectic projections.

Part 11D (07_dualities_9_11, referee pending)

Dualities between nine and eleven dimensions, adapted to the spectral triple

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 7.1 to 7.3 are proved here by hand; the string-theory statements of section 1 are recalled, not reread; the readings are proposals. Referee pending.

1. The dualities (recalled)

1. **M-theory on T^2 and type IIB on S^1** (Schwarz, hep-th/9508143; Aspinwall, hep-th/9508154). The modulus tau of the torus is the IIB coupling and SL(2, Z) is IIB S-duality; the IIB radius scales as a negative power of the area A of the torus. M2-branes wrapped on T^2 are IIB momentum states, and momenta on T^2 are (p, q)-strings wound on S^1. Eleven dimensions at A -> infinity, ten (IIB) at A -> 0, nine in between.
2. **Sixteen supercharges, rank 17** (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209). The two heterotic strings on S^1 are T-dual through Wilson lines; their type I' form is an interval with two orientifold planes O8 and sixteen D8-branes whose positions are the Wilson-line eigenvalues; at strong coupling, M-theory on an interval times a circle, with E8 walls.
3. **Sixteen supercharges, lower rank.** The CHL string (rank 9) and M-theory on the Möbius strip; a rank-1 component related to the Klein bottle (Aharony, Komargodski and Patir, hep-th/0702195); further components with discrete theta angles (Montero and Parra de Freitas, arXiv:2209.03361).
4. **Light-cone M-theory.** On a light-like circle with momentum N, M-theory is the U(N) quantum mechanics of nine Hermitian matrices (Banks, Fischler, Shenker and Susskind, hep-th/9610043; Susskind). On T^2 with a light-like component of C_3, the torus becomes noncommutative and SL(2, Z) acts by Morita equivalence (Connes, Douglas and Schwarz, hep-th/9711162).

2. What each one offers the finite geometry

duality	9	11	the finite triple
1	the nine-dimensional theory	large torus	the lowest Landau level of a torus with flux p: algebra M_p(C), J from charge conjugation with doubling (Proposition 7.3); towers = Landau levels; p = wrapped M2-branes = IIB momentum

duality	9	11	the finite triple
2	type I' interval, or heterotic on S^1	Hořava-Witten on S^1	two sheets = two orientifold walls; D_F = Wilson lines = D8 positions (Propositions 6.1, 7.1, 7.2); A_F from Chan-Paton algebras; J from the orientifold
3	rank-reduced nine-dimensional theories	M-theory on non-orientable surfaces	J as an orientation reversal; the reflection with antiperiodicity of Proposition 1.1 is the Möbius-type ingredient
4	nine transverse matrices, Spin(9)	large N	A_F as the commutant of block-diagonal matrices: stacks of 1, 2, 3 give C + M_2(C) + M_3(C); H_F in off-diagonal blocks; D_F = Γ^i ad(X^i), masses = separations

Duality 2 is the one with an exact meeting point with the corpus, through the angle of the top channel.

3. Proposition 7.1 (the angle of a channel lives on an interval)

Statement. In the parent of lemma 22, a channel with singular value σ of the Yukawa block and angle = l σ has the signed spectrum {(n pi +)/l, (n pi −)/l : n in Z} (Theorem 22.1). As a set, this depends only on the class of in R modulo the group generated by -> − and -> + pi, whose fundamental domain is [0, pi/2]. At the endpoints:

1. = 0: the spectrum is n pi/l with a zero mode of multiplicity two (periodic);
2. = pi/2: the spectrum is (n + 1/2) pi/l, each value twice (antiperiodic), two copies of the spectrum of the chiral interval of Proposition 1.1; it is the limit X -> infinity of the wall data X = tan( ), where the lowest eigenspace is degenerate, and in lemma 23 the point of antipodal gradings, where M_W = M_Z = 0 at fixed l with the fermion at pi/(2l).

Proof. The set is invariant under -> − (it exchanges the two progressions) and -> + pi (it shifts n). The orbit of meets [0, pi/2] in one point. The endpoint values are substitutions; for -> pi/2 the wall conditions Q_R(l) = X U_R(l), U_L(l) = −X^* Q_L(l) tend to U_R(l) = Q_L(l) = 0. The statements at = pi/2 are those of lemma 23 (Theorem 23.6 and the reading after Proposition 23.7). ■

So the parameter space of one channel of the finite Dirac operator, for a given parent, is an interval with two distinguished ends, and lemma 22's restriction l ||D_F|| < pi/2 is the choice of the fundamental domain.

4. Proposition 7.2 (the far end is the displaced D8-brane on the other wall)

Statement. Put a = /pi in [0, 1/2]. By Proposition 6.1 the channel spectrum is that of the states (30)*{±1} of SO(32) on a circle of radius R = l/pi with the Wilson line W_a = 1_30 ⊕ R(2 pi a). The interval [0, 1/2] is the moduli interval of one Wilson-line eigenvalue modulo a -> −a and a -> a + 1; the unbroken group is SO(32) at a = 0 and SO(30) x SO(2) for 0 < a <= 1/2; at a = 1/2 the holonomy is −1 on the plane of SO(2), W{1/2}* = diag(1_30, −1_2), and the (30)_{±1} tower is (n + 1/2)/R, the antiperiodic tower of Proposition 1.1. The one-loop minimum of lemma 24, Theorem 9, = pi/2, is the point a = 1/2.

Proof. Proposition 6.1 and Proposition 7.1 with = pi a; the commutant of W_a in SO(32) for 0 < a < 1/2 is SO(30) x SO(2), and at a = 1/2 its identity component is still SO(30) x SO(2), since only two eigenvalues are −1. Lemma 24, Theorem 9, item 1, places the minimum at t = sin^2 = 1. ■

Reading (proposal). In type I' the Wilson line W_a is fifteen D8-branes, with images, on one orientifold plane and the sixteenth at distance proportional to a from it; a = 1/2 puts it on the other orientifold plane (recalled: hep-th/9711098, hep-th/0108135). The dictionary of file 06 then reads: the top angle is the position of the top's D8-brane on the type I' interval, the two ends of the fundamental domain are the two orientifold walls, and the one-loop vacuum of the flat family of lemmas 23 and 24 carries the top brane to the far wall, where the tower of the stack-brane strings is antiperiodic. In supersymmetric type I' the D8 positions are flat directions; the potential of lemma 24 belongs to the non-supersymmetric flat family, the setting in which a force on the brane is expected.

Under the strong-coupling duality of item 2 of section 1, the D8 position is a Wilson-line eigenvalue of the heterotic string on S^1, and in eleven dimensions an E8 holonomy around the circle on a Hořava-Witten wall. So the same datum, the top angle, is a brane position at 9 and a holonomy at 11.

5. Proposition 7.3 (a torus with flux, doubled, has the colour form)

Statement. Let T^2 be a flat torus of area A with a spin structure, L a line bundle of degree p >= 1 with a connection of constant curvature, and Lbar = L^{-1} with the conjugate connection. On H = L^2(S ⊗ L) ⊕ L^2(S ⊗ Lbar) put D = D_L ⊕ D_Lbar, gamma = gamma_S ⊕ gamma_S, and J(,) = (C , −C), where C is the charge conjugation of the spin structure (antiunitary, C^2 = −1, C D = D C, C gamma_S = −gamma_S C, mapping S ⊗ Lbar to S ⊗ L). Let $C^{\infty}(T^2)$ act by multiplication on the first summand only. Then:

1. J^2 = 1, J D = D J and J gamma = −gamma J, the signs of KO-dimension six, and the algebra satisfies order zero;
2. ker D = H^0(L) ⊕ H^1(Lbar) C^p ⊕ conj(C^p), with gamma = +1 on the first summand and −1 on the second; the other eigenvalues are the Landau levels, of order (p/A)^{1/2}, which go to infinity as A -> 0 at fixed p;
3. the compressions of the functions to the kernel generate M_p(C) on C^p ⊕ 0, so the window is the finite triple (M_p(C), C^p ⊕ conj(C^p), 0, J, gamma), with KO-6 signs and order zero: M_p(C) acts on the first summand and, through J, by its opposite on the second.

Proof. (1) J^2(,) = (C(−C), −C(C)) = (,) since C^2 = −1; JD = DJ and J gamma = −gamma J follow from the properties of C; order zero holds because the algebra acts on the first summand and J b J^{-1} on the second. (2) On a torus the spinor bundle is Λ^{0,*} up to a flat factor, so ker D_L = H^0(L) ⊕ H^1(L) = C^p ⊕ 0 by Riemann-Roch in genus one, and ker D_Lbar = 0 ⊕ H^1(Lbar) conj(C^p) by Serre duality; the Landau levels of a constant field of total flux p on area A scale as (p/A)^{1/2}. (3) The compression of a character e_{m,n} to H^0(L) is a nonzero multiple of a clock-and-shift Weyl operator U^m V^n with UV = e^{2 pi i/p} VU (Berezin-Toeplitz quantization of the torus; recalled), and these operators span M_p(C). ■

The construction is the torus analogue of lemma 16 (a doubled parent with restricted algebra and a J covering the flip; Example 16.9 for the twistor case), and its window at p = 3 has the form of the colour sector of the Standard Model triple: M_3(C) on the quarks and its opposite on the antiquarks. Berezin-Toeplitz quantization of the torus is known; the KO-6 doubling is the corpus device.

Reading for duality 1 (proposal). The area A -> 0 at fixed flux keeps the window and sends the tower to infinity, the towerless limit; A -> infinity brings the Landau levels down. In M-theory on T^2, flux and wrapping number of M2-branes on T^2 are IIB momentum states, so the dimension p of the window counts IIB momentum on the nine-dimensional circle, and the SL(2, Z) of the torus acts on the family by Morita equivalence of the noncommutative tori of Connes, Douglas and Schwarz when a C_3 background is on. Whether p = 3 for colour has a reason in this picture is open.

6. A remark on duality 4 (known)

For U(N) matrices X^i (i = 1, …, 9) that are block-diagonal, X^i = ⊕a x_a^i 1{n_a} with distinct points x_a in R^9, the commutant of all X^i is ⊕a M{n_a}(C), and the fermion operator Γ^i ad(X^i) on the adjoint, with the sixteen-component spinor of Spin(9), vanishes on the diagonal blocks and has eigenvalues ±|x_a − x_b|, each with multiplicity eight times the block size n_a n_b, on the off-diagonal blocks (a, b). With stacks of sizes 1, 2, 3 the commutant is C + M_2(C) + M_3(C), the complex form of A_F, the off-diagonal blocks are bifundamentals as in the Krajewski diagrams of the finite geometry, and the masses are the distances between stacks; coincident stacks restore U(N), the analogue of massless W and Z, and large N is the eleven-dimensional limit. A quaternionic factor H needs an orientifold projection on the two-stack, and chirality needs more than the nine Hermitian matrices.

6b. The three labels in type I' variables (proposal)

Hypothesis H has two parameters for the top channel: the interval length l and the angle = l m_t in [0, pi/2] (Proposition 7.1). In the type I' reading of Proposition 7.2 they are the length of the interval and the position of the top's D8-brane as a fraction of it. Then:

- **label 9:** = pi/2 at fixed l, the brane on the far orientifold wall: the top at the tower scale pi/(2l) (lemma 22, Corollary 22.9: W, Z and top at the tower scale in the 10 -> 9 family), the antiperiodic tower, and the one-loop minimum of the flat family (lemma 24);
- **label 10:** 0 < < pi/2, the brane between the walls, a window below the tower;
- **label 11:** -> 0 together with l -> infinity, the joint family of 01 (_epsilon ~ epsilon^{1 − alpha}): the brane returns to the stack, SO(32) is restored on that channel, and the interval opens.

In this reading the 9 and 11 endpoints are the two ends of the brane's path, the far wall and the stack, and the second scale required by Proposition 1.5 must fix how l grows as the brane returns. The flat one-loop potential pushes the brane to the far wall (label 9); reaching label 11 needs a term that pulls it back to the stack while the interval grows.

7. What this adds, and what remains open

Added: the angle of the top channel lives on an interval whose ends are the two orientifold walls of type I' (7.1, 7.2); the corpus's one-loop vacuum puts the top brane on the far wall, the SO(32) Wilson line diag(1_30, −1_2); the same datum is an E8 holonomy in the eleven-dimensional frame; a torus with flux gives the colour form of the finite triple with KO-6 signs (7.3).

Open: - A nine-dimensional spectral triple with sixteen D8 positions as the eigenvalues of D_F on the type I' interval, and its strong-coupling limit with E8 walls; the configuration 15 + 1 lies at the SO(32) end, while E8 needs seven D8-branes at an orientifold plane with divergent coupling. - The quaternionic factor H from a torus with even flux and a real structure of quaternionic type, and the full A_F from several tori or from a rank-three bundle with flux. - The lower-rank nine-dimensional components (duality 3), whose J-like orientation reversal may supply the real structure of the finite triple.

Part 11D (08_orientifold_sign, referee pending)

The sign of lemma 06 as an orientifold type, and E8 walls from the configuration 15 + 1

Part of the 11D line of research (README, PLAN, task C, next step of 07). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 8.1 is proved (it adds a commutant statement to lemma 06); Proposition 8.2 is proved from Proposition 7.1; Remark 8.3 rests on recalled string-theory rules and on the dictionary of 06; readings are proposals. Referee pending.

1. Setting

In type I' the two ends of the interval are orientifold planes. A stack of n Chan-Paton labels (k D8-branes with their images, n = 2k) on an orientifold plane of type O^- carries SO(n), and on one of type O^+ carries USp(n) (recalled; the physical O8 planes of the SO(32) web are O8^-). Lemma 06 factors a KO-6 finite triple into an odd one through a grading-reversing unitary involution U, commuting with the algebra and with D, with U J = s J U; s = −1 gives KO-5 (label 9) and s = +1 gives KO-7 (label 11), and the antiunitary j induced on H_+ satisfies j^2 = s and commutes with d^2 = D^2|{H+} (lemma 06, proof of Result 2 and the check of the electrodynamics data). STRATEGY T2c: with a realistic seesaw sector only s = +1 occurs.

2. Proposition 8.1 (the factorization sign fixes orthogonal or symplectic symmetry)

Statement. Let E be an eigenspace of d^2 in H_+, of dimension n. The unitaries of E that commute with j form O(n) if s = +1, and USp(n) if s = −1, in which case n is even.

Proof. j preserves E because it commutes with d^2. An antiunitary with j^2 = +1 is a real structure: E = E_R ⊗ C with E_R its fixed points, and a unitary commutes with j exactly when it preserves E_R, which gives O(n). An antiunitary with j^2 = −1 is a quaternionic structure: E is a right H-module of quaternionic dimension n/2, and the unitaries commuting with j form USp(n). ■

The even dimension for s = −1 is lemma 06's own statement (Kramers degeneracy of the singular values). The proposition adds the symmetry group.

Reading (proposal). The groups are those of Chan-Paton labels on an orientifold plane: s = +1 with O^- (SO), s = −1 with O^+ (USp). So the two odd factorizations of the modern triple, which carry the labels 11 and 9, have the form of the two kinds of orientifold planes. The seesaw sector selects s = +1 (T2c), which is the O^- type of the SO(32) web used in 06 and 07: the physical neutrino sector and the type I' walls of the dictionary agree in type.

3. Proposition 8.2 (wall channels come with doubled multiplicity)

Statement. In the parent of lemma 22, k channels at a common angle contribute, at the eigenvalue /l of the window, an eigenspace of dimension k if 0 < < pi/2 and of dimension 2k at the ends = 0 and = pi/2.

Proof. By Proposition 7.1 each channel has the spectrum {(n pi ±)/l}; the value /l occurs once (n = 0, sign +) for 0 < < pi/2, and twice at the ends: at = 0 as the zero mode of multiplicity two, at = pi/2 as (0 + pi/2)/l = (pi − pi/2)/l. ■

Reading (proposal). k D8-branes in the bulk carry k Chan-Paton labels and k D8-branes on an orientifold plane carry 2k, with their images; the parent reproduces this count, with the ends of the angle interval as the walls. With Proposition 8.1, a stack of channels at a wall has a symmetry of type SO(2k) or USp(2k), as a stack of D8-branes on an O8^- or O8^+ plane. The rank-reduced nine-dimensional theory with one O8^- and one O8^+ (duality 3 of file 07) would then be a parent whose two walls carry opposite signs s: one sheet of type 11 and the other of type 9.

4. Remark 8.3 (E8 walls from 15 + 1; conditional)

Conditions: the dictionary of 06 (fifteen D8-branes for the five light quarks in three colours, the sixteenth for the top) and the recalled rule that an O8^- plane with n <= 7 D8-branes on it, at infinite coupling, carries E_{n+1} (E_4 = SU(5), E_5 = SO(10), E_6, E_7, E_8), while n >= 8 admits no such point (Seiberg, hep-th/9608111; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; recalled).

1. Dilaton bookkeeping: the dilaton is e^{phi} = z^{−5/6} with z piecewise linear along the interval (conformal coordinate; Polchinski and Witten, hep-th/9510169) and slope proportional to 8 − n_left, where n_left counts the D8-branes to the left (O8^- carrying −8 units), so the coupling grows toward the wall with fewer branes; this reproduces the constant dilaton of 8 + 8, the SU(2) at the empty wall of the SO(32) point (the self-dual radius of the heterotic string) and 7 + 7 for E8 x E8. In the

configuration 15 + 1 the wall with fifteen branes has no E point (15 > 7), and the coupling can diverge only at the far wall. With the top's brane on the far wall (the label-9 configuration of Proposition 7.2, n = 1) the far wall enhances to E_2 = SU(2) x U(1), the electroweak group; with the top's brane in the bulk it is empty and enhances to E_1 = SU(2). The light-quark wall keeps SO(30).

2. If branes move in colour triples (a light flavour is three D8-branes), a wall holds 3j branes, or 3j + 1 with the top's brane. The reachable wall groups are E_1, E_2 (the top's brane alone), E_4 = SU(5) (one flavour), E_5 = SO(10) (one flavour and the top's brane), E_7 (two flavours) and E_8 (two flavours and the top's brane); E_6 (five branes) is not reachable.
3. Hence an E8 wall needs the top's brane together with two light flavours, and two E8 walls are impossible with one top brane: E8 x E8 requires breaking a colour triple.

In this reading the top is what completes E8 at a wall (7 = 2 x 3 + 1), and the Hořava-Witten endpoint E8 x E8 is not available with colour-respecting moves; E8 x E7 is: with seven branes at one wall, six at the other and the remaining flavour at x_b, the slopes are proportional to +1 on (0, x_b) and −2 on (x_b, L), so z vanishes at both walls when x_b = 2L/3. The E_n series also contains E_3 = SU(3) x SU(2) (two branes at a wall), which colour triples do not reach.

Reading (proposal). At strong coupling the wall that holds the top's brane in the label-9 configuration carries SU(2) x U(1), and the wall of the five light quarks in three colours carries SO(30). The electroweak group then appears at the top's wall, where the M-theory circle opens locally, and the quarks' flavour group at the other wall.

5. What remains open

- The bulk symmetry: k D8-branes in the bulk carry U(k), while Proposition 8.1 gives O or USp on every eigenspace of d^2; the relation between the window's degenerate levels in the bulk and U(k) needs the parent's own structures, not only j.
- The dilaton: a spectral-triple counterpart of the type I' dilaton profile (a coupling varying along the interval, with slope set by the brane charge to one side) and the infinite-coupling points of Remark 8.3. Proposition 9.1 of 09 writes the profile as a trace of a function of |D_F|.
- The physical status of E8 x E7 with colour triples, and whether the top's brane at an E8 wall has a counterpart in the finite geometry.

Part 11D (09_dilaton_spectral_and_so32, referee pending)

The type I' dilaton as a spectral function, the SO(32) of the EPJC paper, and two more dualities

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026, after the steer of the same day (the coincident-brane route is overused; the type I' dilaton and the colour-flavour idea are worth pursuing; compare with the SO(32) of the EPJC paper). Proposition 9.1 is proved from the recalled type I' slope rule; Proposition 9.2 is proved from representation theory and the recalled orientifold projections; Proposition 9.3 is proved (linear algebra and Connes's sign table); the comparison of section 1 is read from the arXiv text of the paper through a summarizing fetch; the rest are proposals. Referee pending.

Correction (same day): the first version stated the slope rule for e^{−phi}. The piecewise-linear function is z, with e^{phi} = z^{−5/6}; Proposition 9.1 holds as stated with z in place of e^{−phi}, and infinite coupling is z = 0.

1. The SO(32) of A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397)

That paper uses the adjoint of SO(32) to classify the scalars of a supersymmetric Standard Model with three generations, as composites: pairs of quarks as colour triplets and quark-antiquark pairs as singlets and octets, with three extra scalars of charge +4/3. Its pattern is SO(2n) SO(n_1) x SU(n_2) x SU(3) x U(1) with 2n = n_1 + 6 n_2, which for SO(32) and n_2 = 5 gives SO(2) x SU(5) x SU(3) x U(1): the 30 =

6 n_2 are the five light quarks u, c, d, s, b ("turtles") in three colours with conjugates, and the top is the non-participating "elephant". Its explicit route also passes through SO(32) SU(16) x U(1), with 496 = 1 + 120 + 120bar + 255 (quarks from the 120s, leptons from the 255), and it mentions open strings, orientifolds and an SU(9) for M2 and M5.

Coincidence. The chain SO(32) SO(2) x SO(30) SO(2) x SU(15) x U(1) SO(2) x SU(5) x SU(3) x U(1), the 30 as five light quarks in three colours, and the top as the factor SO(2), are the same as lemma 21 and the dictionary of 06. Lemma 21 records the branching 496 -> (435, 1) + (1, 1) + (30, 2) and leaves SO(32) unexplored; the EPJC paper explores it through composites, and the files 06 to 09 through type I' branes. What the 11D files add: the type I' realization of the SO(2) as the top's D8-brane and of the 30 as the Chan-Paton labels of fifteen D8-branes on an O8^- wall (06); the top angle as the brane's position and the one-loop vacuum on the far wall (Propositions 7.1, 7.2); the orientifold type O^- matched by the seesaw sign (Proposition 8.1); the far wall's enhancement to SU(2) x U(1) at infinite coupling (Remark 8.3); the dilaton (Proposition 9.1); the diquark counts as strings between the stack and its image, which need the O^- type (Proposition 9.2); and the real-structure condition under which these counts arise, as connection components along the interval and not as finite inner fluctuations (Proposition 9.3). In type I' language the paper's other chain, SU(16) x U(1), is the configuration with all sixteen D8-branes coincident in the bulk, and the chain with SO(2) x SO(30) is the configuration with fifteen on a wall and one apart; the second is the one the corpus selects. The SU(9) of M2 and M5 in the paper is the grading of e8 used in 04.

2. Proposition 9.1 (the type I' coupling function is a trace of a function of the finite Dirac operator)

Setting. Type I' on an interval [0, L], in the conformal coordinate x, with sixteen D8-branes at x_1, ..., x_16 and O8^- planes at the ends. Recalled rule (Polchinski and Witten, hep-th/9510169): the dilaton is e^{phi} = z^{−5/6} for a function z that is continuous and piecewise linear in x, with slope c (8 − N(x)), where N(x) is the number of D8-branes to the left of x and c > 0; the coupling diverges where z vanishes. Dictionary (06, 07): x_k = (2L/pi) _k with _k = l m_k the angles of sixteen channels of the finite Dirac operator (fifteen light-quark channels with colour, and the top).

Statement.

1. z(x) = z(0) + c Σ_k min(x/2, x_k − x/2) for x in [0, L], so z is z(0) plus c Tr g_x(l |D_F|) over the sixteen channels, with g_x( ) = min(x/2, (2L/pi) − x/2): a trace of a function of the finite Dirac operator, as the spectral action is.
2. At the far wall, z(L) = z(0) − c (8L − Σ_k x_k).
3. Infinite coupling at the far wall (z(L) = 0) holds exactly when z(0) = (2cL/pi)(4 pi − l Σ_k m_k): the coupling at the light-quark wall is fixed by the sum of the sixteen channel masses, the trace of |D_F| on these channels.

Proof. *0^x N(y) dy = Σ_k (x − x_k)+*, so z(x) = z(0) + c (8x − Σ_k (x − x_k)+) = *z(0) + c Σ_k (x/2 − (x − x_k)+)*, and x/2 − (x − x_k)_+ = min(x/2, x_k − x/2). Item 2 is x = L with Σ_k (L − x_k) = 16L − Σ x_k. Item 3 sets z(L) = 0 and substitutes x_k = (2L/pi) l m_k. ■

Reading (proposal). With the light quarks near = 0, Σ_k m_k m_t and the condition is z(0) (2cL/pi)(4 pi − _t): 7cL at label 9 (_t = pi/2) and 8cL at label 11 (_t -> 0). The coupling therefore varies along the interval on a scale set by the spectrum of D_F, which is the kind of second scale that Proposition 1.5 requires to select a joint family; whether the type I' coupling can play that role in the corpus's flat family is the next question. Since the slope decreases with x, z is concave, so z > 0 on [0, L) once z(0) > 0 and z(L) = 0; positivity of z(0) gives l Σ_k m_k < 4 pi, a bound on the trace of |D_F| in units of the tower scale, met when only the top is heavy (_t <= pi/2).

3. Proposition 9.2 (diquarks as strings between the stack and its image)

Setting. V = C^5 ⊗ C^3, five light flavours in three colours, with Q = 2/3 on u, c and Q = −1/3 on d, s, b; the 30 of lemma 21 is V ⊕ Vbar. Fifteen D8-branes on an O8 plane carry the Chan-Paton space 30

(branes and images), with group SO(30) on an O8^- and USp(30) on an O8^+, and U(15) when the stack is moved off the plane (recalled).

Statement.

1. Under U(15), the adjoint of SO(30) is $\Lambda^2 V \oplus \Lambda^2 \mathrm{Vbar} \oplus (V \otimes \mathrm{Vbar})$ and the adjoint of USp(30) is $S^2 V \oplus S^2 \mathrm{Vbar} \oplus (V \otimes \mathrm{Vbar})$. The summands $\Lambda^2 V$ or $S^2 V$ and their conjugates are the strings between the stack and its image; $V \otimes \mathrm{Vbar}$ are the strings with both ends on the stack.
2. $\Lambda^2 V = (S^2 5, \mathrm{3bar}) \oplus (\Lambda^2 5, 6)$ and $S^2 V = (S^2 5, 6) \oplus (\Lambda^2 5, \mathrm{3bar})$.
3. In the colour antitriplet, $\Lambda^2 V$ has the charge counts ($-2/3$: 6, $+1/3$: 6, $+4/3$: 3) of lemma 21, and $S^2 V$ has ($-2/3$: 3, $+1/3$: 6, $+4/3$: 1). So the antisquark counts of three generations recorded in lemma 21 (6 of charge $-2/3$ and 6 of charge $+1/3$ in the 3bar) occur among the stack-image strings of an O8^- plane and not among those of an O8^+ plane.
4. With the angles _k = l m_k of 06, a string from channel i to the image of channel j has the spectrum (n pi ± (_i + _j))/l, and a string from channel i to channel j has (n pi ± (_i − _j))/l, n in Z.
5. A single brane has no string to its own image on an O8^- plane (Λ^2 of a line vanishes) and one on an O8^+ plane.

Proof. 1: 30 = V ⊕ Vbar as a U(15) module, $\Lambda^2(V \oplus \mathrm{Vbar}) = \Lambda^2 V \oplus \Lambda^2 \mathrm{Vbar} \oplus (V \otimes \mathrm{Vbar})$ and likewise for S^2; the orientifold projection keeps the antisymmetric part for SO and the symmetric part for USp, and the mixed summand $V \otimes \mathrm{Vbar}$ is the U(15) adjoint. 2: $\Lambda^2(A \otimes B) = (S^2 A \otimes \Lambda^2 B) \oplus (\Lambda^2 A \otimes S^2 B)$ and $S^2(A \otimes B) = (S^2 A \otimes S^2 B) \oplus (\Lambda^2 A \otimes \Lambda^2 B)$, with $\Lambda^2 3 = \mathrm{3bar}$. 3: split 5 = 2 ⊕ 3 by charge; $S^2 5 = S^2 2 \oplus (2 \otimes 3) \oplus S^2 3$ has dimensions 3, 6, 6 with charges 4/3, 1/3, $-2/3$, and $\Lambda^2 5 = \Lambda^2 2 \oplus (2 \otimes 3) \oplus \Lambda^2 3$ has dimensions 1, 6, 3 with the same charges. 4: Proposition 6.1 with charges (+1, +1) and (+1, −1) under U(1)_i x U(1)_j, whose holonomy is e^{2 pi i (a_i ± a_j)}, and = pi a. 5: dimensions. ■

Reading (proposal). The diquark counts of lemma 21 and of the EPJC paper are the stack-image strings of an O8^- plane, the type that Proposition 8.1 associates with the factorization sign s = +1 selected by the seesaw sector (T2c); the O8^+ alternative gives the wrong number of up-type antisquarks. So the seesaw sign and the scalar counts select the same orientifold type. Two features follow from the projection. Diquark masses go with the sum _i + _j and meson masses with the difference _i − _j, so when the stack leaves the wall the diquarks become massive and the mesons stay massless: the diquarks are tied to the orientifold. And the top's single brane has no string to its own image, while its strings to the stack, the (30, 2), have masses near m_t (Proposition 6.1); in the EPJC paper the top's exclusion is an input, and here it follows from the O8^- projection with one brane for the top.

4. Proposition 9.3 (the real structure decides between the two diquark counts)

Setting. E = V ⊕ Vbar with V as in Proposition 9.2, an antiunitary J on E that maps V onto Vbar and Vbar onto V (as particles and antiparticles are exchanged), $J^2 =$, and a Hermitian X on E with JX = ' XJ, the condition that inner fluctuations obey in a KO-dimension with signs (, '). Let B be the block of X from Vbar to V, an element of $V \otimes V$.

Statement.

1. B is symmetric (B in $S^2 V$) if ' = +1 and antisymmetric (B in $\Lambda^2 V$) if ' = −1.
2. In the KO-dimensions 5, 6, 7 of the finite part for the labels 9, 10, 11, (, ') = (−1, −1), (+1, +1), (+1, +1), so ' = +1: the diquark block of every J-compatible fluctuation lies in $S^2 V$, whose colour antitriplet ($\Lambda^2 5$, 3bar) has the counts ($-2/3$: 3, $+1/3$: 6, $+4/3$: 1).
3. For a connection on a circle whose real structure is complex conjugation (KO-dimension 1, (, ') = (+1, −1)), the block lies in $\Lambda^2 V$, whose colour antitriplet ($S^2 5$, 3bar) has the counts ($-2/3$: 6, $+1/3$: 6, $+4/3$: 3) of lemma 21. The SO(32) Wilson line of Proposition 6.1 is of this type.

Proof. Choose a basis of V and let K: V -> Vbar be coordinatewise conjugation; J(v, w) = (K^{-1} w, K v) has $J^2 =$. Write X = [[A, B], [B*, D]]. The component of JX = ' XJ that maps V to V gives K^{-1} B* = ' B K, so B = ' K^{-1} B* K^{-1} = ' B^T in coordinates. Under the U(15)-equivariant identification of Hom(Vbar, V) with $V \otimes V$, symmetric matrices form $S^2 V$ and antisymmetric ones $\Lambda^2 V$. Items 2 and 3 use Connes's sign table, = (+, +, −, −, −, −, +, +) and ' = (+, −, +, +, +, −, +, +) for KO-dimensions

0 to 7, and the counts of Proposition 9.2. On the circle, D = −i d/d and J = conjugation give JD = −DJ, and a Hermitian connection A with JA = −AJ is i times a real antisymmetric matrix, an element of so(n), where the generator of R(2 pi a) lies. ■

Item 2 is a negative statement with the following scope: the blocks of J-compatible Hermitian operators between particles and antiparticles, for one multiplet V, in the three labels. It is the finite-geometry form of the symmetry of Majorana mass matrices, and of the Fermi-statistics constraint on a Lorentz-scalar bilinear _i _j of one Weyl multiplet without derivatives (recalled), under which the (15, 3bar) vanishes. Known escapes: item 3 (connection components along a circle or an interval), blocks between two distinct multiplets, which J does not constrain, and derivative couplings.

Reading (proposal). The counts of lemma 21 and of the EPJC paper, (6, 6, 3), belong to the antisymmetric block. In the corpus's setting this block arises as the component of an SO(30) connection along the circle or interval: the Wilson-line moduli of the type I' stack, whose stack-image part is Proposition 9.2. The finite triple alone, in any of the three labels, gives the symmetric block and the counts (3, 6, 1). So the composite scalars with the counts of lemma 21 point to the interval parent of lemma 22 and its gauge field, and the selection of O8^- in Proposition 9.2 concerns these connection components. This narrows the spin gap of Propositions 6.1 and 9.2: the Λ^2V states are scalars as circle components of a nine-dimensional gauge field, and they come with that field's vector and gaugino partners.

5. Two more dualities between 9 and 11 (recalled; proposals)

1. **E-type walls and del Pezzo surfaces.** The E_{n+1} symmetry of an O8^- plane with n D8-branes at infinite coupling is the symmetry of the five-dimensional theory on a D4-brane probe, which M-theory realizes on a Calabi-Yau threefold with a shrinking del Pezzo surface dP_{n+1} (Seiberg, hep-th/9608111; Morrison and Seiberg, hep-th/9609070; Intriligator, Morrison and Seiberg, hep-th/9702198). So the far wall with the top's brane (n = 1, E_2 = SU(2) x U(1)) has an eleven-dimensional description by a local dP_2, and the empty far wall (E_1 = SU(2)) by dP_1 or P^1 x P^1; the light-quark wall (n = 15) has none. The Witten base CP^2 x CP^1 of 02 is built from the same pieces (CP^2 = dP_0, and P^1 factors).
2. **Colour-flavour exchange as a T-duality.** In the heterotic string on S^1, the Weyl group of the gauge group acts on the Wilson lines as part of the T-duality group O(17, 1; Z). The colour-family transposition M of 05 is an element of the Weyl group of E8, so it is a T-duality of the nine-dimensional E8 x E8 string, and through Hořava-Witten a symmetry of the E8 holonomy on a wall in eleven dimensions. As a Weyl element it acts as a global gauge transformation, so by itself it relabels gauge-equivalent vacua; it exchanges colour and family physically only relative to a structure that tells them apart, such as the placement of H_F in 05. The four-family requirement of Propositions 5.1 and 5.4 remains.

6. Open

- Whether the type I' coupling, as a spectral function of D_F (Proposition 9.1), supplies the second scale of Proposition 1.5 in the flat family of lemmas 23 and 24.
- The local dP_2 of the top's wall and its relation to the Witten base CP^2 x CP^1.
- Spin: in type I' the strings of Proposition 9.2 belong to nine-dimensional vector multiplets, with one real scalar each (the relative position), the same gap as in Proposition 6.1. By Proposition 9.3 a finite triple in the labels 9 to 11 gives S^2V, so the Λ^2V scalars must come from the connection along the interval; the next computation is the inner fluctuations of the parent of lemma 22 with an SO(30) algebra, and whether its interval component carries Λ^2V with the counts of lemma 21.
- Whether the (15, 3bar) of the EPJC paper is meant as bilinears of two distinct multiplets (for example left-handed quarks and conjugates of right-handed ones arranged into one 30), which Proposition 9.3 does not constrain.

Part 11D (10_dilaton_and_second_scale, referee pending)

The type I' dilaton against the second scale of Proposition 1.5

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, after the merge of branch 11D into master, to answer the last item left open in 09: whether the dilaton can be the second scale that Proposition 1.5 requires. Sources read from the arXiv LaTeX: J. Polchinski and E. Witten, hep-th/9510169, section 3; O. Bergman, M. Gaberdiel and G. Lifschytz, hep-th/9711098, section 3. Numbers in Python: tools/d11_needed_numbers.py. The Codex quota is spent, so no Terra referee is available.

Outcome

1. **In the scope stated below, the dilaton gradient does not supply the second scale.** Along any path that opens the circle, its relative variation falls as 6/R^2 (Proposition 10.5). It enters as a power of alpha'/l^2, and that can matter only if the leading Casimir coefficient a_0 vanishes (Proposition 10.6).
2. **a_0 is now computed**, which was the open item of 01: 8 pi^2 a_0 = (3 zeta(5)/16)(4 n_f - 3 n_v - n_s - 5 n_g). For the corpus content it is positive, so at v = 0 the interval runs to large length, in the direction H wants. For one generation the count is 64 - 63 = 1, a near-cancellation.
3. **The dictionary of 06 to 09 needs three corrections** (Propositions 10.1 to 10.4). The map from angles to positions is nonlinear. The tower of Proposition 6.1 is a type I' winding state, so the length l of H and the type I' length are T-dual, and l -> infinity shrinks the type I' interval at fixed heterotic coupling. And the configuration 15 + 1 has a Romans mass in the bulk, which bounds length times coupling, so it has no eleven-dimensional regime.
4. **A scale-free action can still separate the window**, with a fixed small ratio instead of a vanishing one (Remark 10.7), which amends the last sentence of Proposition 1.5.

The direction: the second scale must enter through a bulk energy depending on the electroweak order parameter, through a cancellation a_0 = 0, or through a fixed shape with an exponentially small ratio, as in lemma 29. These targets are stated in Proposition 10.6 and Remark 10.7. None is derived.

The dictionary against the source

Polchinski and Witten give, in conformal gauge, g = Omega^2(x) eta with Omega = C z^{-1/6}, e^phi = z^{-5/6}, and z = 3 C (B mu_8 - nu_0 x)/sqrt 2, with nu_0 = (N(x) - 8) mu_8 piecewise constant. This confirms the slope rule recalled in 08 and 09. We write z for this function up to its constant, with slope proportional to 8 - N(x), and x in [0, L].

Proposition 10.1 (the map from angles to positions). Put W(x) = int_0^x z^{-1/3} dy. By Polchinski-Witten, equations (mass) and (wilmap), the Wilson-line eigenvalue of a brane at x_k is lambda_k = W(x_k) / (2 W(L)), that is

```
theta_k = pi lambda_k = (pi/2) W(x_k) / W(L),
```

and with L = 2 pi, in their units alpha' = 2, R^2 = (3/4) int z^{1/3} dx / (int z^{-1/3} dx)^2. So the linear dictionary x_k = (2L/pi) theta_k of Proposition 9.1 holds exactly when z is constant (8 + 8), and asymptotically for large R.

Proof. The string stretched from the wall to x_k has mass (1/(2 pi alpha')) int_0^{x_k} Omega^2 dx, since its area element is Omega^2 dx^0 dx^9, and Omega^2 is proportional to z^{-1/3}. A string winding from 0 to L and back has the mass 1/R. The ratio gives lambda_k. The formula for R is their equation (rad), reduced as in their equation (dil). ■

Consequence for Proposition 9.1: its items 1 and 2 hold as statements about z in terms of the positions x_k. Reading them as a trace of a function of |D_F| needs the map above, which depends on the whole configuration, so the coupling function is a trace only implicitly. Item 3 changes accordingly.

Proposition 10.2 (15 + 1 with infinite coupling at the far wall). Take fifteen branes at x = 0 and one at x_t, with z(L) = 0, and put rho = (z(x_t)/z(0))^{2/3} in [0, 1]. Then, with alpha' = 2,

```
theta_t = (pi/2) 8 (1 - rho) / (8 - rho),        R_*^2 = 14 (8 - rho^2) / (8 - rho)^2.
```

So R_* runs from sqrt 2, for the top's brane on the stack, to sqrt 7 / 2, for the top's brane on the far wall, which are the values (1/2)|n - 8|^{1/2} of Polchinski and Witten at n = 16 and 15. In units of the string length, R_*/sqrt(alpha') lies in [0.935, 1]. In this chamber, R >= R_*.

Proof. Integrate z^{1/3} and z^{-1/3} on the two segments, with slopes -7 and -8, and use z(x_t) = 8 (L - x_t). ■ Python agrees at x_t/L = 0.5, with theta_t = 0.5859 and R^2 = 1.9652, against 0.7854 for the linear dictionary.

So infinite coupling sits at the *smallest* radius of the chamber, at the string scale, and large R is large B, the weakly coupled region. Polchinski and Witten say this directly: "When B is large, the coupling is small everywhere and the Type I' theory is weakly coupled; in this range the heterotic radius R is large".

Proposition 10.3 (T-dual lengths). The tower of Proposition 6.1, n/R, is a type I' winding tower: "a heterotic Kaluza-Klein state which is dual to a Type I' winding state" (Polchinski and Witten). With constant dilaton, the proper length L_p of the type I' interval and the radius satisfy L_p R_I = pi alpha' on the type I side, so the length l = pi R of H and the type I' length satisfy l L_p = pi^2 alpha'.

Hence the walls of H, at distance l, and the O8 planes, at distance L_p, coincide only at the self-dual length. In particular, the reading "the interval opens" in section 6b of 07 has to be restated: at fixed heterotic coupling, l -> infinity shrinks the type I' interval, and it decompactifies the SO(32) circle.

The endpoint of that limit is the ten-dimensional SO(32) theory (heterotic or type I). This is ten dimensions. The eleven-dimensional limit of the rank-17 moduli space is M-theory on an interval times a circle, with E8 x E8 at the walls, which by Remark 8.3 is out of reach with colour-respecting moves. So label 11 of section 6b is not reached by theta -> 0, l -> infinity in the chamber 15 + 1.

Proposition 10.4 (Romans mass, length and coupling). On a segment with N branes to the left, the proper distance is linear in the inverse coupling:

```
L_p = (C_L l_s / |8 - N|) (e^{-phi_1} - e^{-phi_2}),     C_L = 8 pi / 5,
```

with C_L from the standard normalization of the Romans mass, F_0 = (8 - N)/(2 pi l_s). So in the configuration 15 + 1, g(0) L_p <= C_L l_s / 7 up to the small change on the last segment. Near an infinite-coupling wall the coupling is g ~ C_L l_s / (|8 - N| s) at proper distance s, so g » 1 needs s « l_s, and the eleven-dimensional Planck length g^{1/3} l_s exceeds s there. Therefore no region of the 15 + 1 background has both strong coupling and a length resolved in eleven-dimensional units.

Proof. Omega dx = C z^{-1/6} dz / (a C (8 - N)) integrates to a multiple of z^{5/6} = e^{-phi}. ■ This is the known fact that massive type IIA has no eleven-dimensional lift, seen here in the variables of the dictionary. It holds wherever N != 8.

The far-wall electroweak group. Remark 8.3 places SU(2) x U(1) at the far wall at infinite coupling. By Bergman, Gaberdiel and Lifschytz, the states that become massless there are D0-branes at the wall, heterotic winding states. Combining their equations (windingmass) and (D0mass), with A = (B + 2 pi N')^{2/3}, b = B^{2/3} and N' = 8 - n,

```
M_D0 R = (N'/2) b / (A - b),
```

which vanishes at B = 0, where R = R_*, and grows like R^2 in units alpha' = 2. So if the W and Z are these states, they are massless at the string-scale radius, with the tower fixed, and they become heavier as R grows: M_W l is not constant, and it rises with l. This is the fixed-tower path of the audit (its section 2.2), and on the rest of the branch the W moves the wrong way for hypothesis H. The dictionary can place the electroweak group at the far wall, or the opening of H on the circle of Proposition 6.1, and it cannot do both on one path.

The test

Proposition 10.5 (the gradient is a string-scale effect). With L = 2 pi and alpha' = 2, for large R, R^2 ~ 3 B / (8 pi), and the relative variation of z across the interval is at most 16 pi / B ~ 6 / R^2. So

along a joint family, R_eps ~ eps^{-alpha}, this variation falls as eps^{2 alpha}, the map of Proposition 10.1 linearizes, and the dilaton is constant in the limit.

Proof. z is B plus a function of size at most 8 L = 16 pi. Insert in the formula for R. ■

So the corrections to any potential from the gradient are powers of alpha'/l^2. In the supersymmetric type I' background there is no potential at all: D8 positions, B and C are flat directions. A potential needs broken supersymmetry, as in the flat family of lemmas 23 and 24.

The coefficient a_0. Write 8 pi^2 l^4 V = _0 + 3 F(2u) + (3/2) F((16/5) u) - n_f F(t), the potential of lemma 24 with its theta-independent part restored, where n_f counts twice the Dirac species at frequency one (lemma 26). The relative terms vanish at theta = 0, and each tower at theta = 0 is the half of a periodic circle tower:

```
_0 = (3 zeta(5) / 16) (4 n_D' - 3 n_v - n_s - 5 n_g),
```

with n_D' = 2 N_D counting twice all Dirac species, of any frequency (for frequency zero the tower is nothing but the theta = 0 tower), n_v the five-dimensional gauge generators, n_s the real bulk scalars with integer towers, and n_g = 1 if the five-dimensional graviton is included. This assumes that the bulk vacuum energy and the wall tensions are tuned away, since they carry the terms of order Lambda^5 l and Lambda^4.

Proof. A real degree of freedom on a periodic circle of length 2 l has V = -3 zeta(5)/(4 pi^2 (2 l)^4), and the interval tower n >= 0 gives half of this, -3 zeta(5)/(128 pi^2 l^4). In the normalization of lemma 24 this is -(3/16) zeta(5) per real degree of freedom. For a fermion channel the sign is opposite, with four degrees of freedom per unit of n_f. The check: the shift to theta = pi/2 multiplies a tower by -15/16, so the difference is (31/16)(3/4) zeta(5) = 93 zeta(5)/64 = F(1), which is the value computed in Proposition 1.4. ■

For three generations, with 24 Dirac species and the 21 generators of su(4) + su(2) + su(2), the bracket is 192 - 63 = 129 before gravity and bulk scalars. So a_0 > 0: at v = 0 the length runs to infinity, in the direction of H, and the first half of the modulus target of the audit, a r^{-p} with a > 0 and p = 4, is present. For one generation the bracket is 64 - 63 = 1. We record this count without a reading: equal numbers alone prove nothing, and including the graviton changes it to -4.

Proposition 10.6 (when the gradient can be the second scale). Let V = l^-4 [G_0(theta) + (alpha'/l^2) G_1(theta) + …], theta = l m, with G_0(0) = a_0 and G_0 = a_0 + b theta^2 + … near 0.

1. If a_0 != 0, then at small m the leading balance is the scale-free one of Proposition 1.5: alpha = 1 or a runaway. The term in alpha'/l^2 changes neither.
2. If a_0 = 0, G_1(0) = a_1 > 0 and b < 0, there is a minimum at l^4 = 3 a_1 alpha' / (|b| m^2), so alpha = 1/2: 1/l is the geometric mean of m and the string scale, up to constants, and m « 1/l « 1/l_s.
3. If supersymmetry is exact at theta = 0, then V = 0 at theta = 0 at all orders, a_1 = 0, and V = m^2 l^-2 H(theta, alpha'/l^2). Then l is fixed by H independently of m, or runs away, and no joint family results.

So in the scope of a potential analytic in alpha'/l^2 around the flat family, with corpus content, the gradient is not the second scale. Item 2 would need a Bose-Fermi balance at theta = 0 that is not a supersymmetry at theta = 0, with n_s = 129 real bulk scalars for three generations.

Proof. Differentiate. ■

The flat family with the angle tied. For the top alone, n_f = 6: b is proportional to 54/5 - 6 > 0, and G > 0 on [0, pi/2] (Python: minimum 16.04, with _0 = 25.08). So V has no stationary point with theta < pi/2, and l runs until theta leaves the fundamental domain of Proposition 7.1, where the top joins the tower. For a universal mass at three generations, n_f = 48, G changes sign and there is one stable stationary point, theta_* = 0.877, with M_W/m = 0.876 (Python). For a universal mass at one generation, n_f = 16, theta_* = 0.202 and M_W/m = 1.394 (Python), small because _0 is small. Top-only content therefore has no vacuum in the window in this scope; adding a graviton does not change the classification.

A localized channel. For a channel with bulk mass M = -mu and c = mu l, the spectrum of lemma 9 gives the one-loop term

```
8 pi^2 l^4 V_loc = -2 int_0^infinity x^3 log(1 + ((w + c)/(w - c)) exp(-2 w)) dx,    w = sqrt(x^2 +
↪  c^2),
```

from the secular function cosh(w) - (c/w) sinh(w). At c = 0 it is -(45/64) zeta(5) = -0.729, and for large c it is -4 c^3 exp(-2 c) up to corrections. In between it has a minimum, of depth -1.32, near c = 1 (Python, on a grid). So one fermion channel, by itself, prefers c near 1, where the light pair merges with the tower (lemma 9.7). A large c_*, which is what a separated window needs, must come from other fields, such as the bosonic tower of the scalar that gives mu.

Other second scales

Remark 10.7 (amendment of Proposition 1.5, last sentence). On the scale-free path, m l is a fixed number. Proposition 1.5 says that on it the finite window never separates from the tower. That holds for theta_* of order one. If theta_* is small, or if m is exponentially small in a stabilized shape parameter, the ratio m/Delta is a small fixed number. On the disc bundle of lemma 29, fix M and let R grow: m = lambda_M / R and Delta = delta / R both tend to zero, with lambda_M / delta exponentially small, and the count at fixed energy has seven-dimensional growth. So a potential R^-4 G(M), with M stabilized at a large M_* and G(M_*) > 0, *realizes the hierarchy of lemma 30 with alpha = 1 and a fixed ratio, with no second scale. The cost is that R then has no vacuum at finite value. The physical point at finite v needs one more term, and the stabilization at large M_* is a target*, since one fermion channel prefers c near 1.

Remark 10.8 (what would work). The ansatz V = a r^-p + b eps^2 r^q needs q > 0. On the flat family the angle is a holonomy (Corollary 22.2), which no local bulk operator sees: at large l its effect is the Casimir term, q = -2. So the electroweak order parameter must enter a local bulk energy: for q = 1, V = a/l^4 + b v^2 M^3 l gives l^5 proportional to 1/v^2, so alpha = 2/5, if the bulk energy vanishes at v = 0. That requires the order parameter to be a bulk field with a bulk vacuum energy, and it departs from D_F as a pure boundary condition. Whether the spectral action provides this is open. In the heat cutoff, the dependence on D_F is exponentially small at large l Lambda, by Poisson summation over the tower, so that action has no such term at this order.

Corrections to the earlier files

- 09, Proposition 9.1 and its reading: subject to Proposition 10.1.
- 07, sections 4 and 6b: Proposition 7.2 stands as a statement on the circle; the reading "the interval opens" at label 11 is corrected by Proposition 10.3.
- 08, Remark 8.3 and its reading: the far-wall group is reached at R = R_*, with the tower at the string scale, by Propositions 10.2 and 10.4 and the section on the far wall. There is also a tension of labels: Proposition 8.1 pairs label 9 with O^+, and section 6b of 07 puts label 9 on a far O8^- wall. These are two different meanings of the label, the KO sign and the position of the top's brane.
- 06, section 2: "9, 10, 11" mixes KO labels with metric dimensions. In the string web, nine dimensions is finite R, and R -> 0 is T-dual to R -> infinity, whereas in H the finite interior is labelled 10 and l -> 0 is labelled 9.
- 05, Proposition 5.4: the conclusion is correct on rerun (Python, exact integers, all 192 configurations). The sentence "In every case S - N has d = 19 and d = 20 only" is wrong for the misaligned configurations, which also show d = 13 and d = 17. The sets of values of d differ in every configuration, and that is what the proof needs.

Scope and escapes

Proposition 10.6 assumes a potential with a regular expansion in alpha'/l^2, around the flat family with its towers. It concerns the dilaton *gradient*. The following lie outside it: a dilaton potential from string tree level, in non-supersymmetric strings or with tadpoles, for which a term in positive powers of 1/g could grow with l at fixed v through the law of Proposition 10.4; a different frame relation between l and the four-dimensional Planck mass; wall-localized dynamics; and the other chambers of the moduli space, where

8 + 8 removes the Romans mass and E8 x E8 is available. The string-theory statements are read from the two sources for type I' and recalled for the moduli-space limits; only the first have been checked against the text. By AGENTS.md, a negative result needs an independent check by another model, which this one has not had from a referee of the usual kind.

Part 11D (11_so30_fluctuations, referee pending)

Inner fluctuations of the interval parent with an SO(30) algebra

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, to answer the question left in 09: whether the gauge field along the interval of lemma 22 carries the antisymmetric block with the counts of lemma 21, given that Proposition 9.3 excludes it from the finite triple alone. Numbers in Python: tools/d11_needed_numbers.py. The Codex quota is spent, so no Terra referee is available.

Outcome

1. **Yes, under a condition on the walls.** With a real algebra M_30(R) on the light block, the fluctuation along the interval is an so(30) connection, and its block between particles and antiparticles lies in Λ^2 V, with the counts (6, 6, 3) in the colour antitriplet (Proposition 11.2). Massless scalars in exactly Λ^2 V + conjugate appear when both walls reduce the algebra to M_15(C), the commutant of quark number (Theorem 11.3).
2. **In lemma 22's own parent the massless scalars are absent.** Its algebra is free at x = 0, so the interval component has no zero mode, for any algebra (Theorem 11.3, item 2).
3. **A generator has a massless vector or a massless scalar, never both, on the interval.** So the whole of Λ^2(30) of lemma 21 cannot be massless scalars with colour unbroken. The antisquark counts appear as scalars, while the slepton part (24, 1) + (1, 1) appears as vectors (Theorem 11.3, item 4).
4. **The finite window sees a symmetric shadow.** The connection acts on the window through its commutator with the masses, B_ij -> (m_i - m_j) B_ij, which maps the (S^2 5, 3bar) of Λ^2 V onto the (Λ^2 5, 3bar) of S^2 V, with counts (3, 6, 1) (Proposition 11.4). This reconciles Propositions 9.2 and 9.3. The kernel is the same-flavour diquarks.
5. **SO(30) has a precise meaning in the finite geometry:** it is the commutant of the real structure j = J U of lemma 06 on the massless light block, for the sign s = +1, and USp(30) for s = -1 (Proposition 11.1).

Setting

Let V = C^5 ⊗ C^3, with Q = 2/3 on u, c and Q = -1/3 on d, s, b. In the finite Hilbert space of the five light quarks, with colour, the chirality gamma = +1 subspace is

```
H_+ = {q_L} ⊕ {conj(q_R)} = V ⊕ Vbar,
```

of dimension 30: the left-handed Weyl fields of fifteen Dirac fermions. H_- = {q_R} ⊕ {conj(q_L)}, and J_F exchanges H_+ and H_-, with c = J_F restricted to H_+, as in lemma 22. Let U be a unitary, grading-reversing and J-related as in lemma 06, with U q_L = q_R and U conj(q_R) = s conj(q_L), so U J_F = s J_F U. Then j = U^-1 c is an antiunitary on H_+, with j^2 = s and j q_L = conj(q_R).

The interval parent is that of lemma 22: D_X = E d/dx on L^2([0, l]; C^2) ⊗ (H_+ ⊕ H_-), with Q in H_+, U-components in H_-, and the wall conditions Q_R(0) = U_L(0) = 0, Q_R(l) = X U_R(l), U_L(l) = -X^* Q_L(l).

The block

Proposition 11.1 (the commutant of j). Let s = +1. In the basis V ⊕ Vbar with j(v, w) = (K^-1 w, K v), the skew-Hermitian a with j a j^-1 = a form so(30), and

```
a = [[alpha, B], [conj(B), conj(alpha)]],    alpha^* = -alpha,   B^T = -B,
```

so so(30) = u(15) ⊕ Λ^2 V, with u(15) the commutant of the complex structure I = (i, -i), the quark number. The Hermitian generator H = i a satisfies j H j^-1 = -H. For s = -1 the same computation gives usp(30) = u(15) ⊕ S^2 V.

Proof. Linear algebra, with Proposition 8.1 for the groups. For s = +1, j a j^-1 = a gives the block form, and a^* = -a gives alpha^* = -alpha and B^* = -conj(B), that is B^T = -B. Then j (i a) j^-1 = -i a. ■

So the SO(30) of lemma 21 is the symmetry of the massless block for the sign s = +1, the sign that Proposition 8.1 and STRATEGY T2c associate with label 11 and with the seesaw; and the counts of lemma 21 select the same sign:

	colour antitriplet	charges (-2/3, +1/3, +4/3)
s = +1, Λ^2 V	(S^2 5, 3bar), and (Λ^2 5, 6)	(6, 6, 3)
s = -1, S^2 V	(Λ^2 5, 3bar), and (S^2 5, 6)	(3, 6, 1)

Fluctuations along the interval

Take on the light block D_F = 0, so X = 0, and let C_0, C_l be subalgebras of A = M_30(R), the real elements of End H_+. Let

```
A_l = {f in C^infinity([0, l], A) : f(0) in C_0, f(l) in C_l},    pi(f) = f on Q, 0 on the H_-
↪ components.
```

Proposition 11.2. Assume 1 lies in C_0 and C_l.

1. pi(f) preserves the domain, for every choice of C_0, C_l. Order zero and order one hold on this block.
2. The self-adjoint one-forms sum pi(f_i) [D_X, pi(g_i)] are exactly E a_x on the Q components, with a_x in C^infinity([0, l], so(30)), with no condition at the walls. The fluctuated operator is

   ```
   D_A = E (d/dx + a_x ⊕ U a_x U^-1).
   ```

3. J-hat A J-hat^-1 = E (U a_x U^-1 on H_-), with sign +, as KO-dimension six requires. The block of a_x between Vbar and V is antisymmetric. The two statements agree since E is antisymmetric: the block of the Hermitian operator E a on C^2 ⊗ E is symmetric, as Proposition 9.3, item 2 requires, with B antisymmetric and the factor E carrying the other sign. So the interval component evades Proposition 9.3 through the escape in its item 3.

For D_F != 0 on this block, order one fails: [[D_F, pi(a)], J pi(b) J^-1] = -U b U^-1 M a on H_+, which vanishes for all a, b only if M = 0. So this algebra describes the massless block, and the masses need a treatment without order one, as in lemma 19.

Proof. (1) The conditions Q_R = 0 and U_L = 0 at the walls are preserved by any operator acting on the H_+ index. [D_X, pi(g)] = E g' on Q, and J-hat pi(b) J-hat^-1 acts on the H_- components only, so both commutators vanish. (2) Take f = T, g = x 1 in A_l, which gives any T; self-adjointness of E T, with E skew, requires T skew. (3) J-hat E = E J-hat (direct from the formula for J-hat in Theorem 22.1), and c a c^-1 = U j a j^-1 U^-1 = U a U^-1. ■

Theorem 11.3 (zero modes of the interval component). Let k_0, k_l be the skew parts of C_0, C_l, and assume a five-dimensional Yang-Mills action for the fluctuation on a flat product with M^4, as in Proposition 22.8.

1. The gauge group is {g : I -> O(30), g(0) in K_0, g(l) in K_l}. The holonomy of a_x transforms as P -> g(l) P g(0)^-1, so the moduli are the double coset K_l SO(30) / K_0, and the massless four-dimensional scalars from a_x form

   ```
   (k_0 + k_l)^  in so(30).
   ```

2. In lemma 22's algebra, C_0 = A, so k_0 = so(30), and there are no such scalars, for any algebra of this kind.

3. If C_0 = C_l = M_15(C), the massless scalars are exactly Λ^2 V ⊕ Λ^2 Vbar, with colour-antitriplet counts (6, 6, 3), and the massless vectors are u(15).

4. A generator in k_0 ∩ k_l has a massless vector and no massless scalar; a generator in (k_0 + k_l)^ has a massless scalar and no massless vector; all others have neither. The towers are n pi/l in the first two cases, and (n - 1/2) pi/l in the third, for a generator in exactly one of k_0, k_l.

5. With colour unbroken at both walls, the (1, 8) of Λ^2(30) is a vector, so Λ^2(30) cannot consist entirely of massless scalars. In item 3, the (24, 8) + (24, 1) + (1, 8) + (1, 1) of lemma 21 are vectors, and the scalars are (15, 3bar) + (10, 6) + conjugates.

Proof. (1) A gauge parameter xi(x) with xi(0) in k_0, xi(l) in k_l shifts int a_x by xi(l) - xi(0), so the linear moduli are so(30)/(k_0 + k_l), and a constant a_x in the orthogonal complement is a zero mode of the quadratic form. (2), (3): substitution, with Proposition 11.1. (4) A_mu has Neumann or Dirichlet conditions at each wall according to membership in k_wall, and a_x the opposite, so the mode functions are cos(n pi x/l), sin(n pi x/l) or their half-integer versions. (5) Item 4, and the branching of lemma 21. ■

Note that the choice in item 3 is an input: the domain does not force the wall algebra to M_15(C), since by Proposition 11.2, item 1, every f preserves it at D_F = 0. It says that the walls conserve quark number and the bulk does not. On a circle, without walls, every generator has both a massless vector and a massless scalar, and the whole of Λ^2(30) appears, which is the nine-dimensional adjoint scalar used in 09. A wall condition, or the reflection-even algebra of Proposition 1.1, removes half. So the two readings differ: for the stack off the O8 plane, with holonomy I, diquarks are massive and mesons have massless scalars; on the interval with parity Ad(I), diquarks have massless scalars and mesons have none.

The shadow in the finite window

Proposition 11.4. Let M = U Lambda with Lambda real for j, symmetric, and let a_tot = a ⊕ U a U^-1 be constant. Then:

1. [D_F, a_tot] has the off-diagonal blocks U S and S U^-1, with S = [Lambda, a] real and symmetric, and [D_F, [D_F, a_tot]] = [Lambda, S] ⊕ U [Lambda, S] U^-1.

2. By Theorem 22.4, the compression of E a_tot to the window is -l^-1 int_0^l sin(x ad_{D_F})(a_tot) dx. For Lambda = diag(m_i) on flavours, on both V and Vbar, the block B_ij goes to

   ```
   - ((1 - cos(l (m_i - m_j))) / (l (m_i - m_j))) B_ij,    which is -(l/2)(m_i - m_j) B_ij to
   ↪ leading order,
   ```

   as a change of Lambda. The result is symmetric, as Proposition 9.3 requires of a finite J-compatible fluctuation.

3. For distinct masses, B -> (m_i - m_j) B_ij maps (S^2 5, 3bar) in Λ^2 V onto (Λ^2 5, 3bar) in S^2 V, with kernel the five same-flavour diquarks, and maps (Λ^2 5, 6) into (S^2 5, 6). In charges, (6, 6, 3) - (3, 0, 2) = (3, 6, 1). If k_l is the commutant of Lambda in so(30), then with k_0 = u(15) the massless scalars of Theorem 11.3 are the same off-diagonal blocks.

4. For a universal mass, Lambda = m, the compression vanishes: S = 0, and M = m U is invariant under SO(30). Fifteen Dirac fermions of equal mass are thirty Majorana fermions of equal mass.

Proof. (1) Direct multiplication, using a_tot on H_- equal to U a U^-1. S is real and symmetric since a is real and antisymmetric and Lambda real and symmetric. (2) Item 1 gives sin(x ad)(a_tot) = (sin(x delta)/delta) ad(a_tot) on B_ij, with delta = m_i - m_j; integrate. (3) In the colour antitriplet, B is symmetric in flavour, and (m_i - m_j) B_ij is antisymmetric in flavour, hence total symmetric. Colour sextets are analogous. (4) S = 0. ■ Checked in Python on a small model, with V = C^3: reality, symmetry, the blocks of item 1 and their squares.

Reading. The commutator [D_F, a] is the form of the Higgs field in Connes' finite geometry, and E a_x is a Kaluza-Klein component along the interval. Theorem 22.4 (3), Q_l([D_X, f]) = [D_F, Q_l(f)], says that the second compresses to the first. So in this parent, a finite Higgs one-form is the shadow of a gauge field

along the interval, and it inherits the symmetry type of the finite triple by the commutator with masses. A generator commuting with the masses has no shadow. This explains the difference between the counts (6, 6, 3) of the parent and (3, 6, 1) of the finite triple, and it replaces the alternative O8^- against O8^+ of Proposition 9.2, for these blocks, by a parent and its window.

What remains open

- Whether M_15(C) at the walls can be derived from a boundary condition instead of chosen, for example by a wall term that conserves quark number.
- The treatment of nonzero masses, where order one fails on the block: the quadratic terms of lemma 19 for this algebra.
- The sleptons: on an interval the (24, 1) + (1, 1) are vectors. Scalars in these representations need components along further internal directions, such as the base of 02.
- The colour sextets (10, 6) come with the antitriplets, and nothing removes them in this construction.
- The relation to the top: the (30, 2) of SO(32), with the second factor as the top's SO(2).

Part L (literature check and record of searches)

Literature check for claims A-I

Written by Claude Opus 5.5 on 3 November 2026, as recommendation 1 of evaluation2026-11-03.md. Method: web searches (standard mode, extended mode where results were thin), arXiv abstract pages and full texts, Crossref and INSPIRE metadata for every journal reference below. Where an equation or section number is given, it was read in the full text (arXiv version unless stated). Items that could not be checked are marked **unverified** and collected at the end.

Verdicts: **KNOWN** (same statement in print), **PARTIAL** (a close or weaker version in print; the difference is stated), **NOT FOUND** (no source located after the searches listed).

A. Connes-Lott as the minimal spectral truncation of a chiral interval

Claim (lemma 05, Part 3; lemma 09, Theorem 9.6). The minimal nonzero grading-preserving spectral truncation of the Dirac operator on [0, l] with R(0) = 0, L(l) = 0 is unitarily equivalent to the Connes-Lott two-point triple (C + C, D = [[0, m], [m, 0]]) with |m| = pi/(2l); the compressed operator system P C([0, l]) P equals C + C; the truncated end states are at distance 2l/pi. The deconstructed chain of Part 1 has Connes-Lott as its N = 2 member.

Verdict: PARTIAL. The interpretation of the two-point space as a discrete or Kaluza-Klein fibre, the truncation machinery, the deconstruction picture and the lowest mixed-boundary mass pi/(2l) are all in print. We found no source that identifies the Connes-Lott triple as a spectral truncation of an interval Dirac operator with chiral boundary conditions, computes the compression P C([0, l]) P for such an interval, or relates |m| to the interval length.

Interpretation of the two-point space (in print).

- A. Connes and J. Lott, *Particle models and noncommutative geometry*, Nucl. Phys. B Proc. Suppl. 18B (1991) 29-47, doi:10.1016/0920-5632(91)90120-4; author's copy at math.berkeley.edu/~lott/partmodels.pdf. Section I: "The corresponding geometric space can be thought of as a spacetime of Kaluza-Klein type, where the fiber is formed of a two-point set." Section II.B, item (9): for the two-point space, d(p, p') = L, where L^{-2} is the largest eigenvalue of M*M. Section III (p. 36): the character space of the product is two copies of Z, "Thus the picture is of two copies of Z separated by a distance L. (As one loses information in passing to the character space, this picture should not be taken too literally.)", with (d_Z(p, p')^2 + L^2)^{1/2} <= d(p, p') <= d_Z(p, p') + L.

- N. A. Viet and K. C. Wali, *Noncommutative geometry and a discretized version of Kaluza-Klein theory with a finite field content*, Int. J. Mod. Phys. A 11 (1996) 533-551, doi:10.1142/S0217751X96000249, arXiv:hep-th/9412220; and *A discretized version of Kaluza-Klein theory with torsion and massive fields*, Int. J. Mod. Phys. A 11 (1996) 2403-2418, doi:10.1142/S0217751X96001206, arXiv:hep-th/9508032. The second abstract: "We consider an internal space of two discrete points in the fifth dimension of the Kaluza-Klein theory." The "discrete fifth dimension" reading is therefore explicit in print since 1995.
- A. Dimakis and F. Müller-Hoissen, *Discrete differential calculus, graphs, topologies and gauge theory*, J. Math. Phys. 35 (1994) 6703-6735, doi:10.1063/1.530638, arXiv:hep-th/9404112. Abstract: the formalism "suggests that the [Connes-Lott two-point space] be regarded as an approximation of a manifold and thus opens a way to relate models with an 'internal' discrete space (à la Connes et al.) to models of dimensionally reduced gauge fields". Section III, Example 4: the universal differential algebra on two points has a Hasse diagram approximating the topology of S^2, linked to Manton's six-dimensional Yang-Mills reduction on S^2. Section VII: the Connes-Lott models should be understood "as approximations of higher-dimensional gauge theory models … The details have still to be worked out, however." This is the closest statement of the programme of lemma 05. The approximation there is topological (posets and Hasse diagrams) and the internal space is S^2; the passages we read contain no Dirac operator, boundary condition or mass-length relation.
- P. Martinetti and R. Wulkenhaar, *Discrete Kaluza-Klein from scalar fluctuations in noncommutative geometry*, J. Math. Phys. 43 (2002) 182-204, doi:10.1063/1.1418012, arXiv:hep-th/0104108. Section V, Theorem 4 and eq. (27): the distance between the same point on the two sheets is 1/||M|| (M the off-diagonal block of D_F), with a Pythagorean relation when [D_I, p] = 0. In the proof of Theorem 4 the authors embed the two sheets in a "virtual" five-manifold [0, 1] x M as a proof device and state that "the points between the sheets are not part of the geometry". Section VI: after scalar fluctuations the internal metric component becomes a function on M (the Higgs field); Section VII, Proposition 8, gives this component for the Standard Model. "Discrete Kaluza-Klein" refers to this Higgs-as-g_55 picture. Spectral truncation and boundary conditions do not occur.
- B. Iochum, T. Krajewski and P. Martinetti, *Distances in finite spaces from noncommutative geometry*, J. Geom. Phys. 37 (2001) 100-125, doi:10.1016/S0393-0440(00)00044-9: distances on finite spectral triples, including the two-point case.
- Matrix and discrete Kaluza-Klein precursors (bibliographic data verified; contents known from abstracts and secondary sources only): M. Dubois-Violette, R. Kerner and J. Madore, *Noncommutative differential geometry and new models of gauge theory*, J. Math. Phys. 31 (1990) 323 (Higgs fields as components of connections in matrix directions); J. Madore, *Kaluza-Klein aspects of noncommutative geometry* (1989 proceedings, posted as arXiv:1510.03915); J. Madore and J. Mourad, *Noncommutative Kaluza-Klein theory*, arXiv:hep-th/9601169; R. Coquereaux, G. Esposito-Farèse and G. Vaillant, *Higgs fields as Yang-Mills fields and discrete symmetries*, Nucl. Phys. B 353 (1991) 689 (**unverified** beyond citations); A. H. Chamseddine, J. Fröhlich, B. Schubnel and D. Wyler, *Dimensional reduction without continuous extra dimensions*, J. Math. Phys. 54 (2013) 012302, arXiv:1201.2661.

Spectral truncation (in print, for other spaces).

- A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Commun. Math. Phys. 383 (2021) 2021-2067, doi:10.1007/s00220-020-03825-x, arXiv:2004.14115. Section 1 (Introduction, pp. 2-3 of the arXiv version): "a natural spectral truncation is given simply by a spectral projection P onto eigenspaces of D … The natural truncation of the action of the *-algebra A to the Hilbert space PH is given by the space PAP. Since P does not commute with A, this is not an algebra anymore. Moreover in many interesting examples if one takes the C*-algebra generated by PAP one gets a non-informative full matrix algebra. However, not all is lost: the space PAP is a *-closed subspace in B(H), i.e. it is a so-called operator system". The same passage notes that symmetries of (A, D) induce symmetries of (PAP, PDP). The worked example is the circle (Toeplitz operator systems, Sections 4-5). Relation to lemma 05: the compression map Q of eq. (18) is non-multiplicative, as in this general picture, but for the chiral interval the minimal truncation has range exactly C + C, an algebra; this exceptional case is absent from the paper.
- F. D'Andrea, F. Lizzi and P. Martinetti, *Spectral geometry with a cut-off: topological and metric aspects*, J. Geom. Phys. 82 (2014) 18-45, doi:10.1016/j.geomphys.2014.03.014, arXiv:1305.2605. Cut-off by a

projection on D and/or A; points are approximated by non-pure truncated states at finite distance; on the circle these are Fejér distributions; applications to the Moyal plane and the fuzzy sphere. The interval does not appear.
- Follow-ups on truncations (all circle, tori or groups): E.-M. Hekkelman, *Truncated geometry on the circle*, Lett. Math. Phys. 112 (2022), doi:10.1007/s11005-022-01514-5; L. Glaser and A. Stern, *Reconstructing manifolds from truncations of spectral triples*, J. Geom. Phys. 159 (2021) 103921; M. Leimbach and W. D. van Suijlekom, *Gromov-Hausdorff convergence of spectral truncations for tori*, Adv. Math. 439 (2024) 109496, arXiv:2302.07877.

Deconstruction and lattices.

- N. Arkani-Hamed, A. G. Cohen and H. Georgi, *(De)constructing dimensions*, Phys. Rev. Lett. 86 (2001) 4757-4761, doi:10.1103/PhysRevLett.86.4757, arXiv:hep-th/0104005.
- C. T. Hill, S. Pokorski and J. Wang, *Gauge invariant effective Lagrangian for Kaluza-Klein modes*, Phys. Rev. D 64 (2001) 105005, doi:10.1103/PhysRevD.64.105005, arXiv:hep-th/0104035. The abstract states that the chain of gauge groups linked by bifundamentals "matches the continuum compactified theory truncated in momentum space", and treats chiral fermions. This is the physics statement "deconstruction = momentum truncation of the KK tower".
- H.-C. Cheng, C. T. Hill, S. Pokorski and J. Wang, *The standard model in the latticized bulk*, Phys. Rev. D 64 (2001) 065007, arXiv:hep-th/0104179 (split-generation fermions on a transverse lattice).
- J. Dai and X.-C. Song, *Spontaneous symmetry broken condition in (de)constructing dimensions from noncommutative geometry*, ICFP 2001 proceedings, doi:10.1142/9789812777379_0044, arXiv:hep-ph/0105280. The only paper found that applies NCG (differential calculus on cyclic groups, in the Dimakis-Müller-Hoissen/Sitarz style) to the Arkani-Hamed-Cohen-Georgi model; it treats the bosonic vacuum, without spectral triples or fermion truncation.
- NCG distances on lattices: G. Bimonte, F. Lizzi and G. Sparano, *Distances on a lattice from noncommutative geometry*, Phys. Lett. B 341 (1994) 139-146, arXiv:hep-lat/9404007 (the anomalous lattice distances that lemma 05, Part 1 calls "the familiar lattice-distance anomaly"); E. Atzmon, *Distances on a one-dimensional lattice from noncommutative geometry*, Lett. Math. Phys. 37 (1996) 341-348, doi:10.1007/BF00343197.
- The lowest Dirac (or gauge) mass pi/(2l) for mixed boundary conditions on an interval is textbook interval physics; see C. Csáki et al., *Fermions on an interval* (reference in C), Section 3 and Appendix B.

What remains unmatched. The identification "minimal grading-preserving spectral truncation of the chiral interval = Connes-Lott with |m| = pi/(2l)", the computation P C([0, l]) P = C + C, the truncated end-state distance 2l/pi against the geodesic length l, and the chain family of Part 1 with an exact Connes-Lott member at N = 2 and Gromov-Hausdorff convergence to [0, l].

Searched: "two-point space" with "spectral truncation", "interval", "boundary conditions"; "discrete Kaluza-Klein"; "deconstruction" with "noncommutative geometry", "spectral triple"; "gauge-Higgs unification" with "Connes-Lott"; "spectral truncation" with "interval [0,1]" (extended mode); NCG plus orbifold S^1/Z_2.

B. The two limits of the massive chiral interval

Claim (lemma 09, Theorems 9.8-9.9, Corollary 9.10). As l -> infinity at fixed bulk mass, the light pair becomes two chiral wall modes and the rest of the spectrum obeys a one-dimensional Weyl law above the gap mu. As l -> 0 with mu(l) l ~ log(1/(m_f l)), a Connes-Lott pair of fixed mass m_f survives while the interval collapses; at fixed mu no light fermion survives.

Verdict: PARTIAL. The spectral physics is classical: wall zero modes, exponentially small mixing of the two boundary modes on a finite slab, and the resulting light mass ~ 2 mu e^{-mu l}. Collapse of spectral triples is an active topic in NCG, with a general result for collapse to a point. We found no source stating

either limit as a theorem about families of finite or interval spectral triples, and no source for the tuned collapse mu(l) that keeps a Connes-Lott pair.

References.

- R. Jackiw and C. Rebbi, *Solitons with fermion number 1/2*, Phys. Rev. D 13 (1976) 3398-3409, doi:10.1103/PhysRevD.13.3398: the wall zero mode.
- D. B. Kaplan, *A method for simulating chiral fermions on the lattice*, Phys. Lett. B 288 (1992) 342-347, doi:10.1016/0370-2693(92)91112-M, arXiv:hep-lat/9206013: domain-wall fermions.
- Y. Shamir, *Chiral fermions from lattice boundaries*, Nucl. Phys. B 406 (1993) 90-106, doi:10.1016/0550-3213(93)90162-I, arXiv:hep-lat/9303005. Section 2: a Weyl mode bound to the boundary of a half-infinite fifth direction, profile b^s with |b| < 1 (eq. (14)). Section 3: with a finite fifth direction 0 <= s <= N, "a second chiral fermion appears on the new boundary at s = N … this fermion has the opposite chirality … Strictly speaking, there is a tiny mixing between the two chiral modes, which vanishes like (1 - M)^N." This is the lattice version of Lemma 9.5 and Theorem 9.8.
- C. G. Callan and J. A. Harvey (1985), see section H: the continuum domain-wall zero modes and anomaly inflow.
- C. Csáki, C. Grojean, J. Hubisz, Y. Shirman and J. Terning, *Fermions on an interval: quark and lepton masses without a Higgs*, Phys. Rev. D 70 (2004) 015012, doi:10.1103/PhysRevD.70.015012, arXiv:hep-ph/0310355. Section 2: boundary conditions from the variational principle; Section 4.1, eqs. (4.8)-(4.9): a bulk mass profile realizing Dirichlet-Neumann conditions, with a tan(m_n L) quantization; Appendix B: KK decompositions in flat space.
- C. Farsi and F. Latrémolière, *Collapse in noncommutative geometry and spectral continuity*, J. London Math. Soc. (2) 114 (2026) e70628, doi:10.1112/jlms.70628, arXiv:2404.00240. Collapse of the vertical part of spectral triples in the spectral propinquity, with spectral continuity; Section 4.1, Corollary 4.1: collapse of a spectral triple to a point gives a nontrivial limit acting on ker D (read in the published text). The published text (first published 3 July 2026, read on 3 Nov 2026 from the journal HTML) agrees with arXiv v3 in the statements and equations compared (copy-editing, and L_A corrected to L_B in Definition 2.13): Theorem 3.4 (Theorem 2.5 in v1) keeps the same six hypotheses, and the step at issue is printed as eq. (3.11), so Counterexample R1 of the assessment applies to the printed statement. Relation to lemma 09: Corollary 9.10 (fixed mu, condition (c), ker D = 0, no light sector) agrees with this picture. Theorem 9.9 keeps a nonzero light pair because the Dirac operator itself changes with l (mu = mu(l)); that family lies outside the metric-rescaling collapse treated there. **Assessed in detail:** reviews/2026-11-03_farsi_latremoliere_assessment.md. Corollary 4.1 (a fixed triple scaled by 1/eps converges to (C, ker D, 0), and has no limit when ker D = 0) is sound. The printed hypotheses of Theorem 3.4 (Theorem 2.5 in v1) are insufficient; anticommutation of D_h and D_v, or a uniform relative bound, supplies a sufficient additional hypothesis; Counterexample R1 there, checked by hand by Claude and by Terra 6.1, satisfies the six listed hypotheses and contradicts Corollary 3.5. The anticommuting applications avoid this defect; Theorem 4.6 and Corollary 4.7 also need a nonzero vertical kernel for the displayed target to be a spectral triple. The published version keeps the statement without the hypothesis. Validated by Astra 6 (Astra validation, items 2 and 4). Our tuned collapse (Theorem 9.9) lies outside its hypotheses.
- B. Ammann and C. Bär, *The Dirac operator on nilmanifolds and collapsing circle bundles*, Ann. Global Anal. Geom. 16 (1998) 221-253, doi:10.1023/A:1006553302362, arXiv:math/9801091: in commutative geometry, collapse either sends all Dirac eigenvalues to infinity or keeps a convergent part, depending on the spin structure.

What remains unmatched. The statement of both limits for the interval spectral triple with condition (c) as theorems (Theorems 9.8-9.9), the formula c = log(2c/(m_f l)) + o(1) for the bulk mass needed to keep a fixed light mass, and the identification of the surviving pair with Connes-Lott at each l (Theorem 9.6).

Searched: domain-wall fermions on a finite slab (exponentially small mass); "collapse" with "spectral triple"; "decompactification" and "emergent dimension" with "almost-commutative"; Ammann-Bär collapse and its NCG follow-ups.

C. Towerless Yukawa textures

Claim (lemma 12). For bulk fields Q, U on [0, l] with bulk masses and a boundary Yukawa matrix X at x = l: an exact secular equation det(lambda^2 - A X B X^dagger) = 0 (Theorem 12.4); the one-flavour, massless spectrum (n pi +- arctan|X|)/l (Example 12.5); singular-value bounds (Theorem 12.6); in the limit l -> 0 at fixed c and Y_hat = X/l the low spectrum converges to the finite triple with Yukawa block M_F = Phi_Q Y_hat Phi_U (Theorem 12.7); graded singular values of F X G (Lemma 12.8).

Verdict: PARTIAL. The physics is in print, in more elaborate (warped) form: exact KK determinant equations with brane Yukawa matrices, the effective Yukawa F(c_Q) Y F(c_u), masses from minors and mixings F(c_Qi)/F(c_Qj). The one-flavour flat spectrum is Csáki et al. eq. (5.6) (KNOWN). We found no NCG paper that derives the finite Dirac operator D_F, or its Yukawa texture, from localization on an extra dimension.

References.

- C. Csáki et al., *Fermions on an interval* (full data in B). Section 5.1, eqs. (5.4)-(5.6): flat interval, no bulk mass, boundary conditions psi_eL(0) = 0, chi_eR(0) = 0, psi_eL(L) = -M_D L psi_eR(L), chi_eR(L) = M_D L chi_eL(L), spectrum tan(m_n L) = M_D L. With X = M_D L and signed 4D masses, the roots are (n pi +- arctan|X|)/L, which is Example 12.5. Section 2, eqs. (2.8)-(2.9): general boundary conditions with matrices; the spectra are computed one field at a time (Section 7.1: "a much richer spectrum of possibilities for such mixings which we will however not deal with here"). Section 6.2.2, eq. (6.35): secular equation for bulk mass plus brane Majorana mass (warped). Appendix C.2, eqs. (C.9)-(C.14): the boundary parameter equals M_D L, 2 tanh(M_D L/2) or M_D L/(1 + M_D^2 L^2/4) depending on how the brane mass is regularized, so the X of lemma 12 is a boundary-condition parameter whose relation to a brane Lagrangian mass is scheme dependent.
- S. Casagrande, F. Goertz, U. Haisch, M. Neubert and T. Pfoh, *Flavor physics in the Randall-Sundrum model I*, JHEP 10 (2008) 094, doi:10.1088/1126-6708/2008/10/094, arXiv:0807.4937. Section 3.2: IR boundary conditions with the Yukawa matrices (eqs. (55), (63)) and the exact determinant condition for the KK masses, eq. (65), for N generations; eq. (71): profile normalization F(c); eq. (87): Y_eff = F(c_Q) Y F(c_u); eq. (90): zero-mode approximation det(m_n^2 - (v^2/2) Y_eff Y_eff^dagger) = 0; eqs. (94)-(96): masses as F(c_Q) F(c_u) times ratios of minors of Y; eq. (97): (U_q)_ij ~ F(c_Qi)/F(c_Qj). This is the warped analogue of Theorems 12.4, 12.7, Corollary 12.9 and Remark 12.10.
- Flat and warped localization: N. Arkani-Hamed and M. Schmaltz, *Hierarchies without symmetries from extra dimensions*, Phys. Rev. D 61 (2000) 033005, doi:10.1103/PhysRevD.61.033005; T. Gherghetta and A. Pomarol, *Bulk fields and supersymmetry in a slice of AdS*, Nucl. Phys. B 586 (2000) 141-162, doi:10.1016/S0550-3213(00)00392-8; Y. Grossman and M. Neubert, *Neutrino masses and mixings in non-factorizable geometry*, Phys. Lett. B 474 (2000) 361-371, doi:10.1016/S0370-2693(00)00054-X; S. J. Huber and Q. Shafi, *Fermion masses, mixings and proton decay in a Randall-Sundrum model*, Phys. Lett. B 498 (2001) 256-262, doi:10.1016/S0370-2693(00)01399-X; K. Agashe, G. Perez and A. Soni, *Flavor structure of warped extra dimension models*, Phys. Rev. D 71 (2005) 016002, doi:10.1103/PhysRevD.71.016002; D. E. Kaplan and T. M. P. Tait, *New tools for fermion masses from extra dimensions*, JHEP 11 (2001) 051, doi:10.1088/1126-6708/2001/11/051, arXiv:hep-ph/0110126 (flat compact extra dimension, zero modes localized by odd bulk masses, Higgs on a boundary; the flat-space setting closest to lemma 12).
- Graded matrices: C. D. Froggatt and H. B. Nielsen, *Hierarchy of quark masses, Cabibbo angles and CP violation*, Nucl. Phys. B 147 (1979) 277-298, doi:10.1016/0550-3213(79)90316-X; J. Demmel, M. Gu, S. Eisenstat, I. Slapničar, K. Veselić and Z. Drmač, *Computing the singular value decomposition with high relative accuracy*, Linear Algebra Appl. 299 (1999) 21-80 (singular values of D_1 X D_2 with well-conditioned X are determined to high relative accuracy; theorem numbers **unverified**). The leading-order "masses = products of profiles times ratios of minors" is Casagrande et al. eq. (96). We did not find the two-sided compound-matrix bound of Lemma 12.8 in this explicit form; its proof (Cauchy-Binet on the p-th compound) is routine.
- NCG side: Yukawa matrices enter D_F as free moduli. T. Krajewski (1998) and A. Connes (2006, Theorem 2.7) classify the admissible D_F; see section I. No derivation of these moduli from localization

was found.

What remains unmatched. The flat-space exact secular equation for an arbitrary N_Q x N_U boundary matrix with bulk masses, stated with multiplicities; the two-sided bound of Theorem 12.6 (via concavity of alpha, beta in lambda^2); the towerless limit l -> 0 at fixed c and Y_hat as a convergence theorem with error sigma^3 l^2 (1/L_Q^2 + 1/L_U^2) onto a finite even spectral triple; Corollary 12.3 on vectorlike residual theories; and the reading "D_F is the towerless remnant of internal momentum".

Searched: "noncommutative geometry" or "spectral triple" with "split fermions", "fermion localization", "Randall-Sundrum", "Yukawa overlap" (standard and extended); Csáki et al. full text; Casagrande et al. full text; graded-matrix SVD literature.

D. KO-dimension of products: 9, 10 or 11

Claim (lemmas 06, 07, 08). A finite KO-6 real triple is T_n x P_k with (s, n, k) = (+1, 7, 7) or (-1, 5, 1) iff a grading-reversing involution U commutes with A, A°, D and satisfies UJ = sJU (Theorem 06.2); J alone does not fix s; an unpaired nu_L forces the KO-7 reduction (No-go 06, with [U, Q] = 0); in the internal-first product with the interval, the bare reflection has s = -1 (Theorem 08.5).

Verdict: PARTIAL. Products of real spectral triples of all parities, including the doubled odd x odd product on H_1 x H_2 x C^2, the freedom J versus J gamma in even dimension, and the total KO-dimension 10 = 2 mod 8 of the Standard Model are in print. The converse direction, a criterion for when a given even KO-6 triple factors as (odd) x (odd point), was NOT FOUND, nor were the nu_L statement, the reflection sign, or any discussion of 9 or 11 as alternatives to 10.

References.

- L. Dąbrowski and G. Dossena, *Product of real spectral triples*, Int. J. Geom. Methods Mod. Phys. 8 (2011) 1833-1848, doi:10.1142/S021988781100597X, arXiv:1011.4456. Introduction: in even dimensions "there are always two real structure operators J, that differ by multiplication by the grading operator. None of them should be preferred as they are perfectly on the same footing." Section 4.1 (Tables 2-3) and 4.2 (Tables 4-5): even-even and even-odd products. Section 4.3, eq. (25) and Table 6: the odd-odd product H = (H_1 x H_2) x C^2, D = D_1 x 1 x sigma_1 + 1 x D_2 x sigma_2, chi = 1 x 1 x sigma_3, J = J_1 x J_2 x M K. Section 6: metric and KO-dimension need not agree mod 8. Lemma 06.1 already cites eq. (25) and Table 6 for its product convention. The paper treats the forward direction (building products); the question of when a given triple is a product is left open there.
- F. J. Vanhecke, *On the product of real spectral triples*, Lett. Math. Phys. 50 (1999) 157-162, doi:10.1023/A:1007690509512, arXiv:math-ph/9902029: the naive product J may violate the sign table; a corrected product real structure.
- F. J. Vanhecke, A. R. da Silva and C. Sigaud, *On the finite spectral triple of an almost-commutative geometry*, Braz. J. Phys. 42 (2012) 471-474, doi:10.1007/s13538-012-0101-2, arXiv:1011.4925. Section 2.1: with J = J_1 x J_2, a Minkowski factor of signature 2 or 6 forces an odd internal factor (sigma_2 in {1, 5} or {3, 7}) and "the chirality paradigm is lost"; Section 2.2: with J = J_1 x J_2 Omega_2 an even internal factor is allowed and sigma_2 = 6 gives total signature 0 (Barrett's case). This is the closest printed discussion of odd internal factors, framed as a consequence of the product convention.
- S. Farnsworth, *The graded product of real spectral triples*, J. Math. Phys. 58 (2017) 023507, doi:10.1063/1.4975410, arXiv:1605.07035: graded tensor product, commutative and associative, always a spectral triple.
- B. Ćaćić, *Real structures on almost-commutative spectral triples*, Lett. Math. Phys. 103 (2013) 793-816, doi:10.1007/s11005-013-0616-7, arXiv:1209.4832: real reconstruction theorem; compact oriented manifolds carry real (almost-)commutative triples of every KO-dimension.
- A. Connes, *Noncommutative geometry and the standard model with neutrino mixing*, JHEP 11 (2006) 081, doi:10.1088/1126-6708/2006/11/081, arXiv:hep-th/0608226, Section 1: "the KO-dimension of the product space M x F is in fact equal to 10 ~ 2 modulo 8. Of course the above 10 is very reminiscent of string theory".

- J. W. Barrett, *Lorentzian version of the noncommutative geometry of the standard model of particle physics*, J. Math. Phys. 48 (2007) 012303, doi:10.1063/1.2408400, arXiv:hep-th/0608221. Section 1: J^2 = 1 and J gamma = gamma J "hold only if the total signature is equal to zero"; Section 2: the internal space is "determined to be a real spectral triple of signature six".
- A. H. Chamseddine and A. Connes, *Why the Standard Model*, J. Geom. Phys. 58 (2008) 38-47 (section I): the finite geometry raises "the K-theoretic dimension from four to ten (modulo eight)".
- K. van den Dungen and W. D. van Suijlekom, *Electrodynamics from noncommutative geometry*, J. Noncommut. Geom. 7 (2013) 433-456, doi:10.4171/JNCG/122: the KO-6 two-point triple for electrodynamics, the natural reference for Counterexample 06.3.
- Background: the complex analogue of Theorem 06.2 (an even triple with a commuting, grading-reversing Clifford generator is an odd triple doubled) is the standard identification of odd Fredholm modules with Cl_1-graded ones in Kasparov theory; we did not locate a reference stating the real KO-6 version with the sign s of UJ = sJU.
- Terminology: "doubling" in lemma 06 (H = H_0 x C^2) differs from "fermion doubling" in the NCG literature (F. Lizzi, G. Mangano, G. Miele and G. Sparano, Phys. Rev. D 55 (1997) 6357, section I; resolved in Connes 2006 by the Pfaffian). A paper should use a different word.

What remains unmatched. Theorem 06.2 (criterion), Result 06.3 (both s for electrodynamics), the nu_L no-go under [U, Q] = 0, Theorem 08.5 (s = -1 for the bare reflection in the internal-first product), Counterexample 08.11 and Theorem 08.12.

Searched: "real spectral triple" with "factorization", "product of odd", "Clifford", "KO-dimension 6"; "9" and "11" with "noncommutative standard model"; full texts of Dąbrowski-Dossena, Vanhecke-da Silva-Sigaud, Connes 2006 and Barrett 2007.

E. Spectral action: the W mass relation, the top ratio, heavy-fermion weights

Claim (lemma 11; lemma 03). 8 M_W^2 = sum N_f m_f^2 at unification; m_t/M_W -> sqrt(8/3) when the top dominates; the bounds m_max/(2 sqrt 2) <= M_W <= sqrt(N_g) m_max; gauge kinetic terms weighted by sum_j f(m_j$^{2/\mathrm{Lambda}}$2) tr_{H_j} F^2.

Verdict: KNOWN for the mass relation and the top ratio; Proposition 11.1 and Corollary 11.2 are one-line consequences of it. **NOT FOUND** for the exact heavy-fermion weighting in this form.

References.

- A. H. Chamseddine, A. Connes and M. Marcolli, *Gravity and the standard model with neutrino mixing*, Adv. Theor. Math. Phys. 11 (2007) 991-1089, doi:10.4310/ATMP.2007.v11.n6.a3, arXiv:hep-th/0610241. Eq. (1.5) (Introduction): sum over generations of m_e^2 + m_nu^2 + 3 m_d^2 + 3 m_u^2 = 8 M_W^2. Lemma 4.6, eq. (4.16), proved from the normalization (4.17) Tr(k* k ...) = 2 g^2. Section 5.4 ("The fermion-boson mass relation"): eq. (5.24) m_top = v y_t/sqrt 2 with v = 2M/g; eqs. (5.25)-(5.26): Y_2(S) = 4 g^2; eq. (5.27): neglecting all Yukawas but the top, y_t = (2/sqrt 3) g at unification; eq. (5.28): y_t ~ 0.597. With M_W = g v/2 this is m_t/M_W = sqrt(8/3), which follows at once from (5.24) and (5.27); CCM state the relation in the form y_t = (2/sqrt 3) g. Eq. (5.29): if the tau-neutrino Dirac Yukawa equals the top Yukawa, y_t ~ g, i.e. m_t/M_W = sqrt 2.
- A. H. Chamseddine and A. Connes, *Conceptual explanation for the algebra in the noncommutative approach to the standard model*, Phys. Rev. Lett. 99 (2007) 191601 (abstract: a mass relation linking the fermion masses to the W mass, used to predict the top mass).
- Heavy-fermion weights: CCM Section 3 expands Tr f(D_A/Lambda) with D_F entering polynomially through the coefficients a, b, c, d, e, and Section 5.3 places the Majorana mass at the scale Lambda (Lemma 5.2, eq. (5.20)). The resummed weighting sum_j f(m_j$^{2/\mathrm{Lambda}}$2) of lemma 03 Part A was not found. Checked: M. Eckstein and B. Iochum, *Spectral Action in Noncommutative Geometry*, Springer-Briefs in Mathematical Physics 27 (2018), doi:10.1007/978-3-319-94788-4, arXiv:1902.05306 (abstract and contents only); B. Iochum, C. Levy and D. Vassilevich, *Spectral action beyond the weak-field approximation*, Commun. Math. Phys. 316 (2012) 595-613, doi:10.1007/s00220-012-1587-8 (second

order in the gauge field with full momentum dependence; abstract only). The physics counterpart is decoupling: T. Appelquist and J. Carazzone, *Infrared singularities and massive fields*, Phys. Rev. D 11 (1975) 2856-2861, doi:10.1103/PhysRevD.11.2856. The evaluation already classes lemma 03 as a routine computation; a paper should present it as such.

F. Weinberg angle from a squashed S^3 / lens space and from M^{pqr}

Claim (canonical_research.md §14.3). For the Berger metric with fibre scaled by xi^2, K_W = V R^2 (2 + xi^2)/3, K_Y = V R^2 xi^2, hence cos^2 theta_W = 3 xi^2/(4 xi^2 + 2); the lens order p cancels; 5/8 needs xi^2 = 5/2.

Verdict: KNOWN (same formula, same parameterization).

- A. J. Niemi and S. Slizovskiy, *On relations between electroweak hierarchy problem, fluctuating three branes and general covariance*, arXiv:1004.0212 (no journal version found on INSPIRE; earlier titles "Hiding Higgs with brane" and "Embedding Weinberg-Salam in Kaluza-Klein"). Section IV, eqs. (77)-(78): internal metric dtheta^2 + sin^2 theta dpsi_-^2 + xi^2 (dpsi_+ - cos theta dpsi_-)^2, isometry SU(2)_L x U(1)_R for xi != 1; Section V, eqs. (94)-(95): q^2 = 3/(xi^2 + 2), q'^2 = 1/xi^2; Section VI: sin^2 theta_W = (xi^2 + 2)/(4 xi^2 + 2), i.e. cos^2 theta_W = 3 xi^2/(4 xi^2 + 2). They fix xi by a custodial-symmetry condition and quote sin^2 theta_W = 0.296 (that step, in Section VIII, was not read). The ratios 1/q^2 : 1/q'^2 = (2 + xi^2)/3 : xi^2 coincide with K_W : K_Y of §14.3. The cancellation of the lens order p in the common volume factor is immediate.
- E. Witten, *Search for a realistic Kaluza-Klein theory*, Nucl. Phys. B 186 (1981) 412-428, doi:10.1016/0550-3213(81)90021-3: eleven-dimensional supergravity on the seven-dimensional M^{pqr} with SU(3) x SU(2) x U(1) isometry; gauge couplings determined by the internal metric; the chirality obstruction. Read on 3 Nov 2026 from the PDF supplied by the user, and checked by Astra 6 (Astra validation, item 1). Eq. (3): gauge fields as components of the metric along Killing vectors. Pp. 416-417: seven is the minimum internal dimension with this symmetry, so eleven in total. P. 418: M^{pqr} = G/H with H = SU(2) x U(1) x U(1) and the deleted generator p lambda_8 + q T_3 + r Y; M^{001} = CP^2 x S^2 x S^1, M^{101} = S^5 x S^2, M^{011} = CP^2 x S^3. Pp. 421-423: "One would inevitably get vector-like rather than V - A weak interactions", for a reason that "depends only on the fact that the number of extra dimensions is odd": gamma_i -> -gamma_i fixes sigma_ij and flips Gamma_7, on an orientable manifold, with the gauge group acting by coordinate transformations. Escapes considered: non-orientable internal spaces (none suitable for this symmetry, p. 423) and torsion (left open, pp. 423, 426-427). Pp. 425-426: the three parameters of the invariant metric would determine the three gauge couplings. The paper computes no weak angle, so section F cites it for the method only.
- L. Castellani, R. D'Auria and P. Fré, *SU(3) x SU(2) x U(1) from D = 11 supergravity*, Nucl. Phys. B 239 (1984) 610-652 (INSPIRE): the Einstein metrics on M^{pqr}, with parameters depending on q/(3p). Content from abstract and secondary sources.

G. The protected zero mode on S^5 (requested by the review)

Claim (connes_kk_research_state.md, around the S^5 zero mode): for the nontrivial quaternionic line V -> S^5, dim ker D_{S^5, V} is odd (real mod-two index); for SU(3)-invariant data the invariant line is a zero mode; the text attributes the mechanism to Witten 1982, footnote 2.

Verdict: KNOWN, for the topological statement (a nontrivial SU(2) bundle on S^5 has odd mod-two index) and for the attribution. Footnote 2 of Witten 1982 (p. 327, read on 3 Nov 2026 from the PDF supplied by the user) reads: "Actually, in a special case one can easily find the zero mode. If one conformally compactifies S^4 x R to the five-sphere S^5, then on S^5 one can choose the instanton field to be invariant (up to a gauge transformation) under an SU(3) subgroup of the symmetry group O(6) of the five-sphere.

The fermion zero mode is then the unique SU(3) invariant spinor field that can be defined." The wording of the corpus matches it.

- E. Witten, *An SU(2) anomaly*, Phys. Lett. B 117 (1982) 324-328, doi:10.1016/0370-2693(82)90728-6. As summarized by Wang, Wen and Witten (2019), Section 2: "The main result of [1] is that the fermion path integral for a fermion field on M in a given representation of SU(2) is phi-invariant if and only if the mod 2 index of the Dirac operator on Y – in that representation – vanishes. Here the mod 2 index is defined as the number of zero modes of the five-dimensional Dirac operator, reduced mod 2", where Y is the mapping torus of the gauge transformation phi (the mapping torus S^4 x S^1, with phi in the nontrivial class of pi_4(SU(2)) = Z_2). The original, read on 3 Nov 2026, gives the criterion for other isospins on p. 328: the fermion integration is odd exactly when tr T^2, with tr T^2 = 1/2 on the doublet, is a half-integer.
- I. García-Etxebarria and M. Montero, *Dai-Freed anomalies in particle physics*, JHEP 08 (2019) 003, doi:10.1007/JHEP08(2019)003, arXiv:1808.00009. Section 3.1, eq. (3.4): Omega_5^Spin(BSU(2)) = Z_2, detected by the mod-two index; immediately after: "A simple example with non-vanishing mod-2 index was constructed in [2] [Witten 1982]. While S^5 itself is trivial in Omega_5^Spin …, there is a bundle over it such that the total space is no longer null-bordant in Omega_5^Spin(BSU(2)) = Z_2." Section 3.1.2: SU(3)/SU(2) = S^5, so in five dimensions every SU(3) bundle reduces to an SU(2) bundle.
- Y. Sato, *Majorana fermion zero modes and anomalies in string and M theories* (PhD thesis, Tokyo, 2024), arXiv:2404.18138, Appendix C: the five-dimensional configuration of Witten's argument (S^4 x R with pure-gauge ends) compactifies to S^5; G-bundles on S^5 are classified by pi_4(G), and for G = Sp(N) the mod-two index is an isomorphism onto KO^{-1}(pt) = Z_2.
- J. Wang, X.-G. Wen and E. Witten, *A new SU(2) anomaly*, J. Math. Phys. 60 (2019) 052301, doi:10.1063/1.5082852, arXiv:1810.00844: mod-two index on mapping tori, extended to non-spin manifolds. The corpus cites it for the parity of the index at isospin j = 2k + 1/2. That parity follows from the criterion of Witten 1982: tr T^2 = j(j + 1)(2j + 1)/3 equals (4k + 1)(2k + 1)(4k + 3)/6, a half-integer, at j = 2k + 1/2, and (4k + 3)(k + 1)(4k + 5)/3, an integer, at j = 2k + 3/2. We did not reread the 2019 paper for this statement.

Answer to the review's objection. Witten's original construction uses S^4 x R, with the specified asymptotic gauge fields; footnote 2 discusses its conformal compactification to S^5. The mapping-torus formulation uses S^4 x S^1, with the gauge bundle glued by the gauge transformation. Both that mapping torus (odd index for one doublet) and the nontrivial SU(2) bundle on S^5 (García-Etxebarria-Montero, sentence after eq. (3.4)) are nonzero in Omega_5^Spin(BSU(2)) = Z_2, so they represent the same generator; the mod-two index is a bordism invariant and is odd on both. Sato's Appendix C describes the S^5 bundle as the compactification of the S^4 x R configuration of Witten's argument. García-Etxebarria-Montero state the S^5 version and attribute the example to Witten 1982. What remains specific to the corpus is the identification of V = rho^*Q with this generator via the Hopf map from CP^3 data and the SU(3)-invariant zero-mode check.

H. Anomaly inflow as the five-dimensional parent of chiral boundary modes (requested by the review)

Relevance: claim B (chiral walls of Theorem 9.8) and Corollary 9.3 (equivariant index from boundary types).

Verdict: KNOWN as background; lemma 09 contains no gauge-field inflow, so these references supply context.

- C. G. Callan and J. A. Harvey, *Anomalies and fermion zero modes on strings and domain walls*, Nucl. Phys. B 250 (1985) 427-436, doi:10.1016/0550-3213(85)90489-4. Chiral zero modes of the Dirac equation on domain walls in 2n+1 dimensions and on strings in 2n+2 dimensions; their anomalies are cancelled by inflow from the bulk through an induced Chern-Simons (topological mass) term. Content from the abstract fragment and secondary sources; full text not read.

- S. G. Naculich, *Axionic strings: covariant anomalies and bosonization of chiral zero modes*, Nucl. Phys. B 296 (1988) 837-867, doi:10.1016/0550-3213(88)90400-2: the consistent/covariant anomaly bookkeeping in inflow.
- N. Arkani-Hamed, A. G. Cohen and H. Georgi, *Anomalies on orbifolds*, Phys. Lett. B 516 (2001) 395-402, doi:10.1016/S0370-2693(01)00946-7: anomalies of bulk fermions on S^1/Z_2 localized at the fixed points (content **unverified** beyond title and abstract-level knowledge).
- Kaplan (1992) and Shamir (1993), section B, built lattice chiral fermions on this mechanism.

For the corpus: Theorem 9.2 and Corollary 9.3 are the index-theoretic shadow of Callan-Harvey (boundary type fixes the chiral zero mode); an inflow statement would require bulk gauge fields with a Chern-Simons term on the interval, which lemma 09 does not include.

I. The current Connes et al. framework (requested by the coordinator)

The corpus treats Connes-Lott 1990 as the obsolete layer. The following define the present framework. All bibliographic data verified (Crossref or arXiv) unless marked.

- **A. Connes, *Noncommutative geometry and the standard model with neutrino mixing*, JHEP 11 (2006) 081**, arXiv:hep-th/0608226. Reverses the grading on the antiparticle sector so that the finite space has KO-dimension 6, separating metric dimension (0) from KO-dimension. Theorem 2.7 classifies the Dirac operators D_F (Yukawa matrices, CKM and PMNS data, a Majorana mass); the fermion doubling problem is solved by the Pfaffian in the fermionic action. The total KO-dimension 10 = 2 mod 8 is noted in Section 1.
- **J. W. Barrett, *Lorentzian version ...*, J. Math. Phys. 48 (2007) 012303**, arXiv:hep-th/0608221. Independently arrives at internal signature 6 from the Lorentzian requirement that physical fermions be simultaneous eigenvectors of J and gamma (total signature 0 mod 8), which removes the quadrupling of fermions.
- **A. H. Chamseddine, A. Connes and M. Marcolli, *Gravity and the standard model with neutrino mixing*, Adv. Theor. Math. Phys. 11 (2007) 991-1089**, arXiv:hep-th/0610241. Section 2: the finite geometry, with A_F = C + H + M_3(C), Lemma 2.5 (the irreducible representations of the left-right bimodule algebra have total dimension 32 per generation), Section 2.4 (the order-one condition selecting A_F), Sections 2.6-2.7 (classification and moduli of D_F, i.e. the Yukawa parameters) and Section 2.8 (KO-dimension and Poincaré duality); N = 96 in Section 3.7.1; Section 3: the spectral action on M x F and its heat-kernel coefficients; Section 4: the full Lagrangian; Section 5: predictions at unification (gauge coupling relation, Higgs quartic, the mass relation (4.16)).
- **A. H. Chamseddine and A. Connes, *Conceptual explanation for the algebra in the noncommutative approach to the standard model*, Phys. Rev. Lett. 99 (2007) 191601**, doi:10.1103/PhysRevLett.99.191601, arXiv:0706.3690. Version 1 of arXiv:0706.3690 carried the title *A dress for SM the beggar*; the published version is this PRL (the coordinator's "PNAS 2007" does not match any record we found). Classifies irreducible finite geometries with reality and chirality and no fermion doubling, requiring total K-theoretic dimension 10.
- **A. H. Chamseddine and A. Connes, *Why the Standard Model*, J. Geom. Phys. 58 (2008) 38-47**, doi:10.1016/j.geomphys.2007.09.011, arXiv:0706.3688. Section 2: the order-zero condition; Proposition 2.5 (centre C: dim H = k^2) and Proposition 2.8 (centre C + C: dim H = 2k^2, A_C = M_k(C) + M_k(C)); Lemma 3.1: the KO-6 relation J gamma = -gamma J excludes centre C; eq. (3.1): with a quaternionic, nontrivially graded W of quaternionic dimension 2, A = M_2(H) + M_4(C), k = 4, dim H = 2k^2 = 32 per generation; Theorem 4.1: the order-one condition with an off-diagonal D selects the unique maximal subalgebra A_F = C + H + M_3(C). For the project this is the statement that D_F must connect the two components of the centre: the "two sheets" of the finite space are forced by the classification.
- **A. H. Chamseddine, A. Connes and W. D. van Suijlekom, *Inner fluctuations in noncommutative geometry without the first order condition*, J. Geom. Phys. 73 (2013) 222-234**, doi:10.1016/j.geomphys.2013.06.006, arXiv:1304.7583. Without the first-order condition, inner fluctuations acquire a quadratic term and form a semigroup depending only on A. Relevant to Amendment

2 of lemma 07 (the compressed two-point data fail the first-order condition).

- **A. H. Chamseddine, A. Connes and W. D. van Suijlekom, *Beyond the spectral Standard Model: emergence of Pati-Salam unification*, JHEP 11 (2013) 132**, doi:10.1007/JHEP11(2013)132, arXiv:1304.8050. Dropping the first-order condition, the classification gives an SU(2)_R x SU(2)_L x SU(4) model; the Standard Model is recovered when the vacuum restores the first-order condition.
- **A. Devastato, F. Lizzi and P. Martinetti, *Grand symmetry, spectral action and the Higgs mass*, JHEP 01 (2014) 042**, doi:10.1007/JHEP01(2014)042, arXiv:1304.0415. A larger "grand" algebra mixing gauge and spin degrees of freedom produces the extra scalar sigma that brings the Higgs mass near 126 GeV; requiring an almost-commutative geometry breaks it to the Standard Model.
- **A. H. Chamseddine, A. Connes and V. Mukhanov, *Quanta of Geometry: Noncommutative Aspects*, arXiv:1409.2471** (journal data unverified; read from the LaTeX source, 3 Nov 2026). Clifford algebras C_kappa on n + 1 gammas with kappa = +-1; in dimension 4, C_+ = M_2(H) and C_- = M_4(C), "in perfect agreement with the algebraic constituents of the Standard Model"; two maps Y_+- : M -> S^4 of slashed form, with det e = Omega_+ + Omega_-. Bears on the twistor line of canonical section 4 and on current_framework.md.
- **J. Baptista, *Chiral interactions of fermions and massive gauge fields in Kaluza-Klein models*, JHEP 05 (2026) 008, arXiv:2506.09126** (found by the referee; introduction read from the LaTeX source). The no-go arguments against chiral fermions in Kaluza-Klein models (Witten 1983; Atiyah-Hirzebruch) assume that the gauge fields are linked to exact Killing fields; gauge fields linked to non-Killing fields are massive, can mix fermions of different masses, and can couple asymmetrically to the two chiralities, with examples on even-dimensional K = S^2 and T^2. Bears on the reading of no-go results in the paper and on T4.
- **Lemma 22: the finite Dirac operator as a boundary condition.** Known: the one-flavour formula (Csáki, Grojean, Hubisz, Shirman and Terning, section C here) and the gauge boundary conditions and the form M tan(M pi R) = -V (Csáki, Grojean, Murayama, Pilo and Terning, *Gauge theories on an interval: unitarity without a Higgs*, Phys. Rev. D 69 (2004) 055006, arXiv:hep-ph/0305237, read in the LaTeX source). Not found: the real structure with a finite triple as exact window, the wall algebra C_F, the average formula for the compression. **Searched** (arXiv API, 3 Nov 2026): "noncommutative geometry" with Higgsless; "spectral triple" with "boundary conditions", Higgs, interval; "spectral triple" with Yukawa, "boundary conditions"; "almost-commutative" with interval, boundary; "spectral triple" with "Kaluza-Klein", "boundary conditions": no hits. "noncommutative geometry" with "gauge-Higgs": one hit, arXiv:2511.05909, a survey, abstract only. Absence from these queries does not establish absence. The referee adds operator-system compression in Connes-van Suijlekom (arXiv:2004.14115, sections 1, 3), and A. Chamseddine and A. Connes, *Noncommutative geometric spaces with boundary: spectral action*, J. Geom. Phys. 61 (2011) 317, arXiv:1008.3980 (read in the LaTeX source: sections 3 to 5). There the boundary is that of the manifold in a product with a finite space, with the condition Pi_- Psi = 0, Pi_- = (1 - chi)/2, for a local chirality projector chi, and the spectral action acquires the extrinsic-curvature terms. The internal boundary with finite gradings of lemmas 22 and 23 is a different setting.
- **Lemma 25: the sum rule and the effective dimension.** Known: m_t = 2 M_W at tree level in flat gauge-Higgs models with suitable representations (Cacciapaglia, Csáki and Park, arXiv:hep-ph/0510366, section 5; Adachi and Maru, arXiv:1804.06012, Eq. (3.23)). Not found in the records inspected by the referee: the coefficient (3 - d_eff)/24 and its theta-function crossover. **Searched** by the referee: web-index queries for spectral action, heat kernels, compactification and gauge-Higgs mass relations, including a 2026 query, and the reference chain from Adachi, Kurahashi and Maru (arXiv:1404.4281). A forward-citation search, an author sweep and a broader database search remain open. This does not establish priority.
- **Twisted spectral triples:** A. Connes and H. Moscovici, *Type III and spectral triples*, arXiv:math/0609703 (journal data **unverified**); G. Landi and P. Martinetti, *On twisting real spectral triples by algebra automorphisms*, Lett. Math. Phys. 106 (2016) 1499-1530, doi:10.1007/s11005-016-0880-4, arXiv:1601.00219; A. Devastato and P. Martinetti, *Twisted spectral triple for the Standard Model and spontaneous breaking of the grand symmetry*, Math. Phys. Anal. Geom. 20 (2017) 2, doi:10.1007/s11040-016-9228-7, arXiv:1411.1320. The twist bounds the unbounded vector terms of the

grand-symmetry model and introduces a twisted first-order condition; the breaking to the Standard Model becomes dynamical.

- **Classification of finite triples:** T. Krajewski, *Classification of finite spectral triples*, J. Geom. Phys. 28 (1998) 1-30, doi:10.1016/S0393-0440(97)00068-5, arXiv:hep-th/9701081 (Krajewski diagrams: vertices are chiral gauge multiplets, links are Yukawa couplings); M. Paschke and A. Sitarz, *Discrete spectral triples and their symmetries*, J. Math. Phys. 39 (1998) 6191-6205, doi:10.1063/1.532623, arXiv:q-alg/9612029 (classification of 0-dimensional triples over complex and real algebras). For the project, a Krajewski link is exactly the Yukawa block that lemma 12 derives as Phi_Q Y_hat Phi_U.
- **Fermion doubling:** F. Lizzi, G. Mangano, G. Miele and G. Sparano, *Fermion Hilbert space and fermion doubling in the noncommutative geometry approach to gauge theories*, Phys. Rev. D 55 (1997) 6357-6366, doi:10.1103/PhysRevD.55.6357, arXiv:hep-th/9610035: unphysical degrees of freedom in H = L^2(S) x H_F, analogous to lattice doublers; the 2006-2007 KO-6 framework resolves them (Connes 2006, Barrett 2007).
- **Standard reference:** W. D. van Suijlekom, *Noncommutative Geometry and Particle Physics*, Mathematical Physics Studies, Springer (2015), doi:10.1007/978-94-017-9162-5 (a second edition exists; its data were **not verified**). Covers finite spectral triples and Krajewski diagrams, almost-commutative manifolds, the spectral action, and the electrodynamics and Standard Model examples.

J. Matching in the flat parent (lemma 26)

Claim. Lemma 26: one-loop vacuum with frequency-2 channels; the left-right difference of the couplings; the meeting point; the limit on a top in low multiplets. **Verdict: PARTIAL.** Found by the referee of lemma 26 (report, item 9), with forward citation chains incomplete. We have read none of these beyond what that report quotes.

- Y. Nomura, D. Smith and N. Weiner, *GUT breaking on the brane*, arXiv:hep-ph/0104041: differential logarithmic running, integer versus half-integer towers. M. Carena, T. Tait and C. Wagner, arXiv:hep-ph/0207056: boundary kinetic terms.
- Abe, Goto and Kawamura, PTEP 2018, 103B01: an interval Pati-Salam model with a compactification threshold of order 10^13 GeV, with other intermediate group and matter.
- G. Cacciapaglia, C. Csaki and S. C. Park, arXiv:hep-ph/0510366, section 5: representation-dependent enhancement, including a tree-level m_t = 2 M_W limit. Adachi and Maru, arXiv:1804.06012, Eq. (3.23): a limit approaching 2 M_W with brane mixing, and massless exotics.
- R. Contino, L. Da Rold and A. Pomarol, arXiv:hep-ph/0612048: 5 and 10 embeddings and coloured custodians. Y. Hosotani, S. Oda, T. Ohnuma and Y. Sakamura, arXiv:0806.0480: massless exotics, their lifting by wall masses, and theta_H = pi/2 in a warped model.
- C. S. Lim and N. Maru, arXiv:0904.0304: a finite one-loop violation of gauge-Yukawa universality.

Not found by that search: the classification with n_c, and the pair (M_KK, mu_*). Priority for those exact statements stays open.

K. Fibre collapse in a mapping cylinder and compression (lemma 28)

Claim. Lemma 28: a connected commutative algebra on a mapping cylinder gives a direct sum of two matrix algebras by compression to the two ends. **Verdict: NOT LOCATED.** Found by Astra 6 in a bounded search (report); novelty stays unestablished, and author-by-author citation searches were not done. Identifiers, authors and titles below were confirmed on arXiv by Claude Opus 5.5; the located passages are as reported by Astra 6.

- F. D'Andrea, F. Lizzi and P. Martinetti, *Spectral geometry with a cut-off*, arXiv:1305.2605, section 3.1 and 6: compression of commutative geometries.
- A. Connes and W. D. van Suijlekom, arXiv:2004.14115, section 1 and Proposition 4.2: operator systems from spectral truncation, with the warning on non-informative full matrix algebras, relevant to Counterexample 28.17.

- J. Medina, I. Huet, D. O'Connor and B. Dolan, arXiv:1208.0348: squashed fuzzy CP^3 as a fuzzy S^2 bundle over fuzzy S^4, with suppression of the fibre modes.
- M. Verhoeven, *Spectral truncations of SU(2) and the spectral localizer*, arXiv:2610.19791: full matrix envelopes in a recent truncation example.
- K. Uno and T. Inagaki, *Construction and geometric properties of the fuzzy CP^3 onion*, arXiv:2610.17807: radially coupled fuzzy CP^3 layers.
- C. Bär and W. Ballmann, arXiv:1307.3021: the boundary criterion of Theorem 28.9. C. Kottke, arXiv:0909.5661: the Callias index in families, for Proposition 28.11.
- I. Prokhorenkov and K. Richardson, *Perturbations of Dirac operators*, J. Geom. Phys. 57 (2006) 297-321, arXiv:math/0307251; M. Maridakis, arXiv:1510.07004, Definition 2.1 and Lemma 2.2, and arXiv:2110.11654, section 2: concentration of low modes of D + s A at the zero set of A on closed manifolds. Read by Terra 6.1 in arXiv HTML (report): the operator of lemma 28 is a concentrating pair, and Assumption 3 of the spectral separation theorem fails for the spin-3/2 mass.
- R. Parthasarathy, *Dirac operator and the discrete series*, Ann. of Math. 96 (1972) 1-30, doi:10.2307/1970892, bibliographic data from Crossref: the formula D^2 = Cas_G + R/8 - Cas_K(twist) on a symmetric space, used in lemma 29. Text not read here; the formula is tested on the slice blocks in Python.
- K.-D. Kirchberg, *An estimation for the first eigenvalue of the Dirac operator on closed Kähler manifolds of positive scalar curvature*, Ann. Global Anal. Geom. 4 (1986) 291-325, doi:10.1007/BF00128050, bibliographic data from Crossref: the limit (m + 1) R / (4 m) for odd complex dimension m, which is R/3 on CP^3, used as a check of the normalization in Lemma 29.5. Text not read here.
- Found by the Terra referee of lemma 29, identifiers checked on arXiv by Claude: J. Kaad and W. D. van Suijlekom, *Factorization of Dirac operators on almost-regular fibrations of spin^c manifolds*, arXiv:1710.03182, Proposition 18: a horizontal and vertical factorization with an explicit curvature contraction, under hypotheses that differ from the noncompact model of lemma 29; a close structural ancestor of Proposition 29.1. I. Prokhorenkov and K. Richardson, *Witten deformation and the equivariant index*, arXiv:math/0610160, section 3: equivariant localization and normal models. The referee reports that the search found general identities and did not establish whether the eight-state theorem has appeared elsewhere; it did not exhaust Bismut-Lebeau or the citation chains.

Sources read on 4 November 2026

Read by Claude Opus 5.5 from PDFs supplied by the user (Banks-Casher, Dashen) and from the scan held by INSPIRE (Witten 1983).

- T. Banks and A. Casher, *Chiral symmetry breaking in confining theories*, Nucl. Phys. B 169 (1980) 103-125, doi:10.1016/0550-3213(80)90255-2, INSPIRE 143630. The paper works mainly with a second-order lattice formalism and a strong-coupling picture of trees. The spectral statement is a remark in Section 5, item (iv), p. 122: eq. (49), the integral of over spacetime equals the limit m -> 0 of the average of the integral of d mu rho(mu; A) m/(mu^2 + m^2), with rho the spectral function of D(A); eq. (50), the condition rho(mu; A) -> const as mu -> 0, on a finite fraction of the configurations. The paper states no factor pi, no division by the volume and no order of the limits m -> 0 and V -> infinity. Those belong to later statements, such as Giusti-Lüscher.
- R. Dashen, *Some features of chiral symmetry breaking*, Phys. Rev. D 3 (1971) 1879-1889, doi:10.1103/PhysRevD.3.1879, INSPIRE 61166. Section II.B, pp. 1881-1882: Theorem 1, the vacuum of H_0 + epsilon H' makes the expectation value of the rotated perturbation F(omega) stationary and at least a local minimum at omega = 0; Theorem 2, the same for the epsilon -> 0 limit of the vacuum. Whether omega = 0 is a global minimum is stated as a conjecture, with the existence of a global minimum of F noted separately. The ferromagnet in an external field is the worked example (eq. 15). This is the source for vacuum alignment; the stationarity calculation of lemma 17 is a separate derivation.

- E. Witten, *Fermion quantum numbers in Kaluza-Klein theory*, preprint Princeton, October 1983, 78 pages, in Shelter Island II (1985) 227-277, INSPIRE 193975, scan. Preprint page numbers. Pp. 4-5: bare masses of quarks and leptons are forbidden by gauge invariance, so they cannot have masses of order the Planck mass, and "we don't understand why some of the fermions are so much lighter than M_W". P. 7: if all gauge fields are part of the metric tensor, the Atiyah-Hirzebruch theorem gives vector-like quantum numbers for the Dirac operator on a compact B; with elementary gauge fields, complex representations are "possible but still subtle". P. 8, eq. (2): "We see that D^(n) is the mass operator, in effect. Its eigenvalues are observed in four dimensions as the particle masses. Its zero eigenvalues are the massless fermions." Pp. 43-44: the character-valued index is invariant under equivariant cobordism, so if M bounds a manifold to which the G action extends, it vanishes for any operator on M. This is the statement behind the balance on the boundary CP^3 of lemma 28: the summands of E restrict with indices 4, 0, 0, -4 and total zero. The hypotheses are a compact B without boundary and no mass profile; lemma 28 leaves them through the wall and through the Callias potential, with E as an elementary gauge bundle.

Items not verified in this session

- Castellani-D'Auria-Fré 1984 (section F): equation-level content not read.
- Banks-Casher 1980: read on 4 Nov 2026 from the PDF supplied by the user, in the passages below. The value pi rho(0) and the order of limits in the chiral limit note come from L. Giusti and M. Lüscher, JHEP 03 (2009) 013, arXiv:0812.3638, Section 2 (LaTeX source), which counts the modes of one flavour and states the order of limits lambda -> 0, m -> 0, V -> infinity. The original writes neither.
- Witten 1983: only the passages recorded below were read; the proofs of sections III to VI were not checked.
- Coquereaux-Esposito-Farèse-Vaillant 1991, Dubois-Violette-Kerner-Madore 1990 (section A): bibliographic data from secondary citations, content from secondary sources.
- Callan-Harvey 1985 and Arkani-Hamed-Cohen-Georgi 2001 (section H): full texts not read.
- Demmel et al. 1999: theorem numbers for the D_1 X D_2 results.
- Niemi-Slizovskiy, Section VIII (custodial fixing of xi).
- Connes-Moscovici journal reference; second edition of van Suijlekom's book.

Summary table

Claim	Verdict	Key references
A. Connes-Lott as minimal truncation of the chiral interval,	m	= pi/(2l)
B. l -> infinity (walls, 1D Weyl law) and l -> 0 with mu(l)	PARTIAL: wall modes and exponentially small mixing known; NCG statement and tuned collapse not found	Jackiw-Rebbi 1976; Kaplan 1992; Shamir 1993 §3; Csáki et al. 2004; Farsi-Latrémolière 2026 Cor. 4.1
C. Towerless textures Phi_Q Y_hat Phi_U, secular equation, one-flavour spectrum	PARTIAL: one-flavour spectrum KNOWN (Csáki et al. eq. (5.6)); multi-flavour warped analogue KNOWN (Casagrande et al. eqs. (65), (87), (90), (94)-(97)); NCG derivation of D_F not found	Csáki et al. 2004 §5.1, App. C.2; Casagrande et al. 2008 §3; Kaplan-Tait 2001; Arkani-Hamed-Schmaltz 2000
D. KO-6 = (odd) x (Clifford point) criterion; s from U; nu_L; reflection sign	PARTIAL: products, J vs J gamma, odd x odd product and 10 = 2 mod 8 known; factorization criterion and the 9/11 results not found	Dąbrowski-Dossena 2011 §4.3, eq. (25), Table 6; Vanhecke 1999; Vanhecke-da Silva-Sigaud 2012 §2; Farnsworth 2017; Connes 2006 §1; Barrett 2007

Claim	Verdict	Key references
E. 8 M_W^2 = sum N_f m_f^2; m_t/M_W = sqrt(8/3); heavy weights	KNOWN (mass relation, top ratio); NOT FOUND (exact f($m^{2/\mathrm{Lambda}}$2) weights)	CCM 2007 eq. (1.5), Lemma 4.6 eq. (4.16), §5.4 eqs. (5.24)-(5.29)
F. cos^2 theta_W = 3 xi^2/(4 xi^2 + 2)	KNOWN	Niemi-Slizovskiy 2010 eqs. (94)-(95), §VI; Witten 1981 (method)
G. Odd mod-two index on S^5	KNOWN (topology and footnote attribution)	García-Etxebarria-Montero 2019 §3.1; Witten 1982; Wang-Wen-Witten 2019
H. Anomaly inflow	KNOWN (background)	Callan-Harvey 1985; Naculich 1988; Arkani-Hamed-Cohen-Georgi 2001
I. Current framework	reference list	Connes 2006; Barrett 2007; CCM 2007; Chamseddine-Connes 2007, 2008; CCvS 2013 (two papers); Devastato-Lizzi-Martinetti 2014; Krajewski 1998; Paschke-Sitarz 1998; van Suijlekom 2015

Part R1 (assessment of Farsi-Latremoliere, with the counterexample to their Theorem 3.4)

Farsi-Latrémolière, "Collapse in noncommutative geometry and spectral continuity": referee assessment

Paper. C. Farsi and F. Latrémolière, arXiv:2404.00240: v1 30 Mar 2024, v2 13 Oct 2025, v3 15 Oct 2025. Published as J. London Math. Soc. (2) 114 (2026), no. 1, e70628, 54 pp. (journal reference on the arXiv record).

What was read. The LaTeX sources of v1 and v3, in full. The v2 and v3 sources are byte-identical. The journal page (doi:10.1112/jlms.70628) returned HTTP 403, so we have not seen the published text. Theorem numbers below are those of v3, with v1 numbers in brackets. Read by Claude Opus 5.5 on 3 Nov 2026 for lemma 09 (Theorems 9.8-9.10) and STRATEGY T0. **Update (3 Nov 2026):** the published text was read later that day, from the journal HTML supplied by the user; see Section 3.6.

Summary. - The convergence notion is Latrémolière's spectral propinquity on metric spectral triples (A, H, D). These carry no grading and no real structure. - One collapse is treated: D = D_h + D_v, with only D_v rescaled by 1/eps. The limit is (B, ker D_v, p D_h p), where B ⊆ A and p projects onto ker D_v. - The general Theorem 3.4 is false as stated: it needs a hypothesis that its proof uses but does not list (Counterexample R1). All of the paper's applications satisfy that hypothesis. - Corollary 4.1 (collapse to a point, limit (C, ker D, 0)) is sound. - Our Theorem 9.9 lies outside every theorem of the paper. The paper's method adapts to it (Lemma R2, ours). The limit is (C, C^2, m_f sigma_1), and the Connes-Lott two-point triple stays at distance at least about 1/(2 m_f). - For T0, limits of commutative triples in this metric have commutative algebras (Lemma R3, ours).

1. Setting and main results

1.1 Prerequisites (Section 2)

- **Spectral triple** (Definition 2.1). This is a unital C*-algebra A, a Hilbert space H that is a left A-module, and a self-adjoint D. The Lipschitz algebra A_D = {a : a Dom D ⊆ Dom D, [D, a] bounded}

is dense, and (D + i)^{-1} is compact. No grading, real structure or summability enters anywhere in the paper.

- **Quantum compact metric space** (Definition 2.3). This is a pair (A, L), with L a seminorm on a dense subspace of sa(A), such that:
  - ker L = R 1;
  - the Monge-Kantorovich metric mk_L(phi, psi) = sup{|phi(a) - psi(a)| : L(a) <= 1} metrizes the weak* topology on the state space;
  - the Leibniz inequality holds for Jordan and Lie products;
  - the unit ball {L <= 1} is closed.
- **Metric spectral triple** (the definition after 2.4). A spectral triple is metric when (A, L_D) with L_D(a) = ||[D, a]|| is a quantum compact metric space, i.e. when Connes' distance metrizes the weak* topology.
- **Tunnels, extent, propinquity** (Definitions 2.5-2.10).
  - A quantum isometry is a *-epimorphism pi with L_B(b) = inf{L_A(a) : pi(a) = b}.
  - A tunnel is a quantum compact metric space (D, L_D) with quantum isometries onto both spaces.
  - Its extent is the larger of the two Hausdorff distances, in mk_{L_D}, between S(D) and the pulled-back state spaces.
  - The propinquity is the infimum of extents over tunnels. The paper records that "it induces the same topology as the Gromov-Hausdorff distance on the class of classical metric spaces".
- **Metrical C*-correspondences** (Definition 2.13). A metric spectral triple yields (H, DN, A, L_D, C, 0), where DN(xi) = ||xi|| + ||D xi|| is the graph norm. Metrical tunnels and their extents follow in Definitions 2.16-2.17. Separation and dispersion of operator families are measured through pseudo-states xi -> mu(<omega, xi>) with DN(omega) <= 1 (Definitions 2.20-2.21).
- **Spectral propinquity** (Definition 2.22, taken from Latrémolière, Adv. Math. 404 (2022), and Math. Ann. 389 (2024), Theorem 3.5). The distance Lambda_spec between two metric spectral triples is the infimum of sqrt(2)/2 and of those eps > 0 for which some metrical tunnel has dispersion < eps between the families (exp(i t D_1))*{0 <= t <= 1/eps} and (exp(i t D_2))*{0 <= t <= 1/eps}.
- **Properties listed after Definition 2.22.**
  - "(1) Lambda_spec = 0 if the two metric spectral triples are unitarily equivalent." The paper states only this, the trivial direction. Latrémolière (Adv. Math. 2022) proves both directions: the distance is null exactly on unitarily equivalent pairs, and it dominates the propinquity of the underlying quantum metric spaces.
  - "(2) if Lambda_spec(T_n, T_infinity) -> 0 then for all bounded continuous f, Lambda_op(f(D_n), f(D_infinity)) -> 0, and Sp(D_infinity) = {lim lambda_n : (lambda_n) convergent with lambda_n in Sp(D_n) for all n}."

1.2 The collapse setting and Theorem 3.4 [v1: Theorem 2.5] A unital C*-subalgebra B ⊆ A plays the base, and a positive map E : A -> B plays the bundle projection. The Dirac operator splits as D = D_h + D_v, and only the vertical part is scaled: D_eps = D_h + (1/eps) D_v. The limit object is (B, ker D_v, D_B) with D_B = p D_h p. Quoted, with notation transcribed:

> **Theorem 3.4.** Let (A, H, D) be a metric spectral triple, and let B ⊆ A be a unital C*-subalgebra of A, and such that D = D_h + D_v, where D_v is self-adjoint and such that 0 is isolated in Sp(D), together with the following assumptions. Setting D_eps := D_h + (1/eps) D_v for all eps in (0, 1), the triple (A, H, D_eps) is a spectral triple, such that: 1. for all eps in (0,1), ||[D_h, a]|| <= ||[D_eps, a]||, 2. there exists M > 0 such that for all eps in (0,1), ||[D_v, a]|| <= M eps ||[D_eps, a]||, 3. [D_v, b] = 0 for all b in B, 4. writing p for the projection onto ker D_v, we assume that [p, b] = 0 for all b in B and [D_h, p] = 0, 5. (B, ker D_v, D_B), where D_B := p D_h p, is a metric spectral triple, 6. there exists a positive linear map E : A -> B, whose restriction to B is the identity, and a constant k > 0 such that for all a in A: ||a - E(a)|| <= k ||[D_v, a]||, and ||p [D_h, E(a)] p|| = ||[D_h, E(a)]|| <= ||[D_h, a]||.
>
> Then (A, H, D_eps) is a metric spectral triple, and lim_{eps -> 0+} Lambda_spec((A, H, D_eps), (B, ker D_v, D_B)) = 0.

"0 is isolated in Sp(D)" is a typo for Sp(D_v). The proof uses a gap delta with Sp(D_v) ∩ (-delta, delta) = {0}. The typo is present in v1, in v3 and in the introduction.

Structure of the proof. - The quantum-metric tunnel is A ⊕ B with seminorm max{L_eps(a), L_B(b), ||a - b||/(k M eps)}, and its extent is at most k M eps. - The module tunnel is H ⊕ ker D_v with norm max{DN_eps(xi), DN_B(eta), ||p xi - eta||/K_eps}. Here K_eps = max{k M eps, 4 sqrt(2 eps/delta)}, and the 4 sqrt(2 eps/delta) comes from a Fourier estimate (Lemma 3.3). - Separation is controlled through exp(i t D_eps) p = exp(i t p D_h p) (Lemma 3.1), uniformly in t. - The resulting bound is Lambda_spec <= alpha_eps + K_eps = O(sqrt(eps/delta)).

1.3 Meaning of "spectral continuity" (Corollary 3.5 [v1: 2.7])

> **Corollary 3.5.** Under the same hypothesis as in Theorem 3.4, for any f in C_b(R), and for any sequence (eps_n) in (0, infinity) converging to 0, we have lim_n Lambda_op(f(D_{eps_n}), f(D_B)) = 0, and in particular, Sp(D_B) = {lim_n lambda_n : (lambda_n) convergent with lambda_n in Sp(D_{eps_n}) for all n}.

The corollary has two parts. - The bounded continuous functional calculus converges in Latrémolière's operational propinquity, a metric between operators on different metrical C*-correspondences. - The spectra converge as sets. The statement holds for every sequence eps_n -> 0, so applying it to subsequences gives Kuratowski convergence of Sp(D_{eps_n}) to Sp(D_B).

Multiplicities and eigenvectors are not addressed. The corollary is a direct citation of Latrémolière, Math. Ann. 2024.

1.4 Applications

result (v3 [v1])	data	limit	anticommutation of D_h and D_v
Corollary 4.1 [3.1]	D_h = 0, D_v = D, B = C	(C, ker D, 0)	trivial
Theorem 4.6 [4.5]	(A, H, D) even with grading gamma, (B', J, S) metric, one of them Abelian; D_eps = D x 1 + (1/eps) gamma x S	(A, H x ker S, D x 1)	yes (gamma anticommutes with D)
Corollary 4.7 [4.6]	odd version via two Clifford generators gamma_1, gamma_2 on E	(A, H x ker S x E, D x 1 x gamma_1)	yes
Theorem 5.5 [5.5]	Schwieger-Wagner triple on a free action of compact Lie G on A, B = A^G; D_v = sum_j d_j x gamma_j, D_h = (…) x gamma_0	(B, H_B x H_spin, D_B x gamma_0)	yes (Remark 5.1)
Corollaries 5.6, 5.7	B x_alpha Z^d with equicontinuous alpha; quantum 4-torus collapsing to a quantum 2-torus	as in 5.5	yes

result (v3 [v1])	data	limit	anticommutation of D_h and D_v
Theorem 6.3 [6.3]	M -> N a principal U(1)-bundle (projectable spin structure, constant fibre length, dim N even); D_eps = (1/l_eps) D_v + D_h	(C(N), L^2(Sigma N), D_N)	yes (Ammann-Bär)

2. Section 4 and Corollary 4.1: the shape of the limit

> **Corollary 4.1** [v1: 3.1]. If (A, H, D) is a metric spectral triple with 0 in Sp(D), then lim_{eps -> 0} Lambda_spec((A, H, (1/eps) D), (C, ker D, 0)) = 0.

The proof takes D_h = 0, D_v = D, B = C and E = mu, a state, and uses ||a - mu(a)|| <= diam |||[D, a]|||. The paper follows it with this remark:

> If 0 is not in Sp(D), then there is no bounded sequence (lambda_n) with lambda_n in n Sp(D) […] So by Corollary 3.5, it is not possible for (A, H, n D), and even less (A, H, (1/eps) D), to converge to anything for the spectral propinquity.

The limit object. The limit is a point (algebra C, state space a single point) with Hilbert space ker D, which is finite-dimensional because D has compact resolvent, and with zero Dirac operator. Before the corollary the paper puts it this way: "(C, C^n, 0) […] arise naturally as limits […] of arbitrary metric spectral triples collapsed to a point. Moreover, in that case, n is the dimension of the space of harmonic spinors." This corollary does not depend on the defect in Section 3.1 below, since D_h = 0. A direct proof is short: - the quantum-metric extent is at most eps diam(A, D); - by the spectral theorem, ||xi - p xi|| <= ||D xi||/delta = eps ||(1/eps) D xi||/delta; - so Lambda_spec = O(eps).

The product case (Theorem 4.6). The limit is (A, H x ker S, D x 1), i.e. the base triple with multiplicity dim ker S and no internal Dirac operator. The internal factor survives only as the multiplicity space ker S.

Can a nontrivial finite part survive? - **Finite algebra.** No. The limit algebra is always B ⊆ A, acting on ker D_v, so the finite part never carries an algebra of its own. - **Finite Dirac operator.** Possibly yes. The limit operator is p D_h p, and it can have a nonzero finite piece when D_h contains a bounded term that commutes with A, preserves ker D_v and anticommutes with D_v. Example (ours, not in the paper): - Take A = C(S^1), H = $L^{2(S}$1) x C^2 x C^k, D_v = -i d/dtheta x sigma_1 x 1 and D_h = 1 x sigma_2 x Phi with Phi = Phi* a k x k matrix. - All six hypotheses hold with B = C and E(f) = (1/2pi) integral of f, and D_h anticommutes with D_v. - The limit is (C, C^2 x C^k, sigma_2 x Phi), a finite spectral triple over C with spectrum ±Sp(Phi). - The paper contains no example with p D_h p != 0 and B = C.

3. Skeptical reading

3.1 Theorem 3.4 is missing a hypothesis Counterexample R1. - Let H have orthonormal basis e_0, e_1, e_2, …, and let A = B = C act by scalars. - Set D_v e_n = n e_n and D_h e_n = -n^2 e_n for n >= 0. Then D_eps e_n = (n/eps - n^2) e_n, which has compact resolvent for every eps. - D_v has spectral gap 1 at 0, and D = D_1 has Sp(D) ∩ (-2, 2) = {0}. The projection p onto ker D_v = C e_0 commutes with D_h. - Hypotheses (1)-(6) hold trivially, since commutators with scalars vanish and E = id. (C, C e_0, 0) is a metric spectral triple. - Theorem 3.4 and Corollary 3.5 would therefore give Sp(D_B) = {0}. - Fix x > 0 and let eps_n = 1/(n + x/n) -> 0. Then D_{eps_n} e_n = x e_n, so the constant sequence x lies in Sp(D_{eps_n}), and Corollary 3.5 would put x in Sp(D_B) = {0}.

Corollary 3.5 follows from Theorem 3.4 through Latrémolière's published continuity theorem, so the conclusion of Theorem 3.4 fails for this family.

The failing step is equation (main-eq-1) in the proof (v3 only):

```
||xi - p xi|| <= 4 sqrt(2 eps/delta) (||xi|| + ||(1/eps) D_v xi||) <= 4 sqrt(2 eps/delta)
↪  DN_eps(xi).
```

The second inequality needs ||(1/eps) D_v xi|| <= ||D_eps xi||, which is neither assumed nor proved. For xi = e_n at eps = eps_n, the left-hand side is 1 while the right-hand side is 4 sqrt(2 eps_n)(1 + x) -> 0.

Repair. Add the hypothesis that D_h and D_v anticommute on a common core. Then ||D_eps xi||^2 = ||D_h xi||^2 + eps^{-2} ||D_v xi||^2. A relative bound ||D_v xi|| <= C eps (||xi|| + ||D_eps xi||) also suffices. With anticommutation, the spectral theorem gives ||xi - p xi|| <= ||D_v xi||/delta <= (eps/delta) ||D_eps xi||; with the relative bound it gives ||xi - p xi|| <= (C eps/delta)(||xi|| + ||D_eps xi||). (The estimate for the relative bound is corrected on 3 Nov 2026, as Astra 6 noted.) Either one makes the Fourier Lemma 3.3 unnecessary and improves the rate from O(sqrt(eps)) to O(eps).

All applications in Section 1.4 satisfy anticommutation, so they are not affected by this defect. Theorem 3.4 itself, as printed in both arXiv versions and in the journal (Section 3.6), may not be cited without the extra hypothesis.

3.2 Other points in Section 3

- **Lemma 3.1** [2.1]. v3 added the hypothesis "D and D + S both have discrete spectra". The lemma is applied with D = D_h, and the paper itself says D_h has no compact resolvent in Section 5. The conclusion holds without discreteness: pH reduces both operators, and D_eps p = D_h p. So the proof is fixable and the statement is fine.
- **Inner Leibniz.** Definition 2.13 is ill-typed: it applies L to the non-self-adjoint inner product and names L_A where L_B is meant (the journal text corrects L_A to L_B and keeps the rest). Latrémolière's own definition (arXiv:1811.10843) bounds the real and imaginary parts. Noticed by the referee of lemma 15; the proofs are unaffected.
- **Time window.** The proof concludes with separation over 0 <= t <= M_eps where 1/M_eps is meant. The estimate is uniform in t, so this is harmless. "1/t-covariant reach" is leftover terminology from Latrémolière 2022.
- **Hypotheses that are strong or hard to verify in examples.**
  - (4) requires D_h to preserve ker D_v exactly. A mass or Yukawa term that mixes zero modes with excited modes violates it.
  - The exact kernel with a fixed gap excludes near-zero modes.
  - The fixed D_h excludes horizontal operators that change with eps: the Ammann-Bär term V_eps, curvature terms, non-constant fibre length.
  - (6) requires the horizontal derivative of base elements to be seen entirely on ker D_v.
  - (2) is automatic in Clifford-graded constructions and otherwise needs an argument.

3.3 Sections 4-6: sketched or delegated steps, and asserted examples

- **Theorem 4.6.** The check of hypothesis (6) is garbled; it bounds |phi x psi(c - E(c))| by "mk_D(phi, phi) + eps mk_S(psi, mu)". The needed bound ||c - E(c)|| <= diam(B', S) |||[1 x S, c]|| follows from Lemma 4.4. The statement writes "gamma x S" for gamma ⊗ S.
- **Section 5.** The spectral triple, its compact resolvent and the two sup-bounded commutator conditions are taken from Schwieger-Wagner (Adv. Math. 396 (2022)). Its representation is "a technical matter to check" (Section 5.3).
  - Lemma 5.2 uses |||[D, alpha^g(a)]|| = |||[D, a]|| for D in {D_h, D_v, D_A}.
  - For D_v = sum_j d_{e_j} x gamma_j this needs u^g D_v u^{g-1} to be unitarily equivalent to D_v by a unitary commuting with A.
  - Such a unitary exists when the inner product on the Lie algebra of G is Ad-invariant, but the paper fixes only "some inner product" (a left-invariant metric).
  - All worked examples are tori, where the issue is absent. We have not checked Schwieger-Wagner's conventions to see whether they impose Ad-invariance.
- **Section 6, Theorem 6.3.**

  - (a) The theorem concerns D_eps = (1/l_eps) D_v + D_h. The Dirac operator of the shrunk metric is D_{M_eps} = D_eps + V_eps. The bounded term V_eps tends to 0 under the Ammann-Bär condition, but the passage to D_{M_eps} is not made (the abstract says "a bounded variation").
  - (b) The verification of the hypotheses is sketched and contains errors. Item (0) checks that 0 is isolated in Sp(D_eps) instead of Sp(D_v), and argues self-adjointness from "one of them being bounded", although both summands are unbounded. Item (1) cites the Fourier Lemma 3.3, which is irrelevant, and then Proposition 3.6, whose hypotheses (constant anticommuting unitaries commuting with the D_j) do not literally cover local Clifford frames. Item (2) "proves" [D_v, b] = 0 by displaying [D'_{h,k}, b] = 0, which is false for nonconstant b (it is Clifford multiplication by grad b). Item (3) cites an unrefereed web note (Orduz). Item (5) is delegated ("Verified in the same way as in the proof of Theorem 5.5"), and its displayed inequalities swap D_h and D_v.
  - The conclusions are correct classically, by pointwise Clifford anticommutation and fibre averaging, but the written proof does not establish them.
  - (c) Odd dim N is asserted "mutatis mutandi", and non-constant fibre length is left "to the interested reader".
- **Asserted examples.** The introduction of v1, of v3 and of the journal text lists "homogeneous spaces of compact Lie groups" (the abstract does not; an earlier version of this line said it did). The corresponding passage is commented out in both sources, so no such result is proved.

3.4 Dependencies The proofs rest on published work: - Latrémolière, Adv. Math. 404 (2022) and Math. Ann. 389 (2024); - Farsi-Latrémolière-Packer, Adv. Math. (Paper No. 109442); - Rieffel 1998, 1999, 2005; - Schwieger-Wagner, Adv. Math. 396 (2022); - Ammann 1999 and Ammann-Bär 1998.

No step relies on an unpublished or forthcoming result of the same authors. The preprints Schwieger-Wagner arXiv:2406.08145, Kaad arXiv:2410.03475 and Mesland-Rennie are cited for context only. One proof step (Theorem 6.3, item (3)) cites the web note by Orduz.

3.5 Changes from v1 to v3 v2 and v3 have identical sources. The arXiv comment says: restructured, introduction split, "Section 2 results were cleaned up and some statements corrected", references changed. From the diff:

- **Restructuring.** The background moved from the introduction into a new Section 2. Numbering shifts:
  - main theorem 2.5 -> 3.4;
  - spectral corollary 2.7 -> 3.5;
  - "Collapsing to a point", a separate section in v1 (Corollary 3.1), became subsection 4.1 (Corollary 4.1);
  - product theorem 4.5 -> 4.6;
  - comparison Lemma 2.4 -> Proposition 3.6.
  - Sections 5 and 6 keep their numbers.
- **Theorem 3.4.**
  - The statement is unchanged in substance: the same six hypotheses and the same Sp(D) typo.
  - v1's Remark 2.6 on domains ("Dom D = Dom D_h ∩ Dom D_v") was removed.
  - In the proof, v1's (main-eq-1), ||xi - p xi|| <= 4 sqrt(2 eps/delta), had no norm of xi on the right. v3 added (||xi|| + ||(1/eps) D_v xi||) <= DN_eps(xi), which makes the missing step of Section 3.1 explicit without justifying it.
  - v3 added the argument that p commutes with D_eps, so that DN_B(p xi) <= DN_eps(xi).
  - v3 corrected the circular "M_eps := alpha_eps + M_eps" to alpha_eps + K_eps.
- **Lemma 3.1.** The discrete-spectra hypothesis was added; v1's proof had silently used compact resolvent.
- **Corollary 3.5.** The stray notation D_{x,eps_n} was corrected to D_{eps_n}.
- **Lemma 4.4** [4.3]. v1's proof argued that psi([S, b_j]) = 0 for an extended state, which fails for non-self-adjoint b_j. v3 replaced it with an argument using the grading. The roles of phi and psi in

the statement were corrected.

- **Section 5.** Conjugate representations were corrected (Phi_sigma : Gamma_A(sigma) x conj(V_sigma) -> A(conj sigma)), with a footnote reporting a typo in Schwieger-Wagner Eq. (16). Lemma 5.2 was extended to the L^1 bound for all three operators.
- **Section 6.** v1's Lemma 6.4 was folded into the proof with its errors unchanged. v1's Remark 6.5, which claimed the method covers Roos's weaker collapse condition, is commented out in v3. That claim has been withdrawn.
- The abstract was rewritten and references were added.

3.6 Published version The arXiv record carries the journal reference, and an announcement page lists the paper as accepted. v2/v3 (Oct 2025) is therefore probably the accepted text. We could not see the journal version, so whether the hypothesis missing from Theorem 3.4 was added in print is unknown.

Update (3 Nov 2026). The journal text (J. London Math. Soc. (2) 114 (2026) e70628, first published 3 July 2026) has now been read, and the hypothesis was not added. We compared its prose word by word with the v3 source (Python) and read the following statements directly: Theorem 3.4, the step (main-eq-1), the constant K_eps, Corollaries 3.5 and 4.1, and the remark after 4.1. In those statements and in the prose, the differences are copy-editing: macros written out, step headings in the proof of Theorem 3.4, and L_A corrected to L_B in Definition 2.13. Astra 6 compared the main statements of Sections 3 to 6 and found no further change (validation, item 4); other equations were not compared. The printed Theorem 3.4 has the same six hypotheses and the same typo "0 is isolated in Sp(D)". The step of Section 3.1 is printed as equation (3.11). So Counterexample R1 applies to the printed statement, and the conclusions above on v3 carry over to the statements compared. Anticommutation and the relative bound are sufficient repairs, as Terra 6.1 and Astra 6 both note; the anticommuting applications avoid this defect. Astra 6 also notes that Theorem 4.6 and Corollary 4.7 need a nonzero vertical kernel for the displayed target to be a spectral triple, in v3 and in print.

4. Relevance to lemma 09 and STRATEGY T0

4.1 Our interval as a metric spectral triple **The framework's prerequisites hold.** T_l = (C([0,l]), L^2([0,l]; C^2), D_M), with condition (c), is a metric spectral triple: - the boundary conditions sit in Dom D_M, and multiplication by C^1 functions preserves them; - the mass term M sigma_1 commutes with C([0,l]), so ||[D_M, f]|| = ||f'||_inf and Connes' distance is |x - y|; - compact resolvent is Lemma 9.1.

What the framework does not see. T_l is even with grading Gamma, but the spectral propinquity ignores Gamma. No chirality statement (Theorem 9.2, P_0 Gamma P_0 = sigma_3 in Theorem 9.6) is visible to it, and neither is a real structure or a KO-dimension.

Boundary identity. Write D_M = D_0 + M sigma_1 with D_0 the massless operator. For psi = (L, R) in Dom under condition (c):

```
||D_M psi||^2 = ||D_0 psi||^2 + M^2 ||psi||^2 + M (|R(l)|^2 + |L(0)|^2),
```

since 2 Re <D_0 psi, sigma_1 psi> = [|R|^2]_0^l - [|L|^2]_0^l. After the dilation x = l y, D_M = (1/l) D_0^{[0,1]} + M sigma_1. This is the paper's form with eps = l, D_v = D_0 and D_h = M sigma_1. The anticommutator {D_0, sigma_1} vanishes in the interior, and its quadratic form is the boundary term above. So the hypothesis missing from Theorem 3.4 fails for our interval exactly by the term that creates the wall modes: - For M = -mu the term is negative, and Theorem 9.4.1 gives one eigenvalue below mu. - For M = +mu the identity gives ||D_M psi||^2 >= ((pi/(2l))^2 + mu^2) ||psi||^2. This is a one-line proof of the bound in Theorem 9.4.3.

4.2 Corollary 9.10 (fixed mu, l -> 0) In dilated form this is the paper's setting with D_v = D_0 under condition (c). Hypothesis 0 in Sp(D_v) fails, because ker D_0 = 0 (Theorem 9.2(c)). - D_l is a bounded perturbation, of norm mu, of (1/l) D_0, so every eigenvalue diverges. - The remark after Corollary 4.1, together with Corollary 3.5, then excludes any limit in the spectral propinquity (apart from a degenerate triple on the zero Hilbert space).

This agrees with Corollary 9.10. **The paper covers it** only in this indirect sense: its remark treats pure rescalings, and the bounded-perturbation step is ours.

4.3 Theorem 9.9 (mu(l) tuned, l -> 0) Here D_l = (1/l)(D_0 - c(l) sigma_1) with c(l) = mu(l) l -> infinity. In the paper's form D_h(l) = -mu(l) sigma_1 has norm mu(l) ~ log(1/l)/l -> infinity, and D_v = D_0 has no kernel. Three hypotheses of Theorem 3.4 fail: a fixed D_h, an exact kernel of D_v, and anticommutation. **The paper says nothing about this family**; this confirms the earlier note in LITERATURE.md. Its method does adapt:

Lemma R2 (ours, proof sketch; to be written out and refereed). - **Statement.** Let P_l project onto the light pair ±m_f of Theorem 9.9, and let W : P_l H -> C^2 be the coordinates in the basis (u_L, u_R) of Theorem 9.6. Then Lambda_spec(T_l, (C, C^2, m_f sigma_1)) <= 2l for small l. - **Quantum-metric tunnel.** Take C([0,l]) ⊕ C with seminorm max{||f'||_inf, ||f - z||/l}. Its extent is at most l, since Connes' metric of D_l is the Euclidean one. - **Module tunnel.** Take H ⊕ C^2 with norm max{DN_l(xi), DN(eta), ||W xi - eta||/l}. - **Key estimates.** - P_l is a spectral projection of D_l, and the next eigenvalue satisfies lambda_1(l) >= sqrt(mu^2 + (pi/l)^2) >= pi/l. Hence ||xi - P_l xi|| <= ||D_l xi||/lambda_1 <= l ||D_l xi||/pi, and exp(i t D_l) P_l = W* exp(i t m_f sigma_1) W for all t. - The modular Leibniz inequality needs no commutation of P_l with C([0,l]), because the limit algebra acts by scalars: ||W((f - z) xi)|| <= ||f - z|| ||xi||. - Inner Leibniz and separation follow as in the paper's proof, uniformly in t. - **Consequences.** - Theorem 9.9 becomes a spectral-propinquity limit with rate O(l): the light sector survives as a finite triple over C with D_F = m_f sigma_1. - The limit algebra is C. The two-point algebra C ⊕ C of Theorem 9.6 is the range of the compression f -> P_l f P_l, and the limit object does not contain it. - |diam_A - diam_B| <= 2 Lambda for any tunnel. The Connes-Lott triple (C ⊕ C, C^2, m_f sigma_1) has diameter 1/m_f, and (C, C^2, m_f sigma_1) has diameter 0. Hence Lambda_spec(T_l, Connes-Lott(m_f)) >= 1/(2 m_f) - 2l. - So in this metric the tuned interval converges to a point carrying a two-dimensional massive Dirac block, and it stays at a fixed positive distance from the Connes-Lott triple. - **Wording consequence for lemma 09.** The sentence "the light pair stays Connes-Lott with mass m_f, while the interval collapses to a point" in Theorem 9.9 is correct as a statement about the compression at each l. It should not be read as convergence to the Connes-Lott triple.

4.4 Theorem 9.8 (l -> infinity at fixed mu) Outside the paper's scope. The propinquity concerns compact quantum metric spaces. The diameter l diverges, and |diam_A - diam_B| <= 2 Lambda rules out any limit. Decompactification would need a pointed or locally compact version of the metric, and we did not check whether one exists in Latrémolière's work.

4.5 As the convergence notion for T0 Correction (3 Nov 2026). The conclusions of this subsection on what T0 "needs" overread the metric; see the assessment of the T0 note. Lemmas R2 and R3 are written out in lemma 15, which also corrects the lower bound of R2 (the cap sqrt(2)/2 of the metric was omitted, and the bound improves to (1/m_f - l)/2). The original text follows as a record.

What it offers. - A genuine metric on metric spectral triples, with distance zero exactly at unitary equivalence. - Convergence of the bounded functional calculus, and Kuratowski convergence of spectra. - Corollary 4.1, Theorem 4.6 and Lemma R2 show it handles towerless limits whose finite part is a multiplicity space carrying a finite Dirac block over the base algebra. - Latrémolière, arXiv:2603.19128 (Mar 2026; abstract only read), proves spectral-propinquity continuity for almost-commutative manifolds M x F as the metric and D_F vary. It is the nearest existing treatment of M x F in this metric.

What it cannot do for T0.

Lemma R3 (ours, proof sketch). - **Statement.** If commutative quantum compact metric spaces (A_n, L_n) converge in the propinquity to (A, L), with Leibniz tunnels as in Definition 2.7, then A is commutative. - **Proof sketch.** - Let a_1, a_2 be self-adjoint with finite L. Take a tunnel (D, L_D, pi_A, pi_n) of extent chi_n -> 0, and lifts d_i with pi_A(d_i) = a_i and L_D(d_i) <= L(a_i) + 1. - For self-adjoint e in D, every state of A pulls back to within chi_n of a state of A_n. Hence ||pi_A(e)|| <= ||pi_n(e)|| + chi_n L_D(e), and similarly ||d_i|| <= ||a_i|| + chi_n L_D(d_i). - Apply this to the Lie product e = (d_1

d_2 - d_2 d_1)/2i. Then pi_n(e) = 0 because A_n is commutative, and L_D(e) <= L_D(d_1) ||d_2|| + ||d_1|| L_D(d_2) by the Leibniz rule. - So |||[a_1, a_2]||| <= C chi_n -> 0. Self-adjoint Lipschitz elements are dense, so A is commutative. - **Consequence.** The spectral propinquity dominates the propinquity, so every spectral-propinquity limit of commutative triples has a commutative algebra.

Implications for the T0 family (C^infinity(K_l), L^2(S x E_l), D_l, J_l, gamma_l): - (a) Its limits in this metric can never carry a noncommutative A_F. - (b) With K_l connected and D_l of Dirac type (Clifford symbol plus a zero-order endomorphism), Connes' metric on points is the Riemannian distance. A collapse of K_l therefore yields the base algebra alone, and the finite part appears as (C, H_F, D_F) tensored over the base. Several sheets (C^k) need disconnected K_l or a non-local D_l. - (c) Gamma and J are not part of the data. KO-dimension and chirality would need an extension of the metric, for instance by adding gamma and J to the families whose separation is measured. We found no such extension in this paper. - (d) Noncommutative limits do occur in Section 5, but only when the parent algebra is already noncommutative (the limit is its fixed-point algebra), as in Corollary 5.7.

So spectral propinquity can serve T0 for spectral and metric convergence of the Dirac data. Producing the finite algebra needs another mechanism: compressions or operator systems (Connes-van Suijlekom truncations, STRATEGY T6), or a noncommutative parent algebra.

Verdict

We may cite the paper for: - the framework of vertical collapse in the spectral propinquity (Sections 2-3, as a notion); - Corollary 4.1: a fixed metric spectral triple scaled by 1/eps converges to (C, ker D, 0), and to nothing when ker D = 0. This is sound and independent of the defect below; it supports Corollary 9.10 in the indirect sense of Section 4.2; - Theorems 4.6 and 5.5 (for tori) as statements in their anticommuting settings; - Theorem 6.3 as a statement about D_h + D_v/l_eps, with the caveats of Section 3.3.

For spectral continuity we should cite Latrémolière, Math. Ann. 389 (2024), directly.

We may not cite it for: - Theorem 3.4 in its printed generality, since it needs anticommutation of D_h and D_v or a relative bound (Counterexample R1); - anything about Theorem 9.9, which lies outside all its hypotheses; - l -> infinity (Theorem 9.8); - grading, chirality, real structure or KO-dimension; - Connes-Lott (or any noncommutative finite triple) as a limit; - homogeneous spaces of compact Lie groups; - the actual Dirac operator of the shrunk U(1)-bundle metric.

Open doubts. - We have not seen the published version, which may have repaired Theorem 3.4. *(Answered on 3 Nov 2026: the published version keeps it; see Section 3.6.)* - Lemma R2 (tuned interval -> (C, C^2, m_f sigma_1), at distance at least 1/(2 m_f) - 2l from Connes-Lott) and Lemma R3 (commutative limits are commutative) are our sketches. Both are short, and both bear directly on lemma 09 and on T0, so they should be written out and refereed before use. - The G-invariance step in Lemma 5.2 for nonabelian G is unverified.

Part R2 (referee report, lemmas 08, 13, 14)

Referee report on lemmas 08, 13 and 14 and on the matching parts of the paper

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_08_13_14.md. Reproduced as delivered, with links relativized. The corrections are incorporated in the lemmas and in the paper. Claude checked by hand or with numpy: the counterexample to the unqualified neutrino claim, the invisible profile cos(2 pi x/l), and the ratio 27/26.

We count the 29 numbered results in Lemmas 08, 13 and 14. The paper is assessed separately. Our checks below are analytic; we ran no numerical computation.

The spectral-action calculations use the ordinary trace over the stated finite Hilbert space and the leading four-dimensional a_4 coefficient. The relevant coefficient is in Chamseddine–Connes–Marcolli, arXiv v1, Theorem 6.1, equation (6.10). The KO-product comparison uses the conventions in Dąbrowski–Dossena, arXiv v1, equations (18) and (25).

Lemma 08

08.1 — Classification under pointwise reality **Correct.** Write $J_I = WK$. Pointwise reality makes W a multiplication operator. Comparing derivative and zeroth-order terms in $J_I D_I = tD_I J_I$ gives $WE = tEW$ and $W' = 0$. The endpoint conditions then force W to be diagonal, and $WE = tEW$ gives $\alpha = t\beta$. Thus the signs are $(+, t, +)$.

For $t = -1$, the designation 0_ is an extended convention; the standard Connes KO-0 signs have $t = +1$.

08.2 — Spectral compression **Correct.** The relation $J_I D_I = tD_I J_I$ implies commutation with D_I^2 and its spectral projections. Solving the boundary problem gives eigenvalues $\pm(r - \frac{1}{2})\pi/l$, $r \geq 1$. Direct differentiation of the displayed normalized profiles gives $D_I u_L = m u_R$ and $D_I u_R = m u_L$, with $m = \pi/(2l)$. The diagonal antiunitary compresses as claimed.

08.3 — Reflection sign **Correct.** Reflection reverses the derivative and exchanges the endpoint conditions. For the off-diagonal phase matrix, commutation with $E\partial_x$ requires $a = b$; involutivity then gives $a = b = \pm 1$. The relation $\alpha = t\beta$ gives $\sigma J_I = tJ_I\sigma$, and the lowest profiles are exchanged, so the compressed reflection is $\pm\sigma_1$.

08.4 — Two counterexamples **Correct.** The antiunitary $\sigma_1 R_l K$ preserves the domain, commutes with D_I, squares to 1, and anticommutes with Γ_I. Its opposite action is multiplication by the reflected function, so order zero and first order hold for the scalar interval algebra.

For the compressed full diagonal algebra, $e^o = e$ for $e = \mathrm{diag}(1, 0)$, and

$$[[m\sigma_1, e], e] = m\sigma_1 \neq 0.$$

The first-order failure is therefore exact.

08.5 — Internal-first KO-6 product **Correct.** For the stated grading-twisted tensor structures,

$$J_{a,b}^2 = (-1)^a, \qquad J_{a,b}(\Gamma_0 \otimes D_I) = -t(-1)^b(\Gamma_0 \otimes D_I)J_{a,b}.$$

The standard KO-6 signs therefore require $a = 0$ and $t(-1)^b = -1$. Since $\sigma\Gamma_I = -\Gamma_I\sigma$, the bare reflection has

$$(1 \otimes \sigma)J_F = -J_F(1 \otimes \sigma).$$

This conclusion uses the stated product convention and the specified class of real structures.

08.6 — Grading and lift of reflection **Correct.** The interval grading alone commutes with the displayed tensor real structures, so it cannot supply the standard KO-6 grading. For $U_1 = \Gamma_0 \otimes \sigma$,

$$[U_1, D_F] = [\Gamma_0, D_0] \otimes \sigma = 2\Gamma_0 D_0 \otimes \sigma,$$

which vanishes exactly when $D_0 = 0$. The displayed unitary W transports both D_F and U; consequently it preserves the sign against the transported J.

08.7 — Reconciliation with the odd Clifford point **Correct.** With the scalar interval algebra, the displayed map R is unitary onto the positive-grading space and gives

$$R^*UD_FR = D_0 + m\Gamma_0, \qquad R^*UJ_FR = J_0\Gamma_0.$$

The latter antiunitary squares to -1 and anticommutes with $d_5 = D_0 + m\Gamma_0$. Its opposite action is that of J_0, and $[d_5, a] = [D_0, a]$, so the order conditions descend.

In the resulting two-block basis, the matrices are those of the odd–odd product with $B = -i\sigma_2$. The zero Dirac operator on the point makes the P_1 label a KO-sign convention. The factorization does not extend to the full compressed two-sheet algebra with its inherited real structure: 08.4 shows the first-order failure there.

08.8 — Literal collapse **Correct.** Anticommutation gives

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2 \geq \frac{\pi^2}{4l^2}.$$

Thus every fixed finite energy window is empty for sufficiently small l. This is a conclusion about the stated product, with no compensating bulk mass or modified cutoff prescription.

08.9 — Spacetime grading twists **Correct.** The two summands of $D_M \otimes 1 + \gamma_M \otimes d$ have signs $(-1)^a$ and s against the proposed real structure, so $(-1)^a = s$ gives the stated odd-product convention. Its square is $-s$, giving the claimed KO-1 and KO-3 signs. The displayed unitary V also transports the complete products because $\sigma_1 B_s = s B_s \sigma_1$. Neither calculation selects a finite involution.

08.10 — Euclidean and Lorentzian charge conjugation **Correct.** The stated Euclidean and Lorentzian operators have the displayed squares and chirality relations in the specified matrix basis; in particular $C_L = -i\gamma_E^0 C_E$. Charge conjugation preserves the real Dirac mass while changing the charge, whereas subsequent multiplication by γ_5 reverses the scalar mass sign.

The source identifications are supported by CCM, arXiv v1, §§3.1–3.2 and D'Andrea–Kurkov–Lizzi, arXiv v2, equations (3.13)–(3.15).

08.11 — Fixed real structure with both signs **Correct.** The stated D_F, J_F, Γ_F have the standard KO-6 signs and D_F commutes with the represented algebra. Both $U_- = 1 \otimes \sigma_1$ and $U_+ = \tau_3 \otimes \sigma_1$ commute with D_F, Q and the two algebra actions, reverse the grading, and square to 1. Direct multiplication gives the opposite signs against J_F. The block unitary W gives the displayed electrodynamics matrices.

08.12 — Particle-number pairing **Correct.** Under all the stated hypotheses, $[Q, U] = 0$ makes QU a unitary involution. The commutators of Q with D, Γ, the algebra and its opposite preserve the remaining requirements, while

$$(QU)J = sQJU = -sJ(QU).$$

The final bimodule obstruction also follows: a particle-preserving, grading-reversing U must identify the two particle bimodules.

Lemma 13

13.1 — Inner fluctuations and boundary conditions **Correct.** The commutator with a matrix-valued function has the five displayed Clifford components, and matrix-valued one-forms are generated locally by sums $a\,db$. Self-adjointness gives Hermitian A_μ, A_5. The fluctuation is bounded and self-adjoint, hence preserves self-adjointness on the original Dirac domain without having to map that domain into itself.

Applying the original boundary conditions to $D_A\psi$ yields the two covariant derivative conditions in the proof. They constrain ψ, not endpoint values of the potentials.

13.2 — Fixed-projection compression **Correct.** The diagonal spacetime matrix elements give the two weighted averages, and the vertical off-diagonal element is

$$m + i\int_0^l \frac{\sin(\pi x/l)}{l} A_5(x)\,dx.$$

The proposed cosine lift realizes arbitrary Hermitian A_L, A_R, and $A_5 = (\pi/2)B$ realizes arbitrary Hermitian B. Thus the exact link image is $\phi + \phi^* = 2mI_n$. An unrestricted finite two-sheet fluctuation can in particular set $\phi = 0$, so its image is larger.

13.3 — Failure of gauge covariance of fixed compression **Correct.** For $A_5 = a\cos(2\pi x/l)$, the integral of A_5 vanishes, hence $W(l) = 1$. But

$$B = \frac{a}{\pi}\int_0^\pi \sin t\cos(2t)\,dt = -\frac{2a}{3\pi}.$$

The resulting compressed eigenvalues are $\pm\sqrt{m^2 + 4a^2/(9\pi^2)}$, rather than $\pm m$. Thus even an endpoint-fixed gauge transformation changes the fixed compression. The transported-frame link and the Schur-complement term in the proof follow by direct substitution.

13.4 — Mixed-boundary vector mass **Correct.** Neumann at 0 gives $\cos(kx)$, and Dirichlet at l gives $kl = (r + \frac{1}{2})\pi$. The lowest mass is $\pi/(2l)$. These gauge boundary conditions are additional data; 13.1 does not derive them from the fermion domain.

13.5 — Spectral-action vector-mass formula **Correct.** In the stated convention, the Clifford traces in the proof give

$$\mathcal{L}_{a_4} = \eta\ \mathrm{Tr}\, F_{\mu\nu}F^{\mu\nu} + 3\eta\ \mathrm{Tr}\, D_\mu V D^\mu V, \qquad \eta = \frac{f_0}{24\pi^2}.$$

At V_0, the quadratic terms are $\eta Q_{ab}f^a f^b + 3\eta K_{ab}A^a A^b$. Setting $A_{\mathrm{can}} = 2\sqrt{\eta}\,Q^{1/2}A$ gives

$$\mathcal{M}^2 = \frac{3}{2}Q^{-1/2}KQ^{-1/2}.$$

The coefficient agrees with CCM, arXiv v1, equation (6.10). It is conditional on the stated trace, vacuum and a_4 approximation.

13.6 — Minimal two-sheet action **Correct.** For each generator, the stated trace matrices are

$$Q = \kappa I_2, \qquad K = 2\kappa m^2\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The mass-squared eigenvalues are therefore 0 and $6m^2$, independent of the common multiplicity. For the additional profile restriction, $c_L : c_R = 2 : 1$, and the one-field formula gives $3m^2/5$. That restricted field is not the massive eigenvector of the unrestricted two-sheet action.

13.7 — Conditional CCM comparison **Correct.** The full three-generation trace gives $Q_{11} = 12$, while one massive colour-triplet Dirac fermion gives $K_{11} = 3m^2$. Hence

$$M_W^2 = \frac{3}{2}\frac{3m^2}{12} = \frac{3}{8}m^2.$$

Keeping the massless charged states in Q is essential. This reproduces the conditional relation in CCM, arXiv v1, Lemma 4.6, equation (4.16). The normalized interval matrix element is already m; the separate A_5 overlap does not rescale that background mass.

13.8 — Electroweak doublet and neutral-vector ratio **Correct with amendment.** The doublet calculation gives the displayed W, Z, and ρ formulas. The full trace has $Q_2 = 12$, $Q_Y = 20$, so $g_Y^2/g_2^2 = 3/5$ and $M_W/M_Z = \sqrt{5/8}$.

The last paragraph must say that **the link of the independently gauged finite two-sheet completion** transforms as $g_L\phi g_R^*$. The fixed compression of the original continuum field does not in general obey that endpoint transformation law: 13.3 gives an endpoint-fixed gauge transformation that changes ϕ. The $1 \oplus 3$ decomposition under the diagonal $SU(2)$ remains correct.

Lemma 14

14.1 — Covariant projection and spectral data **Correct with amendment.** The operator identity

$$D_{A^g,\mathsf{M}^g,\mathcal{B}^g} = G_g D_{A,\mathsf{M},\mathcal{B}} G_g^\dagger$$

and the Riesz formula prove covariance of the gapped projection. Conjugation by W gives the stated axial-gauge bulk matrix and domain. In particular, a gauge-invariant bulk mass and the mixed conditions are unchanged, so their spectrum is independent of A_5.

The double-coset statement needs the hypothesis that endpoint transformations act **independently** through H_0 and H_l, with the corresponding extendability understood; it is a local pointwise classification unless global topology is specified. For example, periodic boundary conditions require $g(l) = g(0)$. Their holonomy is classified by conjugation, not by $G\backslash G/G$. For $G = U(1)$, a periodic constant A_5 has a nontrivial gauge-invariant Wilson phase despite both endpoint projections being $U(1)$. The lemma's subsequent discussion of identified endpoints gives the appropriate distinction.

14.2 — Induced connection and finite operator **Correct.** Acting on $U\chi$ gives

$$\nabla^{\mathcal{E}}_\mu \chi = \partial_\mu \chi + U^\dagger \partial_\mu U\, \chi + i U^\dagger A_\mu U\, \chi,$$

which is the displayed $\mathcal{A}_\mu$. Expansion of $P\nabla_\mu P \nabla_\nu P - (\mu \leftrightarrow \nu)$ gives the curvature formula, and $U^g = G_g U h^\dagger$ gives the stated frame transformations.

The sums involving ∇^S and the internal connection are tensor-product connection sums; in a local spin frame the derivative appears once. The Schur-complement correction is the standard result of eliminating the complementary block.

14.3 — Finite algebra and fluctuation image **Error.** The displayed formulas, the compression range, and the **local** fixed-singular-value statement are correct. The unqualified claim that the global continuum link image consists *exactly* of all links with $\Phi^\dagger\Phi = \lambda^2 I$ is too strong.

Take $M_4 = S^1 \times S^3$, $n = 1$, and the globally defined fixed-singular-value link

$$\Phi(\theta, z) = \lambda e^{i\theta}.$$

For any globally smooth real A_5 on the stated trivial bundle,

$$\Phi = \lambda W_l^\dagger = \lambda \exp\left(i \int_0^l A_5(y, x)\, dx \right)$$

has a globally defined real phase and zero winding around S^1. It cannot equal the displayed link. More generally, $W(y, x)$ is a homotopy from the identity to $W_l(y)$.

The corrected statement is that every fixed-singular-value link is obtained **locally in spacetime**; globally, the unitary link must admit the corresponding path from the identity in the chosen trivialization. The reducing-subalgebra criterion is correct: the compression identity in the proof forces both a and a^* to preserve the range. Since p_L is nonzero in the interior, this gives precisely $a(x) = W(x) b W(x)^\dagger$, and conversely such multipliers reduce the light space.

14.4 — Boundary Yukawa data **Correct with amendment.** The transformation law for X, its axial-gauge form, and the limiting texture follow from the boundary equations. In the towerless scaling, the normalized wall values are $l^{-1/2}\Phi_Q$ and $l^{-1/2}\Phi_U$, giving

$$Y_0 = \Phi_Q \widehat{X} \Phi_U.$$

Expanding the transfer functions at fixed bounded λ changes this matrix by $O(l^2)$; after grading-preserving changes of frame the light block converges to Y_0. Positive diagonal Φ_Q, Φ_U give the stated attainability.

Two hypotheses must be made explicit:

1. To obtain the **entire** displayed limiting block, retain the full zero-mode-descended cluster of dimension $N_Q + N_U$, not an arbitrary smaller gapped grading-invariant cluster.
2. The **nonzero** singular values of Y are the retained positive eigenvalues. Zero singular values and possible unmatched chiral zero modes must be included separately.

For the first point, with two massless flavors and $X = l\,\mathrm{diag}(1, 2)$, the cluster containing only the lowest $\pm$ pair is gapped but has a 1×1 limiting block, not the full 2×2 texture.

14.5 — Quadratic matching condition **Correct with amendment.** On a common retained profile space where both kinetic forms are positive, the two mass matrices are the stated generalized eigenvalue problems. If the KK mass matrix is $\kappa^2 I$, equality of all its masses is equivalent to

$$K = \frac{2}{3}\kappa^2 Q.$$

The equality of actions in the same coordinates gives the stronger conditions $G^{(5)} = 4\eta Q$, $H^{(5)} = 6\eta K$. Block multiplication gives the stated C_a, Q, K, and one-vector sum rule.

The comparison must explicitly restrict **both** quadratic forms to the same profile space on which $G^{(5)}$ and Q are positive. A KK profile can be invisible to compression. For the massless mixed fermion pair, $f(x) = \cos(2\pi x/l)$ satisfies Neumann conditions at both endpoints and has positive five-dimensional norm and derivative norm, but

$$\int w_L f = \int w_R f = 0.$$

Thus $Q = K = 0$ for that nonzero KK mode, and the finite action has no corresponding canonically normalized vector.

14.6 — Covariant lowest-mode reduction **Correct.** The displayed gauge transformation closes on the retained Abelian a_μ, b fields. The transported frame gives

$$D\Phi = i\Phi(c_L - c_R)\left(a + \frac{db}{\kappa}\right),$$

including the Berry terms. Canonical normalization then gives

$$M^2_{\mathrm{spec}} = 3\lambda^2 \frac{(c_L - c_R)^2}{c_L^2 + c_R^2}.$$

At $M = 0$, $c_L = 8/(3\pi)$, $c_R = 4/(3\pi)$, and $\lambda = \kappa$, so the ratio is exactly $3/5$.

14.7 — Tuned bulk mass **Correct.** For $M \geq 0$, the lowest root satisfies $M = -k\cot(kl)$, with $t = kl \in [\pi/2, \pi)$, and $\lambda = k/\sin t$. At $t = 3\pi/4$, the displayed normalization integral and overlaps give

$$c_L = \frac{9}{3\pi + 2}, \qquad c_R = \frac{6}{3\pi + 2}, \qquad \frac{M^2_{\mathrm{spec}}}{M^2_{\mathrm{KK}}} = \frac{27}{26}.$$

The ratio is $3/5$ at $M = 0$ and varies continuously, so an M_* exists in the stated open interval. No uniqueness follows from this argument. At such a solution, $\kappa < \lambda < (3/\sqrt{2})\kappa$, so the retained fermion is at the tower scale.

14.8 — Bounded-fermion no-go **Correct.** For any Hermitian linear combination T of retained couplings,

$$\|[T, V_0]\|_{\mathrm{HS}} \leq 2\|V_0\|\,\|T\|_{\mathrm{HS}}.$$

Consequently $K \leq 4\|V_0\|^2 Q$ as quadratic forms, and every finite mass satisfies

$$M^2_{\mathrm{spec}} \leq 6\|V_0\|^2.$$

The constant is sharp: take $V_0 = C\sigma_3$ and $T = \sigma_1$. Thus bounded V_0 cannot match the divergent Neumann–Dirichlet mass within the stated leading-action approximation.

14.9 — Gauge and scalar normalizations **Correct.** With fixed generator and profile normalizations, $\eta q_s F_s^2$ gives $g_{4,s}^{-2} = 4\eta q_s$, whereas integrating the five-dimensional kinetic term gives $J_s/g_{5,s}^2$. Rescaling a five-dimensional coupling changes both of its quadratic forms by the same factor and leaves its KK mass unchanged, so mass matching alone does not determine the relative coupling.

At vanishing ambient A_μ, $D_\mu V = H_i \partial_\mu \varphi^i$, giving $Z_{ij} = 6\eta\, \mathrm{ReTr}(H_i H_j)$. In the eaten-scalar example, imposing both displayed gauge-kinetic and mass equalities gives $Z_b = G$. No corresponding conclusion follows for an independent physical scalar direction.

Comparison with the paper

Section 3: Theorem A and Proposition A.1 Theorem A is **correct with amendment**. Its operator, spectrum, compression range and two-state distance are correct. It should identify the two states as states of the compressed operator system. Their pullbacks are the smeared measures $w_L dx, w_R dx$, not literal endpoint evaluations; for example, $\cos(3\pi x/l)$ compresses to zero but has endpoint values $1, -1$. The theorem also omits that the full diagonal compressed algebra with the inherited J fails first order, as 08.4 proves.

Proposition A.1 correctly states 08.8 for the specified massless product and fixed cutoff.

Section 6: Theorem D

- **D.1 — Correct, with its finite-triple scope made explicit.** The criterion agrees with Lemma 06.2. The point has zero Dirac operator; its KO label is part of the product convention.
- **D.2 — Error as a general statement.** The assertion about *a finite triple with an unpaired left-handed neutrino* is broader than the particular reduced representations treated in Lemma 06. For a counterexample, take

  $$H = \mathbb{C}^2_{\mathrm{gen}} \otimes \mathbb{C}^2, \quad A = \mathbb{C}, \quad D = 0, \quad \Gamma = 1 \otimes \sigma_3, \quad J = (1 \otimes \sigma_1)K.$$

  Regard the two positive-grading states as two unpaired left neutrinos and the negative-grading states as their conjugates. There is no Majorana term. Both

  $$U_+ = 1 \otimes \sigma_1, \qquad U_- = \sigma_2 \otimes \sigma_1$$

  satisfy the factorization criterion, with signs $+1$ and -1, respectively. Conversely, an unpaired neutrino does not alone guarantee the $+1$ factorization: the full chiral algebra can obstruct both, as in Lemma 06.8. The paper must specify the reduced algebra, charged bimodules and neutral multiplicity for its KO-7-only claim.
- **D.3 — Correct with amendment.** The sign calculation is correct. Calling both interval choices "KO-dimension 0" needs the extended 0_+, 0_- convention: under the paper's stated standard convention only $t = +1$ has the standard KO-0 signs. The bare reflection is generally not linear over the full interval algebra.
- **D.4 — Correct with amendment.** The exact factorization is for the **scalar-algebra** interval truncation. With the full compressed two-sheet algebra, the inherited real data fail first order and the reflection does not commute with that algebra.
- **D.5 — Correct with amendment.** The fixed-product example is correct. The general pairing statement also needs Q to be a self-adjoint involution commuting with Γ, the algebra and its opposite, in addition to the stated conditions and particle preservation. The omitted algebra hypothesis matters: with $A = \mathbb{C} \oplus \mathbb{C}$ acting diagonally on a multiplicity factor, $D = 0$, $\Gamma = 1 \otimes \sigma_3$, $J = (1 \otimes \sigma_1)K$, $U = 1 \otimes \sigma_1$, and $Q = \sigma_2 \otimes 1$, the conditions printed in D.5 hold and U preserves Q, but QU does not commute with A.

The paragraph after Theorem D repeats the unqualified neutrino claim and the claim that adding ν_R with Dirac masses gives both signs; both need the reduced-representation hypotheses. Its statement that the Majorana case is open is also less precise than Lemma 06: that lemma gives a one-generation result and a singular-value multiplicity criterion for its specified neutral sector. A broader model can remain open.

Section 7: Propositions E and F and the following paragraph Proposition E states its CCM mass relation as an assumption and correctly limits its use to the matching-scale heat-kernel regime.

Proposition F is **correct with amendment** for its fixed-compression and gauge-covariance claims, but its mass sentence contains a dimensional error: $\mathcal{M}^2$ cannot equal $\sqrt{6}\,m$ or $\sqrt{3/8}\,m$. Those are masses; their squares are $6m^2$ and $3m^2/8$. It must also say that the minimal two-sheet model has a massless diagonal vector, and that the CCM value is for the W under the full three-generation trace, specified vacuum and one-mass assumption. The Neumann–Dirichlet vector boundary conditions are separate inputs.

The following paragraph reports the 3/5 example, tuned mass, and norm bound correctly, but its opening sentence is too general: the two masses can agree in the tuned example described immediately afterward. The 3/5 and M_* claims concern the specified one-fermion Abelian mixed-boundary reduction. The no-go requires **all fermion masses retained in the finite trace** to be uniformly bounded, and applies to the stated leading spectral action. Covariant projection and one mass equality do not establish equality of the full gauge or scalar actions.

The abstract and introduction repeat the overly broad claims that an unpaired neutrino selects KO-7 and that the gauge sector does not reproduce the tower. They need the same qualifications.

Amendments and errors

- **Lemma amendments:** restrict 13.8's $g_L \phi g_R^*$ law to the independently gauged finite completion; add independent endpoint action and extendability to 14.1's double coset; specify the full light cluster and zero modes in 14.4; compare both positive kinetic forms on the same profile space in 14.5.
- **Lemma error:** 14.3's global fixed-singular-value link image omits the homotopy restriction. Its local statement is valid.
- **Paper:** qualify Theorem A's state and real-structure interpretation; restrict Theorem D's neutrino, interval-algebra and particle-pairing claims; correct Proposition F's mass-versus-mass-squared sentence and state the model, trace, vacuum and boundary assumptions; limit the claimed gauge-sector disagreement and bounded-fermion no-go to the cases proved.

Overall: 29 numbered lemma statements checked — 24 correct, 4 correct with amendment, 0 gaps, and 1 error.

Part R3 (referee report, lemmas 15, 16)

Referee report on lemmas 15 and 16 and on the T0 note assessment

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_15_16.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 15, lemma 16 and the assessment. Claude checked the two external citations against the arXiv LaTeX sources: the inner Leibniz condition in arXiv:1811.10843 and the commuting subalgebra C_F in hep-th/0610241, section 2.6.

We use the definitions and notation of Lemma 15 and Lemma 16. All calculations below are analytic; we ran no numerical computation.

The statements involving spectral propinquity require one correction to the supplied Farsi–Latrémolière source. Its displayed inner Leibniz condition is ill-typed as written. We state the correction under Theorem 15.3 and use it throughout.

Part 1. Statements and proofs

Lemma 15.1 Correct.

For self-adjoint e, its norm is the supremum of $|\varphi(e)|$ over states. The definition of extent therefore gives both norm inequalities. The Jordan-Leibniz inequality gives

$$L_E(e^m) \leq m\|e\|^{m-1} L_E(e)$$

by induction, and the polynomial estimate follows by the triangle inequality.

Theorem 15.2 Gap, closed here.

The commutativity argument in item 1 is correct. Bounded-Lipschitz lifts of two self-adjoint elements have uniformly bounded norms. Their Lie product maps to zero in every commutative approximating algebra, so Lemma 15.1 forces the Lie product of the two limit elements to vanish.

In item 2, the stated conditions on g do **not** imply that b_n stays within $3/8$ of an integer. They control neighborhoods of the half-integers, but permit values beyond the outermost integers.

We can repair the proof with the same constants. Choose $g : I \to [0,1]$ such that

$$g = 0 \quad \text{on} \bigcup_{j=0}^{k} [j - \tfrac{1}{8}, j + \tfrac{1}{8}], \qquad g = 1 \quad \text{outside} \bigcup_{j=0}^{k} (j - \tfrac{3}{8}, j + \tfrac{3}{8}).$$

A polynomial p with $\|p - g\|_I < 1/8$ has $\|p(a)\| < 1/8$. Lemma 15.1 then gives $\|p(b_n)\| < 1/4$ for large n, so the range of b_n lies in the union of the disjoint open $3/8$-neighborhoods. Each connected component of X_n maps into one neighborhood. At most k neighborhoods are reached, and a fixed one is missed along a subsequence. The polynomial q constructed in the note then gives the stated contradiction.

The clopen partition used at the start exists because components and quasicomponents agree in a compact Hausdorff space. Both conclusions follow.

Theorem 15.3 Correct with amendment. The estimates hold under the standard inner Leibniz definition. The supplied source's literal definition must be corrected before its tunnels can be checked as stated.

At source lines 602–604 (arXiv:2404.00240v3 LaTeX source), the inner product is required to lie in the domain of L_B, although that domain was defined on self-adjoint elements; the displayed inequality then names L_A. Literally, the first requirement fails for every nonzero complex Hilbert module: $\langle \omega, i\omega \rangle$ is a nonzero skew-adjoint element. The required condition is

$$\max\{L_B(\mathrm{Re}\langle \Theta, \Theta' \rangle), L_B(\mathrm{Im}\langle \Theta, \Theta' \rangle)\} \leq 2\,\mathrm{DN}_P(\Theta)\mathrm{DN}_P(\Theta').$$

This is the condition in Latrémolière's Definition 2.2.

With that correction, the tunnel has the required properties:

- The top seminorm is Leibniz for Jordan and Lie products, has scalar kernel and a closed unit ball. Its centered unit ball is compact. Choosing the midpoint of the spectrum of a proves the quotient condition onto A; scalar lifts prove it onto $\mathbb{C}$. Its extent is at most ρ.
- The base seminorm on $\mathbb{C} \oplus \mathbb{C}$ is an L-seminorm with extent at most s.
- The module D-norm dominates the module norm and has a compact unit ball. The displayed modular estimate is valid without assuming P commutes with A.
- The inner-product calculation proves the stronger complex estimate

  $$|\langle \xi, \xi' \rangle - \langle \eta, \eta' \rangle| \leq 2s\,\mathrm{DN}_P(\xi, \eta)\mathrm{DN}_P(\xi', \eta'),$$

  which implies the corrected real-and-imaginary-parts condition.
- The lifts $\eta = cW\xi$ and $\xi = cW^*\eta$ attain the two D-norm quotients. The coordinate maps also preserve the left and right actions and inner products.

The pseudo-state choices in both directions are valid and are independent of t. Consequently they bound the **Hausdorff distance between families**, as required, by

$$g^{-2} + \rho + \Delta + \Delta/\varepsilon.$$

The conclusion follows by taking ε larger than $g^{-2} + \rho + \Delta + \sqrt{\Delta}$ and then decreasing ρ. The lower bound correctly includes the cap:

$$\Lambda_{\mathrm{spec}} \geq \min\left\{\frac{\sqrt{2}}{2}, \frac{\delta}{2}\right\}.$$

Remark 15.4 **Error.** The uncapped lower bound is asserted for every $\varepsilon > 0$, although spectral propinquity is at most $\sqrt{2}/2$.

The correct statement is

$$\min\left\{\frac{\sqrt{2}}{2}, \frac{\varepsilon\delta_1}{2}\right\} \leq \Lambda_{\text{spec}} \leq \frac{\varepsilon\delta_1}{2} + \frac{\varepsilon^2}{g_1^2}.$$

For example, take the periodic Dirac operator on a circle of length 1. It has zero modes and $\delta_1 = 1/2$. At $\varepsilon = 4$, the asserted uncapped lower bound is $1 > \sqrt{2}/2$. The claimed leading asymptotic remains correct for sufficiently small ε.

Corollary 15.5 **Correct.**

For the stated sufficiently small ℓ, including $\ell < \sqrt{2}$ in the chosen normalization, the cap does not alter the lower bound. The interval diameter is ℓ, the light block is exactly $m_f\sigma_1$, and the next eigenvalue satisfies

$$g^2 \geq \mu(\ell)^2 + \frac{\pi^2}{\ell^2} = \frac{c(\ell)^2 + \pi^2}{\ell^2}.$$

Thus the upper correction is exactly bounded by $\ell^2/(\pi^2 + c^2)$, as displayed. The Connes–Lott lower bound has the correct factor $1/2$ and cap.

Corollary 15.6 **Correct.**

The boundary conditions are preserved by scalar Lipschitz functions, and the commutator seminorm is $\|f'\|_\infty$. The zero modes and positive and negative light modes in Theorem 12.7 have the correct total dimension $N_Q + N_U$. Matching them to the singular vectors of M_F gives $\Delta = O(\ell^2)$, including zero singular values. The gap is $O(\ell^{-1})$, so Theorem 15.3 gives the asserted $O(\ell)$ bound.

Theorem 15.7 **Correct with amendment.** The proof uses the corrected real-and-imaginary-parts inner Leibniz condition stated under Theorem 15.3.

The two base characters have distance below ε_n, since either is a state of the tunnel base algebra and the other is the sole state in its quotient image. Applying the corrected inequality to the real and imaginary parts proves the displayed $4\varepsilon_n$ inner estimate. Indeed, rotating one vector by a phase gives the stronger constant $2\varepsilon_n$; the stated constant is sufficient.

The upper multiplicity argument is valid: the Gram matrix of $N + 1$ target vectors has rank at most N, while the entrywise estimate gives

$$\|I - G\| \leq 16(N + 1)(1 + R)^2\varepsilon_n.$$

For concentration, Hausdorff separation supplies **one** pseudo-state matching the entire time family. Comparing its estimates at t and 0 gives the displayed $8C^2\varepsilon_n$ bound. Averaging over $0 \leq t \leq 1/\varepsilon_n$ gives

$$\int \left(1 - \operatorname{sinc}\frac{x - \lambda}{\varepsilon_n}\right) d\nu(x) \leq 8C^2\varepsilon_n.$$

Since $|\operatorname{sinc} u| \leq 1/2$ for $|u| \geq 2$, the stated tail bound follows.

For each eigenvalue of D_F, projecting the lifted vectors to its $2\varepsilon_n$-window changes them by $o(1)$; their Gram matrix remains invertible. Distinct eigenvalues have disjoint windows eventually. The lower counts and the preceding upper count give the exact total multiplicity N, after which the eigenvectors can be matched with operator error at most $2\varepsilon_n$. The gap to the remaining spectrum tends to infinity. If no spectrum remains, the gap is interpreted as infinite.

Corollary 15.8 **Correct.**

Faithfulness and finite-dimensionality of the limiting Hilbert space make the limiting algebra finite-dimensional. Theorem 15.2 then gives $\mathbb{C}$. On a circle of length ℓ, the constant modes of

$$-i\,\frac{d}{dx} + \operatorname{diag}(\lambda_1, \dots, \lambda_N)$$

carry the prescribed Hermitian matrix, while the other modes have absolute eigenvalue at least $2\pi/\ell - \|D_F\|$. The circle diameter tends to zero, so Theorem 15.7 applies. The construction concerns ungraded triples; additional real or even structures would impose their own compatibility conditions.

Proposition 15.9 **Correct with amendment.**

The symmetric spectral projection commutes with γ and J, and the displayed estimates hold for $D_B = WDW^*$. Two qualifications are needed for the subsequent claim about metric control.

First, if $\Delta > 0$, $W\omega$ need not be a pseudo-state for the graph norm of D_F. The normalized choice $cW\omega$, with $c = (1+\Delta)^{-1}$, is a pseudo-state and adds at most Δ to the stated bounds.

Second, the induced γ_F, J_F satisfy the Dirac relations exactly with D_B, not necessarily with the independently chosen D_F. The defects with D_F are bounded by 2Δ. The case $\Delta = 0$ is exact; when $\Delta \to 0$, the defects vanish.

Finally, composition with an antiunitary produces an antilinear functional. The source's operational propinquity is defined for linear functionals, so convergence of J requires the extension acknowledged in the note.

Example 16(a) **Correct.**

At the lowest massless interval level, the two chiral profiles give

$$Q(f) = \operatorname{diag}(C_0 + C_1, C_0 - C_1).$$

The coefficients C_0, C_1 can be chosen independently, so the range and generated algebra are both $\mathbb{C} \oplus \mathbb{C}$.

Example 16(b) **Correct.**

The product-to-sum identities give the two displayed matrices. Their four coefficients $C_0, \dots, C_3$ are independently recoverable, so the operator system has complex dimension 4.

For the two displayed functions, direct substitution gives

$$Q(f_1) = \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, 0 \right), \qquad Q(f_2) = \left(0, \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right).$$

Each coordinate projection of the generated algebra contains all symmetric 2×2 matrices, hence all of $M_2(\mathbb{C})$. The first displayed element generates a nonzero two-sided ideal supported on the first summand; simplicity of $M_2(\mathbb{C})$ makes that ideal the entire first summand. The second element does the same for the other summand. Therefore

$$A_F = M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

Example 16(c) **Correct.**

For a uniformly distributed rank-one projector P_w on $\mathbb{C}^4$,

$$\mathbb{E}(P_w \otimes P_w) = \frac{I + \mathrm{Flip}}{20}.$$

Together with $\operatorname{tr} \Gamma_a = 0$, this gives $Q(x_a) = \Gamma_a/5$. Four of the gamma matrices generate $M_4(\mathbb{C})$. For the twisted Dolbeault operator, the kernel is $H^0(\mathbb{CP}^3, \mathcal{O}(1))$, of dimension 4; the higher cohomology groups vanish. Thus $d = 0$ on the selected subspace.

Proposition 16.1 **Correct with amendment.**

The order-zero equivalence is correct. The hypotheses imply $J_F Q(a)^* J_F^{-1} = Q(a)$, so order zero forces all compressed generators to commute. Conversely, if their generated algebra is commutative, the opposite image is the same algebra and order zero holds.

For the product assertion, specify that the fixed finite factor itself satisfies order zero and that the product real structure is the compatible one. Also, the hypotheses on J_0 alone do not assert $J_0^2 = \pm 1$ or a grading sign. Those relations must be added if J_F is called a real structure in the full sense.

Theorem 16.2 **Correct with amendment.** The Weyl-exponent assertion requires the original exponent to exist.

The graded flip is well defined on the complex tensor product. On each homogeneous summand it is the ordinary conjugate flip times a sign; it is therefore antiunitary. Direct application to homogeneous tensors gives

$$J^2 = 1, \qquad J\mathcal{D} = \mathcal{D}J, \qquad J\Gamma = \Gamma J.$$

For even a, $a^\circ = 1 \otimes (a^*)^c$. Hence order zero and order one hold, and

$$[\mathcal{D}, \pi(a)] = [D_0, a] \otimes 1.$$

The signs $(+,+,+)$ are the usual KO-0 signs.

An eigenbasis of D_0 decomposes $\mathcal{D}$ into finite-dimensional invariant blocks. On each block,

$$\mathcal{D}^2 = D_0^2 \otimes 1 + 1 \otimes (D_0^c)^2.$$

This proves essential self-adjointness on the algebraic tensor core and compact resolvent of the closure.

Writing $N_0(R)$ and $N_{\mathcal{D}}(R)$ for absolute-eigenvalue counting functions with multiplicity,

$$N_0(R/\sqrt{2})^2 \le N_{\mathcal{D}}(R) \le N_0(R)^2.$$

Thus, if $\lim_{R\to\infty} \log N_0(R)/\log R = p$ exists, the new exponent is $2p$. Compact resolvent alone does not guarantee such a limit: eigenvalue multiplicities can be chosen to make the logarithmic ratio oscillate.

Corollary 16.3 **Correct with amendment.** Assume $k \ge 1$, and distinguish the product projection from a spectral projection of $\mathcal{D}$.

The identities in the corollary and both order conditions hold for every even subalgebra $A_F \subseteq B(V)$. In particular,

$$J_F(d \otimes 1)J_F^{-1} = \gamma_V \otimes d^c, \qquad [D_F, a \otimes 1] = [d, a] \otimes 1.$$

The original compression has range $Q(A_0) \otimes 1$; the corollary uses the **generated** algebra $A_F \otimes 1$.

The projection $P_0 \otimes \overline{P_0}$ reduces $\mathcal{D}$, but need not equal $1_S(\mathcal{D})$ for any set S. For a counterexample, take an even finite D_0 with positive absolute eigenvalues

$$1, \quad 5, \quad 7, \quad \sqrt{73},$$

and let P_0 select the first three absolute values. The selected pair $5, 7$ and the mixed pair $1, \sqrt{73}$ both give squared product eigenvalue 74. Both reducing subspaces contain the eigenvalues $\pm\sqrt{74}$, so the product projection cannot be a spectral projection of $\mathcal{D}$. It is a joint cutoff in the two factors. In the kernel example used below, it *is* a spectral projection.

Corollary 16.4 **Correct.**

The displayed operators have signs $(+,+,-)$, the usual KO-6 signs. Both algebra actions are trivial on the last factor, so order zero and order one persist, and the Hilbert-space dimension is $2k^2$.

The signs alone do not establish the usual zero-dimensional orientability axiom. Every represented zero-cycle built from the left and opposite algebra actions acts as $T \otimes 1_2$, whereas the grading is $\Gamma_F \otimes \sigma_3$. No such zero-cycle can equal that grading.

Caveat 1: commutant **Correct.**

For

$$d = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix},$$

with B invertible, the relation $\operatorname{diag}(x, y)d = d \operatorname{diag}(x, y)$ is equivalent to

$$y = B^{-1}xB, \qquad [x, BB^*] = 0.$$

Invertibility entails equal dimensions of V_+ and V_-. In example (b), $B = \operatorname{diag}(\pi/(2\ell), 3\pi/(2\ell))$; its squared singular values are distinct. The commuting pairs are diagonal with the same entries in both summands, giving the stated $U(1) \times U(1)$.

Caveat 2: spectrum **Correct.**

The square of the doubled Dirac operator is the sum of the two squared factors, with multiplicities counted in the tensor product. In example (a), $D_F^2 = 2m^2 I$, so its eigenvalues are $\pm\sqrt{2}\, m$. Its commutator with the left copy of $\mathbb{C}^2$ still has norm $m|a - b|$, giving distance $1/m$.

Caveat 3: parent **Correct with amendment.**

For an ordinary compact spin example of dimension n, the underlying product Dirac operator has Weyl exponent $2n$, while the restricted algebra has the metric of one n-dimensional factor. The statement about the product manifold requires the charge-conjugation identification and algebra restriction specified in Part 2(iii). The discrepancy also means that the restricted triple is generally not the ordinary finite-rank twisted spin triple over the single factor.

Caveat 4: KO-6 **Correct with amendment.**

The Clifford point is an input to Corollary 16.4. The claim that obtaining the sign from a geometric parent is "open" should be scoped to the work checked here: the cited documents do not establish such a construction or a no-go theorem.

Caveat 5: dimension count **Correct.**

For $k = 4$, the dimension after adding the Clifford point is 32. Neither example produces the stated classified algebra or its representation. Moreover, the C^*-algebras generated in Lemma 16 are complex algebras; a quaternionic real summand requires additional real-algebra data.

Part 2. Interpretation

(i) Unbroken symmetry and the Standard Model **Error, for the unrestricted claim.** A metric spectral triple excludes nonscalar elements of its represented algebra that commute with D. That does not exclude every symmetry implemented on its Hilbert space.

For a counterexample within the spectral-propinquity domain, let $H = M_2(\mathbb{C})$ with Hilbert–Schmidt inner product, let $A = M_2(\mathbb{C})$ act by left multiplication, and let

$$D(x) = x - \tfrac{1}{2} \operatorname{tr}(x) I.$$

If $[D, L_a] = 0$, applying the commutator to I gives a scalar. Finite dimensionality then makes this a metric spectral triple. Yet every conjugation $x \mapsto uxu^*$, $u \in SU(2)$, commutes with D. With $Jx = x^*$, these are the transformations $L_u J L_u J^{-1}$. This example fails order one, which spectral propinquity itself does not require.

A narrower statement does hold for a **finite real first-order** triple. If $uJuJ^{-1}$ commutes with D, order one and $JD = \varepsilon' DJ$ imply that $a = [D, u]u^*$ commutes with u. Thus

$$u^n D u^{-n} = D - na.$$

Since D is bounded, $a = 0$. If the triple is metric, u is scalar and the induced transformation is trivial.

The Standard Model finite triple is outside the metric class. In the usual construction, the color action commutes with the finite Dirac operator; independently, the nonscalar subalgebra C_F is explicitly required to commute with it in Chamseddine–Connes–Marcolli, §2.6. The constant copy of such an element also obstructs the almost-commutative product. The original algebra is a real algebra, so applying a framework formulated for complex C^*-algebras also requires specifying a complexification.

(ii) Morita equivalence Error, as an argument that removes the scalar-algebra restriction. The Morita equivalence $C(K) \sim C(K, M_n(\mathbb{C}))$ is correct, but Morita equivalence does not make the two algebras isomorphic or identify their spectral propinquity limits. Theorem 15.2 remains a restriction on limits of the whole **scalar** algebra $C(K)$.

A vector bundle E can carry matrix-valued connections while $C(K)$ still acts by scalar multiplication on sections. Replacing the represented algebra by $\mathrm{End}(E)$ is an additional change of data. It may be a useful enlargement of T0, but it is not a consequence of the presence of E.

The product-collapse example in Farsi–Latrémolière (arXiv:2404.00240v3 LaTeX source) concerns an already noncommutative finite algebra B in $B \otimes C(K_\ell)$. Its finite triple must be metric, and the limiting Hilbert space has the additional factor $\ker D_K$, which must be nonzero. A one-dimensional kernel recovers the stated finite multiplicity. The example does not produce a noncommutative whole-algebra limit from scalar parent algebras.

(iii) The geometry of the doubled parent Correct with amendment.

For a compact even-dimensional spin manifold, an appropriate charge-conjugation identification of $\overline{H_0}$ with H_0 identifies the displayed $\mathcal{D}$ with the product Dirac operator on $X \times X$. The grading may acquire an overall sign under that identification. The real structure in Lemma 16 covers the exchange of factors and is not the usual pointwise spin real structure.

The spectral triple used in the construction represents only $C(X) \otimes 1$. It is therefore a product-manifold Dirac operator equipped with a restricted observable algebra. If "commutative Kaluza–Klein data" includes that restriction and the flip real structure, the description is accurate. If it means the ordinary spin triple of the total manifold with its full function algebra, or a finite-rank spinor twist over X, it is too broad. With the full product algebra, the generated compression is $A_F \otimes \overline{A_F}$, and the flip generally fails order zero when A_F is noncommutative.

(iv) What Lemma 16 advances Correct with amendment.

For every finite even representation (A, V, d, γ), the construction

$$V \longmapsto V \otimes \overline{V}, \qquad d \longmapsto d \otimes 1 + \gamma \otimes d^c$$

gives a real bimodule satisfying order zero and order one. It uses no pre-existing finite real structure. Consequently, those two order conditions follow formally from the chosen tensor-square representation; the construction does not recover an arbitrary prescribed finite H, D, J, or their multiplicities.

The examples supply more than that formal observation. In particular, the $\mathbb{CP}^3$ kernel example gives a simple tower-decoupling family under the note's generated-compression rule. Put

$$\mathcal{D}_\ell = \ell^{-1}\mathcal{D}$$

on the doubled parent. Its kernel is $V \otimes \overline{V}$, its complementary gap grows like $1/\ell$, and compression followed by algebra generation gives the constant finite data

$$A_F = M_4(\mathbb{C}), \qquad \dim H_F = 16, \qquad D_F = 0.$$

After adding the stated Clifford point, the dimension is 32 and the KO-6 signs hold. Here order one is automatic because the finite Dirac operator vanishes. The whole parent triple has scalar algebra in its spectral-propinquity limit, consistently with Lemma 15.

Thus Lemma 16 advances T0 for a specified truncation procedure, including a noncommutative finite algebra and an explicit bimodule. It does not establish the target under every meaning of ordinary Kaluza–Klein data: the parent algebra is restricted, the real structure flips factors, and algebra generation is an additional operation. For T0b, it gives examples with the requested signs and first-order relations, but leaves the desired algebra and representation, orientability, nonzero mass data, classification, and the beyond-first-order and twisted conditions unestablished. The Clifford point also fails the finite orientability condition described above. The assessment is right to reject an obstruction inferred from one metric; its description of a "concrete origin" should retain these qualifications.

(v) Prior sources **Gap.** We cannot identify a source that states either the full Theorem 15.2, including its component bound, or the exact graded-flip construction of Theorem 16.2. We make no originality claim.

Two constituent results are identifiable in the supplied source: Farsi–Latrémolière state the kernel collapse to $(\mathbb{C}, \ker D, 0)$ at lines 1231–1235 (arXiv:2404.00240v3 LaTeX source), and use the product Dirac formula at lines 1262–1267 (arXiv:2404.00240v3 LaTeX source). Those citations do not establish prior publication of either full statement.

Amendments and errors

- Correct the supplied source's inner Leibniz condition to use L_B on real and imaginary parts; Theorems 15.3 and 15.7 then have the stated estimates.
- Repair the choice of g in Theorem 15.2 as above. The theorem follows.
- Restore the cap in Remark 15.4.
- In Proposition 15.9, normalize the comparison pseudo-state when $\Delta > 0$, and distinguish the exact relations for D_B from the $O(\Delta)$ relations for D_F.
- Add the stated real-structure and product hypotheses to Proposition 16.1; assume a Weyl exponent exists in Theorem 16.2; distinguish a joint cutoff from a spectral projection in Corollary 16.3.
- Qualify the parent geometry and the scope of the claimed open KO-sign problem. Record the orientability limitation of the Clifford-point construction.
- Restrict the unbroken-symmetry claim to the conditions actually proved, and do not use Morita equivalence to erase the distinction between scalar and matrix-valued parent algebras.

Overall: 26 statements and interpretive items checked: 11 correct, 10 correct with amendment, 2 gaps, and 3 errors.

Part R4 (referee report, lemma 18, two sheets, framework note)

Referee report on lemma 18, the two-sheet section of lemma 16, and the background note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_18_16b.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 18, lemma 16, the note and the paper. Claude checked with Python the equations (R1) and (R3), the fully massive kernel and its poles, and read Remark 2.9 of arXiv:0706.3688.

We checked the numbered statements in lemma 18, the two-sheet section of lemma 16, and the claims and table in current_framework.md. We read the relevant passages of the cited papers, identified below. We did not run a computation.

The principal conclusions are:

- The fixed-cutoff quadratic calculation in lemma 18 is correct in its stated top-only family. Its conclusion extends to a broad class of nonnegative smooth cutoffs, including compactly supported cutoffs flat near zero.
- The limiting top-only quadratic action does **not** by itself have unbroken $SU(2) \times U(1)$ acting on the entire light space. It contains an incomplete top-bottom doublet.

- The two-sheet construction reproduces the classified bimodule and real structure as abstract finite data. Example 16.9 does not derive its nontrivial grading as an even spectral truncation of the parent it specifies.
- Corollary 16.8 incorrectly says that its Dirac operator commutes with the whole centre in every case. It always commutes with the two *sheet projections*.

1. Lemma 18, Theorem 1

Correct with amendment. State suitable regularity and decay assumptions on A, for example smooth compactly supported fields, and interpret the trace on $\mathbb{R}^4$ as the stated relative trace or quadratic variation. For several generators, contract the coefficient matrix with the corresponding polarizations.

The Duhamel expansion has the two terms displayed in (5). For transverse a, shifting the loop momentum gives the orbital contribution $2/t$. At the two opposite vertices, the spin traces of the curvature and commutator terms give q^2 and $(m_i - m_j)^2$, respectively; their mixed spin traces vanish. The A^2 vertex contributes

$$-\frac{1}{4\pi^2 t}\sum_{ij}|T_{ij}|^2 e^{-tm_i^2}|a|^2.$$

Symmetrizing this term gives the subtraction in (1), including its factor. Differentiation gives (2)–(3). The divided-difference identities (6)–(8) also follow as written.

The moment assumptions (4) are sufficient for the displayed Euclidean coefficients: $\int s^{-2}\,d\nu$ controls the action density, while the positive moments provide the required differentiability and domination. They do not supply arbitrary timelike analytic continuation.

The trace convention agrees with CCM's bosonic spectral action. Their calculation explicitly counts both particle and antiparticle sectors and four spin components; see CCM, Lemma 3.14 and §3.7.6. The fermionic chirality projection must not be inserted into that bosonic trace.

2. Lemma 18, Lemma 2

Correct. With the stated convention, $A_\mu \mapsto A_\mu - \partial_\mu\alpha$ and $\pi \mapsto \pi + \alpha$, so $A_\mu + \partial_\mu\pi$ is invariant. Conjugating by $e^{-i\pi T}$ proves (12), including the scalar contribution that is missed by using only the linear orbit tangent at a nonstationary background.

The stated longitudinal replacement follows from retaining the longitudinal part of the shifted orbital vertex. For a commuting generator, its integral vanishes by the displayed integration-by-parts identity. Translation invariance and rotation symmetry exclude quadratic mixing of a transverse vector with a scalar.

3. Lemma 18, Corollary 3

Correct with amendment. Formula (13) requires Q to be positive definite on the chosen generator space. Otherwise it applies after quotienting by its null space.

The orbital part of (2) starts at second order in the squared-mass difference, whereas the commutator term starts with d_{ij}^2. Also $\int_0^1 (1-2u(1-u))\,du = 2/3$. This gives the stated C_0, C_1, and factor $3/2$.

For the full Standard Model finite trace, one generation contributes 2 to the particle trace of T_1^2; three generations and charge conjugation give $Q_{11} = 12$. The commutator trace gives (14). This agrees with CCM, equations (4.10) and (4.16), with $F(v) = f(v^2)$. It is a leading expansion, not an identity at fixed m/Λ.

4. Lemma 18, Proposition 4

Correct. In one color of the massive block, let

$$|+\rangle = (t_L + t_R)/\sqrt{2}, \qquad |-\rangle = (t_L - t_R)/\sqrt{2}.$$

Each has squared charged-generator matrix element $1/8$ to b_L, and the reverse ordered entries have the same value. Their total is $1/2$ per color. Thus the massive color triplet contributes $3/2$ to the particle trace, and the other complete doublets contribute $9/2$. The charge-conjugate block supplies the factor two used in (16).

Integration by parts gives $h'_x(0) = 2w(x)/3$ and $g'(0) = 2/3$. Consequently (18) follows. Expanding A and w gives the small-x coefficient $-23/128$, and $A(x) \sim 2/x$, $w(x) \sim 6/x^3$ give the large-x limit. The positive-Laplace extension (23)–(24) follows by dominated convergence under (4).

5. Lemma 18, Theorem 5

Correct. Equation (22) makes h'_x and g' strictly positive for $y \le 0$. Since $F_x(0) > 0$ and $F_x(y) \to -\infty$ as $y \to -\infty$, there is exactly one positive timelike root, and its derivative has the sign giving the stated Euclidean residue.

For $y \le 0$, $g'(y) \ge 2/3$, hence $F'_x(y) \ge 3$. The mean-value theorem gives $z \le A(x)/2$; the Taylor ratio has the same bound. Since $xue^{-xu} \le e^{-1}$, $A(x) \le 2/e$, proving (28). The limiting derivatives are 4 and 3 at small and large x, respectively, yielding (27). The result says nothing about complex poles or Lorentzian unitarity.

6. Lemma 18, Proposition 6

Correct with amendment. The statement about one *positive massive* neutral pole requires $x > 0$. At $x = 0$ there is only the massless neutral system.

For one Dirac species, diagonalizing the left-right mass block gives diagonal neutral charge $(q_L + q_R)/2$ and off-diagonal charge $(q_L - q_R)/2$. In (3), the latter connects eigenvalues $+m$ and $-m$, giving exactly (33). Adding three top colors gives (30). The commuting photon and color traces give (29), including charge conjugates.

The angle in (31) is a sensible **neutral-current definition**: the kinetic-orthogonal massive direction is $K^{-1}(1,-1)^T$, and its current can be written $T_3 - \sin^2\theta_{\mathrm{NC}} Q$. Here

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

so the value $5/8$ is constant. This is a charge-trace identity for this family; it does not make all finite-cutoff definitions of the weak angle coincide.

The neutral mass form is rank one in $v = (1,-1)^T$. Its generalized eigenvalue gives (34), using $v^T B_\infty^{-1} v = 8/27$. For the pole, the left side of (36) has derivative at least $1/2$, so the positive massive root is unique and has positive residue. The substitutions stated for (29)–(33) are valid for (4).

7. Lemma 18, Counterexample 7

Correct. The only nonzero entries of T connect m to $-m$, so (37) follows from (1). Its q^2 coefficient is proportional to $e^{-x}(1-x)$, while its Taylor ratio is $6m^2/(1-x)$.

The pole equation is (36) with $\beta = 4x$. Since $H(-z) > 1$ for $z > 0$, its solution lies strictly between $4x$ and $4x + 2$, and the derivative is positive.

For (39), positive Laplace mixtures give $f(x) + xf'(x)$. The assumptions imply $xf(x) \to 0$, while $xf(x) > 0$ for $x > 0$; therefore its derivative is negative somewhere. A negative Taylor kinetic coefficient does not itself establish a negative-residue pole of the full form factor.

8. Lemma 18, Counterexample 8

Correct. For fixed y, (21) gives $h_x(y) \to 0$, leaving (40), which is positive at fixed positive y. The neutral matrix likewise leaves nonzero kinetic terms. The broken vector directions therefore do not disappear by their masses diverging or their fixed-momentum propagators vanishing.

The narrower decoupling statement is also correct: after restricting the connection to the centralizer, the mass blocks are invariant and the heavy block carries its heat suppression. "Decoupling" here is a statement about the classical bosonic quadratic action, not a quantum effective-action theorem. The symmetry of the unrestricted limiting light space is addressed below.

9. Lemma 18, No-go theorem 9

Correct. Differentiation of the heat-trace density gives (41), which is strictly negative for $c > 0$, nonzero V_*, and the positive measure in (4). The theorem is fairly limited by the phrase "positive Laplace class."

The conclusion can nevertheless be sharpened on flat $\mathbb{R}^4$. For a sufficiently integrable real cutoff f, let λ_j^2 be the eigenvalues of V_*^2. Direct integration of the momentum variable gives

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{R1}$$

For **nonnegative** f, a finite $c > 0$ is stationary exactly when every tail in (R1) associated with a nonzero λ_j vanishes. Thus:

- if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray;
- if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$.

This argument uses the continuum momentum integral; it is not the proof given for a finite torus. Sign-changing cutoffs can have cancellations in (R1), and a radial path with an additional fixed, noncommuting mass matrix need not have the form cV_*.

A nonconstant cutoff flat near zero cannot satisfy (4): a nonzero positive measure on $(0, \infty)$ gives $f'(0) = -\int s\, d\nu(s) < 0$. CCM use positive test functions and their asymptotic moments; profiles flat near zero are allowed and are outside the theorem's stated class. For nonnegative profiles, however, (R1) supplies the separate conclusion above. See CCM, Introduction and §6.2.

10. Lemma 18, Corollary 10

Correct. Equations (19), (27), and (34)–(36) give every entry of (42). At fixed positive μ, all dimensionless pole and Taylor masses are fixed and positive, so their dimensionful values grow with Λ. In the last joint limit, the small-μ expansion gives $M_W \sim \sqrt{3/8}\, m_t$; the neutral masses also diverge, with $M_Z^2 \sim (3/5)m_t^2$.

These are scaling statements for prescribed backgrounds. They do not establish stationary vacua.

11. Lemma 16, reducibility of Corollary 16.4's pair

Correct. The projection

$$P = 1 \otimes \frac{1 + \sigma_1}{2}$$

commutes with the represented algebra and with $J_F \otimes \sigma_1 K$, and is neither zero nor one. It proves reducibility of the **pair** (A, J). The projection need not commute with the grading; irreducibility of the pair is the issue asserted in the text.

12. Lemma 16, Theorem 16.7

Correct. The graded flip is an antiunitary involution. The two direct-summand Dirac operators square to the corresponding sums of squares, and the graded-flip signs give $J\mathcal{D} = \mathcal{D}J$. The minus sign on the second grading gives $J\Gamma = -\Gamma J$.

For even a, b, the opposite action is on the other tensor factor, as displayed. Both order conditions follow from the tensor-factor separation. The product operators have compact resolvent under the spectral-triple hypotheses.

When the two finite spaces are ungraded and have equal dimension, the identification $\xi \otimes \bar{\eta} \leftrightarrow (v \mapsto \xi\langle \eta, v\rangle)$ gives left composition and $J(x, y) = (y^*, x^*)$, exactly as in Chamseddine–Connes, Proposition 2.8 and Remark 2.9.

13. Lemma 16, Corollary 16.8

Error. The KO signs, dimension, order conditions, real orientability formula, and the full-matrix special case are correct. The assertion that D_F commutes with the **whole centre** in every case is false. It always commutes with the two projections distinguishing the sheets.

For a counterexample, take

$$V_X = \mathbb{C}^2, \quad \gamma_X = \sigma_3, \quad d_X = \sigma_1, \quad A_X = \mathbb{C} \oplus \mathbb{C}$$

acting diagonally, and take $V_Y = \mathbb{C}$, $d_Y = 0$. The element $(\mathrm{diag}(1, 0), 0)$ is central in $A_X \oplus A_Y$, but its commutator with D_F is nonzero.

The orientability formula is the usual **real** zero-dimensional relation $\Gamma = \sum_i \pi(a_i) J \pi(b_i) J^{-1}$, with opposite-algebra coefficients. It is not the stronger assertion $\Gamma \in \pi(A)$, which can fail. In the full-matrix, equal-dimension special case, an invertible matrix in each summand is a separating vector; the pair is irreducible because a projection commuting with both left and right full matrix actions is scalar on each summand, and J equates the two scalars.

14. Lemma 16, Example 16.9

Error. The two algebra-generation calculations and the abstract classified bimodule are correct. The claimed application of Corollary 16.8 with the subsequently chosen nontrivial grading is not justified by the specified even parent, and the compressed potential does not repair it.

For the generation calculations:

- The matrices $1, \Gamma_a, \Gamma_a\Gamma_b$ for $a < b$ are 16 real-linearly independent elements commuting with j. Since $\dim_{\mathbb{R}} \mathrm{End}_{\mathbb{H}}(W) = 16$, the real algebra generated by the five Γ_a is $M_2(\mathbb{H})$.
- For Hermitian B, the fourth moment gives $Q(w^*Bw/w^*w) = (B + \mathrm{tr}(B)1)/5$. This ranges over all Hermitian matrices. Their products generate $M_4(\mathbb{C})$ as a real algebra.

The use of two copies of the four-space is mathematically legitimate. It supplies independent direct summands and the 32-dimensional bimodule. For this particular grading of W and trivial grading of V, the graded flip is the ordinary adjoint flip, and

$$\Gamma_F(\xi, \eta) = (\Gamma_5\xi, -\eta\Gamma_5),$$

exactly the graded representation in Chamseddine–Connes, Proposition 3.2. The full algebra is a **graded prealgebra**; its even part is

$$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C}).$$

The obstruction is that an even parent has $[P\gamma_X P, PaP] = 0$ for every compressed parent function. But the proposed grading $\Gamma_5 = 5Q(x_5)$ anticommutes with $Q(x_1) = \Gamma_1/5$. Thus the full compressed $M_2(\mathbb{H})$ and that grading cannot both arise from the stated even parent. They can be chosen **after** truncation as abstract graded-algebra data, and the even subalgebra then gives the real zero-cycle orientability relation.

Similarly, $d = \mu\Gamma_1$ is the compression of a bounded real multiplication operator, but adding that operator to the original parent Dirac operator does not retain the original kernel as a spectral projection: multiplication by the nonconstant real coordinate does not preserve the holomorphic kernel. It also does not preserve the original parent grading as an odd perturbation.

Finally, d commutes with the centre of the full prealgebra, namely $\mathbb{R} \oplus \mathbb{C}$. It does **not** commute with the whole centre after passing to the even algebra: for $e_+ = (1 + \Gamma_5)/2$, $[\Gamma_1, e_+] \neq 0$. It still commutes with the two sheet projections.

15. Framework note: real structure and KO-dimension

Correct with amendment. The KO-degree and antiparticle grading statements are right. The Euclidean Pfaffian construction uses the fermionic action on the appropriate chiral subspace. Barrett's Lorentzian treatment removes the quadrupling through compatible reality and chirality conditions; it should not be presented as the same Euclidean Pfaffian argument. See Barrett, §§1–3 and CCM, §4.4.

16. Framework note: classification of (A, J)

Correct with amendment. The classification is under order zero, a separating vector, irreducibility, and $J^2 = 1$. "Complexified algebra" here means the represented complex span used in the paper. The exclusion of the one-centre case also uses compatibility of the grading with the algebra—normalization in the graded-prealgebra formulation, or commutation in an even triple. Without compatibility, an arbitrary anticommuting grading is not excluded. See Chamseddine–Connes, §§2–3.

17. Framework note: real form and grading

Correct with amendment. The quaternionic grading and smallest choice $\dim_{\mathbb{H}} W = 2$ are hypotheses and choices in the source. The displayed full algebra is graded; $\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$ is its even part. On $\mathrm{Hom}(W, V)$, the grading acts by **right** composition, with the minus sign.

18. Framework note: order one

Correct. Under the hypotheses of Chamseddine–Connes, Theorem 4.1, the stated maximal even subalgebra is selected by the existence of a centre-connecting Dirac operator satisfying order one. The subsequent gauge group uses the represented unimodular lift; the statement is an algebraic selection, not a dynamical derivation of a vacuum.

19. Framework note: unbroken symmetry

Correct with amendment. CCM explicitly impose commutation with the nonscalar $\mathbb{C}_F = \{(\lambda, \lambda, 0)\}$ and identify its physical purpose with the massless photon; see CCM, equation (2.30) and Remark 2.19. The subalgebra is not unital in the full algebra, and the physical electromagnetic lift also uses the unimodularity condition. The nonscalar element in the kernel of $a \mapsto [D_F, a]$ explains the failure of the metric condition cited in the note; a treatment formulated only for complex algebras must specify its complexification.

20. Framework note: beyond first order

Correct with amendment. The quadratic fluctuation, gauge covariance, transitivity, and semigroup statements are supported by Chamseddine–Connes–van Suijlekom, §§I and III. The Pati–Salam construction is in the separate arXiv:1304.8050, §§III and VIII. Its discussion of a vacuum selecting the Standard Model subalgebra is model- and vacuum-dependent: the paper explicitly says that determining all vacua is difficult

and checks a restricted Hessian. It should not be stated as a general theorem that every vacuum restores order one.

21. Framework note: quanta of geometry

Correct. In the conventions of Chamseddine–Connes–Mukhanov, equations (2)–(5), the two Clifford algebras in dimension four are $M_2(\mathbb{H})$ and $M_4(\mathbb{C})$, and the two-sided relation gives the sum of the pulled-back volume densities. This is a quantization condition with two maps, not a truncation result.

22. Framework note: twisted triples and grand symmetry

Correct. The larger algebra mixing spin and internal degrees of freedom and the extra scalar are described in Devastato–Lizzi–Martinetti, §§3–5. A twisted commutator replaces the ordinary bounded-commutator requirement in the later Devastato–Martinetti construction, §3. The note properly says that these constructions have not been studied here.

23. Table: noncommutative finite algebra

Correct. The interval and projective-space examples give the stated generated algebras, with algebra generation an additional operation. The distinction between PAP as an operator system and its generated C^*-algebra is also emphasized in Connes–van Suijlekom, §1.

24. Table: inherited pointwise reality

Correct. Under Proposition 16.1's pointwise-reality and invariant-window hypotheses, order zero for the generated algebra holds exactly when that algebra is commutative. This is scoped to that inherited correspondence.

25. Table: order zero, order one, centre $\mathbb{C}$

Error. Corollary 16.3 gives the order conditions and dimension k^2, but it does not generally give centre $\mathbb{C}$. For its interval example $A_F = \mathbb{C} \oplus \mathbb{C}$, the centre is $\mathbb{C} \oplus \mathbb{C}$. The stated centre requires an additional condition, such as a simple generated algebra.

26. Table: KO-6 signs

Correct. Both constructions give the signs. The first pair is reducible and fails the displayed real orientability relation; the two-sheet input is explicit.

27. Table: centre $\mathbb{C} \oplus \mathbb{C}$, dimension $2k^2$

Correct with amendment. Corollary 16.8 reaches this classified pair when the two spaces have equal dimension and the represented complex spans are full matrix algebras. Arbitrary generated algebras in that corollary can have larger centres and need not give an irreducible pair.

28. Table: orientability

Correct. The displayed zero-cycle works under the stated condition that both finite gradings lie in the even represented algebras. It is real zero-dimensional orientability, in the sense specified in section 13 of this report.

29. Table: $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$

Correct with amendment. The real generated algebras and the 32-dimensional abstract bimodule are reached. The nontrivial grading is added after the stated truncation, and the full algebra with that grading is a graded prealgebra. It is not an even spectral-triple algebra, nor is the grading derived from the specified

even parent. The construction requires a **real** generated algebra on the S^4 side; complex C^*-generation would also include $i1$.

30. Table: selection of A_F

Correct with amendment. An off-diagonal term is missing, but an arbitrary off-diagonal term alone is not enough. One must impose the required order-one condition and the maximal-subalgebra hypothesis of the classification.

31. Table: Yukawa block

Correct with amendment. Theorem 12.7 gives the texture in a towerless limit, up to grading-preserving unitary identifications. An algebra and its representation must also be specified before calling the result a finite spectral triple. It is not embedded in Example 16.9.

32. Table: $[D_F, \mathbb{C}_F] = 0$

Correct with amendment. Proposition 17.3 explains the consequence **when** a chosen subalgebra commutes with D_F; it does not make an arbitrary interval or Yukawa block commute with a Standard Model representation. The appropriate common action on the paired blocks is additional data.

33. Table: bosonic action

Correct with amendment. The leading-order results are correctly described, but "near-cutoff in progress" is stale: lemma 18 now gives the exact quadratic heat-cutoff calculation and the stated positive-Laplace extension. Its scope and limitations are those checked above.

34. Table: beyond first order

Correct. The inherited two-point data supply an order-zero, first-order-failing example to which the generalized-fluctuation question can be posed. The table correctly records that the calculation has not been done and that relaxing order one does not cure order-zero failure.

35. Table: quanta of geometry

Correct with amendment. The same five Clifford matrices occur, and the projector identity is valid. For the quaternionic rank-one projector $\Pi = P_w + P_{jw}$, expansion in the real basis $1, \Gamma_a$ gives $\Pi = (1 + \sum_a x_a \Gamma_a)/2$. This algebraic connection does not derive the two-sided quantization condition from truncation.

36. Table: twisted triples and grand symmetry

Correct. The table records no result and marks the subject untested.

Part 2: scope of the heavy-limit conclusion

Within the three-generation, zero-Majorana, top-only family of Proposition 4, at fixed cutoff and for the heat function, **both** the charged Taylor mass and its unique positive timelike pole tend to zero. The neutral Taylor and pole masses also tend to zero. The result refutes an infinitely heavy W, Z, t limit **for these background data and these quadratic mass definitions**. It also rules out a stationary nonzero radial vacuum in the positive-Laplace class. It does not rule out other finite data, a different action or constraint, a joint $\Lambda \to \infty$ limit, or a quantum effective theory.

The top-only Taylor conclusion is broader than (4). For a nonnegative smooth compactly supported f with $f(0) > 0$, define

$$M_1 = \int_0^\infty z f(z)\, dz, \qquad M_2 = \int_0^\infty z^2 f(z)\, dz.$$

The heat formulas extend by linearity and smooth functional calculus to

$$A_f(x) = \frac{2}{x}\int_0^x zf(z)\,dz, \qquad w_f(x) = \frac{3}{x^3}\int_0^x z^2 f(z)\,dz. \tag{R2}$$

One way to justify the extension is to approximate a smooth compactly supported $f(z)$, with the derivatives needed for the quadratic variation, by finite real linear combinations of e^{-nz}: write $f(z) = e^{-z}g(e^{-z})$ and approximate the smooth g and its derivatives by polynomials. The divided-difference formula for the variation permits passage to the limit. Thus

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{M_1}{f(0)x}.$$

This includes a nonnegative cutoff flat near zero. For a nonanalytic cutoff there need not be a prescribed analytic continuation defining the pole used in Theorem 5.

Several heavy doublets change the answer. Suppose, for the heat cutoff, that P is the total particle-trace weight of complete massless doublets, and heavy doublet i, of weight $d_i > 0$, has one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\left(Pg(y) + \sum_i d_i h_{x_i}(y)\right), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}. \tag{R3}$$

For any fixed finite number of doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, (R3) instead gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{m^4}{2\Lambda^2}.$$

The full heat form factor also has a unique positive pole in this equal-mass case. To find its scale, put $z = cx$, $0 < c < 1$, in (21). Concentration near $u = 0$ gives

$$h_x(-cx) \sim \frac{2(1-2c)}{x(1-c)^2}.$$

Monotonicity then places the root at $z/x \to 1/2$, so $M^2_{W,\mathrm{pole}} \sim m^2/2$. At every fixed Euclidean momentum, however, the whole unnormalized quadratic kernel tends to zero. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory.

There is also a fully massive example. Give **both** members of every Standard Model doublet the same Dirac mass m, with no Majorana masses. The charged form factor is proportional to

$$e^{-x}[g(y) + 2xH(y)].$$

Its positive pole obeys $2x < z_W < 2x+2$. The neutral massive pole obeys $(16/5)x < z_Z < (16/5)x+2$. Both pole masses therefore diverge with m. Yet the fixed-momentum action tends to zero, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1-x/2)$, which is negative for $x > 2$. This is a counterexample to a general fixed-cutoff no-go for diverging pole locations, while leaving the physical quality of that limit open.

The phrase "the heavy fermion leaves the loop" is only a shorthand for the suppression in this **classical spectral trace**. The light complete doublets supply the surviving charged kinetic form, but the top-only light space also retains b_L after removing t_L. If P_{light} is the projection off the top mass block, then on b_L

$$P_{\text{light}} T_1 P_{\text{light}} = P_{\text{light}} T_2 P_{\text{light}} = 0, \qquad P_{\text{light}} T_3 P_{\text{light}} = -\tfrac{1}{2}.$$

These compressed operators do not obey the $SU(2)$ commutation relations. Correspondingly, the limiting particle-trace coefficients have $K_W = 9/2$, $K_{33} = 21/4$, and $K_{3Y} = -1/4$; a linearly realized unbroken electroweak gauge action would require equal weak coefficients and no $3Y$ mixing. Vanishing vector masses therefore do **not** prove unbroken $SU(2) \times U(1)$ on the entire light space.

The remaining **complete** massless doublets do carry the ordinary electroweak representation, and their separate contribution is gauge invariant. If finite data instead make both members of every removed doublet heavy, the retained light subspace can be invariant under the full group; a full limiting theory still requires control of the remaining fields, normalization, interactions, and vacuum.

Part 4: what the two-sheet construction establishes

Example 16.9 is substantive as a concrete compression calculation: real base functions generate a quaternionic matrix algebra, real total-space functions generate a complex matrix algebra, and two copies give the classified 32-dimensional bimodule. The calculation does not select those ingredients dynamically. The restriction to base functions on one sheet, the second independent sheet, real algebra generation, the four-dimensional window, and the grading axis are inputs.

At the finite algebraic level, the two-sheet bimodule is the standard construction from two representations:

$$(W \otimes \bar{V}) \oplus (V \otimes \bar{W}).$$

The graded flip supplies the real structure and the block sign supplies the KO-6 relation. The calculation's additional content is its realization from the particular compressed function systems and its explicit matching to the classified data. The abstract bimodule and its adjoint-flip real structure are already stated in Chamseddine–Connes, Remark 2.9; the report makes no broader novelty claim.

The note appropriately lists several choices and says that no ordinary Kaluza–Klein vacuum is derived. It overstates the reach where it treats the chosen nontrivial grading and nonzero compressed d as consequences of the stated even-parent truncation. It also labels the centre of Corollary 16.3 too generally. Lemma 16's later caveat that its generated algebras are complex and a quaternionic summand still needs additional real data is stale after Example 16.9's real-algebra construction.

For canonical section 4, the example closes the **abstract finite-module** problem after adding a second sheet: it turns the nested actions on one four-space into independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the centre-connecting Dirac operator required for the subsequent algebra selection. The fibre-rank problem remains open.

Amendments and errors

- Add field regularity to Theorem 1, $Q > 0$ or a quotient by its null space to Corollary 3, and $x > 0$ to the massive neutral-pole assertion.
- Keep the neutral-current angle distinct from other finite-cutoff angle definitions.
- Replace Corollary 16.8's claim about the whole centre with commutation with the two sheet projections.
- In Example 16.9, distinguish the correct abstract graded prealgebra from an even spectral truncation; the proposed grading and nonzero compressed perturbation are not derived by the stated parent. The nonzero d also fails to commute with the whole centre of the even algebra.
- Correct the table's unrestricted centre-$\mathbb{C}$ claim for Corollary 16.3, and add the full-matrix and equal-dimension hypotheses to its classified two-sheet row.
- Update the table's near-cutoff entry and qualify the claims about the Yukawa representation, off-diagonal Dirac operator, real-algebra generation, and the vacuum discussion beyond first order.
- Distinguish the Euclidean Pfaffian and Lorentzian fermion reductions, and retain the grading-compatibility hypothesis in the classification.

- On flat $\mathbb{R}^4$, equation (R1) extends the radial analysis to nonnegative non-Laplace cutoffs; compact support permits a stationary plateau.
- The top-only limiting quadratic action does not establish an unbroken electroweak representation on the full light space.

Overall: 36 statements checked: 18 correct, 15 correct with amendment, 0 gaps, and 3 errors.

Part R5 (referee report, main draft and note)

Referee report on the main draft and the note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_paper_note.md. Reproduced as delivered, with links relativized. The corrections are incorporated in the paper, the note and lemmas 17 and 18.

The main draft contains useful exact calculations, but it needs substantial revision before it is ready for a Physical Review journal. The most consequential issues are the identification of the formal KO labels with geometric endpoints, the claim that collapse with a fixed light mass requires a tuned bulk mass, and several statements whose scope changes between the theorems, Table I, the abstract, and the discussion.

The note preserves the corrected reduced-mass formula and the sign of j_-. Its main mathematical defect is that the stated condition on the condensate subtraction does not suffice for the claimed $O(m)$ remainder.

The numbered verdicts below count distinct statements, including separate occurrences in Table I and the abstract. We read the LaTeX sources as text and did not compile or edit any file. The underlying lemmas were consulted for the relevant statements and hypotheses; their proofs were not re-refereed in full.

Part 1. Main draft

Chirality at the odd labels The paragraph is at towerless_intervals.tex:184.

1. Classical objection — Correct with amendment. An irreducible spinor representation in odd spacetime dimension has no intrinsic Weyl decomposition: the Clifford volume element is central, so there is no nontrivial chirality operator that anticommutes with every Clifford generator. In the usual smooth, closed, product compactification of an odd-dimensional bulk, the standard fermion reduction therefore has a vectorlike four-dimensional content.

State the compactification and fermion assumptions. Odd-dimensional bulk theories can have chiral modes localized on defects or boundaries, and "no chiral fermions" should not be read as excluding all localized modes. The full text of the cited Witten 1981 article was inaccessible in this review; its precise attribution remains unchecked. The mathematical amendment above follows directly from the Clifford representation.

2. Full chiral algebra — Correct with amendment. For the *specified Standard Model representation* of $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$, neither proposed odd factorization exists. Let $e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$. Its represented range lies in one grading sector and has no counterpart in the other. A grading-reversing U commuting with the algebra would have to identify those two ranges. This is the argument in lemma 06, Result 8.

The conclusion depends on the representation, not on the abstract algebra alone. Name the Standard Model bimodule in the sentence.

3. Label 9 and the absence of needed chiral fermions — Correct with amendment. The charged surviving sectors are vectorlike under the reduced $SU(3)\times U(1)_{\mathrm{em}}$ representation and each admits the $s=-1$ factorization. This establishes a sectorwise formal label 9.

It does not give that factorization for the strict *whole* light spectrum with one or three unpaired neutrinos. A neutral Weyl fermion has a real gauge representation but remains a chiral spacetime fermion; it may be expressed in Majorana form. State explicitly whether the neutral sector is removed, paired, or treated separately.

4. Neutral representation and parity — Correct. The neutral gauge representation is real in the reduced model. If the descended antiunitary j obeys $j^2 = -1$, its complex representation space has even dimension. Indeed, writing $j = UK$ gives

$$|\det U|^2 = \det(-1) = (-1)^{\dim_{\mathbb{C}} H_0}.$$

For a Majorana block the stronger condition is even multiplicity of **each** singular value, including zero, because j commutes with d^2. Odd total multiplicity is one obstruction.

5. Spectrum at fixed distance from a wall — Error. At fixed $\mu > 0$, the normalized left-wall profile tends to

$$\sqrt{2\mu}\, e^{-\mu y}.$$

It is nonzero at every fixed finite distance y from the wall. Thus a region at fixed distance from a wall still sees the chiral wall mode, even though its amplitude is exponentially small when μy is large.

The gapped bulk statement applies when the observation region recedes from **both** walls as $l \to \infty$. In that limit the translation-invariant bulk Dirac operator has spectral threshold μ, while the two half-line limits retain the chiral zero modes.

6. Assignment of the wall limit to label 11 — Gap. The wall-mode limit and the KO-7 factorization are established separately. The text does not construct an identification that assigns KO label 11 to the decompactifying interval. The interval product in Theorem D also has a stated KO-5 factorization. Present the label-to-endpoint connection as an interpretation, or give the missing compatibility and limiting construction.

7. Hořava–Witten comparison — Correct with amendment. Boundary chirality is a fair qualitative comparison. With a common orientation and common identification of the ten-dimensional tangent spaces, the two Hořava–Witten boundary gaugino sectors have the same ten-dimensional chirality. In the interval model here, the two limiting modes of one type-(c) fermion have opposite Γ-chirality. Boundary-induced orientations reverse between the two ends, so the chirality convention must be stated.

The constructions also have different content: Hořava–Witten has an E_8 vector multiplet on each boundary and anomaly cancellation involving the bulk, whereas the interval calculation follows two modes of a bulk Dirac operator. The comparison should identify that common feature and those limits. See Hořava and Witten, hep-th/9603142, §§1–2 and their preceding orbifold paper, §2.

8. Physical point at finite l in an even interior — Gap. The results do not relate l quantitatively to the measured Fermi constant or weak angle, nor do they identify the physical Standard Model point with a member of this interval family. State this as a proposed interpretation. The finite KO-6 structure does accommodate chirality; its identification with a measured finite-l point remains to be shown.

Labels by sector **9. Charged paired sectors — Correct.** In the reduced representation, each charged species and its conjugate form an invariant sector for the represented algebra, its opposite, D, and J. The displayed paired-sector construction admits both signs, including with flavour mixing within a species.

10. One or three unpaired neutrinos — Correct with amendment. With the stated zero Majorana block and reduced algebra, the strict light triple admits $s = +1$ and excludes $s = -1$. Add that the neutral $s = +1$ involution does **not** preserve particle number. If particle preservation is imposed on the strict unpaired spectrum, neither proposed reduction is available; see lemma 06.

11. Right-handed neutrinos with pure Dirac masses — Correct. Retaining the Dirac partners gives paired singular values in the neutral block, so both signs occur under the stated reduced representation.

12. Majorana masses — Correct with amendment. A Majorana term breaks the particle-number hypothesis used to pair U and QU. Generic nondegenerate neutral Majorana data obstruct $s = -1$, while $s = +1$ remains available under the scalar-character assumptions. State the exact criterion: $s = -1$ exists precisely when every neutral-block singular value has even multiplicity. A nonzero Majorana term alone does not exclude it in several generations.

Weighted sum rule and cutoff limit The claims are at towerless_intervals.tex:202 and towerless_intervals.tex:228; their source is lemma 18, Remark 16.

13. Weighted identity — Correct with amendment. The identity

$$8\kappa M_{W,\mathrm{Taylor}}^2 = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{2[1-(1+x)e^{-x}]}{x^2},$$

follows from the displayed kernel for the heat cutoff $f(z) = e^{-z}$. Its scope is the specified finite collection of doublets, each massive doublet having one massive Dirac member and one massless member, with the trace weights and mass-compatible doublet decomposition used in Proposition 13 of lemma 18. The mass is the **Taylor ratio** $C_W(0)/C_W'(0)$, not the exact pole mass.

Also define the trace on $\mathbb{R}^4$ through a relative heat trace or action density: the unsubtracted vacuum heat trace has infinite volume.

14. Definition of κ — Correct with amendment. In the three-generation normalization, write

$$\kappa = \frac{P + \sum_i d_i w(x_i)}{6}, \qquad N_i = 2d_i, \qquad w(x) = 3\int_0^1 u^2 e^{-xu}\, du,$$

where P is the particle-trace weight of complete massless doublets and $P + \sum_i d_i = 6$. Thus κ is the charged Taylor kinetic coefficient divided by its all-massless value. For the top alone, $P = 9/2$, $d_t = 3/2$, giving $(3+w)/4$.

"Normalization of the kinetic term" without these weights does not define κ for the displayed general sum.

15. Small-mass form — Correct with amendment. If **all** $x_i = m_i^2/\Lambda^2 \to 0$, then $\rho(x_i) \to 1$ and $\kappa \to 1$, giving the CCM relation to leading order. This proves the stated relation for the heat-cutoff Taylor calculation. The CCM asymptotic formula also has its own cutoff and trace conventions; the displayed ρ is not a general-cutoff form factor.

16. Five-percent departure — Correct. For the top alone,

$$\frac{M_W^2/m^2}{3/8} = \frac{\rho(x)}{\kappa(x)}.$$

The expansions

$$\rho(x) = 1 - \frac{2}{3}x + \frac{1}{4}x^2 + O(x^3), \qquad w(x) = 1 - \frac{3}{4}x + \frac{3}{10}x^2 + O(x^3)$$

give a departure of about 5% in m/M_W at $x = 0.45^2$.

17. Peak — Correct. The stated peak is supported by the formula. For an analytic uniqueness check, set

$$B = 1 - (1+x)e^{-x}, \qquad D = x^3 + 2 - (x^2 + 2x + 2)e^{-x}.$$

Apart from a positive factor, the derivative has the sign of

$$T(x) = 4 - x^3 + (x^5 + x^4 - 8x - 8)e^{-x} + (x^3 + 4x^2 + 8x + 4)e^{-2x}.$$

The power series of $e^{2x}T(x)$ has positive coefficients at degrees 5, 6 and 7, and negative coefficients at every degree at least 8. Dividing by x^7 therefore gives a strictly decreasing function on $x > 0$, with one zero. The Taylor mass has one positive maximum and falls thereafter.

One small **Python 3** computation solved this derivative equation and evaluated the stated mass formula. It gave $x = 2.02428$, $m/\Lambda = 1.42277$, and $M_W/\Lambda = 0.52422$. The computation tests the location; the sign argument gives uniqueness.

18. Heavy suppression — Correct with amendment. A massive diagonal contribution to the unperturbed heat trace is exponentially suppressed. In a mixed massive–massless contribution, the form factor instead has the algebraic behavior

$$\rho(x) \sim 2/x^2.$$

The corresponding mass-source contribution behaves as $m^2\rho(m^2/\Lambda^2) \sim 2\Lambda^4/m^2$. State which trace contribution is exponentially suppressed so that the sentence does not suggest exponential suppression of the mixed loop.

19. Necessity of Λ growing faster than m_t — Correct with amendment. It is an exact necessity for the **top-only heat-cutoff Taylor family** if the limit requires $m_t/M_W \to \sqrt{8/3}$. To see this, use

$$\rho(x) = 2\int_0^1 ue^{-xu}\,du, \qquad w(x) = 3\int_0^1 u^2e^{-xu}\,du.$$

The function $3 + w - 4\rho$ vanishes at zero and has derivative $\int_0^1 u^2(8-3u)e^{-xu}\,du > 0$. Hence the exact top-only ratio differs from its CCM value for every $x > 0$; it approaches that value only if $x \to 0$.

The necessity is broader than the calculation if stated without the top-only and cutoff hypotheses. For example, retain one complete massless lepton doublet and give one member of each other doublet the common mass m. Then $\kappa = (1+11w)/12$. Near zero, $\rho - \kappa = x/48 + O(x^2) > 0$, while at infinity it tends to $-1/12$. Continuity gives a positive x_* at which the full weighted identity equals the unweighted CCM sum exactly. Scaling $m = \sqrt{x_*}\Lambda$ preserves that equality with a nonzero m/Λ. This is a different mass family.

Generated algebra and the cited remark **20. Bicommutant — Correct with amendment.** For a unital complex representation on the finite-dimensional V,

$$C^*(Q(A_0)) = Q(A_0)''$$

by the finite-dimensional bicommutant theorem. State that the commutants are taken in $\mathrm{End}_{\mathbb{C}}(V)$, and retain the unital or nondegenerate-representation hypothesis. The quaternionic real algebra in item 4 needs a separate real-algebra statement.

21. Attribution to Connes and van Suijlekom — Correct. Their introduction says that in many interesting examples the C^*-algebra generated by PAP is a "non-informative full matrix algebra." The draft preserves both the scope and the substance of the observation. Their paper also emphasizes the information retained by the operator system itself. Connes and van Suijlekom, arXiv:2004.14115v2, Introduction.

22. Interval block structure — Correct with amendment. The grading commutes with every compressed function, so it forces the generated algebra to be block diagonal. The explicit interval compressions then show that it fills both blocks, giving $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$.

The grading alone gives containment in that block algebra; it does not prove equality. The two matrices displayed in the main proof sketch lie in different blocks and commute, so those two examples alone do not generate both full blocks. Add the independent symmetric matrices within each block and the argument separating the two summands; lemma 16, example (b) supplies it.

23. Quaternionic form on S^4 — Correct with amendment. For **real functions**, the compressed operators commute with the antiunitary j, $j^2 = -1$, and generate the real algebra $M_2(\mathbb{H})$. The five compressed coordinate matrices provide the generators. The complex C^*-algebra generated by the same coordinates is $M_4(\mathbb{C})$.

Thus j identifies a quaternionic real form; it is an antiunitary symmetry and is not an element of the ordinary complex-linear commutant in item 1. State the real-versus-complex distinction at towerless_intervals.tex:296, not only earlier in item 4.

Reading the negative results The directions in towerless_intervals.tex:306 are reasonable research directions when kept within their stated scope.

24. Odd-dimensional chirality → boundaries and an even interior — Correct with amendment. Boundaries are one way to obtain localized chiral modes. The claimed even interior is an additional proposal; singularities, defects and other changes of hypotheses are also possible directions.

25. Bare interval limits → bulk mass — Correct with amendment. A negative bulk mass with the stated localization supplies a light pair. The boundary-Yukawa family supplies another route; see check 34.

26. Fixed-compression covariance failure → covariant projection — Correct. Projecting onto a gauge-transformed spectral subspace addresses the demonstrated covariance defect. Further matching of actions remains a separate question.

27. Bounded-fermion collapse → tower-scale fermion or changed action/order — Correct with amendment. This follows for the leading spectral action, specified trace, common profile setting, and tower matching considered in Proposition F. It is a useful direction beyond those assumptions, not a general theorem about every gauge action.

28. Inherited order-zero failure → doubled parent — Correct. The tensor-square parent supplies commuting left and opposite actions and thereby gives a constructive route. The two-sheet and grading inputs remain explicit.

29. Fixed-cutoff behavior → joint cutoff limit — Correct with amendment. The joint limit is a valid route in the top-only family. At fixed cutoff the lemma also contains fully massive families with diverging pole locations, so the direction must retain the mass-family and mass-definition hypotheses.

30. No finite radial vacuum → a potential or constraint — Correct. Adding a specified radial potential or imposing a constraint could alter the radial stationarity problem. Neither follows from the spectral action already analyzed.

31. Commutative whole limits → truncation — Correct with amendment. Compression followed by algebra generation is a constructive way to obtain noncommutative finite data. The commutative-limit statement concerns the specified spectral propinquity and whole scalar algebras; it does not rule out other notions of convergence or other parent algebras.

Table I The table begins at towerless_intervals.tex:46.

Check	Entry and verdict	Amendment
32	Two-point space — **Correct.**	The stated mass and minimal truncation are the content of Theorem A.
33	10 to 11 — **Correct with amendment.**	Say "in the type-(c) family, at fixed $\mu > 0$." The KO-label identification remains the gap in check 6.
34	10 to 9 — **Error.**	A tuned bulk mass is necessary for the stated one-field type-(c) family, not for every interval collapse. In Theorem C's one-flavour model take zero bulk masses and $X(l) = \tan(ml)$. The light eigenvalue is exactly $\arctan\|X\|/l = m$ for small l, and the higher levels diverge.
35	Yukawa block — **Correct with amendment.**	Replace "D_F is the towerless limit" by "the displayed Dirac Yukawa block of D_F is obtained in the boundary-Yukawa limit." The full finite algebra, Majorana block and real triple are not derived there.

Check	Entry and verdict	Amendment
36	9 and 11 — **Correct with amendment.**	Specify the reduced representation and the particle-number caveat in check 10. The full chiral algebra admits neither proposed factorization.
37	Heavy W — **Correct with amendment.**	The weighted formula is for the heat cutoff and Taylor mass with the stated doublets. The top/W limit needs top dominance in the CCM expansion.
38	W against the tower — **Correct with amendment.**	State the leading action, trace, gauge-boundary input and common-profile matching assumptions.
39	Chirality — **Gap.**	The table promotes the proposed assignment of the two odd labels to limits and of the physical point to an even interior into a result. Supply the identification or label it as an interpretation.
40	Finite algebra — **Correct with amendment.**	Distinguish the complex generated algebra from the real quaternionic form, and retain the two sheets, restricted algebra and added grading as inputs.

Abstract **41. Collapse and boundary Yukawa data — Error.** The abstract repeats the unrestricted "only along a tuned bulk mass" claim and makes the boundary-Yukawa result sound like the continuation of that same tuned family. The counterexample in check 34 applies. State the two different constructions and their scalings separately.

42. KO labels and weighted mass relation — Correct with amendment. State that the charged and neutral factorizations use the reduced representation, and identify the heat cutoff and Taylor mass for the weighted sum. The complete chiral Standard Model algebra has neither factorization.

43. Whole-triple limit and two-sheet parent — Correct with amendment. "Whole triples converge … only to points" needs restriction to the finite limits of the connected commutative metric triples considered in Proposition G. The two-sheet construction reproduces the specified algebraic data with the listed inputs; it does not derive a connected parent or the full finite Dirac operator.

Discussion, literature, and one additional main-text statement **44. Masses as momenta combined in quadrature — Error.** The discussion at towerless_intervals.tex:302 applies the real-momentum formula to the localized light mode. For that mode $k = i\kappa_0$, and

$$\lambda_0^2 = \mu^2 - \kappa_0^2.$$

The formula $\lambda_n^2 = \mu^2 + k_n^2$ with real k_n applies to the excited tower.

45. Localization alone determines the texture — Error. At towerless_intervals.tex:149, order-one **entries** of $\hat{Y}$ do not by themselves give the stated individual mass pattern. Let

$$F = G = \mathrm{diag}(\varepsilon, 1), \qquad \hat{Y} = \begin{pmatrix} 1 & 1 \\ 1 & 1+\eta \end{pmatrix}.$$

All entries are order one and the trailing minors are nonzero for $\eta > 0$, but $\det(F\hat{Y}G) = \varepsilon^2\eta$. The small singular value carries the additional arbitrarily small factor η. Require quantitative lower bounds on the relevant minors, as well as the stated profile hierarchy, or present the formula as a generic numerical expectation.

46. Index and open problems — Correct with amendment. The ordinary index is stable under the boundary coupling in the stated family. The equivariant statement is made for the stabilizer of X: symmetry breaking changes the group on which the representation-valued index is read. The discussion should distinguish those statements and retain its explicit qualifications on the parent, grading, dynamics and action.

47. Claims about what was not found — Gap. The statements about the chiral-interval compression, tuned collapse, localization-derived Yukawa block and factorization criterion may be accurate, but the draft does not say which databases, years, synonyms, or citation chains were searched. We cannot infer a comprehensive absence from the text.

The source check here was limited. Queries included `"noncommutative geometry" "interval" "spectral truncation" two point chiral 2026`, a localization/Yukawa query, and an odd/even real-triple factorization query. We opened the relevant sections of the sources cited in checks 7, 21, 48 and 49. Forward and backward citation chains and author recency searches were not completed, and the full texts of Witten 1981 and Banks–Casher 1980 were unavailable. Consequently this review does not establish priority or absence in the literature.

A relevant recent neighboring paper is Baptista, arXiv:2506.09126v2, §§1 and 5: it studies chiral couplings to massive gauge fields in smooth Kaluza–Klein models. Its examples use even-dimensional internal spaces; it does not refute the odd-dimensional Clifford statement. It is relevant to the broader discussion of routes to chiral interactions.

48. Positive attributions checked — Correct. Csáki et al., hep-ph/0310355v1, Eq. (5.6) give $\tan(m_n L) = M_D L$, yielding the one-flavour arctangent spectrum after identifying the boundary parameter. Chamseddine and Connes, arXiv:0706.3688v1, Remark 2.9, give the two Hom-space bimodule and flip real structure cited in the draft. Check 21 establishes the Connes–van Suijlekom attribution.

49. Farsi–Latrémolière theorem — Correct with amendment. The issue is visible in arXiv:2404.00240v3, Theorem 3.4 and Eq. (3.11). The proof's displayed estimate requires control of the vertical operator by the total rescaled operator. A scalar example shows the need for an additional condition: on e_n, take $D_v e_n = n e_n$, $D_h e_n = -n^2 e_n$, and $\epsilon_n = (n + 1/n)^{-1}$. Then $D_{\epsilon_n} e_n = e_n$ although the proposed limit on $\ker D_v$ has spectrum $\{0\}$.

Anticommutation or a suitable relative bound are **sufficient repairs**, rather than the only conceivable repairs. Identify the checked preprint version; the published text was not examined here.

50. Proposition G converse — Error. The statement at towerless_intervals.tex:263 uses the previously arbitrary unitary W_n in an "if and only if." Take the constant triple

$$T_n = T_F = (\mathbb{C}, \mathbb{C}^2, \operatorname{diag}(1, 2)),$$

with $P = 1$, and let every W_n swap the two basis vectors. The spectral propinquity is zero, while $\Delta_n = \|\operatorname{diag}(2, 1) - \operatorname{diag}(1, 2)\| = 1$.

The converse needs **there exist** appropriate light spectral projections and unitaries W_n for which $\Delta_n \to 0$, with g_n chosen as the actual exterior gap or a lower bound tending to infinity. This is the quantifier stated in lemma 15, Theorem 15.7.

Part 2. Note compared with the corrected lemma

We compared chiral_limit_note.tex against lemma 17 and its referee report.

51. Sign of j_- — Correct. With the standard Pauli matrix,

$$i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The note's $j_- = i\sigma_2 K$ is exactly the matrix in lemma 17. There is no sign discrepancy.

52. Reduced mass — Correct. In the note's basis $H_0 = (q_L, \bar{q}_R)$,

$$d_{\Sigma,s} = (\Sigma M^* \otimes 1_3) \oplus (s\,\overline{\Sigma^* M} \otimes 1_3).$$

On q_L, UD_F gives ΣM^*. On $\bar{q}_R$, the antiparticle block of U contributes the factor s, yielding $s\overline{\Sigma^* M}$. This agrees with the corrected lemma and report.

53. Reduced action and trace — Correct. The theorem retains the trace-class hypothesis, the sign-dependent real structure, the change of independent Grassmann variables in the paired sector, and the warning that the formal chiral rotation has a quantum Jacobian. The Majorana-type unpaired example prevents the Dirac conclusion from being applied to every sector.

54. Commuting subalgebra — Correct with amendment. The note states the no-scalar-fluctuation proposition and the restriction needed before calling the fields photon and gluons. Add the surviving electron pair to the explicit representation check, and state that the conclusion concerns the kept mass blocks and admissible involution. The first paragraph of the abstract should keep the same restriction.

55. Electric charge and flavour — Correct with amendment. The $U(1)^4$ statement needs **two differently charged flavours**, such as u, d. Two flavours of the same electric charge retain a larger flavour symmetry. The note also omits the corrected lemma's corresponding statement for admissible reductions: with charge included, intersect the colour-only admissible set with the charge commutant **and retain** $\Sigma^* M = (\Sigma^* M)^*$. At zero mass for u, d this leaves diagonal phases; positive nonzero diagonal masses leave signs.

56. Stationarity, reduced mass and polar part — Correct. The condition $\Sigma^* M$ Hermitian is equivalent to stationarity of $-\operatorname{Re}\operatorname{tr}(M\Sigma^*)$. The homogeneous-space statement at $M = 0$, the Hermitian involutions for $M = mI > 0$, and the unique polar minimum for invertible M agree with the corrected lemma. "Sign matrices" for positive distinct masses are to be read in a mass eigenbasis.

57. Vacuum alignment and mass signs — Correct. The note says that the stationary points include saddles and maxima, that no physical alignment energy or map to a condensate has been supplied, and that changing a reduction differs from a physical mass-sign change at fixed measure. These preserve the corrections to Remark 17.7.

58. Spacetime-dependent reduction — Correct with amendment. The displayed reduced operator and the statement that it is generally non-self-adjoint agree with Proposition 17.10. The kinetic term belongs to the separately chosen operator $D_M + \gamma_5 d_\Sigma(x)$, with the coefficient tracing both reduced blocks; it vanishes at $M = 0$. State these normalizations beside the formula and emphasize that this does not construct the massless pion effective action.

59. Fixed-background condensate expansion — Error. "A subtraction regular at $m = 0$" does not alone imply $c_V = n_0/(Vm) + O(m)$. For a finite pair-preserving spectral cutoff, subtracting a nonzero constant C is regular at zero and gives

$$c_V^{\mathrm{reg}}(m; A) = \frac{n_0}{Vm} - C + O(m).$$

Require a compatible subtraction whose finite remainder vanishes at zero with $O(m)$, for example an odd analytic subtraction in m, together with the spectral pairing. This issue is also present in the corresponding wording of the corrected lemma.

60. Banks–Casher form and determinant — Correct with amendment. The note retains the added assumptions $\rho_m(0) \to \rho_0(0)$, equicontinuity, tail and subtraction control, and the order of limits. Add $m > 0$, state the spectral-density and condensate normalization, and specify a regulator compatible with the $\pm\lambda$ pairing. The positive determinant formula is for $-iD_M + m$, as written.

The original Banks–Casher full text was inaccessible here, so the precise hypotheses should remain explicitly those of the note rather than be attributed to that paper.

61. Interval routes — Correct with amendment. For the localized branch state $\mu > 0$, $\mu l > 1$, and define $\kappa_0 > 0$ by

$$\tanh(\kappa_0 l) = \kappa_0/\mu.$$

The asymptotic $m \sim 2\mu e^{-\mu l}$ is for $\mu l \to \infty$, and the absolute limit quoted in the note follows at fixed μ. The boundary-Yukawa operator is a distinct construction from the scalar-algebra tensor-product truncation; the corrected lemma explicitly warns that identifying them at finite l needs proof. The note should retain that warning.

62. Fixed-background continuity and quantum input — Correct with amendment. The operator $D_M + \gamma_5 m$ and its square vary continuously with m. The regularized condensate itself need not have a finite limit at fixed background: a zero mode gives $n_0/(Vm)$. Specify the object whose continuity is asserted. The measure, volume sequence and order of limits remain additional inputs.

63. Provenance — Correct with amendment. The note says the identities were checked with Python on finite models. The recorded numerical checks in lemma 17 cover particular identities and examples, not every claim in the note. Name those checked identities. The earlier report also records four amendments in addition to five erroneous items; "corrected five items" should not be read as the whole scope of that report.

Part 3. Readiness for a Physical Review journal

We recommend substantial revision before submission.

1. **Separate the mathematical results from the endpoint proposal.** The introduction moves from formal KO signs to "10 to 9" and "10 to 11" limits before a map from those signs to the interval geometries or measured weak parameters has been constructed. Give the physical hypothesis, the exact results, and the unresolved identification in that order.
2. **Repair the cross-surface overclaims.** Correct the collapse statement in Table I and the abstract, the localized-mode mass formula in the discussion, the flavour-texture claim, the propinquity converse, and the condensate subtraction. State heat-cutoff and Taylor-mass scope wherever the weighted rule is summarized.
3. **Make the proof sketches show the decisive steps.** Theorem B's sketch omits the index and wall/Weyl arguments; Theorem C invokes concavity and compound matrices without defining the quantities that make the bounds work; Proposition H's displayed matrices do not alone establish the generated algebra. Proposition G needs its existential matching unitary stated in the theorem itself. The supplemental references are useful, but a reader should be able to see why each main-text conclusion follows.
4. **Define the notation needed to read the results.** In particular, define Φ_Q, Φ_U, the threshold quantities, and the symbols $f_{T_p}, g_{T_p}, D_p, K_p, \epsilon_p^{F,G}$ in Theorem C. Define κ by its trace weights, distinguish $M_{W,\mathrm{Taylor}}$ from a pole mass, and state the relative-trace or density convention on $\mathbb{R}^4$.
5. **State the status of the literature comparison.** Give a reproducible, appropriately scoped account of the searches behind the "we did not find" statements. Keep the established interval, domain-wall, truncation and spectral-propinquity ingredients visibly distinct from the proposed combination and from the still-open physical construction.

Amendments and errors

- **Chirality and labels:** scope the odd-dimensional objection, the Standard Model representation, the neutral parity criterion, particle preservation, and the Hořava–Witten comparison as in checks 1–3, 7, 10 and 12. The finite-l physical interpretation and the assignment of label 11 remain gaps.
- **Wall and collapse limits:** replace the fixed-distance bulk claim and the unrestricted tuned-bulk claim as in checks 5, 34 and 41. Keep the type-(c) localization and boundary-Yukawa scalings separate.
- **Spectral action:** define the heat cutoff, Taylor mass, trace weights and κ; restrict the asserted necessity of $m_t/\Lambda \to 0$ to the family proved in check 19.

- **Algebras and limits:** distinguish complex bicommutants from quaternionic real forms, expand the interval generation argument, and insert the existential unitary in Proposition G.
- **Discussion and note:** correct the localized-mode momentum formula, add minor bounds to the texture claim, require an $O(m)$-compatible condensate subtraction, and restore the interval-branch, charge, normalization and provenance qualifications in checks 54–63.

Overall: 63 statements checked: 16 correct, 36 correct with amendment, 4 gaps, and 7 errors.

Part R6 (referee report, no-go results and lemma 19)

Referee report on the two-model no-go results and lemma 19

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_nogo_19.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemmas 05, 06 and 19.

We checked the statements below by direct mathematical reasoning. No numerical computation was run. Each numbered verdict contributes one count to the final line.

Part 1. Bare limits and odd factorizations

1. The limit $m \to 0$ in Lemma 05, Part 4 Verdict: Correct with amendment. The fixed-rank statement is correct. The phrase "exact continuum limit throughout a fixed positive energy window" should specify the mode of convergence.

At fixed N, $D_N = mT_N$, with T_N independent of m. Hence, for fixed $t > 0$,

$$K_N(t) = \operatorname{Tr} e^{-tm^2 T_N^2} \longrightarrow 2(N-1).$$

The ultraviolet heat dimension is zero for each finite N. The distances between distinct sites are proportional to $1/m$ and diverge.

For $l \to \infty$, the condition $N/l \to \infty$ gives uniform convergence of the finite-chain dispersion to the continuum dispersion on every fixed bounded energy window. It also gives convergence of normalized heat traces and counting densities. It does not by itself give convergence of the *unnormalized difference* $K_N(t) - K_l(t)$ to zero.

To see the distinction, put

$$b = \frac{l}{2N-1}, \qquad k_r = \frac{(r-\frac{1}{2})\pi}{l}, \qquad \theta = \frac{\pi}{4N-2}.$$

Then

$$\lambda_{N,r} = \frac{\theta}{\sin\theta}\frac{\sin(bk_r)}{b}.$$

For example, taking $N = \lfloor l^{5/4} \rfloor$, expansion of the Gaussian sums gives, for fixed $t > 0$,

$$K_N(t) - K_l(t) \sim \frac{l^3}{16\sqrt{\pi}\, t^{3/2} N^2},$$

which diverges, although $K_N(t)/l \to 1/\sqrt{\pi t}$.

Direction. The finite-rank obstruction concerns the stated bare chain and heat trace. A resolved decompactification requires growing rank; the rate should be chosen for the desired notion of spectral convergence.

2. The limit $m \to \infty$ in Lemma 05, Part 4 Verdict: Correct. Every eigenvalue of the massless chiral interval, and of the calibrated finite chain, has magnitude at least

$$m = \frac{\pi}{2l}.$$

Thus, at every fixed cutoff Λ, the light spectral projection vanishes once $m > \Lambda$. At fixed N,

$$\|(D_N - i)^{-1}\| = \frac{1}{\sqrt{m^2+1}} \longrightarrow 0, \qquad d_N(i,j) \longrightarrow 0.$$

The zero resolvent limit on the original nonzero Hilbert space is not the resolvent of a finite self-adjoint operator there. The metric collapse and the empty fixed-energy sector are consequently compatible.

Direction. A nonempty light sector in a collapse calls for altered data, such as a tuned bulk mass or a scaled boundary coupling. The conclusion above applies to the stated massless interval and its calibrated truncations.

3. Proposition A.1 Verdict: Correct. For the stated graded product

$$D_F = D_0 \otimes 1 + \Gamma_0 \otimes D_I, \qquad \{D_0, \Gamma_0\} = 0,$$

the cross terms cancel and

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2 \geq \frac{\pi^2}{4l^2}.$$

This holds for any finite KO-6 factor with the stated product operator, including an l-dependent finite Dirac operator. The fixed-cutoff light sector is empty for sufficiently small l.

Direction. The proposition points to modifications of the interval operator, boundary data, product coupling, or limiting prescription. It does not exclude a light sector in those modified families.

4. Result 3: electrodynamics Verdict: Correct. With the displayed matrices, both

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1)$$

are grading-reversing unitary involutions. They commute with the algebra, its opposite, and D, while

$$U_+ J = J U_+, \qquad U_- J = -J U_-.$$

Thus the same D, J, γ admits both reductions, including at $m = 0$. The displayed representation and Dirac matrix agree, after the stated phase choice, with the electrodynamics data in van den Dungen–van Suijlekom, Eqs. (4.1)–(4.2).

Direction. A selection of one factorization needs additional input beyond the displayed real structure and the listed doubling conditions.

5. Lemma 4: neutral-sector criterion Verdict: Correct. In bases related by J, the KO-6 relations make the off-diagonal block M complex symmetric. Takagi factorization gives $M = \Sigma$, with nonnegative diagonal entries.

For $s = +1$, $V = I$ satisfies the criterion. For $s = -1$, the descended antiunitary j has $j^2 = -1$ and commutes with $d^2 = M^* M$. Each eigenspace of d^2, including its kernel, therefore has even complex dimension. Conversely, on each pair of equal entries of Σ, $V = \sigma_2$ is a Hermitian unitary and gives

$$(V^* K)^2 = -1, \qquad V^* \Sigma = (V^* \Sigma)^*.$$

This proves the stated equivalence, including zero singular values.

Direction. A KO-5 reduction can be recovered by neutral-sector pairing or by a symmetry enforcing the required even singular-value multiplicities. The result assumes the same scalar algebra character on the whole sector.

6. Proposition 5: the strict one-generation heavy limit **Verdict: Correct.** In the specified reduced representation, the surviving electron and down-quark sectors have both involutions from Result 3. The neutral sector has basis $(\nu_L, J\nu_L)$, zero Dirac block, and

$$\gamma = \sigma_3, \qquad J = \sigma_1 K, \qquad U = \sigma_1.$$

It supplies a KO-7 reduction. The full light space has dimension 18; a KO-5 reduction would require an antiunitary squaring to -1 on a complex space of dimension 9, which is impossible. The neutral bimodule gives the same obstruction directly.

Direction. The conclusion concerns the stated projection and reduced algebra. Adding an appropriate neutral partner can change the result; imposing particle-number preservation changes the admissible reductions, as in item 9 below.

7. Result 7: three generations **Verdict: Correct with amendment.** State that "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen as a projection onto the specified top Dirac sector. Mixing between inequivalent species is outside the displayed Dirac operator.

The dimension counts are

$$96 - 6 - 12 = 78, \qquad 96 - 6 - 36 = 54.$$

The surviving three left neutrinos give a zero neutral block of size 3, so Lemma 4 gives KO-7 and excludes KO-5. Charged square flavor blocks can be brought to singular-value form by bimodule unitaries.

With three right neutrinos and a purely Dirac matrix Y_ν, the symmetric neutral block

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & 0 \end{pmatrix}$$

has every singular value of Y_ν twice, including zero. For a general symmetric Majorana block, the even-multiplicity criterion of Lemma 4 applies.

Direction. The obstruction is controlled by the neutral multiplicities. Paired neutral content or protected degeneracies provide directions to investigate.

8. No-go theorem 8: the full chiral representation **Verdict: Correct.** For the full displayed representation with $g \geq 1$,

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$$

has range of dimension $8g$ in H_+ and zero range in H_-. An algebra-linear grading-reversing unitary would identify these spaces. No such unitary exists, independently of the admissible D and of the sign against J.

This conclusion concerns the full quaternionic algebra, the displayed representation and grading, and the doubling criterion of Result 2. A projection that removes only one member of a weak doublet is generally not invariant under the full quaternionic algebra.

Direction. Possible directions include changing the algebra or representation, retaining the even description, and studying boundary or defect realizations of chirality.

9. No-go theorem with $[U, Q] = 0$ **Verdict: Correct.** In the stated strict light representation, an algebra- and opposite-algebra-linear U preserving particle number must map the ν_L particle subspace to an opposite-grading particle subspace with the same bimodule. No such partner is present. The KO-7 involution in Proposition 5 instead maps ν_L to its conjugate and fails $[U, Q] = 0$.

Direction. A particle-preserving reduction requires a matching particle bimodule of opposite grading, or a change in the particle content or imposed conditions. The conclusion holds for either sign against J.

Part 2. Fluctuations and the parent comparison

10. Comparison with Chamseddine–Connes–van Suijlekom Verdict: Correct. The formulas in Lemma 19 agree with arXiv:1304.7583v2, §III.2, Eqs. (10)–(11). That paper takes a "self-adjoint, universal one-form" and gives

$$A_{(2)} = \sum_j \hat{a}_j[A_{(1)}, \hat{b}_j], \qquad U = uJuJ^{-1}, \qquad UD(\omega)U^* = D(u\delta(u^*) + u\omega u^*).$$

Here $\hat{x} = JxJ^{-1}$. The formula for $A_{(2)}$ uses the same universal one-form as $A_{(1)}$; replacing it by an independently chosen one-form would change the construction. The algebraic calculation in Theorem 19.1 also extends to universal forms that are not self-adjoint.

For $J = \sigma_3 K$, the alternative pointwise choice, $JDJ^{-1} = -D$. The factor $\epsilon = -1$ in

$$\widetilde{A}_{(1)} = \epsilon J A_{(1)} J^{-1}$$

then gives the same displayed two-point formulas.

11. Theorem 19.1 Verdict: Correct. Direct evaluation gives

$$\frac{A_{(2)}}{m} = \begin{cases} \begin{pmatrix} 0 & |\phi_1|^2 \\ |\phi_2|^2 & 0 \end{pmatrix}, & J_0 = K, \\ \begin{pmatrix} 0 & \phi_1\overline{\phi_2} \\ \phi_2\overline{\phi_1} & 0 \end{pmatrix}, & J_r = \sigma_1 K. \end{cases}$$

Adding the two linear terms and D gives both factorizations in the theorem.

The represented self-adjointness condition is $\phi_2 = \overline{\phi_1}$. In this two-point example the universal one-forms are spanned by $p_1\delta p_2$ and $p_2\delta p_1$, which are adjoints and are represented faithfully for $m > 0$. Thus the same condition characterizes self-adjoint universal forms. Setting $\Phi = 1 + \phi_1$ gives

$$D'_{J_0} = m|\Phi|^2\sigma_1, \qquad D'_{J_r} = m\begin{pmatrix} 0 & \Phi^2 \\ \overline{\Phi}^2 & 0 \end{pmatrix}.$$

12. Corollary 19.2 Verdict: Error. The gauge actions and full fluctuated images in items 1–2 are correct. The common linear-only formula in item 3 is incorrect for J_0.

Let $z = e^{i\psi}$. On the gauge orbit of the zero universal form,

$$A_{(1)} = m\begin{pmatrix} 0 & z-1 \\ \bar{z}-1 & 0 \end{pmatrix}.$$

Removing $A_{(2)}$ gives the upper links

	upper link	modulus near $\psi = 0$
J_0	$m(2\cos\psi - 1)$	$m(1 - \psi^2 + O(\psi^4))$
J_r	$m(2e^{i\psi} - 1)$	$m(1 + \psi^2 + O(\psi^4))$.

For example, at $\psi = \pi/3$ the J_0 linear-only operator is zero, while the stated J_r expression has link $i\sqrt{3}\,m$. Adding $A_{(2)}$ restores the stated full results: $D' = D$ for J_0, and the link is mz^2 for J_r.

Two wording amendments accompany the correction. The set $\Phi = e^{i\psi}$ is the orbit in the potential parameter; the J_0 operator orbit consists only of D. The map $\Phi \mapsto \Phi^2$ is two-to-one away from zero and one-to-one at zero. The image statement is pointwise; a globally prescribed smooth link need not have a global smooth square root.

13. Reflection real structure and compression **Verdict: Correct.** In Counterexample 08.4,

$$J_{\mathrm{ref}} = \sigma_1 R_l K.$$

For the real lowest-pair basis

$$u_L = \sqrt{2/l}\,(\cos mx, 0), \qquad u_R = \sqrt{2/l}\,(0, \sin mx), \qquad m = \pi/(2l),$$

the identities $\cos(m(l-x)) = \sin(mx)$ and $\sin(m(l-x)) = \cos(mx)$ give

$$J_{\mathrm{ref}} u_L = u_R, \qquad J_{\mathrm{ref}} u_R = u_L.$$

The subspace is J_{ref}-invariant, and its compressed antiunitary is exactly $\sigma_1 K$. Its signs are $(J^2, JD/DJ, J\gamma/\gamma J) = (+,+,-)$.

For $e = \mathrm{diag}(1,0)$, the two compressed real structures give

$$[[D,e],\hat{e}] = m\sigma_1 \quad (J_0), \qquad [[D,e],\hat{e}] = -m\sigma_1 \quad (J_r).$$

Both two-point data satisfy order zero and fail order one.

14. Proposition 19.3, item 1 **Verdict: Correct.** Without a real structure, self-adjoint finite one-forms give every upper link $m(1+\phi_1) \in \mathbb{C}$. For fixed compression of the real parent gauge coefficient A_5, the link is

$$m + iB, \qquad B = \frac{1}{l}\int_0^l \sin\frac{\pi x}{l}\, A_5(x)\, dx \in \mathbb{R}.$$

Every real B is reached, so the fixed-compression image is the stated vertical line.

15. Proposition 19.3, item 2 **Verdict: Correct with amendment.** "A neutral radial field with no parent scalar" is fair when it means: *the finite pointwise-real fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation.* It does not establish a dynamical scalar field, its action, or the absence of parent fields in other models.

Let $E = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. A represented parent one-form is $Eh(x)$. Self-adjointness requires $h = ia$ with a real, and

$$K(iEa)K^{-1} = -iEa.$$

Thus the parent real fluctuation cancels. By contrast, the finite gauge-invariant link is

$$t = m|\Phi|^2 \geq 0,$$

so t is a function of the radial variable $|\Phi|$, and every $t \geq 0$ occurs.

The proof at line 71 needs this self-adjointness step. An individual commutator does not generally change sign under K: for $f = x$, $K[D,f]K^{-1} = [D,f] = E$. The sign reversal holds for the self-adjoint represented sum iEa.

The parent does possess an A_5 one-form before real symmetrization; the cancellation concerns the resulting real fluctuation. Compression of the algebra is also nonmultiplicative, so the construction gives no canonical identification of all finite potentials with parent potentials.

16. Proposition 19.3, item 3 **Verdict: Correct.** The reflected opposite algebra consists of multiplication operators, so the parent satisfies order one. For real A_5,

$$J_r D_{A_5} J_r^{-1} = D_{A_5 \circ R}.$$

Consequently its real scalar fluctuation has coefficient

$$A_5^s = A_5 + A_5 \circ R.$$

Since the compression weight is reflection-even, $B^s = 2B$, where B is calculated from the unsymmetrized A_5.

17. The table in Proposition 19.3 Verdict: Correct with amendment. Label the real structure and the meaning of B in each row. The covariant projection has fixed singular value m, and the full finite J_r orbit has link modulus m. The linear-only row has the stated modulus for J_r, locally growing at second order.

The fixed-compression row

$$\sqrt{m^2 + B^2}$$

is the formula of Lemma 13 for the unsymmetrized coefficient. For the reflection-real parent of item 3, if B continues to denote the integral of the original A_5, the corresponding expression is

$$\sqrt{m^2 + (B^s)^2} = \sqrt{m^2 + 4B^2}.$$

Equivalently, the table may retain its formula after defining its B to be the integral of the *symmetrized* coefficient. The increase is quadratic along paths for which the relevant B has a nonzero linear term; a nontrivial gauge path can also have $B = 0$.

18. Wilson phase in the paragraph following the table Verdict: Error. For a constant unsymmetrized $A_5 = a$, the transported link and its expansion are

$$mW_l^* = me^{ial} = m + i\,mla + O(a^2) = m + i\frac{\pi a}{2} + O(a^2).$$

Thus the Wilson **phase** is al; $mla = \pi a/2$ is the coefficient of the first-order imaginary change in the link. The fixed-compression value is $m + i(2a/\pi)$.

For the reflection-real fluctuation starting from the same a, the symmetrized coefficient is $2a$: the transported link is me^{2ial}, and the fixed-compression link is $m + i(4a/\pi)$.

19. Reading at the end of Lemma 19 Verdict: Correct with amendment. Replace "truncation generically breaks order one" by the demonstrated statement that spectral compression *can* break it, and state the hypotheses needed to apply the cited framework: a represented algebra, a preserved real structure, and order zero.

For example, on the circle take $D = -i\,d/dx$, $J = K$, and the full multiplication algebra. The parent satisfies order zero and order one. Compression to the zero Fourier mode leaves only scalar operators and preserves both conditions. Compression instead to the modes $-1, 0, 1$ sends e^{ix} to a truncated shift S; its conjugate is S^*, and

$$[S, S^*] \neq 0.$$

Thus this compression fails order zero. It also illustrates why the compressed operator system and the algebra it generates require separate attention.

The proposed relation to a singlet remains an analogy. In the cited 1304.7583v2, Proposition 8 and §IV, the toy model uses three complex fields, and the authors state that its quadratic term "does not generate new scalar fields"; it is expressed in fields already present in $A_{(1)}$. The present $A_{(2)}$ likewise depends on the same two-point parameters, but that fact gives no identification of the models. Devastato–Lizzi–Martinetti, arXiv:1304.0415v3, §4.3, obtain their σ from a Majorana-operator fluctuation in a larger algebra. An identification would require a map of algebras, representations, fields and gauge actions, together with an action comparison.

Amendments and errors

- Specify the convergence asserted in the $m \to 0$ continuum limit; $N/l \to \infty$ controls dispersion and normalized densities.
- Restrict the three-generation mixing statement to the displayed species blocks and specified heavy projection.
- In the parent comparison, use self-adjointness to prove the pointwise-real cancellation, and distinguish the original A_5 from its reflection symmetrization.

- Correct Corollary 19.2's J_0 linear-only link and the Wilson phase following the table.
- Qualify the general reading of truncation and retain the singlet comparison as an open analogy.

Overall: 12 Correct; 5 Correct with amendment; 0 Gap; 2 Error.

Part R7 (referee report, lemmas 20 and 21, Proposition I)

Referee report on lemmas 20 and 21, the comparison in the steer note, and Proposition I

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_20_21.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemmas 20 and 21, the steer note and the paper.

The review finds errors in the marked-point reading, the radial statement, parts of the scale comparison, Proposition I, and the no-go table. The connecting-operator calculations and the three consensus checks otherwise hold in their stated scopes.

The checks below are analytic; no numerical code was run. Source references identify the arXiv versions read.

Part 1. Connecting operators **1. Correct.** In Result 1, block multiplication gives $T = cT^*c$ and $hT + Tg = 0$. The lower off-diagonal block of order one is (2); the other follows using adjoints and the involutive algebra. In c-paired bases, the first condition is complex symmetry.

The averaging proof gives $T = R + \tau(R)$, with R intertwining the left actions and $\tau(R)$ the opposite actions. Conversely, such a decomposition gives order one on $\mathcal{A}(R)$, establishing (4). The final example has one arrow preserving the right character and the other preserving the left character; it satisfies order one although B is not contained in $\mathcal{A}(T)$. No correction is needed.

2. Correct. For the displayed rank-one T_κ, with $\kappa \neq 0$, the two selected lines must reduce their respective actions and carry the same scalar. This gives (6), of real dimension $2 + 4 + 18 = 24$. The self-adjoint off-diagonal operator has complex rank 2.

Chamseddine–Connes, §4, equations (4.6)–(4.7) use an auxiliary intertwiner, also called T there. Its compression $K : \mathbb{C}^2 \to \mathbb{C}^4$ has rank one or two. Rank two gives dimension at most $4 + 4 + 8 = 16$; rank one gives dimension 24, uniquely up to automorphism. For the displayed block, $T_\kappa x = \kappa KxK$. For a general admissible connector, the paper's K is obtained from an auxiliary intertwiner and need not equal the Dirac block. No correction is needed.

3. Correct. Result 3 and Counterexample 4 have the stated domain-preservation equations. In the scalar example, the parent has signs $(+, +, -)$ and satisfies both order conditions. The two displayed chiral vectors form the window for

$$\theta/l < \Lambda_0 < (\pi - \theta)/l.$$

Their compressed algebra is the full diagonal $\mathbb{C}^2$, and for $p = \mathrm{diag}(1, 0)$,

$$[[m\sigma_1, p], p^\circ] = -m\sigma_1.$$

The negative conclusion concerns compression of the full smooth scalar gluing algebra in this window. Restricting the parent algebra, or changing the window or representation, changes the problem; Result 5 gives a positive constant-algebra construction.

4. Correct. In Result 5, (18) and (19) preserve the boundary domain. The equality $cX = X^*c^{-1}$ is the required compatibility for J. The window contains $2n - 2$ uncoupled zero modes and the pair $\pm m$, giving (20). Constant $B = \mathcal{A}(S)$ multipliers commute with the parent Dirac operator, so the stated order-one claim follows.

With the displayed conventions, $U(x) = e^{mxF}$ carries the uncoupled domain to the coupled one, and differentiation gives (21), including its sign. Its compression gives $m(S + S^*)$. The construction assumes the Hom fibres, grading, constant multipliers and coupled channel.

5. Correct with amendment. The rank bound in Result 6 follows from the factorization $T = \mathrm{ev}_2^* X \,\mathrm{ev}_1$. A point with N fibre channels can give rank N. An exact change of boundary domain does not itself establish that factorization.

The separation argument in Result 7 is valid for unchanged finite windows when independent test functions can be supported away from the gluing sets. It applies to the stated point and fibre examples, including S^4 pullbacks after fibre integration. For a general subset upstairs in the twistor space, add the hypothesis that its image in S^4 has empty interior, so that the required independent base test functions exist. A new coupled window, boundary degrees of freedom, or a changed observable representation lies outside the result. The direction is to construct and analyze such a coupled parent.

6. Correct. Hořava–Witten, §2, states $\Gamma_{11}\chi = \chi$ for the intrinsic boundary gaugino. With a common ten-dimensional chirality convention and the standard common orbifold lift, the same-chirality comparison uses two type-(a), or two type-(b), fields. Opposite localization masses can place them near opposite ends.

For unbroken $E_8 \times E_8$, the displayed Hom space vanishes, and an operator odd for the common chirality has no matrix elements between the two retained same-chirality modes. Compactification, symmetry breaking, or additional bulk fields may give different effective couplings; those require a separate construction.

7. Correct. The three checks in Results 9–11 hold:

- The assumed weighted mass relation gives $m_{\max}/(2\sqrt{2}) \le M_W \le \sqrt{N_g}\, m_{\max}$. It forces the maximum to grow, with a fixed species necessarily unbounded only along some subsequence. The relation belongs to the specified asymptotic matching action; a full finite-cutoff calculation is a direction beyond it.
- With pointwise reality and a J_0-invariant window, inherited order zero holds exactly when the algebra generated by the compressed functions is commutative. A separate opposite action, as in the coarse correspondence, changes that hypothesis.
- If $P\gamma P$ is an actual grading, its square forces $[P, \gamma] = 0$. Thus it commutes with compressed even observables, whereas Γ_5 fails to commute with $Q(x_1)$. An added finite grading and restriction to its even algebra evade that particular inheritance claim.

The cited repository reports record earlier checks by another model. This review is a further solo check of the displayed arguments.

Part 2. Readings 8. Correct with amendment. In the CCM preprint, §2.5–2.6, especially (2.34), the right-handed up lepton is ν_R, and the off-diagonal block maps it to its conjugate through a symmetric Majorana matrix. After the stated identification, and an overall grading convention if needed, the one-generation connector has that form.

The interval construction gives one mass $m = \arctan(t)/l < \pi/(2l)$, with next positive absolute level $(\pi - \arctan t)/l$. Three identical copies give three equal masses. A general three-generation CCM Majorana matrix requires additional flavour data; it is not supplied by the single rank-one construction.

9. Error. The projector identity and

$$2\Pi_p - 1 = p_a \Gamma_a = 5Q(p \cdot x)$$

are correct. The pair of projective points determines the two lines, but it does **not** determine the connector's magnitude. Rephasing w and v also changes KxK by a phase. At fixed nonzero $|\kappa|$, the line choices are conjugate under the stated groups; $|\kappa|$ remains a modulus, and other finite data can introduce relative moduli.

The exact correction is to specify the lines **and** a coupling, together with an evaluation and normalization prescription. Moreover, $5Q(p \cdot x)$ fails to commute with the other coordinate compressions. It can be chosen

as additional finite grading data after restricting to its even algebra. That evades Result 11's inherited-grading hypothesis; it does not derive the grading from the even parent or construct the proposed point interface.

10. Correct with amendment. Equation (21) gives a constant matrix connection after a unitary change of domain, with relative endpoint rotation angle $\theta = ml$. On an interval, the open-path transport depends on endpoint frames; the physical quantity is the transport together with the specified boundary conditions. A closed Wilson loop or a dynamical gauge field needs further structure.

The gauge-Higgs comparison is an analogy at this stage: the interval calculation supplies a fixed matrix connection and no Higgs fluctuation or effective potential. The reference is Yutaka Hosotani, "Dynamical mass generation by compact extra dimensions," *Physics Letters B* **126** (1983), 309–313. For the dynamical Wilson-line and gauge-Higgs construction in an orbifold setting, see Hosotani, Noda and Takenaga, §1.

Part 3. Counting 11. Correct with amendment. In Proposition 21.1, $(q_L - q_R)/2 = (T_3/2)(1, -1)$, so the additional term in lemma 18's (33) is annihilated by the photon vector $(1, 1)$. The stated sums and removal entries reproduce:

removed	$\cos^2\theta_{\mathrm{NC}}$	$\kappa_{\mathrm{em}}/\kappa_s$
u-type	5/8	8/3
d-type	59/92	46/15
charged lepton	17/28	7/3

The table assumes the stated three-generation charges and that the other species have the mass assignments used in the removal calculation. At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished: read its angle as the limit from finite mass, as lemma 18 states. Other weak-angle definitions need not have these values.

12. Error. The trace $84 + 12e^{-x}$ is correct for a top-only Dirac mass and no Majorana mass. With the positive heat cutoff and $m > 0$, its radial derivative is strictly negative. The text says there is "no finite stationary point"; at $m = 0$ the derivative is zero.

Replace that sentence by: **there is no finite nonzero stationary point on this free radial ray**. This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints, and paths with an additional fixed noncommuting mass lie outside the strict conclusion. The result points to analyzing those changed actions or prescribed backgrounds.

13. Correct with amendment. Complexifying the real 30 gives

$$(\mathbf{5}, \mathbf{3}) \oplus (\bar{\mathbf{5}}, \bar{\mathbf{3}}),$$

and the displayed exterior-square branching has dimensions

$$192 + 24 + 8 + 1 + 60 + 45 + 60 + 45 = 435.$$

The charge tables and the charged scalar-partner counts follow. The charge operator includes the $U(1)$ commuting with $SU(5)$:

$$Q = \mathrm{diag}(3/5, 3/5, -2/5, -2/5, -2/5) + \tfrac{1}{15} I_5.$$

Also,

$$\mathbf{496} \to (\mathbf{435}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{1}) \oplus (\mathbf{30}, \mathbf{2})$$

under $SO(30) \times SO(2)$.

These are representation and charge counts. Identifying the entries with physical scalar partners requires a field construction, spin and chirality assignments, interactions and dynamics. Rivero's preprint, Tables 1–2, contains the stated diquark and meson counting; its proposed particle interpretation remains a proposal.

14. Correct with amendment. Both decompositions and the overlap of 72 follow by counting four internal states per Dirac species and colour. Without ν_R, the first family has 78, and the charged family

remains 84. A massless three-form in eleven dimensions and its six-form dual each have $\binom{9}{3} = \binom{9}{6} = 84$ on-shell polarizations.

The 84 in H_F is a complex finite-Hilbert-space dimension; the three-form count is a real bosonic polarization count. A physical identification needs a map of representations and degrees of freedom.

15. Correct with amendment. Seiberg, "Electric-Magnetic Duality in Supersymmetric Non-Abelian Gauge Theories," §1–3, *Nuclear Physics B* **435** (1995), 129–146, gives the four-dimensional $\mathcal{N} = 1$ supersymmetric result. For $N_c = 3, N_f = 5$, $9/2 < 5 < 9$; the magnetic group is $SU(2)$, with five flavours and a 5×5 gauge-singlet meson chiral superfield coupled by $W = Mq\tilde{q}$.

To turn readings 1–3 into statements, one needs:

- An $\mathcal{N} = 1$ supersymmetric theory with the required massless matter and dynamics, plus an operator and anomaly map. Five-flavour ordinary QCD alone does not supply Seiberg duality, and the magnetic $SU(2)$ needs a separate identification with any electroweak group.
- A representation and dynamics relating the finite fermionic states to the bosonic form or brane degrees of freedom; equality of the number 84 is insufficient.
- For the KO-label reading, the required involution on the stated finite sector, including its action on D, J, the algebra and its opposite. The vectorlike $SU(3) \times U(1)_{\text{em}}$ charges alone do not establish it, and neutral states still require treatment.

Part 4. Comparison with repository results **16. Correct with amendment.** Example 12.5 gives

$$\tan^2(kl) = |X|^2, \qquad k_{\text{light}} = \frac{\arctan|X|}{l}.$$

For small X, $k_{\text{light}} = |X|/l + O(|X|^3/l)$; the half-integer spectrum occurs as $|X| \to \infty$. At $X = 0$, zero has multiplicity two and must be omitted from the positive spectrum.

With $X(l) = \tan(ml)$, the light mass is exactly m and the remaining levels diverge as $l \to 0$. Identifying X with a physical Yukawa coupling, or $1/l$ with v, remains an extra hypothesis. The Wilson-line wording needs the qualification in item 10.

17. Error. CCM, (4.12) gives

$$\mu_0^2 = 2\frac{f_2}{f_0}\Lambda^2 - \frac{e}{a}.$$

Top dominance in the Dirac Yukawa trace alone does not force $e/a = 0$: the Majorana–neutrino term can remain relevant. Under exact top-only data, or the additional assumption $e = 0$, the stated truncated-potential relations give

$$\frac{m_t^2}{\Lambda^2} = 2\frac{f_2}{f_0}.$$

For the heat profile $F(u) = e^{-u^2}$, $f_2/f_0 = 1/2$, giving $x = 1$. This is a stationary point of the **truncated** potential; lemma 18's literal positive heat action has no finite nonzero stationary point on the free top-only radial ray. Its exact quadratic-background calculation at $x = 1$ gives the stated departure from the small-mass ratio.

18. Error. As written, "finite low-energy electromagnetic coupling" includes a coupling tending to zero. The trajectory in the transcript provides that case: take $\Lambda = \mu_* = v$, keep $\Lambda_5 > 0$ fixed through one-loop matching, and choose

$$\frac{8\pi^2}{g_2^2} = \frac{23}{3}\log\frac{v}{\Lambda_5} - \frac{1}{3}\log\log\frac{v}{\Lambda_5} + O(1).$$

Then $g_2 \to 0$, $m_t/\Lambda \to 0$, the top-only Taylor ratio tends to $\sqrt{8/3}$, and the low-energy electromagnetic coupling tends to zero. The text itself lists that endpoint as an escape.

The intended no-go can be stated with **nonzero limiting** α_{em} and positive finite limiting Λ_5, a matching scale above the relevant thresholds, top dominance at matching, the stated one-loop running, and the top-only heat-cutoff Taylor mass. In a step-threshold regime with $\mu_0 < m_t < \mu_*$ and nonincreasing $\alpha_{\mathrm{em}}^{-1}$ toward μ_*,

$$\alpha_{\mathrm{em}}^{-1}(\mu_0) \geq \frac{28}{3\pi} \log \frac{\mu_*}{\Lambda_5}.$$

This follows from $b_6 = 7$, $b_5 = 23/3$, and $\alpha_{\mathrm{em}}^{-1}(\mu_*) = \frac{8}{3}\alpha_3^{-1}(\mu_*)$. It bounds μ_*/Λ_5; matching then bounds g_2 away from zero, so $m_t \asymp g_2 v$ eventually conflicts with $m_t < \mu_*$.

The signs quoted in the note are correct in their respective active-field ranges: $\sum Q^2 = 20/3$ for five active quarks and three charged leptons, and $b_2 + b_Y = -11/3$ above electroweak thresholds. The thresholds are set by masses, rather than simply by v. The written route $m_t/\Lambda \to \infty$ cannot simultaneously use six-flavour matching at $\mu_* = \Lambda$ above the top. Extra thresholds or gauge mixing, altered matching, a different action or mass definition, and an $e \to 0$ endpoint are directions outside the amended no-go.

19. Error. The one-loop threshold relation

$$\Lambda_5 = \Lambda_6^{21/23} m_t^{2/23}$$

is correct under the transcript's matching assumptions. Its $n_f = 5$ and Seiberg's $N_f = 5$ count different theories; the equality of the numbers establishes no duality.

The transcript says pions *do not have to be* elementary KK modes or finite-algebra fluctuations. The comparison changes that into a stronger exclusion. Proposition 17.10 establishes absence of a kinetic term for the specified coordinate change of a fixed operator; it leaves a new operator or quantum effective action available. Proposition 17.8 gives the stated Banks–Casher order of limits under its measure and regularity assumptions.

For $T^3 = \mathrm{diag}(1/2, -1/2)$ and $Q = \mathrm{diag}(2/3, -1/3)$, the algebraic coefficient is $2N_c \operatorname{tr}(T^3 Q^2) = 1$. The reduced action supplies a starting point; a regulated fermion measure and anomaly calculation remain to be supplied.

Part 5. Paper 20. Error. Proposition I omits $\kappa \neq 0$ in item 2. At $\kappa = 0$, the block has rank zero and $\mathcal{A}(T_\kappa) = \mathcal{A}^{\mathrm{ev}}$, of real dimension 40. Add $\kappa \neq 0$.

Item 1 must also assume that the sheet-diagonal D_F already has the stated signs and order one. In item 3, specify a symmetric cutoff satisfying

$$m < \Lambda_0 < (\pi - \theta)/l.$$

The Majorana identification in the following paragraph needs the basis and flavour qualifications in item 8. The marked-pair paragraph needs the coupling modulus and grading qualification in item 9.

The abstract's wall-coupling sentence is supported for a **nonzero coupling on the chosen decomposable channel**, with the constant matrix observable algebra and the stated fibres and grading supplied as inputs.

21. Error. The table at lines 330–349 needs the following row-by-row corrections.

Row	Assessment and correction
Odd bulk	Say **no net chiral spectrum** under the specified standard smooth closed reduction. A periodic massless Dirac field on a circle has left- and right-handed zero modes. Boundaries and defects change the hypotheses. Baptista, §1, studies chiral interactions of massive non-Killing gauge fields; that is a distinct escape from an exact-isometry assumption.

Row	Assessment and correction
Odd factorization	State the full chiral Standard Model representation and the particular involution criterion of Theorem D. Reducing to invariant vectorlike sectors changes the algebra and representation.
Only KO-7	State the scalar-character neutral sector and the odd multiplicity, including zero singular values, in Theorem D. Even multiplicity and appropriate Dirac pairing are the stated alternatives.
Empty light sector	Specify the **massless type-(c)** interval and fixed spectral cutoff. A massless type-(a) interval has a zero mode for every l. Tuned bulk mass and scaled boundary coupling give the stated alternative families.
Fixed compression	Specify the local gauge transformations and fixed light projection of Proposition F. A projection invariant under a given transformation does not have this defect. The covariant projection changes with the fluctuated operator.
First order	Specify the two-point compression and inherited pointwise J. Fluctuations without order one relax that axiom; the doubled parent changes the data.
Order zero	Add J_0-invariance of the window and the inherited pointwise reality. A doubled or otherwise changed opposite action lies outside those hypotheses.
Bounded-fermion matching	Keep the leading spectral-action order, positive kinetic form, stated trace and common profile space. A tower-scale fermion removes the bound's obstruction; matching the full actions needs more.
$M_W \to 0$	Specify **massless complete doublet**, the stated heavy-doublet family, the Taylor definition, and the cutoff hypotheses including $f(0) > 0$. Keeping a doublet with a fixed nonzero mass can leave a positive residual mass even for the heat cutoff. Moreover, for the nonnegative $f(z) = ze^{-z}$, $f(0) = 0$, the top-only Taylor mass instead has $M_W^2/\Lambda^2 \sim x^2/3$. The heat-cutoff pole result has its own analytic-continuation hypotheses. A joint cutoff limit, altered light sector, or quantum effective action changes the problem.
Radial vacuum	Specify an allowed nonzero ray $V = cV_*$, $c > 0$, flat background, adequate integrability, and no added term or constraint. For nonnegative cutoff of unbounded essential support the derivative is negative. Compact support can give a plateau; other listed changes are possible directions.
Grading	Replace "grading not inherited" by "the proposed Γ_5 grading is not inherited" under the four hypotheses of Result 11. The Dolbeault window itself has an inherited scalar grading. A compressed height function can be added as finite data with a restricted even algebra.

Row	Assessment and correction
Scalar gluing	Add closed empty-interior gluing sets and independent functions supported away from them, with unchanged separate windows. Matrix-valued multipliers alone need not evade the same separation argument; a restricted multiplier algebra, changed representation, or coupled window can.
Finite limits	Specify finite limits of **connected scalar algebras** in the stated spectral-propinquity class. Truncation, matrix-valued parent algebras, or another convergence notion changes the question and needs a separate construction.

The vector-mass row points to exact calculations with the desired light masses and cutoff, while the grading and gluing rows point to an explicitly constructed coupled parent.

Amendments and errors

- Amend the pullback scope in **5**, the flavour and grading conventions in **8**, and the interval transport terminology in **10**.
- Amend the endpoint and degree-of-freedom interpretations in **11, 13–16**.
- Correct the missing connector data in **9**, the stationary-origin statement in **12**, and the missing Majorana term in **17**.
- Replace **18** by the explicitly nonzero-coupling, threshold-matched no-go, with the $e \to 0$ trajectory retained as an escape.
- Correct the transcript attribution in **19**, the hypotheses and marked-point summary in **20**, and the individual no-go rows in **21**.

Overall: 6 Correct; 8 Correct with amendment; 0 Gap; 7 Error.

Part R8 (referee report, lemma 22)

Referee report on lemma 22

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_22.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 22.

We find two errors in the stated claims: the assignment of zero modes at nonzero wall coupling, and the identification of fixed compression with the first derivative of transport. The main interval construction and its compression formulas hold, subject to the amendments below. The numerical collider-exclusion claim remains unsupported.

Throughout, let $\ell > 0$, and write $D = D_F$, $\Gamma = \Gamma_F$, and $J = J_F$ when no confusion can arise.

1. Setting — Correct.

In bases related by c, the condition on M says that its matrix is complex symmetric. Let its Takagi factorization be

$$M = U\Sigma U^T.$$

Then

$$|M| = \overline{U}\,\Sigma U^T, \qquad V = UU^T$$

give $M = V|M|$, with V unitary and $V = cV^*c$. This construction includes the kernel: zero entries of Σ still have corresponding columns in the unitary U. The extension of the polar factor on the kernel need not be unique.

The formula

$$X = \tan(\ell|M|)V^*$$

and the reality condition $cX = X^*c^{-1}$ follow. Conversely, for every X satisfying that condition, the principal functional calculus gives the unique operator

$$D_F = \ell^{-1} \arctan \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad \ell\|D_F\| < \pi/2.$$

Thus the claimed bijection is valid. The expansion $X = \ell M^* + O(\ell^3)$ assumes fixed M.

2. Theorem 22.1 — Correct with amendment.

The boundary subspaces are Lagrangian for the first-order boundary form, giving self-adjointness; compactness of the H^1-to-L^2 embedding gives compact resolvent. Direct substitution verifies the stated signs and all three intertwining relations. The L- and R-components are orthogonal, while $R(x) = e^{x\Gamma_F D_F}$ is unitary, so W is an isometry.

For each singular value σ_k, with $\theta_k = \ell\sigma_k$, the signed spectrum is

$$\frac{n\pi \pm \theta_k}{\ell}, \qquad n \in \mathbb{Z},$$

with multiplicities added when values coincide. This gives the stated spectrum of $|D_X|$, the window, and the lower bound on the next level. The amendment is to state multiplicities, particularly when $\sigma_k = 0$: the zero mode then has multiplicity two per zero singular channel, and the two excited branches coincide.

For arbitrary signed momentum k, the scalar equation is

$$|\tan(k\ell)| = \sigma_k(X).$$

For the positive light mass $k = \sigma_k(D_F)$, its tangent is nonnegative and equals the corresponding singular value of X.

3. Corollary 22.2 — Correct.

On the uncoupled domain, multiplication by R maps the endpoint conditions to those of D_X, and

$$R^* D_X R = D_0 + E\Gamma_F D_F.$$

For one real channel, $\Gamma_F D_F = mF$, with F as defined in equation (21) of lemma 20. The sign and the placement of the spin matrix E agree.

4. Standard Model form and boundary types — Error.

The type assignment holds **at** $X = 0$: Q has type (a) and U has type (b). For $X \neq 0$, the wall conditions couple the fields, and the corresponding zero modes generally lift. For example, one channel with $X = t > 0$ has lowest masses $\pm\arctan(t)/\ell$ and no zero mode.

The correction is to say that the coupled modes continue from the type-(a) and type-(b) zero modes of the uncoupled problem.

One matrix X does encode the full finite operator, including Yukawa and Majorana data. Its entries need not have only the original block pattern: the tangent introduces higher powers of D_F. For example, with real symmetric

$$M = \begin{pmatrix} a & b \\ b & 0 \end{pmatrix},$$

the corresponding entry of $X = \tan(\ell M)$ is

$$X_{22} = \frac{\ell^3}{3} ab^2 + O(\ell^5).$$

Thus the original zero block can acquire an induced entry.

5. Theorem 22.3 — Correct with amendment.

Assume explicitly that the finite algebra and its faithful even representation are unital. The two endpoint intertwining equations are equivalent to

$$[\pi(f(\ell)), D_F] = 0.$$

The commutant of a self-adjoint operator is closed under products and adjoints. Hence C_F, and consequently A_ℓ, are unital $*$-algebras.

For every $f \in A_\ell$, multiplication preserves the whole H^1 domain and

$$[D_X, \pi(f)] = E\pi(f')$$

there. The right side extends to a bounded operator on all of H_ℓ. The opposite action also preserves the domain: this follows from domain preservation by g^* and $J^\wedge D_X = D_X J^\wedge$. Finite order zero then gives

$$[[D_X, \pi(f)], \pi^\circ(g)] = E[\pi(f'), \pi^\circ(g)] = 0.$$

No finite order-one hypothesis is needed.

This agrees with Counterexample 4 of lemma 20: its parent has order one, while its two-point compression fails finite order one. The statement concerns this parent and its pointwise algebra; it does not assert that arbitrary compressions preserve either order condition.

The Standard Model identification of C_F also needs the usual nonzero Yukawa data and the specified representation. For example, if all Yukawa blocks vanish and only an algebra-commuting Majorana block remains, the commutant is larger.

6. Theorem 22.4 — Correct with amendment.

Both compression formulas, including the minus sign in $Q_\ell(ET)$, and the symmetric exponential average for constant a are correct. The four items hold with the stated constants. In particular, interpreting $\|a\|$ as $\|\pi(a)\|$,

$$\|Q_\ell(a) - \pi(a)\| \le \frac{\ell^2}{6}\|[D,[D,\pi(a)]]\| \le \frac{2}{3}\ell^2\|D\|^2\|\pi(a)\|.$$

The constants in item 4 are respectively $8/3$ and $16/3$.

For item 1, compactly supported bulk functions suffice to test every smooth matrix-entry density. Finite-dimensional linear algebra then gives equality of the two images. The constant function a itself need not belong to A_ℓ; the theorem supplies a domain-preserving function with the same compression.

The general warning about generated algebras is sound. It also has a simple analytic example. Let

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}, \quad \pi(\mathbb{C}^3) = \{\mathrm{diag}(a,a) : a \text{ diagonal}\},$$

with $J = \left(\begin{smallmatrix} 0 & K \\ K & 0 \end{smallmatrix}\right)$. If P_1, P_2 are the first two diagonal channel projections, then on either graded block

$$[Q_\ell(P_1), Q_\ell(P_2)]_{12} = -\frac{\ell^2}{3} + O(\ell^4).$$

The original projections commute, whereas their compressed images generate a noncommutative algebra for sufficiently small nonzero ℓ.

The specific dimensions 15 and 30 remain **numerical claims**: the matrices and closure calculation were not retained, and the reported tolerance alone does not establish an exact rank. They should be accompanied by reproducible data and an exact or certified rank calculation. The general role of operator systems under spectral truncation is already developed in Connes and van Suijlekom, §1 and §3, arXiv:2004.14115v2.

7. Theorem 22.5 — Correct with amendment.

Every represented self-adjoint one-form is $E\pi(h)$, with $h^* = -h$. In fact **every smooth** such h arises, including those with nonzero boundary value: set

$$b(x) = \int_\ell^x h(t)\,dt.$$

Then $b(\ell) = 0$, so $b \in A_\ell$, and $[D_X, b] = E\pi(h)$.

Thus imposing compact support would exclude some connection profiles, although it loses no endpoint holonomy in the identity component: any smooth path from 1 to such a unitary can be made constant near both endpoints.

The transport formula and wall condition are correct. The proof should separate its two cases. For a general w, $\pi(w)^* D_F \pi(w)$ need not commute with J_F, so the non-real part uses the spectral and isometry proof of Theorem 22.1 **without** its real-structure hypothesis. For the real fluctuation, $U_w = \pi(w) J_F \pi(w) J_F^{-1}$ commutes with J_F, and the claimed real window follows.

There is a useful stronger statement. Put

$$\tilde{u}(x) = u(x) w^*.$$

Then $\tilde{u}(\ell) = 1$, so $\tilde{u} \in A_\ell$. With

$$\widetilde{U} = \pi(\tilde{u}) J_F \pi(\tilde{u}) J_F^{-1},$$

direct differentiation and the endpoint condition give

$$D_X + A + J^\wedge A (J^\wedge)^{-1} = \widetilde{U} D_X \widetilde{U}^*.$$

Thus every scalar connection in this interval model is gauge equivalent to the original parent under its allowed endpoint gauge group.

8. Corollary 22.6 — Error.

The gauge-orbit comparison itself is correct. Chamseddine–Connes–van Suijlekom, §III.2, equations (10)–(11), arXiv:1304.7583v2 give the quadratic fluctuation and its gauge transformation. Here the correct substitution is $v = w^*$, so the moving window carries

$$U_v D_F U_v^* = U_w^* D_F U_w.$$

For the two-point example, if $w = \mathrm{diag}(e^{i\alpha}, e^{i\beta})$, the upper link is $me^{-2i(\alpha-\beta)}$. It is $me^{2i\psi}$ in lemma 19 when ψ is defined using $v = w^*$.

The error is the assertion that fixed compression is the first derivative of the transported window for the **same** connection. Let h be constant and replace it by εh. At fixed D,

$$\frac{d}{d\varepsilon}\Big|_0 D_{\text{transported}} = -\ell[D, h],$$

$$Q_\ell(Eh) = -\frac{\ell}{2}[D, h] + O(\ell^3).$$

For example, take $D = m\sigma_1$ and $h = i\sigma_3$. Their leading terms differ by a factor of two. The real fluctuation doubles both expressions in this example and retains the discrepancy.

The correction is to state that the displayed fixed-compression expression is a tangent commutator with its own weighted generator. The moving spectral projection contributes to the derivative of the transported operator. The comparison with lemma 14 is a common transport principle, with its global path qualification; lemma 13 computes a distinct fixed compression. The first open point of lemma 19 is answered only on the specified gauge orbits and with the moving window.

9. No-go theorem 22.7 — Correct with amendment.

For the stated one-interval, zero-bulk-mass parent, fixed X, even scalar inner one-forms, and the **moving spectral window**, the finite operator is unitarily conjugate to D_F. Its singular values are therefore fixed. The author's argument and the direct endpoint-preserving gauge transformation in item 7 give the same conclusion in this scope. A fixed compression can show changes in singular values at second order, so the window must be named.

For the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through unit quaternions, a three-real-dimensional gauge orbit. These are gauge-orbit directions. Calling them physical or eaten modes requires a spacetime gauge action and its boundary conditions; the interval calculation alone supplies no such mode count. Degenerate data, including vanishing Yukawas, can have a smaller orbit.

The listed changes of wall coupling, admissible odd bulk terms, bulk masses, or internal dimension lie outside the theorem's hypotheses and are directions for further constructions. Curvature in higher dimensions alone does not prove a radial mode. Likewise, a quantum effective action on the same fixed configuration space does not by itself add a radial coordinate; an additional composite or other degree of freedom would have to be identified.

10. Proposition 22.8 — Correct with amendment.

Evaluation shows that the unsymmetrized gauge one-form has boundary value in C_F. For the displayed masses, assume a flat product interval, a positive Yang–Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Variation then gives Neumann conditions at 0, and Neumann on the wall Lie algebra and Dirichlet on its orthogonal complement at ℓ. The stated integer and half-integer towers follow.

The physical gauge Lie algebra is obtained from skew-adjoint algebra elements through the real gauge representation, with its kernel and any imposed unimodularity condition accounted for. For the usual Standard Model representation, $u(C_F)$ has dimension 10 before unimodularity, whereas photon plus gluons have dimension 9. The extra abelian direction must be addressed. The role of unimodularity and the hypercharge embedding is explicit in Chamseddine–Connes–Marcolli, §2.5, arXiv:hep-th/0610241.

The literature comparison is valid at the level of the boundary mechanism. The Neumann/Dirichlet towers appear in Csáki et al., §5.1, arXiv:hep-ph/0305237v1, and the fermion equation $\tan(m_n L) = M_D L$ is equation (5.6) of Csáki et al., arXiv:hep-ph/0310355v1. Their full gauge models have additional structure.

11. Corollary 22.9 — Correct with amendment.

Items 1, 2, 4 and 5 hold for fixed, orthogonal, J_F- and grading-compatible light and heavy channels, with $0 < \Theta < \pi/2$. For a general non-normal light block, write the upper wall block as

$$X_0 = \tan(\ell|M_0|)V_0^*,$$

or state the formula in terms of singular values. The light-channel coupling is $\ell M_0^* + O(\ell^3)$. Identifying it with a four-dimensional Yukawa coupling remains an additional hypothesis.

For item 3, let P_L reduce $D_F(\ell)$, and suppose $D_L = D_F(\ell)|_{P_L H_F}$ is fixed. Then the precise statement for a constant observable is

$$P_L Q_\ell(a) P_L = \operatorname{sinc}(\ell \operatorname{ad}_{D_L})(P_L \pi(a) P_L),$$

and

$$\|P_L Q_\ell(a) P_L - P_L \pi(a) P_L\| \le \frac{2}{3}\ell^2 \|D_L\|^2 \|\pi(a)\|.$$

This remains true when $\pi(a)$ mixes light and heavy channels. The limiting light compression need not be a representation: products can pass through the heavy subspace.

The full observable estimate does **not** follow from Theorem 22.4, since $\ell\|D_F(\ell)\| \to \Theta$. An explicit counterexample uses

$$D_F(\ell) = \sigma_1 \otimes \operatorname{diag}(m, \Theta/\ell), \qquad \pi(a) = 1 \otimes \sigma_1,$$

with the commutative algebra generated by 1 and $\pi(a)$ and the compatible real structure $J = (\sigma_1 \otimes 1)K$. Then

$$Q_\ell(a) = \operatorname{sinc}(\Theta - \ell m)\, \pi(a),$$

which tends to $\operatorname{sinc}(\Theta)\pi(a)$, rather than $\pi(a)$.

The ratio $m_t/M_W = 2\Theta/\pi < 1$ is a result for the stated lowest-level, zero-bulk-mass fermion channel and strict-wall gauge spectrum. Bulk fermion masses or a changed gauge wall condition are possible directions beyond that family.

12. Numerical comparison for the interior — Gap.

For a separate flat model with Neumann conditions at 0 and a finite Robin wall mass,

$$t \tan t = s, \qquad s = \frac{g_4^2 v^2 \ell^2}{4},$$

where $g_4^2 = g_5^2/\ell$ in the usual flat normalization. The Z equation has $s/\cos^2\theta_0$. This agrees with the form and sign convention of Csáki et al., §5.2, equations (5.13) and (5.22).

For $g'^2/g^2 = 3/5$, the lowest-mode squared mass ratio increases from 5/8 to 1. A small **Python** calculation evaluated the equations at the stated rounded values: $M_W\ell = 1.03$ is consistent with the quoted ratio to the displayed precision, and the next W root lies between $3.58/\ell$ and $3.59/\ell$, about 280 GeV when $1/\ell \approx 78$ GeV. This is a numerical check; the monotonicity follows separately from the equation.

The top constraints are strict:

$$M_W\ell < \frac{\pi}{2\sqrt{8/3}} \simeq 0.962, \qquad M_W\ell < \frac{\pi}{2(m_t/M_W)} \simeq 0.732$$

for the measured ratio. The latter estimate is consistent with the 2025 PDG W, Z values and top-mass table.

The collider-exclusion sentence is a **gap**: a mass alone does not establish a search exclusion without the model's couplings, branching fractions, and an applicable experimental limit. It should be removed or supplied with that analysis. The finite wall scalar also changes the strict wall boundary problem; it is an extended model outside Proposition 22.8 and No-go theorem 22.7. Within the stated flat toy model, the incompatible top and gauge mass requirements support the limited conclusion; bulk masses, warping, and running are directions to investigate.

13. Input and derived statements — Correct with amendment.

The finite algebra, representation, operator, grading, and real structure are inputs. The construction derives the endpoint commutant for that input, the interval operator and its window, the stated compression map, and the transport formulas. The Yang–Mills spectrum additionally depends on the action and boundary assumptions in item 10. The exact window identifies D_F, J_F, Γ_F; the represented algebra is obtained through compression and need not be an exact representation of A_F.

Several ingredients have broader precedents. For any reducing finite spectral subspace, the identity

$$[D_F, W^*TW] = W^*[D_X, T]W$$

holds whenever T preserves the operator domain. Operator-system compression is treated generally in Connes–van Suijlekom, §3. Boundary spectral geometry for the Standard Model was also studied by Chamseddine and Connes, §5, arXiv:1008.3980v1, and the interval fermion and gauge mechanisms occur in the Higgsless papers cited in item 10. The particular combination and formulas of lemma 22 can be stated explicitly; a wider novelty claim would need a fuller literature comparison.

Amendments and errors

- **Errors:** correct the boundary-type zero-mode statement at nonzero X and the first-derivative comparison between fixed compression and transport.
- **Gap:** support or remove the collider-exclusion claim.

- **Amendments:** state spectral multiplicities and unitality; qualify the Standard Model data, gauge quotient, and unimodularity; distinguish profiles from holonomies and real from non-real transport; scope the radial result to the moving window; provide reproducible data for the numerical algebra dimensions; state the light-block estimate and the additional assumptions of the Yang–Mills and finite-wall-scalar models.

Overall: 2 Correct; 8 Correct with amendment; 1 Gap; 2 Error.

Part R9 (referee report, lemma 23)

Referee report on lemma 23

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_23.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 23 and the note.

The boundary-spectrum formulas are correct under the stated Yang–Mills boundary conditions. The principal corrections concern the aligned-wall compression, the comparison with twisted spectral triples, the order of the holonomy double coset, and the bulk-mass extension of the one-loop potential.

The verdicts below count the lettered entries. The numerical checks use Python 3 with SciPy. They supplement the analytic arguments and do not replace them.

1. Theorem 23.1 (statement and proof)

1a. Item 1 — Correct. Each wall condition is a maximal isotropic subspace for the boundary form. The resulting first-order operator is self-adjoint with compact resolvent, and the spin grading preserves its domain and anticommutes with it.

1b. Item 2 — Correct. Under the equal-dimension and no-−1-eigenvalue hypotheses, the principal logarithm gives an anti-Hermitian K satisfying $\Gamma_0 K \Gamma_0 = -K$. The stated $D_F = \Gamma_0 K$, its singular values, and

$$\Gamma_1 = e^{lK} \Gamma_0 e^{-lK}$$

follow from the two-projection decomposition.

1c. Item 3 — Correct with amendment. The graph boundary condition and the converse hold on the stated branch $l\|D_F\| < \pi/2$. Define X explicitly as the $H_- \to H_+$ block of $\tan(lD_F)$. The spectral part of Theorem 22.1 applies without a real structure; its real-structure conclusion additionally needs the hypothesis in item 4. At an angle $\pi/2$, the principal logarithm and the isolated window used here cease to be unique.

1d. Item 4 — Correct. The antiunitary J_1 is an additional hypothesis. With $\widehat{J} = EJ_1$, the signs are $(+,+,-)$, and $J_F = \Gamma_0 J_1$ has the stated finite signs. In particular,

$$J_F \Gamma_1 J_F^{-1} = \Gamma_0 (J_1 \Gamma_1 J_1^{-1}) \Gamma_0 = -\Gamma_0 \Gamma_1 \Gamma_0.$$

The antilinearity of J_1 is compatible with $J_1 K J_1^{-1} = K$ on the principal branch.

2. Theorem 23.2 (statement and proof)

2a. Items 1 and 2 — Correct with amendment. Domain preservation is equivalent to commuting with the respective wall grading, and $[D, f] = Ef'$. The compression formula in item 2 requires the equal-dimension and branch hypotheses under which the isometry W of Theorem 22.1 exists. State those hypotheses at the item.

2b. Item 3, aligned walls — Error as stated for an arbitrary $*$-subalgebra P. A constant odd $a \in P$ generally fails the endpoint conditions and is an ambient multiplier, rather than an element of the parent algebra. More seriously, $E_0(P)$ need not equal $P \cap \{\Gamma_0\}'$.

Let $H = \mathbb{C}^2$, $\Gamma_0 = \sigma_z$, and let P be the unital $*$-algebra generated by $\sigma_x + \sigma_z$. Then

$$E_0(P) = \operatorname{span}\{1, \sigma_z\}, \qquad P \cap \{\Gamma_0\}' = \mathbb{C}1.$$

The first space is the compression image, while the second is the even part of P. Compactly supported scalar profiles show that the image of the actual boundary algebra is $E_0(P)$.

The correction is to assume $\Gamma_0 P \Gamma_0 = P$. Then $E_0(P) = P^{\mathrm{ev}}$, and the aligned-wall conclusion holds. Continue to distinguish constant ambient multipliers from parent-algebra elements.

2c. Item 4 — Correct. Under the stated J_1 hypothesis and $[P, P^\circ] = 0$, the parent has a real structure and satisfies order zero and order one, even when P contains elements odd for Γ_0. The parent grading is $\hat{\Gamma}$, which commutes with those elements. Pointwise,

$$[\pi(f), J\pi(g)^* J^{-1}] = 0, \qquad [[D, \pi(f)], J\pi(g)^* J^{-1}] = E[f', g^\circ] = 0.$$

2d. Comparison with Result 11 — Correct with amendment. The aligned-wall construction evades hypothesis 4 of Result 11: odd coordinate matrices are sent to zero rather than retained as $\Gamma_a/5$. Result 11 assumes an even parent, an inherited involutive grading, and retention of all those coordinate compressions simultaneously. Its stated escapes include restricting to the even algebra or changing the grading construction. The present construction points toward a matrix-valued parent with an inherited grading; it leaves the commutative-parent and retained-coordinate questions open.

3. Remark 23.3 (remark)

3a. Opposite action — Correct with amendment. For an odd **anti-Hermitian** h, the displayed signs for the unstarred conjugations $J_1 h J_1^{-1}$ and JhJ^{-1} are correct. The opposite representation used in order zero is $b^\circ = J_1 b^* J_1^{-1}$; state the star and the anti-Hermiticity assumption when giving signs. On the W-factor it incorporates $\sigma(b) = \gamma_W b \gamma_W$.

3b. Grand-symmetry twist — Error in the assertion that it is the same automorphism. The relevant references are:

- Agostino Devastato, Fedele Lizzi and Pierre Martinetti, "Grand symmetry, spectral action and the Higgs mass," *JHEP* 01 (2014) 042. The larger algebra and its reduction appear in §§3–5.
- Agostino Devastato and Pierre Martinetti, "Twisted spectral triple for the Standard Model and spontaneous breaking of the Grand Symmetry," *Mathematical Physics, Analysis and Geometry* **20** (2017), article 2. Its Eqs. (1.8)–(1.11) define the twisted commutator, first-order condition and automorphism; the publisher record confirms the publication details.

The latter authors call their ρ the "exchange of the left/right spinorial indices." It exchanges algebra summands. Conjugation by γ_W is inner on $M_2(\mathbb{H})$ and fixes its centre. Thus the full automorphisms have different algebraic actions. The correction is to describe a **structural analogy** between involutive automorphisms and compare the actual representations, opposites and first-order conditions before claiming an identification.

4. Theorem 23.4 and Example 23.5 (fluctuations and example)

4a. Connection removal — Correct with amendment. The displayed unitary identity follows from $u' = -hu$. It describes the perturbation $D + Eh$. For a fluctuation of a real triple, include the conjugate one-form:

$$D + Eh + \widehat{J}(Eh)\widehat{J}^{-1}.$$

In the example, the resulting generator has the form (L_F, R_F), as required for the stated action on $E \oplus E^*$. This also preserves the J_1 condition. The odd directions lie outside No-go theorem 22.7, whose hypotheses include an even algebra and a moving window.

4b. Holonomy and double cosets — Error in the order of the factors. With the convention $u(0) = 1$, $u(l) = w$, let $H_i = \mathrm{Stab}(\Gamma_i)$. A gauge transformation removing h has the form $u(x)c$, so it preserves both endpoints precisely when

$$c \in H_0, \qquad wc \in H_1, \qquad \text{equivalently} \qquad w \in H_1 H_0.$$

The holonomy quotient is $H_1 \backslash G / H_0$, or $H_0 \backslash G / H_1$ after replacing w by w^{-1}. The order matters for the stated w. Also restrict G to the unitary component reachable by a smooth path from the identity.

4c. Orbit, angles and polar coordinates — Correct with amendment. The orbit of quaternionic self-adjoint involutions with one-dimensional quaternionic $+1$ and -1 eigenspaces is

$$\mathbb{H}P^1 \simeq Sp(2)/(Sp(1) \times Sp(1)) \simeq S^4.$$

The wording "each grading with quaternionic eigenspaces" also includes ± 1; exclude those two cases. For $\Gamma_p = p_a \Gamma_a$, the eigenvalues of $\Gamma_q \Gamma_p$ are $e^{\pm i d(p,q)}$, so the principal angles are $d(p,q)/2$.

The formula on E^* is correct: p' is the reflection of p through p_0, because

$$\Gamma_{p'} = \gamma \Gamma_p \gamma.$$

The quaternion direction q supplies three angular coordinates away from the poles. It is undefined at both poles, and at $\theta = \pi/2$ the finite logarithm and lowest-window choice are degenerate. The unquotiented orbit is S^4; quotienting the relative orientation by the stabilizer leaves the radial interval.

4d. Bulk-mass texture — Correct with amendment. For $X = \tan\theta\, 1$, the light-mass approximation from Theorem 12.6 has the scale

$$M_{\text{light}} \simeq \frac{\tan\theta}{l}\, \Phi_Q \Phi_U$$

when the light eigenvalues are small compared with the relevant tower thresholds. In the convention of Theorem 12.7, $\widehat{Y} = X/l = (\tan\theta/l)1$; a dimensionless identity matrix can be stated after factoring out that common scale. Positive localization parameters suppress the wall profiles. General finite-angle eigenvalues require the secular equation.

4e. Minimal composite Higgs comparison — Correct with amendment. The coset and the four real Higgs coordinates agree with those in Agashe, Contino and Pomarol, "The Minimal Composite Higgs Model," *Nuclear Physics B* **719** (2005), 165–187, §§2–3. Their Eq. (6) identifies h/f_π with the sphere angle, corresponding here to 2θ. Their construction uses a warped AdS interval and additional fermion and boundary data. The shared coset alone does not identify the actions or potentials.

5. Theorem 23.6 (boundary conditions and theorem)

5a. Connector at one wall — Correct with amendment. The stated gauge boundary conditions form a consistent boundary-value problem with the connector treated as prescribed wall data. A real fermion connector can also be placed at one wall. For example, choose a unit vector e in the $+1$ eigenspace of Γ_1, put $f = J_1 e$, and set

$$K_c = |f\rangle\langle e| - |e\rangle\langle f|, \qquad \Gamma_{1,c} = e^{\eta K_c} \Gamma_1 e^{-\eta K_c}.$$

Then K_c anticommutes with Γ_1, commutes with J_1, and J_1 anticommutes with $\Gamma_{1,c}$. Restricting the wall algebra to the common stabilizer gives the connector conditions and retains the parent order conditions.

The amendment is to specify whether the connector is prescribed gauge-breaking data or a term in the fermion boundary operator. In the latter case, $\Gamma_{1,c}$ changes the affected fermion tower; the universal fermion product later in the lemma must be recalculated for that channel.

5b. Charged and neutral masses — Correct. An independent solution uses principal angles between the Neumann subspaces at the two walls. Put $s = \sin(2\theta)$. For each charged direction the squared sine of the angle is $s^2/2$. In the neutral sector the two Neumann directions are L' and $\sqrt{3/5}\, R' + \sqrt{2/5}\, B$; their projection onto the Dirichlet complement at the first wall has one nonzero squared singular value $4s^2/5$. Consequently

$$\sin^2(M_W l) = \frac{s^2}{2}, \qquad \sin^2(M_Z l) = \frac{4s^2}{5},$$

which give the stated tangent equations. This derivation also covers $\cos(2\theta) = 0$, where the divisions in the written proof need a limiting interpretation.

5c. Massless modes — Correct with amendment. For $0 < \theta < \pi/2$, the intersection has dimension nine: the photon and eight gluons. At each endpoint it has dimension twelve, with three additional electroweak zero modes. The phrase "massless for all θ" is correct; state the endpoint enhancement explicitly.

5d. Normalization — Correct. With the stated fundamental trace forms, equal five-dimensional coefficients, flat profiles and no boundary kinetic terms,

$$g_3 = g_2 = \sqrt{\frac{5}{3}}\, g_Y, \qquad \frac{1}{e^2} = \frac{8}{3g_2^2}.$$

Indeed the hypercharge generator has squared trace norm $1 + 2/3 = 5/3$, while the photon generator has squared norm $1 + 1 + 2/3 = 8/3$. Here g_2 is the normalized four-dimensional zero-mode coupling.

5e. Items 1 and 2 — Correct. Expanding the sine equations gives $M_W^2 \sim 2m^2$, $M_Z^2 \sim (16/5)m^2$ as $\theta \to 0$. At $\theta \to \pi/2$, both displayed branches tend to zero and their squared ratio tends to $5/8$. The ratio itself is undefined at the endpoints.

5f. Item 3 — Correct. Let $a = \sin(2\theta)$. Then

$$\frac{M_W^2}{M_Z^2} = \left[\frac{\arcsin(a/\sqrt{2})}{\arcsin(2a/\sqrt{5})}\right]^2.$$

For $0 < a \le 1$, this decreases with a, since $x/(\sqrt{1-x^2}\arcsin x)$ increases. Hence the midpoint is the global minimum, with exact value

$$\left(\frac{\pi/4}{\arctan 2}\right)^2 = 0.5032315\ldots.$$

The upper value $5/8$ is approached at the endpoints. This supplies the proof missing from the grid check.

5g. Item 4 — Correct. The ratio m/M_W increases from $1/\sqrt{2}$ to infinity. Python 3 with SciPy gives $\theta = 1.05888846\ldots$ for $m/M_W = \sqrt{8/3}$.

6. Comparison with lemma 18 and Proposition F

6a. Constants 2 and 16/5 — Correct with amendment. These are the coefficients in the lower bounds for the fully massive finite-action pole masses in Proposition 14 of lemma 18. They are not the small-x pole coefficients there: those are 3 and 24/5. The common 5/8 limit is a valid comparison, with different limiting behaviour: the finite-action poles grow in the lemma 18 heavy limit, while the interval vector masses tend to zero at the antipodal endpoint.

Specify the trace or Hilbert–Schmidt norm in the commutator-norm claim. With the ordinary operator norm, the charged example gives m^2, rather than $2m^2$.

6b. Factor 3/2 — Correct with amendment. Proposition F gives the factor 3/2 for the stated leading finite spectral-action mass matrix. Comparing it with the assumed five-dimensional Yang–Mills quadratic form gives $3m^2$ versus $2m^2$ in the universal small-mass example. A calculation of the full parent spectral action, with its tower and boundary conditions, remains necessary for a direct matching claim.

7. Proposition 23.7 (heat trace)

7a. Heat trace and monotonicity — Correct. With $s = l\Lambda$,

$$\mathrm{Tr}_{\mathrm{channel}}\, e^{-D^2/\Lambda^2} = \frac{2s}{\sqrt{\pi}}\,\vartheta_3(\theta, e^{-s^2}).$$

The Jacobi product proves strict decrease on the interval, with zero derivative at the endpoints. At fixed $x = \theta^2/s^2$, its towerless limit is $2e^{-x}$. At $\theta = \pi/2$, the full lowest eigenspace is degenerate even though the heat trace is well defined.

7b. Escape from No-go theorem 9 — Correct with amendment. No-go theorem 9 assumes finite internal data, the positive-Laplace cutoff class, the literal flat-space spectral action, $A = 0$, and an allowed unconstrained ray $V = cV_*$, $c > 0$, without an added scalar potential. Its conclusion is strict decrease along that ray. The interval family changes the operator family and replaces the free ray by a compact relative-grading orbit with a tower. It therefore lies outside those hypotheses.

For the stated heat action, the antipodal point is a minimum of this family. Added potentials, constraints, other cutoff classes and quantum effects remain separate possibilities. The result points toward analysing the full tower and any additional scalar terms.

8. One-loop potential (products and potential)

8a. Three products — Correct. The product for a principal angle α, normalized at $\alpha = 0$, is

$$1 + \frac{\sin^2 \alpha}{\sinh^2 x}.$$

Using the fermion angle θ and the two gauge angles from 5b gives all three displayed products.

8b. Fermion coefficient — Correct with amendment. With the conventional four-dimensional Dirac determinant and each physical species counted once, $n_f = 2N_D$; one colour-triplet Dirac top contributes six. The universal example gives masses to both members of each weak doublet. For one complete generation it contains eight physical Dirac species, hence $n_f = 16$, before any connector modification; three identical generations give $n_f = 48$. A top-only $n_f = 6$ potential therefore needs a separately specified fermion sector or a complete accounting of spectator channels.

8c. Potential and endpoint curvatures — Correct with amendment. The logarithmic integrand and the vector multiplicities are correct under the stated one-loop assumptions. The curvature sentence omits the prefactor on its left-hand side. Let

$$I = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x} = \frac{3}{2}\zeta(3), \qquad J = \int_0^\infty \frac{x^3\,dx}{\cosh^2 x} = \frac{9}{8}\zeta(3).$$

The precise expansions are

$$8\pi^2 l^4[V(\theta) - V(0)] = (10.8 - n_f) I\,\theta^2 + o(\theta^2),$$

and, with $\epsilon = \pi/2 - \theta$,

$$8\pi^2 l^4[V(\tfrac{\pi}{2} - \epsilon) - V(\tfrac{\pi}{2})] = (10.8 I + n_f J)\epsilon^2 + o(\epsilon^2).$$

For $n_f \geq 0$, the latter curvature is positive.

8d. Numerical absence of an interior minimum — Gap in any conclusion beyond the sampled grid. The recorded computation is explicitly a grid statement, and its original code was not retained. An independent Python 3/SciPy calculation of the derivative found one sign change for each tested $n_f = 1, 2, 3, 4, 6, 8$; for $n_f = 8$, the stationary angle is approximately 0.3382. The known integral $I = 3\zeta(3)/2$ agreed to 5×10^{-16} in absolute error, and a 1% perturbation failed the same check. A grid does not exclude additional stationary points between samples, so the claim about absence of an interior **local** minimum for all parameters in the displayed massless family remains open in the range $0 < n_f < 10.8$.

Two stronger statements follow analytically from the displayed potential:

- For every $n_f > 0$, $\theta = \pi/2$ is the unique global minimum on $[0, \pi/2]$. The gauge logarithms are nonnegative, and the negative fermion logarithm is minimized when $\sin^2 \theta = 1$.
- For $n_f \geq 10.8$, the potential is strictly decreasing throughout the interior. To see this, let $t = \sin^2 \theta$ and

  $$\mathcal{J}(z) = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x + z}.$$

  The derivative, apart from the positive overall factor, is

  $$(1 - 2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\left(\frac{16}{5}t(1-t)\right)\right] - n_f \mathcal{J}(t).$$

  For $t \leq 1/2$, both arguments of $\mathcal{J}$ in the bracket are at least t, and $\mathcal{J}$ decreases. For $t > 1/2$, both contributions have the required sign.

These conclusions concern the displayed massless flat products, the stated gauge content and boundary conditions, and one-loop order. They direct attention to changed form factors or additional terms for an interior vacuum.

8e. Leading-logarithm form — Correct with amendment. The formal first term is $B\sin^2(2\theta) - A\sin^2\theta$, with $A, B > 0$, and any interior stationary point of that expression is a maximum. The logarithm expansion is not uniform at $x = 0$; the next term has an infrared-divergent integral. Present the expression as a formal leading approximation and use the full integral for conclusions about its global shape.

8f. Bulk-mass scan — Error as a finite one-loop potential for general scanned masses. Let $a = c_Q + c_U$. The stated fermion factor satisfies

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

so, for $t = \sin^2\theta$,

$$\log(1 + t(G-1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by x^3 gives angle-dependent ultraviolet divergences whenever $a \neq 0$. A finite quadrature value then depends on a regulator and subtraction prescription; the reported scan does not establish a regulator-independent potential.

The exceptional case $c_U = -c_Q$ has a common bulk mass on the two mixed components and

$$G(x) = 1 + \frac{w^2}{x^2\sinh^2 w},$$

which has the required exponential decay. Keeping the flat gauge products is consistent as a **separate tree-level gauge-sector assumption**. For a coupled $Sp(2)$ gauge theory, unequal masses on components mixed by the gauge group require gauge-invariant completion or explicit breaking data. The correction is to specify that completion, the regulator and allowed wall counterterms, then recompute the potential.

8g. Escapes and spinor comparison — Correct with amendment. A frequency-two fermion contribution, modified gauge form factors, wall potentials and wall terms are possible changes to the stated flat calculation; each requires its own model and coefficients. For the comparison with Agashe–Contino–Pomarol, their §2.1, Eqs. (19)–(23), uses spinorial $SO(5)$ fermions and permits an approximate potential

$$\alpha\cos(h/f_\pi) - \beta\sin^2(h/f_\pi).$$

With $h/f_\pi = 2\theta$, the displayed leading flat expression has this angular form with $\alpha = A/2$ and $\beta = -B$, up to a constant. Their fermion form factors can also give the sign and competition needed for an interior minimum. The flat scan therefore has its stated scope; it does not exclude the spinorial composite-Higgs construction.

9. Measured values and open questions (comparison)

9a. Numerical comparison — Correct with amendment. Using the 2026 Particle Data Group values $m_t = 172.60$ GeV, $M_W = 80.3625$ GeV and $M_Z = 91.1879$ GeV gives $m_t/M_W \simeq 2.1478$ and $M_W^2/M_Z^2 \simeq 0.7767$. The top entry is a direct-measurement mass and its definition should be kept distinct from a specified pole-mass scheme. PDG quark table, p. 2; PDG gauge-boson table, pp. 1–2.

Python 3 with SciPy gives $\theta \simeq 1.1626$, $M_W l \simeq 0.5413$, $1/l \simeq 148.5$ GeV and the model ratio $M_W^2/M_Z^2 \simeq 0.5814$. For the separate wall-scalar model of lemma 22, fitting the measured vector ratio gives $M_W l \simeq 1.0282$; its lowest no-bulk-mass fermion channel then has $m/M_W < 1.528$. Thus the stated mismatch holds for those flat, tree-level families with the specified lowest-level and coupling assumptions. Bulk masses, warping, running and a combined wall/Wilson-line model remain directions to examine.

9b. Open questions — Correct with amendment. The listed questions remain appropriate. Add the real connector's contribution to the fermion tower, a gauge-invariant and renormalized bulk-mass potential,

the degenerate antipodal window, and the scalar kinetic normalization needed to identify a canonically normalized Higgs field.

Amendments and errors

- Assume $\Gamma_0 P \Gamma_0 = P$ for the aligned-wall even-algebra claim, and distinguish ambient constant multipliers from elements of the boundary algebra.
- Replace the claimed equality with the grand-symmetry twist by a comparison of the involutions and their first-order conditions.
- Reverse the holonomy factors for the convention used: $w \in H_1 H_0$, with quotient $H_1 \backslash G / H_0$.
- Specify the real fluctuation, the connector's role, the altered fermion tower, endpoint degeneracies and trace norms.
- Add the missing $1/l$ scale and light-mode approximation to the bulk-mass texture; add $1/(8\pi^2 l^4)$ to the potential curvatures.
- Restrict the unproved absence of local minima to the sampled grid, and replace the generic bulk-mass scan by a calculation with a stated regulator, counterterms and gauge completion.

Overall: 11 Correct; 19 Correct with amendment; 1 Gap; 4 Error.

Part R10 (referee report, lemma 25)

Referee report on lemma 25

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_25.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 25 and the note.

The leading constant-profile mass formula is supported after the amendments below. The principal mathematical error is the extra factor of s in the displayed heat-trace derivative. The identification with a physical lowest gauge mode requires additional boundary and spectral assumptions.

The calculations in this report are analytical; no numerical code was run. Numerical values below are taken from the cited records.

1. **Proposition 25.1 — Correct with amendment.**

   The unfolding in lemma 25 is correct after multiplying it by $1/\sqrt{2}$. It maps the H^1 domain of D_0 onto the periodic H^1 domain on the circle. With

   $$D_F = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix},$$

   the connection term satisfies

   $$E\Gamma_F D_F : \quad \Psi_Q \mapsto M^* \Psi_U, \qquad \Psi_U \mapsto M \Psi_Q.$$

   Thus the unfolded operator is $-i\partial_x + D_F$, with signed spectrum $n\pi/l + \mu_i$.

   The amendment is to name the frame. Corollary 22.2 states

   $$D_X = R(D_0 + E\Gamma_F D_F)R^*, \qquad R(x) = e^{x\Gamma_F D_F}.$$

   An even T that is constant **after transport** unfolds to $1 \otimes T$. In the original frame it is $R(x)TR(x)^*$, which generally varies with x. Conversely, a constant original-frame T becomes R^*TR after transport and generally does not unfold to $1 \otimes T$. For broken T, it also need not preserve the original coupled wall domain. The transported field generally has an odd bulk component and therefore requires an admissible enlarged gauge algebra; it is not automatically a field of the even algebra in lemma 22.

   There is a separate error in the holonomy sentence at line 31. With the stated convention, put $U = e^{-i\pi\Gamma_0/4}$. The correct relation is

   $$\Gamma_1 \Gamma_0 = e^{2l\Gamma_0 D_F} = U e^{2ilD_F} U^*.$$

Conjugation by Γ_0 instead sends e^{2ilD_F} to its inverse. The circle holonomy can acquire the inverse sign under the opposite parallel-transport convention.

2. **Theorem 25.2 — Correct with amendment.**

   The exact kernel at line 39 and its quadratic coefficient have the stated signs. Lemma 18 states its theorem for a finite internal trace; an extension to this infinite tower must be supplied. Here it is available: V and T split into finite-dimensional blocks indexed by n. Applying the finite theorem to a symmetric finite set of blocks and passing to the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify differentiation and Taylor expansion under the sum.

   Let $t = 1$, $d = \mu_i - \mu_j$, $b = u(1-u)$, and

   $$E(u) = e^{-[(1-u)(k+\mu_i)^2 + u(k+\mu_j)^2]}.$$

   An exact integration-by-parts form of the $q = 0$ bracket is

   $$d^2 \int_0^1 E(u)\big[1 - b(2k + \mu_i + \mu_j)^2\big]\, du.$$

   Its expansion begins

   $$d^2 e^{-k^2}\left[1 - \frac{2}{3}k^2 + \frac{2k^3 - 5k}{3}(\mu_i + \mu_j) + O(\|\mu\|^2)\right].$$

   The displayed cubic term is odd in k and cancels between n and $-n$. Similarly, the q^2 bracket begins

   $$e^{-k^2}\left[\frac{2}{3} - \frac{2k}{3}(\mu_i + \mu_j) + O(\|\mu\|^2)\right],$$

   and its linear term cancels. The grading symmetry gives the same evenness for the summed expression.

   The proof should include these cancellations and remainder terms. At fixed $s = l\Lambda$, for fixed nonzero Hermitian T, the stated $O(\|D_F\|^3/\Lambda)$ mass remainder is valid but can be sharpened to

   $$M^2_{\text{Taylor}} = \left(\frac{3}{2} - \langle k^2\rangle\right)\frac{\|[D_F, T]\|^2_{\text{HS}}}{\|T\|^2_{\text{HS}}} + O\left(\frac{\|D_F\|^4}{\Lambda^2}\right).$$

   Here $k_n = n\pi/(l\Lambda)$ is dimensionless. Equations presented later without a remainder must be labeled leading-order formulas.

3. **Spectral dimension — Error.**

   The factor at line 60 is wrong. Direct differentiation gives

   $$\boxed{\langle k^2\rangle = \frac{1}{2}\frac{d\log S}{d\log s} = \frac{s}{2}\frac{d\log S}{ds}.}$$

   The extra s multiplying the logarithmic derivative must be deleted. The subsequent coefficient $(3 - d_{\text{eff}})/2$ uses the corrected identity.

   The spectral-dimension interpretation is correct for the **unshifted unfolded circle**, equivalently the paired Neumann-plus-Dirichlet interval trace. It should not be called the spectral dimension of an arbitrary interval boundary problem. Indeed, their separate heat traces are

   $$S_N = \frac{S+1}{2}, \qquad S_D = \frac{S-1}{2},$$

   and have different logarithmic derivatives. For nonzero D_F, the actual parent heat trace contains $e^{-t(k_n+\mu_i)^2}$; S is the unperturbed trace used in the leading expansion.

4. **The two limits and covariance — Correct with amendment.**

   The limiting factors 3/2 and 1 are correct. The exponential corrections have polynomial prefactors. More precisely,

$$d_{\text{eff}}(s) \sim \frac{4\pi^2}{s^2} e^{-\pi^2/s^2} \quad (s \to 0), \qquad d_{\text{eff}}(s) = 1 - 4s^2 e^{-s^2} + O(s^2 e^{-2s^2}) \quad (s \to \infty).$$

   State the order of limits: first take the small-D_F/Λ coefficient at fixed s, then take either limit.

   The five-dimensional covariance argument is valid on the unfolded circle for this operator and these insertions. With Hermitian connections the curvature convention gives $F_{5\mu} = i[D_F, A_\mu]$, up to the sign convention for the covariant derivative; its norm gives the quoted commutator quotient.

   On a general space with boundary, the bulk local coefficient alone does not rule out boundary contributions. In the present calculation the full operator with the specified transported fields is unitarily equivalent to the circle problem, so its trace has the required cancellation. State that specific reason rather than a general boundary assertion.

5. **Corollary 25.3 — Correct with amendment.**

   The coefficients agree with Remark 16 of lemma 18 and with the CCM mass relation, Eq. (4.16) at leading order. With the literal particle-and-antiparticle trace,

$$Q_{11} = 12, \qquad K_{11} = \sum_f N_f m_f^2,$$

   which gives $(3 - d_{\text{eff}})/24$. For the top alone, $K_{11} = 3m_t^2$, giving the displayed ratio and

$$y_t = \frac{2g}{\sqrt{3}} \quad (d_{\text{eff}} = 0), \qquad y_t = \sqrt{2}\, g \quad (d_{\text{eff}} = 1).$$

   The first is also CCM's top-dominance approximation, Eq. (5.27).

   The neutral result $M_W^2/M_Z^2 = 5/8$ follows **to this order**, with the common trace and normalization. "Massless directions are unaffected" should say that their mass coefficient remains zero; their kinetic coefficients can change.

   The twelve doublets have the same tower factor S in the stipulated flat, bulk-mass-free model, including when only the top has nonzero Dirac mass. With bulk masses used to localize light fields, their spectra and overlap matrix elements change. The kinetic coefficient must then be recomputed from all eigenstates and overlaps; it cannot generally be obtained by assigning the same S to every doublet. Light zero modes can still contribute even when their excited modes are suppressed. For an explicit localization mechanism and profiles, see Adachi, Kurahashi and Maru, arXiv:1404.4281v1, Eqs. (2.10)–(2.11).

6. **General cutoffs — Correct with amendment.**

   The weighted formula is correct for the positive Laplace class and moment hypotheses in lemma 18, Eq. (4). It is a leading small-D_F formula. A precise version is

$$S_\nu(s) = \int S(s/\sqrt{\tau})\, d\nu(\tau), \qquad d_\nu = \frac{d \log S_\nu}{d \log s} = 2\langle \tau k^2 \rangle_\nu,$$

   so that the coefficient is $(3 - d_\nu)/2$. Under those moment assumptions, dominated convergence gives the stated endpoints. The crossover and its corrections depend on ν.

   Replace "general cutoffs" by "positive Laplace cutoffs satisfying (4)." A general smooth spectral cutoff, including a nonconstant cutoff flat near zero, need not belong to that class.

7. **Scope and the lowest mode — Gap.**

   The small-angle comparison with Theorem 23.6 is correct. For its *universal* fermion mass $m = \theta/l$, the $d_{\text{eff}} = 1$ formula gives

$$M_W^2 = 2m^2, \qquad M_Z^2 = \frac{16}{5}m^2$$

   at leading order. Expanding the exact Yang–Mills formulas gives, respectively,

$$2m^2 - \frac{4}{3}m^4l^2 + \cdots, \qquad \frac{16}{5}m^2 - \frac{64}{75}m^4l^2 + \cdots.$$

   Lemma 25 directly computes a Taylor quotient for a specified constant profile. For the unrestricted unfolded circle, translation invariance separates its zero Fourier sector from nonzero Fourier sectors; one must still diagonalize the generator matrix within that sector. Calling the result the first nonzero term of a **lowest-mode mass** also requires an admissible gauge field, a positive kinetic form and an isolated lowest branch. Boundary constraints or additional terms can alter the profile problem. Theorem 23.6 establishes the comparison for its own Yang–Mills boundary problem; it does not establish the full finite-cutoff spectral-action gauge spectrum.

   Add these hypotheses to Scope, together with the transported-frame qualification, the distinction between Taylor and pole masses, and the fact that the calculation is about a prescribed background rather than a demonstrated vacuum. "First order in D_F" refers to the mass; the displayed mass **squared** starts at quadratic order.

8. **Reading — Correct with amendment.**

   The factor 3/2 in Proposition F is the finite-action coefficient relative to the Yang–Mills commutator quotient in the stated trace. The crossover explains that coefficient's two limits in the present constant-profile calculation. Proposition F also contains a bounded-fermion versus tower-scale obstruction and overlap conditions; those do not follow solely from the factor 3/2. The reading should retain their separate hypotheses.

   The direct-measurement figure uses the PDG 2026 top listing, p. 1, 172.60 ± 0.27 GeV, and the PDG 2026 *W* listing, 80.3625 ± 0.0077 GeV. The top value is a direct-measurement Monte Carlo mass, with a scheme interpretation discussed by PDG.

   The remembered 163 GeV is supportable with a scheme and scale: Yoon and Peskin, arXiv:1810.12352v3, §3.3 and Eq. (23) use approximately $\overline{m}_t(\overline{m}_t) = 163$ GeV for a 173 GeV pole mass. They also quote about 147 GeV at 2 TeV. Dividing either running mass by an on-shell W mass is only a formal orientation, rather than a common-scale matching test.

   The proposed matching scale of order $30M_{\text{KK}}$ and its running are outside lemma 25. The cited steer note labels its calculation unrefereed, uses remembered tower coefficients, and reports that the three couplings do not meet at one point. Retain the figure only as a conditional estimate from that separate model.

9. **Literature — Gap.**

   The endpoint $m_t = 2M_W$ is already present in flat five-dimensional gauge–Higgs literature. Cacciapaglia, Csáki and Park, *Fully Radiative Electroweak Symmetry Breaking*, arXiv:hep-ph/0510366v2, §5 explicitly state the tree-level relation for a suitable rank-four representation and discuss the assumptions on the top zero mode. Adachi and Maru, arXiv:1804.06012v2, Eq. (3.23) obtain a limit approaching $2M_W$ with their specified brane mixings. These are representation-dependent mechanisms; they do not establish the lemma's spectral-action crossover.

   Ordinary universal-extra-dimension models need separate specification. For example, Cordero-Cid et al., arXiv:1108.2926v4, §2 formulate the five-dimensional Standard Model with separate Higgs and Yukawa sectors; universal propagation alone does not impose $m_t = 2M_W$.

The inspected primary records do not exhibit the particular $(3-d_{\text{eff}})/24$ coefficient or its theta-function crossover. We cannot claim novelty on that basis. The search used web-index queries for spectral action, heat kernels, compactification and gauge–Higgs mass relations, including a 2026 query; it followed the 2014 Adachi–Kurahashi–Maru reference to the 2006 paper. A systematic forward-citation search, author recency sweep and broader database search remain open.

10. **Paper's final sections — Error.**

    The sum-rule section of boundary_condition_note.tex carries the leading result into an unqualified displayed equality. Add the small-mass remainder and the transported-frame, trace and constant-profile assumptions. In particular, the towerless top-only Taylor expansion already contains the nonzero correction

    $$M_W^2 = \frac{3}{8} m_t^2 - \frac{23}{128} \frac{m_t^4}{\Lambda^2} + \cdots$$

    in lemma 18, Eq. (19). Thus the leading equality must not be stated as an exact finite-mass result. Apply the same qualification to the abstract and to "throughout" for the neutral ratio.

    The statement that running "would be a power law" also requires a specified effective theory, thresholds and renormalization prescription. The subsequent **Status** section contains separate claims based on lemma 24 and its literature search; those lie outside lemma 25 and were not audited here. Provenance should be updated if this report is recorded.

Amendments and errors

- Correct the factor in the heat-trace derivative and identify the unperturbed paired interval or circle trace.
- Name the transported frame for T, correct the holonomy conjugation, and state the admissible gauge algebra and wall conditions.
- Add the finite-block convergence argument, odd-k cancellations and leading-order remainders.
- Qualify the covariance, cutoff, neutral-ratio and lowest-mode statements by their hypotheses.
- Cite the top-mass schemes and prior gauge–Higgs literature; treat the crossover's priority as open pending a wider search.
- Amend the paper and abstract to state leading-order results, and keep the separate running and Status claims within their own scopes.

Overall: 0 Correct; 6 Correct with amendment; 2 Gap; 2 Error.

Part R11 (referee report, lemma 24)

Referee report on lemma 24

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_24.md. Reproduced as delivered, with links made relative. The amendments are collected at the end of lemma 24.

The main operator-window construction and the new one-loop argument are correct in their stated families. The principal gap is the statement of convergence of compression maps for general observables as the interval shrinks: the varying source spaces need an identification, and different interval profiles have different limits.

This report gives one verdict for each of the ten requested items. The checks are analytic; no numerical computation was used.

1. **Result 1 — Correct.** In Result 1, $C_i^2 = \Pi$, $(\operatorname{sgn} A)^2 = 1 - \Pi$, and the cross terms vanish. Thus both χ_i are self-adjoint involutions and anticommute with E. They commute with $\hat{\gamma}$, so the boundary domain is preserved by the grading.

   Let $B = SA/r$. At 0, the complementary boundary condition selects $B < 0$; at ℓ, it selects $B > 0$. The reversal makes the boundary term in (3) nonnegative and gives (2). The finite kernel conditions are maximal isotropic and give the interval operator of lemma 23.

The H^1 realization is correct. Decomposition into eigenspaces of A gives self-adjoint finite-dimensional interval problems. On their direct sum, (3) controls both q' and Aq; ellipticity of A identifies the graph domain with the stated H^1 domain. Rellich compactness then gives compact resolvent. The projection Π reduces both the expression and the domain.

Differentiation verifies (4). The principal-logarithm hypothesis gives $\ell\|D_F\| < \pi/2$, and the interval spectrum together with (2) gives (5). Thus, when $\Delta_K/r > \|D_F\|$, the displayed embedding is the isolated finite window.

2. **Result 2 — Correct with amendment.** With the endpoint convention made explicit, condition (6) gives the claimed local self-adjoint elliptic realization. Write G_0 at $x = 0$ and G_ℓ at $x = \ell$, or write G_i at $x = i\ell$ for $i \in \{0, 1\}$. The present "at $x = i$" leaves the index convention implicit.

   For $\chi = ZG$, the adapted tangential principal symbol exchanges the two χ-eigenspaces because $[G, \sigma_A] = 0$. The normal symbol exchanges them as well. Bär–Ballmann, Corollary 4.18 and Theorem 4.11 give the relevant ellipticity and self-adjointness criteria. Their Example 4.20 assumes anticommutation with the full adapted operator; (6) supplies the principal-symbol condition needed for Corollary 4.18.

   Equation (7) is correct for the stated maximal smooth scalar $*$**-algebra**. Preservation by both f and $\bar{f}$ makes the boundary subspace reducing, hence gives commutation with χ_i. Commutation with $C_i^2 = \Pi$ then forces multiplication by f_i to preserve $H^0(\mathbb{CP}^3, \mathcal{O}(1))$. Those sections have no common zero, so f_i is holomorphic and therefore constant.

   The local hypothesis is restrictive. On the unaugmented Dolbeault spinor bundle, the six-dimensional complex Clifford representation is irreducible. A fibre endomorphism commuting with every $\sigma_A(y, \xi)$ is scalar; a smooth self-adjoint involution of this form is consequently $+1$ or -1 on connected K. Additional fibres and other local elliptic boundary projectors are directions outside this sufficient criterion.

3. **Result 3 — Correct with amendment.** Equations (9)–(11) follow as stated. Equation (9) proves that $g^2 = 1$ is equivalent to $[G, \Pi] = 0$. Scalar multiplication commutes with G, giving (10); the complex-linear commutant of the generated $M_4(\mathbb{C})$, or of $M_2(\mathbb{H})$ acting on $\mathbb{C}^4$, is $\mathbb{C}1$. Hence the self-adjoint involution g is ± 1.

   The auxiliary example also checks: $Q_K(y_a) = \Gamma_a^{\text{section}}/5$, and the Clifford traces give $\operatorname{tr} g^2 = 16/5$. Applying (9) and taking the largest eigenvalue gives the displayed $2/\sqrt{5}$ lower bound on boundary-trace mixing.

   Amend the marked-point sentence at lines 236–244 to say that the **natural field displayed there** supplies no separately marked point. A local field can encode a chosen point: with the auxiliary bundle, $G_p(y) = p_a \Gamma_a^{\text{aux}}$ is a local constant field whose coefficients contain p. The restriction (11) concerns the unchanged four-section window and an actual involution obtained by compression. A coupled window, extra fibres, or spectral walls remain directions to examine.

4. **Result 4 — Correct with amendment.** Inserting W_ℓ into the scalar multiplier gives (12), including its independence of r. For real S^4 functions, (15) follows from quaternionic Hermiticity and the six independent matrices $1, \Gamma_a$. When the walls are aligned,

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p,$$

   so direct scalar compression gives (13). Generating $M_2(\mathbb{H})$ first and then taking its Γ_p-centralizer gives (14).

   For (16), state explicitly that the two Hom spaces and the two product-factor scalar algebras of Example 16.9 have been supplied. The V-side left action on the second Hom space commutes with its right grading, giving

$$\mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).$$

   The endpoint argument proves equality of the **finite-dimensional linear images**: interior-supported scalar functions separate the same matrix-entry densities. It does not by itself identify the parent

algebras, their norms, gauge groups, or one-forms. The restriction $\mathbb{R} \oplus \mathbb{R}$ applies to real S^4 scalars, aligned walls, and direct compression. Intermediate algebra generation and other windows remain available constructions.

5. **Result 5 — Gap.** Equations (17)–(20) are correct under their displayed hypotheses. In particular, the complementary resolvent in (17) has norm at most r/Δ_K; at fixed D_F, the interval tower also decouples as $\ell \to 0$. The Schur estimates (18) hold with the stated positive g_r. Equation (19) holds for an x-independent ambient multiplier, and (20) follows from the even part of the conjugation series.

   The final assertion about convergence of **the compression maps** needs a common domain and a specified collapse map. For example, take aligned walls, $D_F = 0$, and the admissible scalar functions

   $$f_\ell(y, x) = 6(x/\ell)(1 - x/\ell).$$

   Both endpoint values are zero, but $\Phi_\ell(f_\ell) = 1$. Thus a general family of observables cannot be replaced by its value at the collapsing endpoint.

   One exact correction is to fix $F(y, s)$ on $K \times [0, 1]$, set $f_\ell(y, x) = F(y, x/\ell)$, and state

   $$\Phi_\ell(f_\ell) \longrightarrow \mathcal{E}_0 Q_K \left(\int_0^1 F(\,\cdot\,, s)\, ds \right).$$

   For x-independent ambient multipliers, the limit is $\mathcal{E}_0 Q_K$, as follows from (19). Nonconstant x-independent functions on K are ambient multipliers; they do not themselves obey the spectral-wall endpoint restriction. Result 4's equality of finite images supplies representatives with the required endpoint values.

   The rate discussion after (18) also needs a proof or a strengthened assumption: the two displayed rv_r conditions alone do not control the denominator g_r. A simple sufficient addition is

   $$r(b_r + L) \le (1 - \varepsilon)\Delta_K$$

   for some fixed $\varepsilon > 0$. The stated $b_r = O(1/\ell)$ case can instead be justified from the stronger complementary estimate

   $$|\mathcal{D}_{r,Q}| \ge \left(\frac{\Delta_K^2}{r^2} + \frac{\pi^2}{4\ell^2} \right)^{1/2},$$

   obtained by applying the one-ended Poincaré inequality to the positive and negative B-spectral components. Together with $g_r > 0$ and fixed L, this supplies a complementary gap of order $1/r$.

6. **Result 6 — Correct.** The table in Result 6 separates the chosen data from the algebraic consequences. The proposed $\widehat{J} = E\mathcal{J}_1$ has the stated signs and induces $J_F = \Gamma_0 J_1$ when all the displayed commutation and anticommutation hypotheses hold. Those are sufficient conditions; the note does not construct the required tangential extension or establish its order-zero property.

   The quaternionic j commutes with the Γ_a, so j alone fails the required anticommutation with Γ_p. At the finite level, if the two gradings are Γ_p, Γ_q, a unit $n \perp p, q$ gives $J_1 = \Gamma_n j$ with the required square and anticommutation. Extending that choice to the stated parent remains a separate task.

   The connector calculation gives the displayed algebra after choosing the two complex lines and nonzero coupling. A marked quaternionic line supplies neither of those complex lines by itself.

7. **Result 7 — Correct.** For No-go theorem 22.7, the endpoint-preserving unitary $v = u\, u(\ell)^*$ implements the fluctuation and preserves the original domain. The corresponding real unitary works under the stated order-zero condition. Consequently the moving spectral window is unitarily equivalent to the original one, with the same singular values. The displayed fixed-compression example also has the stated sign and can change singular values at second order.

   The conclusion concerns one interval, fixed wall coupling X, zero bulk mass, scalar inner fluctuations from the even algebra, and the moving spectral window. It is an exact spectral statement without

an action or loop expansion. Variable wall coupling, odd connection directions, suitable bulk masses, further internal connection data, and additional fields lie outside those hypotheses. The result directs the search for a radial variable toward those constructions; it does not decide the full fluctuation space of the $K \times I_\ell$ parent.

8. **Lemma 8 and Theorem 9 — Correct with amendment.** The proof of Lemma 8 checks. The displayed formula for b'' is positive for $x > 0$, since $\sinh x \cosh x > x$. Log-sum-exp is convex and increasing in each argument, so the integrand in (21) is jointly log-concave. Its integral is finite for every u, and Prékopa's Theorem 6, p. 342 applies. Differentiation under the integral gives $0 < e < 1$ and nondecreasing e.

   There is also a separate analytic check of the key monotonicity, using a **covariance argument**. Let

$$\rho(s) = \frac{(\operatorname{arsinh}\sqrt{s})^3}{2\sqrt{s(1+s)}}, \qquad L(v) = \log \rho(e^v).$$

   The substitution $s = \sinh^2 x$ writes $\mathcal{J}(z) = \int_0^\infty \rho(s)/(s+z)\, ds$. For $v = 2\log\sinh x$,

$$L'(v) = \frac{3\tanh x}{2x} - \frac{1+\tanh^2 x}{2},$$

   which is strictly decreasing: $x\operatorname{sech}^2 x - \tanh x < 0$. With $z = e^u$, let the probability density in y be proportional to $e^y \rho(e^{u+y})/(1+e^y)$. Then

$$e(e^u) = \mathbb{E}\left[\frac{1}{1+e^y}\right], \qquad \frac{d}{du} e(e^u) = \operatorname{Cov}\left(\frac{1}{1+e^y}, L'(u+y)\right) > 0.$$

   Both functions in the covariance are strictly decreasing. This check uses the same integral but does not use the log-concavity or Prékopa argument.

   For Theorem 9, the differential inequality for R_c follows from $a = ct(1-t) \geq t$, $e(a) \geq e(t)$, and $e(t) < 1$. Dominated convergence gives

$$H(0+) = 54/5, \qquad H(1/2-) = 0.$$

   Hence H is strictly decreasing and all three conclusions follow. The independent covariance check above verifies the essential monotonicity; the final classification uses the derivative calculation (23).

   Amend the theorem to state $\theta \in [0, \pi/2]$ explicitly. At $\theta = \pi/2$, the determinant has the stated continuous endpoint value, while the isolated principal-logarithm window is degenerate. The no-interior-minimum conclusion assumes the displayed flat, massless, one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no additional angle-dependent term. Modified form factors, fermion sectors, bulk masses, wall terms, and the uncomputed K-tower effective action remain directions for an interior vacuum.

9. **Result 10 — Correct with amendment.** For real finite opposite masses, identity (17) of this result's discussion—namely equation (10)'s displayed G-identity at lines 658–663—makes the fermion contribution strictly decreasing in t. With the separately assumed flat gauge terms and $n_f > 0$, $t = 1$ is the unique global minimum.

   For $a = c_Q + c_U \neq 0$, the displayed expansion of G and of the logarithm is correct. State that the divergent contribution has nonzero fermion multiplicity and is considered without cancellation by additional sectors. A regulator, subtraction and specified physical completion are needed to draw a conclusion from the general-mass potential.

   The opposite-mass Python scan in lemma 23 covers the listed finite sets of masses and multiplicities and 721 sampled angles. It gives a numerical result for those samples; it does not exclude a local minimum between samples or at unsampled parameters. General masses with a specified renormalized completion and changed gauge form factors remain open directions.

10. **Consequence for T0 and T0b, and the note — Correct with amendment.** The consequence keeps the central questions open: the construction supplies an operator and grading window with prescribed spectral walls, while the coupled local scalar window, compatible reality, second sheet and connector require further data. The no-go statements do not close either T0 or T0b.

In the note, replace "higher modes" in the radius-collapse sentence by **higher K-modes**. At fixed ℓ, the interval tower remains; its decoupling uses the subsequent $\ell \to 0$ limit. Also identify the exact window there as the operator-and-grading window, since the required parent reality structure is conditional.

At the one-loop statement, state $0 < n_f < 54/5$ explicitly for the unique interior maximum and retain the determinant assumptions listed in item 8. When this review is recorded in the repository, update the provenance sentence at line 185 to identify the review and Astra 6's one-loop proof.

Amendments and errors

- Make the local-wall endpoint indices explicit and distinguish Bär–Ballmann's principal-symbol criterion from its stronger full-operator example.
- Qualify the marked-point statement to apply to the displayed natural field.
- State the product-factor inputs for (16) and the scope of equality of compression images.
- Define the source identification and averaging map for interval-collapse convergence; add a denominator estimate to the mixing-rate proof.
- State the angle interval, endpoint degeneracy, and complete determinant hypotheses for Theorem 9 and the paper note.
- Scope the bulk-mass scan to its sampled parameters and state the nonzero-multiplicity assumption for the ultraviolet divergence.
- Update the note's radius-collapse and provenance wording when the review is recorded.

No error was established in the displayed equations. Item 5 has the stated gap in the general compression-map conclusion.

Overall: 3 Correct; 6 Correct with amendment; 1 Gap; 0 Error.

Part R12 (referee report, lemma 26)

Referee report on lemma 26

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_26.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 26.

The potential classification and most of the arithmetic reproduce. The principal corrections concern the physical interpretation: the running calculation omits the frequency-two multiplets used to obtain the vacuum, and the finite-angle mass ratio does not determine the physical Higgs–top coupling.

This is one independent review of lemma 26. It supplies a check by a model other than the author; broader consensus on the negative statements remains open.

For the numerical checks below, we used Python 3 with SciPy 1.16.3: quadrature for J, root finding for the extrema and meeting point, and numerical integration of the one-loop Yukawa equation. We checked the Yukawa integration separately by quadrature after transforming to $w = 1/y^2$. Both routes use the same stated one-loop equations and inputs.

1. **Lemma 26.1 and Theorem 26.2 — Correct with amendment.**

   Equation (1) is the correct extension of the determinant in lemma 23 for an independently specified pure frequency-two Dirac channel: its contribution is $-n_2 F(\sin^2 2\theta) = -n_2 F(4u)$. The conclusion assumes precisely that tower and its stated multiplicity; a full representation can contain additional towers.

   The strict monotonicity proof is valid. In the referee amendment to lemma 24, the covariance is strictly positive: both functions in it are strictly decreasing and the measure has nontrivial support. Thus

$e(z)$ is strictly increasing, making each $J(cu)/J(4u)$, $c < 4$, strictly increasing. The argument remains valid at $u = 1/4$, where the arguments of J are positive and J is smooth. Python quadrature gives

$$n_c = 2.8855033120557163.$$

All cases stated in Theorem 26.2 follow. At $n_2 = n_c,\ n_1 = 0$, the minimum at $\theta = \pi/4$ is degenerate: its leading displacement is quartic. At $n_2 = 27/10,\ n_1 = 0$, the endpoint minima are also degenerate, with leading behavior

$$\mathcal{V}(\theta) - \mathcal{V}(0) = \frac{99}{25}\theta^4 \log(1/\theta^2) + O(\theta^4).$$

In item 2, both factors of $|G| = 4J(4u)(n_2 - R(u))$ decrease as u increases. Hence Ψ increases with t as claimed. At $n_1 = n_*$, the endpoint is a degenerate minimum and the potential still decreases strictly on the open interval. The endpoints are stationary in the θ variable because $dt/d\theta = 0$ there; this does not add interior stationary points.

Amendment: state $n_1, n_2 \geq 0$, distinguish endpoint and interior stationarity, and say that the theorem leaves the general case $n_1 > 0,\ n_2 < n_c$ unclassified. Physical multiplicities have further integrality and representation constraints.

2. **Corollary 26.3 and the numerical table — Correct with amendment.**

The spectrum and monotonicity claims in the corollary follow from Theorem 23.6. In the first item, specify that $\theta = \pi/4$ is the minimum for $n_1 = 0,\ n_2 \geq n_c$. The fixed positive M_W interpretation of M_{KK} excludes the endpoints, where $M_W = 0$.

Independent Python values, using $M_W = 80.36$ GeV, include:

(n_1, n_2)	θ_*	M_{KK}	m_1/M_W	m_2/M_W
$(6, 8)$	0.941089	341.90 GeV	1.27450	1.70561
$(14, 8)$	1.137897	443.92 GeV	2.00086	1.52240
$(16, 6)$	1.445931	1.437 TeV	8.23148	1.42168
$(48, 12)$	1.501951	2.597 TeV	15.45087	1.41646
$(604, 116)$	1.564699	29.278 TeV	181.459	1.41423

Solving $m_1/M_W = 2$ gives $\theta = 1.13774393$ and

$$n_1 = 2.7138653\, n_2 - 7.7172464,$$

in agreement with the rounded line. The $(14, 8)$ row is close to that line. At $(604, 116)$, m_1 is approximately $M_{\mathrm{KK}}/2$; equality holds only in the endpoint limit.

There is an endpoint exception to the broader statement that every minimum found at integer counts has $\theta \geq \pi/4$: for $n_1 = 0$ and $n_2 = 0, 1, 2$, **both** $\theta = 0$ and $\theta = \pi/2$ are global minima.

3. **Proposition 26.3b — Gap.**

The representation and charge calculations are correct under the stated embedding. With multiplicities for eigenvalues of F,

representation	frequencies
4	± 1, each twice
5	0 three times, ± 2 once each
10	0 four times, ± 2 three times each.

For $\mathbf{5} = (2,2)\oplus(1,1)$, a singlet U_R of hypercharge $2/3$ needs $B-L = 4/3$; the smallest $SU(4)$ irreducible representation containing the required color triplet is **15**. The ordinary doublet has $R_3 = -1/2$, and the exotic doublet has $Y = 7/6$.

For $\mathbf{10} = (3,1) \oplus (1,3) \oplus (2,2)$, $U_R = R^+$ needs $B - L = -2/3$, supplied by a color triplet in the smallest choice **6**. Here the ordinary doublet has $R_3 = +1/2$. These are minimality claims for the specified **5** or **10** embeddings and irreducible $SU(4)$ representations. Also,

$$\Lambda^2 E = (\mathbf{10}, \mathbf{6}) \oplus ((\mathbf{1} \oplus \mathbf{5}), \overline{\mathbf{10}}),$$

so the stated $(\mathbf{10}, \mathbf{6})$ inclusion is correct.

Let $\phi = 2\theta$, $a = m\ell$, and use the rotated fields $A' = \cos\phi\, A + \sin\phi\, O$, $O' = -\sin\phi\, A + \cos\phi\, O$. An explicit self-adjoint fermion problem is:

- At $x = 0$, let ψ_R have Neumann conditions on S, A and Dirichlet on O; impose the complementary conditions on ψ_L.
- At $x = \ell$, impose Dirichlet on $\psi_R(O')$ and $\psi_R(S+A')$, and Neumann on $\psi_R(S-A')$; again impose complementary conditions on ψ_L.

Its determinant is proportional to

$$\cos a\,(2\sin^2 a - \sin^2\phi).$$

Thus the angle-dependent tower has the stated equation, and there is also an angle-independent half-integer tower. With both S and A' Neumann at the second wall, S_R is an exact massless mode and the other plane has the pure frequency-two tower. The same construction applies to the appropriate **5** plane with the chirality roles exchanged. These statements require the complementary chiral conditions and, for a gauge-invariant completion, conditions on the complete wall multiplets.

The $n_d \geq 27/5$ argument is correct for the displayed determinant with $n_1 = 0$. Python gives 0.858512 and 1.001144 for the two quoted minima, and found no interior stationary point in the two other quoted cases.

Gap: no finite wall-mass boundary problem has been specified or solved. Replace the claim that the top *lies between* M_W and $2M_W$, and that the exotic reaches the tower scale along the interpolation, by an open calculation. The full multiplet's other modes and their contribution to the potential also remain to be included.

4. **Tower table and Propositions 26.4–26.5 — Error.**

At $\theta = 0$, the listed gauge boundary conditions and coefficients are correct. The table should also show the neutral hypercharge combination as N, N and its orthogonal neutral combination as N, D; both have zero contribution to the displayed unbroken-group traces. The coefficient $-\frac{7}{2}\operatorname{tr} Q^2$ includes the massive vector, its fifth component and ghosts. The A_5 Higgs zero mode is counted in the Standard Model coefficient; its nonzero modes are the gauge partners already counted in the vector towers.

For three complete, massless bulk E generations, $F = 8$ per integer level is correct in the $\theta \to 0$ limit. At nonzero angle, frequency-one masses are shifted to $(n\pi \pm \theta)/\ell$; at finite angle their thresholds and couplings cannot generally be replaced by an exactly degenerate integer tower. Consequently Proposition 26.4 is an algebraic property of equation (3), conditional on its common FI_n term.

Equations (3)–(5), the stated asymptotics, and Proposition 26.5 follow algebraically from the displayed step-threshold model. The cancellation of the power term in (4) concerns the **left–right difference within** $Sp(2)$. The full bulk group is $Sp(2) \times SU(4)$, so differences involving color can retain a power contribution.

There are two material errors in applying these equations:

- Equation (4) itself gives, at $S = 1$, $I_h = \log 2$. With the stated inputs and $M_{\rm KK} = 321.44$ GeV it gives
$$\alpha_R^{-1} - \alpha_2^{-1} = 61.69,$$
rather than 60.2. The latter is the Standard Model extrapolation before adding the half-integer tower.
- The frequency-two multiplet required for the proposed vacuum is omitted from (3). For example, a complete five-dimensional Dirac $(\mathbf{10}, \mathbf{6})$ has leading level contributions
$$\Delta b_i = \left(\frac{296}{15}, 24, \frac{40}{3}\right),$$
before its boundary-dependent threshold splitting and zero-mode subtractions. These are not a universal F.

Replace "the tower of the parent" and the claimed physical independence from bulk localization by statements conditional on the exact spectrum in (3). Recompute the full representation, including its zero modes and split thresholds, before applying the result to the proposed parent. Boundary kinetic counterterms also need a renormalization prescription; they are generally generated by orbifold loops, as shown by Carena, Tait and Wagner, sections 1–2.

5. **Numerical Results 26.6 and 26.7 — Correct with amendment.**

For equation (3) as written, Python gives

$$S = 7.0388131, \qquad \mu_* = 2.21467 \times 10^{14}\ \text{GeV}, \qquad M_{\rm KK} = 3.14636 \times 10^{13}\ \text{GeV}.$$

The common inverse coupling is 52.5541 for $F = 0$ and 46.0161 for $F = 8$. The one-solution statement can be strengthened to a proof for $S \geq 1$. Let N_n and N_h count the integer and half-integer levels below S. Eliminating $\log(\mu/M_Z)$ from the left–right and 2-versus-3 equations leaves a function whose derivative between thresholds is

$$\frac{315N_h - 161N_n}{44S} > 0.$$

It is continuous at thresholds, so it has at most one root; the root above exists in the stated scan.

The Result 26.7 table reproduces to its displayed precision except that, with the stated nominal inputs, the last $r = \sqrt{2}$ entry is **196.46 GeV**, which rounds to 196 GeV. For example, the first row is $63.39, 87.01, 119.45, 136.25, 164.12$ GeV. The crossing values depend slightly on whether one starts from $y_t = 0.936$ or exactly $m_t = 163$ GeV; the quoted values are consistent with the former rounding.

Amendments: describe 2.9–3.4×10^{13} GeV as the effect of the indicated input shifts **one at a time**. Taking all three extreme shifts together gives approximately 2.66–3.70×10^{13} GeV. Both numerical results are conditional on the assumed field content and thresholds.

The missing exotic is already a larger effect: one additional light color-triplet Weyl fermion changes α_3^{-1} by about -1.4 between M_Z and 3.1×10^{13} GeV. As a separate illustration, integrating only the Standard Model gauge part of the two-loop terms along the one-loop trajectory gives shifts approximately $(-0.20, -0.27, +0.39)$ in the three inverse couplings at that scale. This omits the top-Yukawa and KK two-loop terms and is not a full two-loop prediction. The coefficients used are in Mambrini et al., equations (42)–(44). Finite wall terms, split KK thresholds, electroweak and top matching, and the pole-to-running top conversion can also exceed the quoted input shifts.

6. **Gauge-Higgs Yukawa matching and finite angles — Error.**

For the specified flat pure channel, canonical normalization and the small-angle mass formulas give the **tree-level** relation $y_q = (q/2)g$. The same higher-dimensional covariant derivative fixes the corresponding bulk gauge and A_5 vertices. Physical four-dimensional gauge and Yukawa couplings nevertheless

receive different finite corrections and threshold effects. In particular, Lim and Maru, equations (1)–(3) calculate a finite one-loop violation of gauge–Yukawa universality. The claim of no separate running must therefore be restricted to the local bulk coupling and a specified matching scheme.

The first row's finite-angle identification is incorrect. Let v_{ang} be the canonically normalized flat Wilson-line background, normalized at small angle. Then

$$v_{\mathrm{ang}} = \frac{2\sqrt{2}\,\theta}{g\ell}, \qquad y_{\mathrm{physical}} = \sqrt{2}\,\frac{\partial m}{\partial v_{\mathrm{ang}}}.$$

At $\theta = \pi/4$, $m_2 = 2M_W$ gives $\sqrt{2}\,m_2/v_F = \sqrt{2}g$ if v_F is defined by $M_W = gv_F/2$. It does **not** give the Higgs coupling. The pure frequency-two branches have slopes corresponding to $|y| = g$ at tree level. They are degenerate at this angle, so the coupling is a matrix on the degenerate subspace. For the tower with $m = M_W$, the mass derivative vanishes at $\theta = \pi/4$.

There is a second obstruction to interpreting 164 GeV as a prediction: the pure channel has the exact massless colored exotic and a second fermion at $M_{\mathrm{KK}}/2$. Other half-integer thresholds also occur below M_{KK}. Standard Model-only running below 321 GeV is therefore an illustrative boundary-value calculation, not the running of that spectrum. The independent-bulk-Higgs calculation in steer item 11 has a different action and remains conditional on its own content. The finite-angle distinction between a Higgs background and its couplings is also explicit in Hosotani et al., equations (6.1)–(6.2).

7. **Quadratic coefficient and cancellation — Correct with amendment.**

   The coefficient

$$\frac{3}{2}\zeta(3)\left(\frac{54}{5} - n_1 - 4n_2\right)$$

   is correct, and it cannot vanish for integer n_1, n_2 in equation (1). The statement about added bosons needs a multiplicity convention: a real scalar determinant carries a $1/2$ coefficient and a real vector carries $3/2$; mixed boundary projections can also yield non-integer prefactors. Integer nq^2 contributions hold only for an explicitly specified normalization and class of pure channels.

   "Cancellation of the quadratic coefficient" is appropriate. It does not mean cancellation of the one-loop potential. In particular,

$$F(z) = J(0)z + \tfrac{1}{4}z^2 \log z + O(z^2),$$

   so the remaining $\theta^4 \log \theta^2$ term must be retained when discussing a tuned small angle.

8. **"Two regimes" — Error.**

   The numerical estimates describe the simplified equations, but several sentences extend beyond what those equations establish.

   - At the low tower, the quoted hypercharge discrepancy of about 60 is the Standard Model-only extrapolation. In the formal step model (3), it is 61.69 at $S = 1$. At $\theta = \pi/4$, the actual mixed profiles and thresholds must be used. A wall term of order 60 in α_Y^{-1} is therefore an estimate, with normalization and matching conditions specified.
   - Separate bulk couplings are required if the only admitted non-universal correction is the stated hypercharge wall term. General non-universal wall kinetic terms provide another way to fit the couplings, so separate bulk couplings are not a general requirement.
   - For the formal high-scale meeting point, the small-angle relation gives $\theta = 5.67 \times 10^{-12}$. For integer counts, equation (1) has no **global** minimum at a small positive angle: for $n_1 > 0$,

$$V(1-t) - V(t) = -n_1[F(1-t) - F(t)] < 0 \quad (0 < t < 1/2),$$

while the $n_1 = 0$ integer cases follow from Theorem 26.2. This proves the global statement. It does not prove the absence of every metastable small-angle minimum in the cases left to scans.

- The statement that warping leaves the left–right difference unchanged is too broad. The flat threshold sums do not determine warped spectra, boundary counterterms or their matching. Warping, bulk masses, antiperiodic conditions, extra representations, boundary interactions and wall kinetic terms remain model-dependent escapes.

The unqualified sentence that frequency-two fermions give a minimum at $\theta \geq \pi/4$ also misses the $\theta = 0$ endpoint minima noted in item 2. The numerical top predictions require the corrections in items 3–6.

There is no direct contradiction with lemma 25: its $y_t/g = \sqrt{2}$ endpoint is a leading small-mass **spectral-action trace** result with its stated matter normalization; lemma 26 assumes a Yang–Mills gauge action and particular frequency channels. The steer note's universal bulk-Higgs running and the main paper's towerless, classical statements likewise have different hypotheses. They should not be combined into one matching calculation without constructing the common action and spectrum.

9. **Literature — Gap.**

The following sources should be added with their model differences stated:

- Nomura, Smith and Weiner, *GUT Breaking on the Brane*, especially sections 2 and 4: differential logarithmic running and integer versus half-integer towers in a five-dimensional unified model. Carena, Tait and Wagner show the importance of boundary kinetic terms.
- Abe, Goto and Kawamura, section 3.2 and equation (31): an interval Pati–Salam model with a compactification threshold of order 10^{13} GeV, specifically $3.675 \times 10^{13}(1.026)^{\xi}$ GeV in their convention. Their intermediate gauge group and matter content differ from lemma 26.
- Cacciapaglia, Csáki and Park, section 5: flat gauge-Higgs models with representation-dependent enhancement, including a tree-level $m_t = 2M_W$ limit. Adachi and Maru, equation (3.23) obtain a limit approaching $2M_W$ with brane mixing and discuss massless exotics and their removal.
- Contino, Da Rold and Pomarol, sections 1–2 study **5** and **10** $SO(5)$ embeddings and their colored custodians. Hosotani, Oda, Ohnuma and Sakamura, equations (4.23), (4.31) and section 5.3, explicitly exhibit massless exotic zero modes, their lifting by wall masses, and the associated change in the spectral equation.

The angle is prior work in the corresponding Wilson-line normalization: Hosotani et al. find $\theta_H = \pi/2$, corresponding to $2\theta = \pi/2$, in their **warped** model. Their flat limit and matter content differ from equation (1).

The inspected records do not establish prior occurrence of the exact n_c classification or the pair (M_{KK}, μ_*) from equation (3). The search covered orbifold differential running, Pati–Salam interval scales, flat large-representation tops, $SO(5)$ vector and adjoint embeddings, exotics, and Wilson-line minima, including a 2026 keyword sweep. Forward citation chains and author-by-author recent-publication sweeps remain incomplete. Priority for those exact claims is therefore open.

Amendments and errors

- State the endpoint behavior, the unclassified parameter range, and the missing $\theta = 0$ integer-count minima.
- Specify both chiralities and complete wall multiplets in Proposition 26.3b; leave the finite wall-mass interpolation open until its spectrum is solved.
- Correct the $S = 1$ difference to 61.69 within equation (3), and include the frequency-two multiplet and its thresholds before calling the result the parent's running.
- Correct the final $r = \sqrt{2}$ table entry to 196 GeV for the nominal inputs; qualify input-shift ranges and all numerical meeting-point claims by the spectrum and matching prescription.
- Replace the finite-angle mass-ratio identification of the Yukawa coupling by a calculation of the Higgs coupling matrix and threshold matching.
- Restrict the bosonic integer-multiplicity remark, retain the non-analytic terms, and narrow the warping and no-small-vacuum statements to their proved scopes.

- Add the cited prior work and leave exact priority open pending the remaining literature search.

Overall: 0 Correct; 4 Correct with amendment; 2 Gap; 3 Error.

Part R13 (referee report, lemma 27 and additions to lemma 26)

Referee report on lemma 27 and the additions to lemma 26

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_27.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 27 and lemma 26.

The trace identity and the two new boundary calculations reproduce under the stated model assumptions. The material errors are in the interpretation of the heavy-flavour weights, the fermion-frame sign, and the summary's claim that a $2M_W$ top must lie in E.

The numerical checks below used Python 3. SymPy was used for the fermion boundary determinant; NumPy, SciPy and mpmath were used for the neutral-vector roots. The neutral calculation also reproduced the known zero-wall-term formula. A deliberately altered boundary condition failed that control. These computations support the displayed derivations; they do not replace them.

Part A

1. Theorem 27.1 and the index formula — Correct with amendment.
In Theorem 27.1, the nonzero eigenspaces of D_f^2 pair between the two chiralities. Trace class makes the sum of their contributions legitimate, so the stated trace identity holds at every cutoff.

The index and bundle conventions need to be explicit. Let $D_f^+ : S^+ \otimes E_f \to S^- \otimes E_f$, and choose $\gamma_5 = +1$ on S^+. Then $\mathrm{ind}_f = n_{f,+} - n_{f,-}$. With the corresponding orientation and Hermitian-curvature convention, the displayed formula is

$$\mathrm{ind}_f = \frac{1}{8\pi^2}\int_M \mathrm{tr}_{E_f}(F_f \wedge F_f) - \frac{N_c}{24}\langle p_1(TM), [M]\rangle.$$

Reversing the chirality convention reverses the index. The curvature decomposition follows because G is traceless:

$$\mathrm{tr}_{E_f}(F_f^2) = N_c Q_f^2 e^2 F_{\mathrm{em}}^2 + g^2\,\mathrm{tr}_c(G^2).$$

Here the products are wedge products. For topologically nontrivial backgrounds, replace the literal trivial bundle $M \times \mathbb{C}^{N_c N_f}$ by the appropriate colour and electromagnetic bundles, with charges normalized so that the flavour representations are globally defined.

2. Corollary 27.2 — Correct with amendment.
The factor 2 in the regulated Jacobian comes from transforming both independent Grassmann variables. With $T^3 = \frac{1}{2}\,\mathrm{diag}(1,-1)$, $N_c = 3$, and $Q = \mathrm{diag}(2/3, -1/3)$,

$$2N_c\,\mathrm{tr}(T^3Q^2) = 1, \qquad F_{\mathrm{em}} \wedge F_{\mathrm{em}} = \tfrac{1}{2}F_{\mu\nu}\widetilde{F}^{\mu\nu}\,d\mathrm{vol}.$$

Thus the electromagnetic coefficient in Corollary 27.2 is correct. The gluonic and gravitational terms cancel for $\mathrm{tr}\,T = 0$, provided the flavours have the same colour bundle. For $T = 1$, they remain; the integrated gravitational contribution to $2\,\mathrm{Tr}$ is

$$-f(0)\frac{N_c N_f}{12}\langle p_1(TM), [M]\rangle.$$

The limit requires continuity of f at zero. The standard anomaly normalization additionally sets $f(0) = 1$; trace class alone supplies neither condition.

For the local statement, assume a smooth cutoff class for which the heat expansion can be used, and define E, Ω by the Laplace-type square of the Dirac operator. The displayed a_4 terms then give the stated density. The omitted lower-coefficient and other a_4 terms have vanishing γ_5 spin trace in this vector-coupled Dirac problem.

3. Proposition 27.3 — Correct with amendment.
The identity in Proposition 27.3 is correct for $m_f > 0$. Conjugation by γ_5 makes the D_M term odd and hence gives it zero trace; the remaining nonzero eigenspaces cancel by chirality pairing. The surviving zero modes contribute $n_{f,+} - n_{f,-}$. The scalar density in Proposition 17.8 instead involves $n_{f,+} + n_{f,-}$.

This has a Ward-identity interpretation if the Jacobian and the composite mass insertion are given the **same specified regularization**. The trace identity alone does not construct an independently regulated functional measure or a local current. In the present mass convention,

$$\{\gamma_5 T, D_M + \gamma_5 d\} = 2dT,$$

so the mass insertion is $\bar{\psi} dT \psi$; after changing to a conventional scalar-mass basis, the corresponding insertion is pseudoscalar. The local identity also has a current-divergence term.

4. Proposition 27.4 — Error.
The displayed coefficient is correct when $f(0) \neq 0$. For the heat cutoff it is

$$\frac{4e^{-x} - 1}{3},$$

with limits 1 and $-1/3$. The gluonic and gravitational regulated traces are proportional to $f(x) - f(0)$.

The interpretation following Proposition 27.4 goes beyond that calculation. An explicitly massive flavour breaks the axial symmetry, and the trace identity alone does not establish a Wess–Zumino term in its low-energy theory. Such a term can be required after specifying the retained symmetry, its background fields or spurions, and the anomaly-matching prescription.

The weights in this calculation depend on the chosen regulator. Making f part of a finite-scale spectral action does not by itself identify the fermion-measure regulator with that action or make these trace weights observables. The fermion measure and matching calculation must also be specified.

5. Proposition 27.5 — Error.
The chiral pure-gauge form in equation (1) is valid after the paired sector is expressed in independent quark and antiquark variables. More precisely, the operator couples $B = \Sigma\, d\Sigma^\dagger$ to one chirality and is generally non-self-adjoint. On the full $H_0 = (q_L, \bar{q}_R)$, the two blocks of $(U_\Sigma dU_\Sigma)|_{H_0}$ are

$$\Sigma\, d\Sigma^\dagger, \qquad \Sigma^T d\overline{\Sigma}.$$

A nontrivial $\Sigma : S^4 \to SU(2)$ cannot be joined to 1 by a continuous path of such maps. To define the relative Weyl Pfaffian sign, use a path of **connections**, such as $A_t = tB$, with the endpoint fibres compared by the gauge transformation. For one Weyl doublet, Witten's equation (6), p. 325, gives "the opposite sign of the square root"; N_c doublets give $(-1)^{N_c}$.

That is an anomalous relative sign for the chosen Pfaffian transport and endpoint identification. The pointwise R_Σ also changes fermion coordinates. Pulling the original measure through that change preserves the value of the same integral; the measure Jacobian must accompany the transformed action. Accordingly, "the quark integral in that frame changes by $(-1)^{N_c}$" is too strong without specifying the comparison of measures. Likewise, the fixed classical form is unchanged as a geometric form, while its expression in fixed reduced variables acquires the connection in (1).

The quoted two-flavour sigma-model coupling is correctly attributed: Lee, Ohmori and Tachikawa, arXiv:2009.00033v2, §2.5 and equation (2.29) define it as $(-1)^{N_c[\sigma]}$ on spin four-manifolds. They explicitly note that "there is no appropriate five-form" for this case. Matching that term to the quark anomaly requires the sigma-model and its anomaly-matching assumptions; Proposition 17.10 alone constructs neither a pion action nor its measure.

6. Remark 27.6 — Correct with amendment.
For the flow from $A = 0$ to the gauge-equivalent $A = B$, Witten relates the Pfaffian sign to the mod-two index on $S^4 \times \mathbb{R}$. The two ends can be compactified, with the appropriate gauge trivializations, to the nontrivial $SU(2)$ bundle on S^5. Witten's footnote 2 describes an invariant zero mode for a special connection; the general protected statement is the **parity** of the kernel.

The nontrivial bundle is, after choices of identification, the bundle $SU(3) \to S^5$ used in research-state §6.12. Its twisted Dirac operator has the same mod-two index. For isospin j, the parity is that of $2j(j+1)(2j+1)/3$, odd for $j = 2k + \frac{1}{2}$.

The outcome paragraph should distinguish the bordism groups: the sigma field represents a class in $\widetilde{\Omega}_4^{\mathrm{Spin}}(SU(2))$, whereas the five-dimensional anomaly is represented in $\Omega_5^{\mathrm{Spin}}(BSU(2))$. Their signs agree in this construction through the anomaly comparison; they are not literally one element of one group. The abstract bundle identification also does not identify the internal gauge group or dynamics with the flavour construction.

Part B

7. Proposition 26.3c — Correct with amendment.
The boundary equation in Proposition 26.3c follows from a self-adjoint bulk-plus-wall problem. For example, normalize $u = (S + A')/\sqrt{2}$, $v = (S - A')/\sqrt{2}$, and take

$$H(f, g, \chi) = (-g',\, f',\, w g_u(\ell)), \qquad f_u(\ell) = w\chi,$$

along with $g_{O'}(\ell) = 0$, $f_v(\ell) = 0$, and the stated conditions at zero. The boundary form cancels. Eliminating χ for $m \neq 0$ gives $f_u = (w^2/m) g_u$.

Let $a = m\ell$, $s = \sin 2\theta$, and $\Delta = \sin^2 a - s^2$. Direct elimination gives

$$2a\Delta \sin a = \omega \cos a\, (\Delta + \sin^2 a).$$

The two nonzero-mode limits reproduce $\Delta \sin a = 0$ and the first referee's $\cos a(2\sin^2 a - s^2) = 0$. At $w = 0$, the decoupled wall Weyl zero mode must also be counted.

At $\theta = \pi/4$, the equation factorizes as stated. The written proof divides by $\cos 2\theta$, so it misses this angle; use the determinant before division. For every $0 < \theta < \pi/2$, put $u = M_W \ell = \arcsin(s/\sqrt{2})$. On $0 < a < u$,

$$\omega = 2a \tan a\, \frac{s^2 - \sin^2 a}{s^2 - 2\sin^2 a}$$

is continuous and strictly increasing from 0 to $+\infty$. This proves the stated light root, including at $\theta = \pi/4$.

For the **displayed** $(10, 6)$ charge-2/3 channel, a state lies below M_W for every finite positive wall mass; in the infinite-mass limit the lowest root of this isolated channel is M_W. The same construction works for the specified $(5, 15)$ plane with chiralities exchanged and the corresponding wall Weyl field. A complete gauge-invariant wall multiplet, other boundary mixings, bulk masses and the other towers require separate analysis. Their presence can add still lighter states, so equality for the lightest state of a completed model is unproved.

8. Result 26.8 — Correct with amendment.
The numbers in Result 26.8 reproduce for the stated tree-level model. At the connector wall, R in the hypercharge combination must mean the **rotated wall direction**

$$L_\ell = (S + A')/\sqrt{2}, \qquad R_\ell = (S - A')/\sqrt{2}.$$

With $p = \cos\phi$, $q = \sin\phi$, the wall conditions are

$$O' = 0, \quad -qR_\ell + pB = 0, \quad \partial_x L_\ell = 0, \quad \partial_x(pR_\ell + qB) = rm^2(pR_\ell + qB).$$

At $x = 0$, S, A, B are Neumann and O is Dirichlet. Eliminating these four amplitudes gives, for the lowest neutral root $a = M_Z\ell$, with $t = r/\ell$,

$$(1 + t\,a\cot a)\sin^2 a = \frac{\sin^2(2\theta)}{2}(1 + p^2 + t\,a\cot a).$$

The definitions of p^2 and r/ℓ in the result follow from the stated zero-angle coupling prescription. Using its inputs, the two recomputed rows are:

θ	$M_Z\ell$	r/ℓ	M_W^2/M_Z^2	4D reference
$\pi/4$	0.990308	1.6336	0.62898	0.76230
1.13774393	0.675757	1.5957	0.70868	0.76067

The quoted shifts, approximately -0.13 and -0.05, therefore reproduce. The four-dimensional number is a meaningful **reference within this matching prescription**. A physical precision comparison needs the model's coupling definitions, wavefunction overlaps, threshold matching and radiative corrections: those can affect the comparison at the same order as a small finite-interval shift. The "about 1.2 TeV" threshold should remain a numerical statement for this prescription, with its parameter range stated.

9. Revised outcome and two regimes — Error.
The claim in the outcome that a top at $2M_W$ must remain in E does not follow from Proposition 26.3c. The proposition proves a lighter charge-2/3 partner in its displayed channel. It does not exclude identifying a higher eigenstate with the top.

Indeed, let $u = M_W\ell < \pi/4$. Equation (3) has an eigenvalue $m = 2M_W$ when

$$\omega = \frac{4u\sin 2u}{1 + 2\cos 2u}.$$

For example, Python gives $\theta = 1$, $\omega = 2.04350$, a state at $2M_W$, and another at $0.84972M_W$. The higher state has nonzero O_L and R_R components. Its physical identification and the lighter state's couplings still need analysis. Replace the summary by the proved lighter-partner statement.

The low-tower potential, the wall-mass spectrum, the wall kinetic term and the step-threshold running also use different simplified spectra. A combined model needs the altered complete-multiplet determinants and thresholds before its vacuum and matching scale can be asserted together. The qualifications in "The top against the matching scale" correctly treat the Yukawa table as an illustration.

The first referee's requirements do **not** appear as a consolidated list in lemma 26. The first report's amendment list is linked, and several corrections are scattered through the lemma. The opening claim that all are incorporated should distinguish textual corrections from work still needed, including complete wall multiplets, their potential and thresholds, and the finite-angle coupling matrix.

Amendments and errors

- Specify the bundles, chirality convention and cutoff regularity for Theorem 27.1 and Corollary 27.2.
- State the common-regularization assumption behind the integrated Ward identity.
- Correct the Wess–Zumino and regulator interpretation in Proposition 27.4.
- Replace the frame-independent sign claim in Proposition 27.5 by the Pfaffian transport statement and include the measure Jacobian under R_Σ.
- Distinguish the four- and five-dimensional bordism groups and the special S^5 connection.
- Supply the self-adjoint wall-field domain, count the decoupled zero mode, and remove the division by $\cos 2\theta$.
- State the neutral wall boundary conditions and the matching convention behind the four-dimensional reference.
- Replace the claim that a $2M_W$ top must lie in E by the proved lighter-partner result, and separate completed referee corrections from outstanding calculations.

Overall: 0 Correct; 6 Correct with amendment; 0 Gap; 3 Error.

Part R14 (Astra validation of the ledger, citations and steer note)

Validation of the corrections to the ledger, the citations and the steer note

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompts in ../lemmas/prompts/astra_validate_corrections.md and ../lemmas/prompts/astra_validate_corrections_resume.md. The job was cut off once by a usage limit and resumed in the same session. Reproduced as delivered, with links made relative. The amendments are applied in `ledger.pl`, `LITERATURE.md`, the steer note, the main paper, the Farsi-Latrémolière assessment and `README.md`.

We checked the current working tree, including the three subsequent fixes. The replacement of α by φ, the updated Farsi–Latrémolière publication statement, and the repaired relative-bound estimate are correct. They are not outstanding defects.

We worked alone, read the supplied sources and referee reports, and made no repository changes. The numerical verification used **Python 3, NumPy and SciPy**; the ledger tests used **SWI-Prolog**. The computations supplement the proofs below.

1. **Correct — Witten 1981.**

   The quotations and page references in LITERATURE.md agree with the supplied source. The chirality argument is on p. 422, within the cited discussion on pp. 421–423. The discussion of nonorientable internal spaces is on p. 423; torsion remains a possibility to investigate on pp. 423 and 426–427. The discussion of metric parameters determining gauge couplings on pp. 425–426 supplies no calculation of a weak angle.

   The hypotheses matter: the argument concerns the usual smooth compactification with an orientable internal space of odd dimension, the geometric spin connection, and gauge transformations induced by coordinate transformations. Witten applies it to the fermion modes of eleven-dimensional supergravity, including the M^{pqr} candidates. The text correctly presents nonorientability and torsion as possibilities considered, without claiming that Witten constructs a successful model using either.

   The opening Clifford-algebra sentence in the paper is correctly separated from the attribution. On an irreducible complex odd-dimensional Clifford module, the volume element is a nonzero scalar. An operator anticommuting with every Clifford generator would anticommute with that scalar, so no nonzero chirality grading exists on that module.

   Read with the immediately following specification of Witten's model, the negative statement has the required scope. Boundaries, defects, changed connections and gauge fields outside the massless isometry construction remain outside it. **No correction is required.**

2. **Correct with amendment — Witten 1982.**

Footnote 2 is indeed at the foot of **p. 327**, and its quotation is accurate after normalizing typography. Section 6.12 of the research state agrees with it: the special invariant construction gives the unique $SU(3)$**-invariant spinor**, rather than establishing that every possible zero mode is invariant or that the entire kernel always has dimension one.

The criterion involving $\operatorname{tr} T^2$ is on **p. 328**, with the doublet normalized to $1/2$. For one left-handed representation, the anomalous case has half-odd-integral index. Let

$$T(j) = \sum_{m=-j}^{j} m^2 = \frac{j(j+1)(2j+1)}{3}.$$

Then

$$T\left(2k + \frac{1}{2}\right) = \frac{(4k+1)(2k+1)(4k+3)}{6}$$

is half an odd integer: its odd numerator is divisible by three. Also

$$T\left(2k + \frac{3}{2}\right) = \frac{(4k+3)(k+1)(4k+5)}{3}$$

is an integer. Both closed forms are correct.

Correction: the original cylinder and the later mapping-torus formulation must be distinguished. Replace the parenthesis identifying the original construction with the mapping torus, and the later sentence "Witten's criterion is phrased on the mapping torus $S^4 \times S^1$," by: $>$ Witten's original construction uses $S^4 \times \mathbb{R}$, with the specified asymptotic gauge fields; footnote 2 discusses its conformal compactification to S^5. The mapping-torus formulation uses $S^4 \times S^1$, with the gauge bundle glued by the gauge transformation.

3. **Correct — Banks–Casher.**

   Section 2 of Giusti–Lüscher counts the eigenvalues of **one flavour**, including both signs. Its convention gives

   $$\Sigma = \pi \lim_{\lambda \to 0} \lim_{m \to 0} \lim_{V \to \infty} \rho(\lambda, m).$$

   Thus infinite volume is taken first, followed by the chiral limit, followed by the limit in spectral argument.

   The paragraph in chiral_limit_note.tex is consistent with that normalization. Its sum defining c_V includes both signs, and the corresponding density does too. Consequently

   $$\int_{\mathbb{R}} \frac{m\, \rho_m(\lambda)}{\lambda^2 + m^2}\, d\lambda \longrightarrow \pi \rho_0(0).$$

   There is no additional factor two. If only positive eigenvalues were counted, that convention would require the compensating factor two.

   The stated equicontinuity, convergence at zero, and uniform control of tails and subtraction justify the approximate-identity argument. The fixed-volume statement also now requires the compatible $O(m)$ subtraction that the referee requested.

   The citation is honest: it identifies Giusti–Lüscher as the source actually consulted and identifies the analytic hypotheses as those of the note. **No correction is required**, provided "condensate" retains the stated flavour and sign convention.

4. **Correct with amendment — Farsi–Latrémolière.**

   The update correctly identifies the unchanged six hypotheses of Theorem 3.4 and the defective estimate printed as equation (3.11). The corrected relative-bound estimate is

   $$\|(1-p)\xi\| \leq \frac{C\varepsilon}{\delta}(\|\xi\| + \|D_\varepsilon \xi\|).$$

Under the separate anticommutation hypothesis, it becomes

$$\|(1-p)\xi\| \leq \frac{\varepsilon}{\delta}\|D_\varepsilon \xi\|.$$

Counterexample R1 satisfies the hypotheses. Let

$$A = B = \mathbb{C}, \qquad H = \ell^2(\mathbb{N}_0), \qquad D_v e_n = n e_n, \qquad D_h e_n = -n^2 e_n.$$

On their natural diagonal domains,

$$D = D_h + D_v, \qquad D_\varepsilon e_n = (n/\varepsilon - n^2)e_n.$$

The operators are self-adjoint. For every fixed $\varepsilon > 0$,

$$|n/\varepsilon - n^2| \longrightarrow \infty,$$

so D_ε has compact resolvent. Its domain is the quadratic weighted domain of D_h; cancellations at finitely many indices do not change this conclusion. In particular, this verifies compact resolvent for **every** $\varepsilon \in (0, 1)$.

The scalar representation is faithful. Its state space is a singleton and its commutator seminorm vanishes precisely on the real scalars, so (A, H, D) and each (A, H, D_ε) are metric spectral triples. Also

$$\mathrm{Sp}(D) = \{0, -2, -6, -12, ...\},$$

so zero is isolated even under the theorem's printed wording involving D, rather than the intended D_v.

The numbered hypotheses hold as follows:

- **(1), (2):** all commutators vanish; any positive M works.
- **(3):** D_v commutes with $B = \mathbb{C}$.
- **(4):** $p = |e_0\rangle\langle e_0|$ commutes with B and D_h.
- **(5):** the proposed limit is the metric spectral triple $(\mathbb{C}, \mathbb{C}e_0, 0)$.
- **(6):** $E = \mathrm{id}_{\mathbb{C}}$ is positive and fixes B; all required inequalities have both sides zero.

Let $a > 0$ and

$$\varepsilon_n = (n + a/n)^{-1}.$$

Then $\varepsilon_n \to 0$ and $D_{\varepsilon_n} e_n = a e_n$. Corollary 3.5 would place a in the limiting spectrum $\{0\}$, a contradiction. The estimate (3.11) likewise gives

$$1 = \|(1-p)e_n\| \leq 4\sqrt{2\varepsilon_n}(1+a) \longrightarrow 0.$$

This proves the Counterexample analytically.

We compared the main statements in Sections 3–6 with v3. We found no further mathematical change in those statements beyond the reported correction $L_A \to L_B$ in Definition 2.13. That definition still needs real and imaginary parts of the inner product. An apparent pair of negative signs in the extracted Theorem 6.2(3) is an extraction artifact: the journal HTML has an invisible `mphantom` minus, retaining the limits $+\mu_j, -\mu_j$.

Corrections:

- In `fl_R1`, replace "needs anticommutation … or a relative bound" by: > The printed hypotheses are insufficient. Anticommutation, or the stated uniform relative bound, supplies a sufficient additional hypothesis.
- Identify the numbering accurately: **Theorem 3.4 in v3 and the journal; Theorem 2.5 in v1**.

- Replace the blanket "applications … are unaffected" by "the anticommuting applications avoid this cancellation defect." In particular, the product statements in Theorem 4.6 and Corollary 4.7 also require a **nonzero vertical kernel** for their displayed target to be a spectral triple. That omission is present in both versions; it is not a new journal change.
- Limit "is v3 after copy-editing" to the statements and equations actually compared.

5. **Correct — tower gauge coefficients.**

   Let $g_1 = \sqrt{5/3}\, g_Y$. For one Standard Model generation,

$$\sum_{\text{Weyl}} T_1 = \frac{3}{5}\left[6\left(\frac{1}{6}\right)^2 + 3\left(\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 2\left(\frac{1}{2}\right)^2 + 1\right] = 2.$$

   The corresponding $SU(2)$ and $SU(3)$ sums are also 2. Three generations therefore give 6 for each group.

   The four-dimensional Weyl contribution is $(2/3)6 = 4$. At each nonzero orbifold level, each chiral zero-mode fermion has a massive Dirac partner, giving $(4/3)6 = 8$.

   One complex Higgs doublet contributes

$$(1/10,\ 1/6,\ 0).$$

   The nonzero gauge level contributes

$$-\frac{11}{3}C_2 + \frac{1}{6}C_2 = -\frac{7}{2}C_2.$$

   Here the real adjoint A_5 contribution belongs to the gauge-fixed massive-vector system. It accounts for the eaten component; it must not be counted again as an additional physical scalar.

   Thus

$$\tilde{b} = (0, -7, -21/2) + (8, 8, 8) + (1/10, 1/6, 0) = \boxed{(81/10,\ 7/6,\ -5/2)}.$$

   The zero-mode calculation gives

$$b = (0, -22/3, -11) + (4, 4, 4) + (1/10, 1/6, 0) = \boxed{(41/10,\ -19/6,\ -7)}.$$

   The source's $S^1/\mathbb{Z}_2$ field expansions have precisely one complex Higgs doublet per positive level. Item 9's derivation agrees with them. **No correction is required within this field content.**

6. **Correct — top Yukawa equation.**

   In the source's equation labelled `eq:gen` and the following display, retaining only the top Yukawa gives a tower self-coupling coefficient

$$\frac{3}{2} + 2(3) = \frac{15}{2}.$$

   The gauge coefficients are

$$-12g_3^2 - \frac{21}{8}g_2^2 - \frac{9}{8}g_1^2,$$

   exactly as written in the note.

   The Standard Model contribution is also correct:

$$\frac{9}{2}y_t^2 - \frac{17}{20}g_1^2 - \frac{9}{4}g_2^2 - 8g_3^2.$$

   For example, the gauge coefficients follow from the left- and right-handed quadratic charges:

$$3\frac{3}{5}\left[(1/6)^2 + (2/3)^2\right] = \frac{17}{20}, \quad 3(3/4) = 9/4, \quad 3(4/3 + 4/3) = 8.$$

The Yukawa coefficient is $3/2 + 3 = 9/2$.

No equation correction is required. The provenance can now record this source comparison and independent check instead of leaving the Standard Model part supported only by memory.

7. **Correct with amendment — numerical table and ranges.**

The independent **Python 3/SciPy** calculation used analytic gauge running and integrated the linear equation for $w = 1/y_t^2$, splitting the integration at every step threshold. Let $S = \mu/M_{\mathrm{KK}}$. Above the first threshold,

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(M_Z) - \frac{b_i \log(\mu/M_Z) + \tilde{b}_i F(S)}{2\pi},$$

where

$$F_{\mathrm{step}}(S) = n \log S - \log(n!), \quad n = \lfloor S \rfloor, \qquad F_{\mathrm{cont}}(S) = S - 1 - \log S.$$

Root finding determined the crossings and the self-consistent running masses.

The integration passed analytic constant-coefficient controls, a deliberately corrupted-sign control, a tolerance comparison, and a comparison with direct integration of y_t. Tightening tolerances changed the reported outputs by less than 3×10^{-10}.

In the following table, S_{ij} denotes the $g_i = g_j$ crossing; the inverse couplings are evaluated at S_{23}. The ratio columns explicitly give the order $23, 12$.

Count	M_{KK}, TeV	S_{12}, S_{23}	$\alpha_1^{-1}, \alpha_2^{-1}$ at S_{23}	y_t/g_2 at S_{23}, S_{12}
Step	0.235	24.8101, 34.2107	15.4359, 26.0038	0.196469, 0.326215
Step	1	23.3248, 32.7204	16.4118, 26.9837	0.237497, 0.384323
Step	10	20.9742, 30.3537	17.9604, 28.5387	0.302775, 0.471369
Step	100	18.6209, 27.9945	19.5049, 30.0897	0.366415, 0.550320
$S-1$	0.235	26.4934, 36.0556	15.4724, 26.0405	0.200248, 0.332162
$S-1$	1	24.9784, 34.5436	16.4495, 27.0215	0.241186, 0.389913
$S-1$	10	22.5700, 32.1396	18.0001, 28.5787	0.306277, 0.476373
$S-1$	100	20.1622, 29.7359	19.5469, 30.1318	0.369697, 0.554772

The mass pairs below use boundary ratios $\sqrt{2}$, $2/\sqrt{3}$, in that order.

Count	M_{KK}, TeV	m_t from S_{23}, GeV	m_t from S_{12}, GeV
Step	0.235	231.399, 230.197	230.509, 227.195
Step	1	224.766, 223.249	223.731, 219.738
Step	10	217.558, 215.471	216.297, 211.166
Step	100	212.679, 209.955	211.207, 204.910
$S-1$	0.235	230.285, 229.067	229.384, 226.029
$S-1$	1	223.994, 222.457	222.947, 218.907
$S-1$	10	217.081, 214.970	215.807, 210.624
$S-1$	100	212.358, 209.607	210.872, 204.520

These reproduce the stated approximate ranges:

- Through 10 TeV: approximately 21–34 tower spacings for steps and 23–36 for $S-1$.
- At the 23 crossing: approximately 26–29 and 15–18 for the two inverse fine-structure couplings.
- Through 100 TeV: approximately 0.20–0.37 and 0.33–0.56 for the two Yukawa ratios.
- The stated mass ranges 205–231 GeV and 211–231 GeV are rounded numerical ranges.
- At the tabulated scales and either crossing, changing the boundary ratio from 1 to 5 changes the mass by at most 25.465 GeV for steps and 25.708 GeV for $S-1$.

- A boundary ratio 0.5 gives 155.534–216.217 GeV for steps and 154.909–214.918 GeV for $S-1$.

Without the tower, the crossings are approximately 1.0003×10^{13} and 1.0030×10^{17} GeV.

Corrections:

- Replace "inverse g_1, g_2" everywhere by $\alpha_1^{-1}, \alpha_2^{-1}$**, where** $\alpha_i = g_i^2/(4\pi)$.
- Label the Yukawa-ratio column's crossing order explicitly.
- Replace its small numerical discrepancies by the values above. The stated boundary $y_t(m_t) = 0.936$, together with $v = 246.22$ GeV, fixes $m_t \simeq 162.965$ GeV.
- Identify the κ range as the **step-threshold** result: approximately 23.7–132.9 through 1 TeV, reaching 1232.8 at 10 TeV. With $S-1$, these become approximately 25.3–140.3, reaching 1305.3.
- Describe "at most 26 GeV" as a numerical scan result. The computed samples do not constitute an analytic bound over every intermediate M_{KK}.

8. **Correct with amendment — infrared attraction and the scan boundary.**

"Infrared quasi-fixed point" is reasonable terminology if it means the limiting infrared trajectory obtained as the ultraviolet Yukawa boundary value grows. The coefficients depend on S, so a constant fixed point of an autonomous system is not being exhibited.

Let $t < T$, and write

$$w' = \frac{C(t)w - A(t)}{8\pi^2}, \qquad w = y_t^{-2},$$

with $A = 9/2 + 15N/2$ and C the positive gauge combination. Then

$$w(t) = e^{-\int_t^T C/(8\pi^2)} w(T) + \int_t^T e^{-\int_t^s C/(8\pi^2)} \frac{A(s)}{8\pi^2}\, ds.$$

The first term measures dependence on the ultraviolet boundary. As $y_t(T) \to \infty$, it vanishes, leaving a limiting trajectory. This proves the relevant attraction mechanism; it does not prove complete independence from finite boundary values.

The conclusion about $2M_W$ is supported only for the displayed universal bulk content, one-loop equations, threshold prescription, crossing scales and boundary ratios. It directs attention to running and matching in the actual parent. Comparing a running top mass with an on-shell W mass also does not supply a common-scale matching test.

The lower endpoint 235 **GeV is illustrative, not derived by Lemmas 22, 23 or 25.** With the level spacing used in the calculation,

$$M_{\mathrm{KK}} = \frac{\pi}{l}.$$

The strict-wall Yang–Mills family of Lemma 22 gives $M_{\mathrm{KK}} = 2M_W$, approximately 161 GeV. In the flat interior family of Lemma 23,

$$\sin^2(M_W l) = \frac{\sin^2(2\theta)}{2},$$

so $M_W l \le \pi/4$, hence

$$\boxed{M_{\mathrm{KK}} \ge 4M_W \simeq 321 \text{ GeV}.}$$

Lemma 25's leading spectral-action sum rule supplies no universal numerical lower bound on M_{KK}.

Replacement wording: > The scan uses an illustrative lower endpoint of 235 GeV. A scan tied to the flat interior Yang–Mills family must instead impose $M_{\mathrm{KK}} \ge 4M_W$. The downward flow approaches an infrared quasi-fixed trajectory. The resulting masses test the stated universal-tower matching conditions; the corresponding calculation for the actual parent remains open.

9. **Correct with amendment — new checker records.**

The source passages support the new provenance records:

- `l10_generates_m4`: Example 16(c) in the 15–16 report checks the Clifford generators and generation of $M_4(\mathbb{C})$.
- `l05_truncation`: the 08–13–14 report checks Theorem A, with its stated amendments.
- `fl_R1`: item 49 of the paper-note report gives the scalar Counterexample and explicitly describes the proposed repairs as sufficient.
- `l18_sum_rule`: items 13–15 verify the heat-cutoff **Taylor** identity with its trace normalization.

A **Python 3** comparison of the facts found **213 existing `checked_by` facts preserved, none removed, and 13 added**. Thus moving the old facts preserved their set; the complete diff also contains genuine additions.

The five corpus descriptions are accurate: §§3.2, 6.2, 6.9 and 6.13 report the stated quadrature or finite-matrix checks without identifying their language. Recording `corpus, not_recorded(...)` preserves that provenance. It does not supply an independent referee.

Corrections to the associated statements:

- `l05_truncation` should say that the two-state **compressed complex data/operator system** have mass $\pi/(2l)$ and distance $2l/\pi$. The pulled-back states are smeared measures; the full diagonal algebra with inherited J fails first order.
- `fl_R1` needs the replacement in item 4.
- Replace `l18_sum_rule` by: > For the heat cutoff and the specified mass-compatible doublet decomposition, $8\kappa M^2_{W,\mathrm{Taylor}} = \sum_i N_i m_i^2 \rho(x_i)$, where $\rho(x) = 2[1-(1+x)e^{-x}]/x^2$, $\kappa = (P + \sum_i d_i w(x_i))/6$, $N_i = 2d_i$, $P + \sum_i d_i = 6$, and $w(x) = 3\int_0^1 u^2 e^{-xu}\,du$. The CCM relation follows to leading order when every $x_i \to 0$.

10. **Correct — the mixing bound, with the missing proof supplied.**

Let

$$M = FXG, \quad F = \mathrm{diag}(f_1, f_2, f_3), \quad G = \mathrm{diag}(g_1, g_2, g_3),$$

with positive entries in increasing order. Let D_1, D_2 be the largest absolute entry and largest absolute 2×2 minor of X, and let

$$\Delta_1 = |X_{33}|, \qquad \Delta_2 = |\det X_{\{2,3\},\{2,3\}}|.$$

Assume X invertible and $\Delta_1\Delta_2 > 0$. Put

$$a = \frac{\sqrt{3}D_1}{\Delta_1}, \qquad b = \frac{\sqrt{3}D_2}{\Delta_2}, \qquad K = (1+a)(1+b).$$

Let $H = MM^\dagger = U\,\mathrm{diag}(\lambda_1,\lambda_2,\lambda_3)U^\dagger$, with increasing eigenvalues. Then

$$\lambda_3 \geq H_{33} \geq f_3^2 g_3^2 \Delta_1^2, \qquad H_{ii} \leq 3D_1^2 f_i^2 g_3^2.$$

Since $H_{ii} \geq \lambda_3 |U_{i3}|^2$,

$$|U_{i3}| \leq a\,\frac{f_i}{f_3}.$$

Next,

$$H^{-1} = F^{-1}X^{-\dagger}G^{-2}X^{-1}F^{-1}.$$

Cofactors of X give

$$|(X^{-1})_{11}| = \frac{\Delta_2}{|\det X|}, \qquad |(X^{-1})_{ki}| \leq \frac{D_2}{|\det X|}.$$

Therefore

$$\lambda_{\max}(H^{-1}) \geq \frac{\Delta_2^2}{f_1^2 g_1^2 |\det X|^2}, \qquad (H^{-1})_{ii} \leq \frac{3D_2^2}{f_i^2 g_1^2 |\det X|^2}.$$

Applying the same diagonal estimate to its largest eigenvector gives

$$|U_{i1}| \le b\,\frac{f_1}{f_i}.$$

The omitted unitary-cofactor step is explicit. From $U^{-1} = U^\dagger = \mathrm{adj}(U)/\det U$ and $|\det U| = 1$,

$$|U_{12}| = |U_{21}U_{33} - U_{23}U_{31}|, \qquad |U_{32}| = |U_{11}U_{23} - U_{13}U_{21}|.$$

Consequently,

$$|U_{12}| \le b\frac{f_1}{f_2} + ab\frac{f_2}{f_3}\frac{f_1}{f_3} \le b(1+a)\frac{f_1}{f_2},$$
$$|U_{32}| \le a\frac{f_2}{f_3} + ab\frac{f_1}{f_3}\frac{f_1}{f_2} \le a(1+b)\frac{f_2}{f_3}.$$

Together with the four preceding off-diagonal bounds and $|U_{ii}| \le 1$, this proves

$$|U_{ij}| \le K\min(f_i/f_j, f_j/f_i).$$

Apply this separately to U_u, U_d. For every k and $i < j$,

$$\min(f_k/f_i, f_i/f_k)\,\min(f_k/f_j, f_j/f_k) \le f_i/f_j.$$

This follows directly from the ordering, or from the triangle inequality for $\log f_i$. Thus

$$|V_{ij}| \le \sum_{k=1}^{3} |(U_u)_{ki}|\,|(U_d)_{kj}| \le \boxed{3K_uK_d\,f_i/f_j}.$$

Identical sectors permit the same ordered singular basis and give $V = I$; hence no positive universal lower bound follows. This review supplies an **independent analytic check** of `l12_mixing_bound`. The proof does not validate the separately reported sample medians.

11. **Correct — Proposition 14, with a complete proof.**

We take the heat cutoff, three generations with right-handed neutrinos, no Majorana masses, and one common positive Dirac mass m for every species. Let

$$x = m^2/\Lambda^2, \qquad y = q^2/\Lambda^2, \qquad H(y) = \int_0^1 e^{-yu(1-u)}\,du, \qquad g(y) = (2+y)H(y) - 2.$$

Every mass eigenvalue is $+m$ or $-m$. The exact quadratic kernel for a generator entry is therefore proportional to

$$e^{-x}\left[g(y) + \frac{(m_i - m_j)^2}{\Lambda^2}H(y)\right]|T_{ij}|^2.$$

For a charged generator, equal weights connect equal and opposite mass signs. The average squared difference is $2m^2$, giving

$$C_W(y) \propto e^{-x}[g(y) + 2xH(y)].$$

In the neutral particle trace, let $v = (1,-1)^T$. The charge and mass-commutator traces give

$$Q = \mathrm{diag}(6,10), \qquad C_{\mathrm{neutral}}(y) \propto e^{-x}[Qg(y) + 12xH(y)vv^T].$$

The determinant factors as

$$\det Q\; g(y)\,[g(y) + 12x\,v^TQ^{-1}v\,H(y)]\,.$$

Since $v^T Q^{-1} v = 1/6 + 1/10 = 4/15$, the massive neutral factor has coefficient $16x/5$. Charge conjugation multiplies the common normalization and does not change the roots.

Both massive pole equations thus have the form

$$(2 + \beta - z)H(-z) = 2, \qquad \beta_W = 2x, \quad \beta_Z = \frac{16x}{5}.$$

Let $h(z) = H(-z)$ and

$$F(z) = z - 2 + \frac{2}{h(z)}.$$

For $z \geq 0$, $h(z) \geq 1$ and

$$0 < h'(z) = \int_0^1 u(1-u)e^{zu(1-u)}\, du \leq h(z)/4.$$

Hence

$$F'(z) = 1 - \frac{2h'(z)}{h(z)^2} \geq 1 - \frac{1}{2h(z)} \geq \frac{1}{2}.$$

Also $F(0) = 0$ and $F(z) \to \infty$. Therefore each $\beta > 0$ has a unique positive solution. At that solution $h(z) > 1$, so

$$z = \beta + 2 - \frac{2}{h(z)} \quad \Longrightarrow \quad \beta < z < \beta + 2.$$

This proves

$$\boxed{2x < z_W < 2x + 2, \qquad \frac{16x}{5} < z_Z < \frac{16x}{5} + 2.}$$

For small z,

$$h(z) = 1 + \frac{z}{6} + O(z^2), \qquad F(z) = \frac{2}{3}z + O(z^2).$$

Thus $z_W \sim 3x$, $z_Z \sim 24x/5$, and their ratio tends to $5/8$. For $x \to \infty$, the bounds give $z_W = 2x + O(1)$, $z_Z = 16x/5 + O(1)$, again giving $5/8$. Both pole masses diverge at fixed Λ.

At fixed Euclidean momentum, every kernel entry is e^{-x} times a function growing at most linearly in x, and therefore tends to zero. Finally,

$$H'(0) = -1/6, \qquad g'(0) = 2/3,$$

so the charged Taylor kinetic coefficient is proportional to

$$e^{-x}\left(\frac{2}{3} - \frac{x}{3}\right) = \frac{2}{3}e^{-x}(1 - x/2),$$

which is negative for $x > 2$.

A supplementary **Python 3/SciPy** quadrature and root calculation gave

$$x = 3: \quad z_W = 7.4996477083, \quad z_Z = 11.3746146417, \quad z_W/z_Z = 0.6593320253.$$

This supports the recorded "about 0.66." The Proposition establishes pole locations and the Taylor sign in this family; it does not establish a healthy Lorentzian limiting theory. This review supplies the independent analytic checker requested for `l18_fully_massive`.

12. **Error — the ledger still contains statements rejected or qualified by referees.**

The relocation of checker facts is sound, but `show/1` still presents several overbroad or incorrect statements as proved. The following replacements are required in ledger.pl. The narrowed statements can retain `proved`; the stronger interpretations cannot inherit that status merely from a checker name.

Row	Replacement or exact amendment
l05_truncation, l09_lightpair	Specify the compressed complex two-point data. Add that the full diagonal algebra with inherited real structure fails first order; do not imply that all real spectral-triple axioms survive.
l07_real	"Under pointwise reality, the real structures have the displayed diagonal form. Only $\varepsilon' = +1$ has standard KO-0 signs; the other sign uses the extended convention."
l07_reflection	"For the bare interval, the reflection sign equals ε'. This calculation does not select a physical odd factorization of a fixed product triple."
l08_pairing	Add that Q is a self-adjoint involution commuting with Γ, A, A°, D, anticommuting with J, and that the admissible U commutes with Q. Restrict the neutrino obstruction to the specified particle bimodules.
l09_limits	Keep the fixed-μ wall limit, bulk threshold and tuned collapse Theorems. Replace the parenthetical $10 \to 11$, $10 \to 9$ identifications by "proposed interpretations; compatibility with the KO factorizations is not established."
l12_towerless	"In a fixed spectral window whose boundary avoids the limiting spectrum, retain precisely the eigenvalues of D_F lying in that window. The full Yukawa block is retained when $E > \|D_F\|$. The exterior gap is in absolute value." Specify an algebra and representation before asserting convergence of spectral triples.
fl_R1	Use the sufficient-repair and version wording in item 4.
fl_R2	Prefix the displayed uncapped $l/2$ lower bound by "for sufficiently small l, including $l < \sqrt{2}$ in this normalization"; otherwise retain the cap $\min(\sqrt{2}/2, l/2)$.
l15_converse	State that **there exist** suitable light projections and identifying unitaries with light-block error tending to zero and exterior gap tending to infinity.
l15_grading	"The signs and Dirac relations descend exactly for the compressed operator D_B. Relative to an independently chosen D_F, the Dirac-relation defects are at most 2Δ; control of J requires the stated extension to antilinear functionals."
l17_condensate	Replace "has no limit beyond the zero-mode term" by "With a pairing-compatible $O(m)$ subtraction, $c_V = n_0/(Vm) + O(m)$; it tends to zero when $n_0 = 0$, and has the displayed pole when $n_0 > 0$."
l17_no_kinetic	Replace "gives a non-self-adjoint operator" by "generally gives a non-self-adjoint reduced operator." Constant reductions are an immediate exception.

Row	Replacement or exact amendment
`l18_taylor`	Restrict the coefficient-one large-mass asymptotic and the bound Λ^2/e to the heat cutoff. For the positive Laplace class, the Taylor asymptotic is $M_W^2 \sim \dfrac{\int s^{-2} d\nu(s)}{f(0)} \dfrac{\Lambda^4}{m^2}$.
`l18_joint`	State necessity of $m_t/\Lambda \to 0$ for the **top-only heat-cutoff Taylor family**. Other finite data and mass definitions require their own argument.
`l18_several`	Add "heat cutoff, finitely many doublets, each with one massive and one massless member"; the stated $P = 0$ asymptotics additionally require all massive members to have the same m.
`l18_fully_massive`	Add "heat cutoff, common $m > 0$, all Dirac species including neutrinos"; identify the wrong-sign coefficient as the **charged** Taylor coefficient. Record the independent proof in item 11.
`l18_sum_rule`	Use the Taylor mass, weight normalization and simultaneous small-mass limit specified in item 9.
`l18_radial`, `l18_radial_general`	Explicitly require $V_* \neq 0$, together with the stated cutoff hypotheses. The zero ray has identically zero derivative.
`l20_rank_one`	Add $\kappa \neq 0$ and the stated nonzero decomposable $K = vw^*$. At $\kappa = 0$, the block has rank zero and the admissible algebra has dimension 40.
`l20_no_selection`	In the twistor application, require the image of the glued subset in S^4 to have empty interior. Empty interior upstairs alone does not supply independent base test functions.
`l22_no_radial`	Add **moving spectral window**. A fixed compression can change singular values at second order.
`l22_family`	Specify the physical real gauge representation of C_F, its kernel and any unimodularity condition; $u(C_F)$ itself has an extra abelian direction before that restriction.
`l23_aligned`	Say "$Q(a) = E_0(a)$ for constant ambient multipliers; the image is $E_0(P)$. It equals P^{ev} when $\Gamma_0 P \Gamma_0 = P$." This avoids reinstating the rejected algebra-image identification.
`l23_wz`	Qualify the count of nine massless modes by $0 < \theta < \pi/2$; the endpoint symmetry is larger.
`l25_mass`, `l25_sum_rule`	Identify the masses as **leading small-mass Taylor quotients for the specified constant transported profile**, using the unshifted unfolded-circle heat trace. Prefix the displayed identities in `l25_sum_rule` by "to leading order." A lowest physical gauge branch requires the additional admissibility, positivity and isolation hypotheses.

The `l18_taylor` issue is a concrete numerical normalization error, not just missing terminology. For example, $f(z) = e^{-z/10}$ belongs to the positive Laplace class and rescales the effective squared cutoff by ten; the universal bound Λ^2/e cannot survive unchanged.

Conversely, the revised statements about the two sheet projections in `l16_two_sheets`, the added grading in `l16_twistor`, the local/global distinction in `l14_image`, and the conditional top-dominance statement in `l11_top` now respect the corresponding referee corrections.

13. **Correct — Contradictions 8 and 10.**

Canonical §14.3 reads **Proved** for the stated Einstein–Hilbert ansatz and embedding. It explicitly separates that result from the additional postulate giving the $p = 5$ resonance.

In §3, φ now consistently denotes the hypercharge phase, including $\lambda^2 = e^{-i\varphi}$. It collides with no other notation there.

The two staircase parameter choices are separately recorded in `research_checks_knowledge.md`: $b/a = 0.02$ and 0.01 belong to different illustrative figures. Their general dimensional interpretation does not require the figures to use identical radii.

Thus **resolved** is accurate for both entries in `ledger.pl` and `DUPLICATES.md`. **No further correction is required.**

14. **Correct — `rs_E2` as a missing record.**

The `not_reconstructed` status and `open_items(missing_record(rs_E2))` correctly record missing provenance without deciding whether the reported flag value is right or wrong.

We searched the four files in `transcripts/` and repository text, including hidden and ignored text files, for the flag discussion and variants of 1/4. The occurrences found repeat the report of the missing derivation or the separate $SU(3)$ example. **No original flag derivation was located in the repository searched.** This finding makes no claim about unavailable conversations.

The example in §3.3 is correct. Let

$$T_8 = \frac{\operatorname{diag}(1, 1, -2)}{2\sqrt{3}}, \qquad Y = T_8/\sqrt{3}.$$

Then

$$\operatorname{tr} T_3^2 = \frac{1}{2}, \qquad \operatorname{tr} Y^2 = \frac{1}{6}.$$

Equal simple-group normalization gives $g_Y Y = g_2 T_8$, hence $g_Y = \sqrt{3} g_2$, and

$$\cos^2 \theta_W = \frac{g_2^2}{g_2^2 + g_Y^2} = \frac{1}{4}.$$

This proves the stated embedding example. It supplies no reconstruction of the missing flag calculation. **No correction is required.**

15. **Correct with amendment — tests and queries.**

The SWI-Prolog run passed **13 of 14 tests**. The failing test is `self_checks_are_corpus_only`, with **two** nonnumerical, noncorpus causes:

```
l18_fully_massive,   steer_tower_coefficients.
```

Therefore settling item 11 alone does not make the current test pass. Items 5 and 11 now supply independent analytic checks for both rows; those checks still need to be recorded.

The new rules correctly find missing checkers, include numerical claims in `self_checked_only/1`, and expose missing records through `open_items/1`. Their limits should be explicit:

- `rows_follow_the_referees` checks the presence of a Terra checker, Terra authorship, or an exception. It never examines the statement or the report's verdict. Item 12 demonstrates why passing it does not establish mathematical agreement.

- `contradictions_match_duplicates` checks that each ledger number has a matching state at the end of a Markdown row. It does not check the resolution's substance, uniqueness, or the converse correspondence.
- `nogo_models/2` counts distinct recorded agent identifiers, combining authors and checkers. Without a model-identity mapping and an independent-check condition, that is weaker than consensus among independent models.
- `unchecked/1` is not restricted to corpus claims. Likewise, `self_checked_only/1` currently returns more than corpus claims and numerical orientations.

Corrections: rename `rows_follow_the_referees` to something such as `refereed_claims_have_review_bookkeeping`, or add explicit claim-to-report verdict records. In README, remove the restrictive parentheticals "corpus claims only" and "corpus claims and numerical orientations." Describe `nogo_below_consensus/2` as counting identifiers unless genuine model identities and independence are represented.

The outstanding amendments and errors are:

- Distinguish Witten's cylinder from the mapping-torus formulation.
- State the Farsi–Latrémolière repairs as sufficient, correct the version numbering, and qualify the application and comparison claims.
- Correct the running table's coupling notation, crossing order, small ratio discrepancies and threshold-dependent κ ranges.
- Describe infrared attraction as a quasi-fixed trajectory and identify 235 GeV as illustrative; use $M_{\mathrm{KK}} \geq 4M_W$ when invoking Lemma 23's flat interior.
- Apply the ledger replacements in items 9 and 12, especially the spectral-window restriction, omitted hypotheses, Taylor-versus-pole distinction and cutoff-dependent normalization.
- Record the independent analytic checks supplied here for the tower coefficients, mixing bound and fully massive Proposition.
- Make the test and query descriptions match what their predicates actually inspect.

Overall: 8 Correct; 6 Correct with amendment; 0 Gap; 1 Error.

Part R15 (Terra referee, lemma 28)

Referee report on lemma 28: Proposition 28.13 and the corrected numerics

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_28.md. Reproduced as delivered, with links made relative. The corrections are applied in lemma 28 and the script. The three items marked Error concern statements in the file: the identification of the boundary channels with a singular vector, a mislabelled table column, and the scope of the Correction and Outcome paragraphs. The matrix of Proposition 28.13 and its closed form are confirmed.

Count: 3 errors; 3 correct with amendment; 0 correct without amendment. The count refers to the six numbered items.

1. Slice operator — Correct with amendment.

With Hermitian Clifford generators $c_j c_k + c_k c_j = 2\delta_{jk}$, the spin connection is

$$\nabla_i = E_i - \frac{1}{4}\Gamma_{ijk} c_j c_k, \qquad \Gamma_{ijk} = \langle \nabla_{E_i} E_j, E_k \rangle.$$

Indeed, the commutator of the connection matrix with c_j gives $\Gamma_{ijk} c_k$. The Koszul expression in the script has the correct order of indices. The $\mathfrak{h}$-part of a bracket is omitted in these reductive-frame coefficients; the H-action remains necessary in the equivariance condition.

For $s_v(g) = F\lambda(g)^{-1}v$, differentiation along $g\exp(te_i)$ gives $-FV_i$, with the appropriate scale divisor. The pull-back connection contributes $+E_iF/\text{scale}_i$ in the vertical directions. Its sign in the script is consistent with the spin term.

The displayed radial form is correct for the script's $h = 1$ metric after absorbing the half-density. For the general metric in Theorem 28.9, let

$$u = h^2\rho\,\psi.$$

Before this conjugation the radial derivative is

$$iZ\left(\partial_x + 2\frac{h'}{h} + \frac{\rho'}{\rho}\right).$$

After it, there is no extra h'/h or ρ'/ρ term, and the precise operator is

$$T = iZ\partial_x + \frac{A_0}{h(x)} + \frac{A_1}{\rho(x)} + \frac{\rho(x)}{h(x)^2}A_2 + i\phi(x)\tau.$$

A collar modification of the connection adds $-s(x)A_E/\rho(x)$. Thus the notation $D_6(\rho(x))$ in Result 28.5 needs the $h = 1$ qualification. The file should also distinguish its Hermitian Clifford convention from the skew-Clifford $c(dr)$ used in Theorem 28.9.

2. Exact tests — Correct with amendment.

The stated anticommutation and Casimir identities have the right form:

$$\{A_0, A_1\} = 0, \qquad D^2_{S^4,\tau_{\text{tw}}} = \text{Cas}_G(V) + \frac{3}{2} - \text{Cas}_K(\tau_{\text{tw}}).$$

Here τ_{tw} denotes the relevant K-twist; it should be distinguished from the Higgs matrix τ. Python reproduced the printed A_0^2 spectra and anticommutator norms in all four representations tested by the script. The zero in the A_0^2 spectrum on the **20** by itself is not a core zero mode of the full normal problem.

For the Fubini–Study metric with base S^4 of radius 1, the fibre radius is $\rho = 1$. The homogeneous strict nearly Kähler metric has $\rho = 1/\sqrt{2}$. More generally these radii are h and $h/\sqrt{2}$. The squashing conventions and the two special metrics are described in Appendix A.1, especially (A.1).

At $\rho = 1$, $j(j+4)$, $j = 0, 1, \ldots$, is the correct D^2 eigenvalue on $(0,0)$-forms with values in $\mathcal{O}(1)$. It is not the spectrum of the *entire* $\mathcal{O}(3)$-twisted spinor bundle: the **20** also contains $D^2 = 10$. The last sentence of the Correction paragraph needs that restriction. In addition, the script prints the Casimir and Fubini–Study spectra; it does not assert their expected numerical values.

The tests do not separately determine all three coefficients. For example, as an algebraic test of the listed identities, replacing

$$A_1 \mapsto tA_1, \qquad A_2 \mapsto A_2 + (1-t)A_1$$

preserves $\{A_0, A_1\} = 0$ and the tested line-connection operator at $\rho = 1$. The curvature construction supplies an additional test of A_2. With $a, b = 0, \ldots, 3$ and $\alpha = 4, 5$, Koszul's formula gives

$$A_2 = -\frac{i}{8}\sum_{a,b,\alpha} C^{\alpha}_{ab}c_ac_bc_{\alpha}.$$

The nonzero C_{ab}^{α} for $a < b$ are $C_{02}^{5} = -1$, $C_{03}^{4} = -1$, $C_{12}^{4} = 1$, and $C_{13}^{5} = -1$. An in-memory Python calculation compared this contraction on the full eight-dimensional spinor space with the script's ρ-coefficient. The Frobenius residual was 4.8×10^{-16}; reversing the coefficient gave residual 2.83. This check shares the script's Lie-bracket input, while using a separate curvature contraction. It is a numerical check of the displayed algebraic derivation.

3. Proposition 28.13 — Error.

The finite wall matrix, its kernel, and its trace calculation are correct for the channel convention used in that calculation. The error is its identification of those channels with the right singular vector for the boundary condition stated in Theorem 28.9.

For the stated $\hat{\Gamma} = \mathrm{diag}(1, -1)$ and $\hat{\Gamma} Z = -1$, the boundary domains are

$$Zu = -u \quad \text{for } T, \qquad Zv = +v \quad \text{for } T^{*}.$$

Let $T_{\eta} = iZ\partial_x + D_6 + i\eta\phi\tau$, with $\phi > 0$, and write a wall profile as $e^{q(x-\ell)}$, $q > 0$. The leading equations give

$$q = -\eta\phi Z\tau \quad \text{for } T_{\eta}, \qquad q = +\eta\phi Z\tau \quad \text{for } T_{\eta}^{*}.$$

Consequently, with the **positive mass and boundary condition of Theorem 28.9**, the right wall channels have $Z = -1, \tau > 0$, and their fibre trace is $+w$ in the script's $\hat{t}$ convention. The $Z = +1, \tau < 0$ channels of Proposition 28.13 give $-w$; they describe the adjoint wall problem for the reversed mass, or a problem with the opposite wall condition. Python reproduced $+2/3$ and $-2/3$, respectively, at $\rho = 1$. Changing the mass sign alone cannot make a right T-domain vector satisfy $Z = +1$.

For the matrix calculation, let $E_{s,t;j}$ be the rank-one map from input basis vector j to spinor basis vector s tensored with twist basis vector t, using the zero-based bases of the script. Choose

$$(f_0, f_1, f_2) = (E_{2,2;2}, E_{2,3;3}, E_{4,2;3}),$$
$$(g_0, g_1) = \left(E_{3,2;3}, \frac{E_{0,2;0} + E_{6,2;1}}{\sqrt{2}} \right).$$

Direct Clifford contraction, with the stable-profile overlaps, gives

$$\begin{pmatrix} 1/\rho & -3/(2\rho) & -(\rho + 1/\rho) \\ -i\sqrt{2} & 0 & -i\sqrt{2} \end{pmatrix}.$$

Reordering and changing unit phases gives the displayed $A(\rho)$. The entry $3/(2\rho)$, for example, contains the $\sqrt{3}/\rho$ vertical connection matrix element and the $\sqrt{3}/2$ overlap. Python/SymPy reproduced this matrix using exact symbolic arithmetic from the bracket and Clifford data.

Its kernel is as stated. The $-3/2$ channel and one $-1/2$ channel have input support $(1)_{-1}$, while the remaining channel has support $(1)_{+1}$; their output spaces are orthogonal. This yields

$$w(\rho) = \frac{2(\rho^2 + 2)^2}{9 + 2(\rho^2 + 2)^2},$$

including $w(0) = 8/17$, $w(1) = 2/3$, and $w(\infty) = 1$. For horizontal radius h, replace ρ^2 in this formula by ρ^2/h^2.

The line-preserving connection replaces the first entry of the first row by zero. Its remaining block has the stated sum and product of squared singular values, is invertible for $\rho > 0$, and has squared singular values $3, 6$ at $\rho = 1$. The Fubini–Study cohomology statement in the proposition has the stated scope.

For fixed positive radius, the overlap matrix is the correct **leading formal wall operator**. Since its kernel on the **4** is one-dimensional and separated from the other two singular values, the next correction is $O(M^{-1})$ in the fixed-block expansion and does not change its limiting kernel. A uniform global reduction still needs proof, as the file acknowledges.

The sentence calling 1.863 and 2.789 "the next values on the 4" needs to say **wall values**. In a Python radial calculation at $\phi_0 = 40$, $\ell = 3$, the levels were approximately 1.865 at the wall, 2.000 at the core, and 2.789 at the wall.

4. Result 28.5 — Error.

The Python rerun reproduced the main tables. The origin exponents are

$$\pm 1, \quad \pm 2 \text{ with multiplicity three}, \quad \pm 3.$$

For $\phi_0 = 3, 4, 5$, the measured decay rates in the first table agree with the corresponding constant-coefficient values $0.8557, 1.1514, 1.4762$ to the stated precision. At fixed ρ, the slow eigenvalues have the reported expansion

$$\kappa = \frac{\phi}{2} - \frac{2}{\rho} + O(\phi^{-1}).$$

For example, Python gave 30.0663 at $(\phi, \rho) = (64, 1)$ and 31.0625 at $(64, 2)$. The exponent comparison is numerical evidence for the displayed profiles; it is not a general tunnelling theorem.

The table has a numerical mesh floor, but its size depends on parameters and the mesh. At $(\phi_0, \ell) = (6, 4)$, Python gave lowest singular values $0.00548, 0.00401, 0.00355$ for $N = 2500, 5000, 10000$. These values do not establish a zero continuum value. Also, the script uses three-point quadrature for non-polynomial coefficients. Rayleigh quotients would give rigorous upper bounds with the exact quadratic form; the quadrature results are numerical approximations to those bounds.

The substantive table error is the heading **"sup of the density ratio."** The script computes

$$\frac{\max_x |\operatorname{tr}(B(x)\widehat{t})|}{\max_x \operatorname{tr} B(x)},$$

not the supremum of the pointwise ratio. At $\ell = 3$, Python gave:

ϕ_0	Ratio of maxima reported	Pointwise maximum of $\lvert\operatorname{tr}(B\widehat{t})\rvert/\operatorname{tr} B$
4	0.00648	0.849
5	0.00197	0.856
6	0.000473	0.858

The pointwise maximum occurs at the outer boundary, where the core density is small. The integrated density values -0.005707, -0.001836, and -0.000455 reproduce the table. Their ratios to λ^2 are empirical observations for these parameters, not a proved $O(\lambda^2)$ asymptotic. They support the hypothesis used by Proposition 28.16 for the numerical modes; numerical nonzero values alone do not prove the exact finite-mass algebra. At $\phi_0 = 16, 40$, the reported integrated values are around 10^{-11}, so their signs in these calculations do not establish nonvanishing.

For the other representations, the zero multiplicities on $\mathbf{1}, \mathbf{5}, \mathbf{10}, \mathbf{14}, \mathbf{35}$ follow from the central character. Python reproduced the reported wall/core identifications on $\mathbf{16}, \mathbf{20}, \mathbf{40}$, including core values close to $\sqrt{5}, \sqrt{8}, \sqrt{12}$. For **64**, an additional in-memory Python construction gave a small-radius wall value approaching $\sqrt{15}$. A scan of **401 radii** in $[0.1, 20]$ found no zero on any of the four representations; this supports a sampled statement and does not prove nonvanishing over the whole interval.

Two profile qualifications matter. The reported $\rho(x) = \tanh x$ metric is not exactly a product near the finite wall. The displayed tanh collar switch is nonzero, though tiny, at $x = 0$; an actual smooth extension with the pull-back connection near the core needs a cutoff that vanishes on a neighbourhood of zero. The rounded collar values therefore support the limiting model and do not establish exact finite-mass equality.

5. Correction, Outcome, and Conjecture — Error.

In the Correction paragraph, the old sign error and the withdrawal of the affected numerical claims are supported. The statement that **every number** was wrong is too broad: the previous Result 28.5 already listed the same origin exponents, and some displayed values agree after rounding. The Astra report says that its referee did not reproduce the old boundary-value numbers and that its proofs did not use them; this review did not repeat those proofs. The final sentence also needs the $(0, 0)$-form restriction described in item 2.

For the four Outcome items:

1. The compression-map mechanism has the stated scope for the **prescribed** windows of Theorem 28.21.
2. The four core states are established for the normal model and its induced core operator. The displayed boundary weight is a finite wall-model result; convergence of an actual spectral mode remains conditional on the missing analysis. The boundary-sign and h-scale amendments above also apply.
3. The isolated window remains a Conjecture. Besides coercivity, the normal reductions, quasimode errors, and matching must be controlled.
4. The asserted exact finite-mass algebra is stronger than the numerical evidence. The table supports quaternionic-symmetry breaking at the resolved sample points, conditional on the continuum nonvanishing required by Proposition 28.16. It does not establish the exact algebra at every mass reported.

Conjecture 28.6 is compatible with Counterexample 28.17 when it asserts convergence of **compression maps**: convergence of those maps does not imply convergence of their generated algebras. The conjecture needs the boundary-sign and horizontal-scale conventions above. Its exponential statement in ℓ also needs a specified family of metrics and mass profiles with a uniform positive decay rate; Theorem 28.9 permits more general profiles.

6. Concentration literature — Correct with amendment.

Let $\mathcal{A} = i\Phi$ and let $\sigma(\xi)$ be the symbol of the self-adjoint Dirac operator. Since Φ acts on the twist factor, it commutes with the Clifford symbol, and

$$\mathcal{A}^*\sigma(\xi) = \sigma(\xi)^*\mathcal{A}.$$

Thus this is a concentrating pair in the sense of Maridakis, arXiv:1510.07004v1, Definition 2.1 and Lemma 2.2. The commutator $[D, \Phi]$ is order zero, and the doubled self-adjoint operator satisfies the corresponding anticommutation condition. On a closed manifold, the concentration principle applies to suitable low modes near the singular set of Φ.

The stronger spectral-separation theorem of Maridakis, arXiv:2110.11654v2, §2, Assumption 3 and the Spectral Separation Theorem has an additional hypothesis that fails for this spin-3/2 mass. At the core, its required direction-independent leading endomorphism would have to represent $(n \cdot J)^2$ for every normal unit vector n. These matrices depend on n: in a $J_3 = 3/2$ state, the expectations of J_3^2 and J_1^2 are $9/4$ and $3/4$. An anisotropic extension of that theorem, or a separate core analysis, is therefore needed. Prokhorenkov–Richardson, arXiv:math/0307251v3, §§2–3 supplies the related symbol condition and treats its stated localization setting.

Adding the wall requires boundary estimates and the stable boundary model. The closed-manifold energy argument acquires a boundary term: on a half-line,

$$\|(\partial_t + M)f\|^2 = \|f'\|^2 + M^2\|f\|^2 - M|f(0)|^2,$$

and e^{-Mt} is a boundary-localized zero mode although the mass has no zero there. A proof for this problem must combine the core analysis with that boundary layer, control the $O(M^{-1})$ reduction errors, and establish coercivity on the complement uniformly across representations.

For the source order, the arXiv LaTeX-source endpoints were requested first, but the available reader could not open their compressed responses and direct network retrieval failed. We then read the cited sections of the three arXiv HTML versions identified above. We did **not** read the LaTeX sources or PDFs of those three papers. The separate Fubini–Study-radius citation above was checked in its PDF appendix.

Part R16 (Terra referee, lemma 29)

Referee report on lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_29.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. Item 10: correct, the definition of A_eff as a chiral map made K(rho) impossible as written, while the displayed operator was the intended one; corrected. Sign of the fibre part: correct, -2/3 from the kernel vector (-2, 1, 1), with channel weights 4, 1, 1 and fibre values -1, -1, +1; the squared values (45 +- 3 sqrt 33)/8 were recomputed by hand. Items 2, 4 to 9, 11, 12: applied. Item 9 also supplies the density B_w, checked at w = 1 and against the trace. Item 13: the identifiers were checked on arXiv. A referee that ran no computation and reports no counterexample has found no error in case (b); that is weaker than a proof by a second author, and the corrected case (a) has not been read again.

The principal error is in case (a): the displayed definition of A_{eff} makes Condition $K(1)$ impossible as written. There is also a sign error in the reported fibre part: at $\rho = 1$ it is $-2/3$. Both have identifiable corrections; neither supplies a counterexample to the proposed eight-state count.

The case-(b) coercivity argument survives the main cutoff and projection checks, subject to the domain, gauge and constant clarifications below. No Python calculation was run. The table was compared with the stated limiting values, but its computations were not independently reproduced.

1. **CONFIRMED — Proposition 29.1.** For the connection metric with $h = 1$ and $\rho = r$, the normal fibres are flat and totally geodesic. In a horizontal/vertical orthonormal frame, the difference between the Levi-Civita connection and the split connection is the O'Neill tensor. Its curvature component is linear in y, and its spin contraction gives, in the convention of Proposition 29.1,

$$R = -\frac{i}{8} \sum_{a,b,\alpha} \langle [\tilde{e}_a, \tilde{e}_b], f_\alpha \rangle c(e_a) c(e_b) c(f_\alpha).$$

   Thus $T_M = T_0 + R$ exactly on the tube. There is no additional first-order term for this metric and the pull-back connection. In the moving radial frame this is rA_2; the apparent polar mean-curvature term belongs to the ordinary vertical Dirac operator.

   The bracket normalization used in the radial system reduces A_2, on the nontrivial horizontal chirality, to a unitary conjugate of

$$\tfrac{1}{2}(s_1 \otimes \sigma_1 + s_2 \otimes \sigma_2),$$

   with s_i Pauli matrices. Its norm is 1; it vanishes on the other horizontal chirality. Hence $\|R\| \le |y|$.

   The common principal connection causes no extra derivative of $\widehat{Q}$. In the equivariant-map realization, the horizontal fields differentiate the G argument and $\widehat{Q}$ acts by the same operator on the fibre at every g. Diagonal K-invariance of the actions on y, the normal spinors and E makes this construction descend. Individual horizontal derivatives have a frame index that transforms under K; the contracted D_H is equivariant.

With $c(f_\alpha) = \gamma_5 \sigma_\alpha$, the relevant algebra is

$$D_H \widehat{Q} = \widehat{Q}^* D_H, \qquad D_H \widehat{Q}^* = \widehat{Q} D_H.$$

It gives both displayed sums of squares and commutation of D_H^2 with the two normal squares. Finally,

$$\Sigma_7|_K = ((2,1) \oplus (1,2)) \otimes (1,2) = (2,2) \oplus (1,3) \oplus (1,1),$$

so the stated spin representation is correct.

2. **AMENDMENT — Proposition 29.2.** The spectrum and kernel conclusions follow, with a distinction between two projections. The projection P_W onto the normal K-representation $W_0 = (1,3) \oplus (1,1)$, extended over *both* base chiralities, commutes with D_H. The projection onto the actual fibre kernel of $\widehat{Q}$ also selects the chirality $\gamma = -\epsilon$:

$$P_0 = P_{\gamma=-\epsilon} \otimes P_W.$$

This projection need not commute with D_H. Rather, $D_H P_0 = P_1 D_H$, where P_1 projects onto the fibre kernel of $\widehat{Q}^*$, and P_0 commutes with D_H^2. The proof should use P_0 and the latter commutation for the normal spectral gap.

For the stated normalization, the twisted round-S^4 formula is

$$D_H^2 = C_2^G(V) + \tfrac{3}{2} - C_2^K(j), \qquad C_2^K(j) = 2j(j+1).$$

The twist has $j = 0, 1$. Only representations with odd central character occur. The least G-Casimir is $5/2$ on the 4; the next is $15/2$ on the 16. The $j = 1$ zero on the 4 occurs in the base chirality of type $(1,2)$, exactly the chirality selected by (H4). The adjoint normal kernel selects the opposite base chirality, where that zero is absent. The $j = 0$ part on the 4 gives the next squared value 4.

On noncompact N, the calculation should first be stated on compactly supported smooth sections and extended through the closed graph forms. The doubled model is essentially self-adjoint on that core. The normal estimate $(y \cdot J)^2 \geq |y|^2/4$ gives compact normal resolvent; the compact base and the commuting squared decomposition give the required discrete model spectrum. No boundary condition is imposed at infinity.

3. **CONFIRMED — Lemma 29.3.** Coupling spin 1/2 to spin 3/2 gives

$$\sigma \cdot J = F^2 - \tfrac{3}{4} - \tfrac{15}{4},$$

so its eigenvalues are $3/2$ and $-5/2$. For each y, rotational invariance gives eigenvalues $\pm|y|/2, \pm 3|y|/2$ for $y \cdot J$; hence the squared lower bound is pointwise valid. Consequently

$$\frac{M^2}{4}\||y|u\|^2 \leq \|T_0 u\|^2 + \frac{5M}{2}\|u\|^2,$$

which gives exactly the stated constants after taking square roots. The adjoint estimate follows in the same way. The positive μ_1, rather than its numerical value, is all that the subsequent proof needs.

4. **AMENDMENT — Lemma 29.4 and IMS.** Define C_Φ explicitly as a bound on the *Clifford contraction*

$$\left\| \sum_i c(e_i) \nabla_{e_i} (f\, n \cdot J) \right\|_{\mathrm{op}}.$$

This removes the ambiguity in writing only $|\nabla(f\, n \cdot J)|$. With that convention the bulk estimate follows from expansion of $(D + i\Phi)^*(D + i\Phi)$.

The IMS step is valid. The derivatives of all three cutoffs are supported in a compact annulus away from both ends; the bulk estimate controls their error by $CM^{-2}(\|T_M u\|^2 + \|u\|^2)$. Dropping the nonnegative $\|T_M \chi_b u\|^2$ is allowed because the same bulk estimate separately controls $\|\chi_b u\|$. Scalar radial cutoffs preserve both wall domains.

The comparison with the model is also correct: extension by zero puts $\chi_c u$ in the model domain, and

$$u \perp \chi_c \ker T_0 \quad \Longrightarrow \quad \Pi(\chi_c u) = 0.$$

The analogous wall orthogonality is exact.

5. **AMENDMENT — Lemma 29.5.** Parthasarathy's formula gives the displayed result. In the standard $SU(4)$ normalization, the central weight of the $\mathcal{O}(k)$ twist is represented by

$$\tfrac{k}{4}(1,1,1,-3),$$

whose squared length contributes $3k^2/8$ to the Casimir. The scalar curvature of the stated Fubini–Study metric is 12, giving $R/8 = 3/2$. Thus

$$D_k^2 = C_2(L) + \tfrac{3}{2} - \tfrac{3}{8}k^2.$$

The invariant unitary connection on each line is the Chern connection: the difference of two invariant line connections is an invariant one-form, and the H-isotropy representation has no trivial summand.

The eligibility argument works. Each of the two $SU(4)$ representations of dimension 20 below the relevant Casimir threshold restricts to $Sp(2)$ as $4 \oplus 16$. The $\mathcal{O}(3)$ slice contains the $Sp(2)$ representation 4 only once, and that occurrence is already the cohomological kernel; neither 20 can therefore occur. In the charge convention of the earlier report, $\mathcal{O}(3)$ has charge -3. The slice weights give its 4 once in $Z = +1$ and zero times in $Z = -1$, proving the kernel chirality without a numerical mode-position argument.

The correction is to "$\kappa_w = \sqrt{3}$, for all V": $\sqrt{3}$ is the **global uniform gap**, attained on the 4. It is not the lowest nonzero value on every isotypic component; the stated lowest wall value on the 16, for example, is $\sqrt{5}$.

6. **AMENDMENT — Lemma 29.6.** In radial parallel gauge the sum of squares and Robin analysis are correct. With $q = -\tau$, the free Robin zero channel is

$$(Z = -1, q > 0) \quad \text{for } T_M, \qquad (Z = +1, q > 0) \quad \text{for } T_M^*.$$

The first-order form $d - m|q|$ has the normalized zero profile $\sqrt{2m|q|}\, e^{m|q|(x-l)}$, and its orthogonal complement has spectrum starting at m^2q^2. The boundary condition kills the cross-term boundary pairing.

The collar gauge should be stated. For example, add $itq\,dr$, with a cutoff supported in the collar, to a line-preserving connection. This is invariant, unitary and satisfies (H3). In the original radial trivialization it changes the Dirac operator by $-tZq$; on the $q = 3/2$, $Z = +1$ line,

$$T^*_{M,\mathrm{new}}(pg) = -\tfrac{3}{2}t\,pg.$$

Radial parallel transport removes this term and restores the displayed profile. Thus the geometric estimate persists, while the displayed formula and trial state require that gauge or the corresponding parallel-transport factor.

The charge calculation in item 5 establishes the relative core/wall sign. The index discussion in Remark 29.12 gives a consistency check; it does not by itself identify the local chirality. If the $\mathcal{O}(3)$ zero mode instead lay in $Z = -1$ with $q > 0$, pg would be a wall zero mode for T_M, contradicting the positive lower bound in part 1 of Lemma 29.6.

7. **AMENDMENT — Theorem 29.7 and Lemma 28.14.** Lemma 28.14 is correct. If the trial range has dimension eight and $\|\widehat{D}U_M\| < \delta$, its spectral projection onto $(-\delta, \delta)$ has dimension at least eight. If that projection had dimension greater than eight, it would contain a nonzero vector perpendicular to the trial range, contradicting the complementary bound. The spectral theorem then gives the stated angle estimate.

With the projection and collar-gauge clarifications above, the case-(b) IMS argument supplies the hypotheses for each fixed $0 < \delta < \sqrt{3}$. The core trial defect is $O(M^{-1/2})$: Lemma 29.3 gives the required first moment even without invoking Gaussian decay. The wall defect is exponentially small in radial parallel gauge. No counterexample to the case-(b) count was found.

8. **AMENDMENT — Proposition 29.8.** The tunnelling identity is correct for a singular pair with real nonnegative λ. For a real scalar weight preserving the domain, the quadratic-form identity

$$\|T_M(\xi u)\|^2 = \lambda^2\|\xi u\|^2 + \|c(d\xi)u\|^2$$

is also correct.

The Agmon proof needs two explicit choices. First, choose the IMS cutoffs so that the region where the exponential weight varies is covered by the bulk estimate, including the transition near the collar. Second, the final cutoff loses the integral across its support. Apply the weight with a smaller parameter, for example $\eta' = \eta/2$, and choose the support width s so that

$$(1-\eta')\left(I - \int_{r_*}^{r_*+s} f\,dr\right) \geq (1-\eta)I, \qquad I = \int_{\varepsilon}^{l-a} f\,dr.$$

This supplies the exponent stated in the proposition. The absorption uses $\lambda \to 0$, supplied by the trial states.

The factor $1/2$ is justified uniformly across channels by $(n\cdot J)^2 \geq 1/4$; it does not require every channel to have eigenvalue $1/2$. For the subsequent $l \to \infty$ statement, require $M \geq M_\eta$ and uniform bounds on the endpoint geometry, cutoffs, connection and positive bulk mass. Translation invariance in the bulk alone should not stand in for those uniform hypotheses.

9. **AMENDMENT — Theorem 29.9.** The compression estimate follows from Theorem 28.15 after choosing equivariant unitary identifications of the varying window spaces with the model copies of 4, and after the applicable window theorem is established.

With the coordinate convention of lemma 28,

$$Q_B(a) = \int_{S^4} a(x)(1 - x\cdot\Gamma)\,d\mu(x), \qquad Q_B(x_a) = -\Gamma_a/5.$$

In case (b), $B(z) = 4p_z$, with $p_z \leq (1 - x\cdot\Gamma)/2$. This base-coordinate convention accounts for the sign when comparing with the coordinates in lemma 10.

More explicitly, if p_z is the coherent line selected by the $q = 3/2$ channel and p_{jz} its quaternionic partner, the case-(a) limiting density is

$$B_w(z) = 2((1+w)p_z + (1-w)p_{jz}).$$

It gives $Q_Y(x_a) = -\Gamma_a/5$ and the stated factor w on the y_T components. Since $w > 0$, the 5- and 10-components are present; together with independent endpoint restrictions they give the claimed real algebra. This is a limit-map statement.

10. **ERROR — Proposition 29.10 and Theorem 29.11 as written.** At the definition of A_{eff}, P contains only $Z = -1$ states and Y only $Z = +1$ states. Therefore

$$F = \Pi_Y(A_d + A_o)P : P \longrightarrow Y$$

is a chiral operator. It is not the self-adjoint operator subsequently described. If regarded as an operator on the ambient space, it kills all of Y; if regarded as $P \to Y$, its kernel lies in $Z = -1$. In either reading, the stated Condition $K(\rho)$, requiring its kernel to be exactly a 4 in $Z = +1$, cannot hold.

The correction is to define the full operator on $P \oplus Y$ by

$$\mathcal{A}_{\text{eff}} = \begin{pmatrix} 0 & F^* \\ F & 0 \end{pmatrix},$$

and impose $K(\rho)$ on this operator. The matrix $A(\rho)$ of Proposition 28.13 represents $F^* : Y \to P$ on the 4; F is its adjoint.

With this correction, the proposed block argument has the required structure. The profile overlap is $\sqrt{3}/2$. The stable-to-stable block has norm $\sqrt{3}/\rho$, so its contribution has norm $3/(2\rho)$; the full perturbation instead has $\|A_o\| = 2/\rho$, as required in the elimination estimate. The complementary block has inverse norm at most $(m/2 - \|A_o\|)^{-1}$ for large m, and its Schur correction is bounded by

$$\frac{\|A_o\|^2}{m/2 - \|A_o\|}.$$

On $V \neq 4$, the corrected effective operator at $\rho = 1$ is bounded below by $\sqrt{5} - 3/2$. On the 4, the nonzero squared singular values of $A(1)$ are

$$\frac{45 \pm 3\sqrt{33}}{8},$$

and its one-dimensional multiplicity kernel lies in $Z = +1$. This proves the intended corrected $K(1)$. The quoted 1.863 is the value on the 4; the displayed argument gives the smaller uniform lower bound $\sqrt{5} - 3/2$ off that representation. The cut-off trial state must use the exact corrected half-cylinder kernel; its finite 4-block equation gives exponential decay.

There is a separate sign error in the reported fibre part. At $\rho = 1$, the kernel vector is $(-2, 1, 1)$, and the signed fibre trace is

$$\frac{-4 - 1 + 1}{6} = -\frac{2}{3}.$$

In general it is $-w(\rho)$. The positive $w(\rho)$ is its magnitude.

11. **AMENDMENT — Compatibility of (H1)–(H4).** The hypotheses are compatible. A smooth positive ρ can equal r near zero and be constant on a disjoint collar; take $h = 1$. Likewise, $f = r$ near zero and $f = f_0 > 0$ on the collar gives $f(r) = r\,b(r^2)$ for a smooth positive b. The metric and mass extend smoothly across the zero section. Since X deformation retracts to S^4, its relevant $H^2(-; \mathbb{Z}_2)$ vanishes and the oriented tangent bundle is spin. Invariant Hermitian connections can be interpolated with a radial cutoff.

    The chirality convention should be explicit. If $\tau = n \cdot J$, then (H4) gives $q = -\gamma_b \tau$; the displayed convention $q = -\tau$ uses $\gamma_b = +1$. The collar expression also uses the radial gauge described in item 6.

12. **ERROR — Outcome and scope.** Outcome item 2 and Theorem 29.11 give the signed fibre part as $+2/3$; it is $-2/3$. The assertion that $K(1)$ holds also requires the corrected effective-operator definition in item 10. The nearest supported statement is the intended case-(a) result after that correction and the accompanying Schur-complement argument.

    Remark 29.12 should say that the opposite-sign **end-model modes** lie in the indicated kernels. At finite mass the full operator can have a nonzero tunnelling singular value, so those localized states need not lie in the exact kernels of T_M or T_M^*.

    The scope paragraph otherwise identifies the restrictions on metrics, the connector and finite-mass algebras. "Uniformity over representations" follows here from the stated full-space estimates and ellipticity of the corrected fixed boundary operator; it should not be attributed to the fixed-manifold setting alone. Away from $\rho = 1$, finiteness of the possible kernel representations follows from ellipticity, while an explicit representation cutoff for a practical finite check remains unwritten.

    The displayed **Python** table is consistent, to its shown precision, with the stated limits: in particular 2.2361, 2.8284 and 3.4641 agree with $\sqrt{5}, \sqrt{8}, \sqrt{12}$, and the larger $M = 32$ wall discrepancies are of the

stated mesh scale. The table was not rerun. The quoted $\mu_1 \approx 1.02$ is a Python approximation; the two supplied tools do not themselves reproduce that oscillator calculation.

13. **AMENDMENT — Prior work.** General forms of two key identities already appear in the literature:

    - Maridakis, arXiv:1510.07004v1, Lemma 2.2 gives the concentrating-pair square and its zeroth-order cross term, which includes the bulk identity used here. Prokhorenkov–Richardson, arXiv:math/0307251v3, §§2–3 treats the earlier Witten-deformation localization framework for the stated isolated-singularity setting.
    - Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18 gives a horizontal/vertical Dirac factorization with an explicit curvature contraction. Its stated proper-submersion and dimension hypotheses differ from this seven-dimensional noncompact model; it is a close structural ancestor of Proposition 29.1.
    - Bismut–Lebeau, *Complex immersions and Quillen metrics*, §§8–9, DOI 10.1007/BF02699352, develops rescaled normal operators and block estimates for complex immersions. Kottke, arXiv:0909.5661v2, Theorem 11 gives the families Callias index in terms of the positive eigenbundle; it supplies an index, rather than this boundary spectral window.
    - Maridakis, arXiv:2110.11654v2, §2 proves a spectral-separation theorem under additional assumptions. Its Assumption 3 requires a direction-independent leading endomorphism; the spin-3/2 mass here has direction-dependent $(n \cdot J)^2$. Prokhorenkov–Richardson, arXiv:math/0610160v2, §3 treats equivariant Witten localization and normal models.

    The targeted search covered Callias operators on bundles, submersion factorization, Bismut–Lebeau localization, equivariant Witten deformation, boundary chirality and a 2026 recency query. The cited sections were opened. Taubes's 1996 paper, DOI 10.4310/jdg/1214459411, was identified through Maridakis's reference chain, but its original full text was inaccessible in this search. Citation chains and the 298-page Bismut–Lebeau paper were not exhaustively reviewed. The search found prior general identities; it does not establish whether the specific eight-state theorem has appeared elsewhere.

The corrections in order of severity are:

1. Replace the definition of A_{eff} and $K(\rho)$ by the full chiral block, then restate the case-(a) proof.
2. Correct the signed case-(a) fibre part to $-w(\rho)$, in particular $-2/3$ at $\rho = 1$.
3. Specify radial parallel gauge, the model domains and the distinction between P_W and the one-chirality fibre-kernel projection.
4. Complete the Agmon cutoff argument with an η-margin and state the uniform hypotheses for varying l.

Part R17 (Terra referee, lemma 30 and the joint-limit note)

Referee report on lemma 30, the note and the corrected case (a) of lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_30_note.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. All items applied, with these checks. Item 7: correct; the bulk interpolation allowed by (H3) need not keep j_E parallel. The fix was checked by hand: j_E (n . J) j_E^-1 = -n . J gives j_E P_q j_E^-1 = P_{-q}, so j_E is parallel for the line-preserving connection and for real affine interpolation with the pull-back connection, which is the interpolation in tools/lemma28_radial.py, and C then commutes with i Phi_M as well as with D. The unconditional bound eta_t = O(M^{-1/2}) is adopted, with its proof written out. Item 4: correct; lemma 24 has V = C^4 with gradings of type (2, 2). Item 9, first point: the counterexample to the implication from the hierarchy alone is correct. Item 5: the sign in Proposition 29.10 and the missing projection estimate are applied; the Weyl constant has the rank 64 = 8 x 4 x 2. Item 1: the gap is an equality, since the next level is (pi - theta_max)/l.

Item 13: the identifiers not already in the record were checked on arXiv. The source-level attributions to Parthasarathy, Bismut-Lebeau and Kirchberg stay unread, as the referee says. This was the last Codex job available in the weekly quota, so these corrections have had no further independent reading.

The main mathematical gap is in Proposition 30.7: the allowed case-(b) connections need not preserve the charge conjugation used in the proof. With an added compatibility hypothesis, the weighted argument gives the displayed exponential bound. A fresh reading of the corrected case (a) also found a remaining sign error in Proposition 29.10.

The interval and disc-bundle counting arguments otherwise support the stated joint limits, subject to the amendments below. No Python calculation was run in this review.

Part A: lemma 30

1. **AMENDMENT — Theorem 30.1.** The compression constant $2/3$, the gap bound and both counts agree with Theorems 22.1 and 22.4. Add $l_* > 0$, $E_* > 0$, and the algebra and representation assumptions needed to define $Q_\varepsilon(a)$ and $\pi(a)$. Alternatively, state the compression estimate for a bounded even operator a, using its operator norm.

   There are $r/2$ singular values counted with multiplicity, and each gives two signed arithmetic progressions. These remain r **progressions with multiplicity** when singular values coincide or vanish. In particular, a zero singular value contributes two zero modes. Thus the moving-window count is r, and the fixed-energy leading term is

   $$\frac{2rl_*E}{\pi}\varepsilon^{-\alpha}.$$

   For small ε, the displayed lower bound for the first discarded level is in fact an equality:

   $$\Delta_\varepsilon = \frac{\pi}{l_\varepsilon} - \varepsilon\|D_*\|.$$

2. **AMENDMENT — Corollary 30.2.** The two endpoint pairs are correct for the **universal fully massive family**, with both members of every Standard Model doublet assigned the same Dirac mass and no Majorana mass. In that family the two commutator quotients are $2m^2$ and $16m^2/5$; multiplying them by $(3 - d_{\text{eff}})/2$ gives $(3, 24/5)m^2$ as $d_{\text{eff}} \to 0$ and $(2, 16/5)m^2$ as $d_{\text{eff}} \to 1$.

   Lemma 25 concerns the leading **Taylor quotient** for the heat-cutoff spectral action, with an even vector generator constant in the transported frame and an enlarged admissible gauge algebra. The interval poles belong to the separately specified five-dimensional Yang–Mills problem. The coefficient comparison holds at leading order; it does not identify the full spectral-action pole problem with that Yang–Mills problem.

   The path is correct:

   $$s = l_\varepsilon E_\varepsilon = l_* E_* \varepsilon^{\beta-\alpha} \longrightarrow 0, \qquad \frac{m_\varepsilon}{E_\varepsilon} \longrightarrow 0.$$

   At fixed $\Lambda > 0$, $s \to \infty$ and $m_\varepsilon/\Lambda \to 0$. Lemma 25 states its expansion first at fixed s; a joint-limit use should add a uniform remainder argument, or state explicitly that the displayed endpoints concern the leading coefficient. Varying the action cutoff is an additional prescription, as the corollary says.

3. **AMENDMENT — Proposition 30.3.** The root and asymptotics follow from Theorem 9.4 and Lemma 9.5. For $c > 1$, the first oscillatory root satisfies

   $$\pi < k_1 l < \frac{3\pi}{2}, \qquad \frac{\Delta}{\mu} = \sqrt{1 + \left(\frac{k_1 l}{c}\right)^2} \longrightarrow 1.$$

   The two tail identities follow from $\mu = \kappa \coth s$ and $m = \kappa/\sinh s$.

The localized family needs a **new length**, since its length cannot also be the $l_*\varepsilon^{-\alpha}$ of Theorem 30.1. State

$$l_\varepsilon^{\mathrm{loc}} = \frac{(1-\alpha)\log(1/\varepsilon) + c_0}{\mu_*\varepsilon^\alpha},$$

with $\mu_* > 0$ and $0 < \alpha < 1$. The reuse of l_ε is also misleading in the note.

4. **ERROR — Corollary 30.4 for arbitrary finite data.** The cylinder in lemma 24 has $\ker A \simeq \mathbb{C}^4$, with wall gradings having two positive and two negative directions. It does not realize an arbitrary dimension r from Theorem 30.1. For example, finite data with $H_+ = H_- = \mathbb{C}^3$ and $D_* = 0$ satisfy the interval hypotheses and have $r = 6$; the stated cylinder has a four-dimensional tangential kernel.

   Restrict the corollary to the compatible $r = 4$ data, or construct and state an enlarged tangential bundle for other dimensions. The gap estimate then follows from lemma 24. The seven-dimensional count needs an additional Weyl argument: Dirichlet and Neumann bracketing of the squared product operator give the same leading term as $l_\varepsilon, r_\varepsilon \to \infty$, proportional to $l_\varepsilon r_\varepsilon^6 E^7$. Lemma 24's gap estimate alone does not prove that asymptotic.

5. **AMENDMENT — Theorem 30.5 and corrected case (a).** For a fixed shape, it is legitimate to substitute $M = M_\varepsilon \to \infty$ into Theorem 29.7 or 29.11. Proposition 29.8 supplies M_η, C_η independent of M after the shape and η are fixed. Substitution gives $\lambda_M/R_\varepsilon \le C_\eta\varepsilon$. No uniformity in a varying shape is needed for this family.

   The domain of $\widehat{D}_M$ is independent of M: the wall projector is fixed and MV is bounded, with $\|V\| \le (3/2)\sup f$. Min–max applied to the singular values gives the displayed comparison of the counting functions of $|\widehat{D}_M|$ and $|\widehat{D}_0|$. The local self-adjoint elliptic boundary problem has the usual leading Weyl law. Counting both signs and complex multiplicity,

$$C_X = \frac{64\,\omega_7\,\mathrm{Vol}_g(X)}{(2\pi)^7},$$

   since the seven-dimensional spinor bundle has complex rank 8, E rank 4, and the operator is doubled. The estimate $M_\varepsilon/R_\varepsilon \to 0$ makes the bounded perturbation negligible at the counting threshold. The physical first discarded level also tends to zero: its shape-unit value is at most $s_9(\widehat{D}_0) + cM_\varepsilon$.

   Dilation is exact under the canonical spinor identification and the L^2 unitary factor $R^{-7/2}$. With physical radial coordinate $s = Rr$, the potential is

$$\frac{M}{R}\, f(s/R)\, n\cdot J.$$

   A radial half-density normalized with respect to ds acquires the corresponding $R^{-1/2}$ factor. The compression estimate uses C^1 norms in fixed shape coordinates. Add $0 < \delta$, $E_* > 0$, and state that if $m_\varepsilon = 0$, zero has total multiplicity eight.

   Fresh check of case (a). The corrected chiral definition of A_{eff}, the profile overlap $\sqrt{3}/2$, the perturbation bound $3/2$, and the squared nonzero singular values

$$\frac{45 \pm 3\sqrt{33}}{8}$$

   are consistent. The displayed matrix at $\rho = 1$ has kernel $(-2, 1, 1)$, giving signed fibre part $(-4 - 1 + 1)/6 = -2/3$. However, Proposition 29.10 still says that the fibre part tends to $+w(\rho)$. It must say $-w(\rho)$. Theorem 29.11 has the correct sign.

   The adjoint estimate in Proposition 29.10 also needs one explicit step. Orthogonality to the *exact* half-cylinder kernel does not immediately give orthogonality to $\ker F^*$. The Schur-complement graph estimate gives $\|P_{\ker T_M^*} - P_{\ker F^*}\| = O(1/m)$; inserting this projection estimate supplies the required $O(1/m)$ loss. With that addition, no further error was found in the corrected case-(a) block argument.

6. **CONFIRMED — Proposition 30.6.** Normalized angular Haar measure and Schur's lemma give

$$Q(f) = \frac{4}{\int \operatorname{tr} B} \int\int f(r,g)\lambda(g)^{-1} B(r) \lambda(g)\, dg\, dr.$$

Quaternionic conjugation exchanges b_+ and b_-, so $\|B - P_{\mathbb{H}} B\| = |b_+ - b_-|/2$. This gives $2\eta_t$. Each compression of a real f with $\|f\|_\infty \le 1$ is a contraction; telescoping against its j-conjugate gives $2k\eta_t$ for a product of k such compressions.

7. **ERROR — Proposition 30.7 as stated for all allowed connections.** The proof assumes $CT_M = T_M C$ throughout the interior. Setting 29.0 permits any invariant unitary interpolation in the bulk; it does not require that interpolation to preserve j_E.

This is a failure of a stated premise in the proof. For an allowed connection that is pull-back on a bulk annulus, put $q = n \cdot J$ there and add the invariant skew-Hermitian connection form

$$i\, h(r)\, \nabla^0(q^2),$$

with real h supported inside the annulus. It has no radial component and leaves both end hypotheses intact. Since $j_E q^2 j_E^{-1} = q^2$, antiunitarity sends this added form to its negative. Consequently j_E is not parallel and C need not commute with the resulting twisted Dirac operator. This example shows the failure of the asserted symmetry; it does not by itself establish a counterexample to the exponential estimate.

Add $\nabla^E j_E = 0$, or another explicit hypothesis proving the same global symmetry. Under that hypothesis, the other checks work as follows.

- $C^2 = j^2 = -1$, so $F \mapsto CFj^{-1}$ is an antilinear involution on $\operatorname{Hom}_G(\mathbf{4}, -)$. On the core it preserves D_H, the full fibre term $i\widehat{Q}$, and R. With the conventions of Proposition 29.1, it anticommutes with $\widehat{Q}$ itself, because it is antilinear.
- The phase can make $\langle \chi_c F_0, u\rangle$ real, and then $\Pi(\chi_c \zeta) = 0$. The qualification by χ_c is essential.
- On $r \le l - a$, $g + \tilde{g} = G$. The cut-off $\xi\zeta$ vanishes near the wall and belongs to the required domain. The bulk first-order term has a positive η-margin; the core estimate uses the projection condition. The derivative of the wall cut-off is supported where $g = G$, so its contribution is $O(e^{-G})$. Thus the proposed argument gives $\|\xi\zeta\| = O(e^{-G})$.
- Pointwise,

$$|b_+ - b_-| \le (\|F\| + \|CF\|)\|\zeta\|.$$

Weighted Cauchy–Schwarz on $r \le l - a$, and the L^2 tail bound beyond $l - a$, both give $O(e^{-2G})$.

Therefore, **conditional on global C-preservation**, the proof gives

$$\eta_t \le C_\eta \exp\left[-(1-\eta) M \int_{2e}^{l-a} f\, dr\right].$$

The integral starts at $2e$, whereas Proposition 29.8 displays the tunnelling bound starting at e; the stated exponent is therefore not literally twice that displayed tunnelling exponent. A smaller core cut-off can be chosen if the comparison is to use the same starting radius.

Under the original general connection hypotheses, the window estimate still gives $\eta_t = O(M^{-1/2})$: choose an equivariant normalized core model fixed by the local real structure and use the L^2 distance to that model from Theorem 29.7. Finally, an exponential upper bound expressed in M does not imply $O(\lambda^2)$ without a lower bound on λ.

The Python samples cited at lemma 30, line 106 vary the length at fixed mass and use profiles outside Setting 29.0. They support a qualitative comparison and do not test the stated $M \to \infty$ theorem.

8. **AMENDMENT — Remark 30.8.** The vector-sector and connector obligations are accurately identified. The gauge formula assumes a fixed microscopic coupling, a constant internal Yang–Mills profile and the stated normalization; then $g_4 \propto R^{-7/2}$.

For the toy potential, require $a, b, p, q > 0$. Its unique minimum is

$$r_\varepsilon = \left(\frac{ap}{bq}\right)^{1/(p+q)} \varepsilon^{-2/(p+q)},$$

and $p + q > 2$ gives $0 < \alpha < 1$. The complete-manifold sentence just before the remark also needs qualification: global charge conjugation gives a real structure on the multiplicity space, but the provided argument does not establish that this space is one-dimensional for **every** M or for every completion. Exact quaternionicity of a selected core copy follows if that multiplicity is one.

Part B: the note

9. **ERROR — Several statements exceed their supplied hypotheses.** The following corrections are needed in joint_limit_note.tex.

   - **Hierarchy, lines 40–44.** The hierarchy alone does not imply that the fixed-energy count diverges. For a compact-resolvent counterexample, take signed spectrum

     $$0, \quad \pm\varepsilon^\alpha, \quad \pm 1, \quad \pm 2, \ldots$$

     and $E_\varepsilon = \varepsilon^\beta$, $\alpha < \beta < 1$. The hierarchy holds with one retained zero mode, while $N_\varepsilon(E) = 3$ for fixed $0 < E < 1$ and small ε. State divergence as a separate property proved for the specific families.
   - **Interval and cylinder, lines 50–79.** Add the representation assumptions for the compression bound, separate the localized length from the angle-family length, and restrict the unmodified cylinder to compatible four-dimensional kernel data, as in items 1, 3 and 4. State the scope of the heat-cutoff Taylor quotient from item 2.
   - **Disc-bundle hypotheses, lines 83–90.** Define $S = G \times_K (1, 2)$ and its pull-back in $E = S^3(S)$; state the round S^4 radius, smoothness and positivity of ρ and f, $f_0 > 0$, and $Sp(2)$-invariance of the connection. The displayed metric with $\rho = r$ does encode flat normal fibres near the core; the note also explicitly states constant f and the Fubini–Study radius on the collar. Case (a) in lemma 29 assumes the pull-back connection throughout, whereas the note only specifies it near the two ends. A bulk interpolation may be treatable by the bulk estimate, but that extension needs to be stated and proved.
   - **Sign and localization, lines 87–90.** The sign selects the core kernel of the **end model**. At finite M, the full T_M can have nonzero tunnelling singular value. Describe the two four-dimensional subspaces by their proved proximity to the cut-off core and wall models, rather than as exact states at the two ends.
   - **Core proof, line 99.** The gap of the first-order fibre operator is of order $\sqrt{M}$; the gap of its square is of order M. D_H exchanges the relevant base chiralities and need not preserve the actual one-chirality fibre kernel. The argument uses the normal-mode projection extended over both chiralities, or the commutation of the actual kernel projection with D_H^2.
   - **Bulk and wall, lines 101–110.** Replace the variable-f lower bound by an integral bound or a bound using the minimum of f on the support. Restrict the asserted wall gap to the relevant stable $q > 0$ channels: the full line sum also includes the $\mathcal{O}(-3)$ zero modes. The case-(a) value $-w(\rho)$ is the limiting effective-wall value, with a finite-M correction.
   - **Tunnelling and joint limit, lines 128–162.** Define the integration endpoints and the fixed cut-offs e, a, rather than writing only "over the bulk." State the equivariant identifications and the shape-coordinate C^1 norm in the compression estimate. Include the zero-mass multiplicity qualification from item 5.
   - **Defect, lines 166–175.** Add the global charge-conjugation compatibility from item 7. Replace "no component on the model kernel" by $\Pi(\chi_c \zeta) = 0$, and distinguish the two action integrals in the tunnelling and defect estimates. The assertion that only the wall breaks C also needs the added connection hypothesis.
   - **Bimodule, line 185.** Specify "32 complex-dimensional bimodule with a real structure," equivalently 64 real dimensions before taking a real fixed subspace.

- **Status, line 189.** Lemma 29 case (b) had a Terra referee report; the corrected case-(a) text had not had a second reading. Proposition 30.7 and Theorem 30.5 were new.

10. **AMENDMENT — Physics framing and named sources.** The abstract acknowledges that the disc-bundle vector poles remain open, and the body says that **4** is pseudoreal. Put both qualifications next to the headline claim: the disc-bundle result treats a **fermion** window, and no calculation identifies its tunnelling scale with M_W or M_Z. The title should describe a target electroweak limit or a fermion joint limit. The auxiliary $\hat{\Gamma}$-grading and the zero virtual $Sp(2)$-index do not establish the chiral Standard Model spectrum.

The reading of Witten's October 1983 preprint, pp. 7–8 and 43–44, is consistent at the stated level. Witten discusses vectorlike quantum numbers under the specified geometric gauge assumptions and states equivariant cobordism invariance of the character-valued index. Here the extending object is the **sum** of the four line summands; the individual $\mathcal{O}(k)$ do not all extend. The signed indices are $(4, 0, 0, -4)$. This index permits the spectral pairing and does not prove the window.

Maridakis, arXiv:2110.11654v2, §2, Assumption 3 requires a direction-independent leading endomorphism. For spin $3/2$, $(n\cdot J)^2$ depends on n as an endomorphism, even though its eigenvalues are direction-independent. The paper also works on a closed manifold, so the boundary layer requires separate analysis.

The Bryant–Salamon qualification is correct. Karigiannis–Lotay, arXiv:2002.06444v3, Eq. (3.36) and Theorem 3.6 gives a complete asymptotically conical G_2 metric on the same total bundle, with a radial horizontal scale and non-flat fibre metric outside Setting 29.0. Homothetic enlargement of the fixed shape has a flat pointed limit at a fixed interior point; a fixed boundary point gives a half-space limit.

11. **AMENDMENT — Citations and companion drafts.** The internal interval references support the exact spectrum and the qualified Taylor coefficient. The note should preserve the companion draft's explicit distinction between a leading Taylor quotient and a pole calculation. towerless_intervals.tex treats the relation between the KO labels and geometric limits as a proposal; the joint note's introduction presents that relation more definitively and should retain the proposal status.

Connes–van Suijlekom, §1 supports the operator-system discussion. Replace "generated algebras do not converge when compression maps do" at line 146 by "need not converge"; constant families already show that convergence can occur.

The Kirchberg citation is apt for the **untwisted** $R/3$ consistency check in complex dimension three. It does not by itself prove the twisted $\mathcal{O}(k)$ spectrum. Kirchberg's 1990 paper, introduction and Theorem 2, explicitly gives the odd-dimensional bound and identifies $\mathbb{CP}^3$ as a limiting case. The Parthasarathy and Bismut–Lebeau entries have matching bibliographic identities in the Annals record and Numdam record, respectively; their full original texts could not be checked directly in this review, so the precise source-level attribution remains unchecked here. The numerical normalization in lemma 29 must be justified in the argument.

Baptista, arXiv:2506.09126v2, §1 and Eq. (1.2), supports the distinction between Killing and non-Killing metric modes in its stated Kaluza–Klein setting. It does not calculate the vector sector on this disc bundle.

12. **AMENDMENT — Table I.** Every displayed $M = 32$ row matches lemma 29's Python table. The first **16** entry is the core value and the second is the wall value; on the **20**, the first is the wall value and the second is the core value. Adding those labels would make the comparison easier to assess.

The smallest computed **4** value is floor-dominated at the reported meshes. Its mesh dependence does not determine whether the continuum value vanishes. The stated 0.005 wall discrepancy is a Python mesh observation from the source, not an independently established error bound. The caption should attribute the limit values to the end models and their spectral comparison: the window theorem by itself gives the low-window count and gap, not all individual excited limits.

13. **AMENDMENT — Prior results and comparison.** Several mechanisms in the note have established precedents that an expert would expect to see distinguished from the new combination.

- Kaplan, hep-lat/9206013v1, Eqs. (0) and surrounding discussion, derives chiral defect localization. Shamir, hep-lat/9303005v1, §§1–3, treats boundary fermions and the finite fifth direction. These precede the interval localization mechanism.
- Csáki et al., hep-ph/0305237v1, §§2 and 6, derives gauge symmetry breaking by interval boundary conditions. Cacciapaglia–Csáki–Park, hep-ph/0510366v2, §5, obtains $m_t = 2M_W$ at tree level for a suitable rank-four representation. Neither source establishes the lemma-25 crossover.
- Maridakis, arXiv:1510.07004v1, Lemma 2.2, gives the concentrating-pair square identity. Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18, gives a horizontal–vertical Dirac factorization with a curvature contraction. Their settings differ from the stated seven-dimensional boundary problem.
- Kottke, arXiv:0909.5661v2, §5, gives a families Callias index theorem under asymptotically conic hypotheses. It supplies index information under those hypotheses; the finite boundary-window estimate is a separate question.
- Iso–Kitazawa–Suyama, arXiv:2111.06508v2, §§1 and 4–5, follows massive Kaluza–Klein gauge modes that become massless during symmetry restoration in flux compactifications. The mechanism and geometry differ from the present joint family.

Witten's closed-geometry index restrictions and the boundary construction have different hypotheses; no contradiction was found in the checked records. These targeted checks do not establish priority for the eight-state theorem.

Corrections in order of severity

1. Add global charge-conjugation compatibility to Proposition 30.7 and the note, or supply a different proof for the allowed case-(b) interpolations. Retain the unconditional $O(M^{-1/2})$ defect estimate until then.
2. Remove the claimed implication from the hierarchy alone to divergent fixed-energy counting; retain the separate counts proved for the named families.
3. Correct Proposition 29.10's signed fibre part to $-w(\rho)$, and add the projection estimate for the adjoint wall bound.
4. Restore the disc-bundle hypotheses in the note, restrict the cylinder data, and give the localized interval family its own length.
5. Qualify the electroweak, chirality, Taylor-mass and finite-versus-limit compression claims in the title, abstract and theorem statements.

Part R18 (assessment of the Astra audit)

Assessment of the audit of commit b2ddf98

The user supplied on 4 November 2026 an outside audit of the repository at b2ddf98, titled "conneskk: audit and a route from the massless electroweak limit to eleven dimensions", kept verbatim in 2026-11-03_astra_audit_d11.md. The file does not state its authorship; the 11D line, merged on 4 November, attributes it to Astra 6, and we follow that. This note records what was checked and what was done. Provenance: Claude Sonnet 5.5 recomputed all items, read-only, in Python and with pdftotext, and Claude Opus 5.5 rederived by hand the items marked below, against the source lemmas. The user asked for attention to detail with the reports of smaller models, so the column "by hand" says which verdicts do not rest on such a report alone.

Verdicts

Audit section	Claim	Verdict	By hand
1	Withdrawal of the alleged normalization error at boundary_condition_note.tex:53	Correct: both components have the factor	Sonnet only

Audit section	Claim	Verdict	By hand
2.1	D^2 = 2 lambda^2 on the 64-dimensional product, P_- D P_- = 0,		[P_-, D]
2.2	The massless limit must specify a path; theta -> 0 at fixed l opens no tower	Correct, and adopted as STRATEGY T14	Yes
2.3	ind D_{O(k)} = (k^3 - k)/6, values 4, 0, 0, -4, total zero	Correct	Yes; Witten 1983, pp. 43-44, gives the reason
2.4	Supplement omits lemma 17	Correct; lemmas 01, 02, 04 are also omitted, and 17 is added	Yes
2.4	docs/ last refreshed at 4ae5b43; T13 steers only to D = 9; docs/index.html has no seven-dimensional entry	Correct	Sonnet only
2.4	PDF overflow, 81 of 478 and 13 of 263 pages	Depends on an unstated tolerance: Sonnet found 113 and 15 at tolerance 0, 83 and 15 at 2 pt, 12 mobile pages above 5 pt, with maximum overhang 8.7 pt, through pdftotext -bbox	Sonnet only
3	Finite window and opening tower on the interval, equations (1) to (4)	Correct; the statement needs Delta_eps defined, and the references are Theorems 22.1, 22.4. Now Theorem 30.1	Yes
3.1	Vector poles and M_W^2 = 2 m^2, M_Z^2 = (16/5) m^2; coefficients 3 and 24/5	Correct, the latter for the fully massive family of lemma 18, Proposition 14. Moreover M^2 = ((3 - d_eff)/2)	
4	Localized family, equations (6) to (8), and the table	Correct. Now Proposition 30.3	Yes; table rows by Sonnet
5.1	Equation (9), Weyl exponent on the cylinder	Correct. Now Corollary 30.4	Yes
5.2	Dilation, conditional scaling of X	Correct; the condition is removed in Theorem 30.5 for the class of lemma 29, pending its referee	Yes
6	Bound (13) with constant 2, and the telescoping	Correct, once “defect” means	

Audit section	Claim	Verdict	By hand
6, (14)	Target: eta_t <= C (lambda/delta)^2 + errors	Stated as a target there. Proposition 30.7 proves the form exp(-2 G) for the upper bound exp(-G) on lambda; a bound in terms of lambda itself would need a lower bound on lambda, which is open	Yes
6.1	Table of eta_t at L = 2, 3, 4, 5	Rows reproduced by Sonnet (Python, same solver, so a reproduction with a new observable)	Sonnet only
7.1	Toy potential and its minimum	Correct as an elementary statement; underived, as the audit says	Yes
Appendix A	M^{111} data: c_1 = 3h + 2u, dim H^0(O(3,2)) = 30; six arXiv identifiers	Correct on the items checked	Sonnet only

No error in a displayed equation of the audit was found.

What was done

- Lemma 30 takes the propositions that are proved, with proofs redone, and adds Theorem 30.5 and Proposition 30.7.
- Item 2 of the audit's list (the shape-unit estimate for X, starting from the line-preserving collar) is lemma 29, which was already in progress when the audit arrived, and goes further: case (a) at the Fubini-Study radius as well.
- STRATEGY gains T14, the joint limit, parallel to T13, as the audit asks.
- paper/build_supplement.sh gains lemma 17, and lemmas 28 to 30 once their referees are applied.

What was not adopted

The note proposed in section 8 is being written at the request of the user, as paper/joint_limit_note.tex. The M^{111} comparison stays a comparison, and no W, Z -> 0 family links it to the finite geometry. The layout repair of the supplement is recorded and has not been done.